

SingleRAN

Multi-RAT Coordinated Energy Saving Feature Parameter Description

Issue

Draft B

Date

2026-01-30



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1 Change History

SRAN22.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes

Change Description	Parameter Change	Base Station Model
Added support for UL coordinated channel shutdown by the UBBPh. For details, see 5.2.3.3 Hardware .	None	5900 series base stations (macro base stations)
Added support for adaptive BBU deep dormancy when the working modes of the RF modules associated with the baseband processing unit include GSM. For details, see: • 7.7.3.4 Networking • 7.7.4.3 Activation Verification	None	• 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R • DBS3900 LampSite and DBS5900 LampSite (only for LN)
Enhanced maintenance and test for pRRU deep dormancy in intelligent mode. For details, see 7.6.4.2 Activation Verification .	None	DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	Base Station Model
Added support for dormancy functions in RRU single-level ring networking. For details, see: • 7.4.3.4 Networking • 7.4.4.2 Activation Verification • 7.7.3.4 Networking • 7.7.4.3 Activation Verification	None	• 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Enhanced RF module dormancy functions in cells where UMTS and LTE Spectrum Sharing Based on DC-HSDPA is enabled. For details, see: • 7.2.3.5 Cells • 7.3.3.5 Cells • 7.4.3.5 Cells	None	3900 and 5900 series base stations (macro base stations)

- Editorial changes

Removed descriptions of the impact relationships of RF module super dormancy with inter-module RF frequency hopping, inter-module power sharing, and Intelligent Shutdown of TRX Due to PSU Failure from [7.4.3.2.3 Function Impacts](#).

Revised descriptions of activation verification for RF module super dormancy in GSM. For details, see [7.4.4.2 Activation Verification](#).

Revised descriptions of license reliability in reliability protection for deep dormancy and super dormancy. For details, see [7.11.1 License Reliability](#).

Revised descriptions of the principles, configuration suggestions, and MML command examples related to the **RRU.CUSTOMDORMANCYSTDSW (LTE eNodeB, 5G gNodeB)** parameter.

Revised descriptions of the impacts of RF module deep dormancy and RF module super dormancy in UMTS. For details, see [7.2.2.2 Impacts](#) and [7.4.2.2 Impacts](#).

Revised descriptions of networking requirements of RF module super dormancy. For details, see [7.4.3.4 Networking](#).

Removed redundant descriptions of the **NRDUCellPowerSaving.SulUserNumOffset** parameter.

Added descriptions of "Neighboring Cell Unavailable", a possible reason why NR load-based RF module fast deep dormancy does not take effect. For details, see [7.3.4.2 Activation Verification](#).

Removed descriptions of the fan temperature control mode from descriptions of the possible reasons why adaptive BBU deep dormancy does not take effect in [7.7.4.3 Activation Verification](#).

Added descriptions of references for lur-g interface configuration for GUL joint carrier shutdown in areas covered by GU networks. For details, see [6.3.1.1.1 GU Cell Joint Intelligent Shutdown](#) and [6.3.4.2.2 Using MML Commands](#).

Revised descriptions of reliability protection for deep dormancy and super dormancy. For details, see [7.11 Reliability Protection for Deep Dormancy and Super Dormancy](#).

Revised descriptions of the benefits of RF module deep dormancy, RF module fast deep dormancy, and RF module super dormancy. For details, see [7.2.2.1 Benefits](#), [7.3.2.1 Benefits](#) and [7.4.2.1 Benefits](#).

Added descriptions of the RRU5527et to the hardware requirements of RF module super dormancy. For details, see [7.4.3.3 Hardware](#).

Revised descriptions of pRRU data sampling and processing in pRRU deep dormancy in intelligent mode. For details, see [7.6.1.2.1 pRRU Data Sampling and Processing](#).

Revised descriptions of MML-based configurations for pRRU deep dormancy. For details, see [7.6.4.1.2 Using MML Commands](#).

Revised descriptions of activation verification for RF module fast deep dormancy. For details, see [7.3.4.2 Activation Verification](#).

Added descriptions of impact relationships of dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios with RF module deep dormancy and RF module super dormancy. For details, see [7.2.3.2.3 Function Impacts](#) and [7.4.3.2.3 Function Impacts](#).

Revised the change description in Draft A.

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 07 (2025-12-14).

- Technical changes

Change Description	Parameter Change	Base Station Model
Added main control board dynamic energy saving. For details, see: 2.4 Differences 3 Overview 7.1.1 Differences in Dormancy Types 7.8 Main Control Board Dynamic Energy Saving	Modified parameter on the Node side: Added the MPT_DYN_ENG_SAV_FO RBID_SW option to the EQUIPMENT.EQMDORO PTFEATSW (LTE eNodeB, 5G gNodeB) parameter.	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R - DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	Base Station Model
Removed the impact relationship between UL coordinated channel shutdown and OCNS from 5.2.3.2.3 Function Impacts .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Removed the mutually exclusive relationship between the Cell Deployment Policy parameter set to POWERSAVING and adaptive BBU deep dormancy from 7.7.3.2.2 Mutually Exclusive Functions .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R - DBS3900 LampSite and DBS5900 LampSite
Added incompatibility of BTS5929R with LTE and NR coordinated symbol power saving. For details, see 4.1.3.3 Hardware .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added incompatibility of BTS5929R with LTE and NR bundled scheduling. For details, see 4.2.3.3 Hardware .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added incompatibility of BTS5929R with LNR coordinated channel shutdown. For details, see 5.1.3.3 Hardware .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added incompatibility of BTS5929R with UL coordinated channel shutdown. For details, see 5.2.3.3 Hardware .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added incompatibility of BTS5929R with LTE and NR intelligent carrier shutdown. For details, see 6.1.3.3 Hardware .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

Change Description	Parameter Change	Base Station Model
Added incompatibility of BTS5929R with joint carrier shutdown in areas covered by UL or GUL networks. For details, see 6.3.3.3 Hardware .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added incompatibility of BTS5929R with RF module deep dormancy. For details, see 7.2.3.3 Hardware .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added incompatibility of BTS5929R with RF module super dormancy. For details, see 7.4.3.3 Hardware .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added incompatibility of BTS5929R with frequency band-based energy saving of wideband modules. For details, see 7.5.3.3 Hardware .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added incompatibility of BTS5929R with adaptive BBU deep dormancy. For details, see 7.7.3.3 Hardware .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

Change Description	Parameter Change	Base Station Model
Considered the SUL UE quantity for entering or exiting LTE and NR intelligent carrier shutdown and RF module dormancy. For details, see: • 6.1.1.2 Entering the Carrier Shutdown State • 6.1.4.2.1 Data Preparation • 6.1.4.2.2 Using MML Commands • 7.2.1.1 Scheduled RF Module Deep Dormancy • 7.2.1.2 NR Load-based RF Module Deep Dormancy • 7.2.1.4 LNR Load-based RF Module Deep Dormancy • 7.2.4.1.1 Data Preparation • 7.2.4.1.2 Using MML Commands • 7.4.1.1 Scheduled RF Module Super Dormancy • 7.4.1.2 NR Load-based RF Module Super Dormancy • 7.4.1.4 LNR Load-based RF Module Super Dormancy • 7.4.4.1.1 Data Preparation • 7.4.4.1.2 Using MML Commands	Added parameters on the NR side: • NRDUCellPowerSaving.SulUserNumThld • NRDUCellPowerSaving.SulUserNumOffset	• 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	Base Station Model
Added support for NR local-cell-load-based RF module dormancy. For details, see: • 7.1 Introduction to Dormancy • 7.2.1 Principles • 7.2.1.3 NR Local-Cell-Load-based RF Module Deep Dormancy • 7.2.3.2.2 Mutually Exclusive Functions • 7.2.3.5 Cells • 7.2.4.1.1 Data Preparation • 7.2.4.1.2 Using MML Commands • 7.2.4.2 Activation Verification • 7.4.1 Principles • 7.4.1.3 NR Local-Cell-Load-based RF Module Super Dormancy • 7.4.3.2.2 Mutually Exclusive Functions • 7.4.3.5 Cells • 7.4.4.1.1 Data Preparation • 7.4.4.1.2 Using MML Commands • 7.4.4.2 Activation Verification	Modified parameters on the NR side: • Added the NR_DORMANCY_JUDGE_THLD_OPT_SW option to the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter. • Added the NR_DORMANCY_JUDGE_THLD_OPT option to the NRDUCellPowerSaving.PowerSavingType parameter.	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

Change Description	Parameter Change	Base Station Model
Added support for RF module deep dormancy and RF module super dormancy in UMTS. For details, see: • 2.3 Features in This Document • 7.2.1 Principles • 7.2.1.1 Scheduled RF Module Deep Dormancy • 7.2.1.6 UMTS Load-based RF Module Deep Dormancy • 7.2.1.7 GU Load-based RF Module Deep Dormancy • 7.2.2.2 Impacts • 7.2.3.1 Licenses • 7.2.2.1 Benefits • 7.2.3.2.3 Function Impacts • 7.2.3.3 Hardware • 7.2.3.4 Networking • 7.2.3.5 Cells • 7.2.3.6 Others • 7.2.4.1.1 Data Preparation • 7.2.4.1.2 Using MML Commands • 7.2.4.2 Activation Verification • 7.2.4.3 Network Monitoring • 7.4.1 Principles • 7.4.1.1 Scheduled RF Module Super Dormancy • 7.4.1.7 UMTS Load-based RF Module Super Dormancy	• Modified parameter on the Node side: Added the GU_DC_SW option to the EQUIPMENT.EQMDO ROPTFEATSW (LTE eNodeB, 5G gNodeB) parameter. • Added MOs on the RNC side: – UCELLDORMANCY POLICY – URNCCELLDORMANCYPARA • Added the UCELL.DormancyCellType parameter on the RNC side. • Added the ULOCCELLALGPARA.DORMANCYALGOSW parameter on the NodeB side.	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	Base Station Model
• 7.4.1.8 GU Load-based RF Module Super Dormancy • 7.4.2.1 Benefits • 7.4.2.2 Impacts • 7.4.3.1 Licenses • 7.4.3.2.2 Mutually Exclusive Functions • 7.4.3.2.3 Function Impacts • 7.4.3.4 Networking • 7.4.3.5 Cells • 7.4.3.6 Others • 7.4.4.1.1 Data Preparation • 7.4.4.1.2 Using MML Commands • 7.4.4.2 Activation Verification • 7.4.4.3 Network Monitoring • 7.7.3.4 Networking • 7.7.4.3 Activation Verification • 7.10 Dormancy Pending and Pending Cancellation • 7.11.2 Configuration Reliability • 7.11.3 Reliability in Fault Scenarios		

Change Description	Parameter Change	Base Station Model
Added support for RF module super dormancy in GSM. For details, see: • 2.3 Features in This Document • 7.4.1 Principles • 7.4.1.1 Scheduled RF Module Super Dormancy • 7.4.1.6 GSM Load-based RF Module Super Dormancy • 7.4.2.1 Benefits • 7.4.2.2 Impacts • 7.4.3.1 Licenses • 7.4.3.2.2 Mutually Exclusive Functions • 7.4.3.2.3 Function Impacts • 7.4.3.6 Others • 7.4.4.1.1 Data Preparation • 7.4.4.1.2 Using MML Commands • 7.4.4.2 Activation Verification • 7.4.4.3 Network Monitoring • 7.7.4.3 Activation Verification • 7.10 Dormancy Pending and Pending Cancellation • 7.11.2 Configuration Reliability	• Modified parameters on the Node side: – Added the GU_DC_SW option to the EQUIPMENT.EQM DOROPTFEATSW (LTE eNodeB, 5G gNodeB) parameter. – Added the value GSM to the RRU.DORMANCYS TD (5G gNodeB, LTE eNodeB) parameter. • Added the GCELLDORMANCY MO on the BSC side. • Added parameters on the BSC side: – GCELLOTHPARA.DormancyEnable – GCELLOTHPARA.C oCoverageCellId1 – GCELLOTHPARA.C oCoverageCellId2 – GCELLOTHPARA.C oCoverageCellId3 – GCELLOTHPARA.C oCoverageCellId4 – BSEXSOFTPARA.I gnoreCbcSwitch – BSEXSOFTPARA.CbcServiceHoldDuration	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

Change Description	Parameter Change	Base Station Model
Added support for adaptive BBU deep dormancy in UMTS. For details, see: • 2.3 Features in This Document • 7.7.1 Principles • 7.7.2.2 Impacts • 7.7.3.1 Licenses • 7.7.3.2.1 Prerequisite Functions • 7.7.3.2.2 Mutually Exclusive Functions • 7.7.3.3 Hardware • 7.7.3.5 Cells • 7.7.3.6 Others • 7.7.4.2.1 Data Preparation • 7.7.4.2.2 Using MML Commands • 7.7.4.3 Activation Verification • 7.7.4.4 Network Monitoring	Added the NODEBALGPARA.DORMANCYSWITCH parameter on the NodeB side.	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	Base Station Model
Added support for RF module fast deep dormancy. For details, see: • 2.3 Features in This Document • 2.4 Differences • 3 Overview • 7.1 Introduction to Dormancy • 7.2.3.2.3 Function Impacts • 7.3 RF Module Fast Deep Dormancy • 7.4.3.2.3 Function Impacts • 7.10 Dormancy Pending and Pending Cancellation • 7.12 Reliability Protection During Fast Deep Dormancy	• Modified the RRUEXTINFO.RRUDOROPTFEATSW (5G gNodeB, LTE eNodeB) parameter on the Node side. • Added the NRDUCellPowerSaving.PwrSaveServiceAssurLevel parameter on the NR side. • Modified parameters on the NR side: – Added the NR_FAST_DORMANCY_BY_LOAD_SW, LNR_FAST_DORMANCY_BY_LOAD_SW, NR_FAST_DORM_SERVICE_ASSUR_SW, PWR_SAVE_SERVICE_ASSUR_SW, and NR_FAST_DORM_LOAD_ASSUR_SW options to the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter. – Added the values NR_FAST_DORMANCY_BY_LOAD and LNR_FAST_DORMANCY_BY_LOAD to the NRDUCellPowerSaving.PowerSavingType parameter. Added support for NR and LNR load-based fast dormancy by the following parameters:	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

Change Description	Parameter Change	Base Station Model
	– NRDUCellPowerS aving.<i>UlPrbThld</i> – NRDUCellPowerS aving.<i>DIPrbThld</i> – NRDUCellPowerS aving.<i>UserNumThld</i> – NRDUCellPowerS aving.<i>PwrSavingPenaltyTime</i> – NRDUCellPowerS aving.<i>UserNumOffset</i> – NRDUCellPowerS aving.<i>DIPrbOffset</i> – NRDUCellPowerS aving.<i>NCellUlPrbOffset</i> Added support for LNR load-based fast dormancy by the following parameters and option: – NRDUCellPowerS aving.<i>EutranCarrierShutdownExitOfs</i> – NRDUCellPowerS aving.<i>EutranNrUserDIPrbThldOfs</i> – NRDUCellPowerS aving.<i>EutranNrUserUlPrbThldOfs</i> – NRDUCellPowerS aving.<i>EutranDIPrbOffset</i> – NRDUCellPowerS aving.<i>EutranNrUserUlPrbThld</i> – NRDUCellPowerS aving.<i>EutranCarrierShutdownRatio</i> – NRDUCellPowerS aving.<i>EutranNrUserDIPrbThld</i>	

Change Description	Parameter Change	Base Station Model
	– NRDUCellPowerSaving.EutranDlPrbThld – NRDUCellPowerSaving.EutranNrUserNumThld – NRDUCellPowerSaving.EutranNrUserNumThldOfs – NRDUCellPowerSaving.NCellUlPrbThld – NRCellEutraNRelation.PowerSavingOverlapInd – LNR_LOAD_DORM_PRIM_OP_RECOG_SW option of the gNB PwrSavingAlgo. PwrSavingAlgoSwitch parameter – NRCellRelation.PowerSavingCellFlag – NRDUCellMobility.CarrShutdFastEvalPrbThld – NRDUCellMobility.CarrShdnEventEvalPrbThld • Modified parameters on the LTE side: – Added the LTE_FAST_DORMANCY_BY_LOAD_SW option to the CellPwrSavingAlgo. PwrSavingAlgoSwitch parameter. – Added the value LTE_FAST_DORMANCY_BY_LOAD to the CellPwrSavingPolicy.PowerSavingType parameter.	

Change Description	Parameter Change	Base Station Model
	Added support for LTE load-based fast dormancy by the following parameters: – CellPwrSavingPolicy.TriggerJudgePeriod – CellPwrSavingPolicy.DlPrbThld – CellPwrSavingPolicy.DlPrbOffset – CellPwrSavingPolicy.UlPrbThld – CellPwrSavingPolicy.UlPrbOffset – CellPwrSavingPolicy.UEnumThld – CellPwrSavingPolicy.ForcibleTriggerUeNumThld – CellPwrSavingPolicy.ExitJudgePeriod – CellPwrSavingPolicy.PenaltyTime	
Added support for pending and pending cancellation for UMTS pRRU deep dormancy. For details, see 7.10 Dormancy Pending and Pending Cancellation.	None	DBS3900 LampSite and DBS5900 LampSite
Added wake-up algorithm optimization for RF module super dormancy. For details, see: • 7.4.1 Principles • 7.4.2.2 Impacts • 7.4.3.3 Hardware • 7.4.4.1.1 Data Preparation • 7.4.4.1.2 Using MML Commands	Modified parameter on the Node side: Added the RF_SDWAO_SW option to the EQUIPMENT.EQMDORO_PTFEATSW (LTE eNodeB, 5G gNodeB) parameter.	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

Change Description	Parameter Change	Base Station Model
Removed the requirement on the temperature control mode of the BBU fan when adaptive BBU deep dormancy is enabled from: • 7.7.3.2.2 Mutually Exclusive Functions • 7.7.3.6 Others • 7.7.4.2.2 Using MML Commands	None	• 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R • DBS3900 LampSite and DBS5900 LampSite
Added RHUB intelligent shutdown. For details, see: • 2.4 Differences • 3.1 Categories of Energy Saving Functions • 7.9 RHUB Intelligent Shutdown	Modified parameter on the Node side: Added the RHUB_SHUTDOWN_SW option to the EQUIPMENT.EQMDORO_PTFEATSW (LTE eNodeB, 5G gNodeB) parameter.	DBS5900 LampSite
Modified the function switch for LTE-NR multi-dimensional scheduling optimization. For details, see: • 6.1.3.2.3 Function Impacts • 7.2.3.2.3 Function Impacts • 7.4.3.2.3 Function Impacts • 7.6.3.2.3 Function Impacts	None	• 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R • DBS3900 LampSite and DBS5900 LampSite
Added the impact relationships of NCR cells with UL coordinated channel shutdown and GUL joint carrier shutdown. For details, see: • 5.2.3.2.3 Function Impacts • 6.3.3.2.3 Function Impacts	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

Change Description	Parameter Change	Base Station Model
Added the impact relationships of port sparseness with RF module deep dormancy and RF module super dormancy. For details, see: • 7.2.3.2.3 Function Impacts • 7.4.3.2.3 Function Impacts	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added the impact relationships of initial transmission power reduction and retransmission power reduction with RF module deep dormancy and RF module super dormancy in LTE FDD. For details, see: • 7.2.3.2.3 Function Impacts • 7.4.3.2.3 Function Impacts	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added the impact relationships of initial transmission power reduction and retransmission power reduction with RF module deep dormancy and RF module super dormancy in NR. For details, see: • 7.2.3.2.3 Function Impacts • 7.4.3.2.3 Function Impacts	None	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	Base Station Model
Added the impact relationships of adaptive BBU deep dormancy with adaptive baseband shutdown, power saving first, and baseband shutdown for energy saving. For details, see 7.7.3.2.3 Function Impacts .	None	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
Removed the mutually exclusive relationship between baseband processor shutdown and adaptive BBU deep dormancy from 7.7.3.2.2 Mutually Exclusive Functions .	None	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
Added the mutually exclusive relationships of Static Multiple Beam with LNR coordinated channel shutdown, LTE and NR bundled scheduling, and UL coordinated channel shutdown. For details, see: 4.2.3.2.2 Mutually Exclusive Functions 5.1.3.2.2 Mutually Exclusive Functions 5.2.3.2.2 Mutually Exclusive Functions	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

Change Description	Parameter Change	Base Station Model
Added the mutually exclusive relationships of dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios with LTE and NR bundled scheduling, adaptive BBU deep dormancy, LTE and NR coordinated symbol power saving, LNR coordinated channel shutdown, and LTE and NR intelligent carrier shutdown. For details, see: • 4.1.3.2.2 Mutually Exclusive Functions • 4.2.3.2.2 Mutually Exclusive Functions • 5.1.3.2.2 Mutually Exclusive Functions • 6.1.3.2.2 Mutually Exclusive Functions • 7.7.3.2.2 Mutually Exclusive Functions	None	5900 series base stations (macro base stations)
Added the mutually exclusive relationships of dual-BBU backup in dual-network redundancy networking with LTE and NR coordinated symbol power saving and LTE and NR intelligent carrier shutdown. For details, see: • 4.1.3.2.2 Mutually Exclusive Functions • 6.1.3.2.2 Mutually Exclusive Functions	None	DBS5900 LampSite

Change Description	Parameter Change	Base Station Model
Added the mutually exclusive relationships of out-of-band relay enhancement with RF module deep dormancy and RF module super dormancy. For details, see: 7.2.3.2.2 Mutually Exclusive Functions 7.4.3.2.2 Mutually Exclusive Functions	None	3900 and 5900 series base stations (macro base stations)
Added support for RF module super dormancy by the AAU5785t. For details, see 7.4.3.3 Hardware .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

- Editorial changes

Added descriptions of the impact relationship between UL coordinated channel shutdown and OCNS. For details, see [5.2.3.2.3 Function Impacts](#).

Added descriptions of the restriction on activating RF module super dormancy when RF module deep dormancy is in effect. For details, see [7.4.3.2.3 Function Impacts](#).

Moved descriptions of UMTS pRRU deep dormancy from *Green BTS* to *Multi-RAT Coordinated Energy Saving*. For details, see [2.3 Features in This Document](#), [7.6 pRRU Deep Dormancy](#), [7.6.1.1 pRRU Deep Dormancy in Scheduled Mode](#), [7.6.2.2 Impacts](#), [7.6.3.1 Licenses](#), [7.6.3.3 Hardware](#), [7.6.3.4 Networking](#), [7.6.4.1.1 Data Preparation](#), and [7.6.4.3 Network Monitoring](#).

Revised descriptions of the setting notes for the **CellPwrSavingPolicy.StopTime** parameter in "Data Preparation" sections throughout this document.

Deleted the descriptions of redundant function impacts.

Added descriptions of the impact relationships of dormancy functions with the following functions: GSM and LTE Buffer Zone Optimization, GSM and LTE Spectrum Concurrency, GSM and LTE Spectrum Concurrency Phase 2, UMTS and LTE Spectrum Sharing Based on DC-HSDPA (LTE FDD), UMTS and LTE Spectrum Sharing (LTE FDD), UMTS and LTE Spectrum Sharing Phase 2 (LTE FDD), and UMTS and LTE Zero Bufferzone (LTE FDD). For details, see [7.2.3.2.3 Function Impacts](#) and [7.4.3.2.3 Function Impacts](#).

Revised descriptions of the networking requirements for LTE and NR intelligent carrier shutdown, RF module deep dormancy, and RF module super dormancy. For details, see [6.1.3.4 Networking](#), [7.2.3.4 Networking](#), and [7.4.3.4 Networking](#).

Revised the function name and reference document of baseband processor shutdown.

Revised descriptions of the networking requirements and activation verification for adaptive BBU deep dormancy. For details, see [7.7.3.4 Networking](#) and [7.7.4.3 Activation Verification](#).

Revised descriptions of activation verification for pRRU deep dormancy. For details, see [7.6.4.2 Activation Verification](#).

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed. Examples of such changes include changes in the sequence of sections.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to GSM, UMTS, LTE, and NR.

For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

2.3 Features in This Document

This document describes the following features.

RAT	Feature ID	Feature Name	Chapter/Section
NR	MRFD-160266	Multi-RAT Coordinated Symbol Power Saving (NR)	4.1 LTE and NR Coordinated Symbol Power Saving 4.2 LTE and NR Bundled Scheduling
LTE FDD	MRFD-161225	Multi-RAT Coordinated Symbol Power Saving (LTE FDD)	
LTE TDD	MRFD-160236	Multi-RAT Coordinated Symbol Power Saving (LTE TDD)	
NB-IoT	MRFD-161255	Multi-RAT Coordinated Symbol Power Saving (NB-IoT)	4.1 LTE and NR Coordinated Symbol Power Saving
NR	MRFD-170264	Multi-RAT Coordinated Channel Shutdown (NR)	5.1 LNR Coordinated Channel Shutdown
LTE FDD	MRFD-161224	Multi-RAT Coordinated Channel Shutdown (LTE FDD)	
LTE TDD	MRFD-170234	Multi-RAT Coordinated Channel Shutdown (LTE TDD)	
LTE FDD	MRFD-161224	Multi-RAT Coordinated Channel Shutdown (LTE FDD)	5.2 UL Coordinated Channel Shutdown
UMTS	MRFD-161214	Multi-RAT Coordinated Channel Shutdown (UMTS)	
NR	MRFD-160263	LTE and NR Intelligent Carrier Shutdown (NR)	6.1 LTE and NR Intelligent Carrier Shutdown
LTE FDD	MRFD-160223	LTE and NR Intelligent Carrier Shutdown (LTE FDD)	
LTE TDD	MRFD-160233	LTE and NR Intelligent Carrier Shutdown (LTE TDD)	
GSM	MRFD-121113	Multi-RAT Carrier Joint Intelligent Shutdown (GBTS)	6.3 GUL Joint Carrier Shutdown

RAT	Feature ID	Feature Name	Chapter/Section
UMTS	MRFD-121123	Multi-RAT Carrier Joint Intelligent Shutdown (NodeB)	
LTE FDD	MRFD-121133	Multi-RAT Carrier Joint Intelligent Shutdown(eNodeB)	
NR	MRFD-161263	RF Module Deep Dormancy (NR)	7.2 RF Module Deep Dormancy
LTE FDD	MRFD-181223	RF Module Deep Dormancy (LTE FDD)	
LTE TDD	MRFD-161233	RF Module Deep Dormancy (LTE TDD)	
UMTS	MRFD-221213	RF Module Deep Dormancy (UMTS)	
NR	MRFD-221261	RF Module Fast Deep Dormancy (NR)	7.3 RF Module Fast Deep Dormancy
LTE FDD	MRFD-221221	RF Module Fast Deep Dormancy (LTE FDD)	
LTE TDD	MRFD-221231	RF Module Fast Deep Dormancy (LTE TDD)	
NR	MRFD-200261	RF Module Super Deep Dormancy (NR)	7.4 RF Module Super Dormancy
LTE FDD	MRFD-201221	RF Module Super Deep Dormancy (LTE FDD)	
LTE TDD	MRFD-200231	RF Module Super Deep Dormancy (LTE TDD)	
UMTS	MRFD-211211	RF Module Super Deep Dormancy (UMTS)	
GSM	MRFD-211201	RF Module Super Deep Dormancy (GSM)	
UMTS	WRFD-200205	pRRU Deep Dormancy	7.6 pRRU Deep Dormancy
LTE FDD	LOFD-130202	pRRU Deep Dormancy	
LTE TDD	TDLOFD-131202	pRRU Deep Dormancy	
NR	FOFD-030216	pRRU Deep Dormancy	
UMTS	MRFD-221212	BBU Deep Dormancy (UMTS)	7.7 Adaptive BBU Deep Dormancy

RAT	Feature ID	Feature Name	Chapter/Section
LTE TDD	MRFD-181233	BBU Deep Dormancy (LTE TDD)	
LTE FDD	MRFD-211221	BBU Deep Dormancy (LTE FDD)	
NR	MRFD-181263	BBU Deep Dormancy (NR)	

2.4 Differences

Differences Between LTE FDD and LTE TDD

Function Name	Difference	Chapter/Section
LTE and NR coordinated symbol power saving	None	4.1 LTE and NR Coordinated Symbol Power Saving
LTE and NR bundled scheduling	None	4.2 LTE and NR Bundled Scheduling
LNR coordinated channel shutdown	FDD: This function is supported only in 2T and 4T scenarios. TDD: This function is supported only in 8T, 32T, and 64T scenarios.	5.1 LNR Coordinated Channel Shutdown
UL coordinated channel shutdown	Supported only in LTE FDD	5.2 UL Coordinated Channel Shutdown
LTE and NR intelligent carrier shutdown	None	6.1 LTE and NR Intelligent Carrier Shutdown
GUL joint carrier shutdown	Supported only in LTE FDD	6.3 GUL Joint Carrier Shutdown
RF module deep dormancy	None	7.2 RF Module Deep Dormancy
RF module fast deep dormancy	None	7.3 RF Module Fast Deep Dormancy
RF module super dormancy	None	7.4 RF Module Super Dormancy
Frequency band-based energy saving of wideband modules	None	7.5 Frequency Band-based Energy Saving of Wideband Modules

Function Name	Difference	Chapter/Section
pRRU deep dormancy	None	7.6 pRRU Deep Dormancy
Adaptive BBU deep dormancy	None	7.7 Adaptive BBU Deep Dormancy
Main control board dynamic energy saving	None	7.8 Main Control Board Dynamic Energy Saving
RHUB intelligent shutdown	None	7.9 RHUB Intelligent Shutdown

Differences Between NR FDD and NR TDD

Function Name	Difference	Chapter/Section
LTE and NR coordinated symbol power saving	None	4.1 LTE and NR Coordinated Symbol Power Saving
LTE and NR bundled scheduling	None	4.2 LTE and NR Bundled Scheduling
LNR coordinated channel shutdown	NR FDD: This function is supported only in 2T and 4T scenarios. NR TDD: This function is supported only in 8T, 32T, and 64T scenarios.	5.1 LNR Coordinated Channel Shutdown
LTE and NR intelligent carrier shutdown	None	6.1 LTE and NR Intelligent Carrier Shutdown
RF module deep dormancy	None	7.2 RF Module Deep Dormancy
RF module fast deep dormancy	None	7.3 RF Module Fast Deep Dormancy
RF module super dormancy	None	7.4 RF Module Super Dormancy
Frequency band-based energy saving of wideband modules	None	7.5 Frequency Band-based Energy Saving of Wideband Modules
pRRU deep dormancy	None	7.6 pRRU Deep Dormancy
Adaptive BBU deep dormancy	None	7.7 Adaptive BBU Deep Dormancy

Function Name	Difference	Chapter/Section
Main control board dynamic energy saving	None	7.8 Main Control Board Dynamic Energy Saving
RHUB intelligent shutdown	None	7.9 RHUB Intelligent Shutdown

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
LTE and NR coordinated symbol power saving	None	4.1 LTE and NR Coordinated Symbol Power Saving
LTE and NR bundled scheduling	None	4.2 LTE and NR Bundled Scheduling
LNR coordinated channel shutdown	None	5.1 LNR Coordinated Channel Shutdown
LTE and NR intelligent carrier shutdown	None	6.1 LTE and NR Intelligent Carrier Shutdown
RF module deep dormancy	None	7.2 RF Module Deep Dormancy
RF module fast deep dormancy	None	7.3 RF Module Fast Deep Dormancy
RF module super dormancy	None	7.4 RF Module Super Dormancy
Frequency band-based energy saving of wideband modules	None	7.5 Frequency Band-based Energy Saving of Wideband Modules
pRRU deep dormancy	None	7.6 pRRU Deep Dormancy
Adaptive BBU deep dormancy	None	7.7 Adaptive BBU Deep Dormancy
Main control board dynamic energy saving	None	7.8 Main Control Board Dynamic Energy Saving
RHUB intelligent shutdown	None	7.9 RHUB Intelligent Shutdown

Differences Between High Frequency Bands and Low Frequency Bands

Function Name	Difference	Chapter/Section
LTE and NR coordinated symbol power saving	Supported only in low frequency bands	4.1 LTE and NR Coordinated Symbol Power Saving
LTE and NR bundled scheduling	Supported only in low frequency bands	4.2 LTE and NR Bundled Scheduling
LNR coordinated channel shutdown	Supported only in low frequency bands	5.1 LNR Coordinated Channel Shutdown
LTE and NR intelligent carrier shutdown	Supported only in low frequency bands	6.1 LTE and NR Intelligent Carrier Shutdown
RF module deep dormancy	Supported in both low and high frequency bands. In high frequency bands, only scheduled RF module deep dormancy and NR load-based RF module deep dormancy are supported.	7.2 RF Module Deep Dormancy
RF module fast deep dormancy	Supported only in low frequency bands	7.3 RF Module Fast Deep Dormancy
RF module super dormancy	Supported in both low and high frequency bands. In high frequency bands, only scheduled RF module super dormancy and NR load-based RF module super dormancy are supported.	7.4 RF Module Super Dormancy
Frequency band-based energy saving of wideband modules	Supported only in low frequency bands	7.5 Frequency Band-based Energy Saving of Wideband Modules
pRRU deep dormancy	Supported only in low frequency bands	7.6 pRRU Deep Dormancy
Adaptive BBU deep dormancy	Macro base stations: Supported in both low and high frequency bands. LampSite base stations: Supported only in low frequency bands.	7.7 Adaptive BBU Deep Dormancy
Main control board dynamic energy saving	None	7.8 Main Control Board Dynamic Energy Saving

Function Name	Difference	Chapter/Section
RHUB intelligent shutdown	Supported only in low frequency bands	7.9 RHUB Intelligent Shutdown

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

3 Overview

3.1 Categories of Energy Saving Functions

The expanding wireless services and increasing network complexity lead to high overall equipment energy consumption. The resulting growth of base station energy consumption is a major contributor to the increasing operating expense (OPEX) of telecom operators. Huawei's energy saving solutions are launched against this background, and they help operators improve equipment operating efficiency, reduce equipment power consumption and OPEX, as well as enhance product competitiveness. The solutions can be implemented at the device, network, and site levels, depending on the objects involved.

- Device level

Among all base station devices, RF modules consume the most power, with baseband units (BBUs) not far behind. Device-level energy saving reduces the power consumption of RF modules in the following ways:

- Time domain

When there is no valid information to be transmitted in symbols, components such as power amplifiers (PAs) are shut down.

- Space domain

In light-load scenarios, components such as PAs of certain RF channels are shut down.

- Frequency domain

In light-load scenarios, most components of the RF modules or PAs corresponding to certain carriers are shut down.

- Network level

The energy saving solutions implemented at this level involve inter-base-station and inter-band coordination.

- Site level

The energy saving solutions implemented at this level are related to the coordination between main equipment of base stations and their auxiliary equipment (such as the power supply). For details about the energy saving functions for power equipment, see *Green Site*.

Multi-RAT coordinated energy saving involves energy saving solutions requiring coordination among NR, LTE, UMTS, and GSM. [Table 3-1](#) describes the multi-RAT coordinated energy saving functions of different categories.

Table 3-1 Categories of energy saving functions

Category	Sub-Category	Related Function
Device level	Time domain	4.1 LTE and NR Coordinated Symbol Power Saving 4.2 LTE and NR Bundled Scheduling
	Space domain	5.1 LNR Coordinated Channel Shutdown 5.2 UL Coordinated Channel Shutdown
	Frequency domain	6.1 LTE and NR Intelligent Carrier Shutdown 6.2 Multi-Carrier Joint Carrier Shutdown 6.3 GUL Joint Carrier Shutdown
Network level	N/A	None
Site level	N/A	7.2 RF Module Deep Dormancy 7.3 RF Module Fast Deep Dormancy 7.4 RF Module Super Dormancy 7.5 Frequency Band-based Energy Saving of Wideband Modules 7.6 pRRU Deep Dormancy 7.7 Adaptive BBU Deep Dormancy 7.8 Main Control Board Dynamic Energy Saving 7.9 RHUB Intelligent Shutdown

 NOTE

- An energy saving function may fall into multiple categories. [Table 3-1](#) describes only the major categories of energy saving functions.
- When carriers of different RATs share a PA, energy saving gains of PA shutdown can be obtained only after carriers of all the RATs on the relevant RF channels enter the energy saving state.

4 Time-Domain Energy Saving

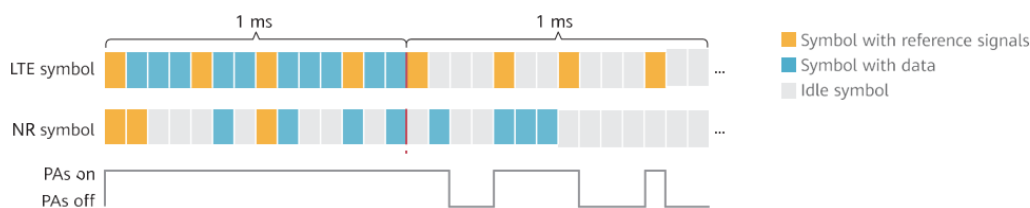
4.1 LTE and NR Coordinated Symbol Power Saving

4.1.1 Principles

Power amplifiers (PAs) consume most energy in a base station and have static energy consumption even when there is no signal output. **Multi-RAT coordinated symbol power saving is introduced to reduce system energy consumption without compromising data transmission integrity.** As shown in [Figure 4-1](#), when LTE and NR share RF channels:

- After detecting that no data is carried in LTE or NR symbols, the base stations shut down the PAs in real time to reduce system energy consumption.
- After detecting that there is data to be carried in LTE or NR symbols, the base stations turn on the PAs in real time to ensure data transmission integrity.

Figure 4-1 Principles of Multi-RAT coordinated symbol power saving



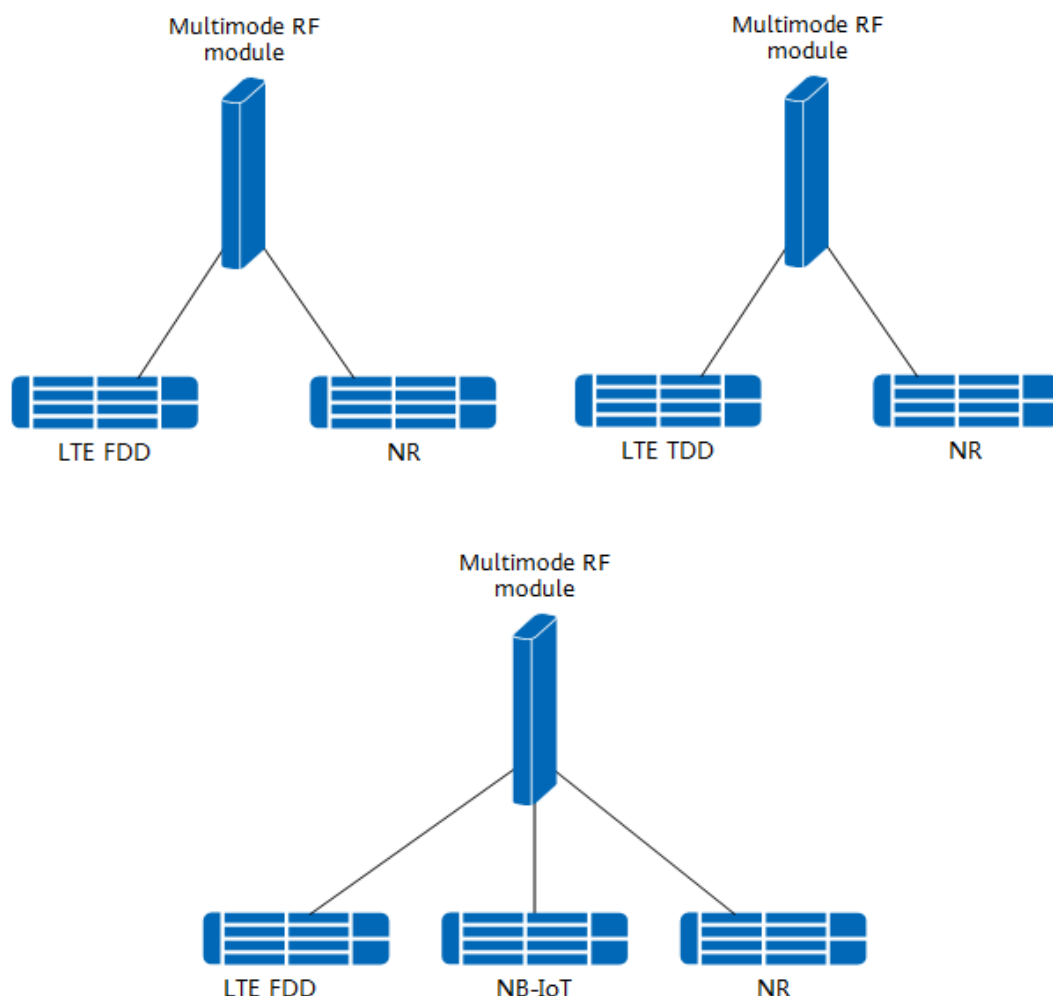
To enable this function, the following switches must be turned on:

- On the LTE side: **SymbolPwrSaving.MultiRatSymbolShutdownSw**
- On the NR side: **MULTI_RAT_SYMBOL_SHUTDOWN_SW** option of **NRDUCellAlgoSwitch.PowerSavingSwitch**

LTE and NR coordinated symbol power saving applies to RF modules working in LN, TN, or LMN mode and requires that these RATs share RF channels, as shown in [Figure 4-2](#). When GSM carriers are configured on RF channels, LTE and NR coordinated symbol power saving takes effect only after GSM carriers are blocked. When UMTS carriers are configured on RF channels, UMTS RF channel dynamic

muting must be enabled by setting the **ULOCCELLALGPARA.RFCHNDYNMUTINGMODE** parameter to **BASICPWRSV** or **DEEPPWRSV**, to allow LTE and NR coordinated symbol power saving to take effect. For details about UMTS RF channel dynamic muting, see *Green BTS in RAN Feature Documentation*.

Figure 4-2 Working modes of a multimode RF module



4.1.2 Network Analysis

4.1.2.1 Benefits

If LTE and NR share RF modules and no data is transmitted in LTE or NR symbols, PAs or RoC systems can be dynamically shut down to reduce system energy consumption. If the temperature difference of RF modules is excessively large within 24 hours, the energy saving gains may decrease after this function is enabled.

For details about the counters used to evaluate the gains of this function, see [4.1.4.3 Network Monitoring](#).

4.1.2.2 Impacts

The impacts between this function and other functions are described in [4.1.3.2.3 Function Impacts](#).

The function has no impact on the network.

4.1.3 Requirements

4.1.3.1 Licenses

The following table lists the license requirements for enabling multi-RAT coordinated symbol power saving.

RAT	Feature ID	Feature Name	Model	NE	Sales Unit
LTE FDD	MRFD-161225	Multi-RAT Coordinated Symbol Power Saving (LTE FDD)	LT1SMRJSPS00	eNodeB	per Cell
LTE FDD	MRFD-161225	Multi-RAT Coordinated Symbol Power Saving (LTE FDD)	LT1SFDMRCS00	LampSite eNodeB	per Carrier per pRRU
NB-IoT	MRFD-161255	Multi-RAT Coordinated Symbol Power Saving (NB-IoT)	ML1SMRCSPS00	eNodeB	per Cell
NB-IoT	MRFD-161255	Multi-RAT Coordinated Symbol Power Saving (NB-IoT)	ML1SMRCSPS01	LampSite eNodeB	per Carrier per pRRU

RAT	Feature ID	Feature Name	Model	NE	Sales Unit
LTE TDD	MRFD-160236	Multi-RAT Coordinated Symbol Power Saving (LTE TDD)	LT4SMRCSPTDD	Macro eNodeB	per Cell
LTE TDD	MRFD-160236	Multi-RAT Coordinated Symbol Power Saving (LTE TDD)	LT1STDMRCS00	LampSite eNodeB	per Carrier per pRRU
NR	MRFD-160266	Multi-RAT Coordinated Symbol Power Saving (NR)	NR0SMRCSPTS00	gNodeB	per Cell
NR	MRFD-160266	Multi-RAT Coordinated Symbol Power Saving (NR)	NR0SMRCSPTS01	LampSite gNodeB	per Carrier per pRRU

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

NOTICE

- Before enabling multi-RAT coordinated symbol power saving, prepare the licenses for multi-RAT coordinated symbol power saving and single-mode symbol power saving for the corresponding RAT.
- For details about required licenses of a single RAT, see *Energy Conservation and Emission Reduction* in *eRAN Feature Documentation* and *Energy Conservation and Emission Reduction* in *5G RAN Feature Documentation*.

4.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.1.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference
UMTS	UMTS RF channel dynamic muting	ULOCCELLALGPARA.RFCHNDY NMUTINGMODE	<i>Green BTS in RAN Feature Documentation</i>
LTE FDD LTE TDD	Symbol power saving	FDD: SymbolShutdownSwitch option of the ENodeBAlgoSwitch.PowerSaveSwitch parameter TDD: TddSymbolShutdownSwitch option of the ENodeBAlgoSwitch.PowerSaveSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>
NR FDD Low-frequency NR TDD	Symbol power saving	NRDUCellAlgoSwitch.SymbolShutdownSwitch	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>

4.1.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD Low-frequency NR TDD	Dual-BBU backup in dual-network redundancy networking	NR: DUAL_BBU_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter LTE: LTE_PASSIVE_DUAL_BBU_BACKUP_SW option of the eNBRedundancy.eNBReliabilitySwitch parameter	None	"Dual-BBU backup in dual-network redundancy networking" is mutually exclusive with "LTE and NR coordinated symbol power saving".
NR FDD Low-frequency NR TDD	Primary/Backup function in intra-frequency dual-network redundancy networking	PRIMARY_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	"Primary/Backup function in intra-frequency dual-network redundancy networking" is mutually exclusive with "LTE and NR coordinated symbol power saving".
NR FDD	Dual-plane function in intra-frequency dual-network redundancy networking	DUAL_PLANE_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	"Dual-plane function in intra-frequency dual-network redundancy networking" is mutually exclusive with "LTE and NR coordinated symbol power saving".

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios	DUAL_PLANE_TOB_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	<i>Base Station Equipment Reliability</i>	"Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" is mutually exclusive with "LTE and NR coordinated symbol power saving".

4.1.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD LTE FDD LTE TDD	RF module deep dormancy	RRU.DORMANCY_SW (5G gNodeB, LTE eNodeB)	<i>Multi-RAT Coordinated Energy Saving</i>	RF module deep dormancy takes precedence over multi-RAT coordinated symbol power saving when both functions take effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When timing carrier shutdown takes effect: • If the RF module is not serving any other NR carriers, multi-RAT coordinated symbol power saving does not take effect. • If the RF module is serving other NR carriers, the status of multi-RAT coordinated symbol power saving for these carriers is not affected.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Intelligent carrier shutdown in precise mode	• INTRA_GNB_MULT I_CARR_S D_SW or INTER_GNB_MULT I_CARR_S D_SW option of the NRDUCell AlgoSwitch.Power SavingSwitch parameter • MULTI_CARR_PRECISE_SHDN_SW option of the NRDUCell AlgoSwitch.Power SavingSwitch parameter selected	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	When intelligent carrier shutdown takes effect: • If the RF module is not serving any other NR carriers, multi-RAT coordinated symbol power saving does not take effect. • If the RF module is serving other NR carriers, the status of multi-RAT coordinated symbol power saving for these carriers is not affected.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Intelligent carrier shutdown in non-precise mode	• INTRA_GNB_MULT I_CARR_S D_SW or INTER_GNB_MULT I_CARR_S D_SW option of the NRDUCell AlgoSwitch.Power SavingSwitch parameter • MULTI_CARR_PRECISE_SHDN_SW option of the NRDUCell AlgoSwitch.Power SavingSwitch parameter deselected	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	When intelligent carrier shutdown takes effect: • If the RF module is not serving any other NR carriers, multi-RAT coordinated symbol power saving does not take effect. • If the RF module is serving other NR carriers, the status of multi-RAT coordinated symbol power saving for these carriers is not affected.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Fast carrier shutdown	For capacity-layer cells: • Intra-base-station multi-carrier scenarios: INTRA_GNB_MULT I_CARR_S D_SW option of the NRDUCell AlgoSwitch.Power SavingSwitch parameter selected Inter-base-station multi-carrier scenarios: INTER_GNB_MULT I_CARR_S D_SW option of the NRDUCell AlgoSwitch.Power SavingSwitch parameter selected • FAST_CARRIER_SHUTDOWN_SW option of the NRDUCell AlgoSwitch	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	When fast carrier shutdown takes effect: If the RF module is not serving any other NR carriers, LTE and NR coordinated symbol power saving does not take effect. If the RF module is serving other NR carriers, the status of LTE and NR coordinated symbol power saving for these carriers is not affected.

RAT	Function Name	Function Switch	Reference	Description
		ch.PowerSavingExtSwitch parameter For basic-layer cells: FAST_CARR_SHUT_SERV_ASSUR_SW option of the NRDUCellAI goSwitch.PowerSavingExtSwitch parameter		
LTE FDD	NB-IoT cell setup in massive MIMO scenarios	Cell.FddTddI nd set to CELL_FDD Cell.TxRxMode de set to 32T32R Cell.NbCellFlag ag set to TRUE	<i>NB-IoT Basics (FDD)</i> in <i>eRAN Feature Documentation</i>	In FDD massive MIMO scenarios, the presence of NB-IoT cells may reduce the proportion of occasions where symbol power saving takes effect in LTE FDD cells, affecting energy saving gains.

4.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- LTE side:
 - 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
 - DBS3900 LampSite and DBS5900 LampSite
- NR side:
 - 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
 - DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

- LTE side:
 - Main control board: All UMPT boards support this function.
 - Baseband processing unit: All LTE-capable baseband processing units support this function.
- NR side:
 - Main control board: All UMPT boards support this function.
 - Baseband processing unit: All NR-capable baseband processing units support this function.

RF Modules

RF module requirements are as follows:

- AAUs and RRUs that support RF module sharing between LTE and NR support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.
- AAUs and RRUs that support LTE and NR symbol power saving support this function. For details, see the following sections:
 - Section "Symbol Power Saving" in *Energy Conservation and Emission Reduction in eRAN Feature Documentation*
 - Section "Symbol Power Saving" in *Energy Conservation and Emission Reduction in 5G RAN Feature Documentation*
- All pRRUs that support RF module sharing between LTE and NR support this function. For details, see *LampSite pRRU&RRU Technical Specifications* in *3900 & 5900 Series Base Station Product Documentation*.

NOTE

The RRU3959 operating in the 900 MHz frequency band does not support this function.

4.1.3.4 Others

None

4.1.4 Operation and Maintenance

4.1.4.1 Data Configuration

Before enabling this function, collect statistics on the counters used for gain analysis. For details, see [4.1.4.3 Network Monitoring](#).

4.1.4.1.1 Data Preparation

Table 4-1 Parameters used for configuration on the LTE side

Parameter Name	Parameter ID	Setting Notes
Multi-RAT Symbol Shutdown Switch	SymbolPwrSaving.MultiRatSymbolShutdownSw	Set this parameter to ON .

Table 4-2 Parameters used for configuration on the NR side

Parameter Name	Parameter ID	Setting Notes
Power Saving Switch	NRDUCellAlgoSwitch.PowerSavingSwitch	Select the MULTI_RAT_SYMBOL_SHUTDOWN_SW option of this parameter.

NOTICE

Multi-RAT coordinated symbol power saving takes effect only when it is enabled for all activated cells of all RATs on a multi-band multi-carrier co-channel.

4.1.4.1.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [4.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

```
//LTE configurations
//Enabling multi-RAT coordinated symbol power saving
MOD SYMBOLPWRSAVING: LocalCellId=0, MultiRatSymbolShutdownSw=ON;
//NR configurations
//Enabling multi-RAT coordinated symbol power saving
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingSwitch=MULTI_RAT_SYMBOL_SHUTDOWN_SW-1;
```

Deactivation Command Examples

```
//LTE configurations
//Disabling multi-RAT coordinated symbol power saving
MOD SYMBOLPWRSAVING: LocalCellId=0, MultiRatSymbolShutdownSw=OFF;
//NR configurations
//Disabling multi-RAT coordinated symbol power saving
```

```
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,  
PowerSavingSwitch=MULTI_RAT_SYMBOL_SHUTDOWN_SW-0;
```

NOTICE

After multi-RAT coordinated symbol power saving is disabled, single-mode symbol power saving also becomes invalid.

4.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.1.4.2 Activation Verification

- To check whether the Multi-RAT Coordinated Symbol Power Saving feature has taken effect in an LTE cell, perform the following steps:
Run the **DSP SYMBOLPWRSAVING** command on the eNodeB side. If the value of **Multi-RAT Symbol Shutdown Status** is **STARTED**, this feature has taken effect.
- To check whether the Multi-RAT Coordinated Symbol Power Saving feature has taken effect in an NR cell, perform the following steps:
Run the **DSP NRDUCELL** command on the gNodeB side. If the value of **Power Saving State** is **MULTI_RAT_SYMBOL_SHUTDOWN: Enabled**, this feature has taken effect.

4.1.4.3 Network Monitoring

Compare the values of the following counters one week before and after the Multi-RAT Coordinated Symbol Power Saving feature is enabled to evaluate the energy saving gains.

- Counters measuring the energy consumption of the entire base station

Counter ID	Counter Name
1593835637	VS.EnergyCons.BTS.Adding.LTE
1593835959	VS.EnergyCons.BTS.Adding.Service.LTE
1593835804	VS.EnergyCons.BTS.Adding.NR
1593835961	VS.EnergyCons.BTS.Adding.Service.NR

- Counter measuring the energy consumption of RF modules

Counter ID	Counter Name
1593835836	VS.EnergyCons.BTSBoard

- Counter measuring symbol power saving duration

Counter ID	Counter Name
1911823033	N.PowerSaving.SymbolShutdown.Dur

NOTE

The policy of counter measurement upon carrier shutdown is determined by the function of energy saving counter optimization upon carrier shutdown, which is controlled by the **COUNTER_OPT_IN_CARROFF_SW** option of the **gNBPowerSavingAlgo.PowerSavingAlgoSwitch** parameter. Assume that symbol power saving takes place on a carrier that has already entered the shutdown state. If this option is deselected, the duration of symbol power saving is counted in the **N.PowerSaving.SymbolShutdown.Dur** counter. If this option is selected, this duration is not counted in this counter.

4.2 LTE and NR Bundled Scheduling

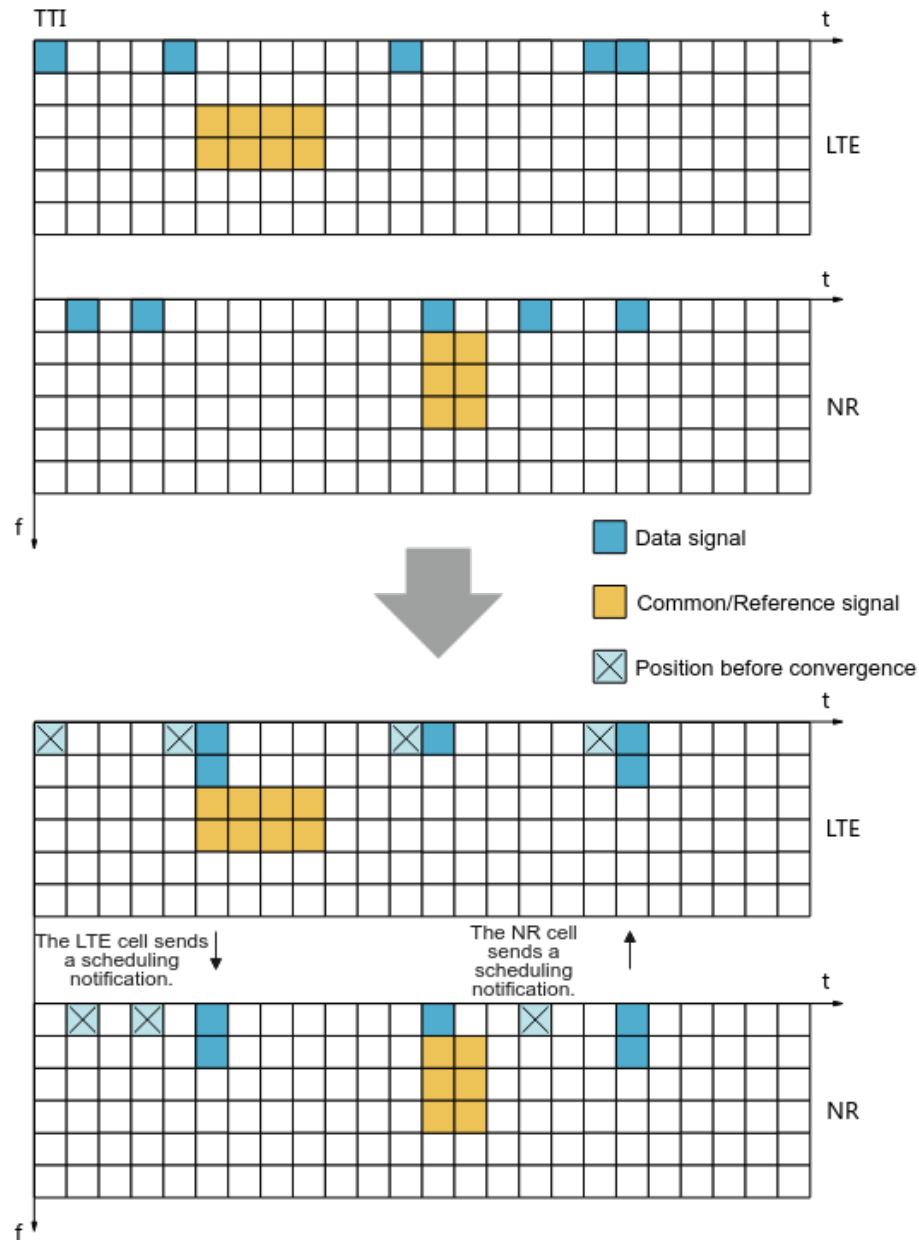
4.2.1 Principles

To further save energy, the LTE and NR bundled scheduling function is introduced. **This function adjusts LTE and NR services based on the delay and load. It delays scheduling of small-packet services that can be transmitted within one TTI in the downlink and converges the services to the TTI for data services of the peer RAT or common messages. In this way, more time windows without data transmission are obtained, and symbol power saving can be performed within these time windows,** as shown in [Figure 4-3](#). Currently, bundled scheduling is supported only between LTE TDD and NR, and between LTE FDD and NR.

The "LTE and NR coordination for data convergence in slots" function is an enhancement to LTE and NR bundled scheduling, and can be enabled only after LTE and NR bundled scheduling is enabled.

- When LTE and NR coordination for data convergence in slots is disabled, only the services of the NR cell are converged and then scheduled in the TTIs corresponding to the common channels and data services of the LTE cell.
- When LTE and NR coordination for data convergence in slots is enabled, mutual scheduling coordination takes effect between inter-RAT cells, where the services of the NR or LTE cell are converged and then scheduled in the TTIs corresponding to the common channels and data services of the other cell. The LTE cell and the NR cell align scheduled slots with each other, increasing the scheduling alignment probability and reducing system power consumption.

Figure 4-3 LTE and NR bundled scheduling



To enable LTE and NR bundled scheduling, the following switches must be turned on:

- LTE side: **LNR_BUNDLED_SCHEDULING_SW** option of the **SymbolPwrSaving.SymbolShutdownEnhancedSw** parameter
 - NR side: **LNR_BUNDLED_SCHEDULING_SW** option of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter
- When GSM or UMTS carriers are configured on all channels of the RF module, LTE and NR bundled scheduling does not take effect. When GSM or UMTS carriers are configured on only some channels of the RF module, LTE and NR bundled scheduling can still take effect.

To enable LTE and NR coordination for data convergence in slots, the following switches must be turned on in addition to LTE and NR bundled scheduling:

- LTE side: **LNR_SLOT_DATA_CONV_CO_SW** option of the **SymbolPwrSaving.SymbolShutdownEnhancedSw** parameter
- NR side:
 - NR single-carrier scenarios: **LNR_SLOT_DATA_CONV_COORD_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter
 - NR multi-carrier scenarios: **LNR_SLOT_DATA_CONV_COORD_SW** and **SLOT_DATA_CONVERGENCE_COORD_SW** options

LTE and NR bundled scheduling or LTE and NR coordination for data convergence in slots takes effect when all of the following conditions are met:

- The time falls within the LTE and NR bundled scheduling period specified by the **NRDUCellPowerSaving.StartTime** and **NRDUCellPowerSaving.StopTime** parameters.
- The average downlink PRB usage of the LTE cell is less than the value of **SymbolPwrSaving.TrigBndlSchDLAvgPrbThld**. (When LTE and NR bundled scheduling is enabled, the value of this parameter cannot be 0.)
- The average downlink PRB usage of the NR cell is less than the value of **NRDUCellPowerSaving.DlPrbThld**.

LTE and NR bundled scheduling or LTE and NR coordination for data convergence in slots stops taking effect when any of the following conditions is met:

- The time is beyond the LTE and NR bundled scheduling period specified by the **NRDUCellPowerSaving.StartTime** and **NRDUCellPowerSaving.StopTime** parameters.
- The average downlink PRB usage of the LTE cell is greater than or equal to **SymbolPwrSaving.TrigBndlSchDLAvgPrbThld** + PRB threshold offset.
- The average downlink PRB usage of the NR cell is greater than or equal to **NRDUCellPowerSaving.DlPrbThld** + PRB threshold offset.

Multi-Carrier Bundled Scheduling Group

Bundled scheduling is performed for cells in the same multi-carrier bundled scheduling group. All available cells served by an RF module are configured as one multi-carrier bundled scheduling group if they meet the following conditions:

- The cells share an RF channel.
- Bundled scheduling is triggered.
- The RF module is configured only with LTE carriers and NR carriers.
- For TDD, all cells have the same subframe configuration.

When a carrier is added, deleted, blocked, or unblocked in a multi-carrier bundled scheduling group, the multi-carrier bundled scheduling group needs to be re-established. A multi-carrier bundled scheduling group supports a maximum of five LTE cells and four NR cells. When the number of LTE cells is greater than five or the number of NR cells is greater than four, bundled scheduling does not take effect.

4.2.2 Network Analysis

4.2.2.1 Benefits

When LTE and NR share an RF module, enabling the LTE and NR bundled scheduling function increases the proportion of idle time during LTE and NR scheduling, reducing power consumption.

Additionally, enabling LTE and NR coordination for data convergence in slots further saves power.

For details about the counters used to evaluate the gains of this function, see [4.2.4.3 Network Monitoring](#).

4.2.2.2 Impacts

The impacts between this function and other functions are described in [4.2.3.2.3 Function Impacts](#).

- Impact on the LTE side
 - The values of the counters related to the downlink data-plane delay increase.
 - The value of the **L.Traffic.SUNPT.DL.Max** counter increases.
 - The value of **User Downlink Average Throughput** decreases.
 - The PUCCH resource allocation for CQI reporting changes, triggering PUCCH RB range extension earlier than expected. This increases uplink PRB usage. In addition, the HARQ feedback on downlink transmission may converge on the PUCCH, leading to increased uplink interference.
 - The values of uplink access indicators (such as the random access success rate), **User Uplink Average Throughput**, uplink MCS index, and uplink IBLER/RBLER may fluctuate.
 - When LTE and NR coordination for data convergence in slots takes effect, the CPU usage increases because scheduling information needs to be exchanged in real time between cells sharing the same RF channel.
- Impact on the NR side
 - After the LTE and NR bundled scheduling function is enabled in a cell, the access delay of UEs in idle mode camping on this cell increases slightly, the maximum waiting duration of UEs in connected mode in the cell increases slightly, and the cell-level and UE-level data rates decrease slightly.
 - After this function is enabled, downlink scheduling of NR cells is aligned with that of LTE cells, increasing the proportion of slots in which NR services are scheduled. As a result, the average downlink throughput of NR cells (**Cell Downlink Average Throughput (DU)**) decreases, and the downlink user-perceived rate may decrease.
 - When LTE and NR coordination for data convergence in slots takes effect, the CPU usage increases because scheduling information needs to be exchanged in real time between cells sharing the same RF channel.

4.2.3 Requirements

4.2.3.1 Licenses

None

4.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.2.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Multi-RAT coordinated symbol power saving (LTE FDD)	SymbolPwrS aving.MultiR atSymbolShu tdownSw	<i>Multi-RAT Coordinated Energy Saving</i>	None
LTE TDD	Multi-RAT coordinated symbol power saving (LTE TDD)	SymbolPwrS aving.MultiR atSymbolShu tdownSw	<i>Multi-RAT Coordinated Energy Saving</i>	None
LTE FDD LTE TDD	Energy saving based on proactive scheduling	SymbolPwrS aving.TrigBn dlSchDLAvgPr bThld	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentatio n</i>	None
NR FDD Low- frequency NR TDD	Multi-RAT coordinated symbol power saving (NR)	MULTI_RAT_S YMBOL_SHU TDOWN_SW option of the NRDUCellAlg oSwitch.Pow erSavingSwit ch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	None

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Power saving scheduling	POWER_SAVING_SCHEDULING_SW option of the NRDUCellAlgorithmSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	The time period set for the power saving scheduling function must include the time period set for LTE and NR bundled scheduling.

4.2.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgorithmSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	The period configured for the timing carrier shutdown function cannot overlap that configured for the LTE and NR bundled scheduling function. Set the period for the LTE and NR bundled scheduling function based on the network plan.
NR FDD Low-frequency NR TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell in 5G RAN Feature Documentation</i>	LTE and NR bundled scheduling cannot be enabled together with Hyper Cell.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Cell Combination	- NRDUCell.NrDuCellNetworking Mode set to HYPER_CELL_COMBINE_MODE - NRDUCellMultiTrp.DataTransMode set to BASIC_MODE or DPS_MODE	<i>Cell Combination in 5G RAN Feature Documentation</i>	LTE and NR bundled scheduling cannot be enabled together with Cell Combination.
NR FDD Low-frequency NR TDD	Joint Transmission Cell Combination	- NRDUCell.NrDuCellNetworking Mode set to HYPER_CELL_COMBINE_MODE - NRDUCellMultiTrp.DataTransMode set to JOINT_MODE	<i>Cell Combination in 5G RAN Feature Documentation</i>	LTE and NR bundled scheduling cannot be enabled together with Joint Transmission Cell Combination.
Low-frequency NR TDD	Dual-BBU backup in dual-network redundancy networking	DUAL_BBU_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	"Dual-BBU backup in dual-network redundancy networking" is mutually exclusive with LTE and NR bundled scheduling.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Primary/Backup function in intra-frequency dual-network redundancy networking	PRIMARY_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	"Primary/Backup function in intra-frequency dual-network redundancy networking" is mutually exclusive with LTE and NR bundled scheduling.
NR FDD	Dual-plane function in intra-frequency dual-network redundancy networking	DUAL_PLANE_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	"Dual-plane function in intra-frequency dual-network redundancy networking" is mutually exclusive with LTE and NR bundled scheduling.
Low-frequency NR TDD	Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios	DUAL_PLANE_TOB_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	<i>Base Station Equipment Reliability</i>	"Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" is mutually exclusive with "LTE and NR bundled scheduling".

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Static Multiple Beam	SectorSplitGroup.BeamTemplateId set to a value greater than 1	<i>Massive MIMO Basics (FDD)</i> in <i>eRAN Feature Documentation</i>	LTE and NR bundled scheduling is not supported in FDD 16T16R massive MIMO scenarios where Static Multiple Beam is enabled.

4.2.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Basic functions of downlink scheduling	None	<i>Downlink Scheduling</i> in <i>5G RAN Feature Documentation</i>	LTE and NR bundled scheduling delays scheduling when certain conditions are met, which is different from the original scheduling mechanism.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Intelligent carrier shutdown in precise mode	• INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter • MULTI_CARR_PRECISE_SHDN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter selected	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	When intelligent carrier shutdown takes effect: • If the RF module is not serving any other NR carriers, multi-RAT coordinated symbol power saving does not take effect. • If the RF module is serving other NR carriers, the status of multi-RAT coordinated symbol power saving for these carriers is not affected.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Intelligent carrier shutdown in non-precise mode	• INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter • MULTI_CARR_PRECISE_SHDN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter deselected	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	When intelligent carrier shutdown takes effect: • If the RF module is not serving any other NR carriers, multi-RAT coordinated symbol power saving does not take effect. • If the RF module is serving other NR carriers, the status of multi-RAT coordinated symbol power saving for these carriers is not affected.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-freque ncy NR TDD	Fast carrier shutdown	For capacity-layer cells: • Intra-base-station multi-carrier scenarios: INTRA_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter selected; • Inter-base-station multi-carrier scenarios: INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter selected • FAST_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter For basic-layer cells:	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	When fast carrier shutdown takes effect: If the RF module is not serving any other NR carrier, LTE and NR bundled scheduling does not take effect. If the RF module is serving other NR carriers, the status of LTE and NR bundled scheduling is not affected.

RAT	Function Name	Function Switch	Reference	Description
		FAST_CARR_SHUT_SERV_ASSUR_SW option of the NRDUCellAlgorithmSwitch.PowerSavingExtensionSwitch parameter		
NR FDD Low-frequency NR TDD	Power saving scheduling	POWER_SAVING_SCHEDULING_SW option of the NRDUCellAlgorithmSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	When both power saving scheduling and LTE and NR bundled scheduling are enabled, it is recommended that the NRDUCellPowerSaving.DLPrbThld or NRDUCellPowerSaving.UserNumThld parameter be set to a larger value with a larger load or number of UEs. Otherwise, the gains of LTE and NR bundled scheduling may be affected.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequen cy NR TDD	Speed turbo UE identificatio n	gNodeBParam.StUelidentSw	None	LTE and NR bundled scheduling fails to take effect in cells served by a base station, resulting in a decrease in the energy saving gains, when any of the following configuration conditions is met and the configured function takes effect:
NR FDD Low-frequen cy NR TDD	gNodeB acceleration service application	• gNBAsAppParam.AppDomainName • gNBAsAppParam.Appplpv4 • gNBAsAppParam.Appplpv6	None	• Any option of the gNodeBParam.StUelidentSw parameter is selected. • The gNBAsAppParam.AppDomainName parameter is set. • The gNBAsAppParam.Appplpv4 parameter is set. • The gNBAsAppParam.Appplpv6 parameter is set. The following methods can be used to address the preceding issue: • Select the SPEED_TURBO_PWR_SAVING_SW option of the gNBPwrSavingAlgo.PwrSavingAlgoSwitch parameter. LTE and NR bundled scheduling can take effect for cells served by the base station in some of the preceding scenarios

RAT	Function Name	Function Switch	Reference	Description
				within the period specified by the gNBWPwrSavingAlgo.SpeedTurboPwrSavStartTime and gNBWPwrSavingAlgo.SpeedTurboPwrSavEndTime parameters, increasing the energy saving gains. • Select the STATIC_ST_PWR_SAVING_SW option of the gNBWPwrSavingAlgo.PwrSavingAlgorithm parameter. LTE and NR bundled scheduling can take effect for cells served by the base station in some of the preceding scenarios within the period specified by the gNBWPwrSavingAlgo.StaticSTPwrSavStartTime and gNBWPwrSavingAlgo.StaticSTPwrSavEndTime parameters, increasing the energy saving gains.

4.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- LTE: 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

- NR: 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

LampSite base stations do not support this function.

Boards

Main control board: All UMPT boards support this function.

Baseband processing unit: All UBBP boards support this function.

RF Modules

RF module requirements are as follows:

- AAUs and RRUs that support RF module sharing between LTE and NR support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.
- AAUs and RRUs that support LTE and NR symbol power saving support this function. For details, see the following sections:
 - Section "Symbol Power Saving" in *Energy Conservation and Emission Reduction in eRAN Feature Documentation*
 - Section "Symbol Power Saving" in *Energy Conservation and Emission Reduction in 5G RAN Feature Documentation*

pRRUs do not support this function.

4.2.3.4 Cells

This function is not supported in two-RRU combination scenarios.

4.2.3.5 Others

None

4.2.4 Operation and Maintenance

4.2.4.1 Data Configuration

4.2.4.1.1 Data Preparation

Table 4-3 Parameters used for configuration on the LTE side

Parameter Name	Parameter ID	Setting Notes
Symbol Shutdown Enhanced Switch	SymbolPwrSaving.SymbolShutdownEnhancedSw	Select the LNR_BUNDLED_SCHEDULING_SW option of this parameter to enable LTE and NR bundled scheduling.

Parameter Name	Parameter ID	Setting Notes
Trig Bundled Scheduling DL Avg PRB Thld	SymbolPwrSaving.TrigBndlSchDLAvgPrbThld	Set this parameter to its recommended value.
Stop Bundling DL TTI-Level PRB Thld	SymbolPwrSaving.StopBundlingDLTtiPrbThld	Set this parameter to its recommended value. It is used for evaluation of whether scheduling should be delayed in the next TTI for an LTE cell. For details, see <i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i> .
Symbol Shutdown Enhanced Switch	SymbolPwrSaving.SymbolShutdownEnhancedSw	Select the LNR_SLOT_DATA_CONV_CO_SW option of this parameter to enable LTE and NR coordination for data convergence in slots.

Table 4-4 Parameters used for configuration on the NR side

Parameter Name	Parameter ID	Setting Notes
Power Saving Switch	NRDUCellAlgoSwitch.PowerSavingSwitch	Select the LNR_BUNDLED_SCHEDULING_SW option of this parameter.
Power Saving Extension Switch	NRDUCellAlgoSwitch.PowerSavingExtSwitch	- In NR single-carrier scenarios, select the LNR_SLOT_DATA_CONV_COORD_SW option of this parameter to enable LTE and NR coordination for data convergence in slots. - In NR multi-carrier scenarios, select both the LNR_SLOT_DATA_CONV_COORD_SW and SLOT_DATA_CONVERGENCE_COORD_SW options to enable LTE and NR coordination for data convergence in slots.

Parameter Name	Parameter ID	Setting Notes
Start Time	NRDUCellPowerSaving.StartTime	Set this parameter based on the network plan. The LTE and NR bundled scheduling period must be a subset of the power saving scheduling period on the NR side. The duration for LTE and NR bundled scheduling should not overlap with the duration specified when the NRDUCellPowerSaving.PowerSavingType parameter is set to TIMING_CARRIER_SHUTDOWN for the local cell.
Stop Time	NRDUCellPowerSaving.StopTime	
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	Set this parameter to LNR_BUNDLED_SCHEDULING .
Downlink PRB Threshold	NRDUCellPowerSaving.DlPrbThld	Set this parameter to its recommended value.

NOTICE

1.

The LTE and NR bundled scheduling function takes effect only when the switch for this function is turned on for all activated cells of all RATs that share the same channel.

2.

The LTE and NR coordination for data convergence in slots function takes effect only when this function is enabled for all activated cells of all RATs that share the same channel.

4.2.4.1.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [4.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

```
//LTE configurations
//Enabling LTE and NR bundled scheduling
```

```
MOD SYMBOLPWRSAVING: LocalCellId=0,  
SymbolShutdownEnhancedSw=LNR_BUNDLED_SCHEDULING_SW-1, TrigBndlSchDIAvgPrbThld=50,  
StopBundlingDlTtiPrbThld=60;  
//Enabling LTE and NR coordination for data convergence in slots  
MOD SYMBOLPWRSAVING: LocalCellId=0,  
SymbolShutdownEnhancedSw=LNR_SLOT_DATA_CONV_CO_SW-1;  
//NR single-carrier scenarios  
//Enabling LTE and NR bundled scheduling  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=LNR_BUNDLED_SCHEDULING_SW-1;  
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0, StartTime=00&00&00,  
StopTime=06&00&00, PowerSavingType=LNR_BUNDLED_SCHEDULING, DlPrbThld=10;  
//Enabling LTE and NR coordination for data convergence in slots  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,  
PowerSavingExtSwitch=LNR_SLOT_DATA_CONV_COORD_SW-1;  
//NR multi-carrier scenarios  
//Enabling LTE and NR bundled scheduling  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=LNR_BUNDLED_SCHEDULING_SW-1;  
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0, StartTime=00&00&00,  
StopTime=06&00&00, PowerSavingType=LNR_BUNDLED_SCHEDULING, DlPrbThld=10;  
//Enabling LTE and NR coordination for data convergence in slots  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,  
PowerSavingExtSwitch=LNR_SLOT_DATA_CONV_COORD_SW-1&SLOT_DATA_CONVERGENCE_COORD_SW-1;
```

Deactivation Command Examples

```
//LTE configurations  
//Disabling LTE and NR coordination for data convergence in slots  
MOD SYMBOLPWRSAVING: LocalCellId=0,  
SymbolShutdownEnhancedSw=LNR_SLOT_DATA_CONV_CO_SW-0;  
//Disabling LTE and NR bundled scheduling  
MOD SYMBOLPWRSAVING: LocalCellId=0,  
SymbolShutdownEnhancedSw=LNR_BUNDLED_SCHEDULING_SW-0, TrigBndlSchDIAvgPrbThld=0;  
//NR single-carrier scenarios  
//Disabling LTE and NR coordination for data convergence in slots  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,  
PowerSavingExtSwitch=LNR_SLOT_DATA_CONV_COORD_SW-0;  
//Disabling LTE and NR bundled scheduling  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=LNR_BUNDLED_SCHEDULING_SW-0;  
//Removing the LTE and NR bundled scheduling policy  
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0;  
//NR multi-carrier scenarios  
//Disabling LTE and NR coordination for data convergence in slots  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,  
PowerSavingExtSwitch=SLOT_DATA_CONVERGENCE_COORD_SW-0&LNR_SLOT_DATA_CONV_COORD_SW-0;  
//Disabling LTE and NR bundled scheduling  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=LNR_BUNDLED_SCHEDULING_SW-0;  
//Removing the LTE and NR bundled scheduling policy  
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0;
```

4.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.2.4.2 Activation Verification

Perform the following operations to check whether the LTE and NR bundled scheduling mode has taken effect:

LTE side:

1. Run the **DSP SYMBOLPWRSAVING** command on the eNodeB when this function is being implemented.
2. Check the status of symbol power saving. If **LNR Bundled Scheduling Status** is **STARTED** in the command output, LTE and NR bundled scheduling has taken effect in the cell. If **LNR Bundled Scheduling Status** is **NOT_STARTED** in the command output, LTE and NR bundled scheduling has not taken effect in the cell. You can check the value of **LNR Bundled Scheduling Disable Cause** to determine the causes. [Table 4-5](#) lists the possible causes.
3. Check the status of symbol power saving. If **LNR Bundled Scheduling Status** is **LNR_SLOT_DATA_CONV_CO_STARTED** in the command output, LTE and NR coordination for data convergence in slots has taken effect in the cell. Otherwise, this function does not take effect. For possible causes, see [Table 4-5](#).

NR side:

- Method 1:
 - a. Run the **DSP NRDUCELL** command on the gNodeB when this function is being implemented.
 - b. Query the cell status. If the value of **Power Saving State** is **LTE and NR Bundled Scheduling:Enabled**, LTE and NR bundled scheduling has taken effect. If the value of **Power Saving State** is **LTE and NR Bundled Scheduling:Disabled**, LTE and NR bundled scheduling has not taken effect. For possible causes, see [Table 4-5](#).
 - c. Query the cell status. If the value of **Power Saving State** is **LNR Data Convergence in Slots Coord:Enabled**, the LTE and NR coordination for data convergence in slots function has taken effect. If the value of **Power Saving State** is **Data Convergence in Slots Coordination:Enabled**, the coordination for data convergence in slots function in NR multi-carrier scenarios has taken effect. If the preceding functions do not take effect, the possible causes are listed in [Table 4-5](#).
- Method 2:

Run the **DSP NRDUCELLPOWERSAVING** command to query the status of the LTE and NR Bundled Scheduling feature.

 - If the value of **Power Saving Type** parameter is **LTE and NR Bundled Scheduling** and that of **Current State** is **Enabled**, LTE and NR bundled scheduling has taken effect.
 - If the value of **Power Saving Type** parameter is **LTE and NR Bundled Scheduling** and that of **Current State** is **Disabled**, LTE and NR bundled scheduling has not taken effect. You can check the value of **Current Disabled Cause** to determine the cause. For possible causes, see [Table 4-5](#).

Table 4-5 Possible cause values why LTE and NR bundled scheduling and LTE and NR coordination for data convergence in slots fail to take effect

Cause Value	Description
SWITCH_OFF	The function switch for the cell has not been turned on.

Cause Value	Description
CELL_UNAVAILABLE	The cell is blocked or deactivated, fails to be set up, or is unavailable due to other reasons.
NOT_IN_SHUTDOWN_TIME	The current time is not within the time range specified for this function. This cause applies only to NR.
IMPACTED_FEATURE_TAKEN_EFFECT	This cause applies only to NR.
DEPENDENT_FEATURE_DISABLED	- Multi-RAT coordinated symbol power saving does not take effect. - Energy saving based on proactive scheduling does not take effect. - Power saving scheduling does not take effect. - LTE and NR bundled scheduling is not enabled. As a result, LTE and NR coordination for data convergence in slots does not take effect.
NOT_IN_DL_PRB_THRESHOLD	The downlink PRB usage of the cell does not reach the threshold for triggering this function.
OTHERS	- The downlink PRB usage of other co-channel cells does not reach the threshold for triggering this function. - The time for triggering this function for other co-channel cells is not reached. - The function switch is not turned on for other co-channel cells. - Internal software or other reasons cause the function not to take effect.

4.2.4.3 Network Monitoring

Compare the values of the following counters one week before and after the LTE and NR Bundled Scheduling feature is enabled to evaluate the energy saving gains.

- Counters measuring the energy consumption of the entire base station

Counter ID	Counter Name
1593835637	VS.EnergyCons.BTS.Adding.LTE
1593835959	VS.EnergyCons.BTS.Adding.Service.LTE
1593835804	VS.EnergyCons.BTS.Adding.NR
1593835961	VS.EnergyCons.BTS.Adding.Service.NR

- Counter measuring the energy consumption of RF modules

Counter ID	Counter Name
1593835836	VS.EnergyCons.BTSBoard

- Counter measuring symbol power saving duration

Counter ID	Counter Name
1911823033	N.PowerSaving.SymbolShutdown.Dur

 NOTE

- It is expected that the symbol power saving duration increases after LTE and NR bundled scheduling is enabled.
- The policy of counter measurement upon carrier shutdown is determined by the function of energy saving counter optimization upon carrier shutdown, which is controlled by the **COUNTER_OPT_IN_CARROFF_SW** option of the **gNBPwrSavingAlgo.PwrSavingAlgoSwitch** parameter. Assume that symbol power saving takes place on a carrier that has already entered the shutdown state. If this option is deselected, the duration of symbol power saving is counted in the **N.PowerSaving.SymbolShutdown.Dur** counter. If this option is selected, this duration is not counted in this counter.

5 Space-Domain Energy Saving

5.1 LNR Coordinated Channel Shutdown

LOFD-001039/TDLOFD-001039 RF Channel Intelligent Shutdown and FOFD-021203 gNodeB Power Saving, which support RF channel shutdown, take effect only on their respective single-mode RF modules. This chapter describes LTE and NR coordinated channel shutdown, which is referred to as LNR coordinated channel shutdown hereinafter. In the following sections, LTE refers to LTE FDD and LTE TDD, and NR refers to NR FDD, NR SDL, and NR TDD.

5.1.1 Principles

LNR coordinated channel shutdown enables LTE and NR cells to negotiate the RF channel combinations with the maximum number of PAs that can be shut down, and to shut down their respective RF channels when they have no services or a low service load during a specified period.

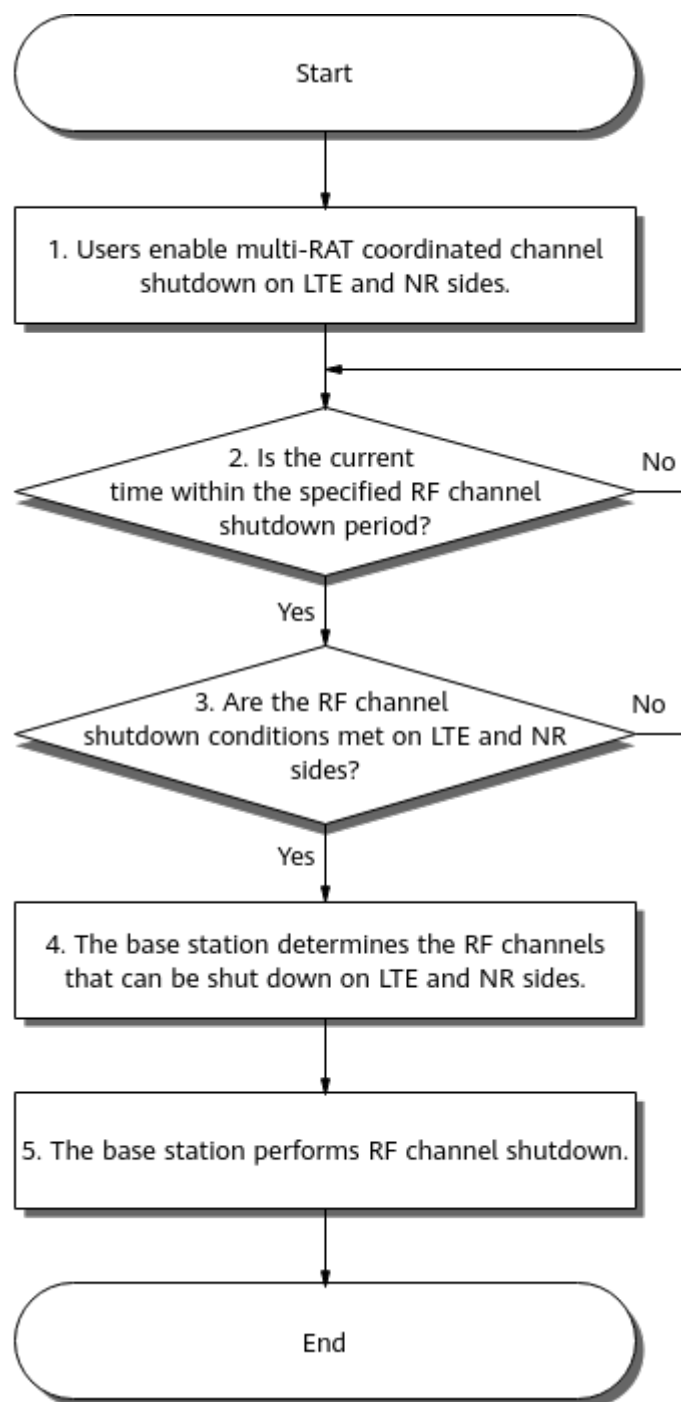
- On the NR side, this feature is controlled by the **MULTI_RAT_RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter.
- On the LTE side, this feature is controlled by the **CellRfShutdown.MultiRatJointChnShutdownSw** parameter.

5.1.1.1 Entering LNR Coordinated Channel Shutdown

LNR Coordinated Channel Shutdown Procedure

Figure 5-1 shows the LNR coordinated channel shutdown procedure.

Figure 5-1 LNR coordinated channel shutdown procedure



The procedure is described as follows:

1. The **CellRfShutdown.MultiRatJointChnShutdownSw** parameter on the LTE side is set to **ON** and the **MULTI_RAT_RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter on the NR side is selected.
2. The base station checks whether the current time is within the shutdown period for the LTE and NR cells.

If so, the base station proceeds with 3. If not, the base station continues to check whether the time is within the shutdown period for the cells.

- On the LTE side, the start time and end time are specified by the **CellRfShutdown.StartTime** and **CellRfShutdown.StopTime** parameters, respectively.
 - On the NR side, the start time and end time are specified by the **NRDUCellPowerSaving.StartTime** and **NRDUCellPowerSaving.StopTime** parameters, respectively. The **NRDUCellPowerSaving.PowerSavingType** parameter needs to be set to **MULTI_RAT_RF_SHUTDOWN**.
3. The base station checks whether the following shutdown conditions are met on the LTE and NR sides.
- a. The base station checks whether all the following conditions are met in the LTE cell. If so, the base station proceeds with 4. If not, the base station continues to check whether all the following conditions are met in the LTE cell.
- The cell and the number of UEs in the cell meet the following conditions:
 - The cell is a cell served by a macro base station with a downlink bandwidth greater than 5 MHz.
 - The number of RRC_CONNECTED UEs in the cell is less than or equal to the **CellRfShutdown.UENumThd** parameter value.
 - The PRB usage of the cell meets all of the following conditions:
 - The uplink PRB usage of the cell is less than or equal to the **CellRfShutdown.UlPrbThd** parameter value.
 - The downlink PRB usage of the cell is less than or equal to the **CellRfShutdown.DlPrbThd** parameter value.
 - The CRS transmit power can be increased by 3 dB.
After RF channels are shut down, LTE CRS transmit power needs to be increased to ensure LTE coverage.
 - The cell is not within a penalty period.
To prevent ping-pong RF channel shutdown, the base station starts a penalty timer on the LTE side if the cell meets specific conditions upon exit from the RF channel shutdown state. The cell will not re-enter the RF channel shutdown state during this penalty period, even if the preceding triggering conditions are met. The penalty period is specified by the **CellRfShutdown.PenaltyDuration** parameter. It is recommended that this parameter be set to **1** for LNR coordinated channel shutdown.
- b. The granularity of the evaluation period on the NR side is specified by the **NRDUCellPowerSaving.EvalPeriodGranularity** parameter.
- If this parameter is set to **MINUTE_LEVEL**, the granularity is minute-level. In this case, the evaluation period is specified by the **NRDUCellPowerSaving.ShutdownTrigJudgePeriod** parameter.

- If this parameter is set to **SECOND_LEVEL**, the granularity is second-level. In this case, the evaluation period is specified by the **NRDUCellPowerSaving.SecondLevelTrigEvalPrd** parameter. For LNR coordinated channel shutdown, the **NRDUCellPowerSaving.SecondLevelTrigEvalPrd** parameter must be set to a value greater than or equal to 3.

The base station periodically checks whether an NR cell enters the low load state based on the preceding parameter value. If all the following conditions are met, the base station proceeds with 4. If not, the base station continues to check whether the following conditions are met in the NR cell.

- Average downlink cell PRB usage $\leq$ **NRDUCellPowerSaving.DlPrbThld**
- Number of RRC_CONNECTED UEs in the cell $\leq$ **NRDUCellPowerSaving.UserNumThld**
 - The number of RRC_CONNECTED UEs in an SUL cell with co-deployment is the sum of the **N.User.RRCCConn.Avg** and **N.User.SUL.UL.Avg** values.
 - The number of RRC_CONNECTED UEs in any other cell is indicated by **N.User.RRCCConn.Avg**.

4. The base station determines the RF channels that can be shut down for both the LTE and NR sides.

When both LTE and NR meet the RF channel shutdown conditions, the RF channel combinations with the maximum number of PAs that can be shut down are selected from the RF channel combinations of the cells served by the same RF modules used by the two RATs. In this way, the number of RF channels shutdown reaches the maximum value, achieving maximum energy saving gains.

5. The base station shuts down the RF channels.

When a cell enters the RF channel shutdown state, transmit power compensation is performed for the common channels of the cell to ensure the base station coverage and service provisioning. The transmit power compensation value can be automatically determined or manually configured, depending on the setting of the **NRDUCellPowerSaving.CovCompensationMode** parameter.

- If this parameter is set to **AUTO**, a macro base station automatically performs transmit power compensation for common channels in the cell, but a LampSite base station does not.
- If this parameter is set to **MANUAL**, the transmit power compensation value for the common channels of the cell served by a macro or LampSite base station is manually configured using the **NRDUCellPowerSaving.PwrCompensationOffset** parameter. However, the power compensation value cannot exceed the RF module's maximum capability of aggregating the cell common channel transmit power. If the configured power compensation value exceeds the RF module's maximum capability, the latter takes effect.

LNR Coordinated Channel Shutdown Example

For example, a 64T LTE TDD massive MIMO cell (SCENARIO_3) and a 64T NR cell (SCENARIO_3) are deployed on a 64T64R AAU.

Table 5-1 Example of combinations of RF channels to be shut down for LTE and NR cells

Combination ^a	Tx0 to Tx3 Tx16 to Tx19 Tx32 to Tx35 Tx48 to Tx51	Tx4 to Tx7 Tx20 to Tx23 Tx36 to Tx39 Tx52 to Tx55	Tx8 to Tx11 Tx24 to Tx27 Tx40 to Tx43 Tx56 to Tx59	Tx12 to Tx15 Tx28 to Tx31 Tx44 to Tx47 Tx60 to Tx63
LTE combination 1	Not shut down	Shut down	Not shut down	Shut down
NR combination 1	Shut down	Not shut down	Shut down	Not shut down
NR combination 2	Not shut down	Shut down	Not shut down	Shut down
a: According to the RF channel shutdown combination principles for the LTE cell, combination 1 is available on the LTE side. According to the RF channel shutdown combination principles for the NR cell, combinations 1 and 2 are available on the NR side.				

According to the coordinated RF channel shutdown principles, LTE combination 1 and NR combination 2 are selected.

If a RAT in the RF channel shutdown state receives a request with new combinations for RF channel shutdown, the RAT exits this state and then enters this state again using the new combination.

5.1.1.2 Exiting LNR Coordinated Channel Shutdown

LNR coordinated channel shutdown is exited when any of the following conditions is met on the LTE or NR side, and services of an involved RAT are restored after the PRB usage for the services exceeds the sum of the corresponding PRB usage threshold and offset.

NOTE

When the coordinated channel shutdown switch on one side is turned off, the shutdown state on the other side is not affected.

Conditions for LTE to Exit RF Channel Shutdown

- The energy saving period specified by the **CellRfShutdown.StartTime** and **CellRfShutdown.StopTime** parameters has elapsed.

- The eNodeB detects that the common channel transmit power is insufficient.
- Faults occur on RF channels not shut down.
- The number of RRC_CONNECTED UEs in the cell is greater than the **CellRfShutdown.UENumThd** parameter value.
- The uplink PRB usage of the cell is greater than the sum of the **CellRfShutdown.UlPrbThd** and **CellRfShutdown.UlPrbOffset** parameter values. The **CellRfShutdown.UlPrbOffset** parameter indicates the offset to the uplink PRB usage threshold and is used to prevent a cell from frequently entering and exiting the RF channel intelligent shutdown state.
- The downlink PRB usage of the cell is greater than the sum of the **CellRfShutdown.DlPrbThd** and **CellRfShutdown.DlPrbOffset** parameter values. The **CellRfShutdown.DlPrbOffset** parameter indicates the offset to the downlink PRB usage threshold and is used to prevent a cell from frequently entering and exiting the RF channel intelligent shutdown state.

Conditions for NR to Exit RF Channel Shutdown

The **NRDUCellPowerSaving.EvalPeriodGranularity** parameter specifies the granularity of the exit condition evaluation period for a cell.

- If this parameter is set to **MINUTE_LEVEL**, the granularity is minute-level. In this case, the evaluation period is specified by the **NRDUCellPowerSaving.ShutdownTrigJudgePeriod** parameter.
- If this parameter is set to **SECOND_LEVEL**, the granularity is second-level. In this case, the evaluation period is specified by the **NRDUCellPowerSaving.SecondLevelExitEvalPrd** parameter.

The base station periodically checks whether the cell meets any of the conditions listed below. If so, the cell exits the LNR coordinated channel shutdown state.

- The energy saving period specified by the **NRDUCellPowerSaving.StartTime** and **NRDUCellPowerSaving.StopTime** parameters has elapsed.
- The downlink PRB usage of the cell is greater than the sum of the **NRDUCellPowerSaving.DlPrbThld** and **NRDUCellPowerSaving.DlPrbOffset** parameter values. The **NRDUCellPowerSaving.DlPrbOffset** parameter indicates the offset to the downlink PRB usage threshold and is used to prevent a cell from frequently entering and exiting the RF channel intelligent shutdown state.
- The number of RRC_CONNECTED UEs in the cell is greater than the sum of the **NRDUCellPowerSaving.UserNumThld** and **NRDUCellPowerSaving.UserNumOffset** parameter values. The **NRDUCellPowerSaving.UserNumOffset** parameter indicates the offset to the number of RRC_CONNECTED UEs and is used to prevent a cell from frequently entering and exiting the RF channel intelligent shutdown state.

After a cell exits the Multi-RAT Coordinated Channel Shutdown state, it cannot enter this state again within the penalty period specified by the **NRDUCellPowerSaving.PwrSavingPenaltyTime** parameter. This prevents the cell from frequently entering and exiting the state.

5.1.2 Network Analysis

5.1.2.1 Benefits

Some RF channels can be shut down for both LTE and NR cells sharing RF modules, to reduce power consumption when the traffic volume is low in the energy saving period.

Compared with LTE-only or NR-only RF channel intelligent shutdown, LNR coordinated channel shutdown enables the same RF channels to be shut down for multiple RATs, maximizing energy saving gains.

To prevent energy saving-triggered shutdown from affecting the lifespan of RF modules, the RF modules may exit the energy saving state when their own temperature differential exceeds a specified threshold within 24 hours. Energy saving gains decrease in this case.

5.1.2.2 Impacts

The impacts between this function and other functions are described in [5.1.3.2.3 Function Impacts](#).

- Impact on the LTE network: For details, see section "RF Channel Intelligent Shutdown" in *Energy Conservation and Emission Reduction in eRAN Feature Documentation*.
- Impact on the NR network: For details, see section "RF Channel Intelligent Shutdown" in *Energy Conservation and Emission Reduction in 5G RAN Feature Documentation*.
- After LNR coordinated channel shutdown mutes some transmit channels in 2T or 4T scenarios where Hybrid DSS Based on Asymmetric Bandwidth is enabled, the PDCCH allocation success rate and downlink UE throughput on the NR side may slightly decrease, and the uplink and downlink scheduling delays may slightly increase.

5.1.3 Requirements

5.1.3.1 Licenses

RAT	Feature ID	Feature Name	Model	Sales Unit
LTE FDD	MRFD-161224	Multi-RAT Coordinated Channel Shutdown (LTE FDD)	LT1S0MRJCS00	Per Cell
LTE TDD	MRFD-170234	Multi-RAT Coordinated Channel Shutdown (LTE TDD)	LT4SMRCCSTD	Per Cell

RAT	Feature ID	Feature Name	Model	Sales Unit
NR FDD NR TDD	MRFD-170264	Multi-RAT Coordinated Channel Shutdown (NR)	NR0SMRCCSD 00	Per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

5.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.1.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	RF channel intelligent shutdown	CellRfShutdown.RfShutdownSwitch	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	For LTE cells that support LNR coordinated channel shutdown, LTE RF channel intelligent shutdown must also be enabled for LNR coordinated channel shutdown to take effect.
LTE FDD LTE TDD	BBU-RRU joint clipping	BBU_RRU_JOINT_CLIP_SW option of the ENodeBAalgoSwitch.HwCapabilityExtSwitch parameter	None	The BBU-RRU joint clipping function must be enabled before the LNR coordinated channel shutdown function is enabled.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-freque ncy NR TDD	RF channel intelligent shutdown	RF_SHUTDOWN_S W option of the NRDUCellAlgoSwit ch.PowerSavingSwi tch parameter	<i>Energy Conservat ion and Emission Reduction in 5G RAN Feature Document ation</i>	For NR cells that support LNR coordinated channel shutdown, NR RF channel intelligent shutdown must also be enabled for LNR coordinated channel shutdown to take effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	PDCCH power check	CellPdcchAlgo.PdccbPowerCheckSwitch set to SYMBOL_0_POWER_CHECK	None	To enable LNR coordinated channel shutdown in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, you should also enable PDCCH power check. Enabling PDCCH power check affects the CCE resource allocation success rate for uplink and downlink scheduling in each TTI. As a result, the number of UEs scheduled in each TTI decreases, and the cell traffic volume and spectral efficiency decrease. In this case, set the CellPdcchAlgo.PdccbPowerCheckOptSwitch parameter to SYMBOL_0_POWER_OPT_CHECK to enable PDCCH symbol 0 power check optimization, which reduces the failure rate of CCE resource allocation for uplink and downlink scheduling in each TTI.

5.1.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	OTDOA-based positioning	OtdoaSwitch option of the ENodeBAlgoSwitch . LcsSwitch parameter and the NB_OTDOA_SWITCH option of the CellLcsCfg . CellLcsSwitch parameter	<i>LCS in eRAN Feature Documentation</i>	None
LTE FDD LTE TDD	Load simulation	CellSimuLoad MO configured	None	When at least one CellSimuLoad MO is configured, LNR coordinated channel shutdown does not take effect.
LTE FDD LTE TDD	Service guarantee for UEs running unlimited services	UsUeGuaranteeSwitch , SfCtrlAsScell , and SfCtrlAsInterNcell options of the CellUSParaCfg . UsAlgoSwitch parameter	None	When the UsUeGuaranteeSwitch option is selected and either the SfCtrlAsScell or SfCtrlAsInterNcell option is selected, LNR coordinated channel shutdown does not take effect.

RAT	Function Name	Function Switch	Reference	Description
LTE TDD	Massive MIMO	None	<i>Massive MIMO Basics (TDD) in eRAN Feature Documentation</i>	When LNR coordinated channel shutdown is enabled, the BfAnt.CoverageScenario parameter cannot be set to SCENARIO_13, EXPANDSCENARIO_1, EXPANDSCENARIO_2, or EXPANDSCENARIO_3 in 32T32R massive MIMO scenarios. When LNR coordinated channel shutdown is enabled, the BfAnt.CoverageScenario parameter cannot be set to EXPANDSCENARIO_1, EXPANDSCENARIO_2, or EXPANDSCENARIO_3 in 64T64R massive MIMO scenarios.
LTE TDD	Zero Guard Band Between Contiguous Intra-Band Carriers	CONTIG_INTRA_BAND_CARR_SW option of the ContigIntraBandCarr.ContigIntraBandCarrSw parameter	<i>Seamless Intra-Band Carrier Joining (TDD) in eRAN Feature Documentation</i>	None

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	On-demand TX power allocation under EME	PDSCHCfg.EmfPowerLimitSwitch	<i>On-Demand TX Power Allocation Under EME in eRAN Feature Documentation</i>	If the PDSCHCfg.CellPowerLimit parameter is set to a value greater than 0, the maximum cell transmit power is limited. In this case, on-demand TX power allocation under EME is incompatible with LNR coordinated channel shutdown.
LTE FDD	Hybrid DSS Based on Asymmetric Bandwidth	SpectrumCloud.SpectrumCloudSwitch set to LTE_NR_SPECTRUM_SHR , and the LNR_SPECTRUM_SHR_ASYM_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter selected	<i>LTE FDD and NR Hybrid Dynamic Spectrum Sharing</i>	When Hybrid DSS Based on Asymmetric Bandwidth is enabled, LNR coordinated channel shutdown can take effect in 2T and 4T scenarios (only when the versions of NR and LTE base stations are V100R020C10 or later), but cannot take effect in 8T or massive MIMO scenarios. For more requirements, see 5.1.3.4 Cells .
LTE FDD	Static Multiple Beam	SectorSplitGroup.BeamTemplateId set to a value greater than 1	<i>Massive MIMO Basics (FDD) in eRAN Feature Documentation</i>	LNR coordinated channel shutdown is not supported in FDD 16T16R massive MIMO scenarios where Static Multiple Beam is enabled.

RAT	Function Name	Function Switch	Reference	Description
NR FDD	Hybrid DSS Based on Asymmetric Bandwidth	LTE_NR_FDD_SPCT_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter selected, and LTE_NR_FDD_SPCT_SHR_ASYM_SW option of the NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch parameter selected	<i>LTE FDD and NR Hybrid Dynamic Spectrum Sharing</i>	When Hybrid DSS Based on Asymmetric Bandwidth is enabled, LNR coordinated channel shutdown can take effect in 2T and 4T scenarios (only when the versions of NR and LTE base stations are V100R020C10 or later), but cannot take effect in 8T or massive MIMO scenarios. For more requirements, see 5.1.3.4 Cells .
NR FDD Low-frequency NR TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	The period configured for the LNR coordinated channel shutdown function cannot overlap with that configured for the timing carrier shutdown function of the local cell. Set the period for the LNR coordinated channel shutdown function based on the network plan.
Low-frequency NR TDD	Supplementary Cell Indicator	NRDUCell.SupplementaryCellInd	None	LNR coordinated channel shutdown is not supported when NRDUCell.Duplex Mode is set to CELL_TDD and the value of the function parameter is INTRA_FREQ_ASSISTANT_CELL .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	When distributed massive MIMO is enabled, LNR coordinated channel shutdown cannot be enabled.
Low-frequency NR TDD	Dual-BBU backup in dual-network redundancy networking	DUAL_BBU_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	"Dual-BBU backup in dual-network redundancy networking" is mutually exclusive with "LNR coordinated channel shutdown".
NR FDD Low-frequency NR TDD	Primary/Backup function in intra-frequency dual-network redundancy networking	PRIMARY_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	"Primary/Backup function in intra-frequency dual-network redundancy networking" is mutually exclusive with "LNR coordinated channel shutdown".
NR FDD	Dual-plane function in intra-frequency dual-network redundancy networking	DUAL_PLANE_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	"Dual-plane function in intra-frequency dual-network redundancy networking" is mutually exclusive with "LNR coordinated channel shutdown".

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios	DUAL_PLANE_TOB_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	<i>Base Station Equipment Reliability</i>	"Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" is mutually exclusive with "LNR coordinated channel shutdown".

5.1.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Power control	None	<i>Power Control in eRAN Feature Documentation</i>	LNR coordinated channel shutdown increases the RS transmit power.
LTE FDD LTE TDD	MIMO	None	<i>MIMO in eRAN Feature Documentation</i>	The base station does not adjust the transmission mode for UEs whose transmission mode has already been adjusted after LNR coordinated channel shutdown is triggered, and keeps the transmission mode until the RF channels are shut down.

RA T	Function Name	Function Switch	Reference	Description
LTE FD D LTE TD D	Intelligent power-off of carriers in the same coverage	CellShutdown.Cel lShutdownSwitch set to ON	<i>Energy Conservatio n and Emission Reduction in eRAN Feature Documenta tion</i>	A cell in the LNR coordinated channel shutdown state will exit this state and enter the intelligent power-off of carriers in the same coverage state when the cell fulfills the conditions for triggering intelligent power-off of carriers in the same coverage.
LTE FD D LTE TD D	Multi-carrier coordinated energy saving	CellShutdown.Cel lShutdownSwitch set to ON_MULTI_CARRI ER_HIER_SHUTDO WN or ON_MUL_CARR_H IER_MANUAL_SH UTDN	<i>Energy Conservatio n and Emission Reduction in eRAN Feature Documenta tion</i>	A cell in the LNR coordinated channel shutdown state will exit this state and enter the multi-carrier coordinated energy saving state when the cell fulfills the conditions for triggering multi-carrier coordinated energy saving.
LTE FD D LTE TD D	Cell Switch-off Based on Automatic Co-coverage Identification	CellShutdown.Cel lShutdownSwitch set to ON_CO_COV_AUT O_IDENT	<i>Energy Conservatio n and Emission Reduction in eRAN Feature Documenta tion</i>	A cell in the LNR coordinated channel shutdown state will exit this state and enter the Cell Switch-off Based on Automatic Co-coverage Identification state when the cell fulfills the conditions for triggering Cell Switch-off Based on Automatic Co-coverage Identification.
LTE FD D LTE TD D	pRRU deep dormancy in intelligent mode	EnodebMpruEs.M pruDormancyDIE arfcn	<i>Multi-RAT Coordinated Energy Saving</i>	When the triggering conditions for both pRRU deep dormancy in intelligent mode and LNR coordinated channel shutdown are met, only pRRU deep dormancy takes effect.

RA T	Function Name	Function Switch	Reference	Description
LTE FD D LTE TD D	ETWS	None	<i>ETWS in eRAN Feature Documentation</i>	If the CellRfShutdown.RsPwrAdjOffset parameter is set to a value other than 0, the number of RBs available for downlink scheduling decreases because the RS transmit power increases. In this case, the maximum allowed size of warning notifications that the cell can broadcast will be reduced. Any warning notification larger than the maximum allowed size will fail to be broadcast to UEs.
LTE FD D LTE TD D	CMAS	CellAlgoSwitch.CmasSwitch	<i>CMAS in eRAN Feature Documentation</i>	
LTE FD D LTE TD D	Smart carrier selection based on virtual grids	SMART_CARRIER_SELECTION_SW option of the MultiCarrUnifiedSch.MultiCarrierUnifiedSchSw parameter	<i>Multi-carrier Unified Scheduling in eRAN Feature Documentation</i>	When LNR coordinated channel shutdown is triggered in the local cell, sample collection of virtual grids for neighboring cells is affected.
LTE FD D	Channel calibration	Joint channel calibration: RruJointCalParaCfg.TxChnCalSwitch	<i>MIMO in eRAN Feature Documentation</i>	For a cell in the LNR coordinated channel shutdown state, if channel calibration fails, other functions that depend on channel calibration may not be properly provided. In addition, a message indicating a failure will be displayed on the OM interface for channel calibration.

RA T	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	eMTC introduction	CellEmtcAlgo.EmtcAlgoSwitch	<i>eMTC in eRAN Feature Documentation</i>	When eMTC is enabled, the overhead of eMTC system information will increase the power insufficiency probability. LNR coordinated channel shutdown is less likely to be triggered, affecting the energy saving gains.
LTE FDD LTE TDD	Carrier aggregation	CaMgtCfg.CellCaAlgoSwitch	<i>Carrier Aggregation in eRAN Feature Documentation</i>	If carrier aggregation is enabled after LNR coordinated channel shutdown takes effect, the cell is likely to exit the RF channel shutdown state. As a result, the energy saving duration and energy saving gains decrease.
LTE FDD NR FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	LTE: SpectrumCloud.SpectrumCloudSwitch set to LTE_NR_SPECTRUM_SHR NR: LTE_NR_FDD_SPECTRUM_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>	For FDD massive MIMO modules, LNR coordinated channel shutdown does not take effect after LTE FDD and NR Flash Dynamic Spectrum Sharing is enabled.
Low-frequency NR TDD	Remote interference adaptive avoidance	RIM_ADAPT_AVOID_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD) in 5G RAN Feature Documentation</i>	LNR coordinated channel shutdown does not take effect when remote interference adaptive avoidance is in effect.

RA T	Function Name	Function Switch	Reference	Description
NR FD D Lo w- freq uen cy NR TD D	Intra-base- station DL CoMP	INTRA_GNB_DL_J T_SW option of the NRDUCellAlgoSwi tch.CompSwitch parameter	<i>CoMP in 5G RAN Feature Documenta tion</i>	When LNR coordinated channel shutdown takes effect in a cell, intra- base-station DL CoMP stops for UEs that have been involved in intra- base-station DL CoMP in the cell and will not take effect on UEs that are newly admitted to this cell. After the cell exits the LNR coordinated channel shutdown state, intra-base-station DL CoMP takes effect again.
NR FD D Lo w- freq uen cy NR TD D	3D coverage pattern	None	<i>Beam Manageme nt in 5G RAN Feature Documenta tion</i>	In customized beam scenarios, if the NRDUCellTrpCust- Beam.SsbBeamHorizWi dth parameter is set to DEFAULT , the coverage cannot be completely aligned before and after RF channel shutdown takes effect. This affects the number of UEs in a cell, RRC Setup Success Rate , handover success rate, and Service Drop Rate .

RA T	Function Name	Function Switch	Reference	Description
NR FD D	8T8R cell managem ent	NRDUCellTrp.TxRxMode set to 8T8R	<i>Smart 8T8R (FDD) in 5G RAN Feature Documenta tion</i>	LNR coordinated channel shutdown does not take effect in 8T8R cells if antennas of either of the following types are used: (1) SDAs; (2) SBS antennas for which the equivalent array spacing for the working frequencies is 0.5 times the wavelength. LNR coordinated channel shutdown does not take effect in 8T8R cells if both of the following conditions are met: (1) SBS antennas for which the equivalent array spacing for antenna working frequencies is 0.7 times the wavelength are used. (2) Four SSB beams are used in the 8T8R cells.
LTE FD D NR FD D	DSS inter- RAT PDCCH spatial multiplexi ng	LTE: DSS_INTER_RAT_P DCCH_SPC_MUX_ SW option of the SpectrumCloud.S pectrumCloudEnh Switch parameter NR: DSS_INTER_RAT_P DCCH_SPC_MUX_ SW option of the NRDUCellAlgoSwi tch.SpectrumClou dEnhSwitch parameter	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>	DSS inter-RAT PDCCH spatial multiplexing takes effect in a lower proportion of cases if it is enabled together with LNR coordinated channel shutdown. As a result, the gains provided by these two functions will decrease.

RA T	Function Name	Function Switch	Reference	Description
Lo w- freq uen cy NR TDD	Co-MM	INTRA_GNB_SU_C OMM_SW option of the NRDUCellCollabs erv.MultiCellMim oSwitch parameter	<i>Co-MM (Low- Frequency TDD)</i>	When LNR coordinated channel shutdown takes effect in a cell, Co-MM stops taking effect for UEs involved in Co-MM in the cell and will not take effect for UEs that are newly admitted to this cell. After the cell exits the LNR coordinated channel shutdown state, Co-MM takes effect again.
LTE FDD LTE TDD	Dynamic Power Sharing Between LTE Carriers	CellDynPowerSha ring.DynamicPow erSharingSwitch	<i>Dynamic Power Sharing Between LTE Carriers in eRAN Feature Documenta tion</i>	Cells that have entered the LNR coordinated channel shutdown state do not participate in Dynamic Power Sharing Between LTE Carriers.
LTE FDD	Channel isolation for PIM interferenc e avoidance	PIM_INTERF_CHN _SHUTDOWN_SW option of the PimInterfSmartA void.PimAlgoSwit ch parameter	<i>Interference Detection and Suppression in eRAN Feature Documenta tion</i>	For LTE FDD, if either LNR coordinated channel shutdown or channel isolation for PIM interference avoidance has taken effect, the other function does not take effect.
LTE FDD	Intelligent PIM interferenc e avoidance	PIM_INTERF_SMA RT_AVOID_SW option of the PimInterfSmartA void.PimAlgoSwit ch parameter	<i>Interference Detection and Suppression in eRAN Feature Documenta tion</i>	For LTE FDD, LNR coordinated channel shutdown is terminated during PIM interference detection.
NR FDD	Channel isolation for PIM interferenc e avoidance	PIM_INTERF_SMA RT_AVOID_SW option of the NRDUCellPimIntr fAvd.PimAlgoSwit ch parameter	<i>Interference Avoidance in 5G RAN Feature Documenta tion</i>	For NR FDD, if either LNR coordinated channel shutdown or channel isolation for PIM interference avoidance has taken effect, the other function does not take effect.

RA T	Function Name	Function Switch	Reference	Description
NR FD D	Intelligent PIM interferenc e avoidance	PIM_INTERF_SMA RT_AVOID_SW option of the NRDUCellPimIntr- fAvd.PimAlgoSwit ch parameter	<i>Interference Avoidance in 5G RAN Feature Documenta tion</i>	For NR FDD, LNR coordinated channel shutdown is terminated when PIM interference detection takes effect.
LTE FD D LTE TD D	Enhanced symbol power saving	MBSFNShutDown Switch option of the CellAlgoSwitch.DI SchSwitch parameter	<i>Energy Conservatio n and Emission Reduction in eRAN Feature Documenta tion</i>	When LNR coordinated channel shutdown takes effect, enhanced symbol power saving does not take effect.
LTE FD D LTE TD D	Dynamic voltage adjustmen t	CellAlgoSwitch.D ynAdjVoltSwitch	<i>Energy Conservatio n and Emission Reduction in eRAN Feature Documenta tion</i>	When the triggering conditions for dynamic voltage adjustment and LNR coordinated channel shutdown are both met, LNR coordinated channel shutdown will take effect but dynamic voltage adjustment will not.
LTE FD D LTE TD D	Low power consumpti on mode (LTE)	CellLowPower.Lo wPwrSwitch	<i>Green Site</i>	When both LNR coordinated channel shutdown and low power consumption mode are enabled, low power consumption mode takes precedence.
NR FD D	Intelligent MCS optimizati on for uplink SU- MIMO	UL_SU_MCS_INTE L_OPT_SW option of the NRDUCellAiAlgo. AiAmcAlgoSwitch parameter	<i>Smart 8T8R (FDD) in 5G RAN Feature Documenta tion</i>	When intelligent MCS optimization for uplink SU-MIMO is enabled together with LNR coordinated channel shutdown, intelligent MCS optimization for uplink SU-MIMO does not take effect or its gains decrease.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD NR FDD	DSS PDCCH resource sharing optimization	- LTE: DSS_PDCCH_SHARING_OPTION option of the SpectrumCloudExtSwitch parameter - NR: DSS_PDCCH_SHARING_OPTION option of the NRDUCellPdcchPdcchAlgorithmSwitch parameter	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>	DSS PDCCH resource sharing optimization takes effect in a lower proportion of cases if it is enabled together with LNR coordinated channel shutdown. As a result, the gains from these two functions will decrease.

RA T	Function Name	Function Switch	Reference	Description
NR FD D Lo w- freq uen cy NR TD D	Speed turbo UE identificati on	gNodeBParam.St UelIdentSw	None	LNR coordinated channel shutdown fails to take effect in cells served by a base station, resulting in a decrease in the energy saving gains, when any of the following configuration conditions is met and the configured function takes effect:
NR FD D Lo w- freq uen cy NR TD D	gNodeB acceleratio n service application	• gNBAsAppParam.AppDomain Name • gNBAsAppParam.AppIpv4 • gNBAsAppParam.AppIpv6	None	• Any option of the gNodeBParam.StUelIdentSw parameter is selected • The gNBAsAppParam.AppDomainName parameter is set. • The gNBAsAppParam.AppIpv4 parameter is set. • The gNBAsAppParam.AppIpv6 parameter is set. The following methods can be used to address the preceding issue: • Select the SPEED_TURBO_PWR_SAVING_SW option of the gNBPwrSavingAlgo.PwrSavingAlgoSwitch parameter. LNR coordinated channel shutdown can take effect for cells served by the base station in some of the preceding scenarios within the period specified by the gNBPwrSavingAlgo.SpeedTurboPwrSav-

RAT	Function Name	Function Switch	Reference	Description
				<i>StartTime</i> and <i>gNBWPwrSavingAlgo.SpeedTurboPwrSavingEndTime</i> parameters, increasing the energy saving gains. • Select the <i>STATIC_ST_PWR_SAVING_SW</i> option of the <i>gNBWPwrSavingAlgo.PwrSavingAlgoSwitch</i> parameter. LNR coordinated channel shutdown can take effect for cells served by the base station in some of the preceding scenarios within the period specified by the <i>gNBWPwrSavingAlgo.StaticSTPwrSavStartTime</i> and <i>gNBWPwrSavingAlgo.StaticSTPwrSavEndTime</i> parameters, increasing the energy saving gains.
Low-frequency NR TDD	Multiple BWP1s for RedCap	<i>NRDUCellRedCap.RedCapUeMaxAvailableBandwidth > 20M</i>	<i>RedCap in 5G RAN Feature Documentation</i>	For a TDD cell, when LNR coordinated channel shutdown is enabled and at least one RedCap BWP1 with the NCD-SSB is activated in the cell, power aggregation is required for each RedCap BWP1. In this case, LNR coordinated channel shutdown does not take effect when SSB power aggregation is insufficient.

RA T	Function Name	Function Switch	Reference	Description
LTE FD D LTE TD D	LTE and NR Flash Dynamic Power Sharing	LTE_NR_DYN_PO WER_SHARING_S W option of the CellDynPowerSha ring.DynamicPow erSharingSwitch parameter	<i>LTE and NR Power Allocation in eRAN Feature Documenta tion</i>	Cells that have entered the LNR coordinated channel shutdown state do not participate in LTE FDD and NR Flash Dynamic Power Sharing.

5.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations) excluding the BTS5929R. 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

For LTE, only the following boards support this function.

Board Name	Board Type
UMPT	Main control board
UBBPd	Baseband processing unit
UBBPe	Baseband processing unit
UBBPf	Baseband processing unit
UBBPg	Baseband processing unit
UBBPh	Baseband processing unit
LBBPd	Baseband processing unit

RF Modules

- All low-frequency massive MIMO AAUs that are capable of both LTE and NR and meet the requirements listed in [Table 5-2](#) support this function. For details, see the technical specifications of AAUs in *3900 & 5900 Series Base Station Product Documentation*.

Table 5-2 Requirements of LNR coordinated channel shutdown for massive MIMO AAUs (with the default power offset used for NR common channels)

Model	TX/RX Mode	Frequency Band	Supported Beam Coverage Scenario ^a
AAU533 _x / AAU563 _x / AAU5831/ AAU5649	64T64R/ 32T32R	No requirements	DEFAULT/SCENARIO_1/ SCENARIO_2/ SCENARIO_3/ SCENARIO_4/ SCENARIO_6/ SCENARIO_7/ SCENARIO_8/ SCENARIO_9/ SCENARIO_12/ SCENARIO_13/ EXPAND_SCENARIO_1
AAU5639	64T64R	n41/n77/n78	DEFAULT/SCENARIO_1/ SCENARIO_2/ SCENARIO_3/ SCENARIO_4/ SCENARIO_6/ SCENARIO_7/ SCENARIO_8/ SCENARIO_9/ SCENARIO_12/ SCENARIO_13/ EXPAND_SCENARIO_1
AAU5639	64T64R	n40/n79	DEFAULT/SCENARIO_2/ SCENARIO_4/ SCENARIO_9
AAU531 _x / AAU561 _x / AAU581 _x	64T64R/ 32T32R	No requirements	SCENARIO_2/ SCENARIO_4/ SCENARIO_6/ SCENARIO_7/ SCENARIO_8/ SCENARIO_9/ SCENARIO_12/ SCENARIO_13/ EXPAND_SCENARIO_1
AAU5711a ^b	32T32R	n3	SCENARIO_210/ SCENARIO_240
AAU5711a ^b / AAU5733	32T32R	n7	SCENARIO_210
AAU5726 ^b	32T32R	n1/n66	DEFAULT/ SCENARIO_210/ SCENARIO_220/ SCENARIO_240

Model	TX/RX Mode	Frequency Band	Supported Beam Coverage Scenario ^a
AAU5726 ^b	32T32R	n2	SCENARIO_210/ SCENARIO_240
AAU5726e ^c	32T32R	n3	DEFAULT/ SCENARIO_210/ SCENARIO_220
AAU5726e ^c	32T32R	n1	DEFAULT/ SCENARIO_210/ SCENARIO_220
AAU5243/ AAU5942/ AAU3940/ AAU5940	4T4R	n1/n3/n41	DEFAULT
a: For NR TDD, the supported beam coverage scenario is specified by the NRDUCellTrpBeam.CoverageScenario parameter. For NR FDD, the supported beam coverage scenario is specified by the NRDUFddCellTrpBeam.CoverageScenario parameter. b: The AAU5711a/AAU5726 supports LNR coordinated channel shutdown only when Dynamic Massive Beam (with SectorSplitGroup.BeamTemplateId set to 1) is used on the LTE side and both the NR and LTE cells operate in 1.8 GHz. c: The AAU5726e does not support LNR coordinated channel shutdown when it is configured with cells operating in band 3, the SECTORSPLITGROUP.BeamDeploymentMode parameter is set to MODE_1, and the SECTORSPLITGROUP.BeamTemplateId parameter is set to 2 on the LTE side. d: For details about the configuration requirements on the LTE side in massive MIMO sector split scenarios, see <i>Energy Conservation and Emission Reduction</i> in <i>eRAN Feature Documentation</i>.			

- All RRUs that meet the requirements listed in [Table 5-3](#) support this function. For details, see the technical specifications of RRUs in *3900 & 5900 Series Base Station Product Documentation*.

Table 5-3 Requirements of LNR coordinated channel shutdown for RRUs

RA T	Model	TX/RX Mode	Supported Beam Coverage Scenario for NR
FD D	RRU5xxx supporting both LTE FDD and NR FDD Note: The NRDUCellPowerSaving.CovCompensation-Mode parameter is set to AUTO .	8T8R ^a / 4T4R/ 2T2R/ 2T4R/4T/ 2T	For FDD and SDL, the NRDUFddCellTrpBeam.CoverageScenario parameter must be set to DEFAULT , SCENARIO_201^b , or SCENARIO_208^e .

RA T	Model	TX/RX Mode	Supported Beam Coverage Scenario for NR
TD D	RRU5xxx ^c supporting both LTE TDD and NR TDD Note: The SSB offset does not exceed the RRU capability.	8T8R/ 4T4R/ 2T2R	For low-frequency TDD, the NRDUCellTrpBeam.Cover ageScenario parameter must be set to DEFAULT^d or EXPAND_SCENARIO_1 .
a: Among cells served by LTE FDD 8T8R non-combined RF modules, only 8T3S and 4T6S cells served by the RRU5862 support this function. b: The requirement of setting the NRDUfddCellTrpBeam.CoverageScenario parameter to SCENARIO_201 applies only to 8T8R RF modules. c: The RRU5254 does not support LNR coordinated channel shutdown. d: LNR coordinated channel shutdown takes effect in the default coverage scenario only when the X_CHN_SHUTDOWN_COMB_OPT_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter is selected. e: This value requires that an 8-channel antenna without a calibration port where the columns in the antenna array are vertically arranged (that is, RETSubunit.ANTENNAARRAYTYPE or ALDSubunit.ANTENNAARRAYTYPE is set to STACK.).			

NOTE

For 5xxx series RRUs, x indicates an integer ranging from 0 to 9.

- Hybrid DSS Based on Asymmetric Bandwidth is only supported by 2T or 4T RF modules.

Certain RF modules support only some scenarios. For details, see the following documents:

- Energy Conservation and Emission Reduction in eRAN Feature Documentation*. The specific section is "RF Channel Intelligent Shutdown."
- Energy Conservation and Emission Reduction in 5G RAN Feature Documentation*. The specific section is "RF Channel Intelligent Shutdown."

5.1.3.4 Cells

- For LTE cells, the transmit channels of the cells must not serve NB-IoT cells. For other requirements, see the cell requirements of the RF channel intelligent shutdown function in *Energy Conservation and Emission Reduction in eRAN Feature Documentation*.
- For details about the requirements for NR cells, see the cell requirements of the RF channel intelligent shutdown function in *Energy Conservation and Emission Reduction in 5G RAN Feature Documentation*. For details about the configuration requirements for RF modules, see [Table 5-4](#).

Table 5-4 RF module configuration requirements for NR cells

RAT	Number of Channels	RF Module Configuration Requirement
NR FDD	2T2R/4T4R/ 2T4R/32T32R	The total number of carriers configured on a single RF module must be less than or equal to 2.
NR SDL	2T/4T	The total number of carriers configured on a single RF module must be less than or equal to 2.
NR TDD	8T8R	The total number of carriers configured on a single RF module must be less than or equal to 2.
NR TDD	32T32R/ 64T64R	The total number of carriers configured on a single RF module must be less than or equal to 4. Among these carriers, the number of NR carriers must be less than or equal to 1, and the number of LTE carriers must be less than or equal to 3.

 NOTE

For an RF module supporting NR TDD, if its working mode is set to LNR but it is not configured with any LTE carrier, NR single-mode channel shutdown, instead of LNR coordinated channel shutdown, is recommended.

- 8T RF modules do not support LNR coordinated channel shutdown when the inter-antenna spacing is 0.5 times the wavelength.
- In scenarios where Hybrid DSS Based on Asymmetric Bandwidth is in effect, LNR coordinated channel shutdown is supported only when any of the bandwidth combinations listed in [Table 5-5](#) is used and the NR PDCCH and LTE PDCCH occupy different symbols (that is, all of the following conditions must be met). For details about the configuration of Hybrid DSS Based on Asymmetric Bandwidth, see *LTE FDD and NR Hybrid Dynamic Spectrum Sharing*.
 - The frequency-domain position of the NR common PDCCH is within the NR shared spectrum.
 - The **UE_PDCCH_FULL_BANDWIDTH_CFG_SW** option of the **NRDUCellPdcch.PdcchAlgoExtSwitch** parameter is selected.
 - When the **PDCCH_RATEMATCH_SW** option of the **NRDUCCELLPDSCH.RateMatchSwitch** parameter is selected, the frequency-domain position of the UE-level PDCCH does not overlap with the NR-dedicated spectrum.

Table 5-5 Cell bandwidth combinations to which Hybrid DSS Based on Asymmetric Bandwidth can be applied

LTE Cell Bandwidth	NR Cell Bandwidth
10	20
15	20
10	15

- For Hybrid DSS Based on Asymmetric Bandwidth, when the downlink bandwidth of the NR cell is set to 10 MHz or 15 MHz and the **NRDUCCELLCORESET.CommonCtrlResRbNum** parameter is set to **RB48**, the configuration requirements for related power parameters are as follows:
 - The value of **NRDUCCellChnPwr.MaxCommonDciPwrOffset** is less than or equal to -3.
 - The value of **NRDUCCellChnPwr.Msg2MaxPwrOffset** is less than or equal to -3.
 - The value of **NRDUCCellChnPwr.Msg4MaxPwrOffset** is less than or equal to -3.
 - The value of **NRDUCCellChnPwr.OsiMaxPwrOffset** is less than or equal to -3.
 - The value of **NRDUCCellChnPwr.PagingMaxPwrOffset** is less than or equal to -3.
 - The value of **NRDUCCellChnPwr.RmsiMaxPwrOffset** is less than or equal to -3.
 - The value of **NRDUCCellTrpCustBeam.MaxSsbPwrOffset** is less than or equal to -3.

5.1.3.5 Others

It is recommended that this function be enabled for all cells served by an RF module.

5.1.4 Operation and Maintenance

5.1.4.1 Data Configuration

5.1.4.1.1 Data Preparation

LTE Side

Parameter Name	Parameter ID	Setting Notes
Multi-RAT Joint Channel Shutdown Switch	CellRfShutdown.MultiRatJointChnShutdownSw	It is recommended that this switch be turned on for cells that share RF modules with an NR cell.

Parameter Name	Parameter ID	Setting Notes
Start time	CellRfShutdown.StartTime	Set this parameter to its recommended value. It is recommended that this parameter be set to the same value as that of the NRDUCellPowerSaving.StartTime parameter on the NR side.
Stop time	CellRfShutdown.StopTime	Set this parameter to its recommended value. It is recommended that this parameter be set to the same value as that of the NRDUCellPowerSaving.StopTime parameter on the NR side.
Reference signal power adjust offset	CellRfShutdown.RsPowerAdjOffset	Set this parameter to its recommended value.
Downlink PRB threshold	CellRfShutdown.DLPrbThd	It is recommended that both parameters be set to values less than or equal to 8 in LNR coordinated channel shutdown scenarios. If a parameter is set to a value greater than 8 , power insufficiency may occur after the LNR coordinated channel shutdown function takes effect, causing this function to stop taking effect.
Uplink PRB threshold	CellRfShutdown.UlPrbThd	

Parameter Name	Parameter ID	Setting Notes
Downlink PRB offset	CellRfShutdown.DlPrbOffset	Set these parameters to their respective recommended values. The exit threshold for LNR coordinated channel shutdown in the uplink or downlink equals the sum of the uplink or downlink PRB usage threshold and the associated offset. It is recommended that the exit thresholds for LNR coordinated channel shutdown be set to values less than or equal to 20. If the thresholds are set to values greater than 20, this function cannot exit because the actual uplink or downlink PRB usage cannot reach the respective exit threshold when the physical power is restricted after RF channels are shut down.
Uplink PRB offset	CellRfShutdown.UlPrbOffset	
UE Number Threshold	CellRfShutdown.UENumThd	- For cells with a bandwidth of 10 MHz or 15 MHz, the default value is recommended. - For cells with a bandwidth greater than 15 MHz: If this parameter is set to a larger value, cells are more likely to enter the RF channel intelligent shutdown state, saving more energy. - If energy saving is a major concern, set this parameter to a value up to 20. A larger value indicates better energy saving results. - If service stability is a major concern, set this parameter to a value no greater than 10. This parameter can be set to 0. In this case, RF channel intelligent shutdown can be triggered only when there are no RRC_CONNECTED UEs in the cell. This minimizes impacts on service stability. However, the energy saving gains are also minimal.

Parameter Name	Parameter ID	Setting Notes
PA for even power distribution	CellDlpcPdschPa.PaPcOff	It is recommended that this parameter be set to a value greater than or equal to -3 dB in LNR coordinated channel shutdown scenarios.
SignalCongregate-Level	CellPdcchAlgo.ComSigCongregLv	You are advised to set this parameter to its recommended value.
Code Rate for Random Access Response and Paging Message	CellDlscAlgo.RarAndPagingCR	It is recommended that this parameter be set to a value greater than or equal to 0.05 in LNR coordinated channel shutdown scenarios.
PCH power	CellChPwrCfg.PchPwr	It is recommended that this parameter be set to the CellDlpcPdschPa.PaPcOff parameter value plus 3 dB.
Downlink power control algorithm switch	CellAlgoSwitch.DlPcAlgoSwitch	It is recommended that the PhichPwrRestrictSwitch option be selected.
Penalty Duration	CellRfShutdown.PenaltyDuration	It is recommended that this parameter be set to 1 minute for LNR coordinated channel shutdown.
PDCCH Power Check Switch	CellPdcchAlgo.PdcchPowerCheckSwitch	To enable LNR coordinated channel shutdown in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, set this parameter to SYMBOL_0_POWER_CHECK .
PDCCH Power Check Optimization Switch	CellPdcchAlgo.PdcchPowerCheckOptSwitch	In LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, if the CellPdcchAlgo.PdcchPowerCheckSwitch parameter is set to SYMBOL_0_POWER_CHECK , set this parameter to SYMBOL_0_POWER_OPT_CHECK .

NR Side

Parameter Name	Parameter ID	Setting Notes
Power Saving Switch	NRDUCellAlgoSwitch. <i>PowerSavingSwitch</i>	Select the MULTI_RAT_RF_SHUTDOWN_SW option.
Start Time	NRDUCellPowerSav- ing.<i>StartTime</i>	You are advised to set this parameter to the time at which the light-traffic period begins. It is recommended that this parameter be set to the same value as that of the CellRfShutdown.<i>StartTime</i> parameter on the LTE side. The time period for LNR coordinated channel shutdown should not overlap with the time period specified when the NRDUCellPowerSav- ing.<i>PowerSavingType</i> parameter is set to TIMING_CARRIER_SHUTDOWN for the local cell.
Stop Time	NRDUCellPowerSav- ing.<i>StopTime</i>	You are advised to set this parameter to the time at which the light-traffic period ends. It is recommended that this parameter be set to the same value as that of the CellRfShutdown.<i>StopTime</i> parameter on the LTE side. The time period for LNR coordinated channel shutdown should not overlap with the time period specified when the NRDUCellPowerSav- ing.<i>PowerSavingType</i> parameter is set to TIMING_CARRIER_SHUTDOWN for the local cell.
Power Saving Type	NRDUCellPowerSav- ing.<i>PowerSavingType</i>	It is recommended that this parameter be set to MULTI_RAT_RF_SHUTDOWN , which indicates that the power saving policy in the corresponding period is LNR coordinated channel shutdown.
Downlink PRB Threshold	NRDUCellPowerSav- ing.<i>DLPrbThld</i>	The default value is recommended.
Downlink PRB Offset	NRDUCellPowerSav- ing.<i>DLPrbOffset</i>	The default value is recommended.

Parameter Name	Parameter ID	Setting Notes
UE Number Threshold	NRDUCellPowerSaving.UserNumThld	Set this parameter to its recommended value.
User Number Offset	NRDUCellPowerSaving.UserNumOffset	Set this parameter to its recommended value.
Coverage Compensation Mode	NRDUCellPowerSaving.CovCompensationMode	Retain the default value.
Power Compensation Offset	NRDUCellPowerSaving.PwrCompensationOffset	When the NRDUCellPowerSaving.CovCompensationMode parameter is set to MANUAL , set this parameter based on the network plan and the maximum aggregation capability of the RF modules for the common channel transmit power of the cell.
Evaluation Period Granularity	NRDUCellPowerSaving.EvalPeriodGranularity	You are advised to set this parameter to its recommended value.
Triggering Evaluation Period	NRDUCellPowerSaving.ShutdownTrigJudgePeriod	You are advised to set this parameter to its recommended value.
Second-Level Exit Evaluation Period	NRDUCellPowerSaving.SecondLevelExitEvalPrd	You are advised to set this parameter to its recommended value.
Second-Level Trigger Evaluation Period	NRDUCellPowerSaving.SecondLevelTrigEvalPrd	You are advised to set this parameter to its recommended value. For LNR coordinated channel shutdown, the NRDUCellPowerSaving.SecondLevelTrigEvalPrd parameter must be set to a value greater than or equal to 3 .

Parameter Name	Parameter ID	Setting Notes
Power Saving Penalty Duration	NRDUCellPowerSaving.PwrSavingPenalty-Time	When Multi-RAT Coordinated Channel Shutdown is enabled, - It is recommended that this parameter be set to its recommended value if the NRDUCellPowerSaving.EvalPeriodGranularity parameter is set to SECOND_LEVEL. - It is recommended that this parameter be set to 0 if the NRDUCellPowerSaving.EvalPeriodGranularity parameter is set to MINUTE_LEVEL.

5.1.4.1.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [5.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

On the LTE side

```
//Configuring the PA value
MOD CELLDLPCPDSCHPA: LocalCellId=0,PaPcOff=DB_3_P_A;
//Configuring the CCE aggregation level for common control signaling
MOD CELLPDCCHALGO: LocalCellId=0, ComSigCongregLv=CONGREG_LV4;
//Configuring the code rate for random access response messages and paging messages
MOD CELLDLSCHALGO: LocalCellId=0, RarAndPagingCR=117;
//Configuring the PCH power (to the PaPcOff parameter value plus 3 dB)
MOD CELLCHPWRCFG: LocalCellId=0, PchPwr=xxx;
//Turning on the PHICH power restriction switch
MOD CELLALGOSWITCH: LocalCellId=0,DlPcAlgoSwitch=PhichPwrRestrictSwitch-1;
//Enabling the BBU-RRU joint clipping function, activating LNR coordinated channel shutdown, and setting the penalty duration to 1 minute
MOD ENODEBALGOSWITCH: HwCapabilityExtSwitch=BBU_RRU_JOINT_CLIP_SW-1;
MOD CELLRFSDOWN: LocalCellId=0, RfShutdownSwitch=ON, StartTime=00&00&00, StopTime=06&00&00, RsPwrAdjOffset=3, DlPrbThd=8, UlPrbThd=8, DlPrbOffset=12, UlPrbOffset=12, UENumThd=5, MultiRatJointChnShutdownSw=ON, PenaltyDuration=1;
//Activating PDCCH power check and PDCCH symbol 0 power check optimization in LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios
MOD CELLPDCCHALGO: LocalCellId=0, PdcchPowerCheckSwitch=SYMBOL_0_POWER_CHECK, PdcchPowerCheckOptSwitch=SYMBOL_0_POWER_OPT_CHECK;
```

On the NR side

Scenario 1: Configure LNR coordinated channel shutdown with a minute-level evaluation period

```
//Activating LNR coordinated channel shutdown
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=RF_SHUTDOWN_SW-1;
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=MULTI_RAT_RF_SHUTDOWN_SW-1;
//Configuring the period for LNR coordinated channel shutdown
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0, StartTime=00&00&00,
StopTime=06&00&00, PowerSavingType=MULTI_RAT_RF_SHUTDOWN, DlPrbThld=10, DlPrbOffset=15,
UserNumThld=20, UserNumOffset=20, CovCompensationMode=AUTO,
EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=1, PwrSavingPenaltyTime=0;
```

(Optional) Scenario 2: Configure LNR coordinated channel shutdown with a second-level evaluation period

```
//Activating LNR coordinated channel shutdown
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=RF_SHUTDOWN_SW-1;
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=MULTI_RAT_RF_SHUTDOWN_SW-1;
//Configuring the period for LNR coordinated channel shutdown
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0, StartTime=00&00&00,
StopTime=06&00&00, PowerSavingType=MULTI_RAT_RF_SHUTDOWN, DlPrbThld=10, DlPrbOffset=15,
UserNumThld=20, UserNumOffset=20, CovCompensationMode=AUTO,
EvalPeriodGranularity=SECOND_LEVEL, SecondLevelTrigEvalPrd=6, SecondLevelExitEvalPrd=12,
PwrSavingPenaltyTime=5;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

On the LTE side

```
//Deactivating PDCCH power check and PDCCH symbol 0 power check optimization
MOD CELLPDCCHALGO: LocalCellId=0, PdcchPowerCheckSwitch=OFF,
PdcchPowerCheckOptSwitch=PDCCH_POWER_OPT_CHECK_OFF;
//Deactivating LNR coordinated channel shutdown
MOD CELLRFSHUTDOWN: LocalCellId=0, MultiRatJointChnShutdownSw=OFF;
```

On the NR side

```
//Deactivating LNR coordinated channel shutdown
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=MULTI_RAT_RF_SHUTDOWN_SW-0;
//Removing the power-saving configurations of the NR DU cell
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0;
```

5.1.4.1.3 Using the MAE-Deployment

- **Fast batch activation**
This function can be batch activated using the Feature Operation and Maintenance function of the MAE-Deployment. For detailed operations, see the following section in the MAE-Deployment product documentation or online help: **MAE-Deployment Operation and Maintenance > MAE-Deployment Guidelines > Enhanced Feature Management > Feature Operation and Maintenance**.
- **Single/Batch configuration**
This function can be activated for a single base station or a batch of base stations on the MAE-Deployment. For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.1.4.2 Activation Verification

Check the RF channel state on the LTE and NR sides to determine whether LNR coordinated channel shutdown has taken effect.

LTE Side

Use the following methods to check whether LNR coordinated channel shutdown has taken effect on the LTE side:

- Method 1: Checking the effective duration indicator
If the value of the **L.ChMeas.DFEE.RFOFF.Dur** counter is greater than 0, LNR coordinated channel shutdown has taken effect.
- Method 2: Using MML commands
Wait for a certain period of time after the channel shutdown conditions are met, and run the **DSP CELLRFSHUTDOWN** command to check the value of the **Current status about RF intelligent shutdown** parameter. If it is **Multi-RAT joint channel shutdown status**, LNR coordinated channel shutdown has taken effect.

The waiting duration is determined based on whether the PRB usage meets the entering conditions for the first time after LNR coordinated channel shutdown is enabled.

- If it does, wait for 5 to 20 minutes.
- If it does not, wait for 1 to 10 minutes.

(Optional) If RF channel shutdown does not take effect in a cell, run the **DSP CELLRFSHUTDOWN** command to query the current status of RF channel shutdown and the reason why it does not take effect. [Table 5-6](#) describes the values of **Current status about RF intelligent shutdown** in the command output, and [Table 5-7](#) describes the values of **Latest exiting cause about RF intelligent shutdown** and **Latest disabled cause about RF intelligent shutdown**.

Latest exiting cause about RF intelligent shutdown and **Latest disabled cause about RF intelligent shutdown** record the reasons for the latest exit and the latest failure for the function to take effect. The reasons can be queried at a time beyond the effective period of the function.

For example, if the effective period is set to 00:00:00 to 06:00:00, **Latest exiting cause about RF intelligent shutdown** and **Latest disabled cause about RF intelligent shutdown** indicate the reasons for the latest exit and the latest failure for the function to take effect during this period. The reasons can be queried from 06:00:00 to 23:59:59. The exact time and failure time are indicated by **Latest exiting time about RF intelligent shutdown** and **Latest disabled time about RF intelligent shutdown**.

Table 5-6 Current status of LNR coordinated channel shutdown

Cause Value	Description
Entering the RF channel intelligent shutdown state	The cell is in the RF channel intelligent shutdown state.

Cause Value	Description
Multi-RAT joint channel shutdown status	The cell is in the multi-RAT coordinated channel shutdown state.
Not entering because the number of UEs in connected mode exceeds threshold	The number of RRC_CONNECTED UEs in the cell is greater than the RRC_CONNECTED UE number threshold.
Not entering because LCS is enabled	The OtdoaSwitch option of the ENodeBAlgoSwitch.LcsSwitch parameter or the NB_OTDOA_SWITCH option of the CellLcsCfg.CellLcsSwitch parameter is selected.
Not entering because the state is delayed.	In some scenarios, when a cell exits the RF channel intelligent shutdown state, a penalty period starts. The eNodeB determines whether the cell can enter the RF channel intelligent shutdown state again only after the penalty period expires.
Not entering because the cell bandwidth, PA value, or CCE aggregation level for common signaling does not meet requirements	The cell bandwidth is less than or equal to 5 MHz. However, the cell bandwidth is required to be greater than 5 MHz. The CellDlpcPdschPa.PaPcOff parameter is set to a value less than -3 dB. The CellPdcchAlgo.ComSigCongregLv parameter is not set to CONGREG_LV4 .
Not entering because the number of antennas does not meet requirements	Fewer than two antennas are working. RF channel intelligent shutdown requires at least two antennas. The channels to be shut down include the current calibration channel. On the alarm management console, check whether ALM-29243 Cell Capability Degraded is reported. If this alarm is reported, clear this alarm.
Not entering because the power is insufficient	The power is limited in the downlink. This occurs when the CellDlschAlgo.RarAndPagingCR parameter is set to a value too small or the system information overhead is too large.
Not entering because the switch is turned off or the license is abnormal	The CellRfShutdown.RfShutdownSwitch parameter is set to OFF , or the licensed resources are insufficient.

Cause Value	Description
Not entering because of other reasons	The cell does not enter the RF channel intelligent shutdown state. The possible causes are as follows: • The current cell is unavailable. • Functions that are mutually exclusive with RF channel intelligent shutdown, such as service guarantee for UEs running unlimited services, intelligent power-off of carriers in the same coverage, and low power consumption mode, are enabled. • The current time is within the time range specified for the RF channel shutdown function, but this function is in its detection period. • Channel calibration fails on the LTE side, and services are switched over to the standby channel. • The RF module is configured to work in a single mode.
Not entering because the existence of QCI=1 UEs	There are UEs using voice services in the cell.
Not entering because the UL/DL PRB ratio exceeds the PRB threshold	• The uplink PRB usage of the cell is greater than the CellRfShutdown.UlPrbThd parameter value. • The downlink PRB usage of the cell is greater than the CellRfShutdown.DlPrbThd parameter value.
RF intelligent shutdown is in the detection mode.	The CellRfShutdown.RfShutdownSwitch parameter is set to DETECTION .
Not entering because co-carrier cell not trigger	The associated NB-IoT cell whose carrier is deployed in LTE in-band mode does not meet the shutdown conditions. Alternatively, other co-carrier LTE cells do not meet the shutdown conditions.
Not entering because cell not in shutdown time	The system time is not within the period of RF channel intelligent shutdown.
Not entering because beam mode not support	The current beam configuration does not support RF channel intelligent shutdown.

Cause Value	Description
Not entering because load simulation is enabled	Load simulation is enabled in the cell.
Not entered because of multi-RAT negotiation failure	There are no RF channels that can be shut down on both the local and peer sides according to the result of negotiation between the two sides.
Not entering because of incompatible hardware capability	The cell does not enter the shutdown state because of incompatible hardware capability.
Not entering because of the existence of CE1 or CE2 UEs	There exist CE1 or CE2 UEs in the cell.

Table 5-7 Causes for the latest exit and the latest failure for RF intelligent shutdown to take effect

Cause Value	Description
The number of UEs in connected mode exceeds threshold	The number of RRC_CONNECTED UEs in the cell is greater than the RRC_CONNECTED UE number threshold.
The power is insufficient	The power is limited in the downlink.
Load simulation is enabled	Load simulation is enabled in the cell.
LCS is enabled	The OtdoaSwitch option of the ENodeBAlgoSwitch.LcsSwitch parameter or the NB_OTDOA_SWITCH option of the CellLcsCfg.CellLcsSwitch parameter is selected.
Other reasons	The possible causes are as follows: • The current cell is unavailable. • Functions that do not work with RF channel intelligent shutdown, for example, unlimited-service UE guarantee, are enabled. • Channel calibration fails on the LTE side, and services are switched over to the standby channel.
The existence of QCI=1 UEs	There are UEs using voice services in the cell.

Cause Value	Description
The UL/DL PRB ratio exceeds the PRB threshold	The PRB usage does not meet the following conditions: - The uplink PRB usage of the cell is less than or equal to the sum of the CellRfShutdown.UlPrbThd and CellRfShutdown.UlPrbOffset parameter values. - The downlink PRB usage of the cell is less than or equal to the sum of the CellRfShutdown.DlPrbThd and CellRfShutdown.DlPrbOffset parameter values.
The co-carrier cell exit or not enter the shutdown state	The associated NB-IoT cell whose carriers are deployed in LTE in-band mode has exited or has not entered the RF channel intelligent shutdown state.
Other carriers exit or not enter the shutdown state	When Multi-RAT Coordinated Channel Shutdown is enabled, the local cell also exits the multi-RAT coordinated channel shutdown state because the peer cell does not meet the Multi-RAT Coordinated Channel Shutdown conditions, or one of the multiple carriers in the local RAT exits the multi-RAT coordinated channel shutdown state.
Delay for punishment	In some scenarios, when a cell exits the RF channel intelligent shutdown state, a penalty period starts. The eNodeB determines whether the cell can enter the RF channel intelligent shutdown state again only after the penalty period expires.
The cell bandwidth, PA value, or CCE aggregation level for common signaling does not meet requirements	- The cell bandwidth is less than or equal to 5 MHz. However, the cell bandwidth is required to be greater than 5 MHz. - The CellDlpcPdschPa.PaPcOff parameter is set to a value less than -3 dB. - The CellPdcchAlgo.ComSigCongregLv parameter is not set to CONGREG_LV4.
The existence of CE1 or CE2 UEs	There exist CE1 or CE2 UEs in the cell.

Cause Value	Description
Beam mode not support	The current beam configuration does not support RF channel intelligent shutdown.
Incompatible hardware capability	The cell does not enter the RF channel intelligent shutdown state because the hardware does not support this function.
Multi-RAT negotiation failure	There are no RF channels that can be shut down on both the local and peer sides according to the result of negotiation between the two sides.
NA	- When this switch is turned off, the cell is deactivated, or no license is available, the cause value is displayed as NA. - The specified effective period is modified when the function is in effect. If the period after modification does not include the current effective period, and there is no cause value for the latest exit, NA is retained as the cause value. - The cell is in the initial state and has never entered the RF channel intelligent shutdown state.

- Method 3: Using signaling tracing
On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**. On the displayed **Signaling Trace Management** tab page, observe the actual transmit power of the channel by tracing signaling messages. The channel whose output power is 0 is the channel that is actually shut down.

NR Side

Use the following methods to check whether LNR coordinated channel shutdown has taken effect on the NR side:

After the LNR coordinated channel shutdown function is activated, use either of the following methods to check whether LNR coordinated channel shutdown has taken effect:

- Method 1:
 - Run the **LST NRDUCELLALGOSWITCH** command. Verify that the **Power Saving Switch** parameter has a value of **RF Channel Power Saving Switch:On** in the command output.
 - Run the **LST NRDUCELLPOWERSAVING** command. Verify that there is at least one record with **Power Saving Type** being **MULTI_RAT_RF_SHUTDOWN**.

 NOTE

If there are two or more records with **Power Saving Type** being **MULTI_RAT_RF_SHUTDOWN**, LNR coordinated channel shutdown has been enabled in two or more time periods.

- c. Run the **DSP TIME** command to check whether the system time is within a period for LNR coordinated channel shutdown to take effect.
 - If it is, go to 5.
 - If it is not, wait until the start time for LNR coordinated channel shutdown arrives.
 - d. On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**. In the navigation tree on the left of the displayed **Signaling Trace Management** tab page, choose **Trace Type > NR > Cell Performance Monitoring > Usage of Radio Resource Monitoring**. Check whether the average downlink PRB usage of the cell is less than or equal to the value of the **NRDUCellPowerSaving.DlPrbThld** parameter.
 - If it is, go to 5.
 - If it is not, wait until the average downlink PRB usage of the cell is less than or equal to the parameter value.
 - e. Wait for a certain period of time after the channel shutdown conditions are met, and run the **DSP NRDUCELL** command to check the value of the **Power Saving State** parameter. If it has a value of **Multi-RAT Coordinated Channel Shutdown:Enabled**, LNR coordinated channel shutdown has taken effect.

The waiting duration is determined based on whether the PRB usage meets the entering conditions for the first time after LNR coordinated channel shutdown is enabled.

 - If it does, wait for 5 to 20 minutes.
 - If it does not, the granularity of the waiting duration is specified by the **NRDUCellPowerSaving.EvalPeriodGranularity** parameter. If this parameter is set to **MINUTE_LEVEL**, the waiting duration is determined by the larger value between **NRDUCellPowerSaving.ShutdownTrigJudgePeriod** and **NRDUCellPowerSaving.PwrSavingPenaltyTime**. If this parameter is set to **SECOND_LEVEL**, the waiting duration is determined by the larger value between **NRDUCellPowerSaving.SecondLevelTrigEvalPrd** and **NRDUCellPowerSaving.PwrSavingPenaltyTime**.
- Method 2:

Run the **DSP NRDUCELLPOWERSAVING** command.

 - If the value of the **Power Saving Type** parameter is **Multi-RAT Coordinated Channel Shutdown** and the value of the **Current State** parameter is **Enabled**, LNR coordinated channel shutdown has taken effect.
 - If the value of the **Power Saving Type** parameter is **Multi-RAT Coordinated Channel Shutdown** and the value of the **Current State**

parameter is **Disabled**, you can determine the possible reasons why LNR coordinated channel shutdown does not take effect within the specified period based on the values of **Current Disabled Cause**, **Latest Exit Cause**, and **Latest Disabled Cause**, as described in [Table 5-8](#).

[Table 5-9](#) describes the meanings of the **Latest Exit Cause**, **Latest Disabled Cause**, **Latest Exit Time**, and **Latest Disabled Time**.

Latest Exit Cause and **Latest Disabled Cause** indicate the reasons for the latest exit and the latest failure for the function to take effect. Such reasons can be queried beyond the effective period. For example, if the effective period is from 00:00:00 to 06:00:00, **Latest Exit Cause** and **Latest Disabled Cause** indicate the reasons for the latest exit and the latest failure for the function to take effect during this period. You can query the reasons at a time from 06:00:00 to 23:59:59.

Table 5-8 Possible reasons why LNR coordinated channel shutdown does not take effect

Cause Value	Description
Insufficient License	No license has been applied for.
Switch Turned Off	None
Cell Unavailable	The cell is blocked or deactivated, fails to be set up, or is unavailable due to other reasons.
Triggering Time Not Reached	The current time is not within the range specified for LNR coordinated channel shutdown.
Impacted Feature Taken Effect	For example, when features related to atmospheric duct take effect, channel shutdown does not take effect.
Downlink PRB Threshold Not Reached	None
UE Number Threshold Not Reached	None
In Penalty State	A 1-minute penalty procedure starts after the function is terminated.
RF Hardware Capability Not Supported	The RF hardware does not support the function.

Cause Value	Description
Power Aggregation Configuration Not Met	The power aggregation capability of the RF module does not meet the function requirements. - The power aggregation capability may be insufficient if the parameters in the NRDUCellTrpBeam and NRDUCellChnPwr MOs are set to excessively large values. For details, contact Huawei engineers. - When the NRDUCellPowerSaving.CovCompensationMode parameter is set to AUTO, LNR coordinated channel shutdown cannot take effect for certain RF modules and power offset settings. For details about these RF modules and power offset settings, contact Huawei engineers. To allow LNR coordinated channel shutdown to take effect, set the NRDUCellPowerSaving.CovCompensationMode parameter to MANUAL. However, the coverage performance of common channels cannot be guaranteed after LNR coordinated channel shutdown takes effect.
Shared Channel Not Coordinated	There are no RF channels that can be shut down on both the local and peer sides according to the result of negotiation between the two sides.
Channel Calibration in Progress	None
Channel Calibration Info Being Queried	None
Detecting for Port Mapping Optimization	None

Cause Value	Description
Parameter Setting Not Supported	The cell does not enter the shutdown state due to the following reasons: - The frequency-domain position of the NR common PDCCH is within the NR-dedicated spectrum. - The UE_PDCCH_FULL_BANDWIDTH_CFG_SW option of the NRDUCellPdcch.PdcchAlgoExtSwitch parameter is deselected. - When the PDCCH_RATEMATCH_SW option of the NRDUCELLPDSCH.RateMatchSwitch parameter is selected, the frequency-domain position of the UE-level PDCCH overlaps with the NR-dedicated spectrum.
High-Speed Train Arrived	None
Xn Heartbeat Packet Not Received	None
Channel Calibration Resource Request Failed	None
Others	- The current time is within the time range specified for the RF channel shutdown function, but this function is in its detection period, which lasts for 5 minutes. - There are internal software or other faults. - A Multi-RAT RRU is used, the Multi-RAT Coordinated Channel Shutdown feature is enabled, and the license meets the requirements. However, the licensed resources for the RF Channel Intelligent Shutdown feature are insufficient for a single RAT.
RF Unit Working Standard Not Met	The RF module is configured to work in NR only.

Table 5-9 Causes for the latest exit and the latest failure for the function to take effect

Command Output of DSP NRDUCELLPOWERSAVING	Description
Latest Exit Cause	Indicates the cause for the latest exit of a feature during its effective period.
Latest Disabled Cause	Indicates the cause for the latest failure for a function to take effect during its effective period.
Latest Exit Time	Indicates the recorded time of the latest exit of a feature during its effective period.
Latest Disabled Time	Indicates the recorded time of the latest failure for a function to take effect during its effective period.

5.1.4.3 Network Monitoring

Benefit Monitoring

Compare the values of the following counters one week before and after LNR coordinated channel shutdown is enabled to evaluate the energy saving gains.

- Counter measuring the energy consumption of the entire base station

Counter ID	Counter Name
1593835637	VS.EnergyCons.BTS.Adding.LTE
1593835959	VS.EnergyCons.BTS.Adding.Service.LTE
1593835804	VS.EnergyCons.BTS.Adding.NR
1593835961	VS.EnergyCons.BTS.Adding.Service.NR

- Counter measuring the energy consumption of RF modules

Counter ID	Counter Name
1593835836	VS.EnergyCons.BTSBoard (LTE eNodeB, 5G gNodeB)

- Counter measuring RF channel shutdown duration

Counter ID	Counter Name
1911822885	N.PowerSaving.RFShutdown.Dur

NOTE

The policy of counter measurement upon carrier shutdown is determined by the function of energy saving counter optimization upon carrier shutdown, which is controlled by the **COUNTER_OPT_IN_CARROFF_SW** option of the **gNBPowerSavingAlgo.PowerSavingAlgoSwitch** parameter. Assume that RF channel shutdown duration occurs on a carrier that is already in the shutdown state. If this option is deselected, the duration of RF channel shutdown for this carrier is counted in the **N.PowerSaving.RFShutdown.Dur** counter. If this option is selected, this duration is not counted in the **N.PowerSaving.RFShutdown.Dur** counter.

Possible Issues

Check whether the RF module is faulty. The cells do not enter the LNR coordinated channel shutdown state if the RF module is faulty.

If an RF module is faulty, the following alarms may be reported:

- ALM-26520 RF Unit TX Channel Gain Out of Range
- ALM-26538 RF Unit Clock Problem
- ALM-26532 RF Unit Hardware Fault
- ALM-26529 RF Unit VSWR Threshold Crossed

NOTE

When the RF module is faulty, related alarms may not be reported due to alarm filtering or alarm correlation reasons. For details about the alarm reporting mechanism, see sections "Alarm Filtering" and "Alarm Correlation" in *Fault Management*.

5.2 UL Coordinated Channel Shutdown

LOFD-001039 RF Channel Intelligent Shutdown and WRFD-190205 VAM Power Saving support RF channel shutdown. However, LOFD-001039 RF Channel Intelligent Shutdown takes effect only on single-mode RF modules. This chapter describes UMTS and LTE FDD coordinated channel shutdown, which is referred to as UL coordinated channel shutdown hereinafter. In the following sections, LTE refers to LTE FDD.

5.2.1 Principles

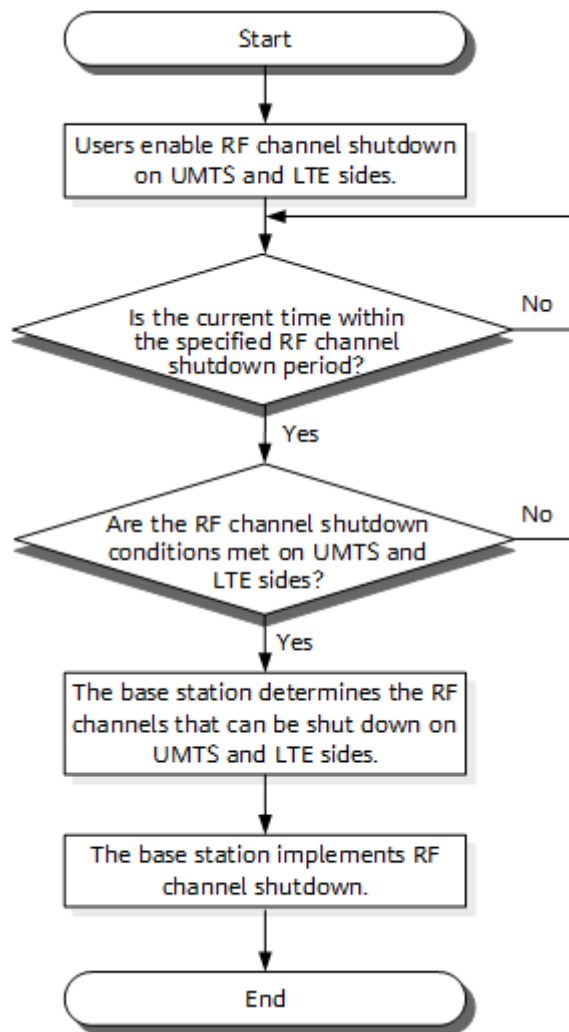
UL coordinated channel shutdown is controlled by the **ULOCCELLALGPARA.MULTIRATCHANNELSHUTDOWN** parameter on the UMTS side and the **CellRfShutdown.MultiRatJointChnShutdownSw** parameter on the LTE side.

5.2.1.1 Entering UL Coordinated Channel Shutdown

UL Coordinated Channel Shutdown Procedure

Figure 5-2 shows the UL coordinated channel shutdown procedure.

Figure 5-2 UL coordinated channel shutdown procedure



The procedure is described as follows:

1. Users set both the **ULOCELLALGPARA.MULTIRATCHANNELSHUTDOWN** parameter on the UMTS side and the **CellRfShutdown.MultiRatJointChnShutdownSw** parameter on the LTE side to **ON**.
2. The base station checks whether the current time is within the RF channel shutdown period for the UMTS and LTE cells.
If so, the base station proceeds with **3** on the UMTS side and proceeds with **4** on the LTE side. If not, the base station continues to check whether the time is within the RF channel shutdown period for the cells.

- On the UMTS side, the start time and end time of the shutdown period are specified by the **ULOCCELLALGPARA.PWRSVSTARTTIME** and **ULOCCELLALGPARA.PWRSVSTOPTIME** parameters.
 - On the LTE side, the start time and end time of the shutdown period are specified by the **CellRfShutdown.StartTime** and **CellRfShutdown.StopTime** parameters.
3. The base station determines whether the UMTS cells have entered the low load state.
- If the downlink power of a cell is continuously lower than the value of the **ULOCCELLALGPARA.PWRSVTPWRLOWTHD** parameter for a period, the cell has entered the low load state and energy saving can be performed. Otherwise, the base station continues to check whether the downlink power of the cell is continuously lower than the value of this parameter.
4. The base station checks whether all the following conditions are met in the LTE cells. If so, the base station proceeds with 5. If not, the base station continues to check whether all the following conditions are met in the LTE cells.
- Each cell and the number of UEs in the cell meet the following conditions:
 - The cell is a macro cell with a downlink bandwidth greater than 5 MHz.
 - The number of RRC_CONNECTED UEs in the cell is less than or equal to the **CellRfShutdown.UENumThd** parameter value. In addition, if the **RF_SHUTDOWN_ACTIVATION_OPT_SW** option of the **CellRfShutdown.RfShutdownAlgoSwitch** parameter is selected, there is no voice UE.
 - The PRB usage of each cell meets all of the following conditions:
 - The uplink PRB usage of the cell is less than or equal to the **CellRfShutdown.UlPrbThd** parameter value.
 - The downlink PRB usage of the cell is less than or equal to the **CellRfShutdown.DlPrbThd** parameter value.
 - The cell-specific reference signal (CRS) transmit power can be increased by 3 dB.

After RF channels are shut down, LTE CRS transmit power needs to be increased to ensure LTE coverage.
 - Each cell is not within a penalty period.

To prevent ping-pong RF channel shutdown, the base station starts a penalty timer on the LTE side if the cell meets specific conditions upon exit from the RF channel shutdown state. The cell will not re-enter the RF channel shutdown state during this penalty period, even if the preceding triggering conditions are met. The penalty period is specified by the **CellRfShutdown.PenaltyDuration** parameter. It is recommended that this parameter be set to 1 for the Multi-RAT Coordinated Channel Shutdown feature.
5. The base station determines the RF channels that can be shut down for both the UMTS and LTE sides and shuts down the channels.

When both the UMTS and LTE sides meet the shutdown conditions, the two sides negotiate and select channels for shutdown for the cells sharing the same RF modules in the multimode base station. In this way, the number of channels for shutdown reaches the maximum, achieving the maximum energy saving gains.

Requirements for UL Coordinated Channel Shutdown

Table 5-10 describes the requirements for coordinated channel shutdown on the UMTS and LTE sides.

Table 5-10 Requirements for coordinated RF channel shutdown on the UMTS and LTE sides

RAT	Cell Band width	Numb er of Trans mit Anten nas	RF Channels that Can Be Shut Down	Type of Cell Supporting Shutdown
UM TS	No require ment	2 or 4	2T cell: Either of the two channels can be shut down. 4T cell: Any two channels in different polarization directions can be shut down.	- Cell providing the same coverage in 0.5/0.5 configuration mode - MIMO Prime cell - Intelligent 2T virtual antenna mapping (VAM) cell 4T Beam Optimization cell (served by integrated 4T RF modules)
LTE	> 5 MHz	2 or 4	2T2P cell: Only the transmit channel of port 1 can be shut down, and the transmit channel of port 0 cannot be shut down. 4T4P cell: The transmit channels of ports 1 and 2 or those of ports 2 and 3 can be shut down, or only the transmit channels of ports 1 and 2 are shutdown^a. 4T2P cell: One transmit channel of each port can be shut down.	No requirement

RAT	Cell Band width	Numb er of Trans mit Anten nas	RF Channels that Can Be Shut Down	Type of Cell Supporting Shutdown
a: In 4-port scenarios, the combination of transmit channels that can be shut down is determined based on the setting of the RF_SHUTDOWN_4PORT_OPT_SW option of the CellRfShutdown.RfShutdownAlgoSwitch parameter. • When this option is deselected, the transmit channels of ports 1 and 2 or those of ports 2 and 3 can be shut down. • When this option is selected, only the transmit channels of ports 1 and 2 can be shut down.				

 NOTE

For an LTE cell with a bandwidth less than or equal to 5 MHz, RF channel shutdown cannot be performed. In multi-carrier scenarios, ensure that this feature is enabled for all LTE cells served by the same RRU.

UL Coordinated Channel Shutdown Example

Assume that an LTE FDD 4T4P cell and a UMTS 4T Beam Optimization cell are deployed on a 4T4R RRU and the **RF_SHUTDOWN_4PORT_OPT_SW** option of the **CellRfShutdown.RfShutdownAlgoSwitch** parameter is deselected. [Table 5-11](#) lists the RF channel relationships between UMTS and LTE cells for RF channel shutdown.

Table 5-11 Example of RF channel relationships between UMTS and LTE cells for RF channel shutdown

-	Tx_A	Tx_B	Tx_C	Tx_D
Polarization	+	-	+	-
Port number for LTE	port_0	port_3	port_2	port_1
According to the RF channel shutdown combination principles for LTE cells, channels of ports 1 and 2 or those of ports 2 and 3 can be shut down.				
LTE cell shutdown combination 1 (ports 1 and 2)	Not shut down	Not shut down	Shut down	Shut down

-	Tx_A	Tx_B	Tx_C	Tx_D
LTE cell shutdown combination 2 (ports 2 and 3)	Not shut down	Shut down	Shut down	Not shut down
According to the RF channel shutdown combination principles for UMTS cells, any two channels in different polarization directions can be shut down. The following four shutdown combinations are available:				
UMTS cell shutdown combination 1	Shut down	Shut down	Not shut down	Not shut down
UMTS cell shutdown combination 2	Shut down	Not shut down	Not shut down	Shut down
UMTS cell shutdown combination 3	Not shut down	Shut down	Shut down	Not shut down
UMTS cell shutdown combination 4	Not shut down	Not shut down	Shut down	Shut down

According to the coordinated RF shutdown principles, LTE cell shutdown combination 1 and UMTS cell shutdown combination 4 are selected, so that the RF channels Tx_C and Tx_D are shut down to maximize energy saving gains.

5.2.1.2 Exiting UL Coordinated Channel Shutdown

The UMTS and LTE sides exit the UL coordinated channel shutdown state when their respective conditions are met.

Conditions for UMTS to Exit RF Channel Shutdown

- The energy saving period specified by the **ULOCCELLALGPARA.PWRSVSTARTTIME** and **ULOCCELLALGPARA.PWRSVSTOPTIME** parameters has elapsed.
- The average output power of the cell for a period of time is greater than the maximum transmit power of the cell multiplied by the **ULOCCELLALGPARA.PWRSVTPWRHIGHTHD** parameter value.
- The transmit power of RF modules is derated due to voltage standing wave ratio (VSWR) or temperature exceptions.

- RF loopback tests have been started by running the **STR NODEBALGSIMLTST** command for one or more cells served by the RF modules.

Conditions for LTE to Exit RF Channel Shutdown

- The energy saving period specified by the **CellRfShutdown.StartTime** and **CellRfShutdown.StopTime** parameters has elapsed.
- The eNodeB detects that the common channel transmit power is limited.
- Faults occur on RF channels not shut down.
- The number of RRC_CONNECTED UEs in the cell is greater than the **CellRfShutdown.UENumThd** parameter value.
- Any of the following conditions is met for a period of time:
 - The uplink PRB usage of the cell is greater than the sum of the **CellRfShutdown.UlPrbThd** and **CellRfShutdown.UlPrbOffset** parameter values.
 - The downlink PRB usage of the cell is greater than the sum of the **CellRfShutdown.DlPrbThd** and **CellRfShutdown.DlPrbOffset** parameter values.

5.2.2 Network Analysis

5.2.2.1 Benefits

Some RF channels can be simultaneously shut down to reduce power consumption when UL coordinated channel shutdown is enabled and the traffic volume is low in scenarios where UMTS and LTE share RF modules. If there are NB-IoT cells, the coordinated channel shutdown switch must also be turned on for the NB-IoT cells to maximize the energy saving gains.

5.2.2.2 Impacts

The impacts between this function and other functions are described in [5.2.3.2.3 Function Impacts](#).

- Impacts on the LTE network
 - When the **CellRfShutdown.UENumThd** parameter value is greater than 0, the **Cell Downlink Average Throughput** and peak throughput of the cell may decrease. This is because there may be a small number of RRC_CONNECTED UEs in the cell when the base station shuts down some of its RF channels. In such a case, these RRC_CONNECTED UEs have to roll back to TM2, which decreases their downlink diversity gains.
 - When RF channels are shut down, the eNodeB changes the number of PDCCH symbols to 3. As a result, more RBs are occupied by the PUCCH, leading to a larger **L.ChMeas.PRBUlUsedAvg** counter value, which indicates increased uplink RB usage. In addition, the number of RBs available for the PUSCH decreases because the number of RBs occupied by the PUCCH increases. As a result, the uplink peak rate slightly decreases.

- System information will be updated when cells enter or exit the RF channel shutdown state, increasing the value of the **L.Paging.UU.SIUpdate** counter.
- Impacts on the UMTS network
 - After the cell enters the RF channel shutdown state, the HSDPA UE throughput slightly decreases because the available power of the cell decreases.
 - The counters related to downlink scheduling change as follows:
 - The distribution of CQIs received from HSDPA UEs in the cell (**VS.CQI.0** to **VS.CQI.30**) changes.
 - The distribution of CQIs sent to HSDPA UEs in the cell (**VS.UsedCQI0** to **VS.UsedCQI30**) changes.
 - The distribution of the numbers of times HS-PDSCH codes in varied quantities are used in the cell (**VS.HSDPA.PdschCodeUsed.0** to **VS.HSDPA.PdschCodeUsed.15**) changes.
 - The HS-PDSCH transmit power in the cell (**VS.PdschPwrRatio.Mean**) increases.
 - The power usage of free UEs in the cell (**VS.HSDPAPwrRatio.Mean.FreeUser**) increases.
 - The power available for HSDPA in the cell (**VS.AvaiHSDPAPwrRatio.Mean**) changes.
 - The average HS-PDSCH transmit power of all E-FACH UEs in the cell (**VS.EFACH.PdschPwrRatio.Mean**) increases.

5.2.3 Requirements

5.2.3.1 Licenses

RAT	Feature ID	Feature Name	Model	Sales Unit
UMTS	MRFD-1612 14	Multi-RAT Coordinated Channel Shutdown (UMTS)	LQW9MULCH SD01	Per Cell
LTE FDD	MRFD-1612 24	Multi-RAT Coordinated Channel Shutdown (LTE FDD)	LT1S0MRJCS0 0	Per Cell

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

5.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.2.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	RF channel intelligent shutdown	CellRfShutdown.RfShutdownSwitch	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	For LTE FDD cells that support Multi-RAT Coordinated Channel Shutdown, LTE FDD RF channel intelligent shutdown must also be enabled for Multi-RAT Coordinated Channel Shutdown to take effect.

5.2.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	OTDOA-based positioning	OtdoaSwitch option of the ENodeBAlgoSwitch . LcsSwitch parameter and the NB_OTDOA_SWITCH option of the CellLcsCfg . CellLcsSwitch parameter	<i>LCS in eRAN Feature Documentation</i>	None
LTE FDD	Load simulation	CellSimuLoad MO configured	None	When at least one CellSimuLoad MO is configured, Multi-RAT Coordinated Channel Shutdown does not take effect.
LTE FDD	Service guarantee for UEs running unlimited services	UsUeGuaranteeSwitch , SfCtrlAsScell , and SfCtrlAsInterNcell options of the CellUSParaCfg . UsAlgoSwitch parameter	None	When the UsUeGuaranteeSwitch option is selected, Multi-RAT Coordinated Channel Shutdown does not take effect if either the SfCtrlAsScell or SfCtrlAsInterNcell option is selected.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Massive MIMO	None	<i>Massive MIMO Basics (TDD) in eRAN Feature Documentation</i>	When UL coordinated channel shutdown is enabled, the BfAnt.CoverageScenario parameter cannot be set to SCENARIO_13, EXPANDSCENARIO_1, EXPANDSCENARIO_2, or EXPANDSCENARIO_3 in 32T32R massive MIMO scenarios. When UL coordinated channel shutdown is enabled, the BfAnt.CoverageScenario parameter cannot be set to EXPANDSCENARIO_1, EXPANDSCENARIO_2, or EXPANDSCENARIO_3 in 64T64R massive MIMO scenarios.
LTE FDD	Zero Guard Band Between Contiguous Intra-Band Carriers	CONTIG_INTRA_BAND_CARR_SW option of the ContigIntraBandCarr.ContigIntraBandCarrSw parameter	<i>Seamless Intra-Band Carrier Joining (TDD) in eRAN Feature Documentation</i>	None

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	On-demand TX power allocation under EME	PDSCHCfg.EmfPowerLimitSwitch	<i>On-Demand TX Power Allocation Under EME in eRAN Feature Documentation</i>	If the PDSCHCfg.CellPowerLimit parameter is set to a value greater than 0 , the maximum cell transmit power is limited. In this case, on-demand TX power allocation under EME is incompatible with Multi-RAT Coordinated Channel Shutdown.
LTE FDD	SFN cell	Cell.MultiRruCellMode	<i>SFN in eRAN Feature Documentation</i>	• SFN cells (Cell.MultiRruCellMode parameter set to SFN) do not support Multi-RAT Coordinated Channel Shutdown. • MPRU aggregation cells (Cell.MultiRruCellMode parameter set to MPRU_AGGREGATION) do not support Multi-RAT Coordinated Channel Shutdown.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Static Multiple Beam	SectorSplitGroup.BeamTemplateId set to a value greater than 1	<i>Massive MIMO Basics (FDD) in eRAN Feature Documentation</i>	UL coordinated channel shutdown is not supported in FDD 16T16R massive MIMO scenarios where Static Multiple Beam is enabled.
UMTS	RuralLink antenna solution (UMTS)	SECTOR.ANTNUM set to 0 SECTOREQM.ANTCFGMODE set to BEAM ULOCCELL.LOCELLTYPE set to NORMAL_CELL BASEBANDEQM.UMTSDEMMODE set to DEM_4_CHAN or DEM_ADPT_CHAN ULOCCELL.BFMODE set to FOURTXBEAMOPT ULOCCELL.FOURTXDEPLOYMENTMODE set to MODE_2 BEAMCELLGROUP.BeamCellGroupType set to FOURTXBEAMOPT	<i>RuralLink Solution</i>	UMTS three-sector split cells do not support Multi-RAT Coordinated Channel Shutdown.
UMTS	RF Channel Muting Based Power Saving	ULOCCELLALGPARA.RFCHANNELMUTPWRSVSW	<i>Green BTS in RAN Feature Documentation</i>	UL coordinated channel shutdown cannot be enabled together with RF Channel Muting Based Power Saving.
UMTS	UMTS RF channel dynamic muting	ULOCCELLALGPARA.RFCHNDYNMUTINGMODE set to BASICPWRSV or DEEPPWRSV	<i>Green BTS in RAN Feature Documentation</i>	UL coordinated channel shutdown cannot be used together with UMTS RF channel dynamic muting.

5.2.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
Power control	None	<i>Power Control in eRAN Feature Documentation</i>	Multi-RAT Coordinated Channel Shutdown increases the reference signal transmit power.
Channel calibration	Joint channel calibration: RruJointCalParaCfg.TxChnCalSwitch Downlink-only module channel calibration (FDD): RruJointCalParaCfg.AauPassivePortCalibPeriod	<i>MIMO in eRAN Feature Documentation</i>	For a cell in the Multi-RAT Coordinated Channel Shutdown state, if channel calibration fails, other functions that depend on channel calibration may not be properly provided. In addition, a message indicating a failure will be displayed on the OM interface for channel calibration.
Intelligent power-off of carriers in the same coverage	CellShutdown.CellShutdownSwitch	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	A cell where Multi-RAT Coordinated Channel Shutdown is enabled enters the intelligent power-off of carriers in the same coverage mode when the cell fulfills the conditions for triggering intelligent power-off of carriers in the same coverage.
Multi-carrier coordinated energy saving	CellShutdown.CellShutdownSwitch set to ON_MULTI_CARRIER_HIER_SHUTDOWN or ON_MULTICARRIER_HIER_MANUAL_SHUTDOWN	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	A cell in the UL coordinated channel shutdown state will exit this state and enter the multi-carrier coordinated energy saving state when the cell fulfills the conditions for triggering multi-carrier coordinated energy saving.

Function Name	Function Switch	Reference	Description
pRRU deep dormancy in intelligent mode	EnodebMpruEs.MpruDormancyDLEarfcn	<i>Multi-RAT Coordinated Energy Saving</i>	When the triggering conditions for both pRRU deep dormancy in intelligent mode and Multi-RAT Coordinated Channel Shutdown are met, only pRRU deep dormancy takes effect.
Dynamic voltage adjustment	CellAlgoSwitch.DynAdjVoltSwitch	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	When the triggering conditions for both dynamic voltage adjustment and Multi-RAT Coordinated Channel Shutdown are met, Multi-RAT Coordinated Channel Shutdown will take effect but dynamic voltage adjustment will not.
Low power consumption mode (LTE)	CellLowPower.LowPwrSwitch	<i>Green Site</i>	The low power consumption mode does not take effect in a cell in the Multi-RAT Coordinated Channel Shutdown state.
CMAS	CellAlgoSwitch.CmasSwitch	<i>CMAS in eRAN Feature Documentation</i>	If the CellRfShutdown.RsPwrAdjOffset parameter is set to a value other than 0, the number of RBs available for downlink scheduling decreases because the reference signal transmit power increases. In this case, the maximum allowed size of warning notifications that the cell can broadcast will be reduced. Any warning notification larger than the maximum allowed size will fail to be broadcast to UEs.
ETWS	None	<i>ETWS in eRAN Feature Documentation</i>	

Function Name	Function Switch	Reference	Description
eMTC introduction	CellEmtcAlgo.EmtcAlgoSwitch	<i>eMTC in eRAN Feature Documentation</i>	When eMTC is enabled, the overhead of eMTC system information will increase the power insufficiency probability. Multi-RAT Coordinated Channel Shutdown is less likely to be triggered, affecting the energy saving gains.
Dynamic Power Sharing Between LTE Carriers	CellDynPowerSharing.Dynami cPowerSharing Switch	<i>Dynamic Power Sharing Between LTE Carriers in eRAN Feature Documentation</i>	Cells that have entered the UL coordinated channel shutdown state do not participate in Dynamic Power Sharing Between LTE Carriers.
Smart carrier selection based on virtual grids	SMART_CARRIER_SELECTION_SW option of the MultiCarrUnifiedSch.MultiCarrierUnifiedSch Sw parameter	<i>Multi-carrier Unified Scheduling in eRAN Feature Documentation</i>	When Multi-RAT Coordinated Channel Shutdown is triggered in the local cell, sample collection of virtual grids for neighboring cells is affected.
Carrier aggregation	CaMgtCfg.CellCaAlgoSwitch	<i>Carrier Aggregation in eRAN Feature Documentation</i>	If carrier aggregation is enabled after Multi-RAT Coordinated Channel Shutdown takes effect, the cell is likely to exit the RF channel shutdown state. As a result, the energy saving duration and energy saving gains decrease.
Enhanced symbol power saving	MBSFNShutdownSwitch option of the CellAlgoSwitch.DLSchSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	When Multi-RAT Coordinated Channel Shutdown takes effect, enhanced symbol power saving does not take effect.

Function Name	Function Switch	Reference	Description
LTE and NR Flash Dynamic Power Sharing	LTE_NR_DYN_POWER_SHARING_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter	<i>LTE and NR Power Allocation in eRAN Feature Documentation</i>	Cells that have entered the UL coordinated channel shutdown state do not participate in LTE FDD and NR Flash Dynamic Power Sharing.
Dynamic Power Sharing in Multi-Carriers	ULOCELLALGPARA.R99TOHPWHSW	<i>Power Sharing in RAN Feature Documentation</i>	When the Multi-RAT Coordinated Channel Shutdown feature takes effect, the power sharing function does not take effect.
HSDPA Inter-Cell Power Sharing	ULOCELLALGPARA.HTOHPWHSW	<i>Power Sharing in RAN Feature Documentation</i>	When the Multi-RAT Coordinated Channel Shutdown feature takes effect, the power sharing function does not take effect.
UMTS and LTE Dynamic Power Sharing (UMTS)	ULOCELL.ULPOWERSHARE	<i>UMTS and LTE Dynamic Power Sharing in RAN Feature Documentation</i>	When the Multi-RAT Coordinated Channel Shutdown feature takes effect, the power sharing function does not take effect.
Energy Efficiency Improved	NODEBOPTDYNADJPARA.DYNADJSW	<i>Green BTS in RAN Feature Documentation</i>	When both the Multi-RAT Coordinated Channel Shutdown and Energy Efficiency Improved functions are enabled, Multi-RAT Coordinated Channel Shutdown preferentially takes effect and Energy Efficiency Improved is less likely to take effect.

Function Name	Function Switch	Reference	Description
VAM Power Saving	NODEBALGPAR A.VAMPWRSA VSW	<i>Green BTS in RAN Feature Documentation</i>	When both the Multi-RAT Coordinated Channel Shutdown and VAM Power Saving functions are enabled, VAM Power Saving preferentially takes effect and Multi-RAT Coordinated Channel Shutdown does not take effect.
OCNS	DISim.LR	<i>OCNS in RAN Feature Documentation</i>	UL coordinated channel shutdown and OCNS cannot take effect together for cells with downlink services established on the WBBP board.
UMTS	NCR cells	ULOCCELL.N CRCELL	After the NCR cells served by the DeNB enter the UL coordinated channel shutdown state, UMTS services within the coverage of the RuralCow site are affected.

5.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

Boards

For UMTS, only the following boards support this function.

Board Name	Board Type
UMPT	Main control board
UMDU	Main control board
UBBPd	Baseband processing unit
UBBPc	Baseband processing unit

Board Name	Board Type
UBBP _{ei}	Baseband processing unit
UBBP _g	Baseband processing unit
UBBP _h	Baseband processing unit
WBBP _f	Baseband processing unit
WBBP _d	Baseband processing unit
WBBP _b	Baseband processing unit

For LTE FDD, only the following boards support this function.

Board Name	Board Type
UMPT	Main control board
UMDU	Main control board
UBBP _d	Baseband processing unit
UBBP _e	Baseband processing unit
UBBP _{ei}	Baseband processing unit
UBBP _{em}	Baseband processing unit
UBBP _{ep}	Baseband processing unit
UBBP _g	Baseband processing unit
UBBP _h	Baseband processing unit
LBBP _d	Baseband processing unit

RF Modules

- AAU
Only the AAU3910, AAU3911, AAU3920, AAU3940, and AAU5940 support this function.
- RFU
Only the MRFU_c and MRFU_d support this function.
- RRU
Only the following RRUs support this function: RRU3808, RRU3832, RRU3928, RRU3929, RRU3930E, RRU3938, RRU3942, RRU3952, RRU3953, RRU3953w, RRU3958, RRU3959, RRU3961, RRU3962, RRU3971, RRU5501, RRU5502, RRU5507, RRU5508, RRU5903w, RRU5904, RRU5904w, RRU5909s, RRU5909, and RRU5920.

5.2.3.4 Cells

- LTE FDD

The cell can be a 2T cell served by two combined 1T RF modules, or a 4T cell served by two combined 2T RF modules.

The cell bandwidth must be greater than 5 MHz, and two or four transmit antennas must be configured.

If a cell that does not support RF channel shutdown shares an RF module with a multi-antenna UMTS cell, enabling this feature is recommended to ensure the appropriate RF channel is shut down for the UMTS cell, thereby conserving energy.

- UMTS

The cell can be a 2T cell served by two combined 1T RF modules.

A cell providing the same coverage in 0.5/0.5 configuration mode, MIMO Prime cell, intelligent 2T virtual antenna mapping (VAM) cell, or 4T Beam Optimization cell (served by integrated 4T RF modules) supports RF channel shutdown.

Cell Type	Configuration Mode
Cell providing the same coverage in 0.5/0.5 configuration mode	The ULOCCELL.LOCELLTYPE parameter is set to HALFFREQ_CELL , and the ULOCCELL.COVERTYPE parameter is set to SAMEZONE .
MIMO Prime cell	The ULOCCELL.VAM parameter is set to TRUE , the ULOCCELL.TTW parameter is set to FALSE , and the switch specified by the ULOCCELLMACHSPARA.MIMOPRIMESW parameter is turned on.
Intelligent 2T VAM cell	The ULOCCELL.INTEL2TVAM parameter is set to TRUE .
4T Beam Optimization cell	The ULOCCELL.BFMODE parameter is set to FOURTXBEAMOPT .

A common cell with a single transmit channel, cell providing different coverages in 0.5/0.5 configuration mode, MIMO cell, or 4T Beam Optimization cell (served by two combined 2T RF modules) does not support RF channel shutdown. If a cell that does not support RF channel shutdown shares an RF module with a multi-antenna LTE cell supporting RF channel shutdown, enabling Multi-RAT Coordinated Channel Shutdown is recommended to ensure the appropriate RF channel is shut down for the LTE cell, thereby conserving energy.

5.2.3.5 Others

This function must be enabled for all LTE cells served by an RRU.

5.2.4 Operation and Maintenance

5.2.4.1 Data Configuration

5.2.4.1.1 Data Preparation

LTE Side

Parameter Name	Parameter ID	Setting Notes
Multi-RAT Joint Channel Shutdown Switch	CellRfShutdown.MultiRatJointChnShutdownSw	It is recommended that this parameter be set to ON .
RF Shutdown Algorithm Switch	CellRfShutdown.RfShutdownAlgoSwitch	In 4-port scenarios, it is recommended that the RF_SHUTDOWN_4PORT_OPT_SW option be selected.
Start time	CellRfShutdown.StartTime	Set this parameter to its recommended value. It is recommended that this parameter be set to the same value as that of the ULOCCELLALGPARA.PWRSVSTAR TIME parameter on the UMTS side.
Stop time	CellRfShutdown.StopTime	Set this parameter to its recommended value. It is recommended that this parameter be set to the same value as that of the ULOCCELLALGPARA.PWRSVSTOP TIME parameter on the UMTS side.
Reference signal power adjust offset	CellRfShutdown.RsPowerAdjOffset	Set this parameter to its recommended value.
Downlink PRB threshold	CellRfShutdown.DlPrbThd	It is recommended that these parameters be set to values less than or equal to 8 in UL coordinated channel shutdown scenarios.
Uplink PRB threshold	CellRfShutdown.UlPrbThd	

Parameter Name	Parameter ID	Setting Notes
Downlink PRB offset	CellRfShutdown.DlPrbOffset	Set these parameters to their recommended values.
Uplink PRB offset	CellRfShutdown.UlPrbOffset	It is recommended that the feature exit thresholds be set to values less than or equal to 20. If the thresholds are set to a value greater than 20, this feature cannot exit because the actual uplink or downlink PRB usage cannot reach the respective exit threshold when the physical power is restricted after RF channels are shut down. The feature exit threshold for the uplink or downlink equals the sum of the uplink or downlink PRB threshold and the associated offset.
UE Number Threshold	CellRfShutdown.UENumThd	- For cells with a bandwidth of 10 MHz or 15 MHz, the default value is recommended. - For cells with a bandwidth greater than 15 MHz: If this parameter is set to a larger value, cells are more likely to enter the RF channel shutdown state, saving more energy. - If energy saving is a major concern, set this parameter to a value up to 20. A larger value indicates better energy saving results. - If service stability is a major concern, set this parameter to a value no greater than 10. This parameter can be set to 0. In this case, RF channel shutdown can be triggered only when there are no RRC_CONNECTED UEs in the cell. This minimizes impacts on service stability. However, the energy saving gains are also minimal.

Parameter Name	Parameter ID	Setting Notes
PA for even power distribution	CellDlpcPdschPa.PaPcOff	It is recommended that this parameter be set to a value greater than or equal to –3 dB in UL coordinated channel shutdown scenarios.
SignalCongregate-Level	CellPdcchAlgo.ComSigCongregLv	Set this parameter to its recommended value.
Code Rate for Random Access Response and Paging Message	CellDlscAlgo.RarAndPagingCR	It is recommended that this parameter be set to a value greater than or equal to 0.05 in UL coordinated channel shutdown scenarios.
PCH power	CellChPwrCfg.PchPower	It is recommended that this parameter be set to the CellDlpcPdschPa.PaPcOff parameter value plus 3 dB.
Penalty Duration	CellRfShutdown.PenaltyDuration	It is recommended that this parameter be set to 1 minute for UL coordinated channel shutdown.

UMTS Side

Parameter Name	Parameter ID	Setting Notes
Multi-RAT Coordinated Channel Shutdown Switch	ULOCCELLALGPARA.MULTIRATCHANNELSHUTDOWNSW	Set this parameter to ON to activate this feature.
Power Saving Start Time	ULOCCELLALGPARA.PWRSVSTARTTIME	Set this parameter to the time at which the light-traffic period begins. It is recommended that this parameter be set to the same value as that of the CellRfShutdown.StartTime parameter on the LTE side.
Power Saving Stop Time	ULOCCELLALGPARA.PWRSVSTOPTIME	Set this parameter to the time at which the light-traffic period ends. It is recommended that this parameter be set to the same value as that of the CellRfShutdown.StopTime parameter on the LTE side.

Parameter Name	Parameter ID	Setting Notes
Power Saving TX Power High Threshold	ULOCCELLALGPARA.P WRSVATXPWRHIGHT HD	The default value is recommended.
Power Saving TX Power Low Threshold	ULOCCELLALGPARA.P WRSVATXPWRLOWT HD	The default value is recommended.

5.2.4.1.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [5.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

UMTS side

```
//Activating Multi-RAT Coordinated Channel Shutdown
SET ULOCCELLPWRSVAPARA: ULOCCELLID=1, MULTIRATCHANNELSHUTDOWNSW=ON,
PWRSAVSTARTTIME=00&00&00, PWRSAVSTOPTIME=06&00&00, PWRSAVTPWRHIGHTHD=40,
PWRSAVTPWRLOWTHD=30;
```

LTE side

```
//Configuring the PA value
MOD CELLDLPCPDSCHPA: LocalCellId=0, PaPcOff=DB_3_P_A;
//Configuring the CCE aggregation level for common control signaling
MOD CELLPDCCHALGO: LocalCellId=0, ComSigCongregLv=CONGREG_LV4;
//Configuring the code rate for random access response messages and paging messages
MOD CELLDLSCHALGO: LocalCellId=0, RarAndPagingCR=117;
//Configuring the PCH power (to the PaPcOff parameter value plus 3 dB)
MOD CELLCHPWRCFG: LocalCellId=0, PchPwr=xxx;
//Turning on the Multi-RAT Coordinated Channel Shutdown switch and setting the penalty duration to 1
minute
MOD CELLRFSHUTDOWN: LocalCellId=0, RfShutdownSwitch=ON, StartTime=00&00&00,
StopTime=06&00&00, RsPwrAdjOffset=3, DIPrbThd=8, UIPrbThd=8, DIPrbOffset=12, UIPrbOffset=12,
UENumThd=5, MultiRatJointChnShutdownSw=ON, PenaltyDuration=1;
//(Optional) Turning on the Multi-RAT Coordinated Channel Shutdown switch for cells that do not support
RF channel shutdown but share RF modules with UMTS cells to improve the energy saving effect of the
UMTS cells. In this case, the setting of the RF channel shutdown switch does not influence this setting.
MOD CELLRFSHUTDOWN: LocalCellId=1, MultiRatJointChnShutdownSw=ON;
//Turning on the 4-port optimization switch for RF channel shutdown
MOD CELLRFSHUTDOWN: LocalCellId=0, RfShutdownAlgoSwitch=RF_SHUTDOWN_4PORT_OPT_SW-1;
```

Deactivation Command Examples

UMTS side

```
//Deactivating Multi-RAT Coordinated Channel Shutdown  
SET ULOCELLPWRSAPARA: ULOCELLID=1, MULTIRATCHANNELSHUTDOWNSW=OFF;
```

LTE side

```
//Deactivating Multi-RAT Coordinated Channel Shutdown  
MOD CELLRFSHUTDOWN: LocalCellId=0, MultiRatJointChnShutdownSw=OFF;
```

5.2.4.1.3 Using the MAE-Deployment

- Fast batch activation
This function can be batch activated using the Feature Operation and Maintenance function of the MAE-Deployment. For detailed operations, see the following section in the MAE-Deployment product documentation or online help: **MAE-Deployment Operation and Maintenance > MAE-Deployment Guidelines > Enhanced Feature Management > Feature Operation and Maintenance**.
- Single/Batch configuration
This function can be activated for a single base station or a batch of base stations on the MAE-Deployment. For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.2.4.2 Activation Verification

To check whether this feature has taken effect on the UMTS side, operators can use either of the following methods:

- Method 1: Checking the effective duration indicator
If the value of the **VS.MultiRatChannelShutDown.Time** counter is greater than 0, this feature has taken effect.
- Method 2: Using MML commands
Run the **DSP ULOCELLPWRSAP** command on the NodeB to query the value of the **Multi-RAT Coordinated Channel Shutdown State** parameter.
If the value of the **Multi-RAT Coordinated Channel Shutdown State** parameter is **Enable**, the Multi-RAT Coordinated Channel Shutdown feature has taken effect. The value of the **Multi-RAT Coordinated Channel Shutdown Time** parameter is also displayed, indicating the effective duration accumulated since the latest start time of RF channel shutdown.
(Optional) If the value of the **Multi-RAT Coordinated Channel Shutdown State** parameter is **Disable**, check the value of the **Cause of Multi-RAT Coordinated Channel Shutdown Failure** parameter. The value can be **Feature switch set to OFF**, **Not supported by the cell capability**, **Not supported by hardware**, **Insufficient license resource**, **Not within the effective duration**, **Unsatisfied output power**, or **Other reasons**.

To check whether this feature has taken effect on the LTE side, operators can use either of the following methods:

- Method 1: Checking the effective duration indicator

If the value of the **L.ChMeas.DFEE.RFOFF.Dur** counter is greater than 0, this feature has taken effect.

To query the time when RF channel shutdown takes effect, subscribe to the external CHR event **PRIVATE_POWER_SAVING_INFO_LTE_NB** in the MR subscription category on the NIC, and then view the record whose **PowerSavingType** is **RFshutdown** on the NIC.

- Method 2: Using MML commands
Run the **DSP CELL** command to check the value of the **Cell power save state** parameter. If it is **RF Channel Intelligent Shutdown Mode**, this function has taken effect.
(Optional) If RF channel shutdown does not take effect in a cell, run the **DSP CELLRFSHUTDOWN** command to query the current status of RF channel intelligent shutdown and the reason why it does not take effect. [Table 5-12](#) and [Table 5-13](#) describe the values of **Current status about RF intelligent shutdown** and **Latest exiting cause about RF intelligent shutdown** in the command output, respectively.

Table 5-12 Causes indicated by the current status about RF channel intelligent shutdown

Cause Value	Description
Entering the RF channel intelligent shutdown state	The cell is in the RF channel intelligent shutdown state.
Multi-RAT joint channel shutdown status	The cell is in the multi-RAT coordinated channel shutdown state.
Not entering because the number of UEs in connected mode exceeds threshold	The number of RRC_CONNECTED UEs in the cell is greater than the RRC_CONNECTED UE number threshold.
Not entering because LCS is enabled	The OtdoaSwitch option of the ENodeBAlgoSwitch.LcsSwitch parameter or the NB_OTDOA_SWITCH option of the CellLcsCfg.CellLcsSwitch parameter is selected.
Not entering because the state is delayed	In some scenarios, when a cell exits the RF channel intelligent shutdown state, a penalty period starts. The eNodeB determines whether the cell can enter the RF channel intelligent shutdown state again only after the penalty period expires.

Cause Value	Description
Not entering because the cell bandwidth, PA value, or CCE aggregation level for common signaling does not meet requirements	The cell bandwidth is less than or equal to 5 MHz. However, the cell bandwidth is required to be greater than 5 MHz. The CellDlpcPdschPa.PaPcOff parameter is set to a value less than -3 dB. The CellPdcchAlgo.ComSigCongregLv parameter is not set to CONGREG_LV4.
Not entering because the number of antennas does not meet requirements	Fewer than two antennas are working. RF channel intelligent shutdown requires at least two antennas. The channels to be shut down include the current calibration channel. On the alarm management console, check whether ALM-29243 Cell Capability Degraded is reported. If this alarm is reported, clear this alarm.
Not entering because the power is insufficient	The power is limited in the downlink. This occurs when the CellDlschAlgo.RarAndPagingCR parameter is set to a value too small or the system information overhead is too large.
Not entering because the switch is turned off or the license is abnormal	The CellRfShutdown.RfShutdownSwitch parameter is set to OFF.
Not entering because of other reasons	The cell does not enter the RF channel intelligent shutdown state. The possible causes are as follows: • The current cell is unavailable. • Functions that do not work with RF channel intelligent shutdown, for example, unlimited-service UE guarantee, are enabled. • The cell is already in another energy saving state, for example, intelligent power-off of carriers in the same coverage or low power consumption mode.
Not entering because the existence of QCI=1 UEs	There are UEs using voice services in the cell.

Cause Value	Description
Not entering because the UL/DL PRB ratio exceeds the PRB threshold	The uplink PRB usage of the cell is greater than the CellRfShutdown.UlPrbThd parameter value. The downlink PRB usage of the cell is greater than the CellRfShutdown.DlPrbThd parameter value.
RF intelligent shutdown is in the detection mode.	The CellRfShutdown.RfShutdownSwitch parameter is set to DETECTION .
Not entering because co-carrier cell not trigger	The associated NB-IoT cell whose carrier is deployed in LTE in-band mode does not meet the shutdown conditions. Alternatively, other co-carrier LTE cells do not meet the shutdown conditions.
Not entering because cell not in shutdown time	The system time is not within the period of RF channel intelligent shutdown.
Not entering because beam mode not support	The current beam configuration does not support RF channel intelligent shutdown.
Not entering because load simulation is enabled	Load simulation is enabled in the cell.

Table 5-13 Latest exiting causes about RF intelligent shutdown

Cause Value	Description
The switch is turned off or the license is abnormal	The CellRfShutdown.RfShutdownSwitch parameter is set to OFF , or the licensed resources are insufficient.
Time expires	The system time is later than the stop time.
The number of UEs in connected mode exceeds threshold	The number of RRC_CONNECTED UEs in the cell is greater than the RRC_CONNECTED UE number threshold.
The power is insufficient	The power is limited in the downlink.

Cause Value	Description
None (Initial entering)	If the value of the Current status about RF intelligent shutdown parameter is Entering the RF channel intelligent shutdown state and that of the Latest exiting cause about RF intelligent shutdown parameter is None (Initial entering) , the cell is in the RF channel intelligent shutdown state. If the value of the Current status about RF intelligent shutdown parameter is not Entering the RF channel intelligent shutdown state and that of the Latest exiting cause about RF intelligent shutdown parameter is None (Initial entering) , the cell has never entered the RF channel intelligent shutdown state.
Load simulation is enabled	Load simulation is enabled in the cell.
LCS is enabled	The OtdoaSwitch option of the ENodeBAlgoSwitch.LcsSwitch parameter or the NB_OTDOA_SWITCH option of the CellLcsCfg.CellLcsSwitch parameter is selected.
Other reasons	The possible causes are as follows: • The current cell is unavailable. • Functions that do not work with RF channel intelligent shutdown, for example, unlimited-service UE guarantee, are enabled.
The existence of QCI=1 UEs	There are UEs using voice services in the cell.
The UL/DL PRB ratio exceeds the PRB threshold	The PRB usage does not meet the following conditions: • The uplink PRB usage of the cell is less than or equal to the sum of the CellRfShutdown.UlPrbThd and CellRfShutdown.UlPrbOffset parameter values. • The downlink PRB usage of the cell is less than or equal to the sum of the CellRfShutdown.DlPrbThd and CellRfShutdown.DlPrbOffset parameter values.

Cause Value	Description
The co-carrier cell exit	The associated NB-IoT cell whose carrier is deployed in LTE in-band mode exits the RF channel intelligent shutdown state. Alternatively, other co-carrier LTE cells exit the RF channel intelligent shutdown state.

Method 3: Using signaling tracing

Log in to the MAE-Access, and observe the actual transmit power of the channel by tracing signaling messages. The channel whose output power is 0 is the channel that is actually shut down.

On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**. On the displayed **Signaling Trace Management** tab page, observe the actual transmit power of the channel by tracing signaling messages. The channel whose output power is 0 is the channel that is actually shut down.

5.2.4.3 Network Monitoring

Compare the values of the following counters one week before and after the Multi-RAT Coordinated Channel Shutdown feature is enabled to evaluate the energy conservation gains.

- Counters measuring the energy consumption of the entire base station

Counter ID	Counter Name
1593835637	VS.EnergyCons.BTS.Adding.LTE
1593835959	VS.EnergyCons.BTS.Adding.Service.LTE
1593835628	VS.EnergyCons.BTS.Adding.UMTS
1593835958	VS.EnergyCons.BTS.Adding.Service.UMTS

- Counters measuring the energy consumption of RF modules

Counter ID	Counter Name
1593835836	VS.EnergyCons.BTSBoard

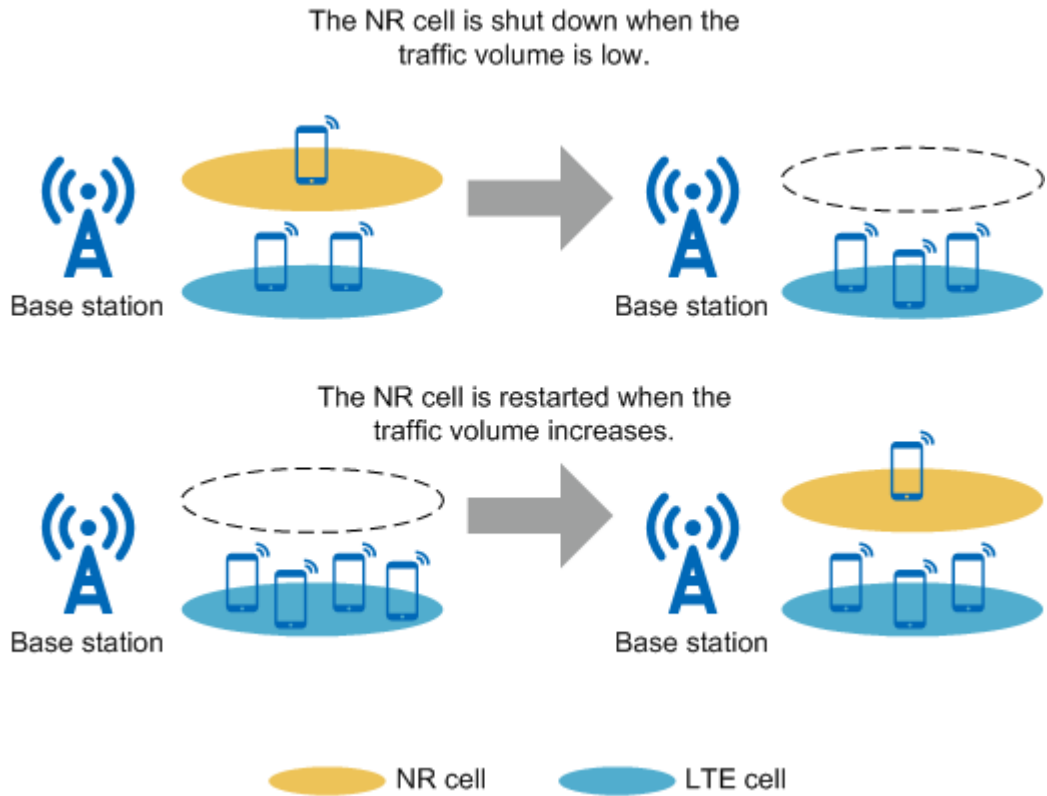
6 Frequency-Domain Energy Saving

6.1 LTE and NR Intelligent Carrier Shutdown

6.1.1 Principles

NR cells in operation still consume energy when there is light or even no traffic in the area covered by both LTE and NR networks. **In this case, the LTE and NR intelligent carrier shutdown function migrates services from NR cells to LTE cells and then shuts down NR carriers. If the power amplifier of the RF module for the disabled carriers does not serve other active carriers, it will also be turned off to reduce overall energy consumption, as shown in [Figure 6-1](#).**

Figure 6-1 Working principles of LTE and NR intelligent carrier shutdown



This function is enabled when the following switches are turned on:

- On the LTE side: the **CellAlgoExtSwitch.LnrSmartCarrierShutdownSw** parameter
- On the NR side: the **LNR_SMART_CARRIER_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter

The energy saving policies of LTE and NR intelligent carrier shutdown are specified by the parameters in the **NRDUCellPowerSaving** MO on the NR side. A specific energy saving policy and its type are determined by the **NRDUCellPowerSaving.PowerSavingPolicyIndex** and **NRDUCellPowerSaving.PowerSavingType** parameters.

6.1.1.1 Identifying Overlapping Neighboring Cells

When LTE and NR networks have overlapping coverage areas, the LTE cells provide continuous coverage for the NR cell. The LTE and NR intelligent carrier shutdown function migrates services from the NR cell to LTE cells and shuts down the NR cell. In this case, the LTE cells are the overlapping neighboring cells of the NR cell.

NOTE

In multi-operator sharing scenarios, if RAN sharing with common carrier is enabled for multiple operators in capacity-layer NR cells, the **CellAlgoExtSwitch.LnrSmartCarrierShutdownSw** parameter must be set to **ON** and the **CARRIER_SHUTDOWN_OVERLAP_FLAG** option of the **NrNRRelationship.AggregationAttribute** parameter must be selected for the overlapping neighboring LTE cells of the primary and secondary operators.

NSA Networking/NSA and SA Hybrid Networking

The overlapping neighboring cells can be identified in any of the following ways:

- Method 1: automatic identification by the system
No separate manual configuration is required. The system automatically identifies the neighbor relationship between NR and LTE cells with overlapping coverage areas through the X2 link. This method is recommended and no separate manual configuration is required.
- Method 2: manual configuration for identification
When configuring a neighbor relationship between the LTE and NR cells, the LTE cell is configured as an overlapping neighboring cell of the target NR cell by selecting the **CARRIER_SHUTDOWN_OVERLAP_FLAG** option of the **NrNRRelationship.AggregationAttribute** parameter. When manually configuring such a neighbor relationship on the LTE side, this option must be set as planned for all cells to be enabled with LTE and NR intelligent carrier shutdown.

NOTE

In NSA networking or NSA and SA hybrid networking, when a cell is configured with the indicator for an overlapping neighboring cell, its neighbor relationship cannot be removed by the inter-RAT ANR function on the LTE side. Instead, the neighbor relationship must be removed by running the **RMV NRNRELATIONSHIP** command on the LTE side.

SA Networking

When configuring a neighbor relationship between the LTE and NR cells, the **LNR_CARR_SHUTDOWN_OVERLAP_IND** option of the **NRCellEutraNRelation.PowerSavingOverlapInd** parameter is selected on the NR side to configure the LTE cell as an overlapping neighboring cell of the NR cell. In inter-base-station SA networking, when configuring such a neighbor relationship, the **CARRIER_SHUTDOWN_OVERLAP_FLAG** option of the **NrNRRelationship.AggregationAttribute** parameter also needs to be selected on the LTE side to configure the LTE cell as an overlapping neighboring cell of the target NR cell.

NOTE

- In SA networking, the automatic identification of the overlapping neighboring cells by the system is implemented on the MAE-Optimization. For details, see the feature parameter description of the MAE-Optimization.
- In SA networking, when a cell is configured with the indicator for an overlapping neighboring cell, its neighbor relationship cannot be removed by the ANR function on the NR side. Instead, the neighbor relationship must be removed by running the **RMV NRCELLEUTRANRELATION** command on the NR side.

For LTE and NR intelligent carrier shutdown to take effect in inter-base-station SA networking, X2 links need to be set up. X2 links can be set up automatically or manually, with automatic setup recommended. For details about X2 configurations, see *X2 Self-Management in SA Networking*. X2 setup is expected to be triggered on the LTE side half an hour later if the following conditions are met:

- The **CARRIER_SHUTDOWN_OVERLAP_FLAG** option of the **NrNRRelationship.AggregationAttribute** parameter is selected for LTE cells.

- LTE and NR intelligent carrier shutdown is enabled.
- The switches for LTE-NR inter-base-station power saving and X2 setup in SA networking are turned on. The specific switches are as follows:
 - LTE-NR SA gNodeB X2 setup switch on the LTE side, controlled by the **LNR_SA_GNB_X2_SETUP_SW** option of the **eNodeBOamParam.InterfaceSonConfigSwitch** parameter
 - LTE-NR SA inter-base-station power saving switch on the LTE side, controlled by the **LNR_INTER_BS_PWR_SAVE_SW** option of the **EnodebPwrSavingAlgo.PwrSavingAlgoSwitch** parameter
 - SA X2 setup switch on the NR side, controlled by the **X2_SETUP_FOR_SA_SW** option of the **gNBInterfaceParam.X2ConfigSw** parameter
 - LTE-NR SA inter-base-station power saving switch on the NR side, controlled by the **SA_LNR_INTER_BS_PWR_SAVE_SW** option of the **gNBPwrSavingAlgo.PwrSavingAlgoSwitch** parameter
- No X2 link is available between the eNodeB and the target gNodeB.

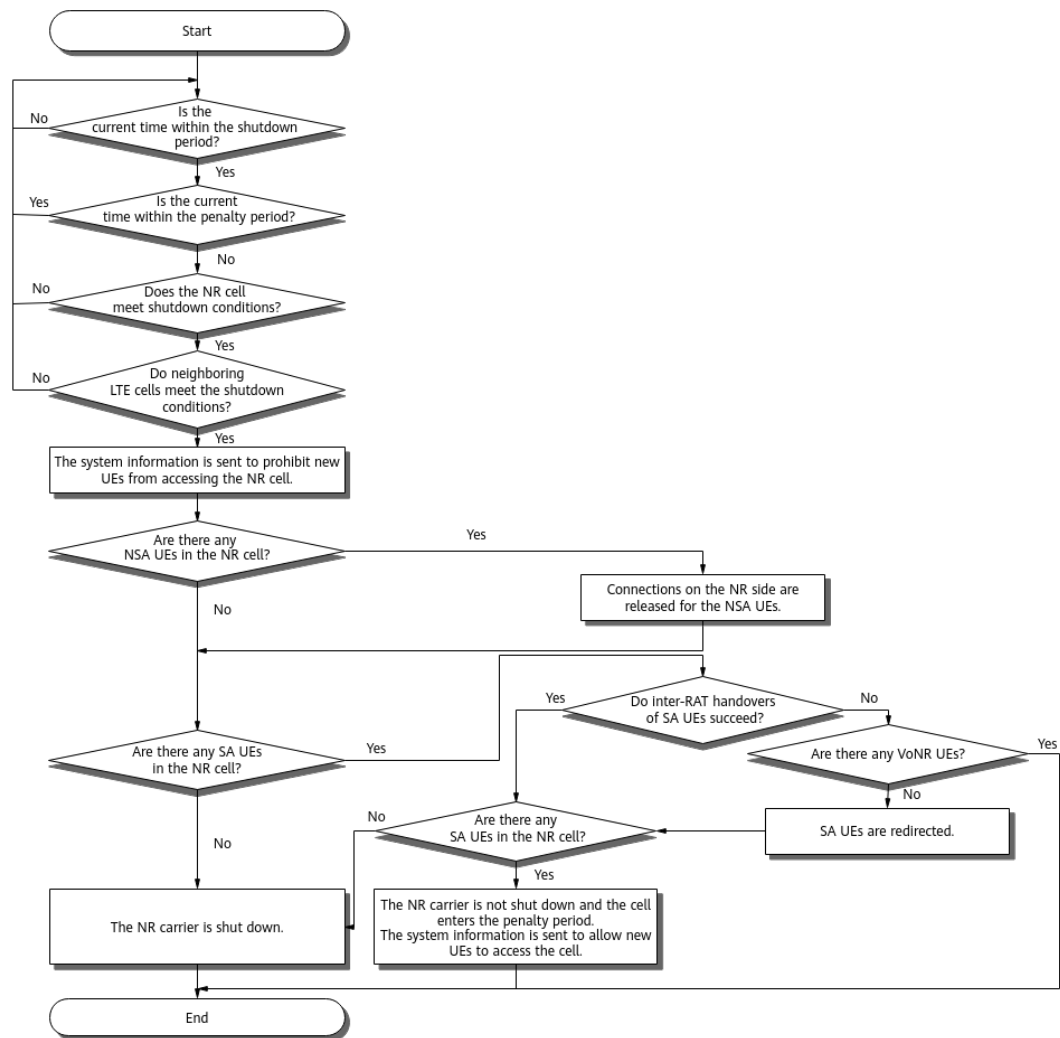
 NOTE

- In inter-base-station SA networking, the X2 transport addresses of either the primary operator of the eNodeB or the operators in the intersection set of operators of the gNodeB and eNodeB must be planned in advance.
- If an X2 link is successfully set up but energy saving does not take effect as the corresponding conditions are not met, the X2 link may be automatically removed. To avoid this issue, it is recommended that the X2 automatic removal timer (specified by the **gNBX2SonConfig.X2SonDeleteTimerForX2Usage** parameter on the NR side and the **GlobalProcSwitch.X2SonDeleteTimerforX2Usage** parameter on the LTE side) be set to the default value **10080** (seven days).

6.1.1.2 Entering the Carrier Shutdown State

Figure 6-2 illustrates the LTE and NR intelligent carrier shutdown procedure, which is initiated when specific switches on the LTE and NR sides are turned on.

Figure 6-2 LTE and NR intelligent carrier shutdown procedure



1. The gNodeB checks whether the current time is within the shutdown period, which is specified by the **NRDUCellPowerSaving.StartTime** and **NRDUCellPowerSaving.StopTime** parameters. If it is, the procedure goes to the next step. Otherwise, the checking process is repeated.
2. The gNodeB checks whether the current time is within the penalty period specified by the **NRDUCellPowerSaving.PwrSavingPenaltyTime** parameter. If it is not, the procedure goes to the next step. If it is, even when the preceding trigger condition is met, the procedure does not go to the next step, and the checking process is repeated.
3. The NR cell periodically checks whether the conditions for entering the intelligent carrier shutdown state are met. The granularity of the evaluation period is specified by the **NRDUCellPowerSaving.EvalPeriodGranularity** parameter.
 - If this parameter is set to **MINUTE_LEVEL**, the granularity is minute-level. In this case, the evaluation period is specified by the **NRDUCellPowerSaving.ShutdownTrigJudgePeriod** parameter.
 - If this parameter is set to **SECOND_LEVEL**, the granularity is second-level. In this case, the evaluation period is specified by the **NRDUCellPowerSaving.SecondLevelTrigEvalPrd** parameter.

The conditions for entering the intelligent carrier shutdown state are listed below:

- Average number of active UEs in the NR cell $\leq$ UE number threshold (specified by the **NRDUCellPowerSaving.UserNumThld** parameter)
- Downlink PRB usage of the NR cell $\leq$ Downlink PRB usage threshold (specified by the **NRDUCellPowerSaving.DlPrbThld** parameter)
- Uplink PRB usage of the NR cell $\leq$ Uplink PRB usage threshold (specified by the **NRDUCellPowerSaving.UlPrbThld** parameter)
- In SUL co-deployment scenarios, the total number of SUL UEs in the NR cell is less than or equal to the value of the **NRDUCellPowerSaving.SulUserNumThld** parameter.

If these conditions are met, the procedure goes to the next step. Otherwise, the checking process is repeated.

 NOTE

- Active UEs are UEs in RRC_CONNECTED mode in an NR cell.
 - The downlink PRB usage of an NR cell is calculated as the average number of PRBs used by PDSCH DRB services divided by the average number of available PRBs in the downlink.
 - The uplink PRB usage of an NR cell is calculated as the average number of PRBs used by PUSCH DRB services divided by the average number of available PRBs in the uplink.
4. The eNodeB sends load information about the neighboring LTE cells of the NR cell to the gNodeB. The gNodeB requires that the neighboring LTE cells be available.
 5. In multi-operator sharing scenarios, operators are divided into normal operators and reserved operators for the NR cell and neighboring LTE cells.
 - The reserved operator of the NR cell is an operator with the **NRExternalNCellPlmn.PlmnReservedFlag** parameter set to **TRUE**.
 - The reserved operator of an LTE cell is an operator with the **CELLOP.CellReservedForOp** parameter set to **CELL_RESERVED_FOR_OP** or **CELL_DYNAMIC_RESERVED_FOR_OP**.

The following requirements must be met:

- The PLMN IDs of the normal operators of the NR cell must be a subset of the PLMN IDs of normal operators of neighboring LTE cells.
 - The PLMN IDs of the reserved operators of the NR cell must be a subset of the PLMN IDs of normal or reserved operators of neighboring LTE cells.
6. The gNodeB checks whether a proportion of available neighboring cells (the proportion is specified by the **NRDUCellPowerSaving.EutranCarrShutdownRatio** parameter) meet all the following conditions throughout the duration specified by the **NRDUCellPowerSaving.ShutdownTrigJudgePeriod** parameter:
 - Downlink PRB usage of the neighboring LTE cell $<$ E-UTRAN downlink PRB usage threshold (specified by the **NRDUCellPowerSaving.EutranDlPrbThld** parameter)
 - Uplink PRB usage of the neighboring LTE cell $<$ Neighboring cell uplink PRB usage threshold (specified by the **NRDUCellPowerSaving.NCellUlPrbThld** parameter)

- Number of NR UEs in the neighboring LTE cell < E-UTRAN NR user number threshold (specified by the **NRDUCellPowerSaving.EutranNrUserNumThld** parameter)
- Downlink PRB usage of NR UEs in the neighboring LTE cell < E-UTRAN NR user downlink PRB usage threshold (specified by the **NRDUCellPowerSaving.EutranNrUserDlPrbThld** parameter)
- Uplink PRB usage of NR UEs in the neighboring LTE cell < E-UTRAN NR user uplink PRB usage threshold (specified by the **NRDUCellPowerSaving.EutranNrUserUlPrbThld** parameter)

The eNodeB also checks whether the proportion of neighboring LTE cells that meet the preceding conditions is greater than or equal to the E-UTRAN carrier shutdown ratio specified by the

NRDUCellPowerSaving.EutranCarrShutdownRatio parameter. If these conditions are met, the procedure goes to the next step. Otherwise, the checking process is repeated.

7. The gNodeB sends system information to prohibit new UEs from accessing the NR cell.
8. The gNodeB checks whether there are NSA UEs in the NR cell. If there are, their connections on the NR side are released.
9. The gNodeB checks whether there are SA UEs in the NR cell. If there are, the procedure goes to the next step; otherwise, it jumps to step 13.
10. The gNodeB performs inter-RAT handovers of SA UEs to LTE cells. If handovers fail:
 - When there are no VoNR UEs, SA UEs are redirected to LTE cells and then the procedure goes to the next step.
 - When there are VoNR UEs, they are not redirected. In the current carrier shutdown procedure, the cell does not enter the LTE and NR intelligent carrier shutdown state. Instead, and the entire carrier shutdown procedure restarts.
11. The gNodeB rechecks whether there are SA UEs in the NR cell. If there are, the procedure goes to the next step; otherwise, it jumps to step 13.
12. The NR carrier is not shut down and the cell enters the penalty period specified by the **NRDUCellPowerSaving.PwrSavingPenaltyTime** parameter. The gNodeB sends system information to allow new UEs to access the NR cell.
13. The gNodeB shuts down the carrier in the NR cell and blocks the NR cell. The NR cell enters the LTE and NR intelligent carrier shutdown state.

If the **INTER_RAT_GRAD_PWR_SAVING_SW** option of the **gNBpwrSavingAlgo.PwrSavingAlgoSwitch** parameter is selected, some neighboring LTE cells can serve as capacity-layer cells for other basic-layer cells and enter the energy saving state due to functions such as intelligent power-off of carriers in the same coverage, Cell Switch-off Based on Automatic Co-coverage Identification, multi-carrier coordinated energy saving, GUL joint carrier shutdown, or dynamic carrier shutdown.

6.1.1.3 Exiting the Carrier Shutdown State

After intelligent carrier shutdown takes effect, the base station periodically checks the conditions for exiting this state. The

NRDUCellPowerSaving.EvalPeriodGranularity parameter specifies the evaluation period granularity for the exit condition.

- If this parameter is set to **MINUTE_LEVEL**, the granularity is minute-level. In this case, the evaluation period is specified by the **NRDUCellPowerSaving.ExitEvaluationPeriod** parameter.
- If this parameter is set to **SECOND_LEVEL**, the granularity is second-level. In this case, the evaluation period is specified by the **NRDUCellPowerSaving.SecondLevelExitEvalPrd** parameter.

After the cell enters the carrier shutdown state, the gNodeB checks whether either of the following conditions is met. If yes, the NR cell exits the carrier shutdown and the NR carrier is enabled again.

- The current time is not within the shutdown period.
- Proportion of neighboring LTE cells that meet one of the following conditions throughout the evaluation period $\geq$ E-UTRAN carrier shutdown ratio (specified by **NRDUCellPowerSaving.EutranCarrShutdownRatio**) – E-UTRAN carrier shutdown exit offset (specified by **NRDUCellPowerSaving.EutranCarrShutdownExitOfs**)
 - Downlink PRB usage of the neighboring LTE cell $\geq$ E-UTRAN downlink PRB threshold (specified by **NRDUCellPowerSaving.EutranDlPrbThld**) + E-UTRAN downlink PRB offset (specified by **NRDUCellPowerSaving.EutranDlPrbOffset**)
 - Uplink PRB usage of the neighboring LTE cell $\geq$ Neighboring cell uplink PRB threshold (specified by **NRDUCellPowerSaving.NCellUlPrbThld**) + Neighboring cell uplink PRB offset (specified by **NRDUCellPowerSaving.NCellUlPrbOffset**)
 - Number of NR UEs in the neighboring LTE cell $\geq$ E-UTRAN NR user number threshold (specified by **NRDUCellPowerSaving.EutranNrUserNumThld**) + E-UTRAN NR user number threshold offset (specified by **NRDUCellPowerSaving.EutranNrUserNumThldOfs**)
 - Downlink PRB usage of NR UEs in the neighboring LTE cell $\geq$ E-UTRAN NR user downlink PRB threshold (specified by **NRDUCellPowerSaving.EutranNrUserDlPrbThld**) + E-UTRAN NR user downlink PRB threshold offset (specified by **NRDUCellPowerSaving.EutranNrUserDlPrbThldOfs**)
 - Uplink PRB usage of NR UEs in the neighboring LTE cell $\geq$ E-UTRAN NR user uplink PRB threshold (specified by **NRDUCellPowerSaving.EutranNrUserUlPrbThld**) + E-UTRAN NR user uplink PRB threshold offset (specified by **NRDUCellPowerSaving.EutranNrUserUlPrbThldOfs**)
 - The PLMN IDs of the normal operators of the NR cell are not a subset of the PLMN IDs of normal operators of neighboring LTE cells.
 - The PLMN ID of the reserved operator of the NR cell is not a subset of the PLMN IDs of normal operators or the PLMN IDs of reserved operators of neighboring LTE cells.

When a neighboring LTE cell, functioning as a capacity-layer cell of other basic-layer cells, enters the energy saving state, whether this cell is considered as a neighboring cell meeting the exit conditions depends on the setting of the

INTER_RAT_GRAD_PWR_SAVING_SW option of the **gNBpwrSavingAlgo.PwrSavingAlgoSwitch**.

- When this option is deselected, this LTE cell is considered as a neighboring cell meeting the exit conditions.
- When this option is selected, this LTE cell is not considered as a neighboring cell meeting the exit conditions and the exit of the carrier shutdown state will not be triggered.

Upon the exit from carrier shutdown,

- A penalty is triggered to prevent the ping-pong carrier shutdown if a cell stays in the carrier shutdown state for less than 15 minutes. The penalty duration is specified by the **NRDUCellPowerSaving.PwrSavingPenaltyTime** parameter.
- No penalty is triggered if the NR cell exits this state because the switch is turned off, the energy saving period ends, the cell is deactivated and then activated, or the base station software is upgraded. After the NR cell is deactivated and then activated or the base station software upgrade is complete, the base station determines whether the cell enters the carrier shutdown state again based on related entering conditions.

 NOTE

- The number of NR UEs in neighboring LTE cells of the NR cell is counted based on the UE capability (that is, whether a UE supports SA or NSA).
- In carrier aggregation (CA) scenarios, only NR UEs on the primary component carrier (PCC) are counted.
- In EN-DC scenarios, after an NR cell exits the carrier shutdown state, the LTE side needs to be informed of the exit through an ENDC_CONFIG_UPDATE message (which is sent at an interval of 3 minutes) over the X2 interface. The NR cell can be added to an SCG through periodic addition only after its status has changed to available on the LTE side.

6.1.2 Network Analysis

6.1.2.1 Benefits

In areas covered by both LTE and NR networks, enabling LTE and NR intelligent carrier shutdown reduces the energy consumption of base stations when the network is under light or no load. However, the energy saving gains are less-than-desirable in the following scenarios:

- Multiple carriers share the same PA of an RF module. The PA cannot be shut down if an NR carrier is shut down but the other carriers are still working.
- An RF module has an excessively large temperature differential within 24 hours before the start time of carrier shutdown. In this case, the RF module does not enter the carrier shutdown state. This is because hardware faults may occur if the temperature differential is excessively large before and after the RF module enters the carrier shutdown state.

6.1.2.2 Impacts

The impacts between this function and other functions are described in [6.1.3.2.3 Function Impacts](#).

- QoS of a single UE
UEs in a to-be-shutdown NR cell will be migrated to neighboring LTE cells in the overlapping coverage areas. Due to less service processing capability in the neighboring LTE cells, the peak rate of a single UE will decrease.
- Number of inter-RAT handovers and cell reselections
This function causes inter-RAT mobility.
 - After an NR cell enters the LTE and NR intelligent carrier shutdown state, multimode UEs are transferred to LTE cells through handovers or cell reselections. The numbers of inter-RAT handovers and cell reselections increase.
 - After an NR cell exits the LTE and NR intelligent carrier shutdown state, multimode UEs are transferred to the NR cell through handovers or cell reselections. The numbers of inter-RAT handovers and cell reselections increase.
- During NR carrier shutdown, the NR cell does not allow UE access. If other cells attempt to hand over UEs to this cell, the handover will fail, causing the handover success rate to deteriorate.
- NSA networking or NSA and SA hybrid networking
Table 6-1 describes the impacts on counters on the LTE side.

Table 6-1 Impacts on counters on the LTE side

Counter Name	Impacts
L.Traffic.User.NsaDc.PCell.Avg	The values of the counters may decrease when an NR cell enters the LTE and NR intelligent carrier shutdown state.
L.NsaDc.SgNB.Add.Att	
L.NsaDc.SgNB.Add.Succ	
L.NsaDc.ScgFailure	
L.NsaDc.SgNB.Rmv.Att	The value of this counter may increase, as UEs are released when an NR cell enters the LTE and NR intelligent carrier shutdown state.
L.NsaDc.E-RAB.Mod.Ind.Att	In Option 3x architecture, the values of the counters may increase because the E-RAB modification procedure is initiated when an NR cell enters the LTE and NR intelligent carrier shutdown state.
L.NsaDc.E-RAB.Mod.Ind.Succ	

In NSA and SA hybrid networking, SA UEs (if any) are handed over to LTE cells, affecting performance counters listed in **Table 6-2**; SA UEs that fail to be handed over are blindly redirected to LTE cells, affecting performance counters listed in **Table 6-3**.

Table 6-2 Impacts on counters related to handovers

RAT	Counter Name	Impact
LTE	L.IRATHO.N2E.PrepareAttIn	The values of the counters may increase if SA UEs are handed over to LTE cells.
LTE	L.IRATHO.N2E.PrepareInSuccess	
LTE	L.IRATHO.N2E.ExecSuccessIn	
NR	N.HO.InterRAT.N2E.PrepareAttOut	
NR	N.HO.InterRAT.N2E.ExecAttOut	
NR	N.HO.InterRAT.N2E.ExecSuccessOut	
NR	N.HO.InterRAT.N2E.PowerSaving.ExecSuccessOut	The values of the counters may increase if SA UEs are handed over to LTE cells for energy saving.
NR	N.HO.InterRAT.N2E.PowerSaving.ExecSuccessOut.PLMN	
NR	N.HO.InterRAT.N2E.PowerSaving.ExecAttOut	
NR	N.HO.InterRAT.N2E.PowerSaving.ExecAttOut.PLMN	
NR	N.HO.InterRAT.N2E.PowerSaving.PrepareAttOut	
NR	N.HO.InterRAT.N2E.PowerSaving.PrepareAttOut.PLMN	
NR	N.HO.InterRAT.N2E.PowerSaving.ExecSuccessOut.Voice	
NR	N.HO.InterRAT.N2E.PowerSaving.ExecSuccessOut.Voice.PLMN	
NR	N.HO.InterRAT.N2E.PowerSaving.ExecAttOut.Voice	
NR	N.HO.InterRAT.N2E.PowerSaving.ExecAttOut.Voice.PLMN	

RAT	Counter Name	Impact
NR	N.HO.InterRAT.N2E.PowerSaving.PrepAttOut.Voice	
NR	N.HO.InterRAT.N2E.PowerSaving.PrepAttOut.Voice.PLMN	

Table 6-3 Impacts on counters related to blind redirections

RAT	Counter Name	Impact
LTE	L.RA.GrpA.Att	The values of the counters may increase if SA UEs are blindly redirected to LTE cells.
LTE	L.RA.GrpB.Att	
NR	N.RRCRedirection.N2E.Blind	
NR	N.RRCRedirection.N2E.Blind.PowerSaving	The values of the counters may increase if SA UEs are blindly redirected to LTE cells for energy saving.
NR	N.RRCRedirection.N2E.Blind.PowerSaving.PLMN	

- SA networking
 - When NR carriers are to be shut down, SA UEs (if any) are handed over to LTE cells, affecting performance counters listed in [Table 6-2](#). SA UEs that fail to be handed over are blindly redirected to LTE cells, affecting performance counters listed in [Table 6-3](#).
 - In inter-base-station SA networking, X2 link setup may be triggered for energy saving purposes. This occupies the specifications of EN-DC X2 links and affects the experience of NSA UEs. The probability of ALM-29225 Maximum Number of X2 Interfaces Reached being reported also increases. Therefore, this function is not recommended when the X2 specifications are less than required.
 - In inter-base-station SA networking, if the **LNR_INTER_BS_PWR_SAVE_SW** option of the **EnodebPwrSavingAlgo.PwrSavingAlgoSwitch** parameter or the **LNR_SA_GNB_X2_SETUP_SW** option of the **eNodeBOamParam.InterfaceSonConfigSwitch** parameter is selected but the **X2_SETUP_FOR_SA_SW** option of the **gNBInterfaceParam.X2ConfigSw** parameter is deselected on the gNodeB side, ALM-29204 X2 Interface Fault and ALM-25888 SCTP Link Fault may be reported.
 - In inter-base-station SA networking, the configurations of the neighboring NR cell with overlapping coverage areas on the LTE side must be the same as those on the NR side, such as the cell ID, base

- station ID, and operator information. Otherwise, the SA X2 link may fail to be set up.
- LTE and NR intelligent carrier shutdown saves energy by shutting down downlink carriers instead of deactivating cells. When carriers are shut down, the counters and KPIs listed in [Table 6-4](#) will still be measured. However, energy saving effects are not affected.

Table 6-4 Related performance counters and KPIs

Function Subset	Counter/KPI	Description
PRB Measurement	• Counters related to the number of available PRBs • Counters related to the number of available physical downlink control channel (PDCCH) control channel elements (CCEs) • PRB usage	None
DU Cell Operator PRB Measurement		
DU Cell Algorithm Measurement	• Counters related to the total number of physical uplink shared channel (PUSCH) transmission time intervals (TTIs) • Counters related to the total number of PDSCH TTIs	None
DU Cell SI Message Measurement	Number of SI transmissions	None

Function Subset	Counter/KPI	Description
Channel Quality Measurement	- Counters related to interference and noise received in the uplink - Counters related to uplink received signal strength indicator (RSSI) - Counters related to uplink and downlink power - Counters related to the average interference and noise detected on a single PRB - Counters related to the maximum interference and noise detected on a single PRB	The PWR_SAVING_COUNTER_CORR_SW option of the gNBOamParam.StatisticsStrategySwitch parameter specifies whether to enable reporting of invalid values of uplink interference-related counters during carrier shutdown. - If this option is selected, invalid uplink interference counter values are reported for carriers shut down. - If this option is deselected, measured uplink interference counter values are reported for carriers shut down.
DU Cell TRP Channel Quality Measurement		
Uplink Interference Measurement on PRB 0 to PRB 99		
Uplink Interference Measurement on PRB 100 to PRB 199		
Uplink Interference Measurement on PRB 200 to PRB 272		

In the case of false physical random access channel (PRACH) detection where the base station mistakenly regards interference or noise as UEs' random access preambles, the following counters and KPIs are still measured:

- Counters related to the number of random preamble receptions
- Counters related to the number of random preamble responses
- Counters related to the number of UEs scheduled in the uplink and downlink in a cell
- Counters related to uplink data transmission duration
- Counters related to the number of transport blocks (TBs) used for uplink data transmission
- Counters related to modulation and coding scheme (MCS) index distribution for scheduling on the PUSCH
- Counters related to the number of RBs used for scheduling on the PUSCH
- PUSCH RB usage
- As the cell in the carrier shutdown state does not send downlink paging messages, the counter listed in [Table 6-5](#) is no longer measured.

Table 6-5 Counter related to paging

Function Subset	Counter
DU Cell Paging Measurement	Number of paged UEs over the Uu interface in a cell

- Assuming that the neighboring base stations are in the carrier shutdown state, CEUs in weak coverage areas may access the cells served by the local base station. This affects KPIs related to the aspects such as access and service drop of the local base station. For example, the value of **N.NsaDc.DRB.AbnormRel** may increase.

6.1.3 Requirements

6.1.3.1 Licenses

RAT	Feature ID	Feature Name	Model	NE	Sales Unit
NR	MRFD-160263	LTE and NR Intelligent Carrier Shutdown (NR)	NR0S0LNICS00	gNodeB	Per Cell
LTE TDD	MRFD-160233	LTE and NR Intelligent Carrier Shutdown (LTE TDD)	LT4SLNRCSTD	eNodeB	Per Cell
LTE FDD	MRFD-160223	LTE and NR Intelligent Carrier Shutdown (LTE FDD)	LT1S0LNICS00	eNodeB	Per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

6.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.1.3.2.1 Prerequisite Functions

- NSA networking or NSA and SA hybrid networking

Function Name	Function Switch	Reference	Description
NSA networking based on EPC	LTE: NSA_DC_CAPABILITY_SWITCH option of the NsaDcMgmtConfig.NsaDcAlgoSwitch parameter	<i>EPC-based NSA Basics</i>	For each NR cell, the NSA_DC_CAPABILITY_SWITCH option of the NsaDcMgmtConfig.NsaDcAlgoSwitch parameter must be selected for at least one overlapping neighboring LTE cell.

- SA networking
None

6.1.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Set the LTE and NR intelligent carrier shutdown period according to the network plan, and ensure it does not overlap with the timing carrier shutdown period.

RA T	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency and high reliability NR TDD	Configuration of low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC in 5G RAN Feature Documentation</i>	LTE and NR intelligent carrier shutdown cannot take effect when low latency and high reliability services are enabled.
LTE FDD LTE TDD	High speed specified policy management	ProcSwitchBasedOnUserSpeed option of the CellAlgoSwitch.HighSpeedSchOptionswitch parameter	<i>High Speed Mobility in eRAN Feature Documentation</i>	The high-speed cell sleeping function in high speed specified policy management does not work with LTE and NR intelligent carrier shutdown.
Low-frequency NR TDD	Supplementary Cell Indicator	NRDUCell.SupplementaryCellInd	None	LTE and NR intelligent carrier shutdown is not supported when this parameter is set to INTRA_FREQ_ASSISTANT_CELL and NRDUCell.DuplexMode is set to CELL_TDD .
NR FDD Low-frequency NR TDD	Multi-carrier joint carrier shutdown	MULTI_CARR_JOINT_CARR_SHDN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	If the NRDUCellPowerSaving.EvalPeriodGranularity parameter is set to SECOND_LEVEL and the NRDUCellPowerSaving.SecondLevelTrigEvalPrd parameter is set to a value less than 6 for LTE and NR intelligent carrier shutdown, multi-carrier joint carrier shutdown must be disabled.

RA T	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD Low- frequency NR TDD	Dual-BBU backup in dual-network redundancy networking	NR: DUAL_BBU_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter LTE: LTE_PASSIVE_DUAL_BBU_BACKUP_SW option of the eNBRedundancy.eNBReliabilitySwitch parameter	None	"Dual-BBU backup in dual-network redundancy networking" is mutually exclusive with "LTE and NR intelligent carrier shutdown".
NR FDD Low- frequency NR TDD	Primary/Backup function in intra-frequency dual-network redundancy networking	PRIMARY_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	"Primary/Backup function in intra-frequency dual-network redundancy networking" is mutually exclusive with "LTE and NR intelligent carrier shutdown".

RA T	Func tion Nam e	Function Switch	Referenc e	Description
NR FD D	Dual-plane function in intra-frequency dual-network redundancy networking	DUAL_PLANE_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	"Dual-plane function in intra-frequency dual-network redundancy networking" is mutually exclusive with "LTE and NR intelligent carrier shutdown".
Lo w- freq uen cy NR TD D	Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios	DUAL_PLANE_TOB_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	<i>Base Station Equipment Reliability</i>	"Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" is mutually exclusive with "LTE and NR intelligent carrier shutdown".

6.1.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Symbol power saving	NRDUCellAlgoSwitch.SymbolShutdownSwitch	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	When LTE and NR intelligent carrier shutdown takes effect: • If the RF module is not serving any other NR carrier, symbol power saving does not take effect. • If the RF module is serving other NR carriers, the status of symbol power saving is not affected.
NR FDD Low-frequency NR TDD	Intelligent carrier shutdown in precise mode	• INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingsSwitch parameter • MULTI_CARR_PRECISE_SHDN_SW option of the NRDUCellAlgoSwitch.PowerSavingsSwitch parameter selected	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	When both intelligent carrier shutdown and LTE and NR intelligent carrier shutdown functions are enabled, a cell can enter the carrier shutdown state only when all conditions for entering both functions are met.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Intelligent carrier shutdown in non-precise mode	• INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAIgoSwitch.PowerSavingsSwitch parameter • MULTI_CARR_PRECISE_SHDN_SW option of the NRDUCellAIgoSwitch.PowerSavingsSwitch parameter deselected	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	When both intelligent carrier shutdown and LTE and NR intelligent carrier shutdown functions are enabled, a cell can enter the carrier shutdown state only when all conditions for entering both functions are met.
NR LTE FDD LTE TDD	RF module deep dormancy	RRU.DORMANCYSW (5G gNodeB, LTE eNodeB)	<i>Multi-RAT Coordinated Energy Saving</i>	RF module deep dormancy takes precedence over LTE and NR intelligent carrier shutdown when the functions are both enabled.
NR FDD Low-frequency NR TDD	Cell load simulation	STRNRDUCELLSIMULATELOAD	None	After LTE and NR intelligent carrier shutdown takes effect, the cell load simulation function cannot be enabled and will be disabled if it has been enabled. It is not recommended that these functions be both enabled.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequ ency NR TDD	CMAS	gNB SibConfig.Si bType set to SIB_TYPE_8	CMAS in 5G RAN Feature Document ation	When an NR carrier is shut down, the corresponding NR cell is blocked and commercial mobile alert system (CMAS) warning notifications cannot be properly sent in this cell. However, UEs can still receive CMAS warning notifications from other carriers.
NR FDD Low-frequ ency NR TDD	Low latency and high reliability	HIGH_RELIABI TY_BASIC_SW option of the NRDUCellAlgoS witch.HighRelia bilitySwitch parameter	URLLC in 5G RAN Feature Document ation	After LTE and NR intelligent carrier shutdown is enabled, the cell enters the carrier shutdown state, which may cause service drops for UEs running high reliability services.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Virtual Grid-based Multi-Frequency Coordination	• RSRP prediction model: VG_MODEL_ALLOW_BUILD_FLAG option of the NRCellFrequencyRelation.AggregationAttribute parameter and NR_IF_HO_MEAS_QTY_PRED_FUN_SW option of the NRCellSmartMultiCarrier.NrMultiCarrierAlgoSwitch parameter • Downlink intra-frequency spectral efficiency prediction model: DL_INTRA_FREQ_E_MODEL_BUILD_ALLOW option of the NRCellSmartMultiCarrier.ModelBuildingSwitch parameter and NR_IF_HO_MEAS_QTY_PRED_FUN_SW option of the NRCellSmartMultiCarrier.NrMultiCarrierAlgoSwitch parameter	<i>Virtual Grid-based Multi-Frequency Coordination in 5G RAN Feature Document</i>	After LTE and NR intelligent carrier shutdown takes effect, the building of virtual grid models will be suspended and existing virtual grid models will be brought offline and cannot be used.

RAT	Function Name	Function Switch	Reference	Description
		Intra-frequency SRS beam prediction model building switch (low-frequency TDD): INTRAF_SRS_MODEL_BUILT_ALLOW option of the NRCellSmartMultiCarrierModelBuildingSwitch parameter		
NR FDD Low-frequency NR TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgorithmSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	When LTE and NR intelligent carrier shutdown is enabled, RF channel intelligent shutdown cannot take effect for some low-frequency NR TDD AAUs and RRUs in automatic coverage compensation mode. For details, see <i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i> .
NR FDD Low-frequency NR TDD	Dynamic RF channel shutdown	RF_CHN_DYN_SHUTDOWN_SW option of the NRDUCellAlgorithmSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	When LTE and NR intelligent carrier shutdown takes effect, dynamic RF channel shutdown will deliver fewer energy saving gains.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Channel calibration	None	None	If channel calibration is being performed when PAs are shut down after LTE and NR intelligent carrier shutdown takes effect, channel calibration fails. Channel calibration is not started when LTE and NR intelligent carrier shutdown takes effect. You can run the DSP NRDUCELLCHNCALIB command to check whether channel calibration is successful.
NR FDD Low-frequency NR TDD	Cell-level MRO	MRO_SCENARIO_IDENT_SWITCH option of the gNB MRO.ScenarioIdentifySwitch parameter and INTRA_FREQ_MRO_SWITCH option of the NRCellMro.MroSwitch parameter	<i>MRO in 5G RAN Feature Documentation</i>	When both LTE and NR intelligent carrier shutdown and intra-frequency MRO are enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, MRO may take effect later than expected.
NR FDD Low-frequency NR TDD	UE-level MRO	UE_MRO_SWITCH option of the NRCellMro.MroSwitch parameter	<i>MRO in 5G RAN Feature Documentation</i>	When both LTE and NR intelligent carrier shutdown and UE-level MRO are enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, MRO may take effect later than expected.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Work mode	Cell.<i>WorkMode</i>	None	An LTE FDD cell with the Cell.<i>WorkMode</i> parameter set to DL_ONLY or LAA cannot be configured as an overlapping neighboring cell of an NR cell. An LTE TDD cell with the Cell.<i>WorkMode</i> parameter set to LAA cannot be configured as an overlapping neighboring cell of an NR cell.
LTE FDD LTE TDD	Cell System Information Switch	CellSiMap.<i>SiSwitch</i>	None	A cell whose ForbidCellSiSwitch option of the CellSiMap.<i>SiSwitch</i> parameter is selected cannot be configured as an overlapping neighboring cell of an NR cell.
LTE FDD LTE TDD	eMTC introduction	CellEmtcAlgo.<i>EmtcAlgoSwitch</i>	<i>eMTC in eRAN Feature Documentation</i>	LTE and NR intelligent carrier shutdown cannot be performed for eMTC-only cells (determined by the EMTC_SWITCH and EMTC_ONLY_CELL_SWITCH options of the CellEmtcAlgo.<i>EmtcAlgoSwitch</i> parameter).
LTE FDD	Static Multiple Beam	Number of SectorSplitCell MOs ≥ 2	<i>Massive MIMO Basics (FDD) in eRAN Feature Documentation</i>	For LTE FDD, LTE and NR intelligent carrier shutdown cannot be performed for Static Multiple Beam cells.

RAT	Function Name	Function Switch	Reference	Description
LTE TDD	Intelligent Multi-Beam of 8T8R	None	<i>Intelligent Multi-Beam of 8T8R (TDD) in eRAN Feature Documentation</i>	For LTE TDD, LTE and NR intelligent carrier shutdown cannot be performed for cells enabled with Intelligent Multi-Beam of 8T8R.
LTE FDD	Static Multiple Beam	None	<i>Smart 8T8R (FDD) in eRAN Feature Documentation</i>	For LTE FDD, LTE and NR intelligent carrier shutdown cannot be performed for Static Multiple Beam cells.
LTE TDD	Intra-frequency split	None	<i>Soft Split Resource Duplex (TDD) in eRAN Feature Documentation</i>	For LTE TDD, LTE and NR intelligent carrier shutdown cannot be performed for cells enabled with intra-frequency split.
NR FDD Low-frequency NR TDD	VoNR	VONR_SW option of the NRCellAlgoSwitch.VonrSwitch parameter	<i>VoNR in 5G RAN Feature Documentation</i>	LTE and NR intelligent carrier shutdown does not take effect when there are VoNR UEs that cannot be handed over from the capacity-layer cell during a UE transfer procedure.

RAT	Function Name	Function Switch	Reference	Description
NR FDD	PIM interference cancellation (FDD) (trial)	PIM_INTERF_CANCELLATION_SW option of the NRDUCellPimIntrfAvd.PimAlgoSwitch parameter	<i>Interference Avoidance in 5G RAN Feature Documentation</i>	After carrier shutdown takes effect in an interfering cell or a victim cell, the gNodeB re-evaluates whether to continue the implementation of PIM interference cancellation in each cell.
NR FDD	RF PIM interference cancellation	RF_PIM_CAN_SW option of the RRUEXTINFO.RruOptlFeatSw parameter	<i>Interference Avoidance in 5G RAN Feature Documentation</i>	• If PIM interference persists after carrier shutdown takes effect, PIM interference cancellation continues. • If PIM interference disappears after carrier shutdown takes effect, PIM interference cancellation is stopped to reduce resource consumption caused by periodic PIM interference characteristic training and interference characteristic extraction. For RF PIM interference cancellation, after carrier shutdown takes effect in an interfering cell or a victim cell, PIM interference characteristic changes and the RF module restarts the PIM interference cancellation training for each cell, causing fluctuations in PIM interference

RAT	Function Name	Function Switch	Reference	Description
				cancellation performance.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	PIM interference cancellation	PIM_INTERF_CANCELLATION_SW option of the PimInterfSmartAvoid.PimAlgoSwitch parameter	<i>Interference Detection and Suppression</i> in <i>eRAN Feature Documentation</i>	For LTE FDD, after carrier shutdown takes effect in an interfering cell or a victim cell, the eNodeB re-evaluates whether to continue the implementation of PIM interference cancellation in each cell.
LTE FDD	RF PIM interference cancellation	RF_PIM_CANCELLATION_SW option of the RRUEXTINFO.RruOptFeatSw (LTE eNodeB, 5G gNodeB) parameter	<i>Interference Detection and Suppression</i> in <i>eRAN Feature Documentation</i>	• If PIM interference persists after carrier shutdown takes effect, PIM interference cancellation continues. • If PIM interference disappears after carrier shutdown takes effect, PIM interference cancellation is stopped to reduce resource consumption caused by periodic PIM interference characteristic training and interference characteristic extraction. For RF PIM interference cancellation, after carrier shutdown takes effect in an interfering cell or a victim cell, PIM interference characteristic changes and the RF module restarts the PIM interference cancellation training for each cell, causing fluctuations in PIM

RAT	Function Name	Function Switch	Reference	Description
				interference cancellation performance.
LTE FDD	Intelligent PIM interference avoidance	PIM_INTERF_SMART_AVOID_SW option of the PimInterfSmartAvoid.PimAlgorithmSwitch parameter	<i>Interference Detection and Suppression in eRAN Feature Documentation</i>	For LTE FDD, LTE and NR intelligent carrier shutdown is terminated during PIM interference detection.
NR FDD	Intelligent PIM interference avoidance	PIM_INTERF_SMART_AVOID_SW option of the NRDUCellPimInterfAvoid.PimAlgorithmSwitch parameter	<i>Interference Avoidance in 5G RAN Feature Documentation</i>	For NR FDD, LTE and NR intelligent carrier shutdown is terminated during PIM interference detection.
NR FDD	Channel isolation for PIM interference avoidance	PIM_INTERF_CHANNEL_SHUTDOWN_SW option of the NRDUCellPimInterfAvoid.PimAlgorithmSwitch parameter	<i>Interference Avoidance in 5G RAN Feature Documentation</i>	For NR FDD, LTE and NR intelligent carrier shutdown is terminated when channel isolation for PIM interference avoidance takes effect.
LTE FDD	Optimization-based scheduling	TIME_FREQ_2D_UNIF_OPT_SCHEDULE_SW and TIME_FREQ_UNIF_OPT_SCHEDULE_ENH_SW options of the CellMultiCarrierUnifiedScheduling.SchedulingSwitch parameter	<i>Multi-carrier Unified Scheduling in eRAN Feature Documentation</i>	For LTE FDD, the optimization-based scheduling function does not take effect for cells in the shutdown state.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Optimization-based carrier selection	OPT_MODEL_BASED_CARR_SELECTION_SW option of the MultiCarrierUnifiedScheduler parameter	<i>Multi-carrier Unified Scheduling in eRAN Feature Documentation</i>	When the optimization-based carrier selection function is enabled, the base station adjusts the UE distribution across cells working in different bands, which leads to changes in the traffic volumes in cells with carrier shutdown enabled. When the traffic volume in a cell with carrier shutdown enabled decreases, the carrier shutdown duration may increase accordingly. As a result, the average downlink UE throughput may decrease.
NR FDD Low-frequency NR TDD	Inter-RAT redirection back to the HPLMN based on roaming UE identification	VOICE_CAPABILITY_TO_HPLMN_SW option of the gNBOperator.HplmnAlgorithm parameter	<i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i>	In LTE and NR intelligent carrier shutdown, only blind redirection can be performed to transfer VoNR-incapable roaming UEs from the visited 5G network to the home 4G network.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Ultra-long holdover for sites with abnormal clock sources	CLK_OTASYNC_PROC_SW option of the TASM.MSCLKDIAGPROCESSSW parameter	<i>Synchronization in 5G RAN Feature Documentation</i>	Assume that the ultra-long hold switch is turned on but the ultra-long hold enhancement switch is turned off. If sequence detection is performed in all cells on a to-be-detected base station during the period when LTE and NR intelligent carrier shutdown takes effect, the sequence detection fails. In this case, the base station cannot enter the ultra-long holdover state even if its clock is faulty, or it exits the ultra-long holdover state as sequence detection cannot be performed for a long time due to energy saving. If the ultra-long hold enhancement switch is turned on, LTE and NR intelligent carrier shutdown does not take effect for the base station in the ultra-long holdover state or its reference neighboring base stations.

RAT	Function Name	Function Switch	Reference	Description
NR LTE FDD	DSS and NR Flexible Refarming	• LTE: DSS_FLEX_RE FARM_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter selected • NR: DSS_FLEX_RE FARM_SW option of the NRDUCellAlgoSwitch.SpectrumCloudEnhSwitch parameter selected	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>	When DSS and NR Flexible Refarming is enabled together with LTE and NR intelligent carrier shutdown, a spectrum sharing LTE cell cannot serve as an overlapping neighboring cell of the associated spectrum sharing NR cell.
NR FDD	Time-frequency two-dimensional unified scheduling optimization	TIME_FREQ_2D_UNIF_OPT_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgoExtSwitch parameter	<i>Multi-Frequency Smart Aggregation in 5G RAN Feature Documentation</i>	Time-frequency two-dimensional unified scheduling optimization does not take effect for cells in the shutdown state.
Low-frequency NR TDD	CA Performance Boost Based on Multi-dimensional Optimization	CA_PRFM_BOOST_BSD_ON_MD_OPT_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgoExtSwitch parameter	<i>Multi-Frequency Smart Aggregation in 5G RAN Feature Documentation</i>	CA Performance Boost Based on Multi-dimensional Optimization does not take effect for cells in the shutdown state.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Fast carrier shutdown	For capacity-layer cells: - Intra-base-station multi-carrier scenarios: INTRA_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingsSwitch parameter selected; - inter-base-station multi-carrier scenarios: INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingsSwitch parameter selected FAST_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter For basic-layer cells: FAST_CARR_SHUT_SERV_ASSUR_SW option of the NRDUCellAlgoSwitch.PowerSavingsSwitch	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	When both fast carrier shutdown and LTE and NR intelligent carrier shutdown functions are enabled, a cell can enter the carrier shutdown state only when all conditions for entering both functions are met.

RAT	Function Name	Function Switch	Reference	Description
		<i>ingExtSwitch</i> parameter		
LTE FDD	Uplink symbol power saving	UL_SYMBOL_SHUTDOWN_SW option of the SymbolPwrSaving.SymbolShutdownEnhancedSw parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	For LTE FDD, uplink symbol power saving and enhanced uplink symbol power saving do not take effect when LTE and NR intelligent carrier shutdown is in effect.
LTE FDD	Enhanced uplink symbol power saving	UL_SYMBOL_SHUTDOWN_ENH_SW option of the SymbolPwrSaving.SymbolShutdownEnhancedSw parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	
NR FDD	Uplink symbol power saving	UL_SYMBOL_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	For NR FDD, uplink symbol power saving does not take effect when LTE and NR intelligent carrier shutdown takes effect.
Low-frequency NR TDD	pRRU-level precise symbol power saving	DL_PRECISE_SYMBOL_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	pRRU-level precise symbol power saving does not take effect when LTE and NR intelligent carrier shutdown is in effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD	Intelligent MCS optimization for uplink SU-MIMO	UL_SU_MCS_INTEL_OPT_SW option of the NRDUCellAiAlgo.o.AiAmcAlgoSwitch parameter	<i>Smart 8T8R (FDD) in 5G RAN Feature Documentation</i>	When intelligent MCS optimization for uplink SU-MIMO is enabled together with LTE and NR intelligent carrier shutdown, intelligent MCS optimization for uplink SU-MIMO does not take effect or its gains decrease.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	NSA anchor carrier shutdown	NSA_ANCHOR_CARR_SHUTDOWN_SW option of the CellShutdown.CarrierShutdownEnhancedSw parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	In NSA or NSA and SA hybrid networking, it is recommended that the INTER_RAT_GRAD_PWR_SAVING_SW option of the gNBpwrSavingAlgo.PwrSavingAlgoSwitch parameter be selected when NSA anchor carrier shutdown is enabled for LTE basic-layer cells and NSA anchor cells and LTE and NR intelligent carrier shutdown is enabled for NR capacity-layer cells and NSA anchor cells. • If NSA anchor carrier shutdown is triggered first, carriers are shut down in NSA anchor cells. If NR capacity-layer cells do not have other neighboring LTE cells, LTE and NR intelligent carrier shutdown does not take effect. • If LTE and NR intelligent carrier shutdown is triggered first, this function can take effect in NR capacity-layer cells.

RAT	Function Name	Function Switch	Reference	Description
NR FDD	Uplink TTI-level RF channel dynamic muting	UL_TTI_CHN_DYN_MUTING_SW option of the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Uplink TTI-level RF channel dynamic muting does not take effect when LTE and NR intelligent carrier shutdown is in effect.
LTE FDD NR FDD	LTE-NR multi-dimensional scheduling optimization	• LTE: LNR_MULTI_D_OPT_SCH_SW option of the GigaBandCell.GigaBandCellAlgoSw parameter • NR: LNR_MULTI_D_OPT_SCH_SW option of the NRDUCellGigaBand.GigaBandCellAlgoSw parameter	<i>LTE FDD and NR Inter-RAT Unified Scheduling</i>	LTE-NR multi-dimensional scheduling optimization does not take effect for cells in the shutdown state.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Speed turbo UE identification	gNodeBParam.StUeIdentSw	None	LTE and NR intelligent carrier shutdown fails to take effect in cells served by a base station, resulting in a decrease in the energy saving gains, when any of the following configuration conditions is met and the configured function takes effect:
NR FDD Low-frequency NR TDD	gNodeB acceleration service application	• gNBAsAppParam.AppDomainName • gNBAsAppParam.Applpv4 • gNBAsAppParam.Applpv6	None	• Any option of the gNodeBParam.StUeIdentSw parameter is selected. • The gNBAsAppParam.AppDomainName parameter is set. • The gNBAsAppParam.Applpv4 parameter is set. • The gNBAsAppParam.Applpv6 parameter is set. The following methods can be used to address the preceding issue: • Select the SPEED_TURBO_PWR_SAVING_SW option of the gNBPwrSavingAlgo.PwrSavingAlgorithm parameter. LTE and NR intelligent carrier shutdown can take effect for cells served by the base station in some of the preceding

RAT	Function Name	Function Switch	Reference	Description
				scenarios within the period specified by the gNBWPwrSavingAlgo.SpeedTurboPwrSavStartTime and gNBWPwrSavingAlgo.SpeedTurboPwrSavEndTime parameters, increasing the energy saving gains. • Select the STATIC_ST_PWR_SAVING_SW option of the gNBWPwrSavingAlgo.PwrSavingAlgorithmSwitch parameter. LTE and NR intelligent carrier shutdown can take effect for cells served by the base station in some of the preceding scenarios within the period specified by the gNBWPwrSavingAlgo.StaticSTPwrSavStartTime and gNBWPwrSavingAlgo.StaticSTPwrSavEndTime parameters, increasing the energy saving gains.

6.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- LTE:
 - 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
 - DBS3900 LampSite and DBS5900 LampSite
- NR:
 - 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
 - DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

- LTE:
 - Main control board: Only the UMPTb, UMPTe, UMPTg, UMPTga, and UMPTha support this function.
 - Baseband processing unit: All UBBP boards support this function.
- NR:

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All low-frequency RF modules support this function.

- For details about low-frequency RRUs, see *RRU Technical Specifications* in *3900 & 5900 Series Base Station Product Documentation*.
- For details about low-frequency AAUs, see *AAU Technical Specifications* in *3900 & 5900 Series Base Station Product Documentation*.

6.1.3.4 Networking

- X2 links need to be set up in inter-base-station SA networking. X2 links can be set up automatically or manually, with automatic setup recommended. For details about X2 configurations, see *X2 Self-Management in SA Networking*.
- In NSA networking, the NSA DC capability must be enabled for all basic-layer cells and capacity-layer cells. This function is controlled by the **NSA_DC_CAPABILITY_SWITCH** option of the **NsaDcMgmtConfig.NsaDcAlgoSwitch** parameter on the eNodeB side and by the **NRCCellAlgoSwitch.NsaDcSwitch** parameter on the gNodeB side. Both of the switches need to be turned on.
- In NSA and SA hybrid networking, the configuration and activation procedure of this function are the same as those in NSA networking. The load information is transmitted between the eNodeB and gNodeB over the X2 interface.

6.1.3.5 Cells

LTE and NR intelligent carrier shutdown is not supported in SDL cells (**NRDUCell.DuplexMode** set to **CELL_SDL**) and SUL cells (**NRDUCell.DuplexMode** set to **CELL_SUL**).

6.1.3.6 Others

- The UE has subscribed to LTE and NR services.
- This function requires that UEs support NSA DC defined in 3GPP Release 15 in NSA networking or NSA and SA hybrid networking.

6.1.4 Operation and Maintenance

6.1.4.1 When to Use

To achieve satisfactory energy-saving gains without impacting network performance, LTE and NR intelligent carrier shutdown is recommended in the following scenarios:

- LTE and NR cells overlap in their coverage areas, and the LTE network provides continuous basic coverage.
 - In SA networking, or NSA and SA hybrid networking, the neighbor relationships with LTE cells must be configured for the NR cell on the gNodeB. In NSA and SA hybrid networking, the neighbor relationship with the NR cell also needs to be configured for the LTE cells on the eNodeB.

NOTE

In multi-operator sharing scenarios, the neighboring LTE cells of the NR cell configured on the gNodeB must be the cells of the LTE primary operator.

- To query the coverage relationship between LTE and NR cells, you can run the **LST NRNRELATIONSHIP** command on the eNodeB in NSA networking or NSA and SA hybrid networking, or run the **LST NRCELLEUTRANRELATION** command on the gNodeB in SA networking or NSA and SA hybrid networking.
- The traffic volume presents a typical tidal effect in some areas such as central business districts (CBDs) and shopping malls.
- The overall traffic volume of LTE and NR services is low, as is often the case in suburban and rural areas.

In overlapping coverage areas, if multiple neighboring LTE cells of the NR cell for which LTE and NR intelligent carrier shutdown is to be enabled are anchor cells:

- When no anchor cell has a small bandwidth, it is recommended that LTE and NR intelligent carrier shutdown be enabled for all anchor cells.
- When there are anchor cells with small bandwidths, LTE and NR intelligent carrier shutdown is not recommended for these small-bandwidth anchor cells.
- When all anchor cells have small bandwidths and LTE and NR intelligent carrier shutdown must be enabled, it is recommended that the UE number threshold (indicated by the **NRDUCellPowerSaving.UserNumThld**

parameter) and downlink PRB usage threshold (indicated by the **NRDUCellPowerSaving.DlPrbThld** parameter) of the NR cell be set to smaller values, and the PRB usage threshold offset (indicated by the **NRDUCellPowerSaving.EutranDlPrbOffset** parameter) of the anchor cells be set to a larger value. In this way, ping-pong migration can be avoided.

 NOTE

An anchor cell with a small bandwidth refers to an anchor cell whose bandwidth is much less than that of an NR cell. (For example, the bandwidth of the NR cell is 100 MHz while that of the anchor cell is 5 MHz.)

This function is not recommended in the following scenario:

There are NR-only UEs. After the NR network is shut down, NR-only UEs cannot access services.

6.1.4.2 Data Configuration

6.1.4.2.1 Data Preparation

Table 6-6 Parameters used for configuration on the LTE side

Parameter Name	Parameter ID	Setting Notes
LNR Intelligent Carrier Shutdown Switch	CellAlgoExtSwitch.LnrSmartCarrierShutdownSw	Set this parameter to ON .
Aggregation Attribute	NrNRRelationship.AggregationAttribute	Set the CARRIER_SHUTDOWN_OVER LAP_FLAG option of this parameter to the recommended value.
Cell reserved for operator	CELLOP.CellReservedForOp	Set this parameter based on the network plan.
Interface SON Configuration Switch	eNodeBOamParam.InterfaceSonConfigSwitch	To enable the eNodeB to set up X2 links with gNodeBs that do not serve NSA cells, select the LNR_SA_GNB_X2_SETUP_SW option.

Parameter Name	Parameter ID	Setting Notes
Power Saving Algorithm Switch	EnodebPwrSavingAlgo.PwrSavingAlgoSwitch	To enable inter-base-station energy saving between LTE and SA NR cells in LTE and NR scenarios, select the LNR_INTER_BS_PWR_SAVE_SW option. The LNR_SA_GNB_X2_SETUP_SW option of the eNodeBOamParam.InterfaceSonConfigSwitch parameter must also be selected.

Table 6-7 Parameters used for configuration on the NR side

Parameter Name	Parameter ID	Setting Notes
Power Saving Switch	NRDUCellAlgoSwitch.PowerSavingSwitch	Select the LNR_SMART_CARRIER_SHUTDOWN_SW option of this parameter.
Start Time	NRDUCellPowerSaving.StartTime	Set this parameter based on the network plan. The time period for LTE and NR intelligent carrier shutdown should not overlap the time period specified when the NRDUCellPowerSaving.PowerSavingType parameter is set to TIMING_CARRIER_SHUTDOWN for the local cell.
Stop Time	NRDUCellPowerSaving.StopTime	Set this parameter based on the network plan. The time period for LTE and NR intelligent carrier shutdown should not overlap the time period specified when the NRDUCellPowerSaving.PowerSavingType parameter is set to TIMING_CARRIER_SHUTDOWN for the local cell.
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	Set this parameter to LNR_SMART_CARRIER_SHUTDOWN .
UE Number Threshold	NRDUCellPowerSaving.UserNumThld	Set this parameter based on the network plan.
Downlink PRB Threshold	NRDUCellPowerSaving.DlPrbThld	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Uplink PRB Threshold	NRDUCellPowerSaving.UlPrbThld	Set this parameter based on the network plan.
E-UTRAN Downlink PRB Threshold	NRDUCellPowerSaving.EutranDlPrbThld	Set this parameter based on the network plan.
Neighboring Cell Uplink PRB Threshold	NRDUCellPowerSaving.NCellUlPrbThld	Set this parameter based on the network plan.
E-UTRAN Downlink PRB Offset	NRDUCellPowerSaving.EutranDlPrbOffset	Set this parameter based on the network plan.
Neighboring Cell Uplink PRB Offset	NRDUCellPowerSaving.NCellUlPrbOffset	Set this parameter based on the network plan.
E-UTRAN Carrier Shutdown Ratio	NRDUCellPowerSaving.EutranCarrShutdownRatio	Set this parameter based on the network plan.
E-UTRAN Carrier Shutdown Offset	NRDUCellPowerSaving.EutranCarrShutdownExitOfs	Set this parameter based on the network plan.
E-UTRAN NR User Number Threshold	NRDUCellPowerSaving.EutranNrUserNumThld	Set this parameter based on the network plan. If this parameter is set to 65535 , this parameter and the NRDUCellPowerSaving.EutranNrUserNumThldOfs parameter are invalid.
E-UTRAN NR User Number Threshold Offset	NRDUCellPowerSaving.EutranNrUserNumThldOfs	Set this parameter based on the network plan.
E-UTRAN NR User Downlink PRB Threshold	NRDUCellPowerSaving.EutranNrUserDlPrbThld	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
E-UTRAN NR User Uplink PRB Threshold	NRDUCellPowerSaving.EutranNrUserUlbThld	Set this parameter based on the network plan.
E-UTRAN NR User Downlink PRB Threshold Offset	NRDUCellPowerSaving.EutranNrUserDlPrbThldOfs	Set this parameter based on the network plan.
E-UTRAN NR User Uplink PRB Threshold Offset	NRDUCellPowerSaving.EutranNrUserUlbThldOfs	Set this parameter based on the network plan.
SUL User Number Threshold	NRDUCellPowerSaving.SulUserNumThld	You are advised to set these parameters to their recommended values. - These parameters are invalid when they are set to 65535. - If NRDUCellPowerSaving.SulUserNumThld is not set to 65535, the sum of NRDUCellPowerSaving.SulUserNumThld and NRDUCellPowerSaving.SulUserNumOffset must be less than 65535, and NRDUCellPowerSaving.SulUserNumThld must be less than NRDUCellPowerSaving.UserNumThld. - If the FLEX_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter is deselected and the cell has a NRDUCellPowerSaving record with power saving type set to LTE and NR intelligent carrier shutdown, optimized evaluation of NR RF module dormancy, NR load-based RF module dormancy, or LN load-based RF module dormancy, then the NRDUCellPowerSaving.SulUserNumThld parameter of this record must be set to 65535.
SUL User Number Offset	NRDUCellPowerSaving.SulUserNumOffset	

Parameter Name	Parameter ID	Setting Notes
Power Saving Penalty Duration	NRDUCellPowerSaving.PwrSavingPenaltyTime	Set this parameter based on the network plan. If the setting of this parameter is too small and some UEs cannot be transferred, other UEs may be frequently transferred. In this case, the carrier frequently attempts to enter the carrier shutdown state, and is not available for UE access for a long time.
Evaluation Period Granularity	NRDUCellPowerSaving.EvalPeriodGranularity	Set this parameter based on the network plan.
Triggering Evaluation Period	NRDUCellPowerSaving.ShutdownTrigJudgePeriod	Set this parameter based on the network plan.
Second-Level Trigger Evaluation Period	NRDUCellPowerSaving.SecondLevelTrigEvalPrd	Set this parameter based on the network plan.
Exit Evaluation Period	NRDUCellPowerSaving.ExitEvaluationPeriod	Set this parameter based on the network plan.
Second-Level Exit Evaluation Period	NRDUCellPowerSaving.SecondLevelExitEvalPrd	Set this parameter based on the network plan.
Power Saving Overlapping Indicator	NRCeUtraNRRela-tion.PowerSavingOverlapInd	Set the LNR_CARR_SHUTDOWN_OVERLAP_IN D option based on the network plan. This parameter is mandatory only in SA networking.
Power Saving Algorithm Switch	gNBpwrSavingAlgo.pwrSavingAlgoSwitch	Select the INTER_RAT_GRAD_PWR_SAVING_SW option of this parameter if inter-RAT level-by-level energy saving is required.
LNR Carrier Shutdown B1 RSRP Threshold	NRCeUtraNRMeaG rp.LnrCarrShutdownB1 RsrpThld	Set this parameter based on the network plan.
PLMN Reserved Flag	NRExternalNCellPlmn.PlmnReservedFlag	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
X2 Config Switch	gNBInterfaceParam.X2ConfigSw	To allow X2 link setup in SA scenarios, select the X2_SETUP_FOR_SA_SW option.
Power Saving Algorithm Switch	gNBPwrSavingAlgo.PwrSavingAlgoSwitch	- To enable energy saving for SA NR cells in LTE-NR inter-base-station energy saving scenarios, select the SA_LNR_INTER_BS_PWR_SAVE_SW option. The X2_SETUP_FOR_SA_SW option of the gNBInterfaceParam.X2ConfigSw parameter must also be selected. - When the SA_LNR_INTER_BS_PWR_SAVE_SW option is deselected, the NRCellEutraNRelation.EnodebId parameter must have the same value in all entries where the LNR_CARR_SHUTDOWN_OVERLAP_IND option of the NRCellEutraNRelation.PowerSavingOverlapInd parameter is set to Yes.

6.1.4.2.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [6.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

On the LTE side:

```
//Turning on the switch for LTE and NR intelligent carrier shutdown
MOD CELLALGOEXTSWITCH: LocalCellId=0, LnrSmartCarrierShutdownSw=ON;
//(Optional) Turning on the LTE-NR SA gNodeB X2 setup switch in inter-base-station SA networking
MOD ENODEBOAMPARAM: InterfaceSonConfigSwitch=LNR_SA_GNB_X2_SETUP_SW-1;
//(Optional) Turning on the LTE-NR SA inter-base-station energy saving switch in inter-base-station SA networking
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=LNR_INTER_BS_PWR_SAVE_SW-1;
//(Optional, only in NSA networking, NSA and SA hybrid networking, or inter-base-station SA networking)
Manually configuring all the neighbor relationships between LTE and NR cells with overlapping coverage areas with the parameters set as follows: mobile country code (MCC) is 460, mobile network code (MNC) is 01, and gNodeB ID is 441233
```

```
MOD NRNRRELATIONSHIP: LocalCellId=0, Mcc="460", Mnc="01", GnodebId=441233, CellId=1,  
AggregationAttribute=CARRIER_SHUTDOWN_OVERLAP_FLAG-1;
```

On the NR side:

Scenario 1: Configure LTE and NR intelligent carrier shutdown with a minute-level evaluation period

```
//Turning on the switch for LTE and NR intelligent carrier shutdown  
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,  
PowerSavingSwitch=LNR_SMART_CARRIER_SHUTDOWN_SW-1;  
//Adding energy saving configurations  
//Setting the evaluation period to minutes  
ADD NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=0, StartTime=00&00&00,  
StopTime=06&00&00, PowerSavingType=LNR_SMART_CARRIER_SHUTDOWN, DlPrbThld=5, UlPrbThld=5,  
UserNumThld=20, PwrSavingPenaltyTime=30, EutranDlPrbThld=30, EutranDlPrbOffset=30,  
NCellUlPrbThld=30, NCellUlPrbOffset=30, EutranCarrShutdownRatio=90, EutranCarrShutdownExitOfs=70,  
EutranNrUserNumThld=20, EutranNrUserNumThldOfs=20, EutranNrUserDlPrbThld=15,  
EutranNrUserDlPrbThldOfs=15, EutranNrUserUlPrbThld=15, EutranNrUserUlPrbThldOfs=15,  
EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=1, ExitEvaluationPeriod=1;  
//(Optional, in SUL co-deployment scenarios) Modifying the energy saving policy  
MOD NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=0,  
PowerSavingType=LNR_SMART_CARRIER_SHUTDOWN, SulUserNumThld=4;  
  
//(Optional) Turning on the inter-RAT level-by-level energy saving switch  
MOD GNBPWRSAVINGALGO: PwrSavingAlgoSwitch=INTER_RAT_GRAD_PWR_SAVING_SW-1;  
//(Optional) Setting the PlmnReservedFlag parameter in the NR external-cell PLMN list for the NR cell  
based on the network plan in multi-operator sharing scenarios  
MOD NREXTERNALNCELLPLMN: Mcc="302", Mnc="220", gNBId=0, CellId=0, SharedMcc="302",  
SharedMnc="610", PlmnReservedFlag=FALSE;  
//(Optional, required only in SA networking) Configuring all neighboring cells in the overlapping coverage  
area with the parameters set as follows: MCC is 460, MNC is 01, and eNodeB ID is 441282  
MOD NRCELLEUTRANRELATION: NrCellId=1, Mcc="460", Mnc="01", EnodebId=441282, CellId=0,  
PowerSavingOverlapInd=LNR_CARR_SHUTDOWN_OVERLAP_IND-1;  
//(Optional) Turning on the SA X2 setup switch in inter-base-station SA networking  
MOD GNBINTERFACEPARAM: X2ConfigSw=X2_SETUP_FOR_SA_SW-1;  
//(Optional) Turning on the LTE-NR SA inter-base-station energy saving switch in inter-base-station SA  
networking  
MOD GNBPWRSAVINGALGO: PwrSavingAlgoSwitch=SA_LNR_INTER_BS_PWR_SAVE_SW-1;
```

(Optional) Scenario 2: Configure LTE and NR intelligent carrier shutdown with a second-level evaluation period

```
//Turning on the switch for LTE and NR intelligent carrier shutdown  
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,  
PowerSavingSwitch=LNR_SMART_CARRIER_SHUTDOWN_SW-1;  
//Adding energy saving configurations  
//Setting the evaluation period to seconds  
ADD NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=0, StartTime=00&00&00,  
StopTime=06&00&00, PowerSavingType=LNR_SMART_CARRIER_SHUTDOWN, DlPrbThld=5, UlPrbThld=5,  
UserNumThld=20, PwrSavingPenaltyTime=30, EutranDlPrbThld=30, EutranDlPrbOffset=30,  
NCellUlPrbThld=30, NCellUlPrbOffset=30, EutranCarrShutdownRatio=90, EutranCarrShutdownExitOfs=70,  
EutranNrUserNumThld=20, EutranNrUserNumThldOfs=20, EutranNrUserDlPrbThld=15,  
EutranNrUserDlPrbThldOfs=15, EutranNrUserUlPrbThld=15, EutranNrUserUlPrbThldOfs=15,  
EvalPeriodGranularity=SECOND_LEVEL, SecondLevelTrigEvalPrd=6, SecondLevelExitEvalPrd=12;  
//(Optional, in SUL co-deployment scenarios) Modifying the energy saving policy  
MOD NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=10,  
PowerSavingType=LNR_SMART_CARRIER_SHUTDOWN, SulUserNumThld=4, SulUserNumOffset=4;  
//(Optional) Turning on the inter-RAT level-by-level energy saving switch  
MOD GNBPWRSAVINGALGO: PwrSavingAlgoSwitch=INTER_RAT_GRAD_PWR_SAVING_SW-1;  
//(Optional) Setting the PlmnReservedFlag parameter in the NR external-cell PLMN list for the NR cell  
based on the network plan in multi-operator sharing scenarios  
MOD NREXTERNALNCELLPLMN: Mcc="302", Mnc="220", gNBId=0, CellId=0, SharedMcc="302",  
SharedMnc="610", PlmnReservedFlag=FALSE;  
//(Optional, required only in SA networking) Configuring all neighboring cells in the overlapping coverage  
area with the parameters set as follows: MCC is 460, MNC is 01, and eNodeB ID is 441282  
MOD NRCELLEUTRANRELATION: NrCellId=1, Mcc="460", Mnc="01", EnodebId=441282, CellId=0,  
PowerSavingOverlapInd=LNR_CARR_SHUTDOWN_OVERLAP_IND-1;
```

```
//(Optional) Turning on the SA X2 setup switch in inter-base-station SA networking  
MOD GNBINTERFACEPARAM: X2ConfigSw=X2_SETUP_FOR_SA_SW-1;  
//(Optional) Turning on the LTE-NR SA inter-base-station energy saving switch in inter-base-station SA  
networking  
MOD GNBPWRSAVINGALGO: PwrSavingAlgoSwitch=SA_LNR_INTER_BS_PWR_SAVE_SW-1;
```

Deactivation Command Examples

On the LTE side:

```
//(Optional) Turning off the LTE-NR SA inter-base-station energy saving switch in inter-base-station SA  
networking  
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=LNR_INTER_BS_PWR_SAVE_SW-0;  
//(Optional) Turning off the LTE-NR SA gNodeB X2 setup switch in inter-base-station SA networking  
MOD ENODEBOAMPARAM: InterfaceSonConfigSwitch=LNR_SA_GNB_X2_SETUP_SW-0;  
//Turning off the switch for LTE and NR intelligent carrier shutdown  
MOD CELLALGOEXTSWITCH: LocalCellId=0, LnrSmartCarrierShutdownSw=OFF;  
//(Optional, required only when the neighbor relationships between LTE and NR cells with overlapping  
coverage areas are manually configured in NSA networking, NSA and SA hybrid networking, or inter-base-  
station SA networking) Removing the neighbor relationships for energy saving in the overlapping coverage  
areas  
MOD NRNRELATIONSHIP: LocalCellId=0, Mcc="460", Mnc="01", GnodebId=441233, CellId=1,  
AggregationAttribute=CARRIER_SHUTDOWN_OVERLAP_FLAG-0;
```

On the NR side:

```
//(Optional) Turning off the LTE-NR SA inter-base-station energy saving switch in inter-base-station SA  
networking  
MOD GNBPWRSAVINGALGO: PwrSavingAlgoSwitch=SA_LNR_INTER_BS_PWR_SAVE_SW-0;  
//(Optional) Turning off the SA X2 setup switch in inter-base-station SA networking  
MOD GNBINTERFACEPARAM: X2ConfigSw=X2_SETUP_FOR_SA_SW-0;  
//Turning off the switch for LTE and NR intelligent carrier shutdown  
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,  
PowerSavingSwitch=LNR_SMART_CARRIER_SHUTDOWN_SW-0;  
//Removing the energy saving policy  
RMV NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=0;  
//(Optional) Turning off the inter-RAT level-by-level energy saving switch  
MOD GNBPWRSAVINGALGO: PwrSavingAlgoSwitch=INTER_RAT_GRAD_PWR_SAVING_SW-0;  
//(Optional, required only in SA networking) Removing all neighbor relationships for energy saving in the  
overlapping coverage areas  
MOD NRCELLEUTRANRELATION: NrCellId=1, Mcc="460", Mnc="01", EnodebId=441282, CellId=0,  
PowerSavingOverlapInd=LNR_CARR_SHUTDOWN_OVERLAP_IND-0;
```

6.1.4.2.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.1.4.3 Activation Verification

Using MML Commands

Use the following methods to check whether LTE and NR intelligent carrier shutdown has taken effect in an NR cell:

- Method 1:
Run the **DSP NRDUCELL** command on the gNodeB during the specified effective period. If the **Power Saving State** parameter has a value of **LNR**

Intelligent Carrier Shutdown:Enabled, the cell has been shut down and this feature has taken effect.

- Method 2:

Run the **DSP NRDUCELLPOWERSAVING** command to query the status of LTE and NR intelligent carrier shutdown.

- If the value of the **Power Saving Type** parameter is **LNR Intelligent Carrier Shutdown** and the value of the **Current State** parameter is **Enabled**, LTE and NR intelligent carrier shutdown has taken effect.
- If the value of the **Power Saving Type** parameter is **LNR Intelligent Carrier Shutdown** and the value of the **Current State** parameter is **DISABLED**, you can determine the possible reasons why this function does not take effect based on the values of **Current Disabled Cause**, **Latest Exit Cause**, and **Latest Disabled Cause** within the specified period. [Table 6-8](#) lists the possible causes.

[Table 6-9](#) describes the meanings of the **Latest Exit Cause**, **Latest Disabled Cause**, **Latest Exit Time**, and **Latest Disabled Time**.

Latest Exit Cause and **Latest Disabled Cause** can be used to query the reasons for the latest exit and disabled causes of the function beyond the effective period. For example, if the effective period is from 00:00:00 to 06:00:00, **Latest Exit Cause** and **Latest Disabled Cause** record the latest exit and disabled causes that happened during 00:00:00 to 06:00:00, which you can query using the two fields from 06:00:00 to 23:59:59.

Table 6-8 Possible reasons why the LTE and NR intelligent carrier shutdown function does not take effect

Cause Value	Description
Insufficient License	The license item has not been applied for.
Switch Turned Off	The function switch for the NR cell has not been turned on.
Obtaining NCell Configurations	• In NSA networking or NSA and SA hybrid networking, the neighbor relationships with LTE cells are not configured for the NR cell, the overlap indicator is not configured for the neighboring LTE cells of the NR cell, the X2 link is faulty, or the NR cell fails to obtain information about its neighboring LTE cells. • In SA networking, neighboring LTE cell information is being subscribed to.

Cause Value	Description
Neighboring Cell Not Configured	• In SA networking, the neighbor relationships with LTE cells are not configured for the NR cell, the overlap indicator is not configured for the neighboring LTE cells of the NR cell, or the NR cell fails to obtain information about its neighboring LTE cells. • In SA multi-operator sharing scenarios: – The PLMN ID of the normal operator of the NR cell is not a subset of the PLMN IDs of normal operators of neighboring LTE cells. – The PLMN ID of the reserved operator of the NR cell is not a subset of the PLMN IDs of normal operators or the PLMN IDs of reserved operators of neighboring LTE cells.
Cell Unavailable	The NR cell is blocked or deactivated, fails to be set up, or is unavailable due to other reasons.
Triggering Time Not Reached	The current time is not within the time range specified for the function.

Cause Value	Description
Neighboring Cell Unavailable	The function switch of the neighboring LTE cell is not turned on, the licenses for the neighboring LTE cell are insufficient, the neighboring LTE cell is in barred state or not activated, or no neighboring LTE cell is anchor cell. If NSA anchor carrier shutdown is enabled in neighboring LTE cells, it is recommended that the INTER_RAT_GRAD_PWR_SAVING_SW option of the gNBPwrSavingAlgo.PwrSavingAlgoSwitch parameter be selected for the NR cell. This will prevent the NR cell from terminating the LTE and NR intelligent carrier shutdown function in this case. For details about NSA anchor carrier shutdown, see <i>Energy Conservation and Emission Reduction</i> in <i>eRAN Feature Documentation</i>.
Impacted Feature Taken Effect	None
Downlink PRB Threshold Not Reached	The user number in the NR cell has reached the trigger threshold, but the downlink PRB usage has not reached the trigger threshold.
UE Number Threshold Not Reached	- The downlink PRB usage in the NR cell has reached the trigger threshold, but the user number has not reached the trigger threshold. - Neither the downlink PRB usage nor the user number in the NR cell has reached the trigger threshold.
Triggering Decision Period Not Met	The current time is within the check period specified by the NRDUCellPowerSaving.ShutdownTrigJudgePeriod parameter, but the thresholds for function triggering have not been met.

Cause Value	Description
In Penalty State	- If this function is in effect for a consecutive period of less than 15 minutes for one time, a penalty is imposed and this function cannot take effect again until the penalty time elapses. - If there are UEs remaining in the cell to be shut down, a penalty is imposed and the carrier shutdown function cannot take effect until the penalty time elapses. The penalty duration is specified by the NRDUCellPowerSaving.PwrSavingPenaltyTime parameter.
Uplink PRB Threshold Not Reached	The current PRB usage does not reach the uplink PRB usage threshold for triggering the function.
Neighboring Cell Uplink PRB Exceeded Threshold	The current uplink PRB usage of the neighboring cell exceeds the configured uplink PRB usage threshold.
Neighboring Cell Downlink PRB Exceeded Threshold	The current downlink PRB usage of the neighboring cell exceeds the configured downlink PRB usage threshold.
Neighboring Cell NR UE Uplink PRB Exceeded Threshold	The current uplink PRB usage of NR UEs exceeds the configured uplink PRB usage threshold.
Neighboring Cell NR UE Downlink PRB Exceeded Threshold	The current downlink PRB usage of NR UEs exceeds the configured downlink PRB usage threshold.
RF Hardware Capability Not Supported	None
BBP Software Capability Not Supported	None
UE Being Transferred or Transfer Failed	Active SA UEs in the NR cell fail to be handed over or redirected.
NR UEs in NCell Exceeding Threshold	The number of NR UEs in a neighboring LTE cell of the NR cell exceeds the threshold.

Cause Value	Description
Ultra-Long Holdover in Progress	None
Others	- The current time is within the time range specified for the LTE and NR intelligent carrier shutdown function, but the gNodeB has not obtained data such as the PRB usage and UE number. - In the current period, the conditions for this feature to take effect are met, but the conditions for NR intelligent carrier shutdown are not met. - Internal software or other reasons cause the function not to take effect. - The configurations are incorrect. For example, in multi-operator sharing scenarios, the neighbor relationships with LTE cells served by the LTE primary operators are not configured for the NR cell.

Table 6-9 Latest exit and disabled causes

Command Output of DSP NRDUCELLPOWERSAVING	Description
Latest Exit Cause	Indicates the cause for the latest exit of a function during its effective period.
Latest Disabled Cause	Indicates the cause for the latest application failure of a function during its latest effective period.
Latest Exit Time	Indicates the recorded time of the latest exit of a function during its effective period.
Latest Disabled Time	Indicates the recorded time of the latest application failure of a function during its effective period.

Observing Counters

Observe the performance counter of the affected NR cell.

Counter Name	Counter Description
N.Cell.Unavail.Dur.EnergySaving	Duration of cell unavailability due to energy saving

Observe the performance counters of the affected gNodeB.

Counter Name	Counter Description
VS.EnergyCons.BTS.Adding.NR	The sum of power consumption of all boards in NR
VS.EnergyCons.BTS.Adding.Service.NR	Sum of power consumption of all boards (NR services)
VS.EnergyCons.BTSBoard	Power consumption of board

Observe the performance counter of the affected eNodeB.

Counter Name	Counter Description
VS.EnergyCons.BTS.Adding.LTE	The sum of power consumption of all boards in the LTE
VS.EnergyCons.BTS.Adding.Service.LTE	Sum of power consumption of all boards (LTE services)

6.1.4.4 Network Monitoring

On the live network, the one-week energy consumption data obtained before and after the power saving feature is enabled can be compared to evaluate the power saving gains. It is expected that the power consumption decreases significantly after this feature is enabled.

- Check the **VS.EnergyCons.BTSBoard (LTE eNodeB, 5G gNodeB)** counter to observe the power consumption of RF modules.
- Check the **VS.EnergyCons.BTS.Adding.Service.NR** or **VS.EnergyCons.BTS.Adding.NR** counter on the MAE-Access to observe the power consumption of the entire gNodeB.
- Check the **VS.EnergyCons.BTS.Adding.Service.LTE** or **VS.EnergyCons.BTS.Adding.LTE** counter on the MAE-Access to observe the power consumption of the entire eNodeB.
- Check the **N.Cell.Unavail.Dur.EnergySaving** counter on the MAE-Access to observe the duration of NR cell unavailability due to energy saving.

- Check the **N.PowerSaving.CarrierShutdown.Dur** counter on the MAE-Access to observe the duration in which an NR cell is in the carrier shutdown state.
- Check the **VS.RADIOEQM.ShallowSleep.Dur (LTE eNodeB, 5G gNodeB)** counter on the MAE-Access to observe the duration in which an RF module is in the carrier shutdown state.

NOTE

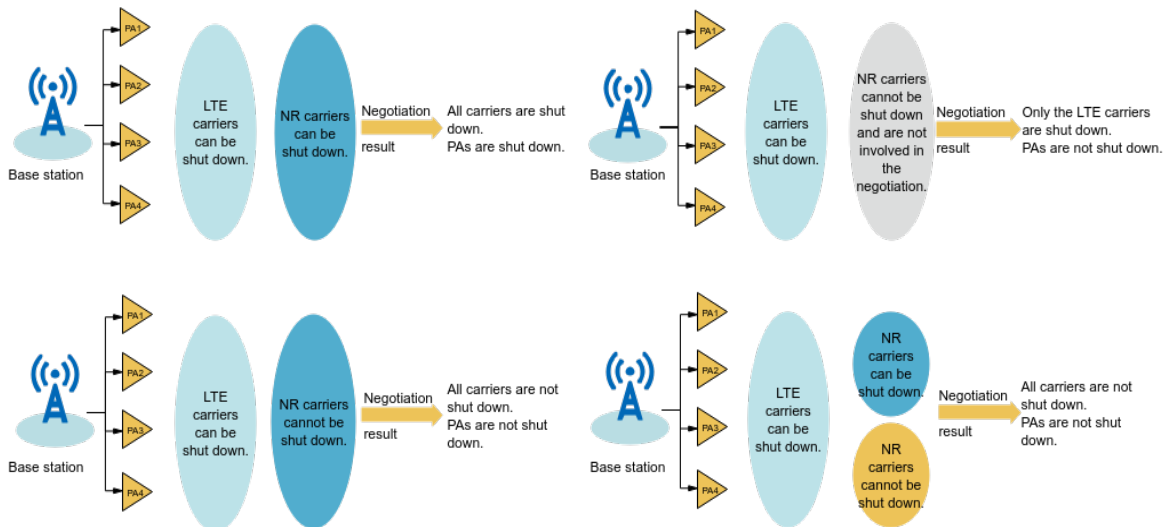
- Evaluation of whether to enter or exit the power saving state takes some time. Therefore, the duration measured by the performance counters has an approximately 1-minute deviation from the actual duration.
- The sampling period is not synchronous with the reporting period, and therefore a deviation of one sampling period may exist.
- If multiple RATs share an RF module using the SDR technology, it is recommended that the power consumption be checked by using module-level counters.

6.2 Multi-Carrier Joint Carrier Shutdown

6.2.1 Principles

When a power amplifier (PA) in an RF module serves more than one carrier, carrier shutdown is evaluated on a per carrier basis. A carrier is shut down when it meets the triggering conditions for a specific carrier shutdown function. The PA is shut down only when all carriers served by the PA are shut down. If the PA keeps working because some of the carriers are not shut down, energy saving gains are relatively low. Against this backdrop, multi-carrier joint carrier shutdown is introduced. [Figure 6-3](#) shows how this function works.

Figure 6-3 Multi-carrier joint carrier shutdown



Multi-carrier joint carrier shutdown is controlled by the **MULTI_CARR_JOINT_CARR_SHDN_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter.

- All cells with this option selected participate in multi-carrier negotiation. If the carriers of these cells all meet the triggering conditions for a carrier shutdown

function and the number of channels configured for the carriers is equal to the negotiated number of channels that can be shut down, all these carriers are shut down. Otherwise, none of these carriers is shut down. Again, the PA is shut down only when all carriers served by the PA are shut down. It is recommended that this option be selected for all cells on the carriers sharing a PA.

- Any cell with this option deselected does not participate in multi-carrier negotiation.

LTE cells, if any, must be enabled with the LTE function multi-carrier joint carrier shutdown so that they can participate in the multi-carrier negotiation. For details about this LTE function, see *Energy Conservation and Emission Reduction* in the *eRAN Feature Documentation*.

The following carrier shutdown functions are involved in the negotiation:

- Intelligent carrier shutdown: See *Energy Conservation and Emission Reduction* in *5G RAN Feature Documentation*.
- Timing carrier shutdown: See *Energy Conservation and Emission Reduction* in *5G RAN Feature Documentation*.
- LTE and NR intelligent carrier shutdown: See [6.1 LTE and NR Intelligent Carrier Shutdown](#). With LTE and NR intelligent carrier shutdown enabled, multi-carrier negotiation takes some time, and therefore the **NRDUCellPowerSaving.PwrSavingPenaltyTime** parameter for intelligent carrier shutdown must be set to a value greater than or equal to 5.
- Intelligent power-off of carriers in the same coverage: See *Energy Conservation and Emission Reduction* in *eRAN Feature Documentation*.
- Cell Switch-off Based on Automatic Co-coverage Identification: See *Energy Conservation and Emission Reduction* in *eRAN Feature Documentation*.
- Multi-carrier coordinated energy saving: See *Energy Conservation and Emission Reduction* in *eRAN Feature Documentation*.

6.2.2 Network Analysis

6.2.2.1 Benefits

Multi-carrier joint carrier shutdown maximizes energy saving gains. However, in case any carrier no longer meets the conditions for the carrier shutdown function involved in the multi-carrier negotiation to take effect, all the carriers exit that function. In this case, the energy saving gains may decrease.

6.2.2.2 Impacts

The impacts between this function and other functions are described in [6.2.3.2.3 Function Impacts](#).

Multi-carrier joint carrier shutdown alters the UE distribution and traffic volume, leading to changes in the performance indicators related to inter-frequency handovers and redirections.

If multi-carrier joint carrier shutdown is enabled for a cell in the energy saving state, the cell exits this state before multi-carrier joint carrier shutdown is evaluated in a coordinated manner.

Assuming that the neighboring base stations are in the carrier shutdown state, CEUs in weak coverage areas may access the available cells served by the local base station. This affects indicators related to the aspects such as access and service drop of the local base station. For example, the value of **N.NsaDc.DRB.AbnormRel** may increase.

6.2.3 Requirements

6.2.3.1 Licenses

None

6.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.2.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD NR FDD Low-frequency NR TDD	LTE and NR intelligent carrier shutdown	- On the LTE side: the CellAlgoExtSwitch.LnrSmartCarrierShutdownSw parameter - On the NR side: the LNR_SMART_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	Multi-carrier joint carrier shutdown depends on LTE and NR intelligent carrier shutdown ^a .

RAT	Function Name	Function Switch	Reference	Description
a: LTE and NR intelligent carrier shutdown can be enabled only when one of the following conditions is met: • The NRDUCellPowerSaving.EvalPeriodGranularity parameter is set to MINUTE_LEVEL. • The NRDUCellPowerSaving.EvalPeriodGranularity parameter is set to SECOND_LEVEL, and the NRDUCellPowerSaving.SecondLevelTrigEvalPrd parameter is set to a value greater than or equal to 6.				

6.2.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Intelligent carrier shutdown in precise mode	• INTRA_GNB_MULT I_CARR_S D_SW or INTER_GNB_MULT I_CARR_S D_SW option of the NRDUCell AlgoSwitch.Power SavingSwitch parameter • MULTI_CARR_PRECISE_SHD N_SW option of the NRDUCell AlgoSwitch.Power SavingSwitch parameter selected	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	If the NRDUCellPowerSaving.EvalPeriodGranularity parameter is set to SECOND_LEVEL and the NRDUCellPowerSaving.SecondLevelTrigEvalPrd parameter is set to a value less than 6 for intelligent carrier shutdown, multi-carrier joint carrier shutdown must be disabled.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Intelligent carrier shutdown in non-precise mode	• INTRA_GNB_MULT I_CARR_S D_SW or INTER_GNB_MULT I_CARR_S D_SW option of the NRDUCellAlgoSwitch.Power SavingSwitch parameter • MULTI_CARR_PRECISE_SHD N_SW option of the NRDUCellAlgoSwitch.Power SavingSwitch parameter deselected	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	If the NRDUCellPowerSaving.EvalPeriod Granularity parameter is set to SECOND_LEVEL and the NRDUCellPowerSaving.SecondLevelTrigEvalPrd parameter is set to a value less than 6 for intelligent carrier shutdown, multi-carrier joint carrier shutdown must be disabled.
NR FDD Low-frequency NR TDD	Fast carrier shutdown	FAST_CARR_SHUT_SERV_ASSUR_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Multi-carrier joint carrier shutdown does not work with fast carrier shutdown.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	LTE and NR intelligent carrier shutdown	LNR_SMART_CARRIER_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	If the NRDUCellPowerSaving.EvalPeriodGranularity parameter is set to SECOND_LEVEL and the NRDUCellPowerSaving.SecondLevelTrigEvalPrd parameter is set to a value less than 6 for LTE and NR intelligent carrier shutdown, multi-carrier joint carrier shutdown must be disabled.
NR FDD Low-frequency NR TDD	Low power consumption mode	LOW_PWR_MODE_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Green Site</i>	Multi-carrier joint carrier shutdown does not work with low power consumption mode.
Low-frequency NR TDD	Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAIgoSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD) in 5G RAN Feature Documentation</i>	Multi-carrier joint carrier shutdown does not work with distributed massive MIMO.

6.2.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Intelligent PIM interference avoidance	PIM_INTERF_SMART_AVOID_SW option of the PimInterfSmartAvoid.PimAlgorithm parameter	<i>Interference Detection and Suppression in eRAN Feature Documentation</i>	For LTE FDD, multi-carrier joint carrier shutdown is terminated during PIM interference detection.
NR FDD	Intelligent PIM interference avoidance	PIM_INTERF_SMART_AVOID_SW option of the NRDUCellPimInterfAvoid.PimAlgorithm parameter	<i>Interference Avoidance in 5G RAN Feature Documentation</i>	For NR FDD, multi-carrier joint carrier shutdown is terminated during PIM interference detection.

6.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- NR: Refer to *Energy Conservation and Emission Reduction in 5G RAN Feature Documentation*.
- LTE: Refer to *Energy Conservation and Emission Reduction in eRAN Feature Documentation*.

Boards

- NR: Refer to *Energy Conservation and Emission Reduction in 5G RAN Feature Documentation*.
- LTE: Refer to *Energy Conservation and Emission Reduction in eRAN Feature Documentation*.

RF Modules

- NR: Refer to *Energy Conservation and Emission Reduction in 5G RAN Feature Documentation*.
- LTE: Refer to *Energy Conservation and Emission Reduction in eRAN Feature Documentation*.

6.2.3.4 Others

None

6.2.4 Operation and Maintenance

6.2.4.1 Data Configuration

6.2.4.1.1 Data Preparation

[Table 6-10](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 6-10 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Power Saving Extension Switch	NRDUCellAlgoSwitch.PowerSavingExtSwitch	It is recommended that the MULTI_CARR_JOINT_CARR_SHDN_SW option be selected for all cells on the carriers sharing a PA.

6.2.4.1.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [6.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

```
//Enabling multi-carrier joint carrier shutdown
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingExtSwitch=MULTI_CARR_JOINT_CARR_SHDN_SW-1;
```

Deactivation Command Examples

```
//Disabling multi-carrier joint carrier shutdown
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
PowerSavingExtSwitch=MULTI_CARR_JOINT_CARR_SHDN_SW-0;
```

6.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.2.4.2 Activation Verification

- Step 1** Run the **LST NRDUCELLALGOSWITCH** command. If the value of the **Power Saving Extension Switch** parameter is **Multi-Carrier Joint Carrier Shutdown Switch:On** in the command output, multi-carrier joint carrier shutdown has been enabled.
- Step 2** Verify the carrier shutdown function involved in the multi-carrier negotiation. For details, see the descriptions of activation verification for that function.
- Step 3** You can run the **DSP NRDUCELLPOWERSAVING** command to query the status of the carrier shutdown function involved in multi-carrier negotiation and the reason why that function did not take effect. If the value of **Current State** is **Disabled** for that function, check the **Current Disabled Cause**, **Latest Exit Cause**, and **Latest Disabled Cause** parameter values to find out the possible reason why that function did not take effect.
- In the case that the carrier shutdown function involved in the negotiation between carriers A and B did not take effect for either of the carriers because carrier A (not B) did not meet the triggering conditions for that function, the **Current Disabled Cause** and **Latest Disabled Cause** parameter values are **Multi-Carrier Not Coordinated** for carrier B.
 - In the case that carriers A and B both exit that carrier shutdown function because carrier A no longer meets the conditions for that function to take effect, the **Current Disabled Cause**, **Latest Exit Cause**, and **Latest Disabled Cause** parameter values are **In Penalty State** for carrier B.

----End

6.2.4.3 Network Monitoring

- Energy consumption data for base stations can be obtained by observing the counter **VS.EnergyCons.BTS.Adding.Service.NR** (for an entire base station), **VS.EnergyCons.BTS.Adding.NR** (for an entire base station), or **VS.EnergyCons.BTSBoard** (for RF modules) on the MAE-Access.
On the live network, the one-week base station energy consumption data obtained before and after multi-carrier joint carrier shutdown is enabled can be compared to assess the energy saving gains of this function. It is expected that the base station energy consumption decreases significantly after this function is enabled.
- The duration of cell unavailability due to multi-carrier joint carrier shutdown can be obtained using the **N.Cell.Unavail.Dur.EnergySaving** counter on the MAE-Access.
- The duration in which the cell stays in the carrier shutdown state due to multi-carrier joint carrier shutdown can be obtained using the **N.PowerSaving.CarrierShutdown.Dur** counter on the MAE-Access.

6.3 GUL Joint Carrier Shutdown

6.3.1 Principles

6.3.1.1 Areas Covered by GU Networks

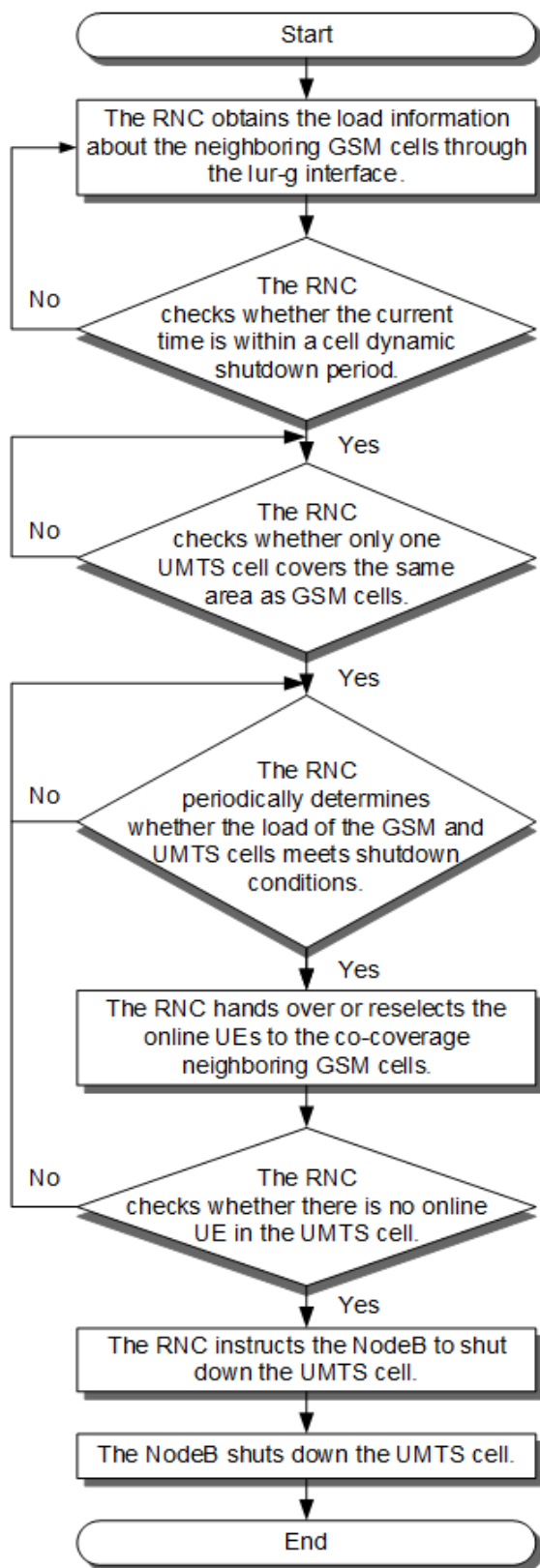
6.3.1.1.1 GU Cell Joint Intelligent Shutdown

If Multi-RAT Carrier Joint Intelligent Shutdown is used in an area covered by GU networks, this feature must be enabled on both the RNC and the BSC sides as follows:

- On the RNC side, **IRATShutDown** of *DynShutDownType* is selected.
- On the BSC side, **IRATShutdownSwitch** is set to **ON**.
- The lur-g interface must also be configured and enabled. For detailed operations, see:
 - BSC6900/BSC6910: section "Configuring the lur-g Interface (IP Transmission Mode)" in *BSC69X0 GU Initial Configuration Guide* in *BSC69X0 GU Product Documentation*.
 - BSC6960: sections "Configuring the lur-g Interface (over IP)" in *BSC6960 GSM Initial Configuration Guide* in *BSC6960 GSM Product Documentation* and "Configuring the lur Interface (in Transmission Resource Pool Mode)" in *BSC6960 UMTS Initial Configuration Guide* in *BSC6960 UMTS Product Documentation*.

Figure 6-4 shows the procedure for GU cell joint shutdown.

Figure 6-4 Procedure for GU cell joint shutdown



1. The RNC starts common measurements (controlled by ***IRATCommonMeasSwitch***) over the Iur-g interface and obtains information about neighboring GSM cells with ***NIRATOverLap*** set to **True**.
2. The RNC checks whether the current time is within a cell dynamic shutdown period. If it is, the RNC proceeds with the subsequent processing. A maximum of three cell dynamic shutdown periods can be configured within a day. They are specified by the following parameters:
 - ***StartTime1*** and ***EndTime1***
 - ***StartTime2*** and ***EndTime2***
 - ***StartTime3*** and ***EndTime3***
3. The RNC checks whether the current UMTS cell has an available inter-frequency co-coverage cell under the same NodeB. If not, the RNC proceeds with the subsequent processing.

 NOTE

If the UMTS cell has inter-frequency same-coverage cells, these UMTS cells must be enabled with either WRFD-020117 Multi-Carrier Switch off Based on Traffic Load or WRFD-020122 Multi-Carrier Switch off Based on QoS. This is to implement intra-RAT carrier shutdown first. Multi-RAT Carrier Joint Intelligent Shutdown is implemented only when operators need to achieve further power-saving performance.

4. The RNC checks whether the current UMTS cell meets shutdown conditions and proceeds with the subsequent processing when the following conditions are met within a period specified by ***IRATShutdownAdjTime***.
 - The number of online users in the cell is less than or equal to the value of ***TotalUserNumThd***.
 - The downlink cell load is less than the value of ***IRATShutdownDLLoadThd***.
 - The uplink cell load is less than the value of ***IRATShutdownULLoadThd***.
 - There are no high-priority users in the cell. The ***HighPriSPI*** parameter specifies whether a user is a high-priority user.
 - ***DynShutDownCchUserSwitch*** is set to **OFF**. Alternatively, ***DynShutDownCchUserSwitch*** is set to **ON**, and no UE is in the CELL_FACH state.
 - The neighboring GSM cells with ***NIRATOverLap*** set to **True** meet the following conditions:
 - The timeslot usage in the cells is less than the value of ***IRATshutdownGSMLoadThd***.
 - The cells support PS services.
 - ***IRATHighPriSwitch*** is set to **OFF**. Alternatively, ***IRATHighPriSwitch*** is set to **ON**, and there are no high-priority users in the cells.
 - ***IRATTBFSwitch*** is set to **OFF**. Alternatively ***IRATTBFSwitch*** is set to **ON**, and the duration of the TBF connection for PS services on a multimode terminal is less than or equal to the value of ***IRATSwitchonTBFFontTimeThd***.
5. The RNC hands over or reselects the UEs in the UMTS cell to neighboring GSM cells. The RNC then instructs the NodeB to shut down the UMTS cell.

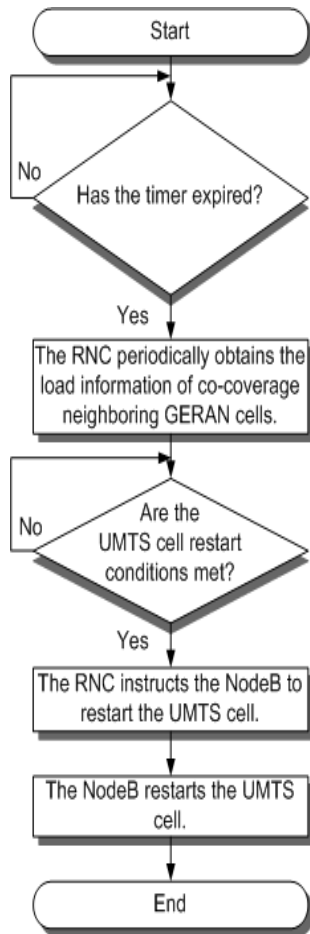
Upon receiving this instruction, the NodeB gradually shuts down the UMTS cell if the Intelligently Out of Service feature (For details, see *Intelligently Out of Service* in *RAN Feature Documentation*) is enabled. Otherwise, the NodeB immediately shuts down the UMTS cell.

6.3.1.1.2 Procedure for GU Cell Joint Intelligent Restart

The RNC intelligently restarts the UMTS cell when the traffic volumes of the GSM cells increase.

Figure 6-5 shows the procedure for GU cell joint intelligent restart.

Figure 6-5 Procedure for GU cell joint intelligent restart



1. After shutting down the UMTS cell, the RNC starts the ***IRATSwitchOnHystTimeLen*** timer. After the timer has expired, the RNC proceeds with the subsequent steps.
2. The RNC checks whether the UMTS cell meets restart conditions based on the load information about the co-coverage neighboring GSM cells. The RNC proceeds with the subsequent steps if the co-coverage neighboring GSM cells meet any of the following conditions:
 - The load of any GSM cell exceeds the sum of ***IRATshutdownGSMLoadThd*** and ***IRATshutdownGSMLoadHyst***.
 - ***IRATHighPriSwitch*** for any GSM cell is set to **ON**, and there are high-priority multimode UEs in the cell.

- ***IRATTBFSwitch*** for any GSM cell is set to **ON**, and the duration of the TBF connection for multimode UEs' PS services is greater than the value of ***IRATSwitchonTBFContTimeThd***.
 - The current cell dynamic shutdown period ends.
3. The RNC instructs the NodeB to restart the UMTS cell. Upon receiving this instruction, the NodeB restarts the UMTS cell.
 4. After the UMTS cell is restarted, it can be shut down again only after the ***DynCellShutdownProtectTimerlen*** timer expires.

 NOTE

The RNC automatically restarts a UMTS cell that has shut down if either of the following conditions is met: (1) The Iur-g interface is faulty and the load information about the co-coverage neighboring GSM cells cannot be obtained. (2) The co-coverage GSM cells are faulty.

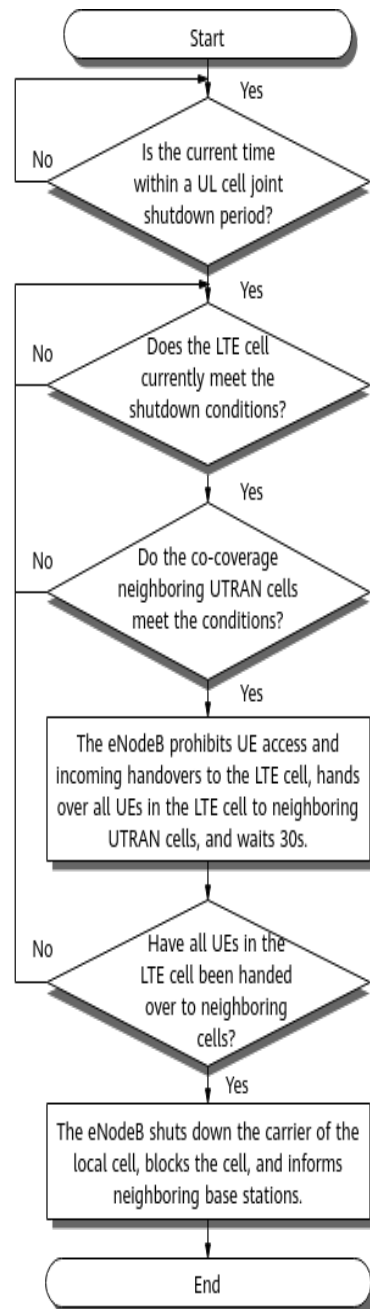
6.3.1.2 Areas Covered by UL Networks

6.3.1.2.1 UL Cell Joint Intelligent Shutdown

If Multi-RAT Carrier Joint Intelligent Shutdown is used in an area covered by UL networks, this feature must be enabled on both the eNodeB and the RNC sides as follows:

- On the eNodeB side, ***ForceShutdownSwitch*** is set to **ON** and ***ShutDownType*** is set to **BY_UTRAN_WITH_LOAD**.
- On the RNC side, the **INTERRAT_LOAD_REPORT_FOR_LTE_SWITCH** option of the ***PROCESSSWITCH*** parameter is selected.
- On the RNC side, the **INTERRAT_LOAD_SINGLE_RPT_FOR_LTE_SWITCH** option of the ***PROCESSSWITCH5*** parameter is selected.

Figure 6-6 Procedure for UL cell joint shutdown



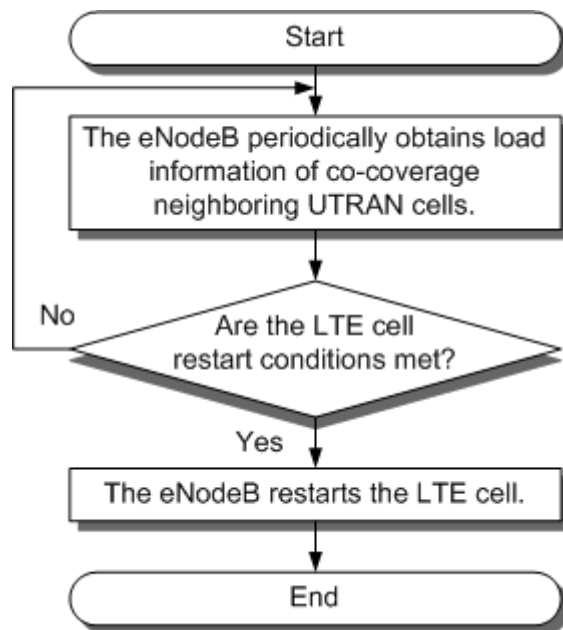
1. The eNodeB checks whether the current time is within a cell dynamic shutdown period specified by **StartTime** and **StopTime**. If it is, the eNodeB proceeds with the subsequent steps.
2. If the eNodeB is deployed with a multi-carrier inter-frequency network, Multi-RAT Carrier Joint Intelligent Shutdown can take effect in basic-layer cells only when all intra-eNodeB capacity-layer cells for which LOFD-001042 Intelligent Power-Off of Carriers in the Same Coverage is enabled are shut down.
3. The eNodeB checks whether the current cell meets the following conditions. If yes, the eNodeB proceeds with the subsequent steps.
 - The uplink PRB usage is less than or equal to the value of **ULPrbThd** and the downlink PRB usage is less than or equal to the value of **DLPrbThd**.

- The number of UEs in connected mode in the cell is less than or equal to the value of ***BearNumThd***.
 - The cell is not carrying any services whose allocation/retention priority (ARP) is less than or equal to the value of ***HighPriArpThd***.
 - More than 30 minutes have elapsed since the last intelligent restart of the LTE cell.
4. The eNodeB obtains load information about neighboring UTRAN cells with ***OverlapInd*** set to **YES** from the RNC through RIM procedures. The eNodeB proceeds with the subsequent steps if all obtained available co-coverage neighboring UTRAN cells meet the following conditions:
 - Uplink load of the co-coverage neighboring UTRAN cell is less than the value of ***UtranCellULLoadThd***.
 - Downlink load of the co-coverage neighboring UTRAN cell is less than the value of ***UtranCellDLLoadThd***.
 5. The eNodeB prohibits UE access and handovers to the LTE cell, and hands all UEs in the LTE cell over to neighboring UTRAN cells within 30s. If the eNodeB fails to do so, the eNodeB removes the prohibition on UE access and incoming handovers and re-evaluates whether the triggering conditions are met.
 6. After handing over all UEs in the LTE cell to UMTS cells, the eNodeB shuts down the RRU serving the LTE cell and thereby the LTE cell is blocked. The eNodeB sends an eNodeB Configuration Update message with the cause value "Deactivation Indication" to the neighboring eNodeB through the X2 interface, signaling that the LTE cell has been deactivated for power-saving purposes.

6.3.1.2.2 Procedure for UL Cell Joint Intelligent Restart

The eNodeB intelligently restarts the LTE cell when the traffic of the co-coverage neighboring UTRAN cells increases.

Figure 6-7 Procedure for UL cell joint intelligent restart



1. After shutting down the LTE cell, the eNodeB periodically obtains load information about co-coverage neighboring UTRAN cells through RIM procedures.
2. The eNodeB periodically determines whether to restart the LTE cell based on the traffic volume of the co-coverage neighboring UTRAN cells. If the eNodeB finds that any of the following conditions is met, the LTE cell stops using Multi-RAT Carrier Joint Intelligent Shutdown and the disabled carriers are enabled.
 - The period specified by **StartTime** and **StopTime** ends.
 - The uplink load of any co-coverage neighboring UTRAN cell of the local cell is greater than or equal to the sum of the uplink load threshold of neighboring UTRAN cells (**UtranCellULLoadThd**) and uplink load offset of neighboring UTRAN cells (**UtranCellULLoadOffset**) throughout the evaluation period specified by **ExitJudgePeriod**.
 - The downlink load of any co-coverage neighboring UTRAN cell of the local cell is greater than or equal to the sum of the downlink load threshold of neighboring UTRAN cells (**UtranCellDLLoadThd**) and downlink load offset of neighboring UTRAN cells (**UtranCellDLLoadOffset**) throughout the evaluation period specified by **ExitJudgePeriod**.
 - The LTE capacity cells and basic cells all exit the intelligent power-off of carriers in the same coverage state.
3. The eNodeB enables the disabled carriers in the LTE cell and sends an eNodeB Configuration Update message through the X2 interface to inform the neighboring eNodeB that the LTE cell has been activated.

NOTE

When Multi-RAT Carrier Joint Shutdown is disabled or load information about co-coverage neighboring UTRAN cells for the LTE cell cannot be obtained, the LTE cell can be restarted automatically.

If an LTE cell involved in UL cell joint intelligent shutdown is also involved in intelligent power-off of carriers in the same coverage, and the LTE cell is a capacity cell, the LTE cell participates only in intelligent power-off of carriers in the same coverage on the LTE side and does not participate in UL cell joint intelligent shutdown. If the LTE cell is a basic cell, it can participate in UL cell joint intelligent shutdown only after all its intra-eNodeB capacity cells are shut down. For details on the intelligent power-off of carriers in the same coverage function on the LTE side, and the definitions of basic and capacity cells, see *Energy Conservation and Emission Reduction* in *eRAN Feature Documentation*.

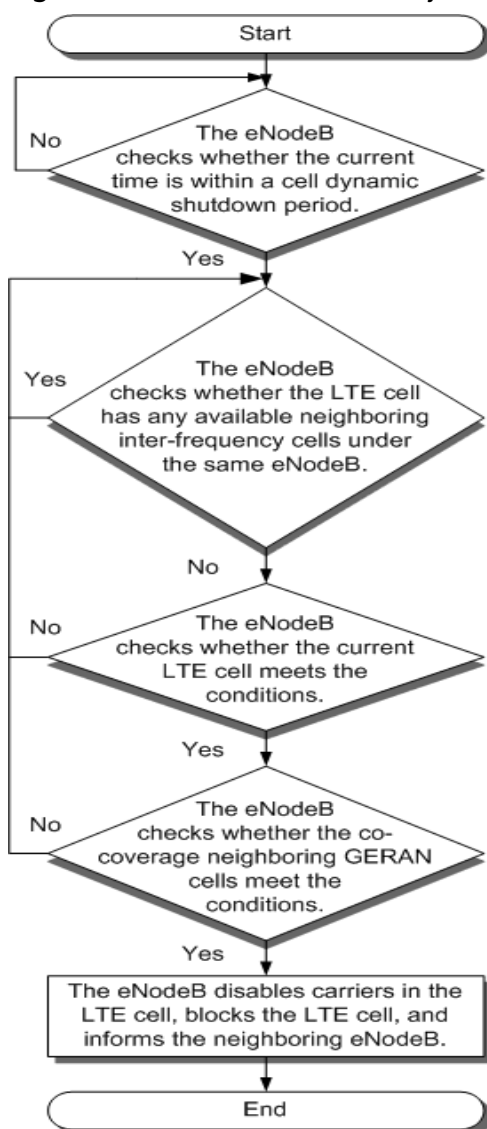
6.3.1.3 Areas Covered by GL Networks

6.3.1.3.1 GL Cell Joint Intelligent Shutdown

If Multi-RAT Carrier Joint Intelligent Shutdown is used in an area covered by GL networks, this feature must be enabled on both the eNodeB and the BSC sides as follows:

- On the eNodeB side, **ForceShutdownSwitch** is set to **ON** and **ShutDownType** is set to **BY_GERAN_WITH_LOAD**.
- On the BSC side, the **RIMSUP** parameter of the Gb interface is set to **YES**.

Figure 6-8 Procedure for GL cell joint shutdown



1. The eNodeB checks whether the current time is within a cell dynamic shutdown period specified by **StartTime** and **StopTime**. If it is, the eNodeB proceeds with the subsequent steps.
2. The eNodeB checks whether the current LTE cell has any available neighboring inter-frequency cells under the same eNodeB. If it does not, the eNodeB proceeds with the subsequent steps.

NOTE

If the eNodeB is deployed with a multi-carrier inter-frequency network, Multi-RAT Carrier Joint Intelligent Shutdown can take effect in basic-layer cells only when all intra-eNodeB capacity-layer cells for which LOFD-001042 Intelligent Power-Off of Carriers in the Same Coverage takes effect are shut down.

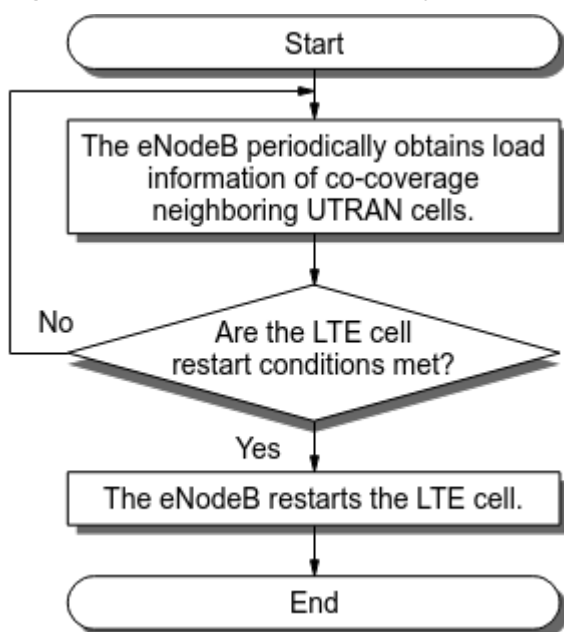
3. The eNodeB checks whether the current LTE cell meets the following conditions. If yes, the eNodeB proceeds with the subsequent steps.
 - The number of UEs in connected mode in the LTE cell is 0.
 - More than 30 minutes have elapsed since the last intelligent restart of the LTE cell.

4. The eNodeB obtains load information about neighboring GERAN cells with **OverlapInd** set to **YES** from the BSC through RIM procedures. The eNodeB proceeds with the subsequent steps if all the load of all obtained available co-coverage neighboring GERAN cells is less than the value of the **GeranCellLoadThd** parameter.
5. The eNodeB disables carriers in the LTE cell and RRUs to block the LTE cell, and sends an eNodeB Configuration Update message with the cause value Deactivation Indication to the neighboring eNodeB through the X2 interface. The cause value Deactivation Indication specifies that the LTE cell has been deactivated to save power.

6.3.1.3.2 Procedure for GL Cell Joint Intelligent Restart

The eNodeB intelligently restarts the LTE cell when the traffic of the co-coverage neighboring GERAN cells increases, as shown in [Figure 6-9](#).

Figure 6-9 Procedure for GL cell joint intelligent restart



1. The eNodeB periodically obtains load information about co-coverage neighboring GERAN cells through RIM procedures.
2. If the eNodeB finds that any of the following conditions is met, the LTE cell stops using Multi-RAT Carrier Joint Intelligent Shutdown and the disabled carriers are enabled.
 - The period specified by **StartTime** and **StopTime** ends.
 - The load of any co-coverage neighboring GERAN cell of the local cell is greater than or equal to the sum of the load threshold of neighboring GERAN cells (**GeranCellLoadThd**) and load offset of neighboring GERAN cells (**GeranCellLoadOffset**) throughout the evaluation period specified by **ExitJudgePeriod**.
3. The eNodeB enables the disabled carriers in the LTE cell and sends an eNodeB Configuration Update message through the X2 interface to inform the neighboring eNodeB that the LTE cell has been activated.

 NOTE

When Multi-RAT Carrier Joint Shutdown is disabled or load information about co-coverage neighboring GERAN cells for the LTE cell cannot be obtained, the LTE cell can be restarted automatically.

6.3.1.4 Areas Covered by GUL Networks

When Multi-RAT Carrier Joint Intelligent Shutdown is used in areas covered by GUL networks, LTE and UMTS cell joint intelligent shutdown and joint intelligent restart must be performed.

The conditions and procedures for LTE cell joint intelligent shutdown and joint intelligent restart in areas covered by GUL networks are the same as those in areas covered by UL networks. For details, see [6.3.1.2 Areas Covered by UL Networks](#). The conditions and procedures for UMTS cell joint intelligent shutdown and joint intelligent restart in areas covered by GUL networks are the same as those in areas covered by GU networks. For details, see [6.3.1.1 Areas Covered by GU Networks](#). In the overlapping coverage area of GSM, UMTS, and LTE, LTE UEs are handed over to UMTS cells and UMTS UEs are handed over to GSM cells. LTE UEs cannot be directly handed over to GSM cells.

6.3.2 Network Analysis

6.3.2.1 Benefits

This feature reduces the overall energy consumption of multi-RAT networks and implements energy saving and emission reduction, decreasing the OPEX.

6.3.2.2 Impacts

The impacts between this function and other functions are described in [6.3.3.2.3 Function Impacts](#).

- **QoS of a single UE**
After an LTE or UMTS cell enters the joint carrier shutdown state in areas covered by GU, UL, GL, or GUL networks, the small number of UEs in the cell will be transferred to inter-RAT co-coverage neighboring cells. As data service processing capability of inter-RAT co-coverage neighboring cells is inferior to that of the LTE or UMTS cell that has been shut down, the peak rate of a single UE is reduced, thereby affecting user experience in data services. For example, data service processing capability of UMTS cells is inferior to that of LTE cells, and data service processing capability of GSM cells is inferior to that of UMTS cells.
- **Service availability to single-mode UEs**
In areas covered by GU, UL, GL, or GUL networks, LTE-only or UMTS-only UEs in idle mode cannot initiate services after the LTE or UMTS cell is shut down.
- **Number of inter-RAT handovers and cell reselections**
This feature triggers the following interoperability procedures between networks of different RATs:

- After the LTE or UMTS cell is shut down, multimode UEs reselect or are handed over to a UMTS or GSM cell. As a result, inter-RAT handovers and cell reselections increase.
- After the LTE or UMTS cell is restarted, the multimode UEs reselect or are handed over to the LTE or UMTS cell. As a result, inter-RAT handovers and cell reselections increase.
- Call drop rate on the FACH
Before shutting down the UMTS cell in areas covered by GU networks, the RNC releases the connections of UEs in the CELL_FACH state so that the UEs reselect a GSM cell if ***DynShutDownCchUserSwitch*** is set to **OFF**. As a result, the call drop rate on the FACH increases.
- Access delay of UEs in idle mode and these UEs' dropping from the network
After the LTE or UMTS cell is shut down, multimode UEs in idle mode reselect a UMTS or GSM cell. After the LTE or UMTS cell is restarted, the multimode UEs in idle mode reselect the LTE or UMTS cell. During the cell reselection, the following possibilities may occur:
 - UEs in idle mode that have radio signal fluctuations or poor performance may drop from the network for a short period of time after cell reselection.
 - The routing areas of cells with different modes may be different, and the cell reselection triggers location area (LA) updates. UEs in idle mode occupy access channel resources during LA updates, thereby temporarily affecting access of other UEs.

6.3.3 Requirements

6.3.3.1 Licenses

None

6.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

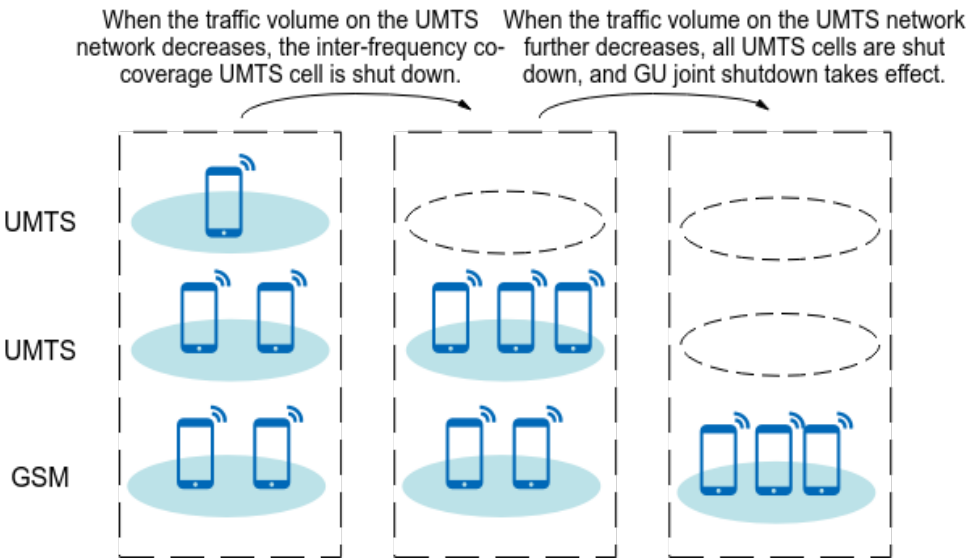
Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.3.2.1 Prerequisite Functions

- Areas covered by GU networks
This feature requires either of the following functions in UMTS multi-carrier scenarios, but does not require any of them in UMTS single-carrier scenarios.

Function Name	Function Switch	Reference	Description
Multi-Carrier Switch off Based on Traffic Load	URNCELLS HUTDOWNP ARA.DynCell ShutDownSwitch	<i>Green BTS in RAN Feature Documentation</i>	When the traffic volume on the UMTS network decreases, the inter-frequency co-coverage UMTS cells for the current UMTS cell are shut down. These features take effect only when the traffic volume on the UMTS network further decreases and the current UMTS cell is the only working UMTS cell, as shown in Figure 6-10 .
Multi-Carrier Switch off Based on QoS			

Figure 6-10 Feature application in UMTS multi-carrier and GSM co-coverage scenarios



- Areas covered by UL networks and covered by GL networks
This feature requires the following function in LTE multi-carrier scenarios, but does not require it in LTE single-carrier scenarios.

Function Name	Function Switch	Reference	Description
Intelligent power-off of carriers in the same coverage	CellShutdownSwitch	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	In a multi-carrier inter-frequency LTE network, it is recommended that intelligent power-off of carriers in the same coverage be enabled in capacity-layer cells and Multi-RAT Carrier Joint Shutdown be enabled in basic-layer cells. When the traffic volume in the LTE network decreases, one or more capacity-layer LTE cells are shut down through intelligent power-off of carriers in the same coverage. When the traffic volume in the LTE network further decreases and LTE basic-layer cells are the only working LTE cells, this feature then takes effect to shut down the LTE basic-layer cells.

RIM Based UMTS Target Cell Selection for LTE must be enabled if it is required on the RNC side.

Function Name	Function Switch	Reference	Description
RIM Based UMTS Target Cell Selection for LTE	INTERRAT_LOAD_REPORT_FOR_LTE_SWITCH option of the URRCTRLSWITCH.PROCESSSWITCH parameter INTERRAT_LOAD_SINGLE_RPT_FOR_LTE_SWITCH option of the URRCTRLSWITCH.PROCESSSWITCH5 parameter CELLLOAD_SHARED_BASED_ON_RIM option of the UCELLICENSE.FuncSwitch1 parameter	Section "WRFD-1502 31 RIM Based UMTS Target Cell Selection for LTE" in chapter "Engineering Guidelines" of <i>Interoperability Between UMTS and LTE Feature Parameter Description in RAN Feature Documentation</i>	In areas covered by UL networks, the eNodeB is informed of the load status of UMTS cells through the RAN information management (RIM) procedure. In this way, the eNodeB can select a target cell based on the obtained load status of UMTS cells before an LTE-to-UMTS handover.

6.3.3.2.2 Mutually Exclusive Functions

- Areas covered by GU networks

Function Name	Function Switch	Reference	Description
GSM and UMTS Dynamic Spectrum Sharing	GDSSPARA.DSSENABLE	<i>GSM and UMTS Dynamic Spectrum Sharing in GBSS Feature Documentation</i>	The application scenarios and purposes of these two features are different. GSM and UMTS Dynamic Spectrum Sharing enables dynamic spectrum sharing (DSS) between a GSM network and a UMTS network based on traffic load, improving data throughput. If both features are enabled, the UMTS cell that uses a GSM frequency through DSS may be shut down, and the traffic volume on the GSM network increases because the GSM cells carry all of the traffic in the overlapping coverage area. The increase in the GSM traffic volume triggers spectrum resource reclamation, and as a result GSM and UMTS Dynamic Spectrum Sharing becomes ineffective.

- Areas covered by UL networks, GL networks, and GUL networks

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	SFN cell	Cell.MultiRruCellMode	<i>SFN</i>	None
LTE FDD	RF channel intelligent shutdown	CellRfShutdown.RfShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	This feature is mutually exclusive with RF channel intelligent shutdown.
LTE FDD	Low power consumption mode (LTE)	CellLowPower.LowPowerSwitch	<i>Green Site</i>	This feature is mutually exclusive with low power consumption mode.

RA T	Function Name	Function Switch	Reference	Description
NB - IoT	LTE in-band deployment	Prb.Deploy Mode	<i>NB-IoT Basics (FDD)</i>	Enabling this energy saving feature will cause an LTE FDD cell and any NB-IoT cell deployed in the FDD band of this cell to be unavailable. Therefore, this feature does not work with LTE in- band deployment.
LTE FD D	Service guarantee for UEs running unlimited services	UsUeGuara nteeSwitch option of the CellUSPara Cfg.UsAlgo Switch parameter	None	None
LTE FD D	Massive MIMO Introduction	MASSIVE_ MIMO_Ph2 option of the eUCellSect orEqm.Cell BeamMode parameter	<i>Massive MIMO Basics (FDD) in eRAN Feature Documenta tion</i>	Massive MIMO cells do not support this feature.
LTE FD D	LTE FDD and NR Flash Dynamic Spectrum Sharing	SpectrumCl oud.Spectr umCloudS witch set to LTE_NR_SP ECTRUM_S HR	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>	The "intelligent power-off of carriers in the same coverage as UMTS networks" function does not take effect when "LTE FDD and NR Flash Dynamic Spectrum Sharing" is in effect. This is because the LTE carrier cannot be independently shut down when LTE FDD and NR share this carrier.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Hybrid DSS Based on Asymmetric Bandwidth	SpectrumCloud.SpectrumCloudSwitch parameter set to LTE_NR_SPECTRUM_SHR_ASYM_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter selected	<i>LTE FDD and NR Hybrid Dynamic Spectrum Sharing</i>	None

6.3.3.2.3 Function Impacts

- Areas covered by GU networks

Function Name	Function Switch	Reference	Description
Intelligently Out of Service	NODEBSMT HPWR.SMT HPWRSWITCH	<i>Intelligently Out of Service in RAN Feature Documentation</i>	It is recommended that operators enable the Multi-RAT Carrier Joint Intelligent Shutdown feature together with this feature and function in areas covered by GU networks to achieve superior performance. • The Intelligently Out of Service feature helps gradually reduce the cell pilot power to enable UEs in the idle state to reselect GSM cells when the UMTS cell is shut down. This avoids a great number of LA update requests caused by the simultaneous cell reselections of UEs in the idle state, minimizing the possibility of congestion on access channels. • The Preferential Allocation of Channels on the BCCH TRX function enables a small number of online users to preferentially perform services on the GSM BCCH TRX, increasing the energy saving gains.
Preferential Allocation of Channels on the BCCH TRX	GTRXDEV.OTL	<i>Intelligent Power Consumption Decrease in GBSS Feature Documentation</i>	
TRX Power Amplifier Intelligent Shutdown	GTRXDEV.CPS	<i>Intelligent Power Consumption Decrease in GBSS Feature Documentation</i>	The Multi-RAT Carrier Joint Intelligent Shutdown feature may affect this feature in areas covered by GU networks. Before shutting down the UMTS cell, a small number of multimode UEs in this UMTS cell are transferred to GSM cells, increasing timeslot usage in the GSM cells. As a result, the possibility that TRX Power Amplifier Intelligent Shutdown is triggered may be lowered.

Function Name	Function Switch	Reference	Description
Dynamic BCCH TRX Timeslot Shutdown	BTSOTHPAR A.BCCHTRXT sShutdownS w	<i>Dynamic BCCH TRX Timeslot Shutdown in GBSS Feature Documentation</i>	When Multi-RAT Carrier Joint Intelligent Shutdown is enabled together with Dynamic BCCH TRX Timeslot Shutdown in areas covered by GU networks, the total energy saving gains decrease.
UMTS	NCR cells	ULOCCELL.NC RCELL	After the NCR cells served by the DeNB enter the GUL joint carrier shutdown state, UMTS services within the coverage of the RuralCow site are affected.

- Areas covered by UL networks

Function Name	Function Switch	Reference	Description
RAN sharing with common carrier	E NodeB S haring M ode. E N o deB S haring M ode	<i>RAN Sharing in eRAN Feature Documentation</i>	In areas covered by UL networks, whether the Multi-RAT Carrier Joint Intelligent Shutdown feature can be used together with the RAN sharing with common carrier function in an LTE cell depends on the co-coverage neighboring UTRAN cells for the LTE cell. • If the co-coverage neighboring UTRAN cells are independent cells, the Multi-RAT Carrier Joint Intelligent Shutdown feature may not take effect when UEs of an operator cannot be handed over to the LTE cell. Therefore, it is recommended that the Multi-RAT Carrier Joint Intelligent Shutdown feature be not used together with the RAN sharing with common carrier function. • If the co-coverage neighboring UTRAN cells are shared cells and have the same operator group as the LTE cell, they can be used together.

Function Name	Function Switch	Reference	Description
UMTS and LTE Dynamic Power Sharing	CellDynPowerSharing.DynamicPowerSharingSwitch	<i>UMTS and LTE Dynamic Power Sharing</i>	- When the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter is deselected, UMTS and LTE Dynamic Power Sharing does not work for cells in the carrier shutdown state. These cells will attempt to participate in power sharing every 10 minutes. - When the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter is selected, dynamic power sharing takes effect normally, but energy saving gains may decrease.
Camp & Handover Based on SPID	None	<i>Flexible User Steering in eRAN Feature Documentation</i>	When the Multi-RAT Carrier Joint Intelligent Shutdown (eNodeB) feature is enabled, the SPID-specific handover policy is used.
Dynamic Power Sharing Between LTE Carriers	CellDynPowerSharing.DynamicPowerSharingSwitch	<i>Dynamic Power Sharing Between LTE Carriers in eRAN Feature Documentation</i>	Cells that have entered the GUL joint carrier shutdown state do not participate in Dynamic Power Sharing Between LTE Carriers.

Function Name	Function Switch	Reference	Description
Automatic management of neighboring E-UTRAN frequency measurement flags	LTE_NFREQ_MEAS_MGMT_SW option of the CellAlgoExtSwitch.AnrOptSwitch parameter	<i>ANR Management in eRAN Feature Documentation</i>	When a carrier is shut down, the FREQ_MEAS_FLAG option of the neighboring frequency may be mistakenly deselected. Therefore, the ANR.NFreqMeasStatPeriod-Num parameter needs to be adjusted to ensure that the active measurement period can include the non-shutdown period of the carrier.
Dynamic carrier shutdown	CARRIER_DYNAMIC_SHUTDOWN_SW option of the CellShutdownEnhancedSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	Multi-RAT Carrier Joint Shutdown does not take effect when dynamic carrier shutdown is in effect.
Intelligent power-off of carriers in the same coverage	CellShutdown.CellShutdownSwitch	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	If intelligent power-off of carriers in the same coverage is enabled for a capacity-layer cell, the UL and GUL carrier joint intelligent shutdown functions do not take effect in this cell.
CRS port mapping detection and reconfiguration	AutoRecfgSwitch parameter in the STR_CRSPORTOPTDET command	<i>MIMO in eRAN Feature Documentation</i>	CRS port mapping detection and reconfiguration are not supported after multi-RAT carrier joint shutdown is triggered, and the detection result indicates that the scenario is not supported. If the conditions for triggering multi-RAT carrier joint shutdown are met during CRS port mapping detection, the shutdown takes effect after the detection is complete.

Function Name	Function Switch	Reference	Description
Dynamic carrier shutdown phase 2	CellPwrSavingsAlgo.CarrierDynShutdownPh2Sw set to CAPACITY_CELL	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	During dynamic carrier shutdown phase 2, joint carrier shutdown does not take effect in the local cell and its co-coverage neighboring cells.
Intelligent PIM interference avoidance	PIM_INTERF_SMART_AVOID_SW option of the PimInterfSmartAvoid.PimAlgoSwitch parameter	<i>Interference Detection and Suppression in eRAN Feature Documentation</i>	For LTE FDD, GUL joint carrier shutdown is terminated during PIM interference detection.
Channel isolation for PIM interference avoidance	PIM_INTERF_CHN_SHUTDOWN_SW option of the PimInterfSmartAvoid.PimAlgoSwitch parameter	<i>Interference Detection and Suppression in eRAN Feature Documentation</i>	For LTE FDD, if either GUL joint carrier shutdown or channel isolation for PIM interference avoidance has taken effect, the other function does not take effect.
Channel isolation for PIM interference avoidance	PIM_INTERF_SMART_AVOID_SW option of the NRDUCellPimIntrfAvd.PimAlgoSwitch parameter	<i>Interference Avoidance in 5G RAN Feature Documentation</i>	If either GUL joint carrier shutdown or channel isolation for PIM interference avoidance has taken effect, the other function does not take effect.

Function Name	Function Switch	Reference	Description
Massive MIMO downlink coordinated power control	CellDynPowerSharing.DynamicPowerSharingSwitch	<i>Massive MIMO Enhancements (FDD) in eRAN Feature Documentation</i>	- When Multi-RAT Carrier Joint Intelligent Shutdown (eNodeB) is in effect, massive MIMO downlink coordinated power control does not take effect. - When the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter is deselected, inter-carrier power sharing does not take effect in a cell where Multi-RAT Carrier Joint Intelligent Shutdown (eNodeB) is in effect. The cell not included in a power sharing group due to its energy saving state will attempt to enter a power sharing group every 10 minutes. - When the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter is selected, both Multi-RAT Carrier Joint Intelligent Shutdown (eNodeB) and inter-carrier power sharing can take effect, but the energy saving gains may decrease.
Extreme power sharing	CellDynPowerSharing.DynamicPowerSharingSwitch	<i>Dynamic Power Sharing Between LTE Carriers in eRAN Feature Documentation</i>	Cells that have entered the GUL joint carrier shutdown state do not participate in extreme power sharing.

Function Name	Function Switch	Reference	Description
LTE and NR Flash Dynamic Power Sharing	CellDynPowerSharing.DynamicPowerSharingSwitch	<i>LTE and NR Power Allocation in eRAN Feature Documentation</i>	Cells that have entered the GUL joint carrier shutdown state do not participate in LTE and NR Flash Dynamic Power Sharing.
CMAS	CellAlgoSwitch.CmasSwitch	<i>CMAS in eRAN Feature Documentation</i>	The following applies when LTE cells enter GUL joint carrier shutdown: • With PWS message-triggered exit of cells from energy saving disabled, CMAS warning notifications cannot be properly sent in LTE cells since carriers in those cells are shut down. However, UEs can still receive CMAS warning notifications from other carriers. • With PWS message-triggered exit of cells from energy saving enabled, cells exit the dormancy state upon receipt of a PWS message.

Function Name	Function Switch	Reference	Description
ETWS	None	<i>ETWS in eRAN Feature Documentation</i>	The following applies when LTE cells enter GUL joint carrier shutdown: • With PWS message-triggered exit of cells from energy saving disabled, ETWS warning notifications cannot be properly sent in LTE cells since carriers in those cells are shut down. However, UEs can still receive ETWS warning notifications from other carriers. • With PWS message-triggered exit of cells from energy saving enabled, cells exit the dormancy state upon receipt of a PWS message.
UMTS	NCR cells	ULOCCELL.NCRCELL	After the NCR cells served by the DeNB enter the GUL joint carrier shutdown state, UMTS services within the coverage of the RuralCell site are affected.

- Areas covered by GL networks

Function Name	Function Switch	Reference	Description
Dynamic Power Sharing Between LTE Carriers	CellDynPowerSharing.DynamicPowerSharingSwitch	<i>Dynamic Power Sharing Between LTE Carriers in eRAN Feature Documentation</i>	Cells that have entered the GUL joint carrier shutdown state do not participate in Dynamic Power Sharing Between LTE Carriers.

Function Name	Function Switch	Reference	Description
Extreme power sharing	CellDynPowerSharing.DynamicPowerSharingSwitch	<i>Dynamic Power Sharing Between LTE Carriers in eRAN Feature Documentation</i>	Cells that have entered the GUL joint carrier shutdown state do not participate in extreme power sharing.
LTE and NR Flash Dynamic Power Sharing	CellDynPowerSharing.DynamicPowerSharingSwitch	<i>LTE and NR Power Allocation in eRAN Feature Documentation</i>	Cells that have entered the GUL joint carrier shutdown state do not participate in LTE and NR Flash Dynamic Power Sharing.
CMAS	CellAlgoSwitch.CmasSwitch	<i>CMAS in eRAN Feature Documentation</i>	The following applies when LTE cells enter GUL joint carrier shutdown: • With PWS message-triggered exit of cells from energy saving disabled, CMAS warning notifications cannot be properly sent in LTE cells since carriers in those cells are shut down. However, UEs can still receive CMAS warning notifications from other carriers. • With PWS message-triggered exit of cells from energy saving enabled, cells exit the dormancy state upon receipt of a PWS message.

Function Name	Function Switch	Reference	Description
ETWS	None	<i>ETWS in eRAN Feature Documentation</i>	The following applies when LTE cells enter GUL joint carrier shutdown: • With PWS message-triggered exit of cells from energy saving disabled, ETWS warning notifications cannot be properly sent in LTE cells since carriers in those cells are shut down. However, UEs can still receive ETWS warning notifications from other carriers. • With PWS message-triggered exit of cells from energy saving enabled, cells exit the dormancy state upon receipt of a PWS message.

6.3.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- Areas covered by GL networks: 3900 and 5900 series base stations (macro base stations)
- Areas covered by GU, UL, or GUL networks: 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

Boards

No requirements

RF Modules

No requirements

6.3.3.4 Networking

- Areas covered by GU networks

This feature requires information exchange between the GSM and UMTS networks. If the base station controllers are a BSC and RNC, operators must add the physical connections for the Iur-g interface between the BSC and the RNC. For details on how to configure and enable the Iur-g interface, see section "Configuring the Iur-g Interface (IP Transmission Mode)" in *BSC6900 GU Initial Configuration Guide* in *BSC6900 GU Product Documentation*.

Operators must also determine the GSM cells that cover the same area as the UMTS cell, configure these GSM cells as the inter-RAT neighboring cells of the UMTS cell, and set **U2GNCELL.NIRATOverLap** for each GSM cell to **TRUE**.

- Areas covered by GL networks or UL networks
In these areas, the eNodeB and the BSC or RNC must be able to exchange cell load information with each other through RIM procedures.
 - If RIM procedures are performed between the eNodeB/BSC/RNC and the CN, the MME and SGSN must support RIM procedures complying with 3GPP Release 9 or later.In areas covered by GL or UL networks, operators must also determine the GSM or UMTS cells that cover the same area as an LTE cell, configure these GSM or UMTS cells as the inter-RAT co-coverage neighboring cells for the LTE cell by setting the **GeranNcell.OverlapInd** or **UtranNCell.OverlapInd** parameter.
- Areas covered by GUL networks
All requirements for the preceding scenarios must be met.

6.3.3.5 Others

- UE
UEs support all the RATs of a multimode network.
- Core network
If RIM procedures are performed between the controller and CN, the MME and SGSN must support RIM procedures when this feature is enabled in areas covered by GL, UL, or GUL networks.

6.3.4 Operation and Maintenance

6.3.4.1 When to Use

To achieve superior power-saving performance without affecting network performance, the Multi-RAT Carrier Joint Shutdown feature is recommended in the following scenarios:

- In areas covered by GU, UL, GL, GUL networks, the GSM or UMTS network can provide continuous basic coverage.
- The traffic volume has a typical tidal effect or the traffic volume significantly changes in a day, for example, in central business districts (CBDs) and shopping malls.
- The penetration rate of UMTS and LTE services is low, for example, in suburban and rural areas.
- Cells of different modes cover the same area and there are no blind spots and coverage holes in the network providing basic coverage.

The Multi-RAT Carrier Joint Shutdown feature is not recommended in the following scenarios:

- There are LTE-only or UMTS-only UEs. After LTE or UMTS network is shut down, LTE-only or UMTS-only UEs cannot be admitted.
- Dual-mode carriers share the same power amplifier (PA). For example, in scenarios where GSM and UMTS carriers share the same PA, the PA cannot be shut down after the UMTS cell is shut down because GSM carriers must use the PA. Therefore, this feature can only produce few gains.

Before feature deployment, collect the following information:

- Site configuration
In areas covered by GU networks, operators must also consider the impact of UMTS traffic transfer on GSM network power consumption. The traffic transfer increases the traffic load on the GSM network. If the GSM base station uses double radio frequency units or multicarrier dual-channel frequency units, the other channel may be enabled, reducing the power-saving gain. If the timeslot usage on the GSM network is low at the time the UMTS cell is shut down, the impact on the power-saving gain can be avoided. Operators can obtain the site configurations of the GSM and UMTS base stations to perform load parameter optimization, achieving superior power-saving performance. In areas covered by UL, GL, or GUL networks, the preceding information is not required.
- Network traffic changes
Operators must obtain this information to determine the cell dynamic shutdown periods. The cell dynamic shutdown periods should be a time when the traffic is low and stable to avoid unnecessary impact on UEs.
- Resource usage
Operators must obtain this information and base station configuration to determine the network power consumption. Based on the network power consumption, operators can estimate the power-saving gain produced by this feature and reconfigure load threshold and cell dynamic shutdown period parameters.
- UE types
Operators must check whether there are LTE-only or UMTS-only UEs on the live network.
If there are LTE-only UEs in areas covered by GL or UL networks, this feature is not recommended.
If there are UMTS-only UEs in areas covered by GU networks, this feature is not recommended.

6.3.4.2 Data Configuration

6.3.4.2.1 Data Preparation

Areas Covered by GU Networks

Table 6-11 Parameter configurations on the RNC side

Parameter Name	Parameter ID	Setting Notes
Cell Dynamic ShutDown Total User Number Threshold	<i>TotalUserNumThd</i>	Set this parameter based on the actual network conditions. The value 2 is recommended. A larger value of this parameter leads to a larger number of users being handed over to the GSM network, increasing the risk of handover failures and affecting user experience. If the inter-RAT handover success rate decreases in the cell dynamic shutdown periods after this feature is enabled, decrease the value of this parameter.
Considering User on Com Channel for Cell Shut	<i>DynShutDownCchUserSwitch</i>	Set this parameter to ON .

Parameter Name	Parameter ID	Setting Notes
Total Trans Power Threshold for Cell Shutdown	<i>IRATShutdownDLLoadThd</i>	This parameter is used to determine whether a UMTS cell is lightly loaded. Set this parameter based on the actual network conditions. The value 30% is recommended. This parameter is used to compare the service volume and power consumption in both GSM and UMTS cells before and after feature deployment within the cell dynamic shutdown period. If the traffic volume and power consumption of the GSM network significantly increase after the UMTS cell is shut down and if the increase affects the overall power-saving gain, decrease the value of this parameter to reduce the load transferred to the GSM network upon UMTS cell shutdown. Operators are advised not to set this parameter to a large value to minimize the impact of this feature.

Parameter Name	Parameter ID	Setting Notes
Total Receive Power Threshold for Cell Shutdown	<i>IRATShutdownULLoadThd</i>	This parameter is used to determine whether a UMTS cell is lightly loaded. Set this parameter based on the actual network conditions. The value 10% is recommended. This parameter is used to compare the service volume and power consumption in both GSM and UMTS cells before and after feature deployment within the cell dynamic shutdown period. If the traffic volume and power consumption of the GSM network significantly increase after the UMTS cell is shut down and if the increase affects the overall power-saving gain, decrease the value of this parameter to reduce the load transferred to the GSM network upon UMTS cell shutdown. Operators are advised not to set this parameter to a large value to minimize the impact of this feature.

Parameter Name	Parameter ID	Setting Notes
Load Thre for Neigh GSM Cell for Cell Shutdown	<i>IRATshutdownGSM-LoadThd</i>	This parameter is used to determine whether a GSM cell is lightly loaded. Set this parameter based on the actual network conditions. The value 25% is recommended. This parameter is used to compare the service volume and power consumption in both GSM and UMTS cells before and after feature deployment within the cell dynamic shutdown period. If the traffic volume and power consumption of the GSM network significantly increase after the UMTS cell is shut down and if the increase affects the overall power-saving gain, decrease the value of this parameter. If the traffic volume of the UMTS cell is low but the UMTS cell is not shut down, the possible reason is that the traffic volume of the GSM cell is relatively high. In this case, you can increase the value of this parameter to achieve a power-saving gain.
Load Hyst Thre for Neigh GSM Cell for Cell Startup	<i>IRATshutdownGSM-LoadHyst</i>	Set this parameter based on the actual network conditions. The value 20% is recommended.
High-Priority Scheduling Indicator	<i>HighPriSPI</i>	Set this parameter based on the actual network conditions so that this feature does not affect the experience of real-time service users or high-priority users.

Parameter Name	Parameter ID	Setting Notes
Period for Restart Inter-RAT Inte Shut UMTS Cell	<i>IRATSwitchOnHyst-TimeLen</i>	Set this parameter based on the actual network conditions. This parameter avoids the UMTS cell restart caused by the drastic increase in the GSM network load after UEs in the idle state are reselected to a GSM cell. The value 60s is recommended.
Delay in Cell Shutdown	<i>IRATshutdownAdj-Time</i>	Set this parameter based on the actual network conditions. This parameter avoids unnecessary UMTS cell shutdowns or restarts caused by GSM or UMTS cell load changes. The value 5 minutes is recommended. If the values of VS.DynClose.CellDel and VS.DynOpen.CellSetup are too large after this feature is enabled, increase the value of this parameter.
Protect Timer Length for Cell Dynamic ShutDown	<i>DynCellShutdown-ProtectTimerlen</i>	Set this parameter based on the traffic changes.
Overlap Flag	<i>NIRATOverLap</i>	Set this parameter for a GSM cell to TRUE when the GSM cell covers the same area as the UMTS cell.
Switch for Inter-RAT Common Measurement	<i>IRATCommonMeasSwitch</i>	Turn on this switch before feature deployment to enable the RNC to obtain GSM cell load information.
First Cell Dynamic ShutDown Interval Start Time	<i>StartTime1</i>	Set this parameter based on the light-traffic hours.
First Cell Dynamic ShutDown Interval End Time	<i>EndTime1</i>	Set this parameter based on the light-traffic hours.
Second Cell Dynamic ShutDown Interval Start Time	<i>StartTime2</i>	Set this parameter based on the light-traffic hours.

Parameter Name	Parameter ID	Setting Notes
Second Cell Dynamic ShutDown Interval End Time	<i>EndTime2</i>	Set this parameter based on the light-traffic hours.
Third Cell Dynamic ShutDown Interval Start Time	<i>StartTime3</i>	Set this parameter based on the light-traffic hours.
Third Cell Dynamic ShutDown Interval End Time	<i>EndTime3</i>	Set this parameter based on the light-traffic hours.
Cell Dynamic ShutDown Switch	<i>DynShutdownSwitch</i>	Set this parameter based on the actual network conditions.
Cell Dynamic ShutDown Type	<i>DynShutdownType</i>	Set this parameter to IRATShutDown for the cell where this feature is required.

Table 6-12 Parameter configurations on the BSC side

Parameter Name	Parameter ID	Setting Notes
High User Priority	<i>IRATHighPriSwitch</i>	Set this parameter based on the UEs on the live network and operators' policy. Turn on this switch if the experience of high-priority users must be ensured. The value OFF is recommended.
Inter-RAT Cell Dynamic Shutdown	<i>IRATShutdownSwitch</i>	Set this parameter based on the actual network conditions.
Inter-RAT Cell TBF Determination	<i>IRATTBFSwitch</i>	If operators must ensure the experience of users performing PS services, set this parameter to ON . The value OFF is recommended.
Inter-RAT Cell TBF Connection Timer	<i>IRATSwitchonTBF-ContTimeThd</i>	Set this parameter based on the duration of the PS services performed by multimode terminals. This parameter avoids the UMTS cell restart caused by multimode terminals' short-time service requirements. The value 60s is recommended.

Areas Covered by UL Networks

Table 6-13 Parameter configurations on the eNodeB side

Parameter Name	Parameter ID	Setting Notes
Inter-RAT cell shutdown switch	<i>ForceShutdownSwitch</i>	Set this parameter to ON .
Shut Down Type	<i>ShutDownType</i>	Set this parameter to BY_UTRAN_WITH_LOAD .
Start time	<i>StartTime</i>	If power saving is a major concern, set these two parameters to values that are far apart. If service stability is a major concern, set the cell shutdown period based on the traffic characteristics (peak hours and off-peak hours). This prevents a cell from frequently entering and exiting Multi-RAT Carrier Joint Shutdown mode.
Stop time	<i>StopTime</i>	
Exit Evaluation Period	<i>ExitJudgePeriod</i>	The default value is recommended.
UE number threshold	<i>BearNumThd</i>	A larger value of the parameter results in a longer cell shutdown period, which potentially saves more power. If service stability is a major concern, set this parameter based on the traffic characteristics. This prevents drastic increase in the load of a neighboring co-coverage UTRAN cell after the LTE cell enters the Multi-RAT Carrier Joint Shutdown state.

Parameter Name	Parameter ID	Setting Notes
Downlink PRB threshold	<i>DLPrbThd</i>	If the values of these two parameters are large, a cell easily enters Multi-RAT Carrier Joint Shutdown mode, potentially saving more power. If the values of these two parameters are small, a cell does not easily enter Multi-RAT Carrier Joint Shutdown mode. In this case, less power can be saved. If service stability is a major concern, set these two parameters based on the traffic characteristics. This prevents drastic increase in the load of a neighboring co-coverage UTRAN cell after the LTE cell enters the Multi-RAT Carrier Joint Shutdown state.
Uplink PRB threshold	<i>ULPrbThd</i>	
UTRAN Cell Downlink Load Threshold	<i>UtranCellDLLoadThd</i>	If the values of these two parameters are large, a cell easily enters Multi-RAT Carrier Joint Shutdown mode, potentially saving more power. If the values of these two parameters are small, a cell does not easily enter Multi-RAT Carrier Joint Shutdown mode. In this case, less power can be saved. If service stability is a major concern, set these two parameters based on the traffic characteristics. This prevents an LTE cell intelligent restart caused by a drastic increase in the load of a co-coverage neighboring UTRAN cell after the LTE cell enters Multi-RAT Carrier Joint Shutdown mode.
UTRAN Cell Uplink Load Threshold	<i>UtranCellULLoadThd</i>	

Parameter Name	Parameter ID	Setting Notes
UTRAN Cell Downlink Load Offset	<i>UtranCellDLLoadOffset</i>	Set these parameters to values less than or equal to the uplink and downlink LDR thresholds for the co-coverage neighboring UTRAN cells minus the downlink load threshold for the neighboring UTRAN cells. If service stability is a major concern, set these two parameters based on the traffic characteristics. This prevents too easy or too difficult triggering of an LTE cell intelligent restart after the cell enters Multi-RAT Carrier Joint Shutdown mode.
UTRAN Cell Uplink Load Offset	<i>UtranCellULLoadOffset</i>	
High Priority ARP Threshold	<i>HighPriArpThd</i>	Set this parameter based on the priority of services that can be affected by Multi-RAT Carrier Joint Shutdown. The value 5 is recommended. A larger value of this parameter results in a higher priority threshold. If the value of this parameter is small, a cell easily enters Multi-RAT Carrier Joint Shutdown mode, potentially saving more power.
Overlap Indicator	<i>OverlapInd</i>	Set this parameter based on the relationships between coverage areas of cells covered by UL networks. Set this parameter to YES for the co-coverage neighboring UTRAN cells of the cell. It is recommended that this parameter be set to YES for at most five co-coverage neighboring UTRAN cells of an LTE cell. For details, contact Huawei technical support.
RIM Coding Policy	<i>RimCodingPolicy</i>	Set this parameter to StandardCoding .

Table 6-14 Parameter configurations on the RNC side

Parameter Name	Parameter ID	Setting Notes
Process Switch	<i>PROCESSSWITCH</i>	Select the INTERRAT_LOAD_REPORT_FOR_LTE_SWITCH option of this parameter.
Process Control Switch 5	<i>PROCESSSWITCH5</i>	Select the INTERRAT_LOAD_SINGLE_REPORT_FOR_LTE_SWITCH option of this parameter.

Areas Covered by GL Networks

Table 6-15 Parameter configurations on the eNodeB side

Parameter Name	Parameter ID	Setting Notes
Inter-RAT cell shutdown switch	<i>ForceShutdownSwitch</i>	Set this parameter to ON .
Shut Down Type	<i>ShutDownType</i>	Set this parameter to BY_GERAN_WITH_LOAD .
Start time	<i>StartTime</i>	A longer period results in a longer duration in which a cell can stay in Multi-RAT Carrier Joint Shutdown mode, potentially saving more power. A shorter period results in a shorter duration in which a cell can stay in Multi-RAT Carrier Joint Shutdown mode. In this case, less power can be saved. If power saving is a major concern, set these two parameters to values that are far apart. If service stability is a major concern, set the cell shutdown period based on the traffic characteristics (peak hours and off-peak hours). This prevents a cell from frequently entering and exiting Intelligent Power-Off of Carriers in the Same Coverage mode.
Stop time	<i>StopTime</i>	

Parameter Name	Parameter ID	Setting Notes
Exit Evaluation Period	<i>ExitJudgePeriod</i>	The default value is recommended.
GERAN Cell Load Threshold	<i>GeranCellLoadThd</i>	Set this parameter based on load during light-traffic hours of the co-coverage neighboring GERAN cells. If service stability is a major concern, set this parameter based on the traffic characteristics. This prevents an LTE cell intelligent restart caused by a drastic increase in the load of a co-coverage neighboring GERAN cell after the LTE cell enters Multi-RAT Carrier Joint Shutdown mode.
GERAN Cell Load Offset	<i>GeranCellLoadOffset</i>	Set this parameter to a value less than or equal to the TCH traffic busy threshold for the co-coverage neighboring GERAN cells minus the load threshold for the neighboring GERAN cells. If service stability is a major concern, set this parameter based on the traffic characteristics. This prevents too easy or too difficult triggering of an LTE cell intelligent restart after the cell enters Multi-RAT Carrier Joint Shutdown mode.
Overlap Indicator	<i>OverlapInd</i>	Set this parameter based on the relationships between coverage areas of cells covered by GL networks. Set this parameter to YES for the co-coverage neighboring GERAN cells of the cell. It is recommended that this parameter be set to YES for at most five co-coverage neighboring GERAN cells of an LTE cell. The description in areas covered by UL networks can be used as a reference. For details, contact Huawei technical support.

Parameter Name	Parameter ID	Setting Notes
RIM Coding Policy	<i>RimCodingPolicy</i>	Set this parameter to StandardCoding .

Table 6-16 Parameter configurations on the BSC side

Parameter Name	Parameter ID	Setting Notes
RIM Support	<i>RIMSUP</i>	Set this parameter to YES .

Areas Covered by GUL Networks

In areas covered by GUL networks, besides the parameter setting notes for the preceding areas, you must pay attention to the following:

- **InterRatCellShutdown.ShutDownType** on the eNodeB side: Set this parameter to **BY_UTRAN_AND_GERAN_WITH_LOAD**.
- The LTE cell shutdown period specified by **InterRatCellShutdown.StartTime** and **InterRatCellShutdown.StopTime** must include at least one of the following cell shutdown periods for the co-coverage neighboring UMTS cells: period specified by **UCELLDYNSHUTDOWN.StartTime1** and **UCELLDYNSHUTDOWN.EndTime1**, period specified by **UCELLDYNSHUTDOWN.StartTime2** and **UCELLDYNSHUTDOWN.EndTime2**, and period specified by **UCELLDYNSHUTDOWN.StartTime3** and **UCELLDYNSHUTDOWN.EndTime3**.

6.3.4.2.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [6.3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples for Areas Covered by GU Networks

RNC side: (For details about the MML commands for configuring and enabling the Iur-g interface, see the initial configuration guide of the corresponding base station controller.)

```
//Configuring co-coverage neighboring GSM cells for a UMTS cell. Ensure that the NIRATOverLap
parameter is set to TRUE for these GSM cells.
ADD U2GNCELL: RncId=1, CellId=1, GSMCellIndex=0, CIOOffset=0, Qoffset1sn=0, Qrxlevmin=-50,
TpenaltyHcsReselect=D0, TempOffset1=D3, BlindHoFlag=FALSE, NPrioFlag=FALSE, NIRATOverLap=TRUE;
MOD U2GNCELL: RncId=1, CellId=1, GSMCellIndex=0, CIOOffset=0, NIRATOverLap=TRUE;
//Enabling the RNC-level cell dynamic shutdown function on the RNC side
```

```
SET URNCELLSHUTDOWNPARA: DynCellShutDownSwitch=ON, DynCellOpenJudgeTimerlen=20,
DynCellShutdownProtectTimerlen=180, IRATSwitchOnHystTimeLen=60, IRATShutdownAdjTime=5;
//Enabling the cell-level inter-RAT cell joint shutdown function on the RNC side
ADD UCELLDYNSHUTDOWN: CellId=1, DynShutdownSwitch=ON_1, StartTime=01&00, EndTime=06&00,
DynShutDownType=IRATSHUTDOWN, TotalUserNumThd=2, DynShutDownCchUserSwitch=ON,
IRATShutdownDLLoadThd=30, IRATShutdownULLoadThd=10, IRATShutdownGSMLoadThd=25,
IRATShutdownGSMLoadHyst=20;
//Enabling the common measurement over the lur-g interface
SET UMBSCCRRM: MbscServiceDiffLdbSwitch=OFF, Mbsc3G2GLdBLCsDeltaThrd=10,
CommonMeasurementPeriod=3, Mbsc2GLoadAdjustCoeff=1, IRATCommonMeasSwitch=On;
//Enabling the Intelligently Out of Service feature for the NodeB
SET NODEBSMTHPWRSW: SMTHPWRSWITCH=OPEN;
SET NODEBSMTHPWRPARA: STEP=10, PERIOD=1;
```

BSC side:

```
//Enabling the function of information exchange on the lur-g interface
SET GCELLOTHEXT: IDTYPE=BYID, CELLID=0, IURGINFOCTRL=YES;
//Enabling the cell-level inter-RAT cell joint shutdown function on the BSC side
SET GCELLSOFT: IDTYPE=BYID, CELLID=0, IRATHighPriSwitch=OFF, IRATShutdownSwitch=ON,
IRATTBFSwitch=OFF;
```

Activation Command Examples for Areas Covered by UL Networks

eNodeB side:

```
//Configuring co-coverage neighboring UTRAN cells for an LTE cell. Ensure that external UTRAN cells and
neighboring UTRAN frequencies have been configured.
ADD UTRANNCCELL: LocalCellId=0, Mcc="460", Mnc="20", RncId=1, CellId=123, OverlapInd=YES;
MOD UTRANNCCELL: LocalCellId=0, Mcc="460", Mnc="20", RncId=1, CellId=123, OverlapInd=YES;
//Enabling cell-level Multi-RAT Carrier Joint Shutdown on the eNodeB side
MOD INTERRATCELLSHUTDOWN: LocalCellId=0, ForceShutdownSwitch=ON, StartTime=03&00&00,
StopTime=05&00&00, BearNumThd=20, DIPrbThd=15, UIPrbThd=15,
ShutDownType=BY_UTRAN_WITH_LOAD, UtranCellDLLoadThd=30, UtranCellULLoadThd=15,
UtranCellDLLoadOffset=40, UtranCellULLoadOffset=40, HighPriArpThd=5, ExitJudgePeriod=4;
//Enabling the eNodeB to use the standard RIM message coding scheme
MOD GLOBALPROCSWITCH: RimCodingPolicy=StandardCoding;
```

RNC side:

```
//Enabling transmission of load information through RIM procedures (single report) on the RNC side
SET URRCTRLSWITCH: PROCESSSWITCH5=INTERRAT_LOAD_SINGLE_RPT_FOR_LTE_SWITCH-1;
```

Activation Command Examples for Areas Covered by GL Networks

eNodeB side:

```
//Configuring co-coverage neighboring GERAN cells for an LTE cell. Ensure that external GERAN cells,
neighboring GERAN ARFCNs, and neighboring GERAN BCCH ARFCNs have been configured.
ADD GERANNCCELL: LocalCellId=0, Mcc="460", Mnc="20", Lac=12, GeranCellId=15, OverlapInd=YES;
MOD GERANNCCELL: LocalCellId=0, Mcc="460", Mnc="20", Lac=12, GeranCellId=15, OverlapInd=YES;
//Enabling cell-level Multi-RAT Carrier Joint Shutdown on the eNodeB side
MOD INTERRATCELLSHUTDOWN: LocalCellId=0, ForceShutdownSwitch=ON, StartTime=03&00&00,
StopTime=05&00&00, ShutDownType=BY_GERAN_WITH_LOAD, GeranCellLoadThd=25,
GeranCellLoadOffset=20, ExitJudgePeriod=4;
//Enabling the eNodeB to use the standard RIM message coding scheme
MOD GLOBALPROCSWITCH: RimCodingPolicy=StandardCoding;
```

BSC side:

```
//Enabling the RIM procedure on the BSC side
MOD NSE: NSEI=9, RIMSUP=YES;
```

Activation Command Examples for Areas Covered by GUL Networks

RNC side:

```
//Configuring co-coverage neighboring GSM cells for a UMTS cell
ADD U2GNCELL: RncId=1, CellId=1, GSMCellIndex=0, CIOffset=0, Qoffset1sn=0, Qrxlevmin=-50,
TpenaltyHcsReselect=D0, TempOffset1=D3, BlindHoFlag=FALSE, NPrioFlag=FALSE, NIRATOverLap=TRUE;
MOD U2GNCELL: RncId=1, CellId=1, GSMCellIndex=0, CIOffset=0, NIRATOverLap=TRUE;
//Enabling the RNC-level cell dynamic shutdown function on the RNC side
SET URNCELLSHUTDOWNPARAM: DynCellShutDownSwitch=ON, DynCellOpenJudgeTimerlen=20,
DynCellShutdownProtectTimerlen=180, IRATSwitchOnHystTimeLen=60, IRATShutdownAdjTime=5;
//Enabling cell-level Multi-RAT Carrier Joint Shutdown on the RNC side
ADD UCELLDYNSHUTDOWN: CellId=1, DynShutdownSwitch=ON_1, StartTime1=01&00, EndTime1=06&00,
DynShutDownType=IRATSHUTDOWN, TotalUserNumThd=2, DynShutDownCchUserSwitch=ON,
IRATShutdownDLLoadThd=30, IRATShutdownULLoadThd=10, IRATShutdownGSMLoadThd=25,
IRATShutdownGSMLoadHyst=20;
//Enabling the common measurement over the lur-g interface
SET UMBSCRRM: MbscServiceDiffLdbSwitch=OFF, Mbsc3G2GLdBLCsDeltaThrd=10,
CommonMeasurementPeriod=3, Mbsc2GLoadAdjustCoeff=1, IRATCommonMeasSwitch=ON;
//Enabling transmission of load information through RIM procedures (single report) on the RNC side
SET URRCTRLSWITCH: PROCESSSWITCH5=INTERRAT_LOAD_SINGLE_RPT_FOR_LTE_SWITCH-1;
//Enabling the Intelligently Out of Service feature for the NodeB
SET NODEBSMTHPWRSW: SMTHPWRSWITCH=OPEN;
SET NODEBSMTHPWRPARAM: STEP=10, PERIOD=1;
```

BSC side:

```
//Enabling the function of information exchange on the lur-g interface
SET GCELLOTEXT: IDTYPE=BYID, CELLID=0, IURGINFOCTRL=YES;
//Enabling the cell-level Inter-RAT Carrier Joint Shutdown feature on the BSC side
SET GCELLSOFT: IDTYPE=BYID, CELLID=0, IRATHighPriSwitch=OFF, IRATShutdownSwitch=ON,
IRATTBFSwitch=OFF;
//Enabling the RIM procedure on the BSC side
MOD NSE: NSEI=9, RIMSUP=YES;
```

eNodeB side:

```
//Configuring co-coverage neighboring UTRAN and GERAN cells for an LTE cell
ADD UTRANNCCELL: LocalCellId=0, Mcc="460", Mnc="20", RncId=1, CellId=1, OverlapInd=YES;
MOD UTRANNCCELL: LocalCellId=0, Mcc="460", Mnc="20", RncId=1, CellId=1, OverlapInd=YES;
ADD GERANNCCELL: LocalCellId=0, Mcc="460", Mnc="20", Lac=12, GeranCellId=15, OverlapInd=YES;
MOD GERANNCCELL: LocalCellId=0, Mcc="460", Mnc="20", Lac=12, GeranCellId=15, OverlapInd=YES;
//Enabling cell-level Multi-RAT Carrier Joint Shutdown on the eNodeB side
MOD INTERRATCELLSHUTDOWN: LocalCellId=0, ForceShutdownSwitch=ON, StartTime=00&00&00,
StopTime=06&00&00, BearNumThd=20, DIPrbThd=15, UIPrbThd=15,
ShutDownType=BY_UTRAN_AND_GERAN_WITH_LOAD, UtranCellDLLoadThd=30, UtranCellULLoadThd=15,
UtranCellDLLoadOffset=40, UtranCellULLoadOffset=40, GeranCellLoadThd=25, GeranCellLoadOffset=20,
HighPriArpThd=5;
//Enabling the eNodeB to use the standard RIM message coding scheme
MOD GLOBALPROCSWITCH: RimCodingPolicy=StandardCoding;
```

Deactivation Command Examples for Areas Covered by GU Networks

```
//Disabling the cell-level inter-RAT cell joint shutdown function on the RNC side
MOD UCELLDYNSHUTDOWN: CellId=1, DynShutdownSwitch=OFF;
//Disabling the cell-level inter-RAT cell joint shutdown function on the BSC side
SET GCELLSOFT: IDTYPE=BYID, CELLID=0, IRATShutdownSwitch=OFF;
```

Deactivation Command Examples for Areas Covered by GL Networks

```
//Disabling the Multi-RAT Carrier Joint Shutdown feature on the eNodeB side
MOD INTERRATCELLSHUTDOWN: LocalCellId=0, ForceShutdownSwitch=OFF;
```

Deactivation Command Examples for Areas Covered by UL Networks

```
//Disabling the Multi-RAT Carrier Joint Shutdown feature on the eNodeB side
MOD INTERRATCELLSHUTDOWN: LocalCellId=0, ForceShutdownSwitch=OFF;
```

Deactivation Command Examples for Areas Covered by GUL Networks

```
//Disabling the Multi-RAT Carrier Joint Shutdown feature on the eNodeB side
MOD INTERRATCELLSHUTDOWN: LocalCellId=0, ForceShutdownSwitch=OFF;
```



```
//Disabling the cell-level inter-RAT cell joint shutdown function on the RNC side  
MOD UCELLDYNSHUTDOWN: CellId=1, DynShutdownSwitch=OFF;  
//Disabling the cell-level inter-RAT cell joint shutdown function on the BSC side  
SET GCELLSOFT: IDTYPE=BYID, CELLID=0, IRATShutdownSwitch=OFF;
```

6.3.4.2.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.3.4.3 Activation Verification

Check whether this feature has taken effect by using MML commands, observing performance counters, or using the MAE-Access.

Using MML Commands

In areas covered by GU or GUL networks, perform the following steps:

- Step 1** Run the **DSP UCELL** command on the RNC side during a shutdown period. For example, if the cell ID is 1, run the following command: **DSP UCELL: DSPT=BYCELL, CellId=1**.
- Step 2** Query the cell status. If the value of **State explanation** is **Cell is setup but dynamic shut down**, it indicates that this feature has taken effect.

----End

In areas covered by GL, UL, or GUL networks, perform the following steps:

- Step 1** Run the **DSP CELL** command on the eNodeB during a shutdown period. For example, if the cell ID is 0, run the following command: **DSP CELL: LocalCellId=0**.
- Step 2** Query the cell status. If the value of **Cell power save state** is **Inter-RAT Cell Shutdown Mode**, it indicates that this feature has taken effect.

----End

Observing Counters

In areas covered by GU or GUL networks, check the values of the counters listed in the following table for a UMTS cell enabled with this feature on the MAE-Access.

Counter ID	Counter Name	Counter Description
67204134	VS.Cell.DynShutDown.Time	Duration of Cell Dynamic Shutdown Due to Automatic Shutdown for Cell
73393802	VS.DynClose.CellDel	Number of Cell Deletions Due to Dynamic Cell Shutdown

Counter ID	Counter Name	Counter Description
73393803	VS.DynOpen.CellSetup	Number of Cell Setup Due to Dynamic Cell Shutdown

In areas covered by GL, UL, or GUL networks, check the values of the counters listed in following table for an LTE cell enabled with this feature on the MAE-Access.

Counter ID	Counter Name	Counter Description
1526728151	L.ChMeas.DFEE.CarrierOFF.Dur	Duration of intelligent power-off of carriers
1526728493	L.Cell.Unavail.Dur.EnergySaving	Duration of cell unavailability due to energy saving

On the MAE-Access, check the values of the counters listed in the following table for the base stations to which cells of different modes belong.

Counter ID	Counter Name	Counter Description
1593835638	VS.EnergyCons.BTS.Measuring.LTE	Power consumption measured by the power system of the LTE
1593835637	VS.EnergyCons.BTS.Adding.LTE	The sum of power consumption of all boards in the LTE
1593835959	VS.EnergyCons.BTS.Adding.Service.LTE	Sum of power consumption of all boards (LTE services)
1593835629	VS.EnergyCons.BTS.Measuring.UMTS	Power consumption measured by the power system of the NodeB
1593835628	VS.EnergyCons.BTS.Adding.UMTS	The sum of power consumption of all boards in the NodeB
1593835958	VS.EnergyCons.BTS.Adding.Service.UMTS	Sum of power consumption of all boards (UMTS services)
1282449257	SITE.EnerConsump.MEASURING	BTS Power Consumption by Measuring Dynamic Power Consumption. This counter applies to base stations other than eGBTs.

Counter ID	Counter Name	Counter Description
1282449256	SITE.EnerConsump.ADDING	BTS Power Consumption by Adding Static Power Consumption. This counter applies to base stations other than eGBTs.
1593835640	VS.EnergyCons.BTS.Measuring.GSM	Power consumption measured by the power system of the GSM. This counter applies to eGBTs.
1593835639	VS.EnergyCons.BTS.Adding.GSM	The sum of power consumption of all boards in the GSM. This counter applies to eGBTs.
1593835957	VS.EnergyCons.BTS.Adding.Service.GSM	Sum of power consumption of all boards (GSM services)

MAE-Access

Monitor the base station to which the cell enabled with this feature belongs on the MAE-Access and compare its traffic volume and power consumption before and after the feature activation. For details, see "Base Station Energy Conservation Management" in *MAE Product Documentation*.

6.3.4.4 Network Monitoring

Performance counters on the MAE-Access are used to monitor and evaluate the performance of this feature. After activating this feature, check the following counters to evaluate gains produced by this feature:

LTE-related Counters

- eNodeB power consumption
It is indicated by the **VS.EnergyCons.BTS.Measuring.LTE**, **VS.EnergyCons.BTS.Adding.Service.LTE**, and **VS.EnergyCons.BTS.Adding.LTE** counter values.
Evaluate the feature gains in scenarios using the following traffic model: the traffic volume is idle for 8 hours, medium for 12 hours, and high for 4 hours in a day. If the eNodeB is disabled for 8 hours every day, the power-saving gain per day may increase by 5% to 8%.
The power-saving gain varies depending on the traffic model and RF module configurations.
- Duration of intelligent power-off of carriers
It is indicated by the **L.ChMeas.DFEE.CarrierOFF.Dur** counter value.
- Duration of cell unavailability due to energy saving
It is indicated by the **L.Cell.Unavail.Dur.EnergySaving** counter value.

UMTS-related Counters

- NodeB power consumption

It is indicated by the **VS.EnergyCons.BTS.Measuring.UMTS**, **VS.EnergyCons.BTS.Adding.Service.UMTS**, or **VS.EnergyCons.BTS.Adding.UMTS** counter value.

Evaluate the feature gains in scenarios using the following traffic model: the traffic volume is idle for 8 hours, medium for 12 hours, and high for 4 hours in a day. If the NodeB is disabled for 8 hours every day, the power-saving gain per day may increase by 6% to 10%.

The power-saving gain varies depending on the traffic model and RF module configurations.

- Duration in which a cell is unavailable due to dynamic cell shutdown

Use the **VS.Cell.DynShutDown.Time** counter to observe the duration of cell unavailability due to dynamic cell shutdown.

- Number of cell deletions and establishments caused by dynamic cell shutdown

Use the **VS.DynClose.CellDel** and **VS.DynOpen.CellSetup** counters to observe the number of cell deletions and establishments caused by dynamic cell shutdown.

6.3.4.5 Possible Issues

Fault Description (1)

In areas covered by GU or GUL networks, a UMTS cell cannot enter Multi-RAT Carrier Joint Shutdown mode.

Troubleshooting Procedure (1)

- Run the **LST U2GNCELL** command to check whether co-coverage neighboring cells are configured correctly.
- Run the **LST URNCCELLSHUTDOWNPARA** command to check whether the RNC-level cell dynamic shutdown function is enabled.
- Run the **LST UCELLDYNSHUTDOWN** command to check whether the inter-RAT cell joint shutdown function is enabled.
- Run the **LST UMBSCRRM** command to check whether the inter-RAT common measurement function is enabled.
- (Optional) In inter-frequency multi-carrier UMTS networks, check whether co-coverage neighboring inter-frequency UMTS cells under the same NodeB as this cell are enabled with the WRFD-020117 Multi-Carrier Switch off Based on Traffic Load and WRFD-020122 Multi-Carrier Switch off Based on QoS features and have entered Multi-RAT Carrier Joint Shutdown mode.

Fault Description (2)

In areas covered by GL, UL, or GUL networks, an LTE cell cannot enter Multi-RAT Carrier Joint Shutdown mode.

Troubleshooting Procedure (2)

- Check whether co-coverage neighboring inter-RAT cells are configured correctly.

- In areas covered by GL or GUL networks, run the **LST GERANNCCELL** command on the eNodeB side and run the **LST GCELL** command on the BSC side to check whether configurations of the co-coverage neighboring GERAN cells on the eNodeB side are consistent with those of the GSM cells on the BSC side.
- In areas covered by UL or GUL networks, run the **LST UTRANNCCELL** command on the eNodeB side and run the **LST UCELL** command on the RNC side to check whether configurations of the co-coverage neighboring UTRAN cells on the eNodeB side are consistent with those of the UTRAN cells on the RNC side.
- Run the **LST INTERRATCELLSHUTDOWN** command to check whether the Multi-RAT Carrier Joint Shutdown feature is enabled and parameters such as **Shut Down Type** are set correctly.
- In areas covered by UL or GUL networks, run the **LST URRCTRLSWITCH** command on the RNC side to check whether the **INTERRAT_LOAD_SINGLE_RPT_FOR_LTE_SWITCH** check box is selected under the **Process Control Switch 5** parameter.
- In inter-frequency multi-carrier LTE networks, check whether neighboring inter-frequency LTE cells under the same eNodeB as this cell are enabled with the LOFD-001042 Intelligent Power-Off of Carriers in the Same Coverage feature and have entered Multi-RAT Carrier Joint Shutdown mode.

7

Site-Level Energy Saving

7.1 Introduction to Dormancy

This section describes the differences in dormancy functions from various dimensions.

7.1.1 Differences in Dormancy Types

To save power, RF modules of a base station are allowed to enter dormancy. The following table outlines the differences between RF module dormancy functions.

Dimension	RF Module Deep Dormancy	RF Module Fast Deep Dormancy	RF Module Super Dormancy
Control mode	• Scheduled • Load-based – Intra-RAT (UMTS/LTE/NR) – Inter-RAT (GU/LNR) • Scheduled+Load-based	Load-based • Intra-RAT (LTE/NR) • Inter-RAT (LNR)	• Scheduled • Load-based – Intra-RAT (GSM/UMTS/LTE/NR) – Inter-RAT (GU/LNR) • Scheduled+Load-based

Dimension	RF Module Deep Dormancy	RF Module Fast Deep Dormancy	RF Module Super Dormancy
Wakeup mode	Wakeup can be realized remotely, for example, by disabling one of these features or modifying configurations.	Wakeup can be realized remotely, for example, by disabling one of these features or modifying configurations. When the network load is high, wakeup from dormancy is expected to be completed more quickly to reduce the impact on user experience.	Wakeup can be realized remotely, for example, by disabling one of these features or modifying configurations.
Energy saving gains: RF module super dormancy saves the most energy, followed by RF module deep dormancy, which saves more energy than RF module fast deep dormancy.			

Main control boards and baseband processing units can also enter dormancy to conserve energy. The following table lists the differences between dormancy functions for these boards.

Dimension	Adaptive BBU Deep Dormancy	Main Control Board Dynamic Energy Saving
Control mode	A baseband processing unit enters deep dormancy when all cells it serves have entered dormancy through scheduled or load-based RF module deep, super, or timing dormancy.	A main control board automatically enters the dynamic energy saving state when the transmission rates over its ports are low.
Wakeup mode	A baseband processing unit starts to exit deep dormancy 5 minutes before the RF modules providing optical fiber connections or the cells it serves are scheduled to exit dormancy.	A main control board automatically exits the dynamic energy saving state when the transmission rates over its ports are high.
Energy saving gains: Adaptive BBU deep dormancy saves more energy than main control board dynamic energy saving.		

7.1.2 Differences in Dormancy Triggering Factors

RF module dormancy functions are classified into multiple sub-functions based on control mode. [Table 7-1](#) lists the factors that trigger the entry and exit of these sub-functions.

Table 7-1 Factors that trigger dormancy entry and exit

Dimension	Factors Triggering Dormancy Entry			Factors Triggering Dormancy Exit	
	Time period	Capacity-layer cell load	Basic-layer cell load	Time period	Basic-layer cell load
Scheduled RF module deep/super dormancy	√	/	/	√	/
NR load-based RF module deep/fast deep/super dormancy	√	√	√	√	√
LNR load-based RF module deep/fast deep/super dormancy					
LTE load-based RF module deep/fast deep/super dormancy					
GSM load-based RF module super dormancy					
UMTS load-based RF module deep/super dormancy					
GU load-based RF module deep/super dormancy					
NR local-cell-load-based RF module deep/super dormancy	√	√	/	√	/

Currently, RF module deep dormancy is supported only for the individual RATs or RAT combinations described above. For RF module deep dormancy in other RAT combinations, refer to the preceding sub-functions.

7.2 RF Module Deep Dormancy

7.2.1 Principles

The RF module deep dormancy function allows RF modules to enter dormancy when the network is under light or no load. This reduces energy consumption without compromising RF module reliability.

[7.1.2 Differences in Dormancy Triggering Factors](#) lists the sub-functions of RF module deep dormancy by control approach.

In addition to the preceding sub-functions, RF module deep dormancy provides the following functions:

- Multi-frequency flexible dormancy
In multi-band networking, cells operating in a specific frequency band of a multi-band AAU will enter dormancy if they meet the conditions for load-based RF module deep dormancy. For details, see [7.2.1.8 Multi-Frequency Flexible Dormancy](#). This function applies only to LTE and NR.

For high-frequency NR TDD, only scheduled RF module deep dormancy and NR load-based RF module deep dormancy are supported.

RF module deep dormancy also provides the following functions:

- Deep dormancy pending and pending cancellation
If operations on the base station (such as upgrades) require all RF modules to remain active, deep dormancy pending can be enabled. For details, see [7.10 Dormancy Pending and Pending Cancellation](#).
- Reliability protection during deep dormancy
The reliability protection involves license reliability protection, configuration reliability protection, reliability protection against faults, and O&M reliability protection during deep dormancy. For details, see [7.11 Reliability Protection for Deep Dormancy and Super Dormancy](#).
- Multi-RRU cell dormancy compatibility
This function is controlled by the **MULTI_RRU_CELL_DORM_COMPAT_SW** option of the **EnodebPwrSavingAlgo.PwrSavingAlgoSwitch** parameter. When this option is selected, RF module deep dormancy can take effect in multi-RRU cells, such as SFN cells.

This function requires that energy saving features take effect and terminate for involved RRUs simultaneously since these RRUs provide services as a whole. Therefore, it is recommended that RF modules be configured with the same dormancy function and dormancy period.

7.2.1.1 Scheduled RF Module Deep Dormancy

The scheduled RF module deep dormancy function allows RF modules to enter dormancy within specified periods when the network is under light or no load. This reduces energy consumption without compromising RF module reliability.

This function is controlled by setting the **RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB)** parameter to **ON** or **ON_DEEP_SUPER**. The value **ON_DEEP_SUPER**

indicates that RF module deep dormancy and RF module super dormancy are both enabled. For details about RF module super dormancy, see [7.4 RF Module Super Dormancy](#).

The dormancy period is specified by the **RRU.DORMANCYSTARTTIME (LTE eNodeB, 5G gNodeB)** and **RRU.DORMANCYSTOPTIME (LTE eNodeB, 5G gNodeB)** parameters.

- For a combined-RRU cell or multi-TRP NR cell (such as an SFN cell, hyper cell, or combined cell):
All RF modules serving the cell need to be configured with the same dormancy period. Otherwise, RF modules will enter deep dormancy only when the current time falls within the overlap of all configured dormancy periods and other dormancy conditions are met.
- In co-RF module scenarios involving UMTS, LTE, or NR, RAT-specific deep dormancy can be enabled by turning on **RRU.CUSTOMDORMANCYSTDSW (LTE eNodeB, 5G gNodeB)** and specifying **RRU.DORMANCYSTD (LTE eNodeB, 5G gNodeB)**. Once RAT-specific dormancy is enabled, only cells of the specified RATs can enter dormancy. If RAT-specific dormancy is not enabled, RF modules can enter dormancy only when dormancy conditions are met across all configured RATs.
- For UMTS inter-frequency handovers, the **HO_INTER_FREQ_HARD_HO_SWITCH** option of the **UCORRMALGOSWITCH.HoSwitch** parameter must be selected.
- For inter-RAT handovers involving UMTS, the **HO_INTER_RAT_CS_OUT_SWITCH** and **HO_INTER_RAT_PS_OUT_SWITCH** options of the **UCORRMALGOSWITCH.HoSwitch** parameter must be selected.
- For dormancy involving UMTS, the **GU_DC_SW** option of the base-station-specific **EQUIPMENT.EQMDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter must be deselected. If the option is deselected and a configuration command is executed to trigger dormancy pending or dormancy pending is enabled on the local end that works in UMTS, then dormancy pending will take effect on the entire base station.

 CAUTION

- You are advised not to perform base station reconfiguration (for example, running the **MOD RETPORT** command to modify RET port configuration), license installation, license activation, or software version download during the specified dormancy period. If any of these operations is required, run the **PND DORMANCY** command and wait 10 minutes before performing the operation.
 - During RF module dormancy, cell-related online detections (such as inter-cell cable sequence detection) and RF module-related detections (such as VSWR test, cross feeder connection detection, and intermodulation interference detection) cannot be performed. If any of these detections is required, run the **PND DORMANCY** command and wait 10 minutes before performing the detection.
-

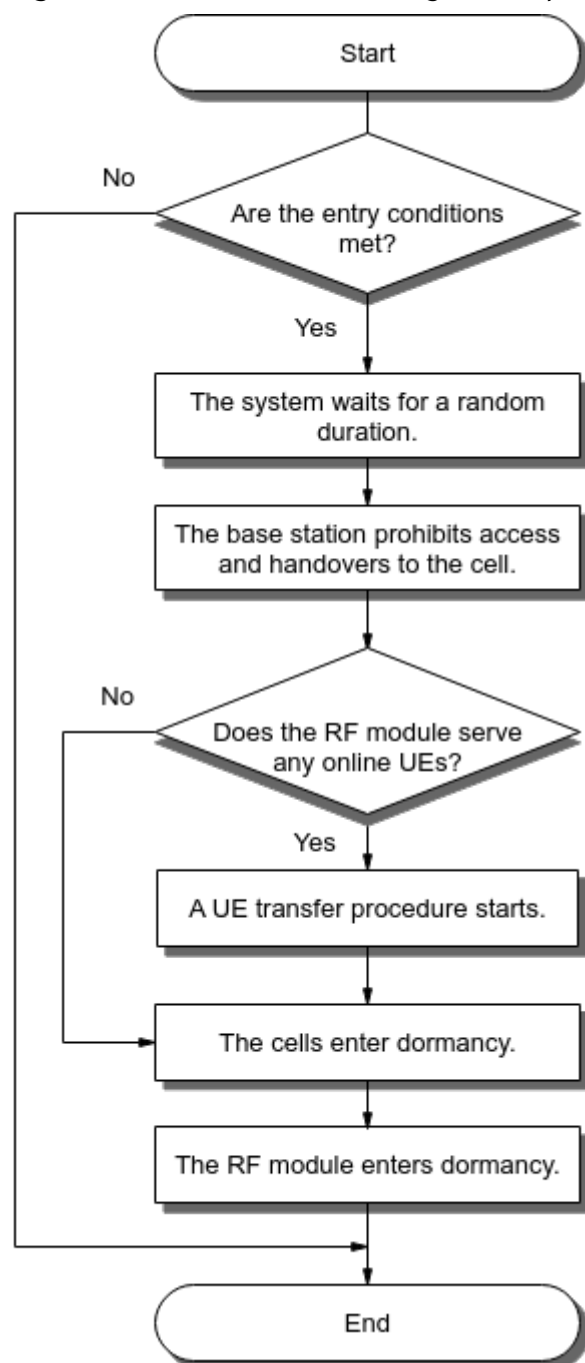
Entering the Deep Dormancy State

The prerequisites for this function to take effect are as follows:

- The RF module works in UMTS, LTE, or NR or is shared by multiple RATs including any of the preceding RATs.
- All activated cells served by the base station are bound to specified BBPs.

Figure 7-1 shows the procedure for entering the deep dormancy state.

Figure 7-1 Procedure for entering the deep dormancy state



1. The base station determines whether the entry conditions are met.
The base station checks whether the current time falls within the effective energy saving period.
 - If the **RRU.PWRSVGPOLICYID (LTE eNodeB, 5G gNodeB)** parameter is set to **0**, the effective period is determined by the **RRU.DORMANCYSTARTTIME (LTE eNodeB, 5G gNodeB)** and **RRU.DORMANCYSTOPTIME (LTE eNodeB, 5G gNodeB)** parameters.
 - If the **RRU.PWRSVGPOLICYID (LTE eNodeB, 5G gNodeB)** parameter is set to a non-zero value, the effective period is determined by the **BRDPWRSVGPOLICYTIME.STARTTIME (5G gNodeB, LTE eNodeB)** and **BRDPWRSVGPOLICYTIME.STOPTIME (LTE eNodeB, 5G gNodeB)** parameters of the policy whose **BRDPWRSVGPOLICYTIME.POLICYID (LTE eNodeB, 5G gNodeB)** is also set to this non-zero value.
2. The system waits for a random duration.
The system waits for a random duration determined by the **EQUIPMENT.MAXRAMDORMANCYWAITTIME (LTE eNodeB, 5G gNodeB)** parameter before entering dormancy. This prevents a large number of sites from entering deep dormancy simultaneously and avoids the resulting impact on core network signaling.
3. The base station prohibits access and handovers to the cell.
4. The base station checks whether the RF module is serving any online UEs.
 - If the cells of all RATs on the RF module are not serving any online UEs, the cells enter dormancy. The RATs on the RF module are indicated by the **RRU.RS (LTE eNodeB, 5G gNodeB)** parameter.
 - If the RF module is serving online UEs, the base station waits until the specified waiting time elapses.
 - If the **RRU.PWRSVGPOLICYID (LTE eNodeB, 5G gNodeB)** parameter is set to **0**, the waiting time is specified by the **RRU.DORMANCYTIMER (LTE eNodeB, 5G gNodeB)** parameter.
 - If the **RRU.PWRSVGPOLICYID (LTE eNodeB, 5G gNodeB)** parameter is set to a non-zero value, the waiting time is specified by the **BRDPWRSVGPOLICY.DORMANCYTIMER (LTE eNodeB, 5G gNodeB)** parameter of the policy whose **BRDPWRSVGPOLICYTIME.POLICYID (LTE eNodeB, 5G gNodeB)** is also set to this non-zero value.
5. A UE transfer procedure starts.
 - If there are online UEs in LTE cells served by the RF module, all UEs in these cells need to be handed over to co-coverage neighboring basic cells. The RATs of the RF module are indicated by the **RRU.RS** parameter. The handover procedure is as follows:
The eNodeB delivers A4 measurement configurations about the operating frequencies of the co-coverage neighboring basic cells to UEs in connected mode. The RSRP threshold for event A4 is specified by the **InterFreqHoGroup.InterFreqHoA4ThdRsrp** parameter. After receiving A4 measurement reports from UEs, the eNodeB initiates inter-frequency handovers of the UEs to the cells on the operating frequencies of the basic-layer cells.

- If there are online UEs in NR cells served by the RF module, a scheduled UE transfer procedure is started. The RATs of the RF module are indicated by the **RRU.RS** parameter.
 - In SA networking, UEs are transferred from the local cell to neighboring cells as follows:
 - Non-CA UEs: UEs without voice services are transferred to inter-frequency or inter-RAT neighboring cells through blind redirections. Voice service UEs are transferred to neighboring cells through inter-frequency or inter-RAT handovers first. If the handovers fail, inter-frequency or inter-RAT redirections are performed.
 - CA UEs: If the local cell serves as a secondary cell (SCell) of a CA UE, the SCell is removed. If the local cell serves as the primary cell (PCell) of a CA UE, the CA UE is transferred to a neighboring cell in the same way as a non-CA UE.
 - In NSA networking, UE connections on the NR side are released as follows:
 - If energy saving takes effect in the PSCell of a UE, all connections on the NR side are released.
 - If energy saving takes effect in an SCell of a UE, its connection to this SCell on the NR side is released but its other connections on the NR side are retained.
- If there are online UEs in UMTS cells served by the RF module, a scheduled UE transfer procedure is started. The RATs of the RF module are indicated by the **RRU.RS (5G gNodeB, LTE eNodeB)** parameter.
 - CELL_DCH UEs are preferentially handed over to inter-band or inter-RAT neighboring cells based on measurement results, and are redirected if the handovers fail.
 - The RNC periodically attempts to perform inter-frequency or inter-RAT handovers for CELL_DCH UEs.
 Inter-frequency handovers: The RNC delivers load-based inter-frequency measurement configurations in which the measurement frequencies are those of inter-band neighboring cells. After receiving a measurement report from the UE, the RNC hands over the UE to an inter-band neighboring cell with signal quality higher than the RSCP (**UCCELLMCLDR.TargetFreqThdRscp**) and Ec/No (**UCCELLMCLDR.TargetFreqThdEcN0**) thresholds of the local cell.
 Inter-RAT handovers: The RNC delivers load-based inter-RAT measurement configurations and hands over a UE to the inter-RAT neighboring cell after receiving a measurement report from the UE.
 - CELL_FACH UEs are redirected.
 - By default, cell broadcast service (CBS) takes precedence over dormancy. Therefore, when CBS is enabled in a cell for which the **URNCCCELLDORMANCYPARA.CellIgnoreCbsSwitch** parameter is set to **OFF** (default value), the cell cannot enter dormancy. If dormancy

is required in scenarios where CBS is enabled, this parameter must be set to **ON**. Once it is set to **ON**:

- If no CBS message is sent and the current time falls outside the CBS protection period (specified by the **URNCELLDORMANCYPARA.CbsServiceHoldDuration** parameter), the cell can enter dormancy.
- If CBS messages are sent, the cell cannot enter dormancy.

 NOTE

After receiving a CBS message, the RNC immediately wakes the dormant cell and processes the CBS message once the cell is successfully set up. Because the cell requires time to wake, CBS messages may not be processed promptly. If CBS has a higher priority, it is recommended that **URNCELLDORMANCYPARA.CellIgnoreCbsSwitch** be set to its default value **OFF**. For details about CBS, see *CBS in RAN Feature Documentation*.

- If the local cell and its inter-band neighboring cells belong to different operators, **URNCELLDORMANCYPARA.CellInterPlmnHoSwitch** must be set to **ON** and **UOPERATORSHARINGMODE.InterPlmnHoAllowedIntraRat** must be set to **YES** to enable UE transfer.
 - If the local cell and its inter-RAT neighboring cells belong to different operators, **UOPERATORSHARINGMODE.InterPlmnHoAllowedInterRat** must be set to **YES** to enable UE transfer.
 - Emergency UEs (if any) are not forcibly released, and the cells enter dormancy only after these UEs are released.
6. The cells enter dormancy.
 7. The RF module enters deep dormancy.

Exiting the Deep Dormancy State

- When the stop time specified by the **RRU.DORMANCYSTOPTIME (LTE eNodeB, 5G gNodeB)** parameter arrives, the RF module starts to exit dormancy.
- After RF module deep dormancy is disabled, the RF module starts to exit dormancy. If the RF module remains disconnected for approximately 15 minutes during the dormancy period, the cells will exit dormancy.

7.2.1.2 NR Load-based RF Module Deep Dormancy

Multi-carrier co-coverage NR networking provides a larger network capacity and, upon a heavy traffic load, a higher data throughput than single-carrier NR networking. However, such networking wastes base station energy when the network is under light or no load. NR load-based RF module deep dormancy is introduced to reduce the waste.

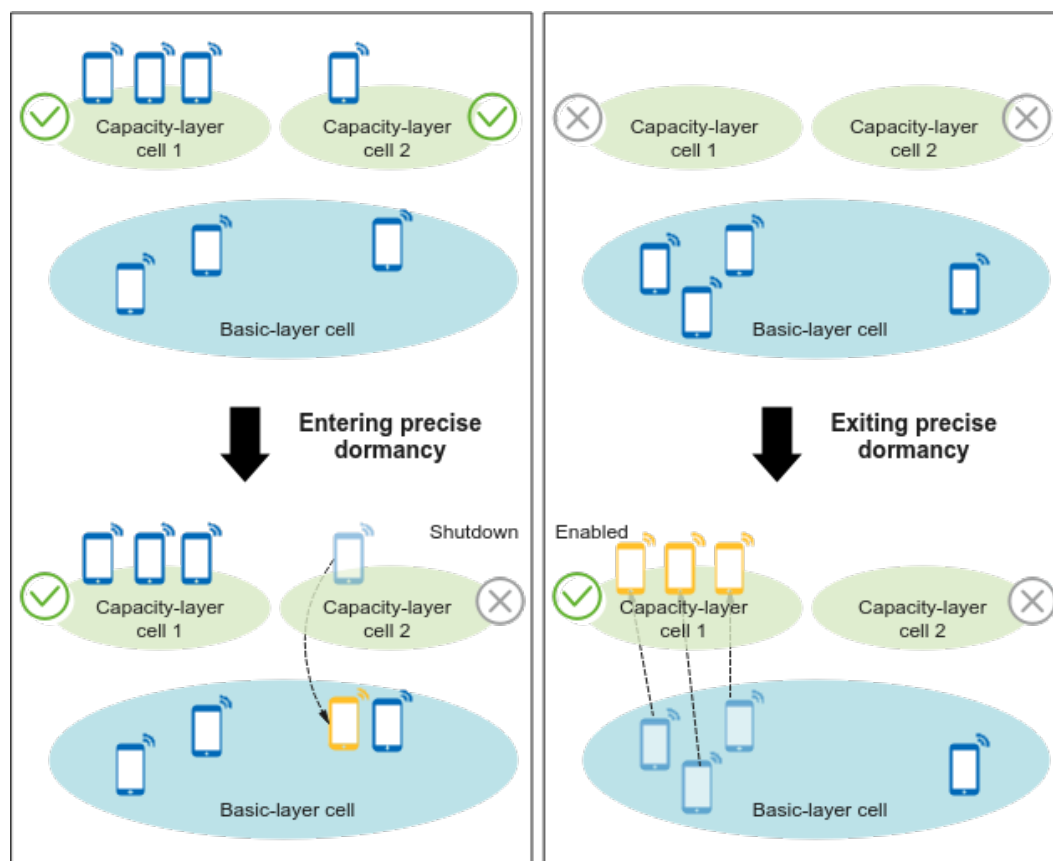
Load-based RF module deep dormancy enables the gNodeB to transfer UEs from capacity-layer cells to basic-layer cells and instruct the RF modules serving the capacity-layer cells to enter deep dormancy within a specified period when the

network is under light or no load. When the load of the basic-layer cells increases to a specified threshold, the dormant RF modules are automatically woken and the capacity-layer cells are restored.

NR load-based RF module deep dormancy works in two modes: non-precise and precise. For high-frequency NR TDD, only the non-precise dormancy mode is supported.

In scenarios where basic-layer cells cover large areas, capacity-layer cells cover small areas, one basic-layer cell corresponds to multiple capacity-layer cells, and UEs are unevenly distributed, the precise dormancy mode enables the base station to accurately determine which cells should enter or exit dormancy based on UE distribution across the basic-layer and capacity-layer cells. This maximizes the probability of cells entering dormancy and minimizes the probability of cells exiting dormancy, thereby extending the dormancy duration, as shown in [Figure 7-2](#).

Figure 7-2 Working principles for precise dormancy mode



This function requires the following switches to be configured:

- RF module-level switch

The **RRU.DORMANCYSW (5G gNodeB, LTE eNodeB)** parameter is set to **ON** or **ON_DEEP_SUPER**, and either of the following conditions is met:

- The **RRU.CUSTOMDORMANCYSTDSW (5G gNodeB, LTE eNodeB)** parameter is set to **OFF**.

- The **RRU.CUSTOMDORMANCYSTDSW (5G gNodeB, LTE eNodeB)** parameter is set to **ON**, and the **NR** option of the **RRU.DORMANCYSTD (5G gNodeB, LTE eNodeB)** parameter is selected.

The value **ON_DEEP_SUPER** of **RRU.DORMANCYSW (5G gNodeB, LTE eNodeB)** indicates that RF module deep dormancy and RF module super dormancy are both enabled. For details about RF module super dormancy, see [7.4 RF Module Super Dormancy](#).

- NR cell-level switch
 - In non-precise dormancy mode, the **INTRA_NR_DORMANCY_BY_LOAD_SW** option of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter is selected.
 - In precise dormancy mode, both the **INTRA_NR_DORMANCY_BY_LOAD_SW** option of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter and the **PRECISE_DORMANCY_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter are selected.

The effective period and threshold parameters for NR load-based RF module deep dormancy can be uniquely determined by using the setting combinations of the **NRDUCellPowerSaving.PowerSavingPolicyIndex** and **NRDUCellPowerSaving.PowerSavingType** parameters in the **NRDUCellPowerSaving** MOs on the NR side. Where,

- The **NRDUCellPowerSaving.PowerSavingPolicyIndex** parameter specifies an energy saving policy index. It uniquely identifies an energy saving policy within the local cell. With the index specified, the corresponding policy can be added, modified, queried, or removed at the local cell level.
- The **NRDUCellPowerSaving.PowerSavingType** parameter specifies the energy saving type indicated by an energy saving policy index. NR load-based RF module deep dormancy requires that this parameter be set to **INTRA_NR_DORMANCY_BY_LOAD**, meaning that under the corresponding energy saving policy, the local cell performs load-based deep dormancy in the specified period under certain load conditions.

Related Concepts

- Basic-layer carriers provide basic coverage for the network. RF modules supporting these carriers do not enter dormancy. Cells served by basic-layer carriers are referred to as basic-layer cells.
- Capacity-layer carriers handle high traffic volumes during peak hours. RF modules supporting these carriers may enter dormancy. Cells served by capacity-layer carriers are referred to as capacity-layer cells.

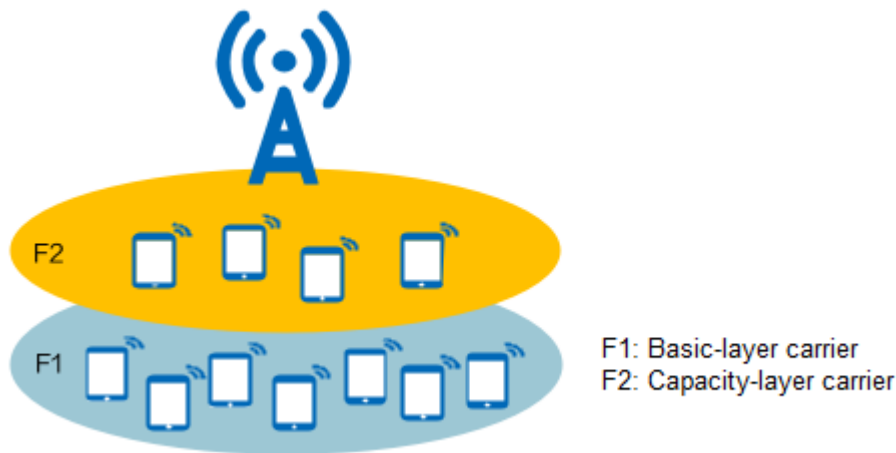
NOTE

Load-based RF module deep dormancy is enabled for capacity-layer cells and their associated RF modules. A capacity-layer cell can be associated with one or more basic-layer cells. Basic-layer cells and capacity-layer cells must be inter-frequency neighboring cells of each other.

Basic-layer cells and capacity-layer cells are relative concepts identified by the **NRCellRelation.PowerSavingCellFlag** parameter during neighbor relationship configuration.

- If this parameter is set to **COVERAGE**, the neighboring cell is a basic-layer cell and the serving cell is a capacity-layer cell.
- If this parameter is set to **CAPACITY**, the neighboring cell is a capacity-layer cell and the serving cell is a basic-layer cell.
- If this parameter is set to **NOT_CONFIG**, there is no energy saving relationship between the neighboring cell and the serving cell. The neighboring cell is not considered in load-based RF module deep dormancy.

Figure 7-3 Basic-layer and capacity-layer cells



For example, the following table describes how to set the Power Saving Cell Flag for cells in the preceding figure.

Table 7-2 Configuring the Power Saving Cell Flag
(NRCellRelation.PowerSavingCellFlag)

Value of NRCellRelation.PowerSavingCellFlag	NRCellRelation.NrCellId	NRCellRelation.CellId
COVERAGE	ID of the cell operating on F2	ID of the cell operating on F1
CAPACITY	ID of the cell operating on F1	ID of the cell operating on F2

In NR multi-carrier networking scenarios, it is recommended that capacity-layer and basic-layer cells be configured as described in the following table.

Table 7-3 Suggestions on designation of capacity-layer and basic-layer cells

Networking	Capacity-Layer Cell	Basic-Layer Cell
NR TDD (low-frequency) and NR FDD coordinated networking	NR TDD (low-frequency) cells	NR FDD cells
NR TDD (low-frequency) sub-6 GHz and NR TDD (low-frequency) sub-3 GHz coordinated networking	NR TDD (low-frequency) sub-6 GHz cells	NR TDD (low-frequency) sub-3 GHz cells
Inter-band NR FDD coordinated networking	Cells on a higher frequency	Cells on a lower frequency
Intra-band multi-carrier networking	Cells on a higher frequency	Cells on a lower frequency
NR TDD (high-frequency) and NR TDD (low-frequency) coordinated networking	NR TDD (high-frequency) cells	NR TDD (low-frequency) cells
NR TDD (high-frequency) and NR FDD (low-frequency) coordinated networking	NR TDD (high-frequency) cells	NR FDD (low-frequency) cells

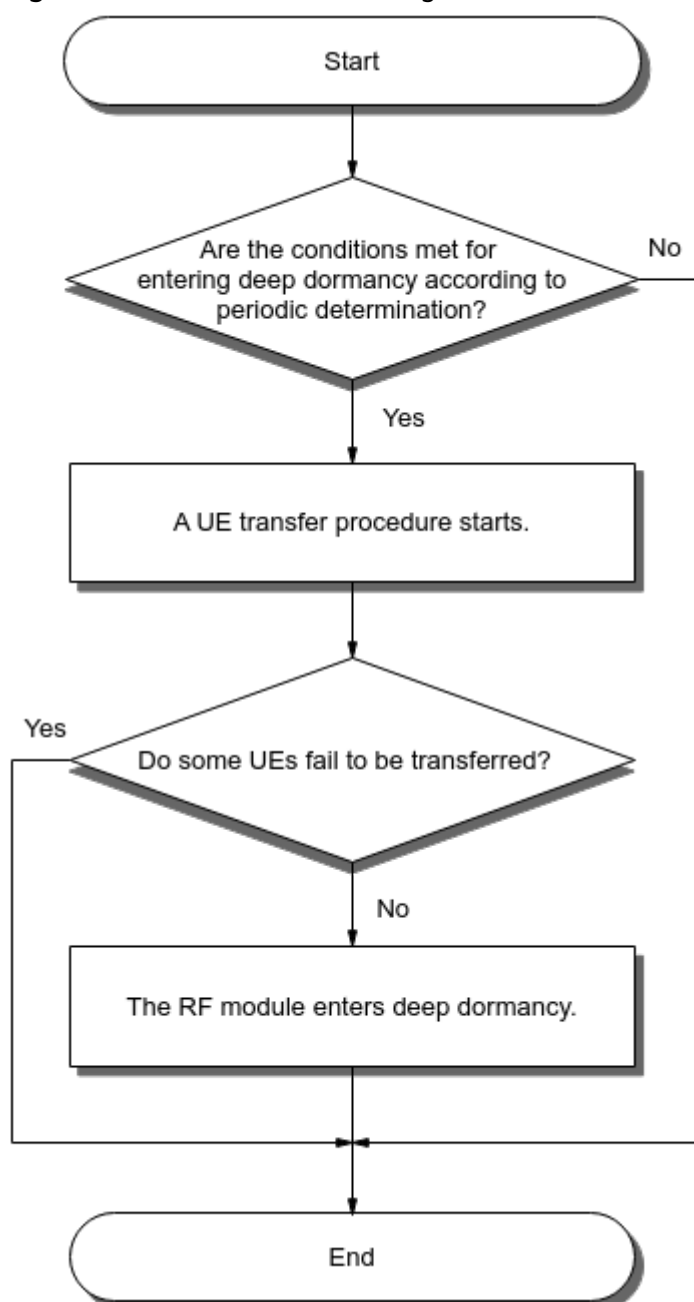
 NOTE

On the MAE, functions are available for automatic identification of co-coverage neighbor relationships (with basic-layer cells) and configuration of the power saving cell flag. For details, contact Huawei engineers.

Entering the Deep Dormancy State

Figure 7-4 shows how the RF module serving a capacity-layer cell enters deep dormancy after NR load-based RF module deep dormancy is enabled.

Figure 7-4 Procedure for entering NR load-based RF module deep dormancy



1. The base station determines whether the entry conditions are met.
The base station periodically checks whether the dormancy entry conditions are met. The evaluation period granularity is specified by the **NRDUCellPowerSaving.EvalPeriodGranularity** parameter.
 - If this parameter is set to **MINUTE_LEVEL**, the evaluation period is expressed in minutes. In this case, the period is specified by the **NRDUCellPowerSaving.ShutdownTrigJudgePeriod** parameter. For high-frequency TDD, the **NRDUCellPowerSaving.EvalPeriodGranularity** parameter must be set to **MINUTE_LEVEL**.

- If this parameter is set to **SECOND_LEVEL**, the evaluation period is expressed in seconds. In this case, the period is specified by the **NRDUCellPowerSaving.SecondLevelTrigEvalPrd** parameter.
- A capacity-layer cell must meet all of the following conditions:
 - The current time falls within the effective energy saving period determined by the **NRDUCellPowerSaving.StartTime** and **NRDUCellPowerSaving.StopTime** parameters.
 - The capacity-layer cell is not in an energy saving penalty state.
- The basic-layer cells associated with the capacity-layer cell must meet one of the following conditions:
 - All basic-layer carriers are available.
 - Some basic-layer carriers are available, while others, functioning as capacity-layer carriers of other basic-layer carriers, are in the energy saving state due to NR intelligent carrier shutdown, LTE and NR intelligent carrier shutdown, NR load-based RF module dormancy, or LNR load-based RF module dormancy.
- The capacity-layer cell and its associated basic-layer cells must meet all the following conditions:
 - The total downlink PRB usage of the capacity-layer and basic-layer carriers is less than or equal to the downlink PRB usage threshold specified by the **NRDUCellPowerSaving.DlPrbThld** parameter.
Where,
NR TDD (low frequency) and NR FDD: Total downlink PRB usage of the capacity-layer and basic-layer carriers = (Number of PRBs used by PDSCH DRBs on the capacity-layer carrier + Number of downlink PRBs used on the basic-layer carriers)/Number of available downlink PRBs on the basic-layer carriers
NR TDD (high frequency): Total downlink PRB usage of the capacity-layer and basic-layer carriers = ((Number of PRBs used by PDSCH DRBs in the high-frequency sector x (Subcarrier spacing in the high-frequency sector/Subcarrier spacing in the basic layer)) + Number of downlink PRBs used on the basic-layer carriers)/Number of available downlink PRBs on the basic-layer carriers
If multiple basic-layer carriers correspond to one capacity-layer carrier, the sum of PRBs of these basic-layer carriers is used in the preceding formula.
Precise dormancy mode: Total downlink PRB usage of the capacity-layer and basic-layer carriers = (Number of PRBs used by PDSCH DRBs on the capacity-layer carrier + Overlap rate of each inter-frequency neighboring cell x Number of downlink PRBs used on each basic-layer carrier)/(Number of available downlink PRBs on the capacity-layer carrier + Overlap rate of each inter-frequency neighboring cell x Number of available downlink PRBs on each basic-layer carrier)

 NOTE

In precise dormancy mode, the overlap rate of each inter-frequency neighboring cell is specified by the **NRCellRelation.OverlapRate** parameter.

- The total uplink PRB usage of the capacity-layer and basic-layer carriers is less than or equal to the uplink PRB usage threshold specified by the **NRDUCellPowerSaving.UlPrbThld** parameter.

Where,

NR TDD (low frequency) and NR FDD: Total uplink PRB usage of the capacity-layer and basic-layer carriers = (Number of uplink PRBs used on the capacity-layer carrier + Number of uplink PRBs used on the basic-layer carriers)/Number of available uplink PRBs on the basic-layer carriers

NR TDD (high frequency): Total uplink PRB usage of the capacity-layer and basic-layer carriers = ((Number of uplink PRBs used in the high-frequency sector x (Subcarrier spacing in the high-frequency sector/Subcarrier spacing in the basic layer)) + Number of uplink PRBs used on the basic-layer carriers)/Number of available uplink PRBs on the basic-layer carriers

Precise dormancy: Total uplink PRB usage of the capacity-layer and basic-layer carriers = (Number of uplink PRBs used on the capacity-layer carrier + Overlap rate of each inter-frequency neighboring cell x Number of uplink PRBs used on each basic-layer carrier)/(Number of available uplink PRBs on the capacity-layer carrier + Overlap rate of each inter-frequency neighboring cell x Number of available uplink PRBs on each basic-layer carrier)

- In precise dormancy mode, the following conditions must also be met:

- Uplink PRB usage of each basic-layer carrier corresponding to the capacity-layer carrier <

**NRDUCellPowerSaving.ProtectUlPrbThld –
NRDUCellPowerSaving.ProtectUlPrbOffset**

Uplink PRB usage of a basic-layer carrier corresponding to the capacity-layer carrier = (Number of uplink PRBs used on the basic-layer carrier + Overlap rate x Number of uplink PRBs used on the capacity-layer carrier)/Total number of available uplink PRBs on the basic-layer carrier

- The downlink PRB usage of each basic-layer carrier corresponding to the capacity-layer carrier <

**NRDUCellPowerSaving.ProtectDlPrbThld –
NRDUCellPowerSaving.ProtectDlPrbOffset**

Downlink PRB usage of a basic-layer carrier corresponding to the capacity-layer carrier = (Number of downlink PRBs used on the basic-layer carrier + Overlap rate x Number of PRBs used by PDSCH DRBs on the capacity-layer carrier)/Total number of available downlink PRBs on the basic-layer carrier

The overlap rate is the proportion of overlapping area in a capacity-layer cell to a basic-layer cell, which is specified by the **NRCellRelation.OverlapRate** parameter at the basic-layer cell.

- The total number of RRC_CONNECTED UEs on the capacity-layer carrier and a basic-layer carrier is less than or equal to the UE number threshold specified by the **NRDUCellPowerSaving.UserNumThld** parameter.
If multiple basic-layer carriers correspond to one capacity-layer carrier, each combination of the basic-layer and capacity-layer carriers must meet this condition.
- In multi-operator sharing scenarios, the local cell and co-coverage neighboring basic cells are classified based on normal operators and reserved operators (whose **NRExternalNCellPlmn.PlmnReservedFlag** is set to **TRUE**).
 - If the operators of the local cell are normal operators, the operators must be a subset of normal operators of the co-coverage neighboring basic cells.
 - If the operators of the local cell are reserved operators, the operators must be a subset of normal or reserved operators of the co-coverage neighboring basic cells.
- In SUL co-deployment scenarios, the total number of SUL UEs in the capacity-layer and basic-layer cells is less than or equal to the value of the **NRDUCellPowerSaving.SulUserNumThld** parameter. If the capacity-layer cell is associated with multiple basic-layer cells, each pair of the basic-layer and capacity-layer cells must meet this condition.
- The Xn interface is normal in multi-carrier inter-base-station scenarios.

 NOTE

If the capacity-layer cell is a hyper cell or combined cell:

- The total downlink or uplink PRB usage of the capacity-layer and basic-layer carriers is calculated based on the maximum number of used PRBs among all TRPs. When load-based deep dormancy takes effect, all TRPs and their carriers enter dormancy.
- It is recommended that basic-layer cells also be hyper cells or combined cells. It is not recommended that common cells be configured as basic-layer cells.

When evaluating the total uplink or downlink PRB usage of capacity-layer and basic-layer cells against the entry threshold, the base station performs conversion on the cell bandwidth, subcarrier spacing, and slot configuration.

2. A UE transfer procedure starts. The gNodeB sends system information to prohibit UE access and handovers to the NR cell and starts the UE transfer procedure with a specified duration.
 - In SA networking, UEs are transferred from the local cell to basic-layer neighboring cells as follows:
 - Non-CA UEs: UEs not performing voice services are transferred to neighboring cells through inter-frequency handovers first. If the handovers fail, the UEs are transferred to inter-frequency neighboring cells through blind redirections. UEs performing voice services are handed over to inter-frequency neighboring cells.

- CA UEs: If the local cell serves as an SCell of a CA UE, the SCell is removed. If the local cell serves as the PCell of a CA UE, the UE is transferred to a neighboring cell in the same way as a non-CA UE.
If UEs cannot be transferred out of the capacity-layer cell within 30 seconds after the UE transfer procedure starts (for example, due to handover failures or unsupported operating bands of the basic-layer cells), the procedure stops and cannot be re-triggered for the capacity-layer cell within 30 minutes.
- In NSA networking, UE connections on the NR side are released as follows:
 - If energy saving takes effect in the PSCell of a UE, all connections on the NR side are released.
 - If energy saving takes effect in an SCell of a UE, its connection to this SCell on the NR side is released but its other connections on the NR side are retained.
- 3. The RF module enters deep dormancy.
Once all UEs are transferred out of the capacity-layer cell, no RRC_CONNECTED UEs remain in the cell, and the cell and its serving RF modules enter deep dormancy. If an RF module serves multiple NR cells, it enters deep dormancy only when load-based deep dormancy is triggered for all these NR cells. If an RF module also serves LTE cells, RF module deep dormancy must also be enabled on the LTE side.

Exiting the Deep Dormancy State

The base station periodically checks whether the dormancy exit conditions are met. The evaluation period granularity is specified by the **NRDUCellPowerSaving.EvalPeriodGranularity** parameter.

- If this parameter is set to **MINUTE_LEVEL**, the evaluation period is expressed in minutes. In this case, the period is specified by the **NRDUCellPowerSaving.ExitEvaluationPeriod** parameter. For high-frequency TDD, the **NRDUCellPowerSaving.EvalPeriodGranularity** parameter must be set to **MINUTE_LEVEL**.
- If this parameter is set to **SECOND_LEVEL**, the evaluation period is expressed in seconds. In this case, the period is specified by the **NRDUCellPowerSaving.SecondLevelExitEvalPrd** parameter.

A capacity-layer cell and its serving RF modules automatically exit dormancy when any of the following conditions is met:

- The current time falls outside the effective energy saving period determined by the **NRDUCellPowerSaving.StartTime** and **NRDUCellPowerSaving.StopTime** parameters.
- The total downlink PRB usage of basic-layer carriers is greater than **NRDUCellPowerSaving.DlPrbThld** plus **NRDUCellPowerSaving.DlPrbOffset**. Additionally for high-frequency NR TDD, the number of high-frequency UEs on the basic-layer carrier is greater than **NRDUCellPowerSaving.MmwaveUserNumThld**.

 NOTE

If the sum of **NRDUCellPowerSaving.DIPrbThld** and **NRDUCellPowerSaving.DIPrbOffset** is greater than 100, RF modules cannot exit deep dormancy based on the PRB usage.

- The total uplink PRB usage of basic-layer carriers is greater than **NRDUCellPowerSaving.UlPrbThld** plus **NRDUCellPowerSaving.NCellUlPrbOffset**. Additionally for high-frequency NR TDD, the number of high-frequency UEs on the basic-layer carrier is greater than **NRDUCellPowerSaving.MmwaveUserNumThld**.

- In precise dormancy mode, the uplink PRB usage of any basic-layer carrier corresponding to the capacity-layer carrier is greater than or equal to the value of the **NRDUCellPowerSaving.ProtectUlPrbThld** parameter.

Uplink PRB usage of a basic-layer carrier corresponding to the capacity-layer carrier = Number of uplink PRBs used on the basic-layer carrier/Total number of available uplink PRBs on the basic-layer carrier

- In precise dormancy mode, the downlink PRB usage of any basic-layer carrier corresponding to the capacity-layer carrier is greater than or equal to the value of the **NRDUCellPowerSaving.ProtectDlPrbThld** parameter.

Downlink PRB usage of a basic-layer carrier corresponding to the capacity-layer carrier = Number of downlink PRBs used on the basic-layer carrier/Total number of available downlink PRBs on the basic-layer carrier

- The number of UEs in connected mode served by the basic-layer carrier is greater than **NRDUCellPowerSaving.UserNumThld** plus **NRDUCellPowerSaving.UserNumOffset**. Additionally for high-frequency NR TDD, the number of high-frequency UEs on the basic-layer carrier is greater than **NRDUCellPowerSaving.MmwaveUserNumThld**.

 NOTE

- If the base station software version of a low-frequency basic-layer cell is V100R020C00 or earlier, the number of high-frequency-capable UEs cannot be reported.
- If the base station software versions of all low-frequency basic-layer cells are V100R020C00 or earlier, the high-frequency capacity-layer cell does not need to check the high-frequency-capable UEs in the basic-layer cells. In this scenario, the cell can exit dormancy if the PRB usage or the number of UEs in the basic-layer cell is greater than the threshold.
- If the base station software version of some low-frequency basic-layer cells is V100R020C00 or earlier:
 - For a low-frequency basic-layer cell that can properly report the number of high-frequency-capable UEs, the high-frequency capacity-layer cell needs to check the number of high-frequency-capable UEs in that low-frequency basic-layer cell. In this scenario, the capacity-layer cell can exit dormancy if the number of high-frequency-capable UEs in the low-frequency basic-layer cell meets the configured threshold and the PRB usage or the number of UEs in connected mode in the basic-layer cell meets the exit conditions.
 - For low-frequency basic-layer cells that do not report the number of high-frequency-capable UEs, this number is 0 by default. In this scenario, the capacity-layer cell cannot exit dormancy, even if the actual number of high-frequency-capable UEs in the low-frequency basic-layer cell meets the configured threshold and the PRB usage or the number of UEs in connected mode meets the exit conditions.
- A basic-layer neighboring cell is added for the capacity-layer cell.

- In multi-operator sharing scenarios:
 - The PLMN IDs of the normal operators of the capacity-layer cell are not a subset of the PLMN IDs of normal operators of basic-layer cells.
 - The PLMN IDs of the reserved operators of the capacity-layer cell are not a subset of the PLMN IDs of normal or reserved operators of basic-layer cells.
- In SUL co-deployment scenarios, the number of SUL UEs in a basic-layer cell is greater than the sum of **NRDUCellPowerSaving.SulUserNumThld** and **NRDUCellPowerSaving.SulUserNumOffset**. If the capacity-layer cell is associated with multiple basic-layer cells, the number of SUL UEs in one of the basic-layer cells must meet the condition.
- The Xn interface is abnormal when basic-layer carriers are provided by different base stations.
- Any of the basic-layer carriers becomes unavailable for a reason other than NR intelligent carrier shutdown, LTE and NR intelligent carrier shutdown, and NR load-based RF module dormancy.

Upon the exit from deep dormancy:

- The capacity-layer cell cannot re-enter deep dormancy within the penalty period specified by the **NRDUCellPowerSaving.PwrSavingPenaltyTime** parameter if it exited dormancy due to uplink/downlink PRB usage or UE number exceptions or unavailability of the basic-layer cell, or Xn interface exceptions.
- A penalty is not imposed if the capacity-layer cell exited deep dormancy because the function switch was turned off, the effective period elapsed, or the base station software was upgraded.

7.2.1.3 NR Local-Cell-Load-based RF Module Deep Dormancy

NR local-cell-load-based RF module deep dormancy allows RF modules to enter dormancy during specified periods based on the local cell load when the network is under light or no load, thereby reducing energy consumption without compromising RF module reliability.

This function is controlled by setting the **RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB)** parameter to **ON** or **ON_DEEP_SUPER**. The value **ON_DEEP_SUPER** indicates that RF module deep dormancy and RF module super dormancy are both enabled. For details about RF module super dormancy, see [7.4 RF Module Super Dormancy](#).

The dormancy period is specified by the **RRU.DORMANCYSTARTTIME (LTE eNodeB, 5G gNodeB)** and **RRU.DORMANCYSTOPTIME (LTE eNodeB, 5G gNodeB)** parameters.

- For a combined-RRU cell or multi-TRP NR cell (such as an SFN cell, hyper cell, or combined cell):

All RF modules serving the cell need to be configured with the same dormancy period. Otherwise, RF modules will enter deep dormancy only when the current time falls within the overlap of all configured dormancy periods and other dormancy conditions are met.
- In co-RF module scenarios involving NR, RAT-specific deep dormancy can be enabled by turning on **RRU.CUSTOMDORMANCYSTDSW (LTE eNodeB, 5G**

gNodeB) and specifying **RRU.DORMANCYSTD (LTE eNodeB, 5G gNodeB)**. Once RAT-specific dormancy is enabled, only cells of the specified RATs can enter dormancy. If RAT-specific dormancy is not enabled, RF modules can enter dormancy only when dormancy conditions are met across all configured RATs.

 CAUTION

- You are advised not to perform base station reconfiguration (for example, running the **MOD RETPORT** command to modify RET port settings), install or activate a license, or download software during the specified dormancy period. If any of these operations is required, run the **PND DORMANCY** command and wait 10 minutes before performing the operation.
 - During RF module dormancy, cell-related online detections (such as inter-cell cable sequence detection) and RF module-related detections (such as VSWR test, cross feeder connection detection, and intermodulation interference detection) cannot be performed. If any of these detections is required, run the **PND DORMANCY** command and wait 10 minutes before performing the detection.
-

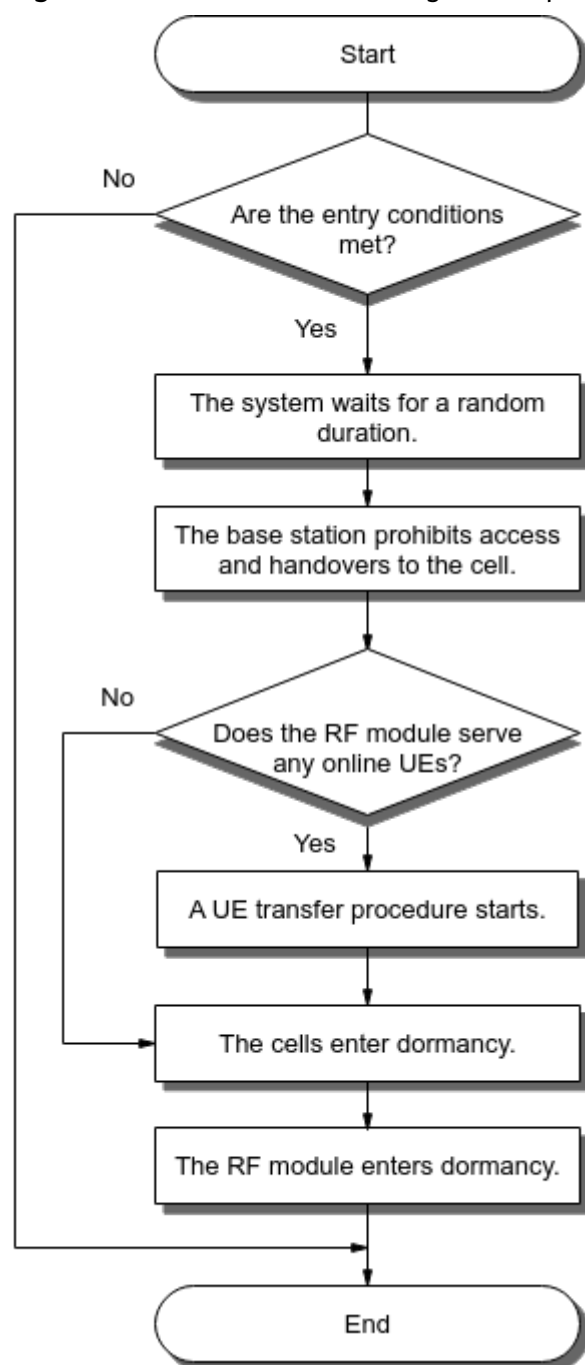
Entering the Deep Dormancy State

The prerequisites for this function to take effect are as follows:

- The RF module works in NR or is shared by multiple RATs including NR.
- All activated cells served by the base station are bound to specified BBPs.

Figure 7-5 shows the procedure for entering the deep dormancy state.

Figure 7-5 Procedure for entering the deep dormancy state



1. The base station determines whether the entry conditions are met.
The base station checks whether the current time falls within the effective energy saving period.
 - If the **RRU.PWRSVGPOLICYID (LTE eNodeB, 5G gNodeB)** parameter is set to **0**, the effective period is determined by the **RRU.DORMANCYSTARTTIME (LTE eNodeB, 5G gNodeB)** and **RRU.DORMANCYSTOPTIME (LTE eNodeB, 5G gNodeB)** parameters.
 - If the **RRU.PWRSVGPOLICYID (LTE eNodeB, 5G gNodeB)** parameter is set to a non-zero value, the effective period is determined by the

BRDPWRSVGPOLICYTIME.STARTTIME (5G gNodeB, LTE eNodeB) and **BRDPWRSVGPOLICYTIME.STOPTIME** (LTE eNodeB, 5G gNodeB) parameters of the policy whose **BRDPWRSVGPOLICYTIME.POLICYID** (LTE eNodeB, 5G gNodeB) is also set to this non-zero value.

If NR is included in the RATs indicated by the **RRU.RS** parameter for the RF module and optimized evaluation of NR RF module dormancy is enabled (by selecting the **NR_DORMANCY_JUDGE_THLD_OPT_SW** option of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter), the base station periodically evaluates whether the local cell can enter scheduled RF module dormancy based on the cell load.

- An energy saving policy is configured for the local cell. The **NRDUCellPowerSaving.PowerSavingPolicyIndex** parameter specifies an energy saving policy index. It uniquely identifies an energy saving policy within the local cell. The **NRDUCellPowerSaving.PowerSavingType** parameter must be set to **NR_DORMANCY_JUDGE_THLD_OPT**. The **NRDUCellPowerSaving.EvalPeriodGranularity** parameter must be set to **MINUTE_LEVEL**.
- An NR cell cannot enter dormancy unless both of the following conditions are met:
 - The current time falls within the effective energy saving period determined by the **NRDUCellPowerSaving.StartTime** and **NRDUCellPowerSaving.StopTime** parameters.
 - The NR cell is not in an energy saving penalty state.
- Regarding cell load, an NR cell cannot enter dormancy unless all of the following conditions are met:
 - Downlink PRB usage of the cell $\leq$ **NRDUCellPowerSaving.DlPrbThld**.
Downlink PRB usage of the cell = $(N.PR.B.DL.DrbUsed.Avg - N.PR.B.DL.ExtendUsed.Avg) / N.PR.B.DL.Actual.Avail.Avg \times 100\%$
 - Uplink PRB usage of the cell $\leq$ **NRDUCellPowerSaving.UlPrbThld**.
Uplink PRB usage of the cell = $N.PR.B.PUSCH.Used.Avg / N.PR.B.UL.PUSCH.Actual.Avail.Avg \times 100\%$
 - Number of RRC_CONNECTED UEs in the cell $\leq$ **NRDUCellPowerSaving.UserNumThld**
 - The number of RRC_CONNECTED UEs in an SUL cell with co-deployment is the sum of the **N.User.RRConn.Avg** and **N.User.SUL.UL.Avg** values.
 - The number of RRC_CONNECTED UEs in any other cell is indicated by **N.User.RRConn.Avg**.
 - In SUL co-deployment scenarios, the total number of SUL UEs in the local cell is less than or equal to the value of the **NRDUCellPowerSaving.SulUserNumThld** parameter.
 - For NR TDD, when the **ELIM_INCR_PWR_SAVING_JUDGE_SW** option of the **gNBPwrSavingAlgo.PwrSavingAlgoSwitch** parameter is selected, the number of UEs in the cell excludes the number of iNCR modules. For details about iNCR, see *LampSite iNCR Product Documentation*.

2. The system waits for a random duration.
The system waits for a random duration determined by the **EQUIPMENT.MAXRAMDORMANCYWAITTIME (LTE eNodeB, 5G gNodeB)** parameter before entering dormancy. This prevents a large number of sites from entering deep dormancy simultaneously and avoids the resulting impact on core network signaling.
3. The base station prohibits access and handovers to the cell.
4. The base station checks whether the RF module is serving any online UEs.
 - If the cells of all RATs on the RF module are not serving any online UEs, the cells enter dormancy. The RATs on the RF module are indicated by the **RRU.RS (LTE eNodeB, 5G gNodeB)** parameter.
 - If the RF module is serving online UEs, the base station waits until the specified waiting time elapses.
 - If the **RRU.PWRSVGPOLICYID (LTE eNodeB, 5G gNodeB)** parameter is set to **0**, the waiting time is specified by the **RRU.DORMANCYTIMER (LTE eNodeB, 5G gNodeB)** parameter.
 - If the **RRU.PWRSVGPOLICYID (LTE eNodeB, 5G gNodeB)** parameter is set to a non-zero value, the waiting time is specified by the **BRDPWRSVGPOLICY.DORMANCYTIMER (LTE eNodeB, 5G gNodeB)** parameter of the policy whose **BRDPWRSVGPOLICYTIME.POLICYID (LTE eNodeB, 5G gNodeB)** is also set to this non-zero value.
5. A UE transfer procedure starts.
If there are online UEs in NR cells served by the RF module, a scheduled UE transfer procedure is started. The RATs of the RF module are indicated by the **RRU.RS** parameter.
 - In SA networking, UEs are transferred from the local cell to neighboring cells as follows:
 - Non-CA UEs: UEs without voice services are transferred to inter-frequency or inter-RAT neighboring cells through blind redirections. Voice service UEs are transferred to neighboring cells through inter-frequency or inter-RAT handovers first. If the handovers fail, inter-frequency or inter-RAT redirections are performed.
 - CA UEs: If the local cell serves as an SCell of a CA UE, the SCell is removed. If the local cell serves as the PCell of a CA UE, the UE is transferred to a neighboring cell in the same way as a non-CA UE.
 - In NSA networking, UE connections on the NR side are released as follows:
 - If energy saving takes effect in the PSCell of a UE, all connections on the NR side are released.
 - If energy saving takes effect in an SCell of a UE, its connection to this SCell on the NR side is released but its other connections on the NR side are retained.
6. The cells enter dormancy.
7. The RF module enters deep dormancy.

Exiting the Deep Dormancy State

- When the stop time specified by the **RRU.DORMANCYSTOPTIME (LTE eNodeB, 5G gNodeB)** parameter arrives, the RF module starts to exit dormancy.
- After RF module deep dormancy is disabled, the RF module starts to exit dormancy. If the RF module remains disconnected for approximately 15 minutes during the dormancy period, the cells will exit dormancy.

7.2.1.4 LNR Load-based RF Module Deep Dormancy

NR cells in operation continue to consume energy even under light or no traffic in areas covered by both LTE and NR, resulting in unnecessary base station energy consumption. To address this, LNR load-based RF module deep dormancy is introduced.

Load-based RF module deep dormancy enables the gNodeB to transfer UEs from NR cells to LTE cells and instruct the RF modules serving the NR cells to enter deep dormancy within a specified period when the network is under light or no load. When the load of the LTE cells increases to a specified threshold, the dormant RF modules are automatically woken and the NR cells are restored.

This function requires the following switches to be configured:

- RF module-level switches
The **RRU.DORMANCYSW (5G gNodeB, LTE eNodeB)** parameter is set to **ON** or **ON_DEEP_SUPER**, and either of the following conditions is met:
 - The **RRU.CUSTOMDORMANCYSTDSW** parameter is set to **OFF**.
 - The **RRU.CUSTOMDORMANCYSTDSW** parameter is set to **ON**, and the **NR** option of the **RRU.DORMANCYSTD** parameter is selected.The value **ON_DEEP_SUPER** of **RRU.DORMANCYSW (5G gNodeB, LTE eNodeB)** indicates that RF module deep dormancy and RF module super dormancy are both enabled. For details about RF module super dormancy, see [7.4 RF Module Super Dormancy](#).
- LTE cell-level switch: **LNR_LOAD_BASED_DORMANCY_SW** of the **CellPwrSavingAlgo.PwrSavingAlgoSwitch** parameter is selected.
- NR cell-level switch: **LNR_DORMANCY_BY_LOAD_SW** of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter is selected.

The effective period for LNR load-based RF module deep dormancy can be uniquely determined by using the setting combinations of the **NRDUCellPowerSaving.PowerSavingPolicyIndex** and **NRDUCellPowerSaving.PowerSavingType** parameters in the **NRDUCellPowerSaving** MOs on the NR side.

Where,

- The **NRDUCellPowerSaving.PowerSavingPolicyIndex** parameter specifies an energy saving policy index. It uniquely identifies an energy saving policy within the local cell. With the index specified, the corresponding policy can be added, modified, queried, or removed at the local cell level.
- The **NRDUCellPowerSaving.PowerSavingType** parameter specifies the energy saving type indicated by an energy saving policy index. LNR load-

based RF module deep dormancy requires that this parameter be set to **LNR_DORMANCY_BY_LOAD**, meaning that under the corresponding energy saving policy, the local cell performs LNR load-based RF module deep dormancy in the specified period under certain load conditions.

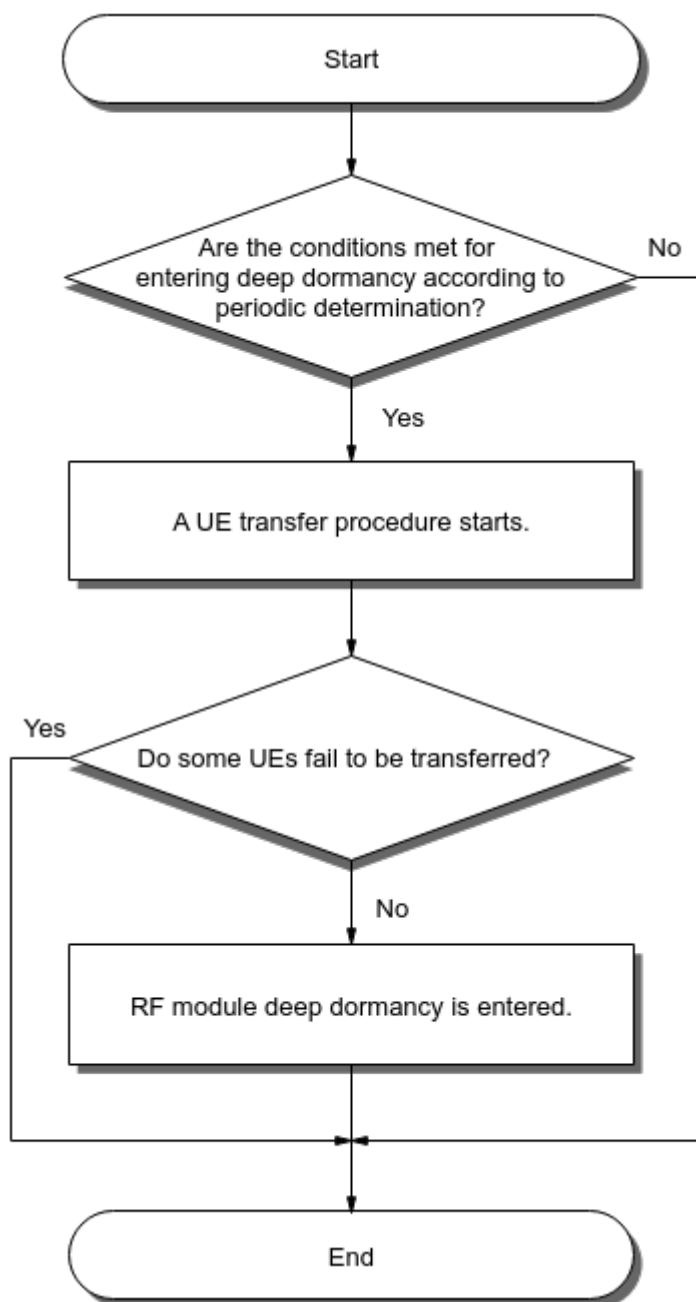
Identifying Overlapping Neighboring Cells

The mechanism of identifying overlapping neighboring cells for LNR load-based RF module deep dormancy is the same as that for LTE and NR intelligent carrier shutdown. For details, see [6.1.1.1 Identifying Overlapping Neighboring Cells](#).

Entering the Deep Dormancy State

[Figure 7-6](#) shows how the RF module serving an NR cell enters deep dormancy after LNR load-based RF module deep dormancy is enabled.

Figure 7-6 Procedure for entering LNR load-based RF module deep dormancy



1. The base station determines whether the entry conditions are met.
The current time falls within the effective energy saving period determined by **NRDUCellPowerSaving.StartTime** and **NRDUCellPowerSaving.StopTime**, and the NR cell is not in an energy saving penalty state.
The base station periodically checks whether the dormancy entry conditions are met. The evaluation period granularity is specified by the **NRDUCellPowerSaving.EvalPeriodGranularity** parameter.
 - If this parameter is set to **MINUTE_LEVEL**, the evaluation period is expressed in minutes. In this case, the period is specified by the **NRDUCellPowerSaving.ShutdownTrigJudgePeriod** parameter.

- If this parameter is set to **SECOND_LEVEL**, the evaluation period is expressed in seconds. In this case, the period is specified by the **NRDUCellPowerSaving.SecondLevelTrigEvalPrd** parameter.
- The NR cell must meet all of the following conditions:
 - Average number of active UEs in the NR cell $\leq$ UE number threshold (specified by the **NRDUCellPowerSaving.UserNumThld** parameter)
 - Downlink PRB usage of the NR cell $\leq$ Downlink PRB usage threshold (specified by the **NRDUCellPowerSaving.DlPrbThld** parameter)
 - Uplink PRB usage of the NR cell $\leq$ Uplink PRB usage threshold (specified by the **NRDUCellPowerSaving.UlPrbThld** parameter)

 NOTE

- If the NR cell is a hyper cell or combined cell, the TRP with the highest PRB usage of the NR cell is used for the calculation.
 - If the NR cell is a DSS cell, the PRB usage equals the number of used RBs divided by the number of available RBs.
 - Active UEs are those in RRC_CONNECTED mode in an NR cell.
 - Downlink PRB usage of an NR cell = Average number of PRBs used by PDSCH DRBs/Average number of available downlink PRBs
 - Uplink PRB usage of an NR cell = Average number of PRBs used by PUSCH DRBs/Average number of available uplink PRBs
- In SUL co-deployment scenarios, the total number of SUL UEs in the NR cell is less than or equal to the value of the **NRDUCellPowerSaving.SulUserNumThld** parameter.
- In multi-operator sharing scenarios, operators are divided into normal operators and reserved operators for the NR cell and neighboring LTE cells.
 - The reserved operator of the NR cell is an operator with the **NRExternalNCellPlmn.PlmnReservedFlag** parameter set to **TRUE**.
 - The reserved operator of the LTE cell is an operator with the **CellOp.CellReservedForOp** parameter set to **CELL_RESERVED_FOR_OP** or **CELL_DYNAMIC_RESERVED_FOR_OP**.
- In multi-operator scenarios, primary operator recognition for LNR load-based dormancy is supported. This function is controlled by the **LNR_LOAD_DORM_PRIM_OP_RECOG_SW** option of the **gNBpwrSavingAlgo.PwrSavingAlgoSwitch** parameter.
 - If this option is deselected, the operator requirements are as follows:
 - The PLMN IDs of the normal operators of an NR cell must be a subset of the PLMN IDs of normal operators of neighboring LTE cells.
 - The PLMN IDs of the reserved operators of an NR cell must be a subset of the PLMN IDs of normal or reserved operators of neighboring LTE cells.
 - If this option is selected, the operator requirements are as follows:

- The PLMN ID of the primary operator of an NR cell served by normal operators must be obtained from the normal operators of neighboring LTE cells.
- The PLMN ID of the primary operator of an NR cell served by reserved operators must be obtained from the normal or reserved operators of neighboring LTE cells.
- The eNodeB sends the load information about the neighboring LTE cells of the NR cell to the gNodeB. The gNodeB requires that the carriers of neighboring cells be available.

If the **INTER_RAT_GRAD_PWR_SAVING_SW** option of the **gNBpwrSavingAlgo.PwrSavingAlgoSwitch** parameter is selected, some neighboring LTE cells can act as capacity-layer cells of other basic-layer carriers and enter the energy saving state due to functions such as intelligent power-off of carriers in the same coverage, Cell Switch-off Based on Automatic Co-coverage Identification, multi-carrier coordinated energy saving, GUL joint carrier shutdown, LTE load-based RF module deep dormancy, or dynamic carrier shutdown.

- The gNodeB checks whether a proportion of available neighboring cells (the proportion is specified by the **NRDUCellPowerSaving.EutranCarrShutdownRatio** parameter) meet the following conditions:
 - Downlink PRB usage of the neighboring LTE cell < E-UTRAN downlink PRB threshold (specified by the **NRDUCellPowerSaving.EutranDlPrbThld** parameter)
 - Uplink PRB usage of the neighboring LTE cell < Neighboring cell uplink PRB usage threshold (specified by the **NRDUCellPowerSaving.NCellUlPrbThld** parameter)
 - Number of NR UEs in the neighboring LTE cell < E-UTRAN NR user number threshold (specified by the **NRDUCellPowerSaving.EutranNrUserNumThld** parameter)
 - Downlink PRB usage of NR UEs in the neighboring LTE cell < E-UTRAN NR user downlink PRB threshold (specified by the **NRDUCellPowerSaving.EutranNrUserDlPrbThld** parameter)
 - Uplink PRB usage of NR UEs in the neighboring LTE cell < E-UTRAN NR user uplink PRB threshold (specified by the **NRDUCellPowerSaving.EutranNrUserUlPrbThld** parameter)

NOTE

Downlink PRB usage of an LTE cell = Average number of used downlink PRBs/Number of available downlink PRBs. The number of RBs for common channels is included in the calculation of downlink PRB usage.

Uplink PRB usage of an LTE cell = Average number of used uplink PRBs/Number of available uplink PRBs

2. A UE transfer procedure starts. The gNodeB sends system information to prohibit UE access and handovers to the NR cell and starts the UE transfer procedure with a specified duration.
 - In SA networking, UEs are transferred from the local cell to basic-layer neighboring cells as follows:

- Non-CA UEs: UEs not performing voice services are transferred to neighboring cells through inter-RAT handovers first. If the handovers fail, the UEs are transferred to inter-RAT neighboring cells through blind redirections. UEs performing voice services are handed over to inter-RAT neighboring cells.
 - CA UEs: If the local cell serves as an SCell of a CA UE, the SCell is removed. If the local cell serves as the PCell of a CA UE, the UE is transferred to a neighboring cell in the same way as a non-CA UE.
- If UEs cannot be transferred out of the NR cell within 30s after the UE transfer procedure starts (for example, due to handover failures or unsupported operating bands of the basic-layer cells), the procedure stops and cannot be re-triggered for the NR cell within the period specified by the **NRDUCellPowerSaving.PwrSavingPenaltyTime** parameter.
- In NSA networking, UE connections on the NR side are released as follows:
 - If energy saving takes effect in the PSCell of a UE, all connections on the NR side are released.
 - If energy saving takes effect in an SCell of a UE, its connection to this SCell on the NR side is released but its other connections on the NR side are retained.
3. The RF module enters deep dormancy.
- Once all UEs are transferred out of the NR cell, no RRC_CONNECTED UEs remain in the cell, and the cell and its serving RF module enter deep dormancy. If the RF module serves multiple NR cells, it enters deep dormancy only when load-based deep dormancy is triggered for all these NR cells. If the RF module also serves LTE cells, RF module deep dormancy must also be enabled on the LTE side.

Exiting the Deep Dormancy State

The base station periodically checks whether the dormancy exit conditions are met. The evaluation period granularity is specified by the **NRDUCellPowerSaving.EvalPeriodGranularity** parameter.

- If this parameter is set to **MINUTE_LEVEL**, the evaluation period is expressed in minutes. In this case, the period is specified by the **NRDUCellPowerSaving.ExitEvaluationPeriod** parameter.
- If this parameter is set to **SECOND_LEVEL**, the evaluation period is expressed in seconds. In this case, the period is specified by the **NRDUCellPowerSaving.SecondLevelExitEvalPrd** parameter.

The dormancy state is exited and the NR carrier is activated when any of the following conditions is met:

- The current time falls outside the shutdown period.
- Proportion of neighboring LTE cells that meet one of the following conditions $\geq$ **NRDUCellPowerSaving.EutranCarrShutdownRatio** – **NRDUCellPowerSaving.EutranCarrShutdownExitOfs**

- Downlink PRB usage of the neighboring LTE cell $\geq$ E-UTRAN downlink PRB threshold (specified by the **NRDUCellPowerSaving.EutranDlPrbThld** parameter) + E-UTRAN downlink PRB offset (specified by the **NRDUCellPowerSaving.EutranDlPrbOffset** parameter)
- Uplink PRB usage of the neighboring LTE cell $\geq$ Neighboring cell uplink PRB threshold (specified by the **NRDUCellPowerSaving.NCellUlPrbThld** parameter) + Neighboring cell uplink PRB offset (specified by the **NRDUCellPowerSaving.NCellUlPrbOffset** parameter)
- Number of NR UEs in the neighboring LTE cell $\geq$ E-UTRAN NR user number threshold (specified by the **NRDUCellPowerSaving.EutranNrUserNumThld** parameter) + E-UTRAN NR user number threshold offset (specified by the **NRDUCellPowerSaving.EutranNrUserNumThldOfs** parameter)
- Downlink PRB usage of NR UEs in the neighboring LTE cell $\geq$ E-UTRAN NR user downlink PRB threshold (specified by the **NRDUCellPowerSaving.EutranNrUserDlPrbThld** parameter) + E-UTRAN NR user downlink PRB threshold offset (specified by the **NRDUCellPowerSaving.EutranNrUserDlPrbThldOfs**)
- Uplink PRB usage of NR UEs in the neighboring LTE cell $\geq$ E-UTRAN NR user uplink PRB threshold (specified by the **NRDUCellPowerSaving.EutranNrUserUlPrbThld** parameter) + E-UTRAN NR user uplink PRB threshold offset (specified by the **NRDUCellPowerSaving.EutranNrUserUlPrbThldOfs** parameter)
- The PLMN IDs of the normal operators of the NR cell are not a subset of the PLMN IDs of normal operators of neighboring LTE cells.
- The PLMN IDs of the reserved operators of the NR cell are not a subset of the PLMN IDs of normal or reserved operators of neighboring LTE cells.
- When a neighboring LTE cell, functioning as a capacity-layer cell of other basic-layer carriers, is in the energy saving state, whether this cell is considered as a neighboring cell meeting the exit conditions depends on the setting of the **INTER_RAT_GRAD_PWR_SAVING_SW** option of the **gNBPwrSavingAlgo.PwrSavingAlgoSwitch** parameter.
 - If this option is deselected, this LTE cell is considered as a cell that meets the exit conditions.
 - If this option is selected, this LTE cell is not considered as a cell meeting the exit conditions and will not trigger the exit from the deep dormancy state.

Upon the exit from deep dormancy:

- The NR cell cannot re-enter deep dormancy within the penalty duration specified by the **NRDUCellPowerSaving.PwrSavingPenaltyTime** parameter if it exited dormancy due to uplink/downlink PRB usage or UE number exceptions, or unavailability of the LTE cell.
- A penalty is not imposed if the NR cell exited deep dormancy because the function switch was turned off, the effective period elapsed, or the base station software was upgraded.

 NOTE

- NR UEs in neighboring LTE cells of the NR cell are counted based on the UE capability (for example, whether a UE supports SA or NSA).
- In CA scenarios, only NR UEs on the primary component carrier (PCC) are counted.

7.2.1.5 LTE Load-based RF Module Deep Dormancy

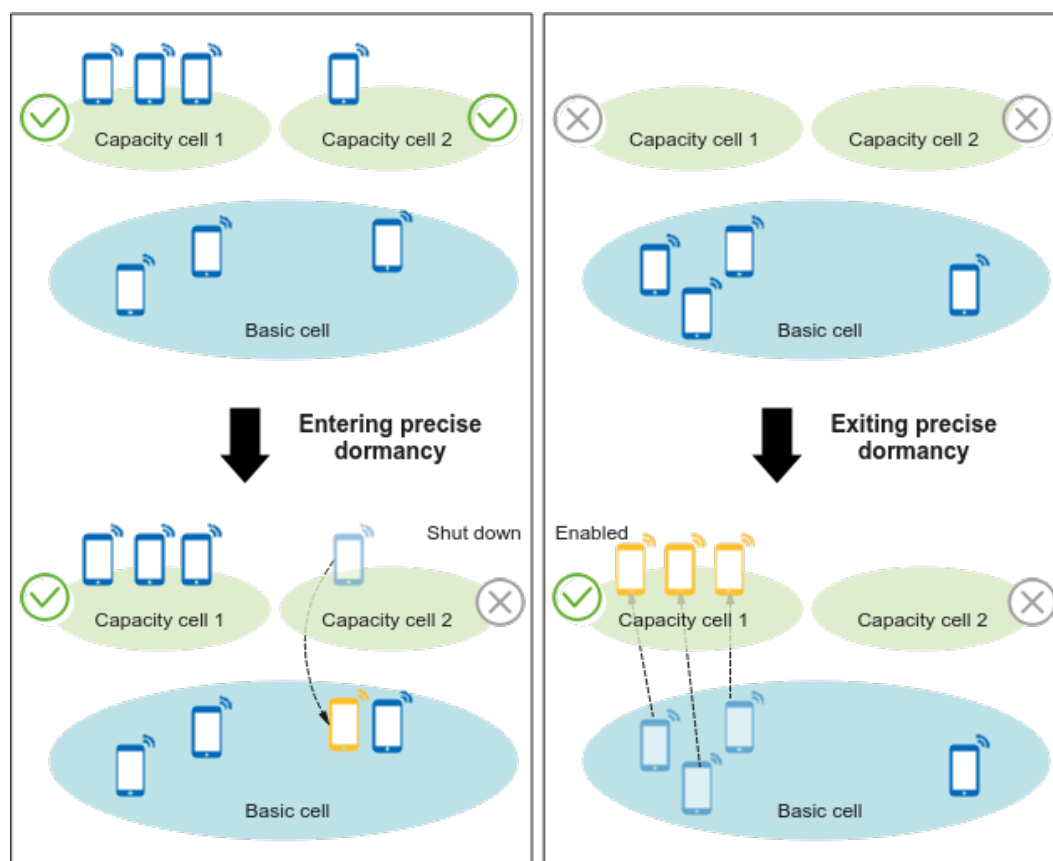
Multi-carrier co-coverage LTE networking provides a larger network capacity and, upon a heavy traffic load, a higher data throughput than single-carrier LTE networking. However, such networking wastes base station energy when the network is under light or no load. LTE load-based RF module deep dormancy is introduced to reduce the waste.

Load-based RF module deep dormancy enables the eNodeB to transfer UEs from capacity cells to basic cells and instruct the RF module serving the capacity cells to enter the deep dormancy state within a specified period when the network is under light or no load. When the load of the basic cells increases and reaches a specified threshold, the RF module in the dormancy state is automatically woken up and the capacity cells are restored.

LTE load-based RF module deep dormancy can be classified into non-precise dormancy mode and precise dormancy mode.

In scenarios where basic cells cover large areas, capacity cells cover small areas, one basic cell corresponds to multiple capacity cells, and UEs are unevenly distributed, the precise dormancy mode enables the base station to precisely identify cells that can enter or exit the dormancy state based on the UE distribution in the basic and capacity cells. This maximizes the probability of cells entering the dormancy state and minimizes the probability of cells exiting this state, prolonging the dormancy duration, as shown in [Figure 7-7](#).

Figure 7-7 Working principles for precise dormancy mode



This function requires the following switches to be configured:

- RF module-level switches:

The **RRU.DORMANCYSW (5G gNodeB, LTE eNodeB)** parameter is set to **ON** or **ON_DEEP_SUPER**, and either of the following conditions is met:

- The **RRU.CUSTOMDORMANCYSTDSW** parameter is set to **OFF**.
- The **RRU.CUSTOMDORMANCYSTDSW** parameter is set to **ON**, and the **LTE_TDD** or **LTE_FDD** option of the **RRU.DORMANCYSTD** parameter is selected.

The value **ON_DEEP_SUPER** of **RRU.DORMANCYSW (5G gNodeB, LTE eNodeB)** indicates that RF module deep dormancy and RF module super dormancy are both enabled. For details about RF module super dormancy, see [7.4 RF Module Super Dormancy](#).

- LTE cell-level switch:

- In non-precise dormancy mode, the **INTRA_LTE_DORMANCY_BY_LOAD_SW** option of the **CellPwrSavingAlgo.PwrSavingAlgoSwitch** parameter is selected.
- In precise dormancy mode, the **LTE_DORMANCY_BY_LOAD_PE_SW** option of the **CellPwrSavingAlgo.PwrSavingAlgoSwitch** parameter is selected.

The effective period and threshold parameters for LTE load-based RF module deep dormancy can be uniquely determined by using the combinations of the

CellPwrSavingPolicy.PolicyIndex and **CellPwrSavingPolicy.PowerSavingType** parameters in the **CellPwrSavingPolicy** MO on the LTE side. Specifically:

- The **CellPwrSavingPolicy.PolicyIndex** parameter specifies an energy-saving policy index. It uniquely identifies an energy-saving policy within the local cell. With the index specified, the corresponding policy can be added, modified, queried, or removed from the local cell.
- The **CellPwrSavingPolicy.PowerSavingType** parameter specifies the energy saving type indicated by an energy-saving policy index. LTE load-based dormancy requires that this parameter be set to **INTRA_LTE_DORMANCY_BY_LOAD**, and LTE load-based precise dormancy requires that this parameter be set to **LTE_PRECISE_DORMANCY_BY_LOAD**, meaning that under the corresponding energy-saving policy, the local cell performs load-based deep dormancy in the specified effective period under certain load conditions.

Related Concepts

- Basic-layer carriers provide basic coverage for the network. RF modules providing such carriers will not enter the dormancy state. The cells served by basic-layer carriers are referred to as basic cells.
- Capacity-layer carriers support large traffic volumes during peak hours. RF modules providing such carriers can enter the dormancy state. The cells served by capacity-layer carriers are referred to as capacity cells.

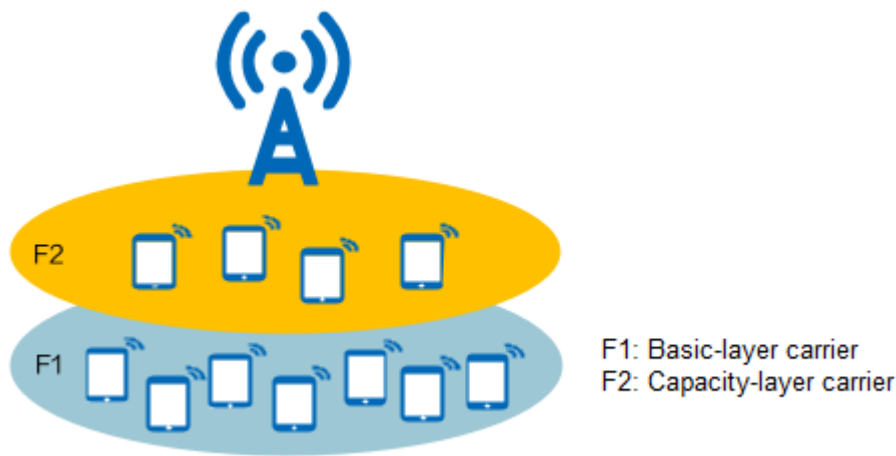
NOTE

Load-based RF module deep dormancy is enabled for capacity cells and corresponding RF modules. One capacity cell corresponds to one or more basic cells. Basic cells and capacity cells must be inter-frequency neighboring cells of each other.

Basic-layer and capacity cells are relative concepts identified by the **POWER_SAVING_OVERLAP_IND** option of the **EutranInterFreqNCell.OverlapIndicatorExtension** parameter during neighbor relationship configuration.

- If this option is deselected, there is no energy-saving relationship between the neighboring cell and the local cell. The neighboring cell is not considered in load-based RF module deep dormancy.
- If this option is selected, the neighboring cell is a basic cell, the local cell is a capacity cell, and the cells have overlapping coverage.

Figure 7-8 Basic cells and capacity cells



For example, on the capacity cell (on F2 in [Figure 7-8](#)) side, the **POWER_SAVING_OVERLAP_IND** option of the **EutranInterFreqNCell.OverlapIndicatorExtension** parameter is selected for the basic cell on F1, which is a neighboring cell.

In precise dormancy mode, independent configurations are used to identify capacity cells and basic cells.

- When the **CellPwrSavingAlgo.DormancyCellType** parameter is set to **BASIC_CELL** for a cell, the cell serves as a basic cell for load-based precise dormancy.
- When the **CellPwrSavingAlgo.DormancyCellType** parameter is set to **CAPACITY_CELL** for a cell, the cell serves as a capacity cell for load-based precise dormancy.

In LTE multi-carrier networking scenarios, it is recommended that capacity and basic cells be configured as described in [Table 7-4](#).

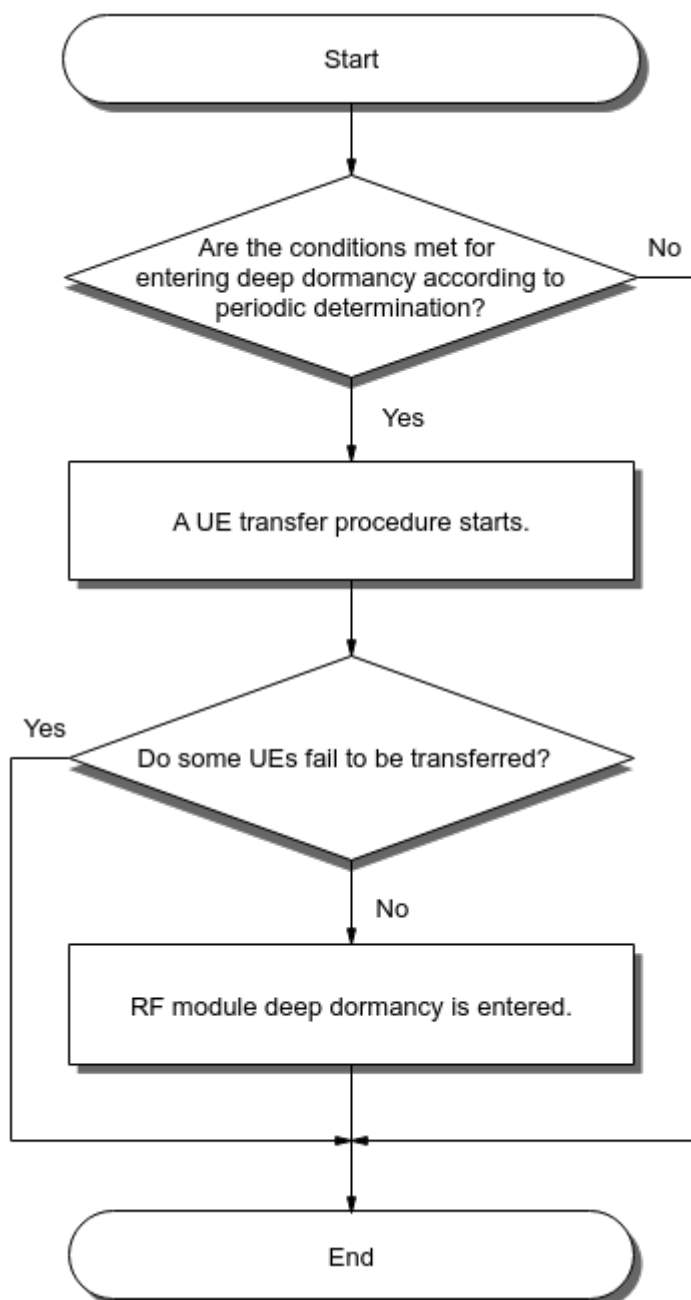
Table 7-4 Suggestions on designation of capacity and basic cells

Networking	Capacity Cell	Basic Cell
LTE TDD and LTE FDD coordinated networking	LTE TDD cells	LTE FDD cells
LTE TDD or LTE FDD intra-band multi-carrier networking	Cells on a higher frequency	Cells on a lower frequency

Entering the Deep Dormancy State

[Figure 7-9](#) shows how the RF module serving a capacity cell enters the deep dormancy state after LTE load-based RF module deep dormancy is enabled.

Figure 7-9 Procedure for entering LTE load-based RF module deep dormancy



1. The base station determines whether entering conditions are met.
The base station periodically checks whether the conditions for entering the deep dormancy state are met within an evaluation period specified by the **CellPwrSavingPolicy.TriggerJudgePeriod** parameter.
 - A capacity cell must meet all of the following conditions:
 - The current time is within the effective period for this function, which is determined by the **CellPwrSavingPolicy.StartTime** and **CellPwrSavingPolicy.StopTime** parameters.
 - The capacity cell is not in any energy saving penalty state.

- The basic cells corresponding to the capacity cell must meet one of the following conditions:
 - All basic cells are available.
 - Some basic cells are available and some basic cells are in the energy saving state. For example, these cells enter the energy saving state when functioning as capacity cells of other basic cells. (Whether this condition applies depends on the setting of the **INTRA_RAT_GRAD_PWR_SAVING_SW** option of the **EnodebPwrSavingAlgo.PwrSavingAlgoSwitch** parameter.)
- For common cells (including multi-RRU cells, FDD Dynamic Massive Beam cells (8T3S scenarios), and TDD cells split through soft split resource duplex (inter-frequency split)), and FDD Dynamic Massive Beam cells in massive MIMO scenarios, a capacity cell and its corresponding basic cell must both meet the following conditions:
 - In non-precise dormancy mode, the following conditions must also be met:
 - The total downlink PRB usage of the capacity and basic cells is less than or equal to the downlink PRB usage threshold specified by the **CellPwrSavingPolicy.DlPrbThld** parameter, where:
Total downlink PRB usage of the capacity cell and basic cells = $[(\text{Downlink PRB usage of the capacity cell} \times \text{System bandwidth of the capacity cell} \times \text{Number of downlink subframes per unit time}) + \sum_{\text{basic cell}} (\text{Downlink PRB usage of the basic cell} \times \text{System bandwidth of the basic cell} \times \text{Number of downlink subframes per unit time})] / \sum_{\text{basic cell}} (\text{System bandwidth of the basic cell} \times \text{Number of downlink subframes per unit time})$
 - The total uplink PRB usage of the capacity and basic cells is less than or equal to the uplink PRB usage threshold specified by the **CellPwrSavingPolicy.UlPrbThld** parameter, where:
Total uplink PRB usage of the capacity cell and basic cells = $[(\text{Uplink PRB usage of the capacity cell} \times \text{System bandwidth of the capacity cell} \times \text{Number of uplink subframes per unit time}) + \sum_{\text{basic cell}} (\text{Uplink PRB usage of the basic cell} \times \text{System bandwidth of the basic cell} \times \text{Number of uplink subframes per unit time})] / \sum_{\text{basic cell}} (\text{System bandwidth of the basic cell} \times \text{Number of uplink subframes per unit time})$
 - In precise dormancy mode, the following conditions must also be met:
 - Uplink PRB usage in the co-coverage area < Maximum uplink PRB usage in the co-coverage area after the local cell is shut down – **CellPwrSavingPolicy.UlPrbOffset** – 5%
 - Downlink PRB usage in the co-coverage area < Maximum downlink PRB usage in the co-coverage area after the local cell is shut down – **CellPwrSavingPolicy.DlPrbOffset** – 5%
 - Uplink PRB usage of each basic-layer carrier corresponding to the capacity-layer carrier < **CellPwrSavingPolicy.ProtectUlPrbThld** – **CellPwrSavingPolicy.ProtectUlPrbOffset**

Uplink PRB usage of each basic-layer carrier corresponding to the capacity-layer carrier = Uplink PRB usage of the basic cell + Overlap rate x Uplink PRB usage of the capacity cell

- Downlink PRB usage of each basic-layer carrier corresponding to the capacity-layer carrier <

CellPwrSavingPolicy.ProtectDlPrbThld – CellPwrSavingPolicy.ProtectDlPrbOffset

Downlink PRB usage of each basic-layer carrier corresponding to the capacity-layer carrier = Downlink PRB usage of the basic cell + Overlap rate x Downlink PRB usage of the capacity cell

The overlap rate is the proportion of overlapping area in a capacity cell to a basic cell, which is specified by the **EutranInterFreqNCell.OverlapRate** parameter at the basic cell side.

NOTE

Downlink PRB usage of an LTE cell = Average number of used downlink PRBs/ Number of available downlink PRBs. The number of RBs for common channels is included in the calculation of downlink PRB usage.

Uplink PRB usage of an LTE cell = Average number of used uplink PRBs/Number of available uplink PRBs

In precise dormancy mode:

- Total downlink PRB usage of the capacity cell and basic cells = [(Downlink PRB usage of the capacity cell x System bandwidth of the capacity cell x Number of downlink subframes per unit time) + $\sum_{\text{basic cell}} (\text{Overlap rate} \times \text{Downlink PRB usage of the basic cell} \times \text{System bandwidth of the basic cell} \times \text{Number of downlink subframes per unit time})$]/[System bandwidth of the capacity cell x Number of downlink subframes per unit time + $\sum_{\text{basic cell}} (\text{Overlap rate} \times \text{System bandwidth of the basic cell} \times \text{Number of downlink subframes per unit time})$]
- Total uplink PRB usage of the capacity cell and basic cells = [(Uplink PRB usage of the capacity cell x System bandwidth of the capacity cell x Number of uplink subframes per unit time) + $\sum_{\text{basic cell}} (\text{Overlap rate} \times \text{Uplink PRB usage of the basic cell} \times \text{System bandwidth of the basic cell} \times \text{Number of uplink subframes per unit time})$]/[System bandwidth of the capacity cell x Number of uplink subframes per unit time + $\sum_{\text{basic cell}} (\text{Overlap rate} \times \text{System bandwidth of the basic cell} \times \text{Number of uplink subframes per unit time})$]

The overlap rate between capacity cells and basic cells for precise dormancy is specified by the **EutranInterFreqNCell.OverlapRate** parameter.

- The number of RRC_CONNECTED UEs in the capacity cell is less than or equal to the UE number threshold specified by the **CellPwrSavingPolicy.UENumThld** parameter.
- In multi-operator sharing scenarios, the PLMN IDs of the operators serving the capacity cell are included in the PLMN ID set of operators of at least one basic cell.
- In multi-operator sharing scenarios, the operators of the local cell must be a subset of the operators of co-coverage neighboring basic cells. If optimization for cell PLMN reservation scenarios is enabled by selecting the **CELL_PLMN_RSVD_OPT_SW** option of the **EnodebPwrSavingAlgo.PwrSavingAlgoSwitch** parameter:

- If the operators of the local cell are non-reserved operators, the operators must be a subset of non-reserved operators of the co-coverage neighboring basic cells.
- If the operators of the local cell are reserved operators, the operators must be a subset of non-reserved operators or reserved operators of the co-coverage neighboring basic cells.

The operator status of the local cell and intra-base-station co-coverage neighboring basic cells is obtained as follows:

- Using the effective operator status of the cells if the **RSVD_FOR_OP_X2_BROADCAST_SW** option of the **eNodeBOamParam.ConfigAdaptOptSwitch** parameter is selected
- Using the value of the **CellOp.CellReservedForOp** parameter if the **RSVD_FOR_OP_X2_BROADCAST_SW** option of the **eNodeBOamParam.ConfigAdaptOptSwitch** parameter is deselected

The operator status of inter-base-station co-coverage neighboring basic cells is obtained as follows:

- Using the value of the **PLMN_RESERVED_FLAG** option of the **EutranExternalCellPlmn.AggregationAttribute** parameter if the **RSVD_FOR_OP_X2_BROADCAST_SW** option of the **eNodeBOamParam.ConfigAdaptOptSwitch** parameter is deselected
- Using the following methods if the **RSVD_FOR_OP_X2_BROADCAST_SW** option of the **eNodeBOamParam.ConfigAdaptOptSwitch** parameter is selected:

Method 1: Using the **PLMN_RESERVED_FLAG** option of the **EutranExternalCellPlmn.AggregationAttribute** parameter

Method 2: The operator status of co-coverage neighboring basic cells is automatically obtained through broadcast over the X2 interface when neighbor relationships are properly configured and the X2 interface between base stations is working properly.

If both of the preceding methods are available, the operator status obtained through these methods must meet the triggering conditions.

- (When the capacity cell and basic cells are provided by different base stations) The X2 interface is normal.
- For sector split cells (excluding FDD Dynamic Massive Beam cells in massive MIMO scenarios), a capacity cell and its corresponding basic cell must meet the following conditions:
 - Uplink PRB usage of the local cell + Uplink PRB usage of co-coverage neighboring basic cells + Uplink PRB usage of other cells in the same sector split group < Uplink PRB threshold for starting intelligent power-off of carriers in the same coverage
 - Downlink PRB usage of the local cell + Downlink PRB usage of co-coverage neighboring basic cells + Downlink PRB usage of other cells

in the same sector split group < Downlink PRB threshold for starting intelligent power-off of carriers in the same coverage

- Uplink PRB usage of the local cell + Uplink PRB usage of a co-coverage neighboring basic cell + Uplink PRB usage of other cells in the same sector split group < Threshold for a co-coverage neighboring basic cell to start load adjustment of inter-frequency load balancing (**CellMLB.InterFreqMlbThd** plus **CellMLB.LoadOffset**)
- Downlink PRB usage of the local cell + Downlink PRB usage of a co-coverage neighboring basic cell + Downlink PRB usage of other cells in the same sector split group < Threshold for a co-coverage neighboring basic cell to start load adjustment of inter-frequency load balancing (**CellMLB.InterFreqMlbThd** plus **CellMLB.LoadOffset**)

All cells in a sector split group are enabled with load-based dormancy and meet the triggering conditions.

NOTE

Sector split cells cannot be configured as capacity cells for LTE load-based RF module dormancy in precise dormancy mode.

2. A UE transfer procedure starts.

- The capacity cell prohibits the access of new UEs and incoming handovers.

The eNodeB broadcasts the barred state of the capacity cell through SIB1. In addition, the eNodeB sends an eNodeB Configuration Update message over the X2 interface to each neighboring eNodeB. The message contains the *Deactivation Indication* IE, indicating that the local cell is unavailable due to energy saving.

- The eNodeB attempts to hand over all RRC_CONNECTED UEs from the capacity cell to the cells on the operating frequencies of the co-coverage neighboring basic cells within 60s.

The eNodeB delivers event A4 measurement configurations about operating frequencies of the co-coverage basic cells to UEs in connected mode. The RSRP threshold for event A4 is specified by the **InterFreqHoGroup.InterFreqHoA4ThdRsrp** parameter. After receiving event A4 measurement reports from UEs, the eNodeB initiates inter-frequency handovers to the cells on those frequencies for the UEs.

During the handover attempt, the eNodeB checks the number of RRC_CONNECTED UEs in the capacity cell every 5s.

- If the number of RRC_CONNECTED UEs is 0, the capacity cell enters the dormancy state.
- If the number of RRC_CONNECTED UEs is not 0, the eNodeB checks whether the number of RRC_CONNECTED UEs is less than or equal to the value of the **CellPwrSavingPolicy.ForcibleTriggerUeNumThld** parameter and whether there are no UEs performing QCI-1 services in the capacity cell after 60s elapses. If so, the capacity cell enters the dormancy state. If not, the eNodeB cancels the barred state of the capacity cell

and notifies all neighboring eNodeBs over the X2 interfaces that the capacity cell has exited the energy saving state. It does not check the entering conditions again until the penalty period specified by the **CellPwrSavingPolicy.PenaltyTime** parameter elapses.

3. The RF module enters the deep dormancy state.

Once all UEs are transferred out of the capacity cell, no RRC_CONNECTED UEs remain in the cell, and the cell and the corresponding RF module enter the deep dormancy state. If an RF module serves multiple LTE cells, it enters the deep dormancy state only when load-based deep dormancy is triggered for all these cells. If an RF module also serves NR cells, RF module deep dormancy must also be enabled on the NR side.

If intra-RAT level-by-level energy saving is enabled by selecting the **INTRA_RAT_GRAD_PWR_SAVING_SW** option of the **EnodebPwrSavingAlgo.PwrSavingAlgoSwitch** parameter, some basic cells can be used as capacity cells of other basic cells and enter the energy saving state due to functions such as intelligent power-off of carriers in the same coverage, Cell Switch-off Based on Automatic Co-coverage Identification, multi-carrier coordinated energy saving, GUL joint carrier shutdown, LTE load-based RF module deep dormancy, or dynamic carrier shutdown.

Exiting the Deep Dormancy State

The base station periodically checks whether the conditions for exiting the dormancy state are met. The exit evaluation period is specified by the **CellPwrSavingPolicy.ExitJudgePeriod** parameter. Within an exit evaluation period, the base station evaluates whether the conditions are met at an interval specified by the **CellPwrSavingPolicy.EvalInterval** parameter. The cell can exit the power saving state only when the conditions are met at each evaluation occasion within the period.

- If the **CellPwrSavingPolicy.EvalInterval** parameter is set to **EVAL_INTERVAL_60S**, the exit evaluation is performed at an interval of 60s in the exit evaluation period. In this case, the **CellPwrSavingPolicy.ExitJudgePeriod** parameter must be set to an integer multiple of 12 (unit: 5s), that is, the exit evaluation period must be an integer multiple of 1 minute.
- If the **CellPwrSavingPolicy.EvalInterval** parameter is set to **EVAL_INTERVAL_5S**, the exit evaluation is performed at an interval of 5s in the exit evaluation period. In this case, the **CellPwrSavingPolicy.ExitJudgePeriod** parameter must be set to a value within the range of 1 to 120 (unit: 5s), that is, the exit evaluation period must be an integer multiple of 5s.

To support second-level evaluation periods, the **EnodebPwrSavingAlgo.InterEnbInfoExchangeMode** parameter has been added to specify how frequently information is exchanged between base stations. This enables base stations to obtain the latest load information more quickly, allowing them to evaluate whether the conditions for exiting the dormancy state are met.

- If this parameter is set to **SLOW_EXCHANGE_MODE**, the slow exchange mode is used, that is, information is exchanged at a long interval.
- If this parameter is set to **FAST_EXCHANGE_MODE**, the fast exchange mode is used, that is, information is exchanged at a short interval.

A capacity cell and the corresponding RF module automatically exit the deep dormancy state when any of the following conditions is met:

- The current time is not within the effective period for this function, which is determined by the **CellPwrSavingPolicy.StartTime** and **CellPwrSavingPolicy.StopTime** parameters.
- The total downlink PRB usage of basic cells is greater than the sum of the **CellPwrSavingPolicy.DlPrbThld** and **CellPwrSavingPolicy.DlPrbOffset** parameter values.
- The total uplink PRB usage of basic cells is greater than the sum of the **CellPwrSavingPolicy.UlPrbThld** and **CellPwrSavingPolicy.UlPrbOffset** parameter values.
- In precise dormancy mode, uplink PRB usage in the co-coverage area $\geq$ Maximum uplink PRB usage in the co-coverage area after the local cell is shut down – 5%
- In precise dormancy mode, downlink PRB usage in the co-coverage area $\geq$ Maximum downlink PRB usage in the co-coverage area after the local cell is shut down – 5%
- In precise dormancy mode, the uplink PRB usage of any basic-layer carrier corresponding to the capacity-layer carrier is greater than or equal to the value of the **CellPwrSavingPolicy.ProtectUlPrbThld** parameter.

Uplink PRB usage of a basic-layer carrier corresponding to the capacity-layer carrier = Number of used uplink PRBs in the basic cell/Total number of available uplink PRBs in the basic cell

- In precise dormancy mode, the downlink PRB usage of any basic-layer carrier corresponding to the capacity-layer carrier is greater than or equal to the value of the **CellPwrSavingPolicy.ProtectDlPrbThld** parameter.

Downlink PRB usage of a basic-layer carrier corresponding to the capacity-layer carrier = Number of used downlink PRBs in the basic cell/Total number of available downlink PRBs in the basic cell

- A basic cell is added for the capacity cell.
- In multi-operator sharing scenarios, the PLMN IDs of the operators serving the capacity cell are not included in the PLMN ID set of operators serving any basic cell.
- In multi-operator sharing scenarios, the operators of the local cell are not a subset of the operators of co-coverage neighboring basic cells. If optimization for cell PLMN reservation scenarios is enabled by selecting the **CELL_PLMN_RSVD_OPT_SW** option of the **ENodebPwrSavingAlgo.PwrSavingAlgoSwitch** parameter:
 - If the operators of the local cell are non-reserved operators, the operators are not a subset of non-reserved operators of the co-coverage neighboring basic cells.
 - If the operators of the local cell are reserved operators, the operators are not a subset of non-reserved operators or reserved operators of the co-coverage neighboring basic cells.

The operator status of the local cell and intra-base-station co-coverage neighboring basic cells is obtained as follows:

- Using the effective operator status of the cells if the **RSVD_FOR_OP_X2_BROADCAST_SW** option of the **eNodeBOamParam.ConfigAdaptOptSwitch** parameter is selected
- Using the value of the **CellOp.CellReservedForOp** parameter if the **RSVD_FOR_OP_X2_BROADCAST_SW** option of the **eNodeBOamParam.ConfigAdaptOptSwitch** parameter is deselected

The operator status of inter-base-station co-coverage neighboring basic cells is obtained as follows:

- Configuring through the **PLMN_RESERVED_FLAG** option of the **EutranExternalCellPlmn.AggregationAttribute** parameter if the **RSVD_FOR_OP_X2_BROADCAST_SW** option of the **eNodeBOamParam.ConfigAdaptOptSwitch** parameter is deselected
- Using the following methods if the **RSVD_FOR_OP_X2_BROADCAST_SW** option of the **eNodeBOamParam.ConfigAdaptOptSwitch** parameter is selected:

Method 1: Using the **PLMN_RESERVED_FLAG** option of the **EutranExternalCellPlmn.AggregationAttribute** parameter

Method 2: The operator status of co-coverage neighboring basic cells is automatically obtained through broadcast over the X2 interface when neighbor relationships are properly configured and the X2 interface between base stations is working properly.

If both of the preceding methods are available, the operator status obtained through these methods must meet the triggering conditions.

- (When the capacity and basic cells are provided by different base stations) The X2 interface is abnormal.
- Any basic cell becomes unavailable due to other reasons, such as cell faults and cell deactivation. This condition does not apply when the **INTRA_RAT_GRAD_PWR_SAVING_SW** option of the **EnodebPwrSavingAlgo.PwrSavingAlgoSwitch** parameter is selected and basic cells become unavailable due to carrier shutdown for energy saving or RF module deep dormancy.

Upon the exit from deep dormancy:

- The capacity cell cannot enter the deep dormancy state again within the penalty period specified by the **CellPwrSavingPolicy.PenaltyTime** parameter if it exited this state because of the uplink or downlink PRB usage, unavailability of a basic cell, or abnormal X2 interface.
- The penalty mechanism is not triggered for the capacity cell if:
 - Deep dormancy was exited because the function switch was turned off or the energy saving period had elapsed.
 - Deep dormancy became pending due to base station software upgrade or base station configuration.

7.2.1.6 UMTS Load-based RF Module Deep Dormancy

Multi-carrier co-coverage UMTS networking provides a larger network capacity and, upon a heavy traffic load, a higher data throughput than single-carrier UMTS networking. However, multi-carrier networking wastes base station energy when the network is under light or no load. To address this, UMTS load-based RF module deep dormancy is introduced.

Load-based RF module deep dormancy enables the RNC to transfer UEs from capacity-layer cells to basic-layer cells and instruct the RF modules serving the capacity-layer cells to enter deep dormancy within a specified period when the network is under light or no load. When the load of the basic-layer cells increases to a specified threshold, the dormant RF modules are automatically woken and the capacity-layer cells are restored.

This function requires the following switches to be configured:

- RF module-level switches

The **RRU.DORMANCYSW (5G gNodeB, LTE eNodeB)** parameter is set to **ON** or **ON_DEEP_SUPER**, and either of the following conditions is met:

- The **RRU.CUSTOMDORMANCYSTDSW (5G gNodeB, LTE eNodeB)** parameter is set to **OFF**.
- The **RRU.CUSTOMDORMANCYSTDSW (5G gNodeB, LTE eNodeB)** parameter is set to **ON**, and the **UMTS** option of the **RRU.DORMANCYSTD (5G gNodeB, LTE eNodeB)** parameter is selected.

The value **ON_DEEP_SUPER** of **RRU.DORMANCYSW (5G gNodeB, LTE eNodeB)** indicates that RF module deep dormancy and RF module super dormancy can be both enabled. For details about RF module super dormancy, see [7.4 RF Module Super Dormancy](#).

- UMTS cell-level switch

- The **DORMANCY_BY_LOAD_SW** option of the **ULOCCELLALGPARA.DORMANCYALGOSW** parameter is selected.
- The effective period and threshold parameters for UMTS load-based RF module deep dormancy can be uniquely determined by using the setting combinations of the **UCELLDORMANCYPOLICY.PolicyIndex** and **UCELLDORMANCYPOLICY.DormancyType** parameters in the **UCELLDORMANCYPOLICY** MOs on the RNC side. Where,
 - The **UCELLDORMANCYPOLICY.PolicyIndex** parameter specifies an energy saving policy index. It uniquely identifies an energy saving policy within the local cell. With the index specified, the corresponding policy can be added, modified, queried, or removed at the local cell level.
 - The **UCELLDORMANCYPOLICY.DormancyType** parameter specifies the energy saving type indicated by an energy saving policy index. UMTS load-based dormancy requires this parameter to be set to **INTRA_DORMANCY_BY_LOAD**. Then under the corresponding energy saving policy index, the local cell performs load-based dormancy in the preset duration based on the load of the local cell and that of inter-band neighboring cells.

- RNC-level switch

The **HO_INTER_FREQ_HARD_HO_SWITCH** option of the **UCORRMALGOSWITCH.HoSwitch** parameter is selected to ensure that inter-frequency handovers can be performed.

- For dormancy involving UMTS, the **GU_DC_SW** option of the base-station-specific **EQUIPMENT.EQMDOROPTFEATSW** parameter must be deselected. If the option is deselected and a configuration command is executed to trigger dormancy pending or dormancy pending is enabled on the local end that

works in UMTS, then dormancy pending will take effect on the entire base station.

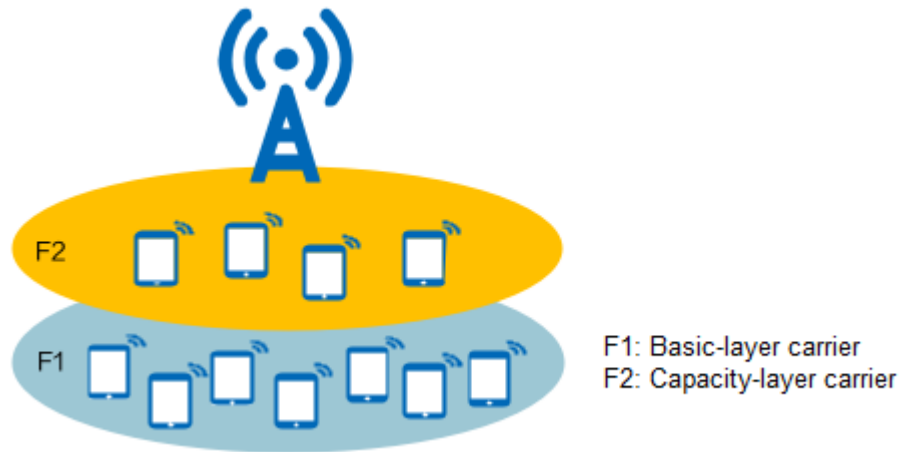
Related Concepts

- Basic-layer carriers provide basic coverage for the network. RF modules supporting these carriers do not enter dormancy. Cells served by basic-layer carriers are referred to as basic-layer cells.
- Capacity-layer carriers handle high traffic volumes during peak hours. RF modules supporting these carriers may enter dormancy. Cells served by capacity-layer carriers are referred to as capacity-layer cells.

NOTE

Load-based RF module dormancy is enabled for capacity-layer cells and their associated RF modules. A capacity-layer cell can be associated with one or more basic-layer cells. Basic-layer cells and capacity-layer cells must be inter-frequency neighboring cells of each other.

Figure 7-10 Basic-layer and capacity-layer cells



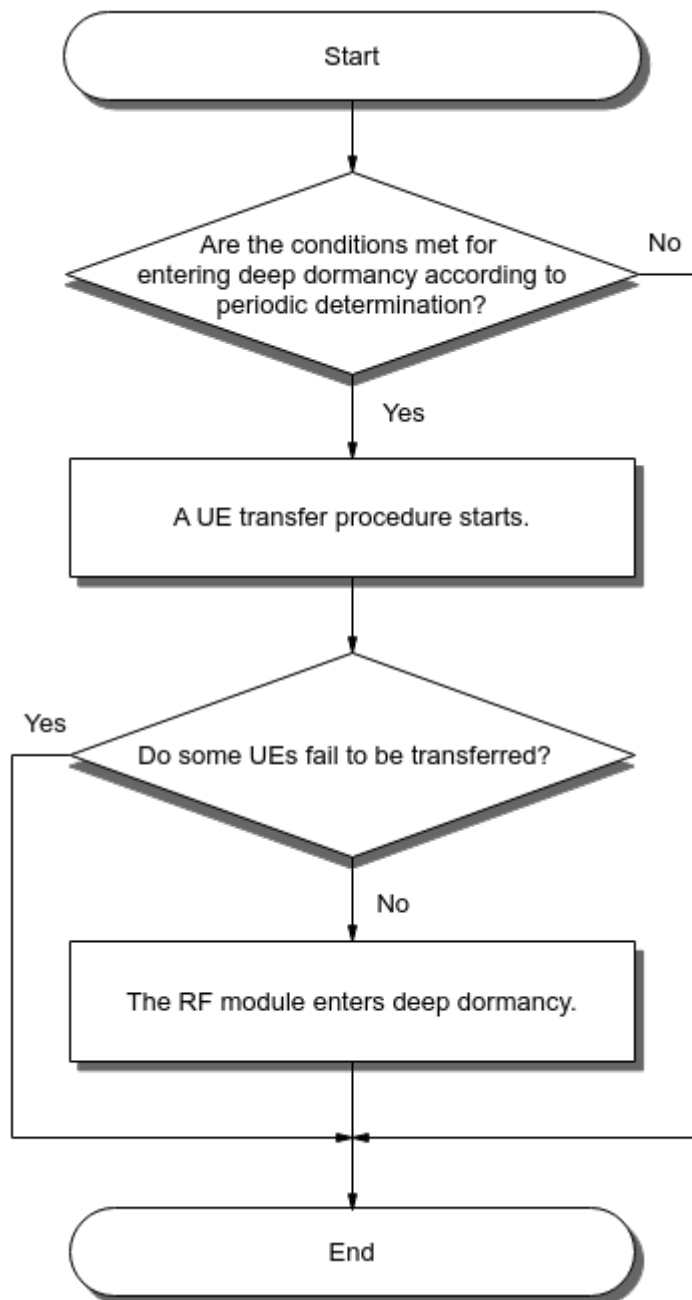
The basic-layer cell must meet the following requirements:

- It has a co-coverage neighbor relationship with the capacity-layer cell. That is, the **UINTERFREQNCCELL.BlindHoFlag** parameter is set to **TRUE** for the basic-layer cell.
- It operates in a different frequency band from the capacity-layer cell.
- It belongs to the same logical RNC as the capacity-layer cell.
- The **UCELL.DormancyCellType** parameter is set to **BASIC_CELL** for this cell.
- In an intra-RAT multi-operator sharing scenario where a capacity-layer cell and its co-coverage neighboring cells belong to different operators, a co-coverage neighboring cell can serve as a basic-layer cell (the target cell for UE handover when the local cell is about to enter the energy saving state) when **URNCCCELLDORMANCYPARA.CellInterPlmnHoSwitch** is set to **ON** and **UOPERATORSHARINGMODE.InterPlmnHoAllowedIntraRat** is set to **YES**.

Entering the Deep Dormancy State

Figure 7-11 shows how the RF module serving a capacity-layer cell enters dormancy after UMTS load-based RF module deep dormancy is enabled.

Figure 7-11 Procedure for entering UMTS load-based RF module deep dormancy



1. The RNC determines whether the entry conditions are met.
The RNC periodically checks whether the dormancy entry conditions are met within an evaluation period specified by the **UCELLDORMANCYPOLICY.TriggerJudgePeriod** parameter.
 - A capacity-layer cell must meet all of the following conditions:
 - The current time falls within the effective energy saving period determined by the **UCELLDORMANCYPOLICY.StartTime** and **UCELLDORMANCYPOLICY.StopTime** parameters.
 - The capacity-layer cell is not in an energy saving penalty state.

- No UEs performing emergency calls exist in the capacity-layer cell.
- By default, cell broadcast service (CBS) takes precedence over dormancy. Therefore, when CBS is enabled in a cell for which the **URNCCCELLDORMANCYPARA.CellIgnoreCbsSwitch** parameter is set to **OFF** (default value), the cell cannot enter dormancy. If dormancy is required in scenarios where CBS is enabled, this parameter must be set to **ON**. Once it is set to **ON**:
 - If no CBS message is sent and the current time falls outside the CBS protection period (specified by the **URNCCCELLDORMANCYPARA.CbsServiceHoldDuration** parameter), the cell can enter dormancy.
 - If CBS messages are sent, the cell cannot enter dormancy.

 NOTE

After receiving a CBS message, the RNC immediately wakes the dormant cell and processes the CBS message once the cell is successfully set up. Because the cell requires time to wake, CBS messages may not be processed promptly. If CBS has a higher priority, it is recommended that **URNCCCELLDORMANCYPARA.CellIgnoreCbsSwitch** be set to its default value **OFF**. For details about CBS, see *CBS* in *RAN Feature Documentation*.

- The number of HSDPA UEs is less than or equal to the value of **UCELLDORMANCYPOLICY.HsdpaUserNumThd**.
- The number of HSUPA UEs is less than or equal to the value of **UCELLDORMANCYPOLICY.HsupaUserNumThd**.
- In multi-carrier HSDPA cell scenarios, the number of multi-carrier HSDPA UEs served by a multi-carrier HSDPA cell is less than or equal to the value of **UCELLDORMANCYPOLICY.McHsdpaUserNumThd**.
- When **UCELLDORMANCYPOLICY.DormancyCchUserSwitch** is set to **ON**, no UEs are in the CELL_FACH state, and the number of UEs in the CELL_DCH state is less than or equal to the value of **UCELLDORMANCYPOLICY.TotalUserNumThd**.
- When **UCELLDORMANCYPOLICY.DormancyCchUserSwitch** is set to **OFF**, the total number of UEs in the CELL_DCH and CELL_FACH states is less than or equal to the value of **UCELLDORMANCYPOLICY.TotalUserNumThd**.
- The basic-layer cells associated with a capacity-layer cell must meet all the following conditions:
 - None of the basic-layer cells are in the LDR or OLC state.
 - All basic-layer cells are free of faults (such as FACH faults) and not manually deactivated.
 - At least one basic-layer cell meets the following load margin conditions:
 - The uplink power load of the neighboring cell is less than the value of **UCELLLDLM.UllDrTrigThd** minus the value of **UCELLDORMANCYPOLICY.NCellLdrRemainThd**.

- The downlink power load of the neighboring cell is less than the value of **UCELLLDM.DLLdrTrigThd** minus the value of **UCELLDORMANCYPOLICY.NCellLdrRemainThd**.
- The value of **UCELLLDM.ULLdrTrigThd** is greater than the value of **UCELLDORMANCYPOLICY.NCellLdrRemainThd**.
- The value of **UCELLLDM.DLLdrTrigThd** is greater than the value of **UCELLDORMANCYPOLICY.NCellLdrRemainThd**.

 NOTE

The uplink and downlink power loads of a cell are related to the cell's admission control algorithm. For details, see *Call Admission Control in RAN Feature Documentation*.

2. A UE transfer procedure starts.

- The capacity-layer cell prohibits the access of new UEs and incoming handovers.

The RNC broadcasts the barred state of the capacity-layer cell in SIB3.

- The RNC attempts to hand over all UEs in the CELL_DCH and CELL_FACH states from the capacity-layer cell to its co-coverage basic-layer cells within 60s.

The RNC selects basic-layer cells that meet the load margin conditions as the candidate set for handover.

- For CELL_DCH UEs, the RNC delivers load-based inter-frequency measurement configurations in which the measurement frequencies are those of inter-band neighboring cells. After receiving a measurement report from a UE, the RNC hands over the UE to an inter-band neighboring cell with signal quality higher than the RSCP (**UCELLMCLDR.TargetFreqThdRscp**) and Ec/No (**UCELLMCLDR.TargetFreqThdEcNo**) thresholds of the local cell.
 - For UEs not performing CS services, measurement-based handovers to inter-band neighboring cells are executed, and if the handovers fail, redirections to inter-band neighboring cells are executed.
 - For UEs performing CS services, measurement-based handovers to inter-band neighboring cells are executed, and if the handovers fail, measurement-based handovers to inter-band neighboring cells are executed again.

- For CELL_FACH UEs, the RNC initiates redirections to the basic-layer cells.

During inter-frequency handover and redirection, the RNC periodically checks the number of UEs in the CELL_DCH and CELL_FACH states in the capacity-layer cell.

- If the number of UEs is 0, the capacity-layer cell enters dormancy.
- If the number of UEs is not 0, the RNC checks the number of UEs in the capacity-layer cell after 60s. If this number is less than or equal to the value of the **UCELLDORMANCYPOLICY.ForcibleTriggerUeNumThld** parameter, the capacity-layer cell enters dormancy. Otherwise, the RNC cancels

the barred state of the capacity-layer cell, and the cell enters the energy saving penalty state and exits the energy saving state. The RNC does not recheck the entry conditions until the penalty period specified by the **UCELLDORMANCYPOLICY.PenaltyTime** parameter has elapsed.

3. The RF module enters deep dormancy.

Once all UEs are transferred out of the capacity-layer cell, the cell and its serving RF module enter deep dormancy. At the same time, the system starts to check the dormancy exit conditions.

Exiting the Deep Dormancy State

The base station periodically checks whether the dormancy exit conditions are met. The exit evaluation period is specified by the **UCELLDORMANCYPOLICY.ExitJudgePeriod** parameter.

A capacity-layer cell and its serving RF module automatically exit dormancy when any of the following conditions is met:

- The current time falls outside the effective energy saving period determined by the **UCELLDORMANCYPOLICY.StartTime** and **UCELLDORMANCYPOLICY.StopTime** parameters.
- The capacity-layer cell receives a CBS message from the CBS server.
- There is no associated inter-band basic-layer cell.
- Any associated basic-layer cell meets either of the following conditions:
 - The basic-layer cell is in the LDR or OLC state.
 - The basic-layer cell is faulty or manually deactivated.

Upon the exit from dormancy:

- The capacity-layer cell wakes and can provide services properly.
- If the wakeup is triggered by a CBS message, the cell is not allowed to enter dormancy during the CBS hold duration (specified by **URNCCCELLDORMANCYPARA.CbsServiceHoldDuration**).

7.2.1.7 GU Load-based RF Module Deep Dormancy

UMTS cells in operation still consume energy when there is light or even no traffic in the area covered by both GSM and UMTS networks, wasting base station energy. To address this, GU load-based RF module deep dormancy is introduced.

Load-based RF module deep dormancy enables the RNC to transfer UEs from UMTS cells to GSM cells and instruct the RF modules serving the UMTS cells to enter deep dormancy within a specified period when the network is under light or no load. When the load of GSM cells increases to a specified threshold, the dormant RF modules are automatically woken and the UMTS cells are restored.

This function requires the following switches to be configured:

- RF module-level switches

The **RRU.DORMANCYSW** parameter is set to **ON** or **ON_DEEP_SUPER**, and either of the following conditions is met:

- The **RRU.CUSTOMDORMANCYSTDSW** parameter is set to **OFF**.
- The **RRU.CUSTOMDORMANCYSTDSW** parameter is set to **ON**, and the **UMTS** option of the **RRU.DORMANCYSTD** parameter is selected.

The value **ON_DEEP_SUPER** of **RRU.DORMANCYSW** indicates that RF module deep dormancy and RF module super dormancy can be both enabled. For details about RF module super dormancy, see [7.4 RF Module Super Dormancy](#).

- UMTS cell-level switch
 - The **DORMANCY_BY_LOAD_SW** option of the **ULOCCELLALGPARA.DORMANCYALGOSW** parameter is selected.
 - The effective period and threshold parameters for GU load-based RF module deep dormancy can be uniquely determined by using the setting combinations of the **UCELLDORMANCYPOLICY.PolicyIndex** and **UCELLDORMANCYPOLICY.DormancyType** parameters in the **UCELLDORMANCYPOLICY** MOs on the RNC side. Where,
 - The **UCELLDORMANCYPOLICY.PolicyIndex** parameter specifies an energy saving policy index. It uniquely identifies an energy saving policy within the local cell. With the index specified, the corresponding policy can be added, modified, queried, or removed at the local cell level.
 - The **UCELLDORMANCYPOLICY.DormancyType** parameter specifies the energy saving type indicated by an energy saving policy index. GU load-based RF module deep dormancy requires this parameter to be set to **INTER_GU_DORMANCY_BY_LOAD**. Then under the corresponding energy saving policy index, the local cell performs load-based dormancy in the preset duration based on the load of the local cell and that of neighboring GSM cells.
- RNC-level switch

The **HO_INTER_RAT_CS_OUT_SWITCH** and **HO_INTER_RAT_PS_OUT_SWITCH** options of the **UCORRMALGOSWITCH.HoSwitch** parameter are selected to ensure that inter-RAT handovers can be performed.
- BSC-level switch
 - The **GCELLOTHEXT.IURGINFOCTRL** parameter is set to **YES** to ensure information exchange on the Iur-g interface, when the BSC6910 is used on the GSM side. This parameter does not need to be set when the BSC6960 is used on the GSM side.
 - The **GCELLSOFT.IRATShutdownSwitch** parameter is set to **ON** to ensure inter-RAT energy saving information exchange.
- For dormancy involving UMTS, the **GU_DC_SW** option of the base-station-specific **EQUIPMENT.EQMDOROPTFEATSW** parameter must be deselected. If the option is deselected and a configuration command is executed to trigger dormancy pending or dormancy pending is enabled on the local end that works in UMTS, then dormancy pending will take effect on the entire base station.

Related Concepts

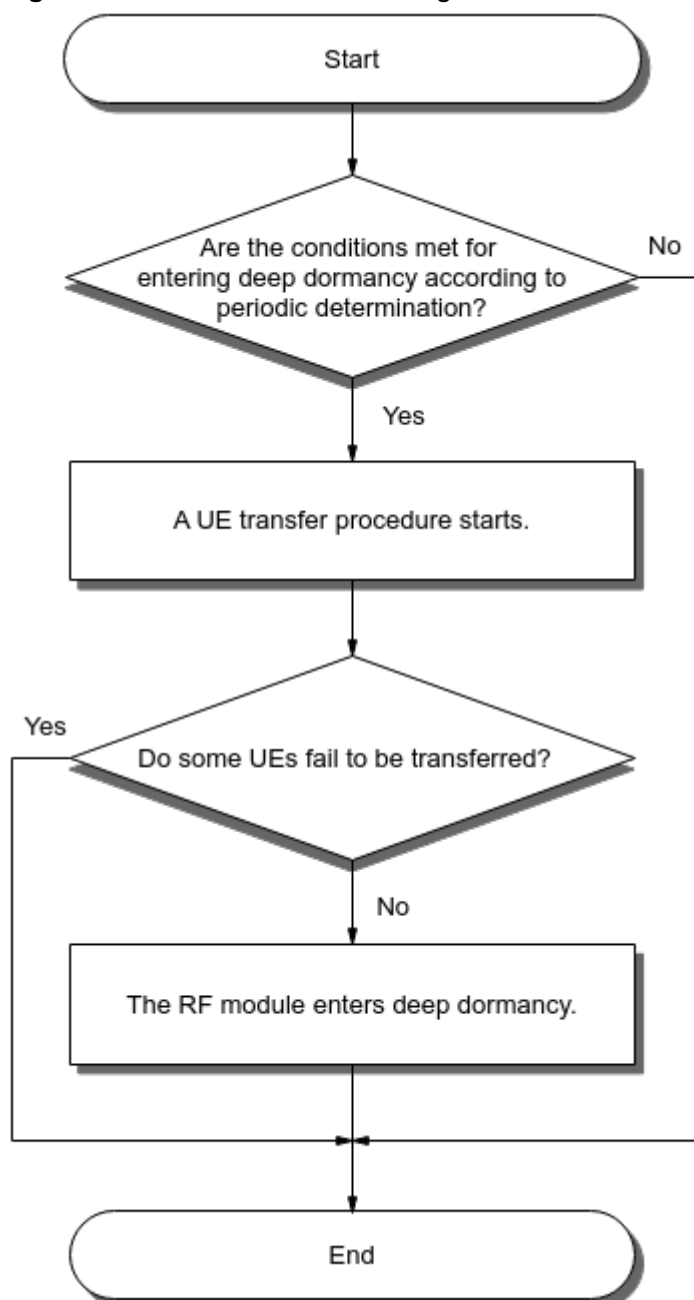
For the concepts of basic-layer carriers, capacity-layer carriers, basic-layer cells, and capacity-layer cells, refer to [Related Concepts](#). In this scenario, the basic-layer and capacity-layer cells must be inter-RAT neighboring cells. In this situation:

- The **U2GNCELL.NIRATOverLap** parameter must be set to **TRUE** for co-coverage neighboring GSM cells (basic-layer cells) of the UMTS cell that acts as the capacity-layer cell. When the UMTS cell enters the energy saving state, neighboring GSM cells with this parameter set to **TRUE** provide coverage.
- If the local cell and its inter-RAT neighboring cells belong to different operators, the **UOPERATORSHARINGMODE.InterPlmnHoAllowedInterRat** parameter must be set to **YES** to enable UE transfer.

Entering the Deep Dormancy State

[Figure 7-12](#) shows how the RF module serving a capacity-layer cell enters deep dormancy after GU load-based RF module deep dormancy is enabled.

Figure 7-12 Procedure for entering GU load-based RF module deep dormancy



1. The RNC determines whether the entry conditions are met.
The RNC periodically checks whether the dormancy entry conditions are met within an evaluation period specified by the **UCELLDORMANCYPOLICY.TriggerJudgePeriod** parameter. The information about the basic-layer cells is obtained from inter-RAT common measurements over the Iur-g interface (**UMBSCRRM.IRATCommonMeasSwitch** must be set to **ON**).
 - A capacity-layer cell must meet all of the following conditions:
 - The current time falls within the effective energy saving period determined by the **UCELLDORMANCYPOLICY.StartTime** and **UCELLDORMANCYPOLICY.StopTime** parameters.

- The capacity-layer cell is not in an energy saving penalty state.
- No UEs performing emergency calls exist in the capacity-layer cell.
- By default, cell broadcast service (CBS) takes precedence over dormancy. Therefore, when CBS is enabled in a cell for which the **URNCELLDORMANCYPARA.CellIgnoreCbsSwitch** parameter is set to **OFF** (default value), the cell cannot enter dormancy. If dormancy is required in scenarios where CBS is enabled, this parameter must be set to **ON**. Once it is set to **ON**:
 - If no CBS message is sent and the current time falls outside the CBS protection period (specified by the **URNCELLDORMANCYPARA.CbsServiceHoldDuration** parameter), the cell can enter dormancy.
 - If CBS messages are sent, the cell cannot enter dormancy.

 NOTE

After receiving a CBS message, the RNC immediately wakes the dormant cell and processes the CBS message once the cell is successfully set up. Because the cell requires time to wake, CBS messages may not be processed promptly. If CBS has a higher priority, it is recommended that **URNCELLDORMANCYPARA.CellIgnoreCbsSwitch** be set to its default value **OFF**. For details about CBS, see *CBS* in *RAN Feature Documentation*.

- The total downlink load of the capacity-layer cell is less than the value of **UCELLDORMANCYPOLICY.IRATShutdownDLLoadThd**.
 - The total uplink load of the capacity-layer cell is less than the value of **UCELLDORMANCYPOLICY.IRATShutdownULLoadThd**.
 - When **UCELLDORMANCYPOLICY.DormancyCchUserSwitch** is set to **ON**, no UEs are in the CELL_FACH state, and the number of UEs in the CELL_DCH state is less than or equal to the value of **UCELLDORMANCYPOLICY.TotalUserNumThd**.
 - When **UCELLDORMANCYPOLICY.DormancyCchUserSwitch** is set to **OFF**, the total number of UEs in the CELL_DCH and CELL_FACH states is less than or equal to the value of **UCELLDORMANCYPOLICY.TotalUserNumThd**.
 - Any basic-layer GSM cell associated with a capacity-layer cell must meet all the following conditions:
 - The timeslot usage in the GSM cell is less than the value of **UCELLDORMANCYPOLICY.IRATShutdownGSMLoadThd**.
 - The GSM cell supports PS services.
 - The GSM cell supports inter-RAT power saving (with **GCELLSOFT.IRATShutdownSwitch** set to **ON**).
2. A UE transfer procedure starts.
- The capacity-layer cell prohibits the access of new UEs and incoming handovers.
- The RNC broadcasts the barred state of the capacity-layer cell in SIB3.

- The RNC attempts to hand over all UEs in the CELL_DCH and CELL_FACH states from the capacity-layer cell to its co-coverage basic-layer GSM cells within 60s.

The RNC selects qualified basic-layer GSM cells as the candidate set for handover.

- For CELL_DCH UEs, the RNC delivers load-based inter-RAT measurement configurations in which the measurement frequencies are those of basic-layer cells. After receiving a measurement report from the UE, the RNC hands over the UE to the corresponding basic-layer cell.
 - For UEs not performing CS services, measurement-based handovers to inter-RAT neighboring cells are executed, and if the handovers fail, redirections to inter-RAT neighboring cells are executed.
 - For UEs performing CS services, measurement-based handovers to inter-RAT neighboring cells are executed, and if the handovers fail, measurement-based handovers to inter-RAT neighboring cells are executed again.
- For CELL_FACH UEs, the RNC initiates redirections to the basic-layer cells.

During inter-RAT handover and redirection, the RNC periodically checks the number of UEs in the CELL_DCH and CELL_FACH states in the capacity-layer cell.

- If the number of UEs is 0, the capacity-layer cell enters dormancy.
 - If the number of UEs is not 0, the RNC checks the number of UEs in the capacity-layer cell after 60s. If this number is less than or equal to the value of the **UCELLDORMANCYPOLICY.ForcibleTriggerUeNumThld** parameter, the capacity-layer cell enters dormancy. Otherwise, the RNC cancels the barred state of the capacity-layer cell, and the cell enters the energy saving penalty state and exits the energy saving state. The RNC does not recheck the entry conditions until the penalty period specified by the **UCELLDORMANCYPOLICY.PenaltyTime** parameter has elapsed.
3. The RF module enters deep dormancy.
- Once all UEs are transferred out of the capacity-layer cell, the cell and its serving RF module enter deep dormancy. At the same time, the system starts to check the dormancy exit conditions.

Exiting the Deep Dormancy State

The base station periodically checks whether the dormancy exit conditions are met. The exit evaluation period is specified by the **UCELLDORMANCYPOLICY.ExitJudgePeriod** parameter.

A capacity-layer cell and its serving RF modules automatically exit dormancy when any of the following conditions is met:

- The current time falls outside the effective energy saving period determined by the **UCELLDORMANCYPOLICY.StartTime** and **UCELLDORMANCYPOLICY.StopTime** parameters.
- There is no associated basic-layer GSM cell.
- The capacity-layer cell receives a CBS message from the CBS server.
- Any associated basic-layer GSM cell meets either of the following conditions:
 - The timeslot usage of the GSM cell is greater than or equal to the sum of **UCELLDORMANCYPOLICY.IRATshutdownGSMLoadThd** and **UCELLDORMANCYPOLICY.IRATshutdownGSMLoadHyst**.
 - The GSM cell does not support inter-RAT power saving (with **GCELLSOFT.IRATShutdownSwitch** set to **OFF**).
 - The load information of the GSM cell is unavailable. For example, the GSM cell is deactivated, deleted, or faulty, or the lur-g transmission is faulty.

Upon the exit from dormancy:

- The capacity-layer cell wakes up and can provide services properly.
- If the wakeup is triggered by a CBS message, the cell is not allowed to enter dormancy during the CBS hold duration (specified by **URNCCCELLDORMANCYPARA.CbsServiceHoldDuration**).

NOTE

When both UMTS load-based RF module deep dormancy and GU load-based RF module deep dormancy are enabled, capacity-layer cells can enter dormancy during the overlap of their respective dormancy periods only if the dormancy conditions of both functions are met.

7.2.1.8 Multi-Frequency Flexible Dormancy

In multi-band networking scenarios, the network energy consumption is in proportional to the number of frequency bands. Multi-frequency flexible dormancy is applicable to multi-band network scenarios where certain multi-band AAUs are used. It allows the multi-band AAUs to enter the dormancy state by frequency band, thereby reducing the equipment power consumption.

When the cells in a specific frequency band of a multi-band AAU meet the conditions for load-based RF module deep dormancy, the RF components that only serve these cells can be shut down to obtain energy saving gains. Load-based RF module deep dormancy can be NR load-based RF module deep dormancy, LNR load-based RF module deep dormancy, or LTE load-based RF module deep dormancy. Multi-frequency flexible dormancy is controlled by the **MULTI_FREQ_FLEX_SLEEP_SW** option of the **RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter. This function can be enabled only when the **RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB)** parameter is set to **ON** or **ON_DEEP_SUPER**. This function is recommended when dormancy is not enabled for cells in some frequency bands of a multi-band AAU. [Figure 7-13](#) illustrates how multi-frequency flexible dormancy works. When this function is enabled and only the cells in cell group 1 meet the conditions for entering the load-based RF module deep dormancy state, the RF components that serve cells in cell group 1 can be shut down to achieve energy saving gains. The procedures for entering and exiting the dormancy state are the same as those of the load-based RF module deep dormancy.

Figure 7-13 Schematic diagram of multi-frequency flexible dormancy

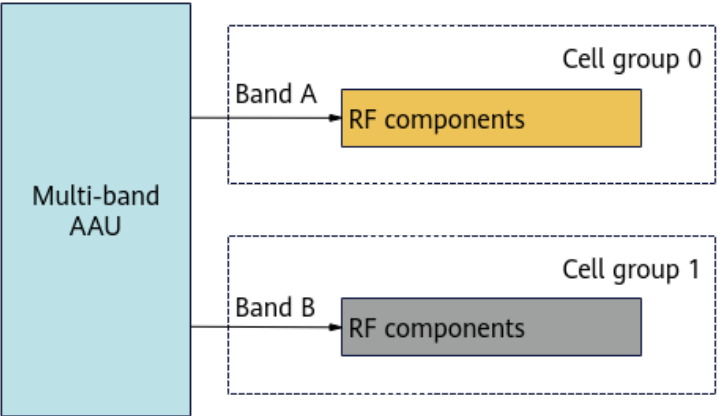


Table 7-5 describes the support for multi-frequency flexible dormancy when the basic-layer and capacity-layer cells are of different RATs.

Table 7-5 Support for multi-frequency flexible dormancy when cells are of different RATs

Cell Combination Scenario	Frequency Band A (Capacity-Layer Cell)	Frequency Band B (Basic-Layer Cell)	Support for Multi-Frequency Flexible Dormancy
Scenario 1	LTE	LTE	Not supported
Scenario 2	LTE	NR	Not supported
Scenario 3	NR	NR	Supported
Scenario 4	NR	LTE	Supported

7.2.2 Network Analysis

7.2.2.1 Benefits

It is recommended that RF module deep dormancy be enabled during a fixed time period with light or no load every day to reduce the power consumption of the base station. To ensure that the actual operating duration of an RF module meets lifecycle management requirements in typical scenarios, the module can enter dormancy when its reliability requirements (such as temperature and humidity) are met. If the RF module is reset during the dormancy period for purposes such as upgrade, the module and its cells will return to the normal state, which leads to reduced energy saving gains.

It is not recommended that the dormancy period be frequently changed or the dormancy feature be frequently enabled or disabled every day. Otherwise, the dormancy period starts about 10 minutes later than it started on the previous day, consequently decreasing energy saving gains.

In a separate-MPT base station where the local RAT is limited to GSM or UMTS, the peer RAT includes LTE or NR, and RF modules are configured on both sides with the working RAT only including LTE or NR, RF module deep dormancy can be enabled to increase energy savings.

RF modules with load-based RF module deep dormancy enabled may exit or fail to enter dormancy when scheduled energy saving functions (such as timing carrier shutdown, scheduled RF module deep dormancy, and scheduled RF module super dormancy) are enabled on neighboring base stations or the neighboring base stations or RF modules are reset or powered off. This is because such situations of the neighboring base stations or RF modules may cause the load of capacity-layer or basic-layer cells served by the RF modules with load-based RF module deep dormancy enabled to increase to above the preset threshold. If this is the case, energy saving gains decrease.

7.2.2.2 Impacts

The impacts between this function and other functions are described in [7.2.3.2.3 Function Impacts](#).

For scheduled RF module deep dormancy, NR load-based RF module deep dormancy, LNR load-based RF module deep dormancy, LTE load-based RF module deep dormancy, UMTS load-based RF module deep dormancy, and GU load-based RF module deep dormancy:

- RF module deep dormancy triggers handovers or redirections for active UEs, increasing the numbers of handover requests and redirections from the cell.
- The signal strength in an area served by an RF module in the deep dormancy state decreases because the RF module does not transmit signals. UEs may experience service drops and access failures.
- After the RF module enters deep dormancy, RAT introduction to the base station and Fast Fourier Transform (FFT) frequency scanning are not supported, and related RF module hardware status checks cannot be started using MML commands such as **STR RRUCHK**, **STR HWTST**, and **STR HWOLTST**.
- After the RF module enters dormancy or is in the process of exiting dormancy, the following alarms caused by the RF module super dormancy are not reported:
 - Node alarms
 - ALM-26235 RF Unit Maintenance Link Failure
 - NR alarms
 - ALM-29840 gNodeB Out of Service
 - ALM-29841 NR Cell Unavailable
 - ALM-29844 NR DU Cell Unavailable
 - ALM-29870 NR DU Cell TRP Unavailable
 - ALM-29871 NR DU Cell TRP Capability Degraded
 - LTE alarms

- ALM-29240 Cell Unavailable
- ALM-29243 Cell Capability Degraded
- ALM-29249 eNodeB Out of Service
- UMTS alarms
 - ALM-28203 Local Cell Unusable
 - ALM-28206 Local Cell Capability Decline
 - ALM-22202 UMTS Cell Unavailable (RNC)
 - ALM-22206 UMTS Cell Setup Failed (RNC)
 - ALM-22214 NodeB Unavailable (RNC)
 - ALM-22215 UMTS Cell Max DL Power Mismatch (RNC)
 - ALM-22203 UMTS Cell Common Channel Faulty (RNC)
 - ALM-22207 UMTS Cell Reconfiguration Failed (RNC)
 - ALM-22208 UMTS Cell Common Channel Setup Failed (RNC)
 - ALM-22209 UMTS Cell Common Channel Reconfiguration Failed (RNC)
 - ALM-22217 UMTS Cell HSDPA Function Fault (RNC)
 - ALM-22218 UMTS Cell HSUPA Function Fault (RNC)
 - ALM-22219 UMTS Cell EFACH Function Fault (RNC)
 - ALM-22220 UMTS Cell MIMO Function Faulty (RNC)
 - ALM-22222 UMTS Cell ERACH Function Fault (RNC)
 - ALM-22223 UMTS Cell EPCH Function Fault (RNC)
 - ALM-22237 UMTS Cell DC-HSUPA Function Fault (RNC)
 - ALM-22241 UMTS Cell MC-HSDPA Function Fault (RNC)

The RF module restarts during the dormancy exit process. After the restart, RF module startup events (for purposes such as power-off reset and self-reset) are reported normally.

- Base station reconfiguration during the specified dormancy period is not recommended. If reconfiguration is required, the dormancy function must be disabled first. Cell reconfigurations performed when the RF module is in the dormancy state take effect only after the RF module exits the dormancy state. If new configurations do not meet the conditions for the RF module to enter the dormancy state, the RF module exits the state.
- In some scenarios, if the dormancy period of an RF module is modified or the dormancy start time is repeatedly moved forward, the changes may not take

effect immediately in order to protect the reliability of the RF module hardware.

- If a batch processing task of the OSS contains NodeBFunction, eNodeBFunction, or gNodeBFunction MML commands that affect cell setup, or Node MML commands, and an NE of the base station is in the deep dormancy state, the OSS executes the related configurations only after the base station exits deep dormancy. This increases the waiting time before configuration execution. On average, the base station exits deep dormancy in less than 10 minutes.
- If a power-related command needs to be executed on a base station where an NE is in deep dormancy, the base station first exits deep dormancy and then executes the command. On average, the base station exits deep dormancy in less than 10 minutes.
- Dormancy pending is triggered in scenarios such as execution of the **PND DORMANCY** command or software download and activation. If the main control board is powered off before the pending duration elapses, dormancy pending resumes after the board is powered on and continues until the pending duration elapses.
- If a switchover occurs between the active and standby main control boards, the pending duration needs to be reconfigured.
- By default, the base station triggers a 30-minute pending period after startup. During this period, dormancy does not take effect.
- When the RF module exits deep dormancy, cell reestablishment will be triggered. Multiple cell setup attempts may be initiated before the cell is successfully established. In this case, the value of the **VS.Cell.UnavailTime.Sys**, **VS.Cell.HSDPA.UnavailTime**, **VS.Cell.HSUPA.UnavailTime**, **VS.Cell.UnavailTime.Sys.lubFail**, **L.Cell.Unavail.Dur.Sys**, or **N.Cell.Unavail.Dur.System** counter is not 0. If cell-related licenses have expired, the dormant cells will fail to be reestablished. If the number of cells configured on a BBP exceeds its specification, the dormant cells will fail to be reestablished when the RF module exits dormancy. Therefore, if new cells are added for capacity expansion or other purposes during the dormancy period, the new cells should be manually bound to specified BBPs.
- If dormancy entry conditions are met during a base station version upgrade, the RF module will enter dormancy only after the upgrade is complete. If an upgrade is performed while the RF module is in deep dormancy, deep dormancy pending will be triggered. Assume that the main control board on one side requires the RF module to be upgraded while the main control board on the other side requires the RF module to exit dormancy, when the RF module is shared between the two sides of a separate-MPT base station using SDR technology. In this case, cells on the side where the RF module is required to exit dormancy may fail to be reestablished, and alarms related to these cells are reported. These cells become normal after the upgrade is complete.
- Alarms reported for a cell before it enters dormancy will not be cleared after the cell enters dormancy.
 - UMTS/LTE cells: If a fault that causes an alarm is rectified during the specified dormancy period, the alarm persists and will be cleared after the cell exits dormancy.

- NR cells: If a cell becomes unavailable due to a CU fault (such as an NG interface fault), the cell will not enter dormancy. If the CU fault is rectified during the specified dormancy period for the RF module, the related alarm will be cleared and the cell will enter dormancy. If the cell becomes unavailable due to a DU fault, the alarm persists after the fault is rectified and will be cleared after the cell exits dormancy.
- If an RF module serving an NR hyper cell or combined cell does not meet the dormancy conditions, all RF modules serving the cell exit dormancy. If this cell is also a sector split cell, then all RF modules serving another NR hyper cell or combined cell that shares this RF module with this cell may exit dormancy accordingly.
- If the local base station is upgraded while its neighboring base stations are not when they are all in the deep dormancy state, then the local base station returns to the normal state after the upgrade and its cells may absorb cell edge users (CEUs) in weak coverage areas. This affects KPIs related to the aspects such as access and service drops of the local base station. It is recommended that the **PND DORMANCY** command be executed on both the base station to be upgraded and its neighboring base stations within the upgrade time window.
- Before an RF module enters deep dormancy, new UEs are prohibited from accessing an NR cell during NR carrier shutdown. If UEs attempt to hand over to the NR cell, the values of the counters related to handover preparation failures (such as **N.HO.IntraFreq.FailOut.PrepFailure**) slightly increase. Such counters are not used for calculating Huawei-defined KPIs outgoing intra- and inter-frequency handover success rates, but may be used for calculating user-defined KPIs. When dormancy is set to take effect on the hour, no UEs can be handed over to the deactivated NR cell. Therefore, even slight increases in the counter values may significantly affect the user-defined outgoing handover success rates. If deep dormancy is scheduled to take effect on the hour, you are advised to set the dormancy start time to 5 minutes earlier and the lengths of penalty timers applied to handover failures to 30s. After penalty timers for handover failures are set, UEs may experience service drops due to delayed handovers. The penalty timers for handover failures are specified by the **gNBMobilityCommParam.NecHoPrepFailPunishTimer** and **gNBMobilityCommParam.OptHoPrepFailPunishTimer** parameters.
- If the neighboring base stations are in the dormancy state, CEUs in weak coverage areas may access the cells served by the local base station. This affects KPIs related to the aspects such as access and service drop of the local base station. For example, the value of **N.NsaDc.DRB.AbnormRel** may increase.
- If the dormancy type is changed while dormancy is in effect, the current dormancy does not exit.
- After RF module dormancy takes effect, if the configured end time is changed to a later value, the modification applies from the next dormancy cycle. If the configured end time is changed to an earlier value, the ongoing dormancy ends at the updated end time.
- When the multi-RRU cell dormancy compatibility function is enabled, RF module dormancy can take effect in multi-RRU cells. As dormancy can take effect in more scenarios, the energy consumption of the entire base station decreases and the user-perceived rate may decrease. If this function is disabled, there will be no such impacts.

- When the multi-RRU cell dormancy compatibility function is enabled and dormancy functions configured for RF modules are different, the specified dormancy functions will take effect based on the configurations.
- When the multi-RRU cell dormancy compatibility function is enabled and the dormancy periods configured for RF modules are different, the specified dormancy functions will take effect only during the period in which multiple RF modules meet the dormancy requirements.

In addition to the preceding impacts, some functions have their respective network impacts:

For scheduled RF module deep dormancy:

- When scheduled RF module deep dormancy takes effect:
 - For NR
 - If NR UEs are transferred to LTE cells, performance counters on the LTE and NR sides are affected. The affected performance counters are the same as those described in [6.1.2.2 Impacts](#) for LTE and NR intelligent carrier shutdown.
 - If NR UEs are transferred to NR cells, see the network impact of intelligent carrier shutdown in *Energy Conservation and Emission Reduction in 5G RAN Feature Documentation* for the impact on performance counters and KPIs.
 - For UMTS
 - UMTS scheduled RF module dormancy causes changes in the values of counters related to inter-frequency handovers, inter-RAT handovers, and redirections.
 - During UE transfer, new UE access and incoming handovers are prohibited, which may cause changes in the values of counters related to access and service drops.
- If the multi-RRU cell dormancy compatibility function is enabled and scheduled RF module deep dormancy is required in multi-RRU cells, scheduled RF module deep dormancy must be enabled for all RRUs for dormancy to take effect.

For NR load-based RF module deep dormancy:

- NR load-based RF module deep dormancy causes changes in the values of counters related to inter-frequency handovers and redirections.
- If a UE attempts to hand over to a capacity-layer cell that is about to enter deep dormancy during UE transfer (lasting up to 30s), the source cell does not initiate handover execution. In this case, the values of the counters related to outgoing and incoming handover attempts increase in the source and capacity-layer cells, respectively, while the values of the counters related to handover executions and successful handovers do not increase. As Huawei-defined KPIs **Inter-Frequency Handover Out Success Rate (CU)** and **Intra-RAT Handover In Success Rate (CU)** are calculated using the numbers of outgoing and incoming handover executions as their denominators, these KPIs are not affected in this scenario. However, if the user-defined KPIs are calculated using the numbers of outgoing and incoming handover attempts as their denominators, these KPIs may be slightly affected in this scenario.

- When precise dormancy is in effect, the base station capacity decreases. When precise dormancy is enabled and overlap rates are calculated based on grids, that is, the **GRID_MODE_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter is selected, the CPU usage slightly increases.

For LNR load-based RF module deep dormancy:

- When LNR load-based RF module deep dormancy takes effect, performance counters on the LTE and NR sides are affected. The affected performance counters are the same as those described in [6.1.2.2 Impacts](#) for LTE and NR intelligent carrier shutdown.
- In inter-base-station SA networking, X2 link setup may be triggered for energy saving purposes. This occupies the specifications of EN-DC X2 links and affects the experience of NSA UEs. The probability of ALM-29225 Maximum Number of X2 Interfaces Reached being reported also increases. Therefore, this function is not recommended when the X2 specifications are less than required.
- In inter-base-station SA networking, if the **LNR_INTER_BS_PWR_SAVE_SW** option of the **EnodebPwrSavingAlgo.PwrSavingAlgoSwitch** parameter or the **LNR_SA_GNB_X2_SETUP_SW** option of the **eNodeBOamParam.InterfaceSonConfigSwitch** parameter is selected but the **X2_SETUP_FOR_SA_SW** option of the **gNBInterfaceParam.X2ConfigSw** parameter is deselected on the gNodeB side, ALM-29204 X2 Interface Fault and ALM-25888 SCTP Link Fault may be reported.
- In inter-base-station SA networking, the configurations of the neighboring NR cell with overlapping coverage areas on the LTE side must be the same as those on the NR side, such as the cell ID, base station ID, and operator information. Otherwise, the SA X2 link may fail to be set up.

For LTE load-based RF module deep dormancy:

- When LTE load-based RF module deep dormancy is triggered, inter-frequency handovers are performed for UEs in the capacity-layer cell, increasing the values of counters listed in [Table 7-6](#) and [Table 7-7](#). In addition, to promptly transfer UEs out of capacity-layer cells, the eNodeB prohibits UEs from accessing the cells and broadcasts the barred state of the cells in SIB1. However, a small number of UEs may still send RRC connection setup requests to access the cells, causing an increase in the **L.RRC.SetupFail.Rej** counter value and a decrease in the **RRC Setup Success Rate** value. The number of such access failures is indicated by the **L.RRC.SetupFail.ResFail.CellBarred** counter.
- When precise dormancy is enabled and cell overlap rates are calculated based on grids, the CPU usage of the main control board slightly increases.

Table 7-6 Performance counters of capacity-layer cells

Networking	Counter Description	Counter Name
Macro networking	Number of intra-eNodeB inter-frequency outgoing handover attempts in a cell	L.HHO.IntraeNB.InterFreq.PrepAttOut

Networking	Counter Description	Counter Name
	Number of intra-eNodeB inter-frequency outgoing handover executions in a cell	L.HHO.IntraeNB.InterFreq.ExecAttOut
	Number of successful intra-eNodeB inter-frequency outgoing handovers in a cell	L.HHO.IntraeNB.InterFreq.ExecSuccOut
Inter-eNodeB networking	Number of inter-eNodeB inter-frequency outgoing handover attempts in a cell	L.HHO.IntereNB.InterFreq.PrepAttOut
	Number of inter-eNodeB inter-frequency outgoing handover executions in a cell	L.HHO.IntereNB.InterFreq.ExecAttOut
	Number of successful inter-eNodeB inter-frequency outgoing handovers in a cell	L.HHO.IntereNB.InterFreq.ExecSuccOut

Table 7-7 Performance counters of basic-layer cells

Networking	Counter Description	Counter Name
Macro networking	Number of intra-eNodeB intra-duplex-mode incoming handover attempts in a cell	L.HHO.IntraeNB.PrepAttIn
	Number of intra-eNodeB intra-duplex-mode incoming handover executions in a cell	L.HHO.IntraeNB.ExecAttIn
	Number of successful intra-eNodeB intra-duplex-mode incoming handovers in a cell	L.HHO.IntraeNB.ExecSuccIn
Inter-eNodeB networking	Number of inter-eNodeB intra-duplex-mode incoming handover attempts in a cell	L.HHO.IntereNB.PrepAttIn

Networking	Counter Description	Counter Name
	Number of inter-eNodeB intra-duplex-mode incoming handover executions in a cell	L.HHO.IntereNB.ExecAttIn
	Number of successful inter-eNodeB intra-duplex-mode incoming handovers in a cell	L.HHO.IntereNB.ExecSucIn

For UMTS load-based RF module deep dormancy:

- UMTS load-based RF module dormancy causes changes in the values of counters related to inter-frequency handovers and redirections.
- During UE transfer, new UE access and incoming handovers are prohibited, which may cause changes in the values of counters related to access and service drops.
- If the **UCELLDORMANCYPOLICY.ForcibleTriggerUeNumThld** parameter is set to a value other than **0** and the cell serving CS services enters the energy saving state during UE transfer, voice services will be interrupted.
- The load-based dormancy algorithm uses the **UCELLDORMANCYPOLICY.StartTime** and **UCELLDORMANCYPOLICY.StopTime** parameters on both the RNC and NodeB. Therefore, the system time of the RNC and that of the NodeB must be the same. If they are different, the corresponding cell will enter load-based dormancy earlier or later than expected.

For GU load-based RF module deep dormancy:

- When GU load-based RF module dormancy is in effect, performance counters on the GSM and UMTS sides are affected. The affected performance counters are the same as those described in [6.3.2.2 Impacts](#) under [6.3 GUL Joint Carrier Shutdown](#).
- It is not recommended that GSM load-based RF module super dormancy be enabled for GSM cells that serve as basic-layer cells for a UMTS cell that covers the same area as these GSM cells. Otherwise, in extreme cases, the UMTS cell and these neighboring GSM cells are in dormancy in the same period, resulting in coverage unavailability and consequently UE access failures in the area in this period.
- During UE transfer, new UE access and incoming handovers are prohibited, which may cause changes in the values of counters related to access and service drops.
- If the **UCELLDORMANCYPOLICY.ForcibleTriggerUeNumThld** parameter is set to a value other than **0** and the cell serving CS services enters the energy saving state during UE transfer, voice services will be interrupted.
- The load-based dormancy algorithm uses the **UCELLDORMANCYPOLICY.StartTime** and **UCELLDORMANCYPOLICY.StopTime** parameters on both the RNC and NodeB. Therefore, the system time of the RNC and that of the NodeB must be

the same. If they are different, the corresponding cell will enter load-based dormancy earlier or later than expected.

For multi-frequency flexible dormancy:

When multi-frequency flexible dormancy is enabled, multi-band AAUs can enter dormancy by frequency band. This affects the performance counters on the LTE and NR sides. The affected performance counters are related to the triggering mode of RF module deep dormancy. For details, see descriptions of the corresponding functions in [7.2.2.2 Impacts](#).

In some scenarios, if the dormancy period of an RF module is modified or the dormancy start time is repeatedly moved forward, the changes may not take effect immediately in order to protect the reliability of the RF module hardware.

7.2.3 Requirements

7.2.3.1 Licenses

Multi-frequency flexible dormancy is not under license control.

RAT	Feature ID	Feature Name	Model	NE	Sales Unit
NR	MRFD-1 61263	RF Module Deep Dormancy (NR)	NR0S00R FDD00	gNodeB	per RU
LTE FDD	MRFD-1 81223	RF Module Deep Dormancy (LTE FDD)	LT1SRFM DEDY1	eNodeB	per RU
LTE TDD	MRFD-1 61233	RF Module Deep Dormancy (LTE TDD)	LT4SRFM DDTDD	eNodeB	per RU
UMTS	MRFD-2 21213	RF Module Deep Dormancy (UMTS)	LQW9RF MDD01	NodeB	per RU

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

 CAUTION

- For an RF module shared by multiple RATs using SDR technology, it can enter deep dormancy only when the licenses for all RATs of the RF module are acquired.
 - It is recommended that the RF unit working mode (specified by the **RRU.RS (LTE eNodeB, 5G gNodeB)** parameter) be set to a single RAT for single-mode RF modules. If this parameter is set to a value indicating multiple RATs for an RF module and RAT introduction has been performed on the base station, the RF module can enter dormancy only when the licenses for the newly introduced RATs are loaded.
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7.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.2.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD High-frequency NR TDD	Virtual grid model building and maintenance	VG_MODEL_ALLOW_BUILD_FLAG option of the NRCellFreqRelation.AggregationAttribute parameter and NR_IF_HOMEBUILD_QTY_PRESERVE_SWITCH option of the NRCellSmartMultiCarrierAlgorithmSwitch parameter	<i>Virtual Grid-based Multi-Frequency Coordination in 5G RAN Feature Documentation</i>	When precise dormancy is enabled and overlap rates need to be calculated based on grids, virtual grid model building and maintenance must be enabled. To further improve the accuracy of grid models, see <i>Virtual Grid-based Multi-Frequency Coordination in 5G RAN Feature Documentation</i> .

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Virtual grid function	SMART_CARRIER_SELECTION_SW option of the MultiCarrierUnifiedSch parameter	<i>Multi-carrier Unified Scheduling in eRAN Feature Documentation</i>	To use grid models to calculate the cell co-coverage proportion for precise dormancy, virtual grid function needs to be enabled. To further improve the accuracy of grid models, see <i>Multi-carrier Unified Scheduling in eRAN Feature Documentation</i>.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	RF module deep dormancy	RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB) set to ON or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	Multi-frequency flexible dormancy (controlled by the MULTI_FREQ_FLEX_SLEEP_SW option of the RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB) parameter) can be enabled only when the RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB) parameter is set to ON or ON_DEEP_SUPER .

7.2.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD High-frequency NR TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	The period configured for timing carrier shutdown cannot overlap with the period configured for NR load-based RF module deep dormancy, LNR load-based RF module deep dormancy, or optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module deep dormancy of the local cell. You are advised to preferentially plan the periods during which timing carrier shutdown takes effect.
LTE FDD LTE TDD	High speed specified policy management	HSCellSleepInterfAvoidSwitch option of the HighSpdAdaptionPara.InterfAvoidMode parameter	<i>High Speed Mobility in eRAN Feature Documentation</i>	The high-speed cell sleeping function in high speed specified policy management does not work with LNR load-based RF module deep dormancy.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Deployment Position	NRDUCell.DeployPosition set to DEPLOY_IAB	None	If this parameter is set to DEPLOY_IAB , the following functions cannot be enabled: NR load-based RF module deep dormancy, LNR load-based RF module deep dormancy, and optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module deep dormancy.
Low-frequency NR TDD	Supplementary Cell Indicator	NRDUCell.SupplementaryCellInd	None	If the NRDUCell.DuplexMode parameter is set to CELL_TDD and the value of supplementary cell indicator is INTRA_FREQ_ASSISTANT_CELL , the following functions cannot be enabled: NR load-based RF module deep dormancy, LNR load-based RF module deep dormancy, and optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module deep dormancy.

RAT	Function Name	Function Switch	Reference	Description
NR FDD	MBS broadcast	MBS_BROADCAST_SWITCH option of the NRCellFeatureSwitch.MbsSwitch parameter	None	When the MBS broadcast switch is on, the following functions cannot be enabled: NR load-based RF module deep dormancy, LNR load-based RF module deep dormancy, and optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module deep dormancy.
LTE FDD LTE TDD	Multi Cell Share Mode	Cell.MultiCellShareMode set to SPLIT	None	Split cells cannot be configured as capacity-layer cells for LTE load-based RF module deep dormancy in precise dormancy mode.
LTE TDD	Intelligent Multi-Beam of 8T8R	None	<i>Intelligent Multi-Beam of 8T8R (TDD) in eRAN Feature Documentation</i>	Cells with Intelligent Multi-Beam of 8T8R enabled cannot be configured as capacity-layer cells for LTE load-based RF module deep dormancy in precise dormancy mode.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Intra-RAT level-by-level energy saving	INTRA_RAT_GRAD_PWR_SAVING_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	If the MULTI_RRU_CELL_DORM_COMPAT_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter is selected, the INTRA_RAT_GRAD_PWR_SAVING_SW option of this parameter must be deselected.
LTE FDD	Static Multiple Beam	Number of SectorSplitCell MOs ≥ 2	<i>Massive MIMO Basics (FDD) in eRAN Feature Documentation</i> <i>Smart 8T8R (FDD) in eRAN Feature Documentation</i>	Cells with Static Multiple Beam enabled in FDD massive MIMO scenarios cannot be configured as capacity-layer cells for LTE load-based RF module deep dormancy in precise dormancy mode. Cells with Static Multiple Beam enabled in FDD Smart 8T8R scenarios cannot be configured as capacity-layer cells for LTE load-based RF module deep dormancy in precise dormancy mode.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	SFN Cell	Cell.MultiRruCellMode set to SFN	<i>SFN in eRAN Feature Documentation</i>	A cell cannot be configured as an SFN cell when it serves as a capacity-layer cell for LTE load-based RF module deep dormancy in precise dormancy mode (that is, when the LTE_DORMANCY_BY_LOAD_PESW option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter is selected and the CellPwrSavingAlgo.DormancyCellType parameter is set to CAPACITY_CELL) and the CellAuxEnbBind or CellAuxDuBind MO is configured for the cell.
LTE FDD LTE TDD	Inter-eNodeB SFN based on eNodeB coordination	Number of CellAuxEnbBind MOs ≥ 1	<i>SFN in eRAN Feature Documentation</i>	Cells with inter-eNodeB SFN based on eNodeB coordination enabled do not support LTE load-based RF module deep dormancy in non-precise dormancy mode.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Inter-eNodeB adaptive SFN/SDMA based on eNodeB coordination	Number of CellAuxEnbBind MOs ≥ 1	<i>SFN in eRAN Feature Documentation</i>	Cells with inter-eNodeB adaptive SFN/SDMA based on eNodeB coordination enabled do not support LTE load-based RF module deep dormancy in non-precise dormancy mode.
LTE FDD LTE TDD	Auxiliary eNodeB Cell Distributed Unit Binding Relationship	Number of CellAuxDuBind MOs ≥ 1	None	Cells with Auxiliary eNodeB Cell Distributed Unit Binding Relationship configured do not support LTE load-based RF module deep dormancy in non-precise dormancy mode.
NR FDD Low-frequency NR TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SWITCH option of the NRDUCellAlgoSwitch . <i>HighReliabilitySwitch</i> parameter	<i>URLLC in 5G RAN Feature Documentation</i>	In low latency and high reliability scenarios, the following functions cannot be enabled: NR load-based RF module deep dormancy, LNR load-based RF module deep dormancy, and optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module deep dormancy.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Dynamic MIMO layer number decrease	DYNAMIC_MIMO_LAYER_DEC_SW option of the gNodeBAlgo.DedicatedNetworkFunctionSw parameter	None	Dynamic MIMO layer number decrease does not work with optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module deep dormancy.
NR FDD Low-frequency NR TDD High-frequency NR TDD	NR load-based RF module deep dormancy	INTRA_NR_DORMANCY_BY_LOAD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	NR load-based RF module deep dormancy does not work with optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module deep dormancy.
LTE TDD LTE FDD NR FDD Low-frequency NR TDD	LNR load-based RF module deep dormancy	LNR_DORMANCY_BY_LOAD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	LNR load-based RF module deep dormancy does not work with optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module deep dormancy.
LTE FDD	Out-of-band relay enhancement	OUT_OF_BAND_RELAY_ENH_SW option of the CellRelayAlgoPara.RelayEnhSw parameter	<i>RuralCell Solution</i>	When out-of-band relay enhancement is enabled, RF module deep dormancy must be disabled.

7.2.3.2.3 Function Impacts

When RF module deep dormancy and another energy saving function related to RF modules (such as symbol power saving, channel shutdown, fast deep dormancy, or carrier shutdown) are both enabled, RF module deep dormancy preferentially takes effect.

RF module super dormancy takes precedence over RF module deep dormancy when both functions are enabled.

When RF module deep dormancy (scheduled RF module deep dormancy, NR load-based RF module deep dormancy, LNR load-based RF module deep dormancy, and LTE load-based RF module deep dormancy) is enabled together with multi-frequency flexible dormancy, RF module deep dormancy preferentially takes effect. The functions that affect multi-frequency flexible dormancy are the same as those that affect load-based RF module deep dormancy.

Cells do not provide services after entering dormancy, and related features do not take effect in these cells.

The following tables list other function impacts.

- **Table 7-8** lists the function impacts of scheduled RF module deep dormancy.
- **Table 7-9** lists the function impacts of NR load-based RF module deep dormancy.
- In addition to the function impacts of NR load-based RF module deep dormancy (except the impact relationship with intra-RAT ANR) listed in **Table 7-9**, LNR load-based RF module deep dormancy has the function impacts listed in **Table 7-10**.
- **Table 7-11** lists the function impacts of LTE load-based RF module deep dormancy.
- The UMTS-related content in **Table 7-8** describes the function impacts of UMTS and GU load-based RF module deep dormancy.

Table 7-8 Function impacts of scheduled RF module deep dormancy

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Carrier aggregation	None	<i>Carrier Aggregation in eRAN Feature Documentation</i>	- Cells in the RF module deep dormancy state do not participate in CA. In CA-group-based configuration mode, after such a cell in a CA group exits this energy saving state, this cell cannot be configured as an SCell for an existing CA UE until the UE accesses the network again. - Cells that exit the RF module deep dormancy state do not participate in CA if the number of routes is limited. The route for CA will be released for a cell in the RF module deep dormancy state. After this cell exits this state, it may not obtain a route if the number of routes is limited and therefore, cannot

RAT	Function Name	Function Switch	Reference	Description
				participate in CA.
LTE FDD LTE TDD	Automatic baseband and resource configuration	eNodeBResModeAlgo.BbpResAutoConfigSw	None	When the eNodeBResModeAlgo.BbpResAutoConfigSw parameter is set to ON , RF module deep dormancy does not take effect.
LTE TDD	Clock out-of-synchronization detection	ClkDetect.ClkAsyncDetectSwitch	<i>Synchronization in eRAN Feature Documentation</i>	If clock out-of-synchronization occurs during the dormancy period, this issue may fail to be detected.
Low-frequency NR TDD	Clock out-of-synchronization detection	CLK_DETECT_SW option of the gNodeBParam.ClkOutofsyncDetSwitch parameter	<i>Synchronization in 5G RAN Feature Documentation</i>	If clock out-of-synchronization occurs during the dormancy period, this issue may fail to be detected.
Low-frequency NR TDD	Air-interface-based network-wide synchronization deviation detection	GAP_BASED_SYNC_DIFF_DET_SW option of the gNodeBParam.SyncDiffDetectSwitch parameter	<i>Synchronization in 5G RAN Feature Documentation</i>	If sequence detection is performed on a to-be-detected base station during the period when RF module deep dormancy takes effect for all cells of the base station, the sequence detection fails. In this case, the possible clock fault of the base station may fail to be detected.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module deep dormancy takes precedence over energy saving based on flexible frequency-domain scheduling when both functions are enabled. Energy saving based on flexible frequency-domain scheduling does not take effect when RF module deep dormancy is in effect.
NR FDD Low-frequency NR TDD	TTI-level channel shutdown	TTI_RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module deep dormancy takes precedence over TTI-level channel shutdown when both functions are enabled. TTI-level channel shutdown does not take effect when RF module deep dormancy is in effect.
NR FDD Low-frequency NR TDD	Intelligent power control energy saving	SMART_PWR_CTRL_ENERGY_SAVE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module deep dormancy takes precedence over intelligent power control energy saving when both functions are enabled. Intelligent power control energy saving does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Low-power RF channel dynamic shutdown	LOW_POWER_RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module deep dormancy takes precedence over low-power RF channel dynamic shutdown when both functions are enabled. Low-power RF channel dynamic shutdown does not take effect when RF module deep dormancy is in effect.
LTE FDD LTE TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCHEDULE_ENERGY_SAVING_SW option of the CellPwrSavingAlgo.PwrSavingAlgorithm parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over energy saving based on flexible frequency-domain scheduling when both functions are enabled. Energy saving based on flexible frequency-domain scheduling does not take effect when RF module deep dormancy is in effect.
NR FDD Low-frequency NR TDD	Cell-level MRO	MRO_SCENARIO_ID option of the gNBMRo.ScenarioId parameter and INTRA_FREQ_MRO_SW option of the NRCellMRo.MroSwitch parameter	<i>MRO in 5G RAN Feature Documentation</i>	When both RF module deep dormancy and intra-frequency MRO are enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, MRO may take effect later than expected.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	UE-level MRO	UE_MRO_SWITCH option of the NRCellMro.MroSwitch parameter	<i>MRO in 5G RAN Feature Documentation</i>	When both RF module deep dormancy and UE-level MRO are enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, the penalty mechanism accuracy of UE-level MRO is affected.
LTE FDD LTE TDD	Intelligent power control energy saving	- Single-cell intelligent power control energy saving: INTEL_INTRA_CELL_PC_SW option of the CellPwrSavingAlgorithm.PwrSavingAlgorithmSwitch parameter - Multi-cell coordinated intelligent power control energy saving: INTEL_INTRA_CELL_PC_SW and INTEL_MULTI_CELL_COOR_PC_SW options of the CellPwrSavingAlgorithm.PwrSavingAlgorithmSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over intelligent power control energy saving when both functions are enabled. Intelligent power control energy saving does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Dynamic carrier shutdown phase 2	CellPwrSavingAlgo.CarrierDynShutdownPh2Sw set to CAPACITY_CELL	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over dynamic carrier shutdown phase 2 when both functions are enabled. Dynamic carrier shutdown phase 2 does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	CMAS	CellAlgoSwitch.CmasSwitch	<i>CMAS in eRAN Feature Documentation</i>	The following applies when the RF module enters deep dormancy: • With PWS message-triggered exit of cells from energy saving disabled, CMAS warning notifications cannot be properly sent in LTE cells in the dormancy state. However, UEs can still receive CMAS warning notifications from other cells. • With PWS message-triggered exit of cells from energy saving enabled, cells exit the dormancy state upon receipt of a PWS message.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	CMAS	gNB SibConfig. <i>SibType</i> set to SIB_TYPE_8	<i>CMAS in 5G RAN Feature Documentation</i>	When an RF module enters the deep dormancy state, the NR cells it serves also enter the dormancy state. In this case, CMAS warning notifications cannot be properly sent in these cells. However, UEs can still receive CMAS warning notifications from other cells.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	ETWS	None	<i>ETWS in eRAN Feature Documentation</i>	The following applies when the RF module enters deep dormancy: • With PWS message-triggered exit of cells from energy saving disabled, ETWS warning notifications cannot be sent in LTE cells in the dormancy state. However, UEs can still receive ETWS warning notifications from other cells. • With PWS message-triggered exit of cells from energy saving enabled, cells exit the dormancy state upon receipt of a PWS message.
LTE FDD	Downlink intelligent joint power control	DL_INTEL_JOINT_P C_SW option of the CellDisSchExtParam.DISchExtSwitch parameter	<i>Smart 8T8R (FDD) in eRAN Feature Documentation</i>	Downlink intelligent joint power control does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Down link intelligent joint AMC and power control	DL_INTEL_JOINT_AMC_AND_PC_SW option of the CellDISchExtParam.DISchExtSwitch parameter	<i>Smart 8T8R (FDD) in eRAN Feature Documentation</i>	For LTE FDD, downlink intelligent joint AMC and power control does not take effect when RF module deep dormancy is in effect.
LTE FDD LTE TDD	RF channel intelligent shutdown	CellRfShutdown.RfShutdownSwitch	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over RF channel intelligent shutdown when both functions are enabled. RF channel intelligent shutdown does not take effect when RF module deep dormancy is in effect.
LTE FDD LTE TDD	Intelligent power-off of carriers in the same coverage	CellShutdown.CellShutdownSwitch set to ON	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over intelligent power-off of carriers in the same coverage when both functions are enabled. Intelligent power-off of carriers in the same coverage does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Multi-carrier coordinated energy saving	CellShutdown.Cells hutdownSwitch set to ON_MULTI_CARRIER_HIER_SHUTDOWN , ON_MULTICARRIER_HIER_MANUAL_SHUTDOWN , or ON_CARRIER_PRECISE_SHUTDOWN	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over multi-carrier coordinated energy saving when both functions are enabled. Multi-carrier coordinated energy saving does not take effect when RF module deep dormancy is in effect.
LTE FDD LTE TDD	Dynamic carrier shutdown	CARRIER_DYN_SHUTDOWN_SW option of the CellShutdown.CarrierShutdownEnhancedSw parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over dynamic carrier shutdown when both functions are enabled. Dynamic carrier shutdown does not take effect when RF module deep dormancy is in effect.
LTE FDD LTE TDD	Dynamic voltage adjustment	CellAlgoSwitch.DynAdjVoltSwitch	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over dynamic voltage adjustment when both functions are enabled. Dynamic voltage adjustment does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Low power consumption mode (LTE)	CellLowPower.LowPwrSwitch	<i>Green Site</i>	RF module deep dormancy takes precedence over low power consumption mode when both functions are enabled. Low power consumption mode does not take effect when RF module deep dormancy is in effect.
LTE FDD LTE TDD	Cell Switch-off Based on Automatic Co-coverage Identification	CellShutdown.CellsShutdownSwitch set to ON_CO_COV_AUTO_IDENT	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over Cell Switch-off Based on Automatic Co-coverage Identification when both functions are enabled. Cell Switch-off Based on Automatic Co-coverage Identification does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Basic symbol power saving	FDD: SymbolShutdownSwitch option of the ENodeBAlgoSwitch.PowerSaveSwitch parameter TDD: TddSymbolShutdownSwitch option of the ENodeBAlgoSwitch.PowerSaveSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over basic symbol power saving when both functions are enabled. Basic symbol power saving does not take effect when RF module deep dormancy is in effect.
LTE FDD LTE TDD	CRS dynamic muting	CRS_DYNAMIC_MUTING_SW option of the SymbolPwrSaving.SymbolShutdownEnhancedSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over CRS dynamic muting when both functions are enabled. CRS dynamic muting does not take effect when RF module deep dormancy is in effect.
LTE FDD LTE TDD	Automatic cell shutdown	CellAutoShutdown.CellAutoShutdownSwitch	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over automatic cell shutdown when both functions are enabled. Automatic cell shutdown does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	RF channel dynamic muting	RF_CHN_DYN_MUTING_SW option of the CellRfChnDynMuting.RfChnDynMutingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	Scheduled RF module deep dormancy takes precedence over RF channel dynamic muting when both functions are enabled. RF channel dynamic muting does not take effect when scheduled RF module deep dormancy is in effect.
LTE FDD	Massive MIMO downlink coordinated power control	MM_DL_COORD_PC_SW option of the SectorSplitGroup.IntelligentBeamSwitch parameter	<i>Massive MIMO Enhancements (FDD) in eRAN Feature Documentation</i>	Massive MIMO downlink coordinated power control does not take effect when RF module deep dormancy is in effect.
NR FDD Low-frequency NR TDD High-frequency NR TDD	Symbol power saving	NRDUCellAlgoSwitch.SymbolShutdownSwitch	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module deep dormancy takes precedence over symbol power saving when both functions are enabled. Symbol power saving does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD High-frequency NR TDD	Power saving scheduling	POWER_SAVING_SCHEDULE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module deep dormancy takes precedence over power saving scheduling when both functions are enabled. Power saving scheduling does not take effect when RF module deep dormancy is in effect.
NR FDD Low-frequency NR TDD High-frequency NR TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module deep dormancy takes precedence over RF channel intelligent shutdown when both functions are enabled. RF channel intelligent shutdown does not take effect when RF module deep dormancy is in effect.
NR FDD Low-frequency NR TDD High-frequency NR TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module deep dormancy takes precedence over timing carrier shutdown when both functions are enabled. Timing carrier shutdown does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Intelligent carrier shutdown in precise mode	• INTRA_GNB_MULTICARRIER_SHUTDOWN or INTER_GNB_MULTICARRIER_SHUTDOWN option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter • MULTICARRIER_PRECISE_SHUTDOWN option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter selected	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module deep dormancy takes precedence over intelligent carrier shutdown when both functions are enabled. Intelligent carrier shutdown does not take effect when RF module deep dormancy is in effect.
NR FDD Low-frequency NR TDD High-frequency NR TDD	Intelligent carrier shutdown in non-precise mode	• INTRA_GNB_MULTICARRIER_SHUTDOWN or INTER_GNB_MULTICARRIER_SHUTDOWN option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter • MULTICARRIER_PRECISE_SHUTDOWN option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter deselected	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module deep dormancy takes precedence over intelligent carrier shutdown when both functions are enabled. Intelligent carrier shutdown does not take effect when RF module deep dormancy is in effect.
NR FDD Low-frequency NR TDD	Uplink symbol power saving	UL_SYMBOL_SHUTDOWN option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	When RF module deep dormancy takes effect, uplink symbol power saving produces no energy saving gains.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation in 5G RAN Feature Documentation</i>	A carrier exits a power sharing group once it enters deep dormancy.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Ultra-long holdover for sites with abnormal clock sources	CLK_OTASYNC_PRO C_SW option of the TASM.MSCLKDIAGPROCESSSW parameter	<i>Synchronization in 5G RAN Feature Documentation</i>	Assume that the ultra-long hold switch is turned on but the ultra-long hold enhancement switch is turned off. If sequence detection is performed on a to-be-detected base station during the period when RF module deep dormancy takes effect for all cells of the base station, the sequence detection fails. In this case, the base station cannot enter the ultra-long holdover state even if its clock is faulty, or it exits the ultra-long holdover state as sequence detection cannot be performed for a long time due to energy saving. If the ultra-long hold enhancement switch is turned on, RF module deep dormancy does not take effect for the base station in the ultra-long holdover state and its reference neighboring base stations.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Channel isolation for PIM interference avoidance	PIM_INTERF_CHN_SHUTDOWN_SW option of the PimInterfSmartAvoid.PimAlgorithm parameter	<i>Interference Detection and Suppression in eRAN Feature Documentation</i>	For LTE FDD, if RF module deep dormancy has taken effect, channel isolation for PIM interference avoidance does not take effect. If channel isolation for PIM interference avoidance has taken effect, RF module deep dormancy does not take effect.
LTE FDD	Intelligent PIM interference avoidance	PIM_INTERF_SMART_AVOID_SW option of the PimInterfSmartAvoid.PimAlgorithm parameter	<i>Interference Detection and Suppression in eRAN Feature Documentation</i>	For LTE FDD, the RF module deep dormancy function is terminated during PIM interference detection.
NR FDD	Channel isolation for PIM interference avoidance	PIM_INTERF_SMART_AVOID_SW option of the NRDUCellPimInterfAvd.PimAlgorithm parameter	<i>Interference Avoidance in 5G RAN Feature Documentation</i>	If RF module deep dormancy has taken effect, channel isolation for PIM interference avoidance does not take effect. If channel isolation for PIM interference avoidance has taken effect, RF module deep dormancy does not take effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD	Intelligent PIM interference avoidance	PIM_INTERF_SMART_AVOID_SW option of the NRDUCellPimInterfAvd.PimAlgoSwitch parameter	<i>Interference Avoidance in 5G RAN Feature Documentation</i>	The RF module deep dormancy function is terminated during PIM interference detection.
NR FDD	Time-frequency two-dimensional unified scheduling optimization	TIME_FREQ_2D_UNIF_OPT_SCH_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgorithmSwitch parameter	<i>Multi-Frequency Smart Aggregation in 5G RAN Feature Documentation</i>	Time-frequency two-dimensional unified scheduling optimization does not take effect for cells in the shutdown state.
NR TDD	CA Performance Boost Based on Multi-dimensional Optimization	CA_PRFM_BOOST_BSD_ON_MD_OPT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgorithmSwitch parameter	<i>Multi-Frequency Smart Aggregation in 5G RAN Feature Documentation</i>	CA Performance Boost Based on Multi-dimensional Optimization does not take effect for cells in the shutdown state.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Fast carrier shutdown	For capacity-layer cells: - Intra-base-station multi-carrier scenarios: INTRA_GNB_MULTICARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter selected; inter-base-station multi-carrier scenarios: INTER_GNB_MULTICARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter selected FAST_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter For basic-layer cells: FAST_CARRIER_SHUTDOWN_ASSUR_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Fast carrier shutdown does not take effect when RF module deep dormancy is in effect.
NR FDD Low-frequency NR TDD	Deep symbol power saving	DEEP_SYMBOL_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Deep symbol power saving does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Deep symbol power saving	DEEP_SYMBOL_SHUTDOWN_SW option of the SymbolPwrSaving.SymbolShutdownEnhancedSw parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	Deep symbol power saving does not take effect when RF module deep dormancy is in effect.
LTE FDD	Uplink symbol power saving	UL_SYMBOL_SHUTDOWN_SW option of the SymbolPwrSaving.SymbolShutdownEnhancedSw parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	For LTE FDD, uplink symbol power saving and enhanced uplink symbol power saving do not take effect when RF module deep dormancy is in effect.
LTE FDD	Enhanced uplink symbol power saving	UL_SYMBOL_SHUTDOWN_ENH_SW option of the SymbolPwrSaving.SymbolShutdownEnhancedSw parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	
NR FDD	Uplink symbol power saving	UL_SYMBOL_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	For NR FDD, uplink symbol power saving does not take effect when RF module deep dormancy is in effect.
Low-frequency NR TDD	NR multi-band power adaptive scheduling	NR_MULTI_BAND_POWER_ADAPT_SWITCH option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation in 5G RAN Feature Documentation</i>	A carrier exits the power adaptive scheduling group once it enters the deep dormancy state.

RAT	Function Name	Function Switch	Reference	Description
NR FDD	Intelligent MCS optimization for uplink SU-MIMO	UL_SU_MCS_INTEL_OPT_SW option of the NRDUCellAiAlgo.AiAmcAlgoSwitch parameter	<i>Smart 8T8R (FDD) in 5G RAN Feature Documentation</i>	Intelligent MCS optimization for uplink SU-MIMO does not take effect or its gains decrease when it is enabled together with RF module deep dormancy.
LTE FDD LTE TDD	SOC level-based low power consumption mode	SOCLVL_BASED_LOW_PWR_SW option of the CellLowPower.LowPwrConsumeModeExtSw parameter	<i>Green Site</i>	RF module deep dormancy takes precedence over SOC level-based low power consumption mode when both functions are enabled. SOC level-based low power consumption mode does not take effect when RF module deep dormancy is in effect.
NR FDD Low-frequency NR TDD	Dynamic RF channel shutdown	RF_CHN_DYN_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Dynamic RF channel shutdown does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD NR FDD	LTE-NR multi-dimensional scheduling optimization	• LTE: LNR_MULTI_D_OPT_SCH_SW option of the GigaBandCell.GigaBandCellAlgoSw parameter • NR: LNR_MULTI_D_OPT_SCH_SW option of the NRDUCellGigaBandCellAlgoSw parameter	<i>LTE FDD and NR Inter-RAT Unified Scheduling</i>	LTE-NR multi-dimensional scheduling optimization does not take effect for cells in the dormancy state.
LTE FDD LTE TDD	Load simulation	CellSimuLoad MO configured	None	Scheduled RF module deep dormancy does not take effect when one or more CellSimuLoad MO instances exist.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD High-frequency NR TDD	Cell load simulation	STRNRDUCELLSIMULATeload	None	- If the SIMU_LOAD_DORMANCY_IMP_SW option of the gNBpwrSavingsAlgo.PwrSavingsAlgoSwitch parameter is selected and cell load simulation takes effect, the RF module serving the local cell is not allowed to enter dormancy, whereas RF modules serving cells not enabled with cell load simulation are allowed to enter dormancy. - If the SIMU_LOAD_DORMANCY_IMP_SW option of the gNBpwrSavingsAlgo.PwrSavingsAlgoSwitch parameter is deselected and cell load simulation takes effect, all RF modules of the base station are not allowed to enter dormancy.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	8T beam mode shift for energy saving	BEAM_MODE_SHIFT_SW option of the SectorSplitGroup.SplitCellGroupOptSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over 8T beam mode shift for energy saving when both functions are enabled.
LTE FDD	32T beam mode shift for energy saving	BEAM_MODE_SHIFT_SW option of the SectorSplitGroup.SplitCellGroupOptSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over 32T beam mode shift for energy saving when both functions are enabled.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	Intelligent Detection for Fronthaul Optical Paths	None	<i>Intelligent Detection for Fronthaul Optical Paths</i>	• If the BBU and RF module are not in the dormancy state when optical path detection is started, the detection process will delay the time at which they enter that state. • If the BBU and RF module are in the dormancy state when optical path detection is started, the following logic applies: – In a scenario where wakeup from dormancy is supported, starting optical path detection will cause the BBU and RF module to automatically exit the dormancy state. They will not re-enter the state within 30 minutes after the detection process

RAT	Function Name	Function Switch	Reference	Description
				starts. Once 30 minutes elapse, they will evaluate whether to re-enter the dormancy state. - In a scenario where wakeup from dormancy is not supported, optical path detection cannot be started.
LTE FDD	Static Multiple Beam 2.0	SMB2_SW option of the SectorSplitGroup.SplitCellGroupOptSwitch parameter	<i>Smart 8T8R (FDD) in eRAN Feature Documentation</i>	RF module deep dormancy preferentially takes effect when it is enabled together with Static Multiple Beam 2.0.
LTE FDD	LNR CRS muting	LNR_CRS_MUTE_SW option of the CellAlgoExtSwitch.LnrZeroBufferzoneSw parameter	<i>LTE and NR Zero Bufferzone</i>	RF module deep dormancy takes precedence over LNR CRS muting when both functions are enabled. LNR CRS muting does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Initial transmission power reduction	INIT_TRANS_POWER_REDUCE_SW option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over initial transmission power reduction when both functions are enabled.
NR FDD Low-frequency NR TDD	Initial transmission power reduction	INIT_TRANS_POWER_REDUCE_SW option of the NRDUCellPwrSavingAlgo.PowerSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Initial transmission power reduction does not take effect when RF module deep dormancy is in effect.
LTE FDD LTE TDD	Retransmission power reduction	RETRANS_POWER_REDUCE_SW option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over retransmission power reduction when both functions are enabled.
NR FDD Low-frequency NR TDD	Retransmission power reduction	RETRANS_POWER_REDUCE_SW option of the NRDUCellPwrSavingAlgo.PowerSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Retransmission power reduction does not take effect when RF module deep dormancy is in effect.
LTE FDD	Uplink TTI-level RF channel dynamic muting	UL_TTI_CHN_DYN_MUTING_SW option of the CellRfChnDynMuting.RfChnDynMutingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over uplink TTI-level RF channel dynamic muting when both functions are enabled.

RAT	Function Name	Function Switch	Reference	Description
NR FDD	Uplink TTI-level RF channel dynamic muting	UL_TTI_CHN_DYN_MUTING_SW option of the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Uplink TTI-level RF channel dynamic muting does not take effect when RF module deep dormancy is in effect.
LTE FDD	Port sparseness	PORT_SPARSENESS_SW option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over port sparseness when both functions are enabled.
LTE FDD	GSM and LTE Buffer Zone Optimization	CellAlgoSwitch.AvoidInterfSwitch	<i>GSM and LTE Buffer Zone Optimization</i>	Cells in the RF module deep dormancy state do not participate in GSM and LTE Buffer Zone Optimization.
LTE FDD	GSM and LTE Spectrum Concurrency	SpectrumCloud.SpectrumCloudSwitch set to GL_SPECTRUM_CONCURRENCY	<i>GSM and LTE Spectrum Concurrency</i>	Cells in the RF module deep dormancy state do not participate in GSM and LTE Spectrum Concurrency.
LTE FDD	GSM and LTE Spectrum Concurrency Phase 2	GL_SPCT_CONCURRENCY_PH2_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter	<i>GSM and LTE Spectrum Concurrency</i>	Cells in the RF module deep dormancy state do not participate in GSM and LTE Spectrum Concurrency Phase 2.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	UMTS and LTE Spectrum Sharing Based on DC-HSDPA (LTE FDD)	SpectrumCloud.SpectrumCloudSwitch set to DC_HSDPA_BASED_UL_SPECTRUM_SHARING	<i>UMTS and LTE Spectrum Sharing Based on DC-HSDPA</i>	RF module deep dormancy cannot take effect in cells where UMTS and LTE Spectrum Sharing Based on DC-HSDPA is enabled.
LTE FDD	UMTS and LTE Spectrum Sharing (LTE FDD)	SpectrumCloud.SpectrumCloudSwitch set to UL_SPECTRUM_SHARING	<i>UMTS and LTE Spectrum Sharing</i>	UMTS and LTE Spectrum Sharing does not take effect when RF module deep dormancy is in effect.
LTE FDD	UMTS and LTE Spectrum Sharing Phase 2 (LTE FDD)	UL_SPECTRUM_SHARING_PH2_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter	<i>UMTS and LTE Spectrum Sharing</i>	UMTS and LTE Spectrum Sharing Phase 2 does not take effect when RF module deep dormancy is in effect.
LTE FDD	UMTS and LTE Zero Bufferzone (LTE FDD)	UMTS_LTE_ZERO_BUFFER_ZONE_SW option of the ULZeroBufferZone.ZeroBufZoneSwitch parameter	<i>UMTS and LTE Zero Bufferzone</i>	UMTS and LTE Zero Bufferzone does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	UMTS and LTE Spectrum Sharing Based on DC-HSDPA (LTE FDD)	SpectrumCloud.SpectrumCloudSwitch set to DC_HSDPA_BASED_UL_SPECTRUM_SHR	<i>UMTS and LTE Spectrum Sharing Based on DC-HSDPA</i>	RF module deep dormancy cannot take effect in cells where UMTS and LTE Spectrum Sharing Based on DC-HSDPA is enabled.
UMTS	Multi-Carrier Switch off Based on Power Backup	NODEBPOWEROUTAGE.ISDSW	<i>Green BTS in RAN Feature Documentation</i>	RF module deep dormancy takes precedence over Multi-Carrier Switch off Based on Power Backup when both functions are enabled. Multi-Carrier Switch off Based on Power Backup does not take effect when RF module deep dormancy is in effect.
UMTS	UMTS low power consumption mode	NODEBALGPARA.LOWPWRCONSUMPTIONMODESW	<i>Green BTS in RAN Feature Documentation</i>	RF module deep dormancy takes precedence over low power consumption mode when both functions are enabled. Low power consumption mode does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
UMTS	VAM Power Saving	NODEBALGPARA.VAMPWRSVSW	<i>Green BTS in RAN Feature Documentation</i>	RF module deep dormancy takes precedence over VAM Power Saving when both functions are enabled. VAM Power Saving does not take effect when RF module deep dormancy is in effect.
UMTS	RF Channel Muting Based Power Saving	ULOCCELLALGPARA.RFCHANNELMUTPWRSVSW	<i>Green BTS in RAN Feature Documentation</i>	RF module deep dormancy takes precedence over RF Channel Muting Based Power Saving when both functions are enabled. RF Channel Muting Based Power Saving does not take effect when RF module deep dormancy is in effect.
UMTS	UMTS RF channel dynamic muting	ULOCCELLALGPARA.RFCHNDYNMUTIN <i>G</i> MODE set to BASICPWRSV or DEEPPWRSV	<i>Green BTS in RAN Feature Documentation</i>	RF module deep dormancy takes precedence over UMTS RF channel dynamic muting when both functions are enabled. UMTS RF channel dynamic muting does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
UMTS	Time Division Carrier Off Based Power Saving	ULOCCELLALGPARA.TDPWRSVSW	<i>Green BTS in RAN Feature Documentation</i>	RF module deep dormancy takes precedence over Time Division Carrier Off Based Power Saving when both functions are enabled. Time Division Carrier Off Based Power Saving does not take effect when RF module deep dormancy is in effect.
UMTS	Dynamic Power Saving Based on Maximum Power Restriction	ULOCCELLALGPARA.PWRRSTRBASEDDY NENRSVVMODE set to BASICPWRSV or DEEPPWRSV	<i>Green BTS in RAN Feature Documentation</i>	RF module deep dormancy takes precedence over Dynamic Power Saving Based on Maximum Power Restriction when both functions are enabled. Dynamic Power Saving Based on Maximum Power Restriction does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
UMTS	UMTS Adaptive Power Saving	ULOCCELLALGPARA.ADPPWRSVSW	<i>Green BTS in RAN Feature Documentation</i>	RF module deep dormancy takes precedence over UMTS Adaptive Power Saving when both functions are enabled. UMTS Adaptive Power Saving does not take effect when RF module deep dormancy is in effect. UMTS Adaptive Power Saving may dynamically adjust power parameters based on air interface changes and update the parameter values on the GUI at around 23:35 each day. If a cell is in dormancy at that time, it exits dormancy, which affects dormancy gains.

RAT	Function Name	Function Switch	Reference	Description
UMTS	Multi-Carrier Switch off Based on Traffic Load	• UCELLDYN SHUT DOWN.DynShut downSwitch set to ON_1 • UCELLDYN SHUT DOWN.DynShut DownType set to CONDITIONALS HUTDOWN	<i>Green BTS in RAN Feature Documentation</i>	It is not recommended that the carrier shutdown period set on the base station controller overlap the RF module deep dormancy period of the local cell. If the periods overlap, RF module deep dormancy preferentially takes effect. Multi-Carrier Switch off Based on Traffic Load does not take effect when RF module deep dormancy is in effect.
UMTS	Multi-Carrier Switch off Based on QoS	• UCELLDYN SHUT DOWN.DynShut downSwitch set to ON_1 • UCELLDYN SHUT DOWN.DynShut DownType set to QoSSHUTDOWN	<i>Green BTS in RAN Feature Documentation</i>	It is not recommended that the carrier shutdown period set on the base station controller overlap the RF module deep dormancy period of the local cell. If the periods overlap, RF module deep dormancy preferentially takes effect. Multi-Carrier Switch off Based on QoS does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
UMTS	Forced cell shutdown	• UCELLDYN SHUT DOWN.DynShut downSwitch set to ON_1 • UCELLDYN SHUT DOWN.DynShut DownType set to FORCESHUTDOWN	<i>Green BTS in RAN Feature Documentation</i>	It is not recommended that the carrier shutdown period set on the base station controller overlap the RF module deep dormancy period of the local cell. If the periods overlap, RF module deep dormancy preferentially takes effect. Forced cell shutdown does not take effect when RF module deep dormancy is in effect.
UMTS	RRU Intelligent Dormancy	NODEBALG PARA.RRUINTELDORFEQB AND	<i>Green BTS in RAN Feature Documentation</i>	It is not recommended that the carrier shutdown period set on the base station controller overlap the RF module deep dormancy period of the local cell. If the periods overlap, RF module deep dormancy preferentially takes effect. RRU Intelligent Dormancy does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
UMTS	Uplink CoMP (Joint Reception)	ULOCCELL.ULCOMPSW	<i>Uplink CoMP in RAN Feature Documentation</i>	After RF module deep dormancy takes effect, coordinating links cannot be set up for the corresponding cells, which affects the gains of Uplink CoMP (Joint Reception).
UMTS	Uplink CoMP (Joint Reception) Based on Coordinated BBU	ULOCCELL.ULCOMPSW	<i>Uplink CoMP in RAN Feature Documentation</i>	After RF module deep dormancy takes effect, coordinating links cannot be set up for the corresponding cells, which affects the gains of Uplink CoMP (Joint Reception) Based on Coordinated BBU.
UMTS	HetNet Uplink CoMP (Joint Reception)	ULOCCELL.ULCOMPSW	<i>Uplink CoMP in RAN Feature Documentation</i>	After RF module deep dormancy takes effect, coordinating links cannot be set up for the corresponding cells, which affects the gains of HetNet Uplink CoMP (Joint Reception).

RAT	Function Name	Function Switch	Reference	Description
UMTS	Coverage Expansion Under Interference	ULOCCELL.ULCOVEXPANSION	<i>Coverage Expansion Under Interference in RAN Feature Documentation</i>	After RF module deep dormancy takes effect, coordinating links cannot be set up for the corresponding cells, which affects the gains of Coverage Expansion Under Interference.
UMTS	Inter-NodeB Uplink CoMP (Joint Reception)	ULOCCELL.INTERNODEBULCOM	<i>Inter-NodeB Uplink CoMP in RAN Feature Documentation</i>	After RF module deep dormancy takes effect, coordinating links cannot be set up for the corresponding cells, which affects the gains of Inter-NodeB Uplink CoMP (Joint Reception).
UMTS	Inter-NodeB HS-DPCCH CoMP (Joint Reception)	ULOCCELL.INTERNODEBHS-DPCCHCOM	<i>Inter-NodeB Uplink CoMP in RAN Feature Documentation</i>	After RF module deep dormancy takes effect, coordinating links cannot be set up for the corresponding cells, which affects the gains of Inter-NodeB HS-DPCCH CoMP (Joint Reception).

RAT	Function Name	Function Switch	Reference	Description
UMTS	Sleeping Cell Detection and Recovery	NODEBALGPARA.SLEEPINGCELLDETECTION	<i>Sleeping Cell Detection and Recovery in RAN Feature Documentation</i>	8:00 to 8:30 is the key monitoring period for differential detection. If the dormancy function is enabled during this period, differential detection for the corresponding cell becomes inaccurate.
UMTS	Adaptive Interference Rejection	NODEBBASEBANDQMPARA.ULBASEBANDQMWORKE set to AIR	<i>Adaptive Interference Rejection in RAN Feature Documentation</i>	When both Adaptive Interference Rejection and dormancy are enabled, Adaptive Interference Rejection is more likely to be triggered.
UMTS	Machine Learning based PCPICH Power Optimization	ULOCCELLALGPARA.SMARTPCPICHOPT SWITCH	<i>Machine Learning based PCPICH Power Optimization in RAN Feature Documentation</i>	Machine Learning based PCPICH Power Optimization may dynamically adjust power parameters based on air interface changes and update the parameter values on the GUI at around 23:35 each day. If a cell is in dormancy at that time, it exits dormancy, which affects dormancy gains.

RAT	Function Name	Function Switch	Reference	Description
UMTS	HSDPA Scheduler Pool	NODEBALGPARA.HSDPASCHPOOLSW	<i>HSDPA Scheduler Pool in RAN Feature Documentation</i>	After RF module dormancy is enabled, HSDPA Scheduler Pool cannot take effect as the baseband equipment can contain only one BBP.
UMTS	HSUPA Scheduler Pool	NODEBALGPARA.HSUPASCHPOOLSW	<i>HSUPA Scheduler Pool in RAN Feature Documentation</i>	After RF module dormancy is enabled, HSUPA Scheduler Pool cannot take effect as the baseband equipment can contain only one BBP.
UMTS	Inter-Dependence of BBU Uplink Resource	NODEBALGPARA.U LRESWORKMODE not set to NORMAL	<i>Inter-Dependence of BBU Uplink Resource in RAN Feature Documentation</i>	After RF module dormancy is enabled, Inter-Dependence of BBU Uplink Resource cannot take effect as the baseband equipment can contain only one BBP.
UMTS	Hitless Cell Rehoming	NODEBALGPARA.D LAUTORESHUFFLES W	<i>Hitless Cell Rehoming in RAN Feature Documentation</i>	After RF module dormancy is enabled, Hitless Cell Rehoming cannot take effect as the baseband equipment can contain only one BBP.

RAT	Function Name	Function Switch	Reference	Description
UMTS	DB-HSDPA	NODEBMULTICELLGRP.MULTICELLGRP <i>TYPE</i> set to HSDPA , and cells added to a group identified by the NODEBMULTICELLGRP MO operating in different frequency bands	<i>Multi-Carrier HSDPA in RAN Feature Documentation</i>	Due to networking restrictions, RF modules operating in mid- and low-frequency bands are not recommended to enter dormancy simultaneously. An RF module serving DB-HSDPA cells cannot enter dormancy.
UMTS	DB-HSDPA+MIMO	ULOCCELL.HSDPADBMIMO	<i>Multi-Carrier HSDPA in RAN Feature Documentation</i>	Due to networking restrictions, RF modules operating in mid- and low-frequency bands are not recommended to enter dormancy simultaneously. An RF module serving DB-HSDPA+MIMO cells cannot enter dormancy.

RAT	Function Name	Function Switch	Reference	Description
UMTS	Inter-Node B DB-HSDPA	INTERNBDB_HSDPA option of the UCELLALGOSWITCH.HspaPlusSwitch parameter	<i>Multi-Carrier HSDPA in RAN Feature Documentation</i>	Due to networking restrictions, RF modules operating in mid- and low-frequency bands are not recommended to enter dormancy simultaneously. An RF module serving inter-NodeB DB-HSDPA cells cannot enter dormancy.
UMTS	UMTS SOC level-based low power consumption mode	NODEBALGPARAM.LOWPWRCONSUMEMODEEXTSW	<i>Green Site</i>	RF module deep dormancy takes precedence over UMTS SOC level-based low power consumption mode when both functions are enabled. UMTS SOC level-based low power consumption mode does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
UMTS	Cell Switch-off Based on Automatic Co-coverage Identification (UMTS)	ULOCCELLALGPARA.CELLSWITCHOFFBS DONCCISW	<i>Cell Switch-off Based on Automatic Co-coverage Identification</i>	It is not recommended that the carrier shutdown period set on the base station controller overlap the RF module deep dormancy period of the local cell. If the periods overlap, RF module deep dormancy preferentially takes effect. Cell Switch-off Based on Automatic Co-coverage Identification (UMTS) does not take effect when RF module deep dormancy is in effect.
UMTS	NCR cell	ULOCCELL.NCRCELL	<i>RuralCow Solution</i>	If NCR cells are configured on an RF module, RF module deep dormancy cannot take effect on that module.
UMTS	UMTS and LTE Spectrum Sharing Based on DC-HSDPA (UMTS)	ULOCCELL.DCHSDPA SHAREDWITHLTE set to DYNAMICBANDWIDTH	<i>UMTS and LTE Spectrum Sharing Based on DC-HSDPA</i>	RF module deep dormancy cannot take effect in cells where UMTS and LTE Spectrum Sharing Based on DC-HSDPA is enabled.

RAT	Function Name	Function Switch	Reference	Description
UMTS	UMTS and LTE Spectrum Sharing (UMTS)	ULOCCELL.ULSPECTRUMSHARINGSWITCH	<i>UMTS and LTE Spectrum Sharing</i>	UMTS and LTE Spectrum Sharing does not take effect when RF module deep dormancy is in effect.
UMTS	UMTS and LTE Spectrum Sharing Phase 2 (UMTS)	ULOCCELL.ULSPECTRUMSHARINGPHASE2	<i>UMTS and LTE Spectrum Sharing</i>	UMTS and LTE Spectrum Sharing Phase 2 does not take effect when RF module deep dormancy is in effect.
UMTS	UMTS and LTE Zero Bufferzone (UMTS)	ULOCCELL.UL0BUFFERZONE	<i>UMTS and LTE Zero Bufferzone</i>	UMTS and LTE Zero Bufferzone does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios	DUAL_PLANE_TOB_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	<i>Base Station Equipment Reliability</i>	When RF module deep dormancy is enabled, RF modules enter the deep dormancy state, which causes an active/standby BBU switchover failure. Therefore, RF module deep dormancy and "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" cannot be both enabled.

Table 7-9 Function impacts of NR load-based RF module deep dormancy

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Clock out-of-synchronization detection	CLK_DETECT_SW option of the gNodeBParam.ClkOutofsyncDetSwitch parameter	<i>Synchronization in 5G RAN Feature Documentation</i>	If clock out-of-synchronization occurs during the dormancy period, this issue may fail to be detected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Air-interface-based network-wide synchronization deviation detection	GAP_BASED_SYNC_DIFF_DET_SW option of the gNodeBParam.SyncDiffDetectSwitch parameter	<i>Synchronization in 5G RAN Feature Documentation</i>	If sequence detection is performed on a to-be-detected base station during the period when RF module deep dormancy takes effect for all cells of the base station, the sequence detection fails. In this case, the possible clock fault of the base station may fail to be detected.
NR FDD Low-frequency NR TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module deep dormancy takes precedence over energy saving based on flexible frequency-domain scheduling when both functions are enabled. Energy saving based on flexible frequency-domain scheduling does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	TTI-level channel shutdown	TTI_RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module deep dormancy takes precedence over TTI-level channel shutdown when both functions are enabled. TTI-level channel shutdown does not take effect when RF module deep dormancy is in effect.
NR FDD Low-frequency NR TDD	Intelligent power control energy saving	SMART_PWR_CTRL_ENERGY_SAVE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module deep dormancy takes precedence over intelligent power control energy saving when both functions are enabled. Intelligent power control energy saving does not take effect when RF module deep dormancy is in effect.
Low-frequency NR TDD	Low-power RF channel dynamic shutdown	LOW_POWER_RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module deep dormancy takes precedence over low-power RF channel dynamic shutdown when both functions are enabled. Low-power RF channel dynamic shutdown does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Intra-RAT ANR	NR_NR_ANR_DEL_SW and NR_NR_ANR_SW options of the NRCellAlgoSwitch.AnrSwitch parameter	<i>ANR in 5G RAN Feature Documentation</i>	When automatic removal of neighboring cells in ANR takes effect and the neighbor relationships with the basic-layer cells are automatically removed for capacity-layer cells, load-based RF module deep dormancy does not take effect.
NR FDD Low-frequency NR TDD	Basic VoNR functions	VONR_SW option of the NRCellAlgoSwitch.VonrSwitch parameter	<i>VoNR in 5G RAN Feature Documentation</i>	Load-based RF module deep dormancy does not take effect if some VoNR UEs fail to be handed over from the capacity-layer cell during UE transfer.
NR FDD Low-frequency NR TDD	Cell-level MRO	MRO_SCENARIO_ID option of the gNBMrro.ScenarioidentifySwitch parameter and INTRA_FREQ_MRO_SW option of the NRCellMrro.MroSwitch parameter	<i>MRO in 5G RAN Feature Documentation</i>	When both RF module deep dormancy and intra-frequency MRO are enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, MRO may take effect later than expected.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	UE-level MRO	UE_MRO_SWITCH option of the NRCellMro.MroSwitch parameter	<i>MRO in 5G RAN Feature Documentation</i>	When both RF module deep dormancy and UE-level MRO are enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, the penalty mechanism accuracy of UE-level MRO is affected.
NR FDD Low-frequency NR TDD	CMAS	gNBsibConfig.SibType set to SIB_TYPE_8	<i>CMAS in 5G RAN Feature Documentation</i>	When an RF module enters the deep dormancy state, the NR cells it serves also enter the dormancy state. In this case, CMAS warning notifications cannot be properly sent in these cells. However, UEs can still receive CMAS warning notifications from other cells.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD High-frequency NR TDD	Symbol power saving	NRDUCellAlgoSwitch.SymbolShutdownSwitch	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module deep dormancy takes precedence over symbol power saving when both functions are enabled. Symbol power saving does not take effect when RF module deep dormancy is in effect.
NR FDD Low-frequency NR TDD High-frequency NR TDD	Power saving scheduling	POWER_SAVING_SCHEDULE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module deep dormancy takes precedence over power saving scheduling when both functions are enabled. Power saving scheduling does not take effect when RF module deep dormancy is in effect.
NR FDD Low-frequency NR TDD High-frequency NR TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module deep dormancy takes precedence over RF channel intelligent shutdown when both functions are enabled. RF channel intelligent shutdown does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Intelligent carrier shutdown in precise mode	• INTRA_GNB_MULTICARR_SD_SW or INTER_GNB_MULTICARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter • MULTICARR_PRECISE_SHDN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter selected	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module deep dormancy takes precedence over intelligent carrier shutdown when both functions are enabled. Intelligent carrier shutdown does not take effect when RF module deep dormancy is in effect.
NR FDD Low-frequency NR TDD High-frequency NR TDD	Intelligent carrier shutdown in non-precise mode	• INTRA_GNB_MULTICARR_SD_SW or INTER_GNB_MULTICARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter • MULTICARR_PRECISE_SHDN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter deselected	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module deep dormancy takes precedence over intelligent carrier shutdown when both functions are enabled. Intelligent carrier shutdown does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Data convergence in slots	DATA_CONVERGENCE_IN_SLOTS_SW option of the NRDuCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module deep dormancy takes precedence over data convergence in slots when both functions are enabled. Data convergence in slots does not take effect when RF module deep dormancy is in effect.
NR FDD Low-frequency NR TDD	Uplink symbol power saving	UL_SYMBOL_SHUT_DOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	When RF module deep dormancy takes effect, uplink symbol power saving produces no energy saving gains.
NR FDD Low-frequency NR TDD	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation in 5G RAN Feature Documentation</i>	A carrier exits a power sharing group once it enters deep dormancy.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Ultra-long holdover for sites with abnormal clock sources	CLK_OTASYNC_PRO C_SW option of the TASM.MSCLKDIAGPROCCESSSW parameter	<i>Synchronization in 5G RAN Feature Documentation</i>	Assume that the ultra-long hold switch is turned on but the ultra-long hold enhancement switch is turned off. If sequence detection is performed on a to-be-detected base station during the period when RF module deep dormancy takes effect for all cells of the base station, the sequence detection fails. In this case, the base station cannot enter the ultra-long holdover state even if its clock is faulty, or it exits the ultra-long holdover state as sequence detection cannot be performed for a long time due to energy saving. If the ultra-long hold enhancement switch is turned on, RF module deep dormancy does not take effect for the base station in the ultra-long holdover state and its reference neighboring base stations.

RAT	Function Name	Function Switch	Reference	Description
NR FDD	Channel isolation for PIM interference avoidance	PIM_INTERF_SMART_AVOID_SW option of the NRDUCellPimInterfAvd.PimAlgoSwitch parameter	<i>Interference Avoidance in 5G RAN Feature Documentation</i>	If RF module deep dormancy has taken effect, channel isolation for PIM interference avoidance does not take effect. If channel isolation for PIM interference avoidance has taken effect, RF module deep dormancy does not take effect.
NR FDD	Intelligent PIM interference avoidance	PIM_INTERF_SMART_AVOID_SW option of the NRDUCellPimInterfAvd.PimAlgoSwitch parameter	<i>Interference Avoidance in 5G RAN Feature Documentation</i>	The RF module deep dormancy function is terminated during PIM interference detection.
NR FDD	Time-frequency two-dimensional unified scheduling optimization	TIME_FREQ_2D_UNIF_OPT_SCH_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgorithmSwitch parameter	<i>Multi-Frequency Smart Aggregation in 5G RAN Feature Documentation</i>	Time-frequency two-dimensional unified scheduling optimization does not take effect for cells in the shutdown state.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	CA Performance Boost Based on Multi-dimensional Optimization	CA_PRFM_BOOST_BSD_ON_MD_OPT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgorithmSwitch parameter	<i>Multi-Frequency Smart Aggregation in 5G RAN Feature Documentation</i>	CA Performance Boost Based on Multi-dimensional Optimization does not take effect for cells in the shutdown state.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Fast carrier shutdown	For capacity-layer cells: - Intra-base-station multi-carrier scenarios: INTRA_GNB_MULTICARRIER_SHUTDOWN_SWITCH.PowerSavingSwitch parameter selected; inter-base-station multi-carrier scenarios: INTER_GNB_MULTICARRIER_SHUTDOWN_SWITCH.PowerSavingSwitch parameter selected FAST_CARRIER_SHUTDOWN_SWITCH option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter For basic-layer cells: FAST_CARRIER_SHUTDOWN_SWITCH option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Fast carrier shutdown does not take effect when RF module deep dormancy is in effect.
NR FDD Low-frequency NR TDD	Deep symbol power saving	DEEP_SYMBOL_SHUTDOWN_SWITCH option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Deep symbol power saving does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD High-frequency NR TDD	Cell load simulation	STRNRDUCELLSIMULATELOAD	None	• If the SIMU_LOAD_DORMANCY_IMP_SW option of the gNBPwrSavingsAlgo.PwrSavingsAlgoSwitch parameter is selected, and cell load simulation takes effect, the RF module serving the local cell is not allowed to enter the dormancy state, but the RF modules serving the cells not enabled with cell load simulation are allowed to enter the dormancy state. • If the SIMU_LOAD_DORMANCY_IMP_SW option of the gNBPwrSavingsAlgo.PwrSavingsAlgoSwitch parameter is deselected, and cell load simulation takes effect, all RF modules of the base station are not allowed to

RAT	Function Name	Function Switch	Reference	Description
				enter the dormancy state.
NR FDD	Uplink symbol power saving	UL_SYMBOL_SHUT_DOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	For NR FDD, uplink symbol power saving does not take effect when RF module deep dormancy is in effect.
NR FDD Low-frequency NR TDD High-frequency NR TDD	Aerial Cell Flag	NRDUCell.AerialCellFlag set to AERIAL_CELL	None	NR load-based RF module deep dormancy cannot take effect in a cell with the NRDUCell.AerialCellFlag parameter set to AERIAL_CELL .
Low-frequency NR TDD	NR multi-band power adaptive scheduling	NR_MULTI_BAND_PWR_ADAPT_SCH_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	A carrier exits the power adaptive scheduling group once it enters the deep dormancy state.
NR FDD	Intelligent MCS optimization for uplink SU-MIMO	UL_SU_MCS_INTEL_OPT_SW option of the NRDUCellAiAlgo.AiAmcAlgoSwitch parameter	<i>Smart 8T8R (FDD) in 5G RAN Feature Documentation</i>	Intelligent MCS optimization for uplink SU-MIMO does not take effect or its gains decrease when it is enabled together with RF module deep dormancy.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Dynamic RF channel shutdown	RF_CHN_DYN_SHUT_DOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Dynamic RF channel shutdown does not take effect when RF module deep dormancy is in effect.
NR FDD	LTE-NR multi-dimensional scheduling optimization	LNR_MULTI_D_OPT_SCH_SW option of the NRDUCellGigaBand.GigaBandCellAlgoSwitch parameter	<i>LTE FDD and NR Inter-RAT Unified Scheduling</i>	LTE-NR multi-dimensional scheduling optimization does not take effect for cells in the dormancy state.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Speed turbo UE identification	gNodeBParam.StUeIdentSw	None	When any of the following configuration operations is performed and the configured function takes effect on the base station, the served cells already in the dormancy state remain in this state but other served cells will not enter this state, resulting in a decrease in the energy saving gains: - Any option of the gNodeBParam.StUeIdentSw parameter is selected. - The gNBAsAppParam.AppDomainName parameter is set. - The gNBAsAppParam.AppIpv4 parameter is set. - The gNBAsAppParam.AppIpv6 parameter is set. The following methods can be used to address the preceding issue:
NR FDD Low-frequency NR TDD	gNodeB acceleration service application	gNBAsAppParam.AppDomainName gNBAsAppParam.AppIpv4 gNBAsAppParam.AppIpv6	None	

RAT	Function Name	Function Switch	Reference	Description
				• Select the SPEED_TURBO_PWR_SAVING_SW option of the gNBPwrSavingsAlgo.PwrSavingsAlgoSwitch parameter. Dormancy can take effect for cells served by the base station in some of the preceding scenarios within the period specified by the gNBPwrSavingsAlgo.SpeedTurboPwrSavingsStartTime and gNBPwrSavingsAlgo.SpeedTurboPwrSavingsEndTime parameters, increasing the energy saving gains. • Select the STATIC_ST_PWR_SAVING_SW option of the gNBPwrSavingsAlgo.PwrSavingsAlgoSwitch parameter. Dormancy can take effect for cells served by the base station in some of the

RAT	Function Name	Function Switch	Reference	Description
				preceding scenarios within the period specified by the gNB PwrSavin gAlgo.StaticST PwrSavStartTime and gNB PwrSavin gAlgo.StaticST PwrSavEndTime parameters, increasing the energy saving gains.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD High-frequency NR TDD	Intelligent Detection for Fronthaul Optical Paths	None	<i>Intelligent Detection for Fronthaul Optical Paths</i>	• If the BBU and RF module are not in the dormancy state when optical path detection is started, the detection process will delay the time at which they enter that state. • If the BBU and RF module are in the dormancy state when optical path detection is started, the following logic applies: – In a scenario where wakeup from dormancy is supported, starting optical path detection will cause the BBU and RF module to automatically exit the dormancy state. They will not re-enter the state within 30 minutes after the detection

RAT	Function Name	Function Switch	Reference	Description
				process starts. Once 30 minutes elapse, they will evaluate whether to re-enter the dormancy state. - In a scenario where wakeup from dormancy is not supported, optical path detection cannot be started.
NR FDD Low-frequency NR TDD	Initial transmission power reduction	INIT_TRANS_POWER_REDUCE_SW option of the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Initial transmission power reduction does not take effect when RF module deep dormancy is in effect.
NR FDD Low-frequency NR TDD	Retransmission power reduction	RETRANS_POWER_REDUCE_SW option of the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Retransmission power reduction does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD	Uplink TTI-level RF channel dynamic muting	UL_TTI_CHN_DYN_MUTING_SW option of the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Uplink TTI-level RF channel dynamic muting does not take effect when RF module deep dormancy is in effect.
NR FDD Low-frequency NR TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag set to HIGH_SPEED	<i>High Speed Mobility in 5G RAN Feature Documentation</i>	When the high-speed cell function in high speed specified policy management is enabled, you are advised not to enable NR load-based RF module deep dormancy or LNR load-based RF module deep dormancy.
Low-frequency NR TDD	Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios	DUAL_PLANE_TOB_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	<i>Base Station Equipment Reliability</i>	When RF module deep dormancy is enabled, RF modules enter the deep dormancy state, which causes an active/standby BBU switchover failure. Therefore, RF module deep dormancy and "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" cannot be both enabled.

In addition to the function impacts of NR load-based RF module deep dormancy (except the impact relationships with intra-RAT ANR and sensing task), LNR load-based RF module deep dormancy has the following function impacts:

Table 7-10 Function impacts of LNR load-based RF module deep dormancy

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Work mode	Cell.WorkMode	None	• An LTE FDD cell with the Cell.WorkMode parameter set to DL_ONLY or LAA cannot be configured as an overlapping neighboring cell of an NR cell. • An LTE TDD cell with the Cell.WorkMode parameter set to LAA cannot be configured as an overlapping neighboring cell of an NR cell.
LTE FDD LTE TDD	Cell System Information Switch	CellSiMap.SiSwitch	None	A cell whose ForbidCellSiSwitch option of the CellSiMap.SiSwitch parameter is selected cannot be configured as a basic-layer cell.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	eMTC introduction	EMTC_SWITCH option of the CellEmtcAlgo.EmtcAlgoSwitch parameter	<i>eMTC in eRAN Feature Documentation</i>	If a capacity-layer cell serves eMTC UEs, these UEs may fail to be handed over, causing a failure for the capacity-layer cell to enter dormancy. An eMTC-only cell (specified by the EMTC_SWITCH and EMTC_ONLY_CELL_SWITCH options of the CellEmtcAlgo.EmtcAlgoSwitch parameter) cannot be configured as a basic-layer cell.
LTE FDD	Static Multiple Beam	SectorSplitGroup.BeamTemplateId set to a value greater than 1	<i>Massive MIMO Basics (FDD) in eRAN Feature Documentation</i>	For LTE FDD, if a Static Multiple Beam cell is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SWITCH option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in NR cells, the user-perceived rate may decrease in the split cell that serves as a basic-layer cell.

RAT	Function Name	Function Switch	Reference	Description
LTE TDD	Intelligent Multi-Beam of 8T8R	None	<i>Intelligent Multi-Beam of 8T8R (TDD) in eRAN Feature Documentation</i>	For LTE TDD, if a cell enabled with Intelligent Multi-Beam of 8T8R is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in NR cells, the user-perceived rate may decrease in the split cell that serves as a basic-layer cell.
LTE FDD	Static Multiple Beam	None	<i>Smart 8T8R (FDD) in eRAN Feature Documentation</i>	For LTE FDD, if a Static Multiple Beam cell is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in NR cells, the user-perceived rate may decrease in the split cell that serves as a basic-layer cell.

RAT	Function Name	Function Switch	Reference	Description
LTE TDD	Intra-frequency split	None	<i>Soft Split Resource Duplex (TDD) in eRAN Feature Documentation</i>	For LTE TDD, if an intra-frequency split cell is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in NR cells, the user-perceived rate may decrease in the split cell that serves as a basic-layer cell.
LTE FDD	Rural Link antenna solution (LTE)	None	<i>RuralLink Solution</i>	For LTE FDD, if a split cell established through the RuralLink antenna solution (LTE) is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in NR cells, the user-perceived rate may decrease in the split cell that serves as the basic-layer cell.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Lumi Link solution (LTE)	• SectorSplitGroup.BeamTemplateId set to 0 • SectorSplitGroup.BeamDeploymentMode set to MODE_5 • SectorSplitCell.CellBeamIndex set to 0, 1, or 2	<i>LumiLink Solution</i>	For LTE FDD, if a split cell established through the LumiLink solution (LTE) is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the ENodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in NR cells, the user-perceived rate may decrease in the split cell that serves as the basic-layer cell.
LTE FDD LTE TDD	RAN sharing	RAN sharing with common carrier: ENodeBSharingMode.ENodeBSharingMode set to SHARED_FREQ RAN sharing with dedicated carrier: ENodeBSharingMode.ENodeBSharingMode set to SEPARATED_FREQ Hybrid RAN sharing: ENodeBSharingMode.ENodeBSharingMode set to HYBRID_SHARED	<i>RAN Sharing in eRAN Feature Documentation</i>	In RAN sharing scenarios, the PLMN IDs of a capacity-layer cell must be a subset of the PLMN IDs of its co-coverage basic-layer cells. Otherwise, the capacity-layer cell cannot enter dormancy.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Fast carrier shutdown	For capacity-layer cells: - Intra-base-station multi-carrier scenarios: INTRA_GNB_MULTICARRIER_SHUTDOWN_SWITCH.PowerSavingSwitch parameter selected; inter-base-station multi-carrier scenarios: INTER_GNB_MULTICARRIER_SHUTDOWN_SWITCH.PowerSavingSwitch parameter selected FAST_CARRIER_SHUTDOWN_SWITCH option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter For basic-layer cells: FAST_CARRIER_SHUTDOWN_SWITCH.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Fast carrier shutdown does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Uplink symbol power saving	UL_SYMBOL_SHUT DOWN_SW option of the SymbolPwrSaving.SymbolShutdownEnhancedSw parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	For LTE FDD, uplink symbol power saving and enhanced uplink symbol power saving do not take effect when RF module deep dormancy is in effect.
LTE FDD	Enhanced uplink symbol power saving	UL_SYMBOL_SHUT DOWN_ENH_SW option of the SymbolPwrSaving.SymbolShutdownEnhancedSw parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	
NR FDD	Uplink symbol power saving	UL_SYMBOL_SHUT DOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	For NR FDD, uplink symbol power saving does not take effect when RF module deep dormancy is in effect.
LTE FDD LTE TDD	Dynamic voltage adjustment	CellAlgoSwitch.Dyn AdjVoltSwitch	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over dynamic voltage adjustment when both functions are enabled. Dynamic voltage adjustment does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	NSA anchor carrier shutdown	NSA_ANCHOR_CARRIER_SHUTDOWN_SW option of the CellShutdownCarrierShutdownEnhancedSw parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	In NSA or NSA and SA hybrid networking, it is recommended that the INTER_RAT_GRAD_PWR_SAVING_SW option of the gNBpwrSavingAlgo.PwrSavingAlgoSwitch parameter be selected when NSA anchor carrier shutdown is enabled for the LTE basic-layer cell and NSA anchor cells and LNR load-based RF module deep dormancy is enabled for NR capacity-layer cells and NSA anchor cells. • If NSA anchor carrier shutdown takes effect first, carriers are shut down in NSA anchor cells. If NR capacity-layer cells do not have other neighboring LTE cells, LNR load-based RF module deep dormancy does not take effect. • If LNR load-based RF module deep

RAT	Function Name	Function Switch	Reference	Description
				dormancy takes effect first, this function can take effect in NR capacity-layer cells.
LTE FDD	LTE-NR multi-dimensional scheduling optimization	LNR_MULTI_D_OPT_SCH_SW option of the GigaBandCell.GigaBandCellAlgoSw parameter	<i>LTE FDD and NR Inter-RAT Unified Scheduling</i>	LTE-NR multi-dimensional scheduling optimization does not take effect for cells in the dormancy state.
LTE FDD	8T beam mode shift for energy saving	BEAM_MODE_SHIFT_SW option of the SectorSplitGroup.SplitCellGroupOptSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over 8T beam mode shift for energy saving when both functions are enabled.
LTE FDD	32T beam mode shift for energy saving	BEAM_MODE_SHIFT_SW option of the SectorSplitGroup.SplitCellGroupOptSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over 32T beam mode shift for energy saving when both functions are enabled.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	Intelligent Detection for Fronthaul Optical Paths	None	<i>Intelligent Detection for Fronthaul Optical Paths</i>	• If the BBU and RF module are not in the dormancy state when optical path detection is started, the detection process will delay the time at which they enter that state. • If the BBU and RF module are in the dormancy state when optical path detection is started, the following logic applies: – In a scenario where wakeup from dormancy is supported, starting optical path detection will cause the BBU and RF module to automatically exit the dormancy state. They will not re-enter the state within 30 minutes after the detection

RAT	Function Name	Function Switch	Reference	Description
				process starts. Once 30 minutes elapse, they will evaluate whether to re-enter the dormancy state. - In a scenario where wakeup from dormancy is not supported, optical path detection cannot be started.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Ultra-long holdover for sites with abnormal clock sources	CLK_OTASYNC_PRO C_SW option of the TASM.MSCLKDIAGPROCCESSW parameter	<i>Synchronization in 5G RAN Feature Documentation</i>	Assume that the ultra-long hold switch is turned on but the ultra-long hold enhancement switch is turned off. If sequence detection is performed on a to-be-detected base station during the period when RF module deep dormancy takes effect for all cells of the base station, the sequence detection fails. In this case, the base station cannot enter the ultra-long holdover state even if its clock is faulty, or it exits the ultra-long holdover state as sequence detection cannot be performed for a long time due to energy saving. If the ultra-long hold enhancement switch is turned on, RF module deep dormancy does not take effect for the base station in the ultra-long holdover state and its reference neighboring base stations.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Static Multiple Beam 2.0	SMB2_SW option of the SectorSplitGroup.SplitCellGroupOptSwitch parameter	<i>Smart 8T8R (FDD) in eRAN Feature Documentation</i>	RF module deep dormancy preferentially takes effect when it is enabled together with Static Multiple Beam 2.0.
LTE FDD	LNR CRS muting	LNR_CRS_MUTE_SW option of the CellAlgoExtSwitch.LnrZeroBufferzoneSwitch parameter	<i>LTE and NR Zero Bufferzone</i>	RF module deep dormancy takes precedence over LNR CRS muting when both functions are enabled. LNR CRS muting does not take effect when RF module deep dormancy is in effect.
LTE FDD	Uplink TTI-level RF channel dynamic muting	UL_TTI_CHN_DYN_MUTING_SW option of the CellRfChnDynMuting.RfChnDynMutingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	When uplink TTI-level RF channel dynamic muting and RF module deep dormancy are both enabled, RF module deep dormancy preferentially takes effect.
NR FDD	Uplink TTI-level RF channel dynamic muting	UL_TTI_CHN_DYN_MUTING_SW option of the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Uplink TTI-level RF channel dynamic muting does not take effect when RF module deep dormancy is in effect.

Table 7-11 Function impacts of LTE load-based RF module deep dormancy

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Carrier aggregation	None	<i>Carrier Aggregation in eRAN Feature Documentation</i>	- Cells in the RF module deep dormancy state do not participate in CA. In CA-group-based configuration mode, after such a cell in a CA group exits this energy saving state, this cell cannot be configured as an SCell for an existing CA UE until the UE accesses the network again. - Cells that exit the RF module deep dormancy state do not participate in CA if the number of routes is limited. The route for CA will be released for a cell in the RF module deep dormancy state. After this cell exits this state, it may not obtain a route if the number of routes is limited and therefore, cannot participate in CA.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Automatic baseband resource configuration	eNodeBResModeAlgo.BbpResAutoConfigSw	None	When the eNodeBResModeAlgo.BbpResAutoConfigSw parameter is set to ON , RF module deep dormancy does not take effect.
LTE TDD	Clock out-of-synchronization detection	ClkDetect.ClkAsyncDetectionSwitch	<i>Synchronization in eRAN Feature Documentation</i>	If clock out-of-synchronization occurs during the dormancy period, this issue may fail to be detected.
LTE FDD LTE TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over energy saving based on flexible frequency-domain scheduling when both functions are enabled. Energy saving based on flexible frequency-domain scheduling does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Work mode	Cell.WorkMode	None	- An LTE FDD cell with the Cell.WorkMode parameter set to DL_ONLY or LAA cannot be configured as a basic-layer cell. - An LTE TDD cell with the Cell.WorkMode parameter set to LAA cannot be configured as a basic-layer cell.
LTE FDD LTE TDD	Cell System Information Switch	CellSiMap.SiSwitch	None	A cell whose ForbidCellSiSwitch option of the CellSiMap.SiSwitch parameter is selected cannot be configured as a basic-layer cell.
LTE FDD LTE TDD	eMTC introduction	EMTC_SWITCH option of the CellEmtcAlgo.EmtcAlgoSwitch parameter	<i>eMTC in eRAN Feature Documentation</i>	If a capacity-layer cell serves eMTC UEs, these UEs may fail to be handed over, causing a failure for the capacity-layer cell to enter dormancy. An eMTC-only cell (specified by the EMTC_SWITCH and EMTC_ONLY_CELL_SWITCH options of the CellEmtcAlgo.EmtcAlgoSwitch parameter) cannot be configured as a basic-layer cell.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Static Multiple Beam	SectorSplitGroup.BeamTemplateId set to a value greater than 1	<i>Massive MIMO Basics (FDD) in eRAN Feature Documentation</i>	For LTE FDD, if a Static Multiple Beam cell is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in LTE cells, the user-perceived rate may decrease in the split cell that serves as a basic-layer cell.
LTE TDD	Intelligent Multi-Beam of 8T8R	None	<i>Intelligent Multi-Beam of 8T8R (TDD) in eRAN Feature Documentation</i>	For LTE TDD, if a cell enabled with Intelligent Multi-Beam of 8T8R is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in LTE cells, the user-perceived rate may decrease in the split cell that serves as a basic-layer cell.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Static Multiple Beam	None	<i>Smart 8T8R (FDD) in eRAN Feature Documentation</i>	For LTE FDD, if a Static Multiple Beam cell is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in LTE cells, the user-perceived rate may decrease in the split cell that serves as a basic-layer cell.
LTE FDD	Downlink intelligent joint power control	DL_INTEL_JOINT_PC_SW option of the CellDisSchExtParam.DisSchExtSwitch parameter	<i>Smart 8T8R (FDD) in eRAN Feature Documentation</i>	Downlink intelligent joint power control does not take effect when RF module deep dormancy is in effect.
LTE FDD	Downlink intelligent joint AMC and power control	DL_INTEL_JOINT_AMC_AND_PC_SW option of the CellDisSchExtParam.DisSchExtSwitch parameter	<i>Smart 8T8R (FDD) in eRAN Feature Documentation</i>	Downlink intelligent joint AMC and power control does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE TDD	Intra-frequency split	None	<i>Soft Split Resource Duplex (TDD) in eRAN Feature Documentation</i>	For LTE TDD, if an intra-frequency split cell is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in LTE cells, the user-perceived rate may decrease in the split cell that serves as the basic-layer cell.
LTE FDD	RuralLink antenna solution (LTE)	None	<i>RuralLink Solution</i>	For LTE FDD, if a split cell established through the RuralLink antenna solution (LTE) is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in LTE cells, the user-perceived rate may decrease in the split cell that serves as the basic-layer cell.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	LumiLink solution (LTE)	• SectorSplitGroup.BeamTemplateId set to 0 • SectorSplitGroup.BeamDeploymentMode set to MODE_5 • SectorSplitCell.CellBeamIndex set to 0, 1, or 2	<i>LumiLink Solution</i>	For LTE FDD, if a split cell established through the LumiLink solution (LTE) is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in LTE cells, the user-perceived rate may decrease in the split cell that serves as the basic-layer cell.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Basic functions of mobility management in connected mode	None	<i>Mobility Management in Connected Mode in eRAN Feature Documentation</i>	LTE load-based RF module deep dormancy triggers inter-frequency handover. An inter-frequency blind handover to a basic-layer cell fails if the cell is in the dormancy state due to energy saving. Therefore, measurement-based inter-frequency handovers are recommended for cells where LTE load-based RF module deep dormancy is enabled.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	RAN sharing	RAN sharing with common carrier: ENodeBSharingMode.ENodeBSharingMode set to SHARED_FREQ RAN sharing with dedicated carrier: ENodeBSharingMode.ENodeBSharingMode set to SEPARATED_FREQ Hybrid RAN sharing: ENodeBSharingMode.ENodeBSharingMode set to HYBRID_SHARED	<i>RAN Sharing in eRAN Feature Documentation</i>	In RAN sharing scenarios, the PLMN IDs of a capacity-layer cell must be a subset of the PLMN IDs of its co-coverage basic-layer cells. Otherwise, the capacity-layer cell cannot enter dormancy.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	CMAS	CellAlgoSwitch.CmasSwitch	<i>CMAS in eRAN Feature Documentation</i>	The following applies when the RF module enters deep dormancy: • With PWS message-triggered exit of cells from energy saving disabled, CMAS warning notifications cannot be properly sent in LTE cells in the dormancy state. However, UEs can still receive CMAS warning notifications from other cells. • With PWS message-triggered exit of cells from energy saving enabled, cells exit the dormancy state upon receipt of a PWS message.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	ETWS	None	<i>ETWS in eRAN Feature Documentation</i>	The following applies when the RF module enters deep dormancy: • With PWS message-triggered exit of cells from energy saving disabled, ETWS warning notifications cannot be sent in LTE cells in the dormancy state. However, UEs can still receive ETWS warning notifications from other cells. • With PWS message-triggered exit of cells from energy saving enabled, cells exit the dormancy state upon receipt of a PWS message.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Intelligent power control energy saving	- Single-cell intelligent power control energy saving: INTEL_INTRA_CELL_PC_SW option of the CellPwrSavingAlgo. <i>PwrSavingAlgoSwitch</i> - Multi-cell coordinated intelligent power control energy saving: INTEL_INTRA_CELL_PC_SW and INTEL_MULTICELL_COOR_PC_SW options of the CellPwrSavingAlgo. <i>PwrSavingAlgoSwitch</i> parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over intelligent power control energy saving when both functions are enabled. Intelligent power control energy saving does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Dynamic carrier shutdown phase 2	CellPwrSavingsAlgo.CarrierDynShutdownPh2Switch set to CAPACITY_CELL	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over dynamic carrier shutdown phase 2 when both functions are enabled. Dynamic carrier shutdown phase 2 does not take effect when RF module deep dormancy is in effect.
LTE FDD LTE TDD	RF channel intelligent shutdown	CellRfShutdown.RfShutdownSwitch	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over RF channel intelligent shutdown when both functions are enabled. RF channel intelligent shutdown does not take effect when RF module deep dormancy is in effect.
LTE FDD LTE TDD	Intelligent power-off of carriers in the same coverage	CellShutdown.CellShutdownSwitch set to ON	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over intelligent power-off of carriers in the same coverage when both functions are enabled. Intelligent power-off of carriers in the same coverage does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Multi-carrier coordinated energy saving	CellShutdown.CellShutdownSwitch set to ON_MULTI_CARRIER_HIER_SHUTDOWN , ON_MULTICARRIER_HIERMANUAL_SHUTDOWN , or ON_CARRIER_PRECISE_SHUTDOWN	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over multi-carrier coordinated energy saving when both functions are enabled. Multi-carrier coordinated energy saving does not take effect when RF module deep dormancy is in effect.
LTE FDD LTE TDD	Dynamic carrier shutdown	CARRIER_DYNAMIC_SHUTDOWN_SWITCH option of the CellShutdown.CarrierShutdownEnhancedSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over dynamic carrier shutdown when both functions are enabled. Dynamic carrier shutdown does not take effect when RF module deep dormancy is in effect.
LTE FDD LTE TDD	Dynamic voltage adjustment	CellAlgorithm.DynAdjustVoltageSwitch	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over dynamic voltage adjustment when both functions are enabled. Dynamic voltage adjustment does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Low power consumption mode (LTE)	CellLowPower.LowPwrSwitch	<i>Green Site</i>	RF module deep dormancy takes precedence over low power consumption mode when both functions are enabled. Low power consumption mode does not take effect when RF module deep dormancy is in effect.
LTE FDD LTE TDD	Cell Switch-off Based on Automatic Co-coverage Identification	CellShutdown.CellShutdownSwitch set to ON_CO_COV_AUTO_IDENT	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over Cell Switch-off Based on Automatic Co-coverage Identification when both functions are enabled. Cell Switch-off Based on Automatic Co-coverage Identification does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Basic symbol power saving	FDD: SymbolShutdownSwitch.PowerSaveSwitch parameter TDD: TddSymbolsShutdownSwitch option of the ENodeBAlgoSwitch.PowerSaveSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over basic symbol power saving when both functions are enabled. Basic symbol power saving does not take effect when RF module deep dormancy is in effect.
LTE FDD LTE TDD	CRS dynamic muting	CRS_DYNAMIC_MUTING_SW option of the SymbolPowerSaving.SymbolShutdownEnhancedSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over CRS dynamic muting when both functions are enabled. CRS dynamic muting does not take effect when RF module deep dormancy is in effect.
LTE FDD LTE TDD	Automatic cell shutdown	CellAutoShutdown.CellAutoShutdownSwitch	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over automatic cell shutdown when both functions are enabled. Automatic cell shutdown does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	RF channel dynamic muting	RF_CHN_DYN_MUTING_SW option of the CellRfChnDynaMuting.RfChannelDynMutingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over RF channel dynamic muting when both functions are enabled. RF channel dynamic muting does not take effect when RF module deep dormancy is in effect.
LTE FDD	Channel isolation for PIM interference avoidance	PIM_INTERF_CHN_SHUT_DOWN_SW option of the PimInterfSmartAvoid.PimAlgoSwitch parameter	<i>Interference Detection and Suppression in eRAN Feature Documentation</i>	If RF module deep dormancy has taken effect, channel isolation for PIM interference avoidance does not take effect. If channel isolation for PIM interference avoidance has taken effect, RF module deep dormancy does not take effect.
LTE FDD	Intelligent PIM interference avoidance	PIM_INTERF_SMART_AVOID_SW option of the PimInterfSmartAvoid.PimAlgoSwitch parameter	<i>Interference Detection and Suppression in eRAN Feature Documentation</i>	The RF module deep dormancy function is terminated during PIM interference detection.
LTE FDD	Massive MIMO downlink coordinated power control	MM_DL_COORD_PC_SW option of the SectorSplitGroup.IntelligentBeamSwitch parameter	<i>Massive MIMO Enhancements (FDD) in eRAN Feature Documentation</i>	Massive MIMO downlink coordinated power control does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Load simulation	CellSimuLoad MO configured	None	LTE load-based RF module deep dormancy does not take effect when at least one CellSimuLoad MO is configured.
LTE FDD	Uplink symbol power saving	UL_SYMBOL_SHUTDOWN_SW option of the SymbolPowerSaving.SymbolsShutdownEnhancedSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	For LTE FDD, uplink symbol power saving and enhanced uplink symbol power saving do not take effect when RF module deep dormancy is in effect.
LTE FDD	Enhanced uplink symbol power saving	UL_SYMBOL_SHUTDOWN_ENH_SW option of the SymbolPowerSaving.SymbolsShutdownEnhancedSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	
LTE FDD	Massive MIMO downlink coordinated power control	MM_DL_COORD_PC_SWITCH option of the SectorSplitGroup.IntelligentBeamSwitch parameter	<i>Massive MIMO Enhancements (FDD) in eRAN Feature Documentation</i>	Massive MIMO downlink coordinated power control does not take effect when RF module deep dormancy is in effect.
LTE FDD LTE TDD	Deep symbol power saving	DEEP_SYMBOL_SHUTDOWN_SW option of the SymbolPowerSaving.SymbolsShutdownEnhancedSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	Deep symbol power saving does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	High speed specified policy management	HSCellSleepInterfAvoidSwitch option of the HighSpdAdaptionPara.InterfAvoidMode parameter	<i>High Speed Mobility in eRAN Feature Documentation</i>	LTE load-based RF module deep dormancy is not recommended for cells where high speed specified policy management is enabled. This is to avoid UE performance loss caused by slow wakeup.
LTE FDD LTE TDD	SOC level-based low power consumption mode	SOCLVL_BASED_LOW_POWER_SW option of the CellLowPower.LowPwrConsumeModeExtSw parameter	<i>Green Site</i>	RF module deep dormancy takes precedence over SOC level-based low power consumption mode when both functions are enabled. SOC level-based low power consumption mode does not take effect when RF module deep dormancy is in effect.
LTE FDD	LTE-NR multi-dimensional scheduling optimization	LNR_MULTI_D_OPT_SCHEDULE_SW option of the GigaBandCellGigaBandCellAlgoSw parameter	<i>LTE FDD and NR Inter-RAT Unified Scheduling</i>	LTE-NR multi-dimensional scheduling optimization does not take effect for cells in the dormancy state.
LTE FDD	8T beam mode shift for energy saving	BEAM_MODE_SHIFT_SW option of the SectorSplitGroup.SplitCellGroupOptSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over 8T beam mode shift for energy saving when both functions are enabled.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	32T beam mode shift for energy saving	BEAM_MODE_SHIFT_SW option of the SectorSplitGroupSplitCellGroupOptSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over 32T beam mode shift for energy saving when both functions are enabled.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Intelligent Detection for Fronthaul Optical Paths	None	<i>Intelligent Detection for Fronthaul Optical Paths</i>	• If the BBU and RF module are not in the dormancy state when optical path detection is started, the detection process will delay the time at which they enter that state. • If the BBU and RF module are in the dormancy state when optical path detection is started, the following logic applies: – In a scenario where wakeup from dormancy is supported, starting optical path detection will cause the BBU and RF module to automatically exit the dormancy state. They will not re-enter the state within 30 minutes after the detection process starts. Once

RAT	Function Name	Function Switch	Reference	Description
				30 minutes elapse, they will evaluate whether to re-enter the dormancy state. - In a scenario where wakeup from dormancy is not supported, optical path detection cannot be started.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Specified service carrier	Cell.SpecificCellFlag	<i>WBB in eRAN Feature Documentation</i>	WBB and non-emergency-call MBB UEs can only be handed over to the corresponding specified-service cells. LTE load-based RF module deep dormancy enables UEs to be transferred out of a capacity-layer cell when the trigger conditions are met. If the number of UEs in the cell after the transfer is greater than the value of the CellPwrSavingPolicy.ForcibleTriggerUeNumThld parameter, LTE load-based RF module deep dormancy cannot take effect; if the number of UEs in the cell after the transfer is less than or equal to the value of this parameter, LTE load-based RF module deep dormancy can take effect and the remaining UEs in the cell are directly released.
LTE FDD	Static Multiple Beam 2.0	SMB2_SW option of the SectorSplitGroup.SplitCellGroupOptSwitch parameter	<i>Smart 8T8R (FDD) in eRAN Feature Documentation</i>	RF module deep dormancy preferentially takes effect when it is enabled together with Static Multiple Beam 2.0.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	LNR CRS muting	LNR_CRS_MUTE_SW option of the CellAlgoExtSwitch.LnrZeroBufferzoneSw parameter	<i>LTE and NR Zero Bufferzone</i>	RF module deep dormancy takes precedence over LNR CRS muting when both functions are enabled. LNR CRS muting does not take effect when RF module deep dormancy is in effect.
LTE FDD	Initial transmission power reduction	INIT_TRANSPowerREDUCE_SW option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over initial transmission power reduction when both functions are enabled.
LTE FDD LTE TDD	Retransmission power reduction	RETRANS_PowerREDUCE_SW option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	When retransmission power reduction and RF module deep dormancy are both enabled, RF module deep dormancy preferentially takes effect.
LTE FDD	Uplink TTI-level RF channel dynamic muting	UL_TTI_ChannelDynamicMUTING_SW option of the CellRfChnDynamicMuting.RfChannelDynMutingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	When uplink TTI-level RF channel dynamic muting and RF module deep dormancy are both enabled, RF module deep dormancy preferentially takes effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Port sparseness	PORT_SPARSENESS_SW option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module deep dormancy takes precedence over port sparseness when both functions are enabled.
LTE FDD	GSM and LTE Buffer Zone Optimization	CellAlgoSwitch.AvoidInterfSwitch	<i>GSM and LTE Buffer Zone Optimization</i>	Cells in the RF module deep dormancy state do not participate in GSM and LTE Buffer Zone Optimization.
LTE FDD	GSM and LTE Spectrum Concurrency	SpectrumCloud.SpectrumCloudSwitch set to GL_SPECTRUM_CONCURRENCY	<i>GSM and LTE Spectrum Concurrency</i>	Cells in the RF module deep dormancy state do not participate in GSM and LTE Spectrum Concurrency.
LTE FDD	GSM and LTE Spectrum Concurrency Phase 2	GL_SPCT_CONCURRENCY_PH2_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter	<i>GSM and LTE Spectrum Concurrency</i>	Cells in the RF module deep dormancy state do not participate in GSM and LTE Spectrum Concurrency Phase 2.
LTE FDD	UMTS and LTE Spectrum Sharing Based on DC-HSDPA (LTE FDD)	SpectrumCloud.SpectrumCloudSwitch set to DC_HSDPA_BASED_UL_SPECTRUM_SHR	<i>UMTS and LTE Spectrum Sharing Based on DC-HSDPA</i>	RF module deep dormancy cannot take effect in cells where UMTS and LTE Spectrum Sharing Based on DC-HSDPA is enabled.
LTE FDD	UMTS and LTE Spectrum Sharing (LTE FDD)	SpectrumCloud.SpectrumCloudSwitch set to UL_SPECTRUM_SHARING	<i>UMTS and LTE Spectrum Sharing</i>	UMTS and LTE Spectrum Sharing does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	UMTS and LTE Spectrum Sharing Phase 2 (LTE FDD)	UL_SPECTRUM_SHR_PH2_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter	<i>UMTS and LTE Spectrum Sharing</i>	UMTS and LTE Spectrum Sharing Phase 2 does not take effect when RF module deep dormancy is in effect.
LTE FDD	UMTS and LTE Zero Bufferzone (LTE FDD)	UMTS_LTE_ZERO_BUFFER_ZONE_SW option of the ULZeroBufferZone.ZeroBufferZoneSwitch parameter	<i>UMTS and LTE Zero Bufferzone</i>	UMTS and LTE Zero Bufferzone does not take effect when RF module deep dormancy is in effect.

7.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

Boards

Only UMPT series main control boards support this function.

Only UBBP series boards support this function.

RF Modules

AAUs that support UMTS (listed in [Table 7-55](#)) can be used.

5000 series 32T or 64T AAUs that support LTE TDD or NR TDD and comply with the eCPRI protocol can be used.

5000 series 32T AAUs that support LTE FDD or NR FDD and comply with the CPRI or eCPRI protocol can be used.

For high-frequency NR TDD:

Only the HAAU5233w and HAAU5235w support scheduled RF module deep dormancy and NR load-based RF module deep dormancy.

Only the AAU5368md and AAU5767t support multi-frequency flexible dormancy.

7.2.3.4 Networking

- An RF module shared by multiple RATs can enter dormancy only when the dormancy conditions of all involved RATs are met.
- In a separate-MPT base station, if two main control boards are configured with different dormancy types, the expected dormancy type does not take effect on the RF module shared by two sides of the base station. You are advised to configure the same dormancy type for the two main control boards.
- In multi-operator scenarios, the procedure of entering this function for both primary and secondary operators is determined by the networking of the primary operator.
- Requirements of NR load-based RF module deep dormancy:
In NSA networking, if the capacity layer and basic layer are deployed at different sites, information is transmitted over the Xn interface. If no Xn interface is deployed between the gNodeBs enabled with this function, IP addresses need to be allocated for Xn setup. Otherwise, NR load-based RF module deep dormancy cannot take effect in inter-base-station scenarios.
- Requirements of LNR load-based RF module deep dormancy:
 - X2 links need to be set up in inter-base-station SA networking. X2 links can be set up automatically or manually. Automatic setup is recommended. For details about X2 configurations, see *X2 Self-Management in SA Networking*.
 - In NSA networking, the NSA DC capability must be enabled for all basic-layer and capacity-layer cells. To do so, select the **NSA_DC_CAPABILITY_SWITCH** option of the **NsaDcMgmtConfig.NsaDcAlgoSwitch** parameter on the eNodeB side and turn on the switch controlled by the **NRCellAlgoSwitch.NsaDcSwitch** parameter on the gNodeB side.
 - In NSA and SA hybrid networking, the configuration and activation procedures of this function are the same as those in NSA networking. The load information is transmitted between the eNodeB and gNodeB over the X2 interface.
- When LNR or NR load-based deep dormancy is enabled and the NR capacity-layer cell is a MOCN-enabled cell, ensure that the operator group configured for the basic-layer cells (co-coverage neighboring LTE or NR cells) contains the operator group of the capacity-layer cell. MOCN refers to multi-operator core network.
- All activated UMTS/LTE/NR cells of the base station have been bound to BBPs.

NOTE

BBP binding is optional for LTE cells in the standby mode (indicated by the **Cell.EuCellStandbyMode** parameter with the value of **STANDBY**).

The following binding methods for LTE/NR are for reference:

- Method 1: automatic binding
 - LTE side: Set the **eNodeBResModeAlgo.BbpResAutoConfigSw** parameter to **ON_BIND** to enable automatic baseband resource configuration.

- NR side: Set the **gNodeBParam.AutoCellBindBasebandEqmSw** parameter to **ON** to enable automatic cell binding to baseband equipment.
- Method 2: manual binding
 - LTE side: Run the **MOD EUCCELLSECTOREQM** command to bind specified BBPs to cells.
 - NR side: Run the **MOD NRDUCELLTRP** command to bind specified BBPs to cells.

Binding method for UMTS: Set the **ULOCCELL.ULBASEBANDEQMID** and **ULOCCELL.DLBASEBANDEQMID** parameters to bind specified BBPs to cells in the uplink and downlink, respectively. Each piece of uplink or downlink baseband equipment can contain only one BBP.

- GU load-based RF module dormancy and GU inter-RAT scheduled RF module dormancy require information exchange between the GSM and UMTS networks. If the base station controllers are a BSC and an RNC, operators must add the physical connections for the lur-g interface between the BSC and the RNC. For details on how to configure and enable the lur-g interface, see the following paths:
 - BSC6910: section "Configuring the lur-g Interface (IP Transmission Mode)" in *BSC6910 GU Initial Configuration Guide* in *BSC6910 GU Product Documentation*.
 - BSC6960: sections "Configuring the lur-g Interface (over IP)" in *BSC6960 GSM Initial Configuration Guide* in *BSC6960 GSM Product Documentation* and "Configuring the lur Interface (in Transmission Resource Pool Mode)" in *BSC6960 UMTS Initial Configuration Guide* in *BSC6960 UMTS Product Documentation*.

7.2.3.5 Cells

- LNR or NR load-based deep dormancy cannot be enabled for NR SUL cells (with the **NRDUCell.DuplexMode** parameter set to **CELL_SUL**) alone. SUL cells enter the dormancy state along with the NR TDD cell.
- NR SUL cells do not support optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module deep dormancy.
- SDL cells (with **NRDUCell.DuplexMode** set to **CELL_SDL**) do not support LNR load-based RF module deep dormancy.
- SDL cells cannot serve as basic-layer cells. If an SDL cell is configured as a basic-layer cell, dormancy-related functions cannot take effect in the corresponding capacity-layer cells.
- Because the RNC and NodeB must collaborate to determine whether a UMTS RF module can enter deep dormancy, all UMTS cells on the RF module must be in the logical-cell active state.
- RF module deep dormancy cannot take effect in cells where UMTS and LTE Spectrum Sharing Based on DC-HSDPA is enabled.

7.2.3.6 Others

- When the **RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB)** parameter is set to **ON**, the working mode of RF modules (specified by the **RRU.RS (5G**

gNodeB, LTE eNodeB) parameter) must be set to **NO, TDL, LO, MO, UO**, or any combination of these values.

- When the **RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB)** parameter is set to **ON_DEEP_SUPER**, the working mode of RF modules (specified by the **RRU.RS (5G gNodeB, LTE eNodeB)** parameter) must be set to **NO, TDL, LO, UO**, or any combination of these values.
- To implement reliability protection during deep dormancy, both the OSS and NEs must be upgraded to versions that support reliability protection.
- In high-frequency NR TDD scenarios, it is recommended that base stations serving low-frequency basic-layer cells be upgraded to V100R020C10 or later. If these base stations run V100R020C00 or an earlier version, some exit conditions will become invalid.
- For UMTS dormancy scenarios, only the BSC6910 and BSC6960 support this function, and the base station controller must be upgraded to the mapping version.

7.2.4 Operation and Maintenance

7.2.4.1 Data Configuration

7.2.4.1.1 Data Preparation

Table 7-12 describes the parameter shared by all sub-functions. The parameters that need to be separately set for each sub-function are as follows:

- **Table 7-13** describes the parameters used for activating scheduled RF module deep dormancy/NR local-cell-load-based RF module deep dormancy.
- **Table 7-14** describes the parameters used for activating NR load-based RF module deep dormancy.
- **Table 7-15** and **Table 7-16** describe the parameters used for activating LNR load-based RF module deep dormancy.
- **Table 7-17** describes the parameters used for activating LTE load-based RF module deep dormancy.
- **Table 7-18** describes the parameters used for activating UMTS load-based RF module deep dormancy.
- **Table 7-19** and **Table 7-20** describe the parameters used for activating GU load-based RF module deep dormancy.
- **Table 7-14, Table 7-16, and Table 7-17** describe the parameters used for activating multi-frequency flexible dormancy.

Table 7-12 Parameter shared by all sub-functions

Parameter Name	Parameter ID	Setting Notes
Equipment Dormancy Optional Feature Switch	EQUIPMENT.EQMDO ROPTFEATSW (5G gNodeB, LTE eNodeB)	Deselect the GU_DC_SW option, which is the default setting. If the option is deselected and a configuration command is executed to trigger dormancy pending or dormancy pending is enabled on the local end that works in UMTS, then dormancy pending will take effect on the entire base station. In this case, the dormancy duration of the entire base station is shortened. This option must also be deselected for the NE to which the RF module belongs if the RRU.RS (LTE eNodeB, 5G gNodeB) parameter is configured to include GSM or UMTS and the RRU.DORMANCYSW (5G gNodeB, LTE eNodeB) parameter is set to ON, ON_SUPER, or ON_DEEP_SUPER.

Table 7-13 Parameters used for activating scheduled RF module deep dormancy/NR local-cell-load-based RF module deep dormancy

Parameter Name	Parameter ID	Setting Notes
Dormancy Switch	RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB)	To enable scheduled RF module deep dormancy, set this parameter to ON or ON_DEEP_SUPER. The former value is recommended. After this switch is turned on, an RF module enters dormancy during a specified period each day. If, on a particular day, the RF module should not enter dormancy during that period due to service fluctuations or other factors, turn off this switch in advance to ensure service continuity.

Parameter Name	Parameter ID	Setting Notes
Dormancy Start Time	<i>RRU.DORMANCYSTARTTIME (LTE eNodeB, 5G gNodeB)</i>	You are advised to set this parameter to the beginning of a period with no services. You are not advised to set the dormancy start time to the top of the hour to avoid significant fluctuations in the N.HO.IntraFreq.FailOut.Prepare Failure counter value. For example, if deep dormancy needs to take effect at 00:00:00, you are advised to set the <i>RRU.DORMANCYSTARTTIME (LTE eNodeB, 5G gNodeB)</i> parameter to 23:55:00.
Dormancy Stop Time	<i>RRU.DORMANCYSTOPTIME (LTE eNodeB, 5G gNodeB)</i>	You are advised to set this parameter to the end of a period with no services. The stop time specified by <i>RRU.DORMANCYSTOPTIME (LTE eNodeB, 5G gNodeB)</i> must be later than the start time specified by <i>RRU.DORMANCYSTARTTIME (LTE eNodeB, 5G gNodeB)</i> plus the waiting time specified by <i>RRU.DORMANCYTIMER (LTE eNodeB, 5G gNodeB)</i>. If the start time is the same as the stop time, scheduled RF module deep dormancy does not take effect. If the stop time is 1s earlier than the start time (for example, when the configured dormancy period is 00:00:00–23:59:59), RF module deep dormancy remains in effect throughout the entire day.
Dormancy Waiting Timer	<i>RRU.DORMANCYTIMER (LTE eNodeB, 5G gNodeB)</i>	The default value is recommended.

Parameter Name	Parameter ID	Setting Notes
Max Random Dormancy Waiting Time	EQUIPMENT.MAXRAMDORMANCYWAITTIME (LTE eNodeB, 5G gNodeB)	The default value is recommended. When the dormancy conditions are met, the system waits for a random period determined by this parameter before entering deep dormancy. This prevents a large number of sites from entering deep dormancy simultaneously and reduces the resulting signaling impact on the core network.
Custom Dormancy Standard Switch	RRU.CUSTOMDORMANCYSTDSW (LTE eNodeB, 5G gNodeB)	The default value is recommended.
Dormancy Standard	RRU.DORMANCYSTD (LTE eNodeB, 5G gNodeB)	The default value is recommended. It is recommended that the current parameter be set to include the RATs requiring dormancy only when the RRU.CUSTOMDORMANCYSTDSW (LTE eNodeB, 5G gNodeB) parameter is set to ON .
Power Saving Policy ID	RRU.PWRSVGPOLICYID (LTE eNodeB, 5G gNodeB)	When the dormancy switch is turned on and this parameter is set to a non-zero value, the dormancy period is determined by the configured power saving policy. If this parameter is set to 0 , the dormancy period is determined by the specified dormancy start time and stop time.
Operation Type	OPERATIONTYPE	The value PENDING is recommended.
Pending Duration	PENDINGDURATION	None
Necessary HO Prepare Fail Punish Timer	gNBMobilityCommParam.NecHoPrepFailPunish-Timer	It is recommended that this parameter be set to 30 in scheduled RF module deep dormancy scenarios.

Parameter Name	Parameter ID	Setting Notes
Optimized HO Prepare Fail Punish Timer	gNBMobilityCommParam.OptHoPrepFailPunish-Timer	It is recommended that this parameter be set to 30 in scheduled RF module deep dormancy scenarios.
Policy ID	BRDPWRSVGPOLICY.POLICYID (LTE eNodeB, 5G gNodeB)	When the dormancy switch is turned on and the RRU.PWRSVGPOLICYID (LTE eNodeB, 5G gNodeB) parameter is set to a non-zero value, use this parameter to associate the specified dormancy type and waiting time with the RF module.
Power Saving Type	BRDPWRSVGPOLICY.POWERSAVINGTYPE (LTE eNodeB, 5G gNodeB)	It is recommended that this parameter be set to RF_DORMANCY .
Dormancy Waiting Timer	BRDPWRSVGPOLICY.DORMANCYTIMER (LTE eNodeB, 5G gNodeB)	The default value is recommended.
Policy ID	BRDPWRSVGPOLICY-TIME.POLICYID (LTE eNodeB, 5G gNodeB)	When the dormancy switch is turned on and the RRU.PWRSVGPOLICYID (LTE eNodeB, 5G gNodeB) parameter is set to a non-zero value, set this parameter to configure the corresponding dormancy period.
Time ID	BRDPWRSVGPOLICY-TIME.TIMEID (LTE eNodeB, 5G gNodeB)	None

Parameter Name	Parameter ID	Setting Notes
Start Time	BRDPWRSVGPOLICY-TIME.STARTTIME (LTE eNodeB, 5G gNodeB)	You are advised to set this parameter to the beginning of a period with no services. You are not advised to set the dormancy start time to the top of the hour to avoid significant fluctuations in the N.HO.IntraFreq.FailOut.Prepare Failure counter value. For example, if deep dormancy needs to take effect at 00:00:00, you are advised to set the BRDPWRSVGPOLICY-TIME.STARTTIME (LTE eNodeB, 5G gNodeB) parameter to 23:55:00.
Stop Time	BRDPWRSVGPOLICY-TIME.STOPTIME (LTE eNodeB, 5G gNodeB)	You are advised to set this parameter to the end of a period with no services. The stop time specified by BRDPWRSVGPOLICY-TIME.STOPTIME (LTE eNodeB, 5G gNodeB) must be later than the start time specified by BRDPWRSVGPOLICY-TIME.STARTTIME (LTE eNodeB, 5G gNodeB) plus the waiting time specified by BRDPWRSVGPOLICY.DORMANCYTIMER (LTE eNodeB, 5G gNodeB). If the start time is the same as the stop time, scheduled RF module deep dormancy does not take effect. If the stop time is 1s earlier than the start time (for example, when the configured dormancy period is 00:00:00–23:59:59), RF module deep dormancy remains in effect throughout the entire day.

Parameter Name	Parameter ID	Setting Notes
Multi Period Power Saving Type	BRDPWRSVGPOLICY-TIME.MPPWRSVGTYPE (LTE eNodeB, 5G gNodeB)	The default value is recommended.
Power Saving Algorithm Switch	EnodebPwrSavingAlgo.PwrSavingAlgoSwitch	Select the MULTI_RRU_CELL_DORM_COMPAT_SW option of this parameter when the multi-RRU cell dormancy compatibility function needs to be enabled for RF module dormancy to take effect in multi-RRU cells, such as SFN cells.
Power Saving Algorithm Switch	gNBPwrSavingAlgo.PwrSavingAlgoSwitch	Select the SIMU_LOAD_DORMANCY_IMP_SW option to prevent RF modules serving cells with cell load simulation enabled from entering dormancy, while allowing RF modules serving cells without cell load simulation enabled to enter dormancy.
Power Saving Algorithm Switch	NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch	Select the NR_DORMANCY_JUDGE_THRESHOLD_OPT_SW option of this parameter if optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module deep dormancy is required.
Power Saving Policy Index	NRDUCellPowerSaving.PowerSavingPolicyIndex	Set this parameter based on the network plan if optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module deep dormancy is required.

Parameter Name	Parameter ID	Setting Notes
Start Time	NRDUCellPowerSaving.StartTime	Set these parameters based on the network plan if optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module deep dormancy is required. In addition, the following requirements must be met: - For the same policy of the same energy saving type, the interval between the stop time and the start time must be longer than 20 minutes. If the stop time is earlier than or equal to the start time, the stop time is interpreted as occurring on the next day. For example: If the start time is 8:00 a.m. and the stop time is 7:00 a.m., the power saving function is enabled from 8:00 a.m. to 7:00 a.m. on the next day. - For different policies of the same energy saving type, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving type needs to be enabled after policy A of this type has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A. - The period of RF module timing dormancy cannot overlap with that specified when the NRDUCellPowerSaving.PowerSavingType
Stop Time	NRDUCellPowerSaving.StopTime	

Parameter Name	Parameter ID	Setting Notes
		parameter is set to TIMING_CARRIER_SHUTDOWN for the local cell. The period must be the same as that specified by the RRU.DORMANCYSTARTTIME (LTE eNodeB, 5G gNodeB) and RRU.DORMANCYSTOPTIME (LTE eNodeB, 5G gNodeB) parameters.
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	Set this parameter to NR_DORMANCY_JUDGE_THRESHOLD_OPT if optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module deep dormancy is required.
Downlink PRB Threshold	NRDUCellPowerSaving.DlPrbThld	You are advised to set this parameter to its recommended value when optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module deep dormancy is enabled.
Uplink PRB Threshold	NRDUCellPowerSaving.UlPrbThld	You are advised to set this parameter to its recommended value when optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module deep dormancy is enabled.
UE Number Threshold	NRDUCellPowerSaving.UserNumThld	You are advised to set this parameter to its recommended value when optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module deep dormancy is enabled. In addition, this parameter must be set to a value less than or equal to 150 .

Parameter Name	Parameter ID	Setting Notes
Evaluation Period Granularity	NRDUCellPowerSaving.EvalPeriodGranularity	Set this parameter to MINUTE_LEVEL if optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module deep dormancy is enabled.
Triggering Evaluation Period	NRDUCellPowerSaving.ShutdownTrigJudgePeriod	You are advised to set this parameter to its recommended value when optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module deep dormancy is enabled.
Cell Load Filtering Mode	gNBPwrSavingAlgo.CellLoadFilterMode	This parameter specifies the filtering mode used in uplink/downlink PRB load measurement when optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module deep dormancy is enabled. You are advised to set this parameter to its recommended value.

Parameter Name	Parameter ID	Setting Notes
SUL User Number Threshold	NRDUCellPowerSaving.SulUserNumThld	You are advised to set this parameter to its recommended value when optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module deep dormancy is enabled. - This parameter is invalid when it is set to 65535. - If the FLEX_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter is deselected and the cell has a NRDUCellPowerSaving record with power saving type set to LTE and NR intelligent carrier shutdown, optimized evaluation of NR RF module dormancy, NR load-based RF module dormancy, or LN load-based RF module dormancy, then the NRDUCellPowerSaving.SulUserNumThld parameter of this record must be set to 65535.
HandOver Switch	UCORRMALGOSWITCH.HoSwitch	- To enable UMTS inter-frequency handovers, select the HO_INTER_FREQ_HARD_HO_SWITCH option. - To enable inter-RAT handovers for UMTS, select the HO_INTER_RAT_CS_OUT_SWITCH and HO_INTER_RAT_PS_OUT_SWITCH options.

Parameter Name	Parameter ID	Setting Notes
GSM Cell Index	U2GNCELL.<i>GSMCellIndex</i>	To enable inter-RAT handovers for UMTS, configure a basic-layer cell as a co-coverage inter-RAT neighboring cell of the capacity-layer cell.
Inter-freq Target Frequency Trigger RSCP THD	UCELLMCLDR.<i>TargetFreqThdRscp</i>	You are advised to set this parameter to its recommended value.
Inter-freq Target Frequency Trigger Ec/No THD	UCELLMCLDR.<i>TargetFreqThdEcNo</i>	You are advised to set this parameter to its recommended value.
Cell Power Saving Inter Plmn Handover Allowed Switch	URNCELLDORMANCY-PARA.<i>CellInterPlmnHoSwitch</i>	Set this parameter to ON .
Intra RAT Inter Plmn Ho Allowed	UOPERATORSHARING-MODE.<i>InterPlmnHoAllowedIntraRat</i>	Set this parameter to YES .
Inter RAT Inter Plmn Ho Allowed	UOPERATORSHARING-MODE.<i>InterPlmnHoAllowedInterRat</i>	Set this parameter to YES .
Cell Power Saving Ignore CBS Switch	URNCELLDORMANCY-PARA.<i>CellIgnoreCbsSwitch</i>	You are advised to set this parameter to its recommended value.
CBS Service Hold Duration	URNCELLDORMANCY-PARA.<i>CbsServiceHoldDuration</i>	You are advised to set this parameter to its recommended value.

Table 7-14 Parameters used for activating NR load-based RF module deep dormancy

Parameter Name	Parameter ID	Setting Notes
Dormancy Switch	RRU.<i>DORMANCYSW (5G gNodeB, LTE eNodeB)</i>	To enable NR load-based RF module deep dormancy, set this parameter to ON or ON_DEEP_SUPER . The former value is recommended.

Parameter Name	Parameter ID	Setting Notes
RRU Dormancy Optional Feature Switch	RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)	To enable multi-frequency flexible dormancy, select the MULTI_FREQ_FLEX_SLEEP_SW option.
Power Saving Switch	NRDUCellAlgoSwitch.PowerSavingSwitch	To enable NR load-based RF module deep dormancy, select the INTRA_NR_DORMANCY_BY_LOAD_SW option of this parameter.
Power Saving Extension Switch	NRDUCellAlgoSwitch.PowerSavingExtSwitch	To enable NR load-based precise dormancy, both the INTRA_NR_DORMANCY_BY_LOAD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter and the PRECISE_DORMANCY_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter must be selected. To calculate the overlap rate based on grids, select the INTRA_NR_DORMANCY_BY_LOAD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter and the PRECISE_DORMANCY_SW and GRID_MODE_SW options of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter. The precise mode is not supported in high-frequency TDD.
Power Saving Policy Index	NRDUCellPowerSaving.PowerSavingPolicyIndex	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Start Time	NRDUCellPowerSaving.StartTime	Set these parameters based on the network plan and the following requirements: - For the same policy of the same energy saving type, the interval between the stop time and the start time must be longer than 20 minutes. If the stop time is earlier than or equal to the start time, the stop time is interpreted as occurring on the next day. For example: If the start time is 8:00 a.m. and the stop time is 7:00 a.m., the power saving function is enabled from 8:00 a.m. to 7:00 a.m. on the next day. - For different policies of the same energy saving type, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving type needs to be enabled after policy A of this type has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A. - The period for RF module deep dormancy should not overlap with the period specified when the NRDUCellPowerSaving.PowerSavingType parameter is set to TIMING_CARRIER_SHUTDOWN for the local cell.
Stop Time	NRDUCellPowerSaving.StopTime	
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	Set this parameter to INTRA_NR_DORMANCY_BY_LOAD .
Downlink PRB Threshold	NRDUCellPowerSaving.DlPrbThld	You are advised to set this parameter to its recommended value.
Downlink PRB Offset	NRDUCellPowerSaving.DlPrbOffset	You are advised to set this parameter to its recommended value.

Parameter Name	Parameter ID	Setting Notes
Uplink PRB Threshold	NRDUCellPowerSaving.UlPrbThld	You are advised to set this parameter to its recommended value.
Neighboring Cell Uplink PRB Offset	NRDUCellPowerSaving.NCellUlPrbOffset	You are advised to set this parameter to its recommended value.
UE Number Threshold	NRDUCellPowerSaving.UserNumThld	You are advised to set this parameter to its recommended value. The maximum value is 150 .
User Number Offset	NRDUCellPowerSaving.UserNumOffset	You are advised to set this parameter to its recommended value.
Evaluation Period Granularity	NRDUCellPowerSaving.EvalPeriodGranularity	You are advised to set this parameter to its recommended value. For high-frequency TDD, this parameter must be set to MINUTE_LEVEL .
Triggering Evaluation Period	NRDUCellPowerSaving.ShutdownTrigJudgePeriod	You are advised to set this parameter to its recommended value.
Exit Evaluation Period	NRDUCellPowerSaving.ExitEvaluationPeriod	You are advised to set this parameter to its recommended value.

Parameter Name	Parameter ID	Setting Notes
Second-Level Trigger Evaluation Period	NRDUCellPowerSaving.SecondLevelTrigEvalPrd	You are advised to set this parameter to its recommended value. - This parameter can be configured and take effect only when the NRDUCellPowerSaving.EvalPeriodGranularity parameter is set to SECOND_LEVEL. - When the NRDUCellPowerSaving.PowerSavingType parameter is set to INTRA_NR_DORMANCY_BY_LOAD and the NRDUCellPowerSaving.EvalPeriodGranularity parameter is set to SECOND_LEVEL, the NRDUCellPowerSaving.SecondLevelTrigEvalPrd parameter must be set to a value greater than or equal to 6. - The smaller the value of this parameter, the higher the probability of triggering power saving. The larger the value of this parameter, the lower the probability.
Second-Level Exit Evaluation Period	NRDUCellPowerSaving.SecondLevelExitEvalPrd	You are advised to set this parameter to its recommended value. - This parameter can be configured and take effect only when the NRDUCellPowerSaving.EvalPeriodGranularity parameter is set to SECOND_LEVEL. - The smaller the value of this parameter, the higher the probability of triggering exit from power saving and the smaller the power saving gains. The larger the value of this parameter, the lower the probability.

Parameter Name	Parameter ID	Setting Notes
UL PRB Protect Thld for Power Saving	NRDUCellPowerSaving.ProtectUlPrbThld	You are advised to set this parameter to its recommended value.
DL PRB Protect Thld for Power Saving	NRDUCellPowerSaving.ProtectDlPrbThld	You are advised to set this parameter to its recommended value.
Uplink PRB Protection Offset	NRDUCellPowerSaving.ProtectUlPrbOffset	You are advised to set this parameter to its recommended value.
Downlink PRB Protection Offset	NRDUCellPowerSaving.ProtectDlPrbOffset	You are advised to set this parameter to its recommended value.
SUL User Number Threshold	NRDUCellPowerSaving.SulUserNumThld	You are advised to set these parameters to their recommended values. - These parameters are invalid when they are set to 65535. - If NRDUCellPowerSaving.SulUserNumThld is not set to 65535, the sum of NRDUCellPowerSaving.SulUserNumThld and NRDUCellPowerSaving.SulUserNumOffset must be less than 65535, and NRDUCellPowerSaving.SulUserNumThld must be less than NRDUCellPowerSaving.UserNumThld. - If the FLEX_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter is deselected and the cell has a NRDUCellPowerSaving record with power saving type set to LTE and NR intelligent carrier shutdown, optimized evaluation of NR RF module dormancy, NR load-based RF module dormancy, or LN load-based RF module dormancy, then the NRDUCellPowerSaving.SulUserNumThld parameter of this record must be set to 65535.
SUL User Number Offset	NRDUCellPowerSaving.SulUserNumOffset	

Parameter Name	Parameter ID	Setting Notes
mmWave User Number Threshold	NRDUCellPowerSaving.MmwaveUserNumThld	You are advised to set this parameter to its recommended value.
Power Saving Penalty Duration	NRDUCellPowerSaving.PwrSavingPenaltyTime	You are advised to set this parameter to its recommended value.
Power Saving Cell Flag	NRCellRelation.PowerSavingCellFlag	This parameter specifies the flag of a neighboring cell for load-based deep dormancy. Set this parameter based on the network plan. When the NRDUCellPowerSaving.PowerSavingType parameter is set to INTRA_NR_DORMANCY_BY_LOAD and the NRDUCellPowerSaving.EvalPeriodGranularity parameter is set to SECOND_LEVEL, the number of corresponding NRCellRelation MOs with NRCellRelation.PowerSavingCellFlag set to COVERAGE must be less than or equal to 16.
Overlap Rate	NRCellRelation.OverlapRate	This parameter specifies the degree of the overlap between the capacity layer and the basic layer. When NR load-based precise dormancy is enabled, set this parameter based on the co-coverage degree of respective neighboring cells.
No Removal Flag	NRCellRelation.NoRmvFlag	When automatic removal of neighboring cells in ANR is enabled (with the NR_NR_ANR_DEL_SW option of the NRCellAlgoSwitch.AnrSwitch parameter selected), it is recommended that the NRCellRelation.NoRmvFlag parameter be set to FORBID_ANR_RMV for the mutual neighbor relationships between capacity-layer and basic-layer cells.
PLMN Reserved Flag	NRExternalNCellPlmn.PlmnReservedFlag	In multi-operator sharing scenarios, set this parameter for capacity-layer cells based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Cell Load Filtering Mode	gNBPwrSavingAlgo.CellLoadFilterMode	This parameter specifies the filtering mode used for uplink and downlink PRB usage in load-based deep dormancy. You are advised to set this parameter to its recommended value.
Power Saving Algorithm Switch	gNBPwrSavingAlgo.PwrSavingAlgoSwitch	Select the SIMU_LOAD_DORMANCY_IMP_SW option to prevent RF modules serving cells with cell load simulation enabled from entering dormancy, while allowing RF modules serving cells without cell load simulation enabled to enter dormancy.

Table 7-15 Parameters on the LTE side used for activating LNR load-based RF module deep dormancy

Parameter Name	Parameter ID	Setting Notes
Power Saving Algorithm Switch	CellPwrSavingAlgo.PwrSavingAlgoSwitch	Select the LNR_LOAD_BASED_DORMANCY_SW option of this parameter if LNR load-based RF module deep dormancy needs to be enabled.
Aggregation Attribute	NrNRRelationship.AggregationAttribute	Set the CARRIER_SHUTDOWN_OVERLAP_FLAG option of this parameter to the recommended value.

Parameter Name	Parameter ID	Setting Notes
Power Saving Algorithm Switch	EnodebPwrSavingAlgo.PwrSavingAlgoSwitch	- Select the MULTI_RRU_CELL_DORM_COMPAT_SW option of this parameter when the multi-RRU cell dormancy compatibility function needs to be enabled for RF module dormancy to take effect in multi-RRU cells, such as SFN cells. - To enable LTE sector split cells to serve as basic-layer cells for load-based dormancy, select the SPLIT_CELL_RF_DORMANCY_SW option.
Interface SON Configuration Switch	eNodeBOamParam.InterfaceSonConfigSwitch	To enable the eNodeB to set up X2 links with gNodeBs that do not have NSA cells, select the LNR_SA_GNB_X2_SETUP_SW option.
Power Saving Algorithm Switch	EnodebPwrSavingAlgo.PwrSavingAlgoSwitch	To enable inter-base-station power saving between LTE and SA NR cells in LTE and NR scenarios, select the LNR_INTER_BS_PWR_SAVE_SW option. The LNR_SA_GNB_X2_SETUP_SW option of the eNodeBOamParam.InterfaceSonConfigSwitch parameter must also be selected.

Table 7-16 Parameters on the NR side used for activating LNR load-based RF module deep dormancy

Parameter Name	Parameter ID	Setting Notes
Dormancy Switch	RRU.DORMANCYSW (5G gNodeB, LTE eNodeB)	To enable LNR load-based RF module deep dormancy, set this parameter to ON or ON_DEEP_SUPER . The former value is recommended.

Parameter Name	Parameter ID	Setting Notes
RRU Dormancy Optional Feature Switch	RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)	To enable multi-frequency flexible dormancy, select the MULTI_FREQ_FLEX_SLEEP_SW option.
Power Saving Switch	NRDUCellAlgoSwitch.PowerSavingSwitch	Select the LNR_DORMANCY_BY_LOAD_SW option of this parameter.
Start Time	NRDUCellPowerSaving.StartTime	Set this parameter based on the network plan.
Stop Time	NRDUCellPowerSaving.StopTime	Set this parameter based on the network plan.
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	Set this parameter to LNR_DORMANCY_BY_LOAD .
UE Number Threshold	NRDUCellPowerSaving.UserNumThld	Set this parameter based on the network plan.
Downlink PRB Threshold	NRDUCellPowerSaving.DlPrbThld	Set this parameter based on the network plan.
Uplink PRB Threshold	NRDUCellPowerSaving.UlPrbThld	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
SUL User Number Threshold	NRDUCellPowerSaving.SulUserNumThld	You are advised to set this parameter to its recommended value. - This parameter is invalid when it is set to 65535. - If the FLEX_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter is deselected and the cell has a NRDUCellPowerSaving record with power saving type set to LTE and NR intelligent carrier shutdown, optimized evaluation of NR RF module dormancy, NR load-based RF module dormancy, or LN load-based RF module dormancy, then the NRDUCellPowerSaving.SulUserNumThld parameter of this record must be set to 65535.
E-UTRAN Downlink PRB Threshold	NRDUCellPowerSaving.EutranDlPrbThld	Set this parameter based on the network plan.
Neighboring Cell Uplink PRB Threshold	NRDUCellPowerSaving.NCellUlPrbThld	Set this parameter based on the network plan.
E-UTRAN Downlink PRB Offset	NRDUCellPowerSaving.EutranDlPrbOffset	Set this parameter based on the network plan.
Neighboring Cell Uplink PRB Offset	NRDUCellPowerSaving.NCellUlPrbOffset	Set this parameter based on the network plan.
E-UTRAN Carrier Shutdown Ratio	NRDUCellPowerSaving.EutranCarrShutdownRatio	Set this parameter based on the network plan.
E-UTRAN Carrier Shutdown Offset	NRDUCellPowerSaving.EutranCarrShutdownExitOfs	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
E-UTRAN NR User Number Threshold	NRDUCellPowerSaving.EutranNrUserNumThld	Set this parameter based on the network plan. If this parameter is set to 65535 , this parameter and the NRDUCellPowerSaving.EutranNrUserNumThldOfs parameter are invalid.
E-UTRAN NR User Number Threshold Offset	NRDUCellPowerSaving.EutranNrUserNumThldOfs	Set this parameter based on the network plan.
E-UTRAN NR User Downlink PRB Threshold	NRDUCellPowerSaving.EutranNrUserDlPrbThld	Set this parameter based on the network plan.
E-UTRAN NR User Uplink PRB Threshold	NRDUCellPowerSaving.EutranNrUserUlPrbThld	Set this parameter based on the network plan.
E-UTRAN NR User Downlink PRB Threshold Offset	NRDUCellPowerSaving.EutranNrUserDlPrbThldOfs	Set this parameter based on the network plan.
E-UTRAN NR User Uplink PRB Threshold Offset	NRDUCellPowerSaving.EutranNrUserUlPrbThldOfs	Set this parameter based on the network plan.
Power Saving Penalty Duration	NRDUCellPowerSaving.PwrSavingPenaltyTime	Set this parameter based on the network plan.
Evaluation Period Granularity	NRDUCellPowerSaving.EvalPeriodGranularity	You are advised to set this parameter to its recommended value.
Triggering Evaluation Period	NRDUCellPowerSaving.ShutdownTrigJudgePeriod	Set this parameter based on the network plan.
Exit Evaluation Period	NRDUCellPowerSaving.ExitEvaluationPeriod	You are advised to set this parameter to its recommended value.

Parameter Name	Parameter ID	Setting Notes
Second-Level Trigger Evaluation Period	NRDUCellPowerSaving.SecondLevelTrigEvalPrd	You are advised to set this parameter to its recommended value. • This parameter can be configured and take effect only when the NRDUCellPowerSaving.EvalPeriodGranularity parameter is set to SECOND_LEVEL. • When the NRDUCellPowerSaving.PowerSavingType parameter is set to LNR_DORMANCY_BY_LOAD and the NRDUCellPowerSaving.EvalPeriodGranularity parameter is set to SECOND_LEVEL, the NRDUCellPowerSaving.SecondLevelTrigEvalPrd parameter must be set to a value greater than or equal to 6. • The smaller the value of this parameter, the higher the probability of triggering power saving. The larger the value of this parameter, the lower the probability.

Parameter Name	Parameter ID	Setting Notes
Second-Level Exit Evaluation Period	NRDUCellPowerSaving.SecondLevelExitEvalPrd	You are advised to set this parameter to its recommended value. - This parameter can be configured and take effect only when the NRDUCellPowerSaving.EvalPeriodGranularity parameter is set to SECOND_LEVEL. - The smaller the value of this parameter, the higher the probability of triggering exit from power saving and the smaller the power saving gains. The larger the value of this parameter, the lower the probability.
Power Saving Overlapping Indicator	NRCellEutraNRelation.PowerSavingOverlapInd	Set the LNR_CARR_SHUTDOWN_OVERLAP_IND option based on the network plan. This parameter is mandatory only in SA networking.
Cell Load Filtering Mode	gNBPwrSavingAlgo.CellLoadFilterMode	This parameter specifies the filtering mode used for uplink and downlink PRB usage in load-based deep dormancy. You are advised to set this parameter to its recommended value.
Power Saving Algorithm Switch	gNBPwrSavingAlgo.PwrSavingAlgoSwitch	To enable inter-RAT level-by-level energy saving, select the INTER_RAT_GRAD_PWR_SAVING_SW option.

Parameter Name	Parameter ID	Setting Notes
Power Saving Algorithm Switch	gNBPwrSavingAlgo.PwrSavingAlgoSwitch	When the LNR_LOAD_DORM_PRIM_OP_RECOG_SW option is selected, primary operator recognition for LNR load-based dormancy in MOCN scenarios is enabled. Only neighboring LTE cells served by the same operator as the primary operator of an NR cell can be configured as basic-layer cells for LNR load-based dormancy in the NR cell.
Power Saving Algorithm Switch	gNBPwrSavingAlgo.PwrSavingAlgoSwitch	Select the SIMU_LOAD_DORMANCY_IMP_SW option to prevent RF modules serving cells with cell load simulation enabled from entering dormancy, while allowing RF modules serving cells without cell load simulation enabled to enter dormancy.
X2 Config Switch	gNBInterfaceParam.X2ConfigSw	To allow X2 link setup in SA scenarios, select the X2_SETUP_FOR_SA_SW option.

Parameter Name	Parameter ID	Setting Notes
Power Saving Algorithm Switch	gNBPwrSavingAlgo.PwrSavingAlgoSwitch	- To enable energy saving for SA NR cells in LTE-NR inter-base-station energy saving scenarios, select the SA_LNR_INTER_BS_PWR_SAVE_SW option. The X2_SETUP_FOR_SA_SW option of the gNBInterfaceParam.X2ConfigSw parameter must also be selected. - When the SA_LNR_INTER_BS_PWR_SAVE_SW option is deselected, the NRCellEutraNRelation.EnodebId parameter must have the same value in all NRCellEutraNRelation MOs where the LNR_CARR_SHUTDOWN_OVERLAP_IND option of the NRCellEutraNRelation.PowerSavingOverlapInd parameter is set to Yes.

 NOTE

If the load-based deep dormancy period configured for an NR DU cell overlaps with the scheduled deep dormancy period configured for the RRU, the base station determines whether to enter dormancy based on the entry conditions of load-based deep dormancy.

Table 7-17 Parameters used for activating LTE load-based RF module deep dormancy

Parameter Name	Parameter ID	Setting Notes
Dormancy Switch	RRU.DORMANCYSW	To enable LTE load-based RF module deep dormancy, set this parameter to ON or ON_DEEP_SUPER . The former value is recommended.

Parameter Name	Parameter ID	Setting Notes
RRU Dormancy Optional Feature Switch	RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)	To enable multi-frequency flexible dormancy, select the MULTI_FREQ_FLEX_SLEEP_SW option.
Power Saving Algorithm Switch	CellPwrSavingAlgo.PwrSavingAlgoSwitch	It is recommended that the INTRA_LTE_DORMANCY_BY_LOAD_SW option of this parameter be selected if LTE load-based dormancy needs to be enabled. To enable LTE load-based precise dormancy, deselect the INTRA_LTE_DORMANCY_BY_LOAD_SW option and then select the LTE_DORMANCY_BY_LOAD_PE_SW option of this parameter. To calculate overlap rates based on grids, select both the LTE_DORMANCY_BY_LOAD_PE_SW and OVERLAP_GRID_MODE_SW options of this parameter.
Policy Index	CellPwrSavingPolicy.PolicyIndex	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Start Time	CellPwrSavingPolicy.StartTime	Set these parameters based on the network plan and the following requirements: - For the same policy of the same energy saving type, the interval between the stop time and the start time must be longer than 20 minutes. If the end time is earlier than the start time, the end time is interpreted as occurring on the next day. For example: If the start time is 8:00 a.m. and the stop time is 7:00 a.m., the power saving function is enabled from 8:00 a.m. to 7:00 a.m. on the next day. - For different policies of the same energy saving type, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving type needs to be enabled after policy A of this type has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A.
Stop Time	CellPwrSavingPolicy.StopTime	
Power Saving Type	CellPwrSavingPolicy.PowerSavingType	- To enable LTE load-based dormancy, set this parameter to INTRA_LTE_DORMANCY_BY_LOAD. - To enable LTE load-based precise dormancy, set this parameter to LTE_PRECISE_DORMANCY_BY_LOAD.

Parameter Name	Parameter ID	Setting Notes
Trigger Judge Period	CellPwrSavingPolicy.TriggerJudgePeriod	You are advised to set this parameter to its recommended value.
Downlink PRB Threshold	CellPwrSavingPolicy.DlPrbThld	You are advised to set this parameter to its recommended value.
Downlink PRB Offset	CellPwrSavingPolicy.DlPrbOffset	You are advised to set this parameter to its recommended value.
Uplink PRB Threshold	CellPwrSavingPolicy.UlPrbThld	You are advised to set this parameter to its recommended value.
Uplink PRB Offset	CellPwrSavingPolicy.UlPrbOffset	You are advised to set this parameter to its recommended value.
UE Number Threshold	CellPwrSavingPolicy.UENumThld	You are advised to set this parameter to its recommended value.
Forcible Trigger UE Number Threshold	CellPwrSavingPolicy.ForcibleTriggerUeNumThld	You are advised to set this parameter to its recommended value.

Parameter Name	Parameter ID	Setting Notes
Exit Judge Period	CellPwrSavingPolicy.ExitJudgePeriod	You are advised to set this parameter to its recommended value. • If the CellPwrSavingPolicy.PowerSavingType parameter is set to INTRA_LTE_DORMANCY_BY_LOAD or LTE_PRECISE_DORMANCY_BY_LOAD and the CellPwrSavingPolicy.EvalInterval parameter is set to EVAL_INTERVAL_60S, the CellPwrSavingPolicy.ExitJudgePeriod parameter must be set to an integer multiple of 12. • The smaller the value of this parameter, the higher the probability of triggering exit from power saving and the smaller the power saving gains. The larger the value of this parameter, the lower the probability.

Parameter Name	Parameter ID	Setting Notes
Evaluation Interval	CellPwrSavingPolicy.EvalInterval	You are advised to set this parameter to its recommended value. - Set this parameter to EVAL_INTERVAL_5S if a second-level evaluation period is required. In this case, the cell may be less likely to exit power saving given the same exit evaluation period (specified by CellPwrSavingPolicy.ExitJudgePeriod), and may be more likely to exit power saving given a shorter exit evaluation period. - Set this parameter to EVAL_INTERVAL_60S if a minute-level evaluation period is required. In this case, the CellPwrSavingPolicy.ExitJudgePeriod parameter must be set to an integer multiple of 12.
Penalty Time	CellPwrSavingPolicy.PenaltyTime	You are advised to set this parameter to its recommended value.
Inter-Frequency Mobility Load Balancing Threshold	CellMLB.InterFreqMlbThd	You are advised to set this parameter to its recommended value.
Load Offset	CellMLB.LoadOffset	You are advised to set this parameter to its recommended value.
Overlap Indicator Extension	EutranInterFreqN-Cell.OverlapIndicatorExtension	Set this parameter based on the network plan. If this parameter is set to POWER_SAVING_OVERLAP_IND , the neighboring cell has overlapping coverage with the serving cell.

Parameter Name	Parameter ID	Setting Notes
Overlap Rate	EutranInterFreqN-Cell.OverlapRate	In precise dormancy mode, set this parameter to the overlap rate between the capacity layer and the basic layer.
Power Saving Algorithm Switch	EnodebPwrSavingAlgo.PwrSavingAlgoSwitch	Select the INTRA_RAT_GRAD_PWR_SAVING_SW option of this parameter if intra-RAT level-by-level energy saving is required.
Cell Load Filtering Mode	EnodebPwrSavingAlgo.CellLoadFilterMode	This parameter specifies the filtering mode used for uplink and downlink PRB usage in load-based deep dormancy. You are advised to set this parameter to its recommended value.
Inter-eNodeB Information Exchange Mode	EnodebPwrSavingAlgo.InterEnbInfoExchangeMode	You are advised to set this parameter to its recommended value. - If this parameter is set to SLOW_EXCHANGE_MODE, the slow exchange mode is used, meaning that information is exchanged at long intervals. - If this parameter is set to FAST_EXCHANGE_MODE, the fast exchange mode is used, meaning that information is exchanged at short intervals. - You are advised to set this parameter to FAST_EXCHANGE_MODE if the CellPwrSavingPolicy.EvalInterval parameter is set to EVAL_INTERVAL_5S.
Dormancy Cell Type	CellPwrSavingAlgo.DormancyCellType	In precise dormancy mode, set this parameter to CAPACITY_CELL for capacity-layer cells and BASIC_CELL for basic-layer cells.

Parameter Name	Parameter ID	Setting Notes
Downlink PRB Protection Threshold	CellPwrSavingPolicy.ProtectDLPrbThld	Use the recommended value for a capacity-layer cell. This parameter does not need to be set for a basic-layer cell.
Uplink PRB Protection Threshold	CellPwrSavingPolicy.ProtectULPrbThld	Use the recommended value for a capacity-layer cell. This parameter does not need to be set for a basic-layer cell.
Uplink PRB Protection Offset	CellPwrSavingPolicy.ProtectULPrbOffset	Use the recommended value for a capacity-layer cell. This parameter does not need to be set for a basic-layer cell.
Downlink PRB Protection Offset	CellPwrSavingPolicy.ProtectDLPrbOffset	Use the recommended value for a capacity-layer cell. This parameter does not need to be set for a basic-layer cell.
Power Saving Algorithm Switch	EnodebPwrSavingAlgo.PwrSavingAlgoSwitch	To enable optimization for cell PLMN reservation scenarios in multi-operator sharing scenarios, select the CELL_PLMN_RSVD_OPT_SW option.
Aggregation Attribute	EutranExternalCellPlmn.AggregationAttribute	In multi-operator sharing scenarios, set this parameter based on the network plan. To set the operator as a reserved operator, select the PLMN_RESERVED_FLAG option of this parameter.

Parameter Name	Parameter ID	Setting Notes
Configuration Adaptation Opt Sw	eNodeBOamParam.ConfigAdaptOptSwitch	In multi-operator sharing scenarios, set this parameter based on the network plan. To enable the cell reservation status to be automatically broadcast to neighboring base stations through X2 signaling messages, select the RSVD_FOR_OP_X2_BROADCAST_SW option of this parameter.
Power Saving Algorithm Switch	EnodebPwrSavingAlgo.PwrSavingAlgoSwitch	- Select the MULTI_RRU_CELL_DORM_COMPAT_SW option of this parameter when the multi-RRU cell dormancy compatibility function needs to be enabled for RF module dormancy to take effect in multi-RRU cells, such as SFN cells. - Select the SPLIT_CELL_RF_DORMANCY_SW option of this parameter when an LTE sector split cell needs to serve as a basic cell for load-based dormancy.

Table 7-18 Parameters used for activating UMTS load-based RF module deep dormancy

Parameter Name	Parameter ID	Setting Notes
Dormancy Switch	RRU.DORMANCYSW	To enable UMTS load-based RF module deep dormancy, set this parameter to ON or ON_DEEP_SUPER . The former value is recommended.

Parameter Name	Parameter ID	Setting Notes
Dormancy Algorithm Switch	UCELLALGPARA.DORMANCYALGOSW	To enable UMTS load-based RF module deep dormancy, select the DORMANCY_BY_LOAD_SW option.
Dormancy Cell Type	UCELL.DormancyCellType	Set this parameter to CAPACITY_CELL for capacity-layer cells. Set this parameter to BASIC_CELL for basic-layer cells.
Blind Handover Flag	UINTERFREQNCCELL.BlindHoFlag	Set this parameter to TRUE when a basic-layer cell is configured as a co-coverage inter-band neighboring cell of the capacity-layer cell.
HandOver Switch	UCORRMALGOSWITCH.HoSwitch	To enable UMTS inter-frequency handovers, select the HO_INTER_FREQ_HARD_HO_SWITCH option.
CELL ID	UCELLDORMANCYPOLICY.CellId	Set this parameter based on the network plan and the following requirements: When UCELL.DormancyCellType is set to BASIC_CELL , UCELLDORMANCYPOLICY.CellId cannot be configured for the cell specified by UCELL.CellId . In other words, no energy saving policy can be configured for basic-layer cells.
Policy Index	UCELLDORMANCYPOLICY.PolicyIndex	Set this parameter based on the network plan.
Dormancy Type	UCELLDORMANCYPOLICY.DormancyType	Set this parameter to INTRA_DORMANCY_BY_LOAD .

Parameter Name	Parameter ID	Setting Notes
Start Time	UCELLDORMANCYPOLICY.StartTime	Set these parameters based on the network plan and the following requirements: - For the same policy of the same energy saving type, the interval between the stop time and the start time must be longer than 20 minutes. If the stop time is earlier than or equal to the start time, the stop time is interpreted as occurring on the next day. For example: If the start time is 8:00 a.m. and the stop time is 7:00 a.m., the power saving function is enabled from 8:00 a.m. to 7:00 a.m. on the next day. - For different policies of the same energy saving type, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving type needs to be enabled after policy A of this type has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A.
Stop Time	UCELLDORMANCYPOLICY.StopTime	
Trigger Judge Period	UCELLDORMANCYPOLICY.TriggerJudgePeriod	Use the recommended value.

Parameter Name	Parameter ID	Setting Notes
Exit Judge Period	UCELLDORMANCYPOLICY.ExitJudgePeriod	Use the recommended value.
Penalty Time	UCELLDORMANCYPOLICY.PenaltyTime	Use the recommended value.
Forcible Trigger UE Number Threshold	UCELLDORMANCYPOLICY.ForcibleTriggerUeNumThld	Use the recommended value.
Cell Power Saving Total User Number Threshold	UCELLDORMANCYPOLICY.TotalUserNumThd	Use the recommended value.
Cell Power Saving Hsdpa User Number Threshold	UCELLDORMANCYPOLICY.HsdpaUserNumThd	Use the recommended value.
Cell Power Saving Hsupa User Number Threshold	UCELLDORMANCYPOLICY.HsupaUserNumThd	Use the recommended value.
Cell Power Saving Neighbour Cell Load Remain Threshold	UCELLDORMANCYPOLICY.NCellLdrRemainThd	Use the recommended value.
Considering User on Common Channel for Dormancy	UCELLDORMANCYPOLICY.DormancyCchUserSwitch	Use the recommended value.
Multi-carrier Cell Power Saving HSDPA User Number Threshold	UCELLDORMANCYPOLICY.McHsdpaUserNumThd	Use the recommended value.
Cell Power Saving Inter Plmn Handover Allowed Switch	URNCCCELLDORMANCYPARA.CellInterPlmnHoSwitch	Set this parameter to ON .
Intra RAT Inter Plmn Ho Allowed	UOPERATORSHARING-MODE.InterPlmnHoAllowedIntraRat	Set this parameter to YES .

Parameter Name	Parameter ID	Setting Notes
Cell Power Saving Ignore CBS Switch	URNCELLDORMANCY-PARA.<i>CellIgnoreCbsSwitch</i>	Use the recommended value. CBS is generally used to send emergency and important information. Therefore, CBS takes precedence over dormancy, and this parameter is set to OFF by default. To increase the dormancy priority, set this parameter to ON . However, with this setting, the cell may already be in the dormancy state when CBS messages need to be sent, and waking it up takes time. As a result, CBS message broadcasting in the cell is delayed.
CBS Service Hold Duration	URNCELLDORMANCY-PARA.<i>CbsServiceHoldDuration</i>	Use the recommended value.

Table 7-19 Parameters on the UMTS side used for activating GU load-based RF module deep dormancy

Parameter Name	Parameter ID	Setting Notes
Dormancy Switch	RRU.<i>DORMANCYSW</i>	To enable GU load-based RF module deep dormancy, set this parameter to ON or ON_DEEP_SUPER . The former value is recommended.
Dormancy Algorithm Switch	ULOCCELLALGPARA.<i>DORMANCYALGOSW</i>	To enable GU load-based RF module deep dormancy, select the DORMANCY_BY_LOAD_SW option.
Policy Index	UCELLDORMANCYPOLICY.<i>PolicyIndex</i>	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Dormancy Type	UCELLDORMANCYPOLICY.DormancyType	Set this parameter to INTER_GU_DORMANCY_BY_LOAD .
Start Time	UCELLDORMANCYPOLICY.StartTime	Set these parameters based on the network plan and the following requirements: - For the same policy of the same energy saving type, the interval between the stop time and the start time must be longer than 20 minutes. If the stop time is earlier than or equal to the start time, the stop time is interpreted as occurring on the next day. For example: If the start time is 8:00 a.m. and the stop time is 7:00 a.m., the power saving function is enabled from 8:00 a.m. to 7:00 a.m. on the next day. - For different policies of the same energy saving type, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving type needs to be enabled after policy A of this type has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A.
Stop Time	UCELLDORMANCYPOLICY.StopTime	

Parameter Name	Parameter ID	Setting Notes
Trigger Judge Period	UCELLDORMANCYPOLICY.TriggerJudgePeriod	Use the recommended value.
Exit Judge Period	UCELLDORMANCYPOLICY.ExitJudgePeriod	Use the recommended value.
Penalty Time	UCELLDORMANCYPOLICY.PenaltyTime	Use the recommended value.
Forcible Trigger UE Number Threshold	UCELLDORMANCYPOLICY.ForcibleTriggerUeNumThld	Use the recommended value.
Cell Power Saving Total User Number Threshold	UCELLDORMANCYPOLICY.TotalUserNumThd	Use the recommended value.
Considering User on Common Channel for Dormancy	UCELLDORMANCYPOLICY.DormancyCchUserSwitch	Use the recommended value.
Overlap Flag	U2GNCCELL.NIRATOverlap	Set this parameter to TRUE when a basic-layer cell is configured as a co-coverage inter-RAT neighboring cell of the capacity-layer cell.
Inter RAT Inter Plmn Ho Allowed	UOPERATORSHARING-MODE.InterPlmnHoAllowedInterRat	Set this parameter to YES .
Switch for Inter-RAT Common Measurement	UMBSCRRM.IRATCommonMeasSwitch	Turn on this switch if inter-RAT common measurement needs to be started.
HandOver Switch	UCORRMALGOSWITCH.HoSwitch	To enable inter-RAT handovers for UMTS, select the HO_INTER_RAT_CS_OUTPUT_SWITCH and HO_INTER_RAT_PS_OUTPUT_SWITCH options.
Total Trans Power Threshold for Cell Power Saving	UCELLDORMANCYPOLICY.IRATShutdownDLLoadThd	You are advised to set this parameter to its recommended value.
Total Receive Power Threshold for Cell Power Saving	UCELLDORMANCYPOLICY.IRATShutdownULLoadThd	You are advised to set this parameter to its recommended value.

Parameter Name	Parameter ID	Setting Notes
Load Thre for Neigh GSM Cell for Cell Power Saving	UCELLDORMANCYPOLICY.IRATshutdownGSM-LoadThd	You are advised to set this parameter to its recommended value.
Load Hyst Thre for Neigh GSM Cell for Cell Power Saving	UCELLDORMANCYPOLICY.IRATshutdownGSM-LoadHyst	You are advised to set this parameter to its recommended value.
Cell Power Saving Ignore CBS Switch	URNCCCELLDORMANCY-PARA.CellIgnoreCbsSwitch	You are advised to set this parameter to its recommended value. CBS is generally used to send emergency and important information. Therefore, CBS takes precedence over dormancy, and this parameter is set to OFF by default. To increase the dormancy priority, set this parameter to ON . However, with this setting, the cell may already be in the dormancy state when CBS messages need to be sent, and waking it up takes time. As a result, CBS message broadcasting in the cell is delayed.
CBS Service Hold Duration	URNCCCELLDORMANCY-PARA.CbsServiceHoldDuration	You are advised to set this parameter to its recommended value.

Table 7-20 Parameters on the GSM side used for activating GU load-based RF module deep dormancy

Parameter Name	Parameter ID	Setting Notes
Information Exchange Switch at lur-g	GCELLOTTEXT.IURGINFCTRL	Set this parameter to YES when the GSM base station controller is BSC6910.
Inter-RAT Cell Dynamic Shutdown	GCELLSOFT.IRATShutdownSwitch	Set this parameter to ON .

7.2.4.1.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [7.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

For details about configuring intra-base-station neighboring cells, see *SA Mobility Management in Connected Mode in 5G RAN Feature Documentation*.

Activating scheduled RF module deep dormancy/NR local-cell-load-based RF module deep dormancy

(Recommended) Scenario 1: Enabling scheduled RF module deep dormancy and specifying a dormancy period

```
//Enabling scheduled RF module deep dormancy
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=ON, DORMANCYSTARTTIME=00&10&00,
DORMANCYSTOPTIME=08&00&00, DORMANCYTIMER=5, CUSTOMDORMANCYSTDSW=OFF;
//(Optional) Turning on the multi-RRU cell dormancy compatibility switch
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=MULTI_RRU_CELL_DORM_COMPAT_SW-1;
//(Optional) Enabling optimized evaluation of NR RF module dormancy
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=0,
PowerSavingAlgoSwitch=NR_DORMANCY_JUDGE_THLD_OPT_SW-1;
//(Optional) Configuring an energy saving policy for optimized evaluation of NR RF module dormancy
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=11,
PowerSavingType=NR_DORMANCY_JUDGE_THLD_OPT, StartTime=00&00&00, StopTime=06&00&00,
DlPrbThld=5, UserNumThld=2, EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=5,
UlPrbThld=5;
//(Optional, in SUL co-deployment scenarios) Modifying the energy saving policy
MOD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=11,
PowerSavingType=NR_DORMANCY_JUDGE_THLD_OPT, SulUserNumThld=4;
```

Scenario 2: Enabling scheduled RF module deep dormancy and specifying multiple dormancy periods

```
//Enabling scheduled RF module deep dormancy
//Configuring a board energy saving policy
ADD BRDPWRSVGPOLICY: POLICYID=1, POWERSAVINGTYPE=RF_DORMANCY;
//Setting the time for the board energy saving policy
ADD BRDPWRSVGPOLICYTIME: POLICYID=1, TIMEID=1, STARTTIME=00&10&00, STOPTIME=03&00&00;
ADD BRDPWRSVGPOLICYTIME: POLICYID=1, TIMEID=2, STARTTIME=04&00&00, STOPTIME=06&00&00;
//Configuring the energy saving policy for an RF module
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=ON, PWRSVGPOLICYID=1;
//(Optional) Turning on the multi-RRU cell dormancy compatibility switch
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=MULTI_RRU_CELL_DORM_COMPAT_SW-1;
//(Optional) Enabling optimized evaluation of NR RF module dormancy
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=0,
PowerSavingAlgoSwitch=NR_DORMANCY_JUDGE_THLD_OPT_SW-1;
//(Optional) Configuring an energy saving policy for optimized evaluation of NR RF module dormancy
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=11,
PowerSavingType=NR_DORMANCY_JUDGE_THLD_OPT, StartTime=00&00&00, StopTime=06&00&00,
DlPrbThld=5, UserNumThld=2, EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=5,
UlPrbThld=5;
//(Optional, in SUL co-deployment scenarios) Modifying the energy saving policy
MOD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=11,
PowerSavingType=NR_DORMANCY_JUDGE_THLD_OPT, SulUserNumThld=4;
```


The following configurations are required in scheduled UMTS cell dormancy scenarios:

```
//RNC side:
//(Optional) Allowing all CBS cells of the RNC to enter scheduled dormancy when CBS exists in UMTS cells
SET URNCELLDORMANCYPARA: LogicRncId=0, CellIgnoreCbsSwitch=ON, CbsServiceHoldDuration=5;

//Inter-frequency handover
//Enabling inter-frequency hard handover
SET UCORRMALGOSWITCH: HoSwitch=HO_INTER_FREQ_HARD_HO_SWITCH-1;
//Supporting measurement-based handovers (from dormant cell 0 to cell 1 operating in a different frequency band)
ADD UINTERFREQNCELL: RNCId=0, CellId=0, NCellRncId=0, NCellId=1;

//Inter-RAT handover
//Selecting the HO_INTER_RAT_CS_OUT_SWITCH and HO_INTER_RAT_PS_OUT_SWITCH options
SET UCORRMALGOSWITCH:
HoSwitch=HO_INTER_RAT_CS_OUT_SWITCH-1&HO_INTER_RAT_PS_OUT_SWITCH-1;
//Supporting measurement-based inter-RAT handovers (from dormant cell 0 to GSM cell 1)
ADD U2GNCELL: RNCId=0, CellId=0, GSMCellIndex=1;

//(Optional) Multi-operator sharing scenarios
//RNC side:

//Inter-frequency handover
//Performing the following configuration if cell 1 is intended to serve as the basic-layer cell of cell 0 when they belong to different operators
SET URNCELLDORMANCYPARA: LogicRncId=0, CellInterPlmnHoSwitch=ON;
SET UOPERATORSHARINGMODE: LogicRncId=0, InterPlmnHoAllowedIntraRat=YES;

//Inter-RAT handover
//Allowing UEs to be handed over between inter-RAT inter-operator cells
SET UOPERATORSHARINGMODE: LogicRncId=0, InterPlmnHoAllowedInterRat=YES;
```

(Optional) UMTS capacity-layer cell with CBS enabled

```
//RNC side:
//Setting CellIgnoreCbsSwitch to ON if all CBS cells under the RNC are allowed to enter load-based dormancy. Otherwise, the default value OFF is recommended.
SET URNCELLDORMANCYPARA: LogicRncId=0, CellIgnoreCbsSwitch=ON, CbsServiceHoldDuration=5;
```

Activating NR load-based RF module deep dormancy

```
//Turning on the RF module-level switches on the NR side. If the start time and stop time are the same, only load-based RF module deep dormancy takes effect, while scheduled RF module deep dormancy does not take effect.
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=ON, DORMANCYSTARTTIME=00&00&00, DORMANCYSTOPTIME=00&00&00, DORMANCYTIMER=5, CUSTOMDORMANCYSTDSW=OFF;
//Turning on the switch on the NR DU cell side
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=INTRA_NR_DORMANCY_BY_LOAD_SW-1;
//Configuring an energy saving policy in NR TDD low-frequency and NR FDD scenarios
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=10, PowerSavingType=INTRA_NR_DORMANCY_BY_LOAD, StartTime=00&00&00, StopTime=06&00&00, DlPrbThld=10, DlPrbOffset=10, UserNumThld=20, UserNumOffset=20, EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=5, ExitEvaluationPeriod=1, PwrSavingPenaltyTime=30, UlPrbThld=25, NCellUlPrbOffset=25;
//Configuring an energy saving policy in NR TDD high-frequency scenarios
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=10, PowerSavingType=INTRA_NR_DORMANCY_BY_LOAD, StartTime=00&00&00, StopTime=06&00&00, DlPrbThld=10, DlPrbOffset=10, UserNumThld=20, UserNumOffset=20, EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=5, ExitEvaluationPeriod=1, PwrSavingPenaltyTime=30, UlPrbThld=25, NCellUlPrbOffset=25, MmwaveUserNumThld=20;
//Setting Power Saving Cell Flag, and (recommended) setting No Removal Flag to FORBID_ANR_RMV for the bidirectional neighbor relationships between capacity-layer and basic-layer cells when automatic removal of neighboring cells in ANR takes effect
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=0, CellId=10, PowerSavingCellFlag=COVERAGE, NoRmvFlag=FORBID_ANR_RMV;
MOD NRCELLRELATION: NrCellId=10, Mcc="302", Mnc="220", gNBId=0, CellId=0, PowerSavingCellFlag=CAPACITY, NoRmvFlag=FORBID_ANR_RMV;
//(Optional) Configuring the precise dormancy switch and energy saving policy. The precise mode is not
```

```
supported in high-frequency TDD.  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingExtSwitch=PRECISE_DORMANCY_SW-1;  
MOD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=10,  
PowerSavingType=INTRA_NR_DORMANCY_BY_LOAD, ProtectDlPrbThld=50, ProtectDlPrbOffset=20,  
ProtectUlPrbThld=50, ProtectUlPrbOffset=20;  
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=0, CellId=10, OverlapRate=40;  
//(Optional) Turning on the grid mode switch  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingExtSwitch=GRID_MODE_SW-1;  
//Setting the cell load filtering mode  
MOD GNBPWRSAVINGALGO: CellLoadFilterMode=FAST_FILTER_MODE;  
//(Optional) Turning on the multi-frequency flexible dormancy switch  
ADD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=MULTI_FREQ_FLEX_SLEEP_SW-1;
```

(Optional) Multi-operator sharing scenarios

```
//Setting the PlmnReservedFlag parameter in the NR external-cell PLMN list for the capacity-layer cell  
based on the network plan  
ADD NREXTERNALNCELLPLMN: Mcc="302", Mnc="220", gNBId=0, CellId=10, SharedMcc=xx, SharedMnc=xx,  
PlmnReservedFlag=FALSE;
```

(Optional) In SUL co-deployment scenarios

```
//Modifying the energy saving policy  
MOD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=10,  
PowerSavingType=INTRA_NR_DORMANCY_BY_LOAD, SulUserNumThld=4, SulUserNumOffset=4;
```

Activating LNR load-based RF module deep dormancy

```
//LTE side:  
//Turning on the LNR load-based RF module deep dormancy switch  
MOD CELLPWRSAVINGALGO: LocalCellId=0, PwrSavingAlgoSwitch=LNR_LOAD_BASED_DORMANCY_SW-1;  
//(Optional) Turning on the LTE-NR SA gNodeB X2 setup switch in inter-base-station SA networking  
MOD ENODEBOAMPARAM: InterfaceSonConfigSwitch=LNR_SA_GNB_X2_SETUP_SW-1;  
//(Optional) Turning on the LTE-NR SA inter-base-station energy saving switch in inter-base-station SA  
networking  
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=LNR_INTER_BS_PWR_SAVE_SW-1;  
//(Optional, only in NSA networking, NSA and SA hybrid networking, or inter-base-station SA networking)  
Manually configuring all the neighbor relationships between LTE and NR cells with overlapping coverage  
areas with the parameters set as follows: MCC is 460, MNC is 01, and gNodeB ID is 441233  
MOD NRNRELATIONSHIP: LocalCellId=0, Mcc="460", Mnc="01", GnodebId=441233, CellId=1,  
AggregationAttribute=CARRIER_SHUTDOWN_OVERLAP_FLAG-1;  
//(Optional) Turning on the multi-RRU cell dormancy compatibility switch  
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=MULTI_RRU_CELL_DORM_COMPAT_SW-1;  
//(Optional) Turning on the RF module dormancy switch for LTE sector split cells that serve as basic-layer  
cells for load-based dormancy  
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=SPLIT_CELL_RF_DORMANCY_SW-1;  
//NR side:  
//Turning on the RF module-level switches on the NR side. If the start time and stop time are the same,  
only load-based RF module deep dormancy takes effect, while scheduled RF module deep dormancy does  
not take effect.  
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=ON, DORMANCYSTARTTIME=00&00&00,  
DORMANCYSTOPTIME=00&00&00, DORMANCYTIMER=5, CUSTOMDORMANCYSTDSW=OFF;  
//Turning on the LNR load-based RF module deep dormancy switch  
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, PowerSavingSwitch=LNR_DORMANCY_BY_LOAD_SW-1;  
//(Optional, required only in SA networking) Configuring all neighbor relationships in the overlapping  
coverage area with the parameters set as follows: MCC is 460, MNC is 01, and eNodeB ID is 441282  
MOD NRCELLEUTRANRELATION: NrCellId=1, Mcc="460", Mnc="01", EnodebId=441282, CellId=0,  
PowerSavingOverlapInd=LNR_CARR_SHUTDOWN_OVERLAP_IND-1;  
//Adding energy saving configurations  
ADD NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=0, StartTime=00&00&00,  
StopTime=06&00&00, PowerSavingType=LNR_DORMANCY_BY_LOAD, DlPrbThld=5, UlPrbThld=5,  
UserNumThld=20, PwrSavingPenaltyTime=30, EutranDlPrbThld=30, EutranDlPrbOffset=30,  
NCellUlPrbThld=30, NCellUlPrbOffset=30, EutranCarrShutdownRatio=90, EutranCarrShutdownExitOfs=70,  
EutranNrUserNumThld=20, EutranNrUserNumThldOfs=20, EutranNrUserDlPrbThld=15,  
EutranNrUserDlPrbThldOfs=15, EutranNrUserUlPrbThld=15, EutranNrUserUlPrbThldOfs=15,  
EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=1, ExitEvaluationPeriod=1;  
//Setting the cell load filtering mode  
MOD GNBPWRSAVINGALGO: CellLoadFilterMode=FAST_FILTER_MODE;  
//(Optional, in SUL co-deployment scenarios) Modifying the energy saving policy  
MOD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0,
```

```
PowerSavingType=LNR_DORMANCY_BY_LOAD, SulUserNumThld=4;  
//(Optional) Turning on the inter-RAT level-by-level energy saving switch  
MOD GNBWPWSAVINGALGO: PwrSavingAlgoSwitch=INTER_RAT_GRAD_PWR_SAVING_SW-1;  
//(Optional) Turning on the switch for primary operator recognition for LNR load-based dormancy  
MOD GNBWPWSAVINGALGO: PwrSavingAlgoSwitch=LNR_LOAD_DORM_PRIM_OP_RECOG_SW-1;  
//(Optional) Turning on the multi-frequency flexible dormancy switch  
ADD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=MULTI_FREQ_FLEX_SLEEP_SW-1;  
//(Optional) Turning on the SA X2 setup switch in inter-base-station SA networking  
MOD GNBINTERFACEPARAM: X2ConfigSw=X2_SETUP_FOR_SA_SW-1;  
//(Optional) Turning on the LTE-NR SA inter-base-station energy saving switch in inter-base-station SA  
networking  
MOD GNBWPWSAVINGALGO: PwrSavingAlgoSwitch=SA_LNR_INTER_BS_PWR_SAVE_SW-1;
```

Activating LTE load-based RF module deep dormancy

```
//Turning on the RF module-level switches on the LTE side. If the start time and stop time are the same,  
only load-based RF module deep dormancy takes effect, while scheduled RF module deep dormancy does  
not take effect.  
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=ON, DORMANCYSTARTTIME=00&00&00,  
DORMANCYSTOPTIME=00&00&00, DORMANCYTIMER=5, CUSTOMDORMANCYSTDSW=OFF;  
//Turning on the switch on the LTE cell side  
MOD CELLPWRSAVINGALGO: LocalCellId=0, PwrSavingAlgoSwitch=INTRA_LTE_DORMANCY_BY_LOAD_SW-1;  
//Adding an energy saving policy  
ADD CELLPWRSAVINGPOLICY: LocalCellId=0, PolicyIndex=0,  
PowerSavingType=INTRA_LTE_DORMANCY_BY_LOAD, StartTime=00&00&00, StopTime=06&00&00,  
DIPrbThld=10, DIPrbOffset=10, UIPrbThld=10, UIPrbOffset=10, UENumThld=10, ExitJudgePeriod=24,  
EvalInterval=EVAL_INTERVAL_60S;  
//Configuring the overlap indicator extension for an inter-frequency neighboring cell  
MOD EUTRANINTERFREQNCCELL: LocalCellId=0, Mcc="460", Mnc="00", eNodeBId=720800, CellId=1,  
OverlapIndicatorExtension=POWER_SAVING_OVERLAP_IND-1;  
//(Optional) Turning on the intra-RAT level-by-level energy saving switch  
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=INTRA_RAT_GRAD_PWR_SAVING_SW-1;  
//(Optional) Setting the inter-eNodeB information exchange mode  
MOD ENODEBPWRSAVINGALGO: InterEnbInfoExchangeMode=SLOW_EXCHANGE_MODE;  
//Setting the cell load filtering mode  
MOD ENODEBPWRSAVINGALGO: CellLoadFilterMode=FAST_FILTER_MODE;  
//(Optional) Turning on the multi-frequency flexible dormancy switch  
ADD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=MULTI_FREQ_FLEX_SLEEP_SW-1;  
//(Optional) Turning on the RF module dormancy switch for LTE sector split cells that serve as basic-layer  
cells for load-based dormancy  
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=SPLIT_CELL_RF_DORMANCY_SW-1;  
//(Optional) Setting the inter-frequency MLB threshold and load offset  
MOD CELLMLB: LocalCellId=0, InterFreqMLbThld=40, LoadOffset=8;
```

Activating LTE load-based precise dormancy

```
//Turning on the RF module-level switches on the LTE side. If the start time and stop time are the same,  
only load-based RF module deep dormancy takes effect, while scheduled RF module deep dormancy does  
not take effect.  
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=ON, DORMANCYSTARTTIME=00&00&00,  
DORMANCYSTOPTIME=00&00&00, DORMANCYTIMER=5, CUSTOMDORMANCYSTDSW=OFF;  
//Turning on the switch for LTE load-based precise dormancy and setting the dormancy cell type  
MOD CELLPWRSAVINGALGO: LocalCellId=0, PwrSavingAlgoSwitch=LTE_DORMANCY_BY_LOAD_PE_SW-1,  
DormancyCellType=CAPACITY_CELL;  
MOD CELLPWRSAVINGALGO: LocalCellId=1, PwrSavingAlgoSwitch=LTE_DORMANCY_BY_LOAD_PE_SW-1,  
DormancyCellType=BASIC_CELL;  
//(Optional) Turning on the overlap grid mode switch  
MOD CELLPWRSAVINGALGO: LocalCellId=0, PwrSavingAlgoSwitch=OVERLAP_GRID_MODE_SW-1;  
MOD CELLPWRSAVINGALGO: LocalCellId=1, PwrSavingAlgoSwitch=OVERLAP_GRID_MODE_SW-1;  
//Adding an energy saving policy  
ADD CELLPWRSAVINGPOLICY: LocalCellId=0, PolicyIndex=0,  
PowerSavingType=LTE_PRECISE_DORMANCY_BY_LOAD, StartTime=00&00&00, StopTime=06&00&00,  
DIPrbThld=10, DIPrbOffset=10, UIPrbThld=10, UIPrbOffset=10, UENumThld=10, ExitJudgePeriod=24,  
EvalInterval=EVAL_INTERVAL_60S, ProtectDIPrbThld=50, ProtectDIPrbOffset=20, ProtectUIPrbThld=50,  
ProtectUIPrbOffset=20;  
//Setting the overlap rate for an inter-frequency neighboring cell  
MOD EUTRANINTERFREQNCCELL: LocalCellId=0, Mcc="460", Mnc="00", eNodeBId=720800, CellId=1,  
OverlapRate=40;  
//(Optional) Turning on the intra-RAT level-by-level energy saving switch  
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=INTRA_RAT_GRAD_PWR_SAVING_SW-1;
```

```
//(Optional) Setting the inter-eNodeB information exchange mode
MOD ENODEBPWRSAVINGALGO: InterEnbInfoExchangeMode=SLOW_EXCHANGE_MODE;
//Setting the cell load filtering mode
MOD ENODEBPWRSAVINGALGO: CellLoadFilterMode=FAST_FILTER_MODE;
//(Optional) Turning on the multi-frequency flexible dormancy switch
ADD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=MULTI_FREQ_FLEX_SLEEP_SW-1;
//(Optional) Turning on the RF module dormancy switch for LTE sector split cells that serve as basic-layer cells for load-based dormancy
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=SPLIT_CELL_RF_DORMANCY_SW-1;
//(Optional) Setting the inter-frequency MLB threshold and load offset
MOD CELLMLB: LocalCellId=0, InterFreqMlbThd=40, LoadOffset=8;
```

(Optional) Multi-operator sharing scenarios

```
//Enabling optimization for cell PLMN reservation scenarios
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=CELL_PLMN_RSVD_OPT_SW-1;
//Setting PLMN_RESERVED_FLAG in the external-cell PLMN list based on the network plan
ADD EUTRANEXTERNALCELLPLMN: Mcc="460", Mnc="11", eNodeBId=222, CellId=1, ShareMcc="460", ShareMnc="20", AggregationAttribute=PLMN_RESERVED_FLAG-1;
//Turning on the switch for broadcast of operator-specific cell reservation status over the X2 interface based on the network plan
MOD ENODEBOAMPARAM: ConfigAdaptOptSwitch=RSVD_FOR_OP_X2_BROADCAST_SW-1;
```

Activating UMTS load-based RF module deep dormancy

```
//NodeB side:
//Turning on the RF module-level switches on the UMTS side. If the start time and stop time are the same, only load-based RF module deep dormancy takes effect, while scheduled RF module deep dormancy does not take effect.
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=ON, DORMANCYSTARTTIME=00&00&00, DORMANCYSTOPTIME=00&00&00, DORMANCYTIMER=5, CUSTOMDORMANCYSTDSW=OFF;
//Turning on the switch for the UMTS cell, after which the base station enters the dormancy-pending state
SET ULOCCELLPWRSVAPARA: ULOCCELLID=0, DORMANCYALGOSW=DORMANCY_BY_LOAD_SW-1;
//Canceling the dormancy-pending state of the base station
PND DORMANCY: OPERATIONTYPE=UNPENDING;

//RNC side:
//Configuring cell 0 as the capacity-layer cell and cell 1 as the basic-layer cell
MOD UCELL: LogicRncId=0, CellId=0, DormancyCellType=CAPACITY_CELL;
MOD UCELL: LogicRncId=0, CellId=1, DormancyCellType=BASIC_CELL;
//Setting cell 1 as an inter-frequency co-coverage neighbor of cell 0 with BlindHoFlag set to TRUE
ADD UINTERFREQNCELL: RNCId=0, CellId=0, NCellRncId=0, NCellId=1, BlindHoFlag=TRUE;
//(Optional) Setting cell 1 as an inter-frequency co-coverage neighbor of cell 0 by using the MOD command if the neighbor relationship has been configured
MOD UINTERFREQNCELL: RNCId=0, CellId=0, NCellRncId=0, NCellId=1, BlindHoFlag=TRUE;
//Enabling inter-frequency hard handover
SET UCORRMALGOSWITCH: HoSwitch=HO_INTER_FREQ_HARD_HO_SWITCH-1;
//Adding an energy saving policy for capacity-layer cell 0
ADD UCELLDORMANCYPOLICY: LogicRncId=0, CellId=0, PolicyIndex=0, DormancyType=INTRA_DORMANCY_BY_LOAD, StartTime=00&00&00, StopTime=06&00&00, TriggerJudgePeriod=5, ExitJudgePeriod=24, PenaltyTime=30, ForcibleTriggerUeNumThld=0, TotalUserNumThd=1, HsdpaUserNumThd=0, HsupaUserNumThd=0, NCellLdrRemainThd=20, DormancyCchUserSwitch=ON, McHsdpaUserNumThd=2;
```

(Optional) Multi-operator sharing scenarios

```
//RNC side:
//Performing the following configuration if cell 1 is intended to serve as the basic-layer cell of cell 0 when they belong to different operators
SET URNCCCELLDORMANCYPARA: LogicRncId=0, CellInterPlmnHoSwitch=ON;
SET UOPERATORSHARINGMODE: LogicRncId=0, InterPlmnHoAllowedIntraRat=YES;
```

(Optional) Capacity-layer cell with CBS enabled

```
//RNC side:
//Setting CellIgnoreCbsSwitch to ON if all CBS cells under the RNC are allowed to enter load-based dormancy. Otherwise, the default value OFF is recommended.
SET URNCCCELLDORMANCYPARA: LogicRncId=0, CellIgnoreCbsSwitch=ON, CbsServiceHoldDuration=5;
```

Activating GU load-based RF module deep dormancy

```
//NodeB side:
//Turning on the RF module-level switches on the UMTS side. If the start time and stop time are the same,
only load-based RF module deep dormancy takes effect, while scheduled RF module deep dormancy does
not take effect.
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=ON, DORMANCYSTARTTIME=00&00&00,
DORMANCYSTOPTIME=00&00&00, DORMANCYTIMER=5, CUSTOMDORMANCYSTDSW=OFF;
//Turning on the switch for the UMTS cell, after which the base station enters the dormancy-pending state
SET ULOCELLPWRSAV PARA: ULOCELLID=0, DORMANCYALGOSW=DORMANCY_BY_LOAD_SW-1;
//Canceling the dormancy-pending state of the base station
PND DORMANCY: OPERATIONTYPE=UNPENDING;

//RNC side:
//Turning on the switch for inter-RAT common measurement
SET UMBSCCRRM: LogicRncId=0, IRATCommonMeasSwitch=ON;
//Selecting the HO_INTER_RAT_CS_OUT_SWITCH and HO_INTER_RAT_PS_OUT_SWITCH options
SET UCORRMALGOSWITCH:
HoSwitch=HO_INTER_RAT_CS_OUT_SWITCH-1&HO_INTER_RAT_PS_OUT_SWITCH-1;
//Setting GSM cell 1 as the basic-layer cell of UMTS cell 0 (capacity-layer cell)
ADD U2GNCELL: RNCId=0, CellId=0, GSMCellIndex=1, NIRATOverLap=TRUE;
//((Optional) Setting GSM cell 1 as the basic-layer cell of UMTS cell 0 (capacity-layer cell) using the MOD
command if the neighbor relationship has been configured
MOD U2GNCELL: RNCId=0, CellId=0, GSMCellIndex=1, NIRATOverLap=TRUE;
//Adding an energy saving policy for capacity-layer cell 0
ADD UCELLDORMANCYPOLICY: LogicRncId=0, CellId=0, PolicyIndex=1,
DormancyType=INTER_GU_DORMANCY_BY_LOAD, StartTime=00&00&00, StopTime=06&00&00,
TriggerJudgePeriod=5, ExitJudgePeriod=24, PenaltyTime=30, ForcibleTriggerUeNumThld=0,
TotalUserNumThd=2, DormancyCchUserSwitch=ON, IRATShutdownDLLoadThd=30,
IRATShutdownULLoadThd=10, IRATShutdownGSMLoadThd=25, IRATShutdownGSMLoadHyst=20;

//BSC side:
//Enabling the function of information exchange on the Iur-g interface (only on the BSC6910, not on the
BSC6960)
SET GCELLOTHEXT: IDTYPE=BYID, CELLID=1, IURGINFOCTRL=YES;
//Enabling inter-RAT cell dynamic shutdown on the BSC side
SET GCELLSOFT: IDTYPE=BYID, CELLID=1, IRATShutdownSwitch=ON;
```

(Optional) Multi-operator sharing scenarios

```
//RNC side:
//Allowing UEs to be handed over between inter-RAT inter-operator cells
SET UOPERATORSHARINGMODE: LogicRncId=0, InterPlmnHoAllowedInterRat=YES;
```

(Optional) Capacity-layer cell with CBS enabled

```
//RNC side:
//Setting CellIgnoreCbsSwitch to ON if all CBS cells under the RNC are allowed to enter load-based
dormancy. Otherwise, the default value OFF is recommended.
SET URNCELLDORMANCYPARA: LogicRncId=0, CellIgnoreCbsSwitch=ON, CbsServiceHoldDuration=5;
```

Setting parameters shared by preceding sub-functions

```
//(Optional) Enabling super dormancy pending
PND DORMANCY: OPERATIONTYPE=PENDING, PENDINGDURATION=30;
```

Optimization Command Examples

```
//NR side:
//Changing the lengths of the penalty timers applied to resource-related handovers to 30s when the
handover success rate decreases due to scheduled RF module deep dormancy
MOD GNBMOBILITYCOMMPARAM: NecHoPrepFailPunishTimer=30, OptHoPrepFailPunishTimer=30;
//Changing the lengths of the penalty timers applied to resource-related handovers to 30s when the
handover success rate decreases due to NR load-based RF module deep dormancy
MOD GNBMOBILITYCOMMPARAM: NecHoPrepFailPunishTimer=30, OptHoPrepFailPunishTimer=30;
//Turning on the simulated load dormancy impact switch
MOD GNBPWRSAVINGALGO: PwrSavingAlgoSwitch=SIMU_LOAD_DORMANCY_IMP_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

Deactivating scheduled RF module deep dormancy

```
//(Optional) Disabling optimized evaluation of NR RF module dormancy
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=0,
PowerSavingAlgoSwitch=NR_DORMANCY_JUDGE_THLD_OPT_SW-0;
//Removing the energy saving configuration of optimized evaluation of NR RF module dormancy
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=11;
//(Optional) Turning off the multi-RRU cell dormancy compatibility switch
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=MULTI_RRU_CELL_DORM_COMPAT_SW-0;
//Disabling RF module deep dormancy
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=OFF;
```

Deactivating NR load-based RF module deep dormancy

```
//Disabling RF module deep dormancy
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=OFF;
//Turning off the switches for precise dormancy. The precise mode is not supported in high-frequency TDD.
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingExtSwitch=GRID_MODE_SW-0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingExtSwitch=PRECISE_DORMANCY_SW-0;
//Turning off the switch for NR load-based RF module deep dormancy on the NR DU cell side
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=INTRA_NR_DORMANCY_BY_LOAD_SW-0;
```

Deactivating LNR load-based RF module deep dormancy

```
//Turning off the multi-frequency flexible dormancy switch
MOD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=MULTI_FREQ_FLEX_SLEEP_SW-0;
//Disabling RF module deep dormancy
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=OFF;
//LTE side:
//(Optional) Turning off the multi-RRU cell dormancy compatibility switch
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=MULTI_RRU_CELL_DORM_COMPAT_SW-0;
//(Optional) Turning off the LTE-NR SA inter-base-station energy saving switch in inter-base-station SA networking
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=LNR_INTER_BS_PWR_SAVE_SW-0;
MOD ENODEBOAMPARAM: InterfaceSonConfigSwitch=LNR_SA_GNB_X2_SETUP_SW-0;
//Turning off the LNR load-based RF module deep dormancy switch
MOD CELLPWRSAVINGALGO: LocalCellId=0, PwrSavingAlgoSwitch=LNR_LOAD_BASED_DORMANCY_SW-0;
//(Optional, required only when the neighbor relationship between LTE and NR cells with overlapping coverage areas is manually configured in NSA networking, NSA and SA hybrid networking, or inter-base-station SA networking) Removing the neighbor relationship for energy saving in the overlapping coverage areas
MOD NRNRELATIONSHIP: LocalCellId=0, Mcc="460", Mnc="01", GnodebId=441233, CellId=1, AggregationAttribute=CARRIER_SHUTDOWN_OVERLAP_FLAG-0;
//NR side:
//(Optional) Turning off the switch for primary operator recognition for LNR load-based dormancy
MOD GNBPWRSAVINGALGO: PwrSavingAlgoSwitch=LNR_LOAD_DORM_PRIM_OP_RECOG_SW-0;
//(Optional) Turning off the LTE-NR SA inter-base-station energy saving switch in inter-base-station SA networking
MOD GNBPWRSAVINGALGO: PwrSavingAlgoSwitch=SA_LNR_INTER_BS_PWR_SAVE_SW-0;
//(Optional) Turning off the SA X2 setup switch in inter-base-station SA networking
MOD GNBINTERFACEPARAM: X2ConfigSw=X2_SETUP_FOR_SA_SW-0;
//Turning off the LNR load-based RF module deep dormancy switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, PowerSavingSwitch=LNR_DORMANCY_BY_LOAD_SW-0;
//(Optional, required only in SA networking) Removing the overlapping relationship (between the local cell and the neighboring LTE cell) for energy saving
MOD NRCELLEUTRANRELATION: NrCellId=1, Mcc="460", Mnc="01", EnodebId=441282, CellId=0, PowerSavingOverlapInd=LNR_CARR_SHUTDOWN_OVERLAP_IND-0;
//Removing the energy-saving policy
RMV NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=0;
```

Deactivating LTE load-based RF module deep dormancy

```
//Turning off the multi-frequency flexible dormancy switch
MOD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=MULTI_FREQ_FLEX_SLEEP_SW-0;
//Disabling RF module deep dormancy
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=OFF;
//Turning off the switch on the LTE cell side
MOD CELLPWRSAVINGALGO: LocalCellId=0, PwrSavingAlgoSwitch=INTRA_LTE_DORMANCY_BY_LOAD_SW-0;
//((Optional) Turning off the intra-RAT level-by-level energy saving switch
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=INTRA_RAT_GRAD_PWR_SAVING_SW-0;
```

Deactivating LTE load-based precise dormancy

```
//Turning off the multi-frequency flexible dormancy switch
MOD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=MULTI_FREQ_FLEX_SLEEP_SW-0;
//Disabling RF module deep dormancy
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=OFF;
//((Optional) Turning off the overlap grid mode switch
MOD CELLPWRSAVINGALGO: LocalCellId=0, PwrSavingAlgoSwitch=OVERLAP_GRID_MODE_SW-0;
MOD CELLPWRSAVINGALGO: LocalCellId=1, PwrSavingAlgoSwitch=OVERLAP_GRID_MODE_SW-0;
//Turning off the switch for LTE load-based precise dormancy
MOD CELLPWRSAVINGALGO: LocalCellId=0, PwrSavingAlgoSwitch=LTE_DORMANCY_BY_LOAD_PE_SW-0,
DormancyCellType=CAPACITY_CELL;
MOD CELLPWRSAVINGALGO: LocalCellId=1, PwrSavingAlgoSwitch=LTE_DORMANCY_BY_LOAD_PE_SW-0,
DormancyCellType=CAPACITY_CELL;
//Removing the energy saving policy
RMV CELLPWRSAVINGPOLICY: LocalCellId=0, PolicyIndex=0;
//((Optional) Turning off the intra-RAT level-by-level energy saving switch
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=INTRA_RAT_GRAD_PWR_SAVING_SW-0;
```

((Optional) Multi-operator sharing scenarios

```
//Turning off the switch for broadcast of operator-specific cell reservation status over the X2 interface based
on the network plan
MOD ENODEBOAMPARAM: ConfigAdaptOptSwitch=RSVD_FOR_OP_X2_BROADCAST_SW-0;
//Disabling optimization for cell PLMN reservation scenarios
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=CELL_PLMN_RSVD_OPT_SW-0;
```

Deactivating UMTS load-based RF module deep dormancy

```
//NodeB side:
//Disabling RF module deep dormancy
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=OFF;
//Turning off the switch on the UMTS cell side
SET ULOCELLPWRSAPARA: ULOCELLID=0, DORMANCYALGOSW=DORMANCY_BY_LOAD_SW-0;

//RNC side:
//Removing the energy saving policy
RMV UCELLDORMANCYPOLICY: LogicRncId=0, CellId=0, PolicyIndex=0;
```

Deactivating GU load-based RF module deep dormancy

```
//NodeB side:
//Disabling RF module deep dormancy
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=OFF;
//Turning off the switch for the UMTS cell, after which the base station enters the dormancy-pending state
SET ULOCELLPWRSAPARA: ULOCELLID=0, DORMANCYALGOSW=DORMANCY_BY_LOAD_SW-0;
//Canceling the dormancy-pending state of the base station
PND DORMANCY: OPERATIONTYPE=UNPENDING;

//RNC side:
//Removing the energy-saving policy
RMV UCELLDORMANCYPOLICY: LogicRncId=0, CellId=0, PolicyIndex=1;
```

7.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.2.4.2 Activation Verification

The activation of this function can be verified during the specified dormancy period using the following common methods:

- You can run the **DSP BRDDORMANCY** command to observe the current or latest RF module dormancy status. The values displayed in the query results are periodically refreshed and may not always reflect the most recent updates.
 - Run the **DSP BRDDORMANCY** command with the **QUERYTYPE** parameter set to **BOARD** and the **QUERYMODE** parameter set to **CURRENT**.
 - If the value of the **Board Dormancy State** parameter in the command output is **Dormancy**, RF module deep dormancy has taken effect. If RF module deep dormancy has taken effect for an eCPRI-capable AAU, the value of the **Baseband Shutdown State** parameter is **Enable**. To learn which AAU supports eCPRI, see the hardware description in the product documentation.
 - If the value of the **Board Dormancy State** parameter is **Multi-Frequency Flexible Dormancy**, multi-frequency flexible dormancy has taken effect.
 - If the value of the **Board Dormancy State** parameter is **Not in the Dormancy State**, follow the operations in either case that suits to check the reason:
 - Check the value of **Reason for Not Entering Deep Dormancy** to determine the possible reason why deep dormancy is not in effect, as described in [Table 7-21](#).
 - Check the value of **Reason for Not Entering Multi-Frequency Flexible Dormancy** to determine the possible reason why multi-frequency flexible dormancy is not in effect, as described in [Table 7-21](#).
 - Run the **DSP BRDDORMANCY** command with the **QUERYTYPE** parameter set to **BOARD**, the **QUERYMODE** parameter set to **HISTORY**, and the **PWRSAVINGFEATURE** parameter set to **DEEP_DORMANCY**. You can determine the possible reasons why RF module deep dormancy does not take effect within the specified period based on the values of **Latest Exit Cause** and **Latest Disabled Cause**. [Table 7-21](#) lists the possible reasons.

[Table 7-22](#) describes the meanings of **Latest Exit Cause**, **Latest Disabled Cause**, **Latest Exit Time**, and **Latest Disabled Time**.

Latest Exit Cause and **Latest Disabled Cause** indicate the reasons for the most recent exit and failure of the function. These reasons can be queried after the effective period has ended. For example, if the effective period is from 00:00:00 to 06:00:00, **Latest Exit Cause** and **Latest Disabled Cause** record the reasons for the most recent exit and failure of

the function from 00:00:00 to 06:00:00. These reasons can be queried by an operation performed within the time from 06:00:00 to 23:59:59.

- Only scheduled RF module deep dormancy supports the query of the **Latest Exit Time**, **Latest Exit Cause**, **Latest Disabled Time**, and **Latest Disabled Cause** fields by running the **DSP BRDDORMANCY** command.
- The collection period is 10 minutes. If the causes change frequently, the latest causes may not be updated in a timely manner.
- In separate-MPT scenarios where the system time is not strictly aligned, the cause for an exit from dormancy is not recorded if the remaining time is less than 1 minute.
- When the **QUERYMODE** parameter is set to **HISTORY**, values of **Process Being Protected**, **Not All NE Dormancy Notifications Received**, and **Carriers Existed on the RF Module** are not included in the command output.

Table 7-21 Possible reasons why board dormancy does not take effect

Cause	Description
Maintenance Link Abnormal	The RF module is disconnected, and the BBU cannot obtain the dormancy state of the RF module.
Board Unavailable	The link is normal, and the RF module is starting up.
Dormancy Capability Not Supported by the Hardware	The RF module does not support deep dormancy or multi-frequency flexible dormancy.
Optical Module Compatibility Requirement Not Met	The optical module capability does not support this function.

Cause	Description
Configuration Conditions Not Met	- The deep dormancy or multi-frequency flexible dormancy switch is turned off. - No NodeB, eNodeB, or gNodeB application is available. - When the customized deep dormancy standard switch is turned on, the deep dormancy RATs do not include all RATs of the local end. - The RF module is powered by the iSDU, and the customized timing power-off switch is turned on for the power supply port on the iSDU (by setting the PDPATTR.CTSSW (LTE eNodeB, 5G gNodeB) parameter to ON). - The RF module is configured as the iSDU manager, and the customized timing power-off switch is turned on for any port on the iSDU (by setting the PDPATTR.CTSSW (LTE eNodeB, 5G gNodeB) parameter to ON).
License Insufficiency	- The deep dormancy license items of an RAT are insufficient. - The license items affecting basic cell service capabilities (such as the cell access capability) are insufficient.
Upgrading	- The base station is being upgraded. - The base station is in the protection period for a successful upgrade. The protection period lasts less than 15 minutes.
Base Station in Dormancy Suspension	- Dormancy pending is triggered due to the execution of the PND DORMANCY command. - Dormancy pending is triggered due to base station reset or configuration verification. - Dormancy pending is triggered due to reliability protection.
Protection Due to Ongoing Device Maintenance and Test	- Device-related maintenance and test operations are being performed. - Protection will be canceled within 30 minutes after device-related maintenance and test operations are completed. Dormancy will be allowed after the protection is canceled.
Reliability Requirements Not Met	- The reliability requirements, such as those for solder joints or temperature difference, are not met. - The current RF module is faulty and does not meet the dormancy entry conditions.

Cause	Description
Service Conditions Not Met	When load-based RF module deep dormancy is enabled, dual-band RF modules do not meet the dormancy entry conditions if one frequency band serves as the basic layer while the other serves as the capacity layer. LTE side: • Some activated cells under the base station are not bound to BBPs. • Service faults exist and the dormancy entry conditions are not met. • The cell is served by combined RRUs, which does not meet the dormancy entry conditions. • For load-based deep dormancy, run the DSP CELLPWRSAVINGPOLICY command to query the reason why the cell does not meet the conditions. For details, see Table 7-27 or Table 7-28. NR side: • A possible reason why scheduled RF module deep dormancy does not take effect is that some activated cells under the base station are not bound to BBPs. • Service faults exist and the dormancy entry conditions are not met. • Different dormancy periods are configured for the RF modules bound to a combined-RRU cell or multi-TRP cell (such as an SFN cell, hyper cell, or combined cell), or the current time falls outside the configured periods. • For load-based deep dormancy, run the DSP NRDUCELLPOWERSAVING command to query the reason why the cell does not meet the conditions. For details, see Table 7-25 or Table 7-26. UMTS side: • Multiple RF modules are configured for a cell, and their dormancy periods are different or the current time falls outside the configured periods. • Multiple RF modules are configured for a cell group, and their dormancy periods are different or the current time falls outside the configured periods. • Run the DSP ULOCELLPWRSAV command to query the reason why the cell does not meet

Cause	Description
	the dormancy entry conditions. For details, see Table 7-23 .
Protection Due to Ongoing Service Maintenance and Test	- Service-related maintenance and test operations are being performed. - Protection will be canceled within 30 minutes after service-related maintenance and test operations are completed. Dormancy will be allowed after the protection is canceled.
Not Within Dormancy Period	- The current time falls outside the configured scheduled dormancy period, and load-based dormancy is not configured for some LTE or NR cells. - The NE does not enter dormancy because its time synchronization fails. You can run the DSP TIMESRC command to check whether a time source is available.
Process Being Protected	- After the RF module link recovers, it takes about 5 minutes for the related alarm to be cleared. The cell will enter the dormancy state after the alarm is cleared. - The base station instructs the cell to enter dormancy, and then instructs the RF module to enter dormancy 30s later. - Random dormancy waiting is ongoing. The random dormancy waiting time can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to SITE. - Less than 5 minutes have passed since the RF module exited dormancy.
Not All NE Dormancy Notifications Received	The dormancy notifications of some NEs are not received. For example, in a separate-MPT base station, the peer NE does not meet the dormancy entry conditions, and therefore the RF module does not enter dormancy.
Carriers Existed on the RF Module	None
Other Reason	- A plug-and-play module is used, which does not meet the function entry conditions. - The internal software is faulty, or the evaluation period defined in the software is not reached.

Table 7-22 Time and causes for the most recent exit and failure of a function

Command Output	Description
Latest Exit Cause	Indicates the cause of the most recent exit of a function during its effective period.
Latest Disabled Cause	Indicates the cause of the most recent failure of a function to take effect during its effective period.
Latest Exit Time	Indicates the recorded time of the latest exit of a function during its effective period.
Latest Disabled Time	Indicates the recorded time of the latest failure for a function to take effect during its effective period.

- Query the cell dormancy status using the following methods if the outputs of the preceding commands indicate that RF module deep dormancy does not take effect:
 - Querying the cell dormancy status on the UMTS side:
 - Run the **DSP ULOCELL** command. If the value of **Local Cell Status** is **Local Cell Unavailable-RF Module Dormancy**, the cell has entered dormancy.
 - Run the **DSP ULOCELLPWRSAV** command to query the reason why the cell does not enter dormancy. The values displayed in the query results are periodically refreshed and may not always reflect the most recent updates. After the reset, the reason why the cell does not enter dormancy within 15 minutes is incorrect. [Table 7-23](#) describes the possible reasons why RF module deep dormancy does not take effect.

You can also run the **DSP ULOCELLPWRSAV** command to query the **Cause of Latest Dormancy Failure**, **Time of Latest Dormancy Failure**, **Cause of Latest Dormancy Exit**, and **Time of Latest Dormancy Exit** in the specified period.

[Table 7-24](#) describes the meanings of **Cause of Latest Dormancy Failure**, **Time of Latest Dormancy Failure**, **Cause of Latest Dormancy Exit**, and **Time of Latest Dormancy Exit**.

Cause of Latest Dormancy Failure and **Cause of Latest Dormancy Exit** indicate the reasons for the most recent failure and exit of the function. These reasons can be queried after the effective period has ended. For example, if the effective period is from 00:00:00 to 06:00:00, **Cause of Latest Dormancy Failure** and **Cause of Latest Dormancy Exit** record the reasons for the most recent failure and exit of the function from 00:00:00 to 06:00:00. These reasons can be

queried by an operation performed within the time from 06:00:00 to 23:59:59.

Table 7-23 Possible reasons why UMTS scheduled RF module deep dormancy and UMTS or GU load-based RF module deep dormancy do not take effect

Cause	Description
RNC Version Not Support	The RNC is not upgraded to a required version.
Suspended Due to RNC Configuration	During dormancy, an RNC MML command affecting dormancy is executed, leading to dormancy suspension. After 30 minutes, dormancy suspension is canceled and dormancy is allowed.
RNC Fault	Alarms that affect dormancy activation exist on the RNC, such as ALM-22202 UMTS Cell Unavailable.
RNC Service Maintenance Protection	- Service-related maintenance and test operations are being performed on the RNC. - Protection will be canceled within 30 minutes after service-related maintenance and test operations on the RNC side are completed. Dormancy will be allowed after the protection is canceled.
RNC High Priority Feature In Effect	A cell with CBS enabled cannot enter dormancy in the following scenarios: - If URNCELLDORMANCY-PARA.CellIgnoreCbsSwitch is set to OFF for a cell, the cell cannot enter dormancy. - If URNCELLDORMANCY-PARA.CellIgnoreCbsSwitch is set to ON for a cell, the cell can enter dormancy. However, when CBS is triggered, it is preferentially guaranteed, and the cell cannot enter dormancy during the duration specified by URNCELLDORMANCY-PARA.CbsServiceHoldDuration.
Service Fault	- Alarms that affect dormancy activation exist on the NodeB, such as ALM-28206 Local Cell Capability Decline. - The Iub interface is faulty. For example, the value of Local Cell Status is Local Cell Available-NCP Fault in the DSP ULOCELL command output.
RF Module Capability Not Support	The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD . Table 7-21 lists the possible reasons.

Cause	Description
Baseband Capability Not Support	The uplink or downlink baseband equipment contains multiple BBPs or contains BBPs that do not support RF module deep dormancy.
Local Cell Unavailable	None
Logical Cell Not Activated for Non-Power-Saving Reasons	None
Mutually Exclusive or Impacted Feature	The cell is enabled with mutually exclusive or impacted features/functions as described in 7.2.3.2.2 Mutually Exclusive Functions or 7.2.3.2.3 Function Impacts , respectively.
Service Maintenance Protection	- Service-related maintenance and test operations are being performed on the NodeB. - Protection will be canceled within 30 minutes after service-related maintenance and test operations on the NodeB side are completed. Dormancy will be allowed after the protection is canceled.
Under Penalty	The cell enters the penalty state when UE transfer is not completed. The cell can enter dormancy only after the penalty state is cleared. The specific penalty time is as follows: 30 minutes for scheduled RF module deep dormancy - Value of the UCELLDORMANCYPOLICY.PenaltyTime parameter on the RNC for UMTS load-based RF module deep dormancy or GU load-based RF module deep dormancy
Cell in Cell Group Not Meet Conditions	A multi-cell group is present. In such situations, all cells in the group must meet the dormancy entry conditions. Multi-cell group configurations are as follows: - The NODEBMULTICELLGRP MO is present. - The NODEBEXSECCELLGRP MO is present. - A local cell for which UCELL.DLASYMFILTER is set to TRUE and cells with UCELL.DLASYMFILTERNFCELLID specified

Cause	Description
Not In Dormancy Period	The current time falls outside the specified energy-saving effective period.
Load Conditions Not Met	The load of the capacity-layer cell does not meet the conditions.
No Configured Power Saving Policy	No energy saving policy for UMTS load-based dormancy or UMTS-GSM load-based dormancy is configured for the capacity-layer cell.
No Configured Co-coverage Neighboring Cell	No inter-band or inter-RAT co-coverage neighboring cells are configured for the capacity-layer cell.
Number of UEs Not Meet Threshold Conditions	The total number of UEs, number of FACH UEs, number of HSDPA UEs, number of HSUPA UEs, or number of multi-carrier UEs in the capacity-layer cell does not meet the energy-saving policy configuration requirements.
No Available Co-coverage Neighboring Cell	Some inter-band or inter-RAT co-coverage neighboring cells are unavailable.
Co-coverage Neighboring Cell Not Meet Load Conditions	The load of inter-band or inter-RAT co-coverage neighboring cells does not meet dormancy entry conditions. Specifically: • The inter-band or inter-RAT co-coverage neighboring cells are heavily loaded or overloaded. • The load margin of the inter-band or inter-RAT co-coverage neighboring cells does not meet the energy-saving policy configuration requirements. • The load of the inter-RAT co-coverage neighboring cells does not meet the energy-saving policy configuration requirements. • The load of inter-RAT co-coverage neighboring cells cannot be obtained.

Cause	Description
Services of Co-coverage Neighboring Cells Not Support	In GU load-based RF module deep dormancy scenarios, data services configured for capacity-layer cells are not supported by inter-RAT co-coverage neighboring cells.
Process Being Protected	• The cell setup procedure is protected to ensure configuration consistency between the base station and the base station controller. • After the logical cell is successfully activated, a 3-minute protection timer is started. Dormancy is allowed only after the protection timer expires. • Other energy saving features are being activated or deactivated.
Carrier Shutdown Reliability Protected	• When RF module dormancy is enabled and the UCELLDYNSHUTDOWN.DynShutdownSwitch parameter is set on the RNC side to enable dynamic cell shutdown, the cell enters the carrier shutdown reliability protection state if any RNC or NodeB configuration affecting dormancy is modified during dynamic cell shutdown. The cell can enter dormancy only after carrier shutdown is exited. • When RF module dormancy is enabled and carrier shutdown features are enabled on the NodeB side, the cell enters the carrier shutdown reliability protection state if any RNC configuration affecting dormancy is modified during cell shutdown. The cell can enter dormancy only after carrier shutdown is exited. The carrier shutdown features on the NodeB side are as follows: – Multi-Carrier Switch off Based on Power Backup (NODEBPOWEROUTAGE.ISDSW) – UMTS Low Power Consumption Mode (NODEBALGPARA.LOWPWRCONSUMPMODESW) – UMTS SOC Level-Based Low Power Consumption Mode (NODEBALGPARA.LOWPWRCONSUMEMODEEXTSW)

Cause	Description
PIM Detection Protected	- When PIM interference detection is enabled for a UMTS cell by setting ULOCCELLALGPARA.PIMDETECTIONSW to ON and the cell is in the dormancy state, it exits dormancy when LTE or NR PIM interference detection is started. - If a massive MIMO module is shared by UMTS and LTE or NR cells and the UMTS cell is in the dormancy state, it exits dormancy when LTE or NR PIM interference characteristic separation training is started.
Other Reasons	- The conditions for entering the energy saving function are not met throughout the triggering evaluation period. - Other reasons, such as internal software issues, cause the function not to take effect.
NULL	Services meet the dormancy conditions or dormancy features are not enabled.

Table 7-24 Time and causes for the most recent exit and failure of the function in the **DSP ULOCCELLPWRSAP** command output

Command Output	Description
Cause of Latest Dormancy Failure	Indicates the cause of the most recent dormancy failure during the effective period.
Time of Latest Dormancy Failure	Indicates the recorded time of the most recent dormancy failure during the effective period.
Cause of Latest Dormancy Exit	Indicates the cause of the most recent dormancy exit during the effective period.
Time of Latest Dormancy Exit	Indicates the recorded time of the most recent dormancy exit during the effective period.

- LTE side: Run the **DSP CELL** command. If the value of the **Cell power save state** parameter is **CELL_PWRSAP_SAVE_RF_DEEP_SLEEP** in the command output, the cell has entered dormancy.
- NR side: Run the **DSP NRDUCELL** command. If the **Power Saving State** parameter has a value of **Dormancy of All RF Modules:Enabled** in the command output, the cell has entered dormancy.

In addition to the preceding observation methods, the following functions can be observed using the **DSP NRDUCELLPOWERSAVING** and **DSP CELLPWRSAVINGPOLICY** commands: optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module deep dormancy (referred to as optimized evaluation of NR RF module dormancy), load-based RF module deep dormancy, and multi-frequency flexible dormancy. The values displayed in the query results are periodically refreshed and may not always reflect the most recent updates.

- The **DSP NRDUCELLPOWERSAVING** command can be used to query the status of NR load-based RF module deep dormancy, multi-frequency flexible dormancy, and optimized evaluation of NR RF module dormancy, as well as the reasons why these functions do not take effect.
 - If the value of the **Power Saving Type** parameter is **Intra-NR Load-based Dormancy** and the value of the **Current State** parameter is **Enabled**, NR load-based RF module deep dormancy or multi-frequency flexible dormancy has taken effect. If the value of the **Power Saving Type** parameter is **NR RF Module Dormancy Evaluation Opt** and the value of the **Current State** parameter is **Enabled**, optimized evaluation of NR RF module dormancy has taken effect.
 - If the value of the **Power Saving Type** parameter is **Intra-NR Load-based Dormancy** or **NR RF Module Dormancy Evaluation Opt** and the value of the **Current State** parameter is **Disabled**, check the **Current Disabled Cause** parameter value to determine the reason why this function does not take effect. [Table 7-25](#) lists the possible reasons.
 - If the value of the **Power Saving Type** parameter is **Intra-NR Load-based Dormancy** or **NR RF Module Dormancy Evaluation Opt**, check the **Latest Exit Cause** and **Latest Disabled Cause** parameter values to determine the reason why this function does not take effect within the specified period. [Table 7-25](#) lists the possible reasons.

[Table 7-22](#) describes the meanings of **Latest Exit Cause**, **Latest Disabled Cause**, **Latest Exit Time**, and **Latest Disabled Time**.

Latest Exit Cause and **Latest Disabled Cause** indicate the reasons for the most recent exit and failure of the function. These reasons can be queried after the effective period has ended. For example, if the effective period is from 00:00:00 to 06:00:00, **Latest Exit Cause** and **Latest Disabled Cause** record the reasons for the most recent exit and failure of the function from 00:00:00 to 06:00:00. These reasons can be queried by an operation performed within the time from 06:00:00 to 23:59:59.

Table 7-25 Possible reasons why NR load-based RF module deep dormancy, optimized evaluation of NR RF module dormancy, or multi-frequency flexible dormancy does not take effect

Cause	Description
Cell in Base Station Not Bound to BBP	None

Cause	Description
Switch Turned Off	None
RF Group Configuration Condition Not Met	The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD . Table 7-21 lists the possible reasons.
Cell Unavailable	The cell is blocked or deactivated, fails to be set up, or is unavailable due to other reasons.
Protection Due to Ongoing Maintenance and Test	- Service-related maintenance and test operations are being performed. - Protection will be canceled within 30 minutes after service-related maintenance and test operations are completed. Dormancy will be allowed after the protection is canceled.
Reliability Requirements Not Met	Alarms that affect the dormancy function exist, such as the High-Risk Function Enable alarm.
Triggering Time Not Reached	The current time falls outside the specified energy-saving effective period.
Parameter Inconsistency Between Cells	In high-frequency TDD, the parameters governing RF module deep dormancy policies are inconsistent across cells.
Ultra-Long Holdover in Progress	None
Impacted Feature Taken Effect	None
Neighboring Cell Not Configured	- No inter-frequency neighboring cells are configured for the serving cell. - Inter-frequency neighboring cells are configured for the serving cell, but there is no neighboring cell for which Power Saving Cell Flag is set to COVERAGE. - In multi-operator sharing scenarios: - The PLMN IDs of the normal operators of the capacity-layer cell are not a subset of the PLMN IDs of normal operators of basic-layer cells. - The PLMN IDs of the reserved operators of the capacity-layer cell are not a subset of the PLMN IDs of normal or reserved operators of basic-layer cells.

Cause	Description
In Penalty State	- A cell enters the penalty state if it exits deep dormancy within the specified effective period. After the penalty ends, it can re-enter deep dormancy. The penalty duration is specified by the NRDUCellPowerSaving.PwrSavingPenaltyTime parameter. - A cell enters the penalty state when some UEs failed to be transferred out of the cell. The penalty lasts 30 minutes. The causes of UE transfer failures include UE capability mismatch and missing configuration of neighboring frequency relationships.
Xn Interface Unavailable	The Xn interface is abnormal in multi-carrier inter-base-station scenarios.
Neighboring Cell Unavailable	- A neighboring cell for which Power Saving Cell Flag is set to COVERAGE is unavailable. - A neighboring cell for which Power Saving Cell Flag is set to COVERAGE is a hyper cell or combined cell, and some TRPs of this cell are unavailable. - A neighboring cell for which Power Saving Cell Flag is set to COVERAGE is an NSA cell.
Downlink PRB Threshold Not Reached	For NR load-based RF module deep dormancy or multi-frequency flexible dormancy, the total downlink PRB usage of the capacity-layer and basic-layer cells does not reach the downlink PRB usage threshold for triggering this function. For optimized evaluation of NR RF module dormancy, the downlink PRB usage of the current NR cell does not reach the downlink PRB usage threshold for triggering this function.
Uplink PRB Threshold Not Reached	For NR load-based RF module deep dormancy or multi-frequency flexible dormancy, the total uplink PRB usage of the capacity-layer and basic-layer cells does not reach the uplink PRB usage threshold for triggering this function. For optimized evaluation of NR RF module dormancy, the uplink PRB usage of the current NR cell does not reach the uplink PRB usage threshold for triggering this function.
Basic Cell PRB Usage Limit Exceeded	For NR load-based precise dormancy, the uplink or downlink PRB usage of a basic-layer cell exceeds the congestion threshold.

Cause	Description
UE Number Threshold Not Reached	None
Triggering Decision Period Not Met	For NR load-based RF module deep dormancy or multi-frequency flexible dormancy, the current time falls within the specified effective period, but the cell does not meet the dormancy entry conditions throughout the period specified by the NRDUCellPowerSaving.ShutdownTrigJudgePeriod parameter.
Multi-Carrier Not Coordinated	Not all the activated cells sharing the same RF module meet the entry conditions for NR load-based RF module deep dormancy or optimized evaluation of NR RF module dormancy.
RF Dormancy Basic Capb Condition Not Met	The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD . Table 7-21 lists the possible reasons.
Others	- The local NR cell meets the dormancy entry conditions but does not receive an instruction to enter dormancy. The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD. Table 7-21 lists the possible reasons. - In the current period, the conditions for this function to take effect are met, but the conditions for LNR load-based RF module deep dormancy are not met. - For optimized evaluation of NR RF module dormancy, the dormancy entry conditions are not met throughout the triggering evaluation period. - Internal software reasons cause the function not to take effect.

- You can also run the **DSP NRDUCELLPOWERSAVING** command to query the status of LNR load-based RF module deep dormancy and multi-frequency flexible dormancy and the reasons why these functions do not take effect. The values displayed in the query results are periodically refreshed and may not always reflect the most recent updates.
 - If the value of the **Power Saving Type** parameter is **LNR Load-based Dormancy** and the value of the **Current State** parameter is **Enabled**, this function has taken effect.
 - If the value of the **Power Saving Type** parameter is **LNR Load-based Dormancy** and the value of the **Current State** parameter is **Disabled**, check the **Current Disabled Cause** parameter value to determine the

reason why this function does not take effect. [Table 7-26](#) lists the possible reasons.

- If the value of the **Power Saving Type** parameter is **LNR Load-based Dormancy**, check the **Latest Exit Cause** and **Latest Disabled Cause** parameter values to determine the reason why this function does not take effect within the specified period. [Table 7-26](#) lists the possible reasons.

[Table 7-22](#) describes the meanings of **Latest Exit Cause**, **Latest Disabled Cause**, **Latest Exit Time**, and **Latest Disabled Time**.

Latest Exit Cause and **Latest Disabled Cause** indicate the reasons for the most recent exit and failure of the function. These reasons can be queried after the effective period has ended. For example, if the effective period is from 00:00:00 to 06:00:00, **Latest Exit Cause** and **Latest Disabled Cause** record the reasons for the most recent exit and failure of the function from 00:00:00 to 06:00:00. These reasons can be queried by an operation performed within the time from 06:00:00 to 23:59:59.

Table 7-26 Possible reasons why LNR load-based RF module deep dormancy or multi-frequency flexible dormancy does not take effect

Cause	Description
Cell in Base Station Not Bound to BBP	None
Switch Turned Off	The function switch for the NR cell is turned off.
RF Group Configuration Condition Not Met	The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD . Table 7-21 lists the possible reasons.
Cell Unavailable	The NR cell is blocked or deactivated, fails to be set up, or is unavailable due to other reasons.
Protection Due to Ongoing Maintenance and Test	• Service-related maintenance and test operations are being performed. • Protection will be canceled within 30 minutes after service-related maintenance and test operations are completed. Dormancy will be allowed after the protection is canceled.
Reliability Requirements Not Met	Alarms that affect the dormancy function exist, such as the High-Risk Function Enable alarm.
Triggering Time Not Reached	The current time falls outside the specified effective period for the function.

Cause	Description
Ultra-Long Holdover in Progress	None
Impacted Feature Taken Effect	None
Neighboring Cell Not Configured	• In SA networking, the neighbor relationships with LTE cells are not configured for the NR cell, the overlap indicator is not configured for the neighboring LTE cells of the NR cell, or the NR cell fails to obtain information about its neighboring LTE cells. • In SA networking: – The PLMN IDs of the normal operators of the NR cell are not a subset of the PLMN IDs of normal operators of neighboring LTE cells. – The PLMN IDs of the reserved operators of the NR cell are not a subset of the PLMN IDs of normal or reserved operators of neighboring LTE cells.
Obtaining NCell Configurations	• In NSA networking or NSA and SA hybrid networking, the neighbor relationships with LTE cells are not configured for the NR cell, the overlap indicator is not configured for the neighboring LTE cells of the NR cell, the X2 link is faulty, or the NR cell fails to obtain information about its neighboring LTE cells. • In SA networking, the system is subscribing to neighboring LTE cell information. • In NSA networking or NSA and SA hybrid networking: – The PLMN IDs of the normal operators of the NR cell are not a subset of the PLMN IDs of normal operators of neighboring LTE cells. – The PLMN IDs of the reserved operators of the NR cell are not a subset of the PLMN IDs of normal or reserved operators of neighboring LTE cells.
In Penalty State	• A cell enters the penalty state if it exits deep dormancy within the specified effective period. After the penalty ends, it can re-enter deep dormancy. The penalty duration is specified by the NRDUCellPowerSaving.PwrSavingPenaltyTime parameter. • A cell enters the penalty state when some UEs failed to be transferred out of the cell. The penalty lasts 30 minutes.

Cause	Description
Neighboring Cell Unavailable	- The function switches of the neighboring LTE cells are not turned on, the licenses for the neighboring LTE cells are insufficient, the neighboring LTE cells are in barred state or not activated, no neighboring LTE cell is an anchor cell, or the LTE cells are unavailable due to other reasons. - If NSA anchor carrier shutdown is enabled in neighboring LTE cells, it is recommended that the INTER_RAT_GRAD_PWR_SAVING_SW option of the gNBpwrSavingAlgo.PwrSavingAlgoSwitch parameter be selected for the NR cell. This will prevent the NR cell from exiting LNR load-based RF module deep dormancy in this case. For details about NSA anchor carrier shutdown, see <i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>. - LTE sector split cells are configured as basic-layer cells, and the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter is deselected.
UE Number Threshold Not Reached	- The downlink PRB usage in the NR cell has reached the trigger threshold, but the number of UEs remains below the trigger threshold. - The downlink PRB usage and the number of UEs in the NR cell have not reached their respective trigger thresholds.
Downlink PRB Threshold Not Reached	The number of UEs in the NR cell has reached the trigger threshold, but the downlink PRB usage remains below the trigger threshold.
Uplink PRB Threshold Not Reached	The number of UEs in the NR cell has reached the trigger threshold, but the uplink PRB usage remains below the trigger threshold.
Neighboring Cell Downlink PRB Exceeded Threshold	The current downlink PRB usage of neighboring cells exceeds the configured downlink PRB usage threshold. That is, the proportion of neighboring LTE cells meeting the PRB usage threshold does not meet the requirement specified by the NRDUCellPowerSaving.EutranCarrShutdownRatio parameter.

Cause	Description
Neighboring Cell Uplink PRB Exceeded Threshold	The current uplink PRB usage of neighboring cells exceeds the configured uplink PRB usage threshold. That is, the proportion of neighboring LTE cells meeting the PRB usage threshold does not meet the requirement specified by the NRDUCellPowerSaving.EutranCarrShutdownRatio parameter.
NR UEs in NCell Exceeding Threshold	The number of NR UEs in neighboring LTE cells of an NR cell exceeds the threshold. That is, the proportion of neighboring LTE cells meeting the NR UE number threshold does not meet the requirement specified by the NRDUCellPowerSaving.EutranCarrShutdownRatio parameter.
Neighboring Cell NR UE Downlink PRB Exceeded Threshold	The current downlink PRB usage of NR UEs exceeds the configured downlink PRB threshold. That is, the proportion of neighboring LTE cells meeting the NR UE downlink PRB usage threshold does not meet the requirement specified by the NRDUCellPowerSaving.EutranCarrShutdownRatio parameter.
Neighboring Cell NR UE Uplink PRB Exceeded Threshold	The current uplink PRB usage of NR UEs exceeds the configured uplink PRB usage threshold. That is, the proportion of neighboring LTE cells meeting the NR UE uplink PRB usage threshold does not meet the requirement specified by the NRDUCellPowerSaving.EutranCarrShutdownRatio parameter.
Triggering Decision Period Not Met	The current time falls within the check period specified by the NRDUCellPowerSaving.ShutdownTrigJudgePeriod parameter, but the thresholds for function triggering have not been met.
UE Being Transferred or Transfer Failed	None
Multi-Carrier Not Coordinated	Not all the activated cells that share the same RF module meet the conditions for entering LNR load-based RF module deep dormancy.
RF Dormancy Basic Capb Condition Not Met	The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD . Table 7-21 lists the possible reasons.

Cause	Description
Others	- In the current period, the conditions for this function to take effect are met, but the conditions for NR load-based RF module deep dormancy are not met. - The current time falls within the effective period specified for the LNR load-based RF module deep dormancy function, but the gNodeB has not obtained data such as the PRB usage and UE number. - The local NR cell meets the dormancy entry conditions but does not receive an instruction to enter dormancy. The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD. Table 7-21 lists the possible reasons. - Other reasons, such as internal software issues, cause the function not to take effect. - The configurations are incorrect. For example, in multi-operator sharing scenarios, the neighbor relationships with LTE cells served by the LTE primary operators are not configured for the NR cell.

- You can also run the **DSP CELLPWRSAVINGPOLICY** command to query the status of LTE load-based RF module deep dormancy and multi-frequency flexible dormancy and the reasons why these functions do not take effect. The values displayed in the query results are periodically refreshed and may not always reflect the most recent updates.
 - If the value of **Power Saving Type** is **INTRA_LTE_DORMANCY_BY_LOAD** or **LTE_PRECISE_DORMANCY_BY_LOAD** and the value of **Current State** is **ENABLED**, this function has taken effect.
 - If the value of **Power Saving Type** is **INTRA_LTE_DORMANCY_BY_LOAD** or **LTE_PRECISE_DORMANCY_BY_LOAD** and the value of **Current State** is other than **ENABLED**, check [Table 7-27](#) and [Table 7-28](#) for the possible reasons.
 - If the value of **Power Saving Type** is **INTRA_LTE_DORMANCY_BY_LOAD** or **LTE_PRECISE_DORMANCY_BY_LOAD**, determine the possible reason why this function does not take effect within the specified period based on the values of **Latest Exit Cause** and **Latest Disabled Cause**, as described in [Table 7-27](#) and [Table 7-28](#).

[Table 7-22](#) describes the meanings of **Latest Exit Cause**, **Latest Disabled Cause**, **Latest Exit Time**, and **Latest Disabled Time**.

Latest Exit Cause and **Latest Disabled Cause** indicate the reasons for the most recent exit and failure of the function. These reasons can be queried after the effective period has ended. For example, if the effective period is from 00:00:00 to 06:00:00, **Latest Exit Cause** and **Latest Disabled Cause** record the reasons for the most recent exit and failure of the function from 00:00:00 to 06:00:00. These reasons can be queried by an operation performed within the time from 06:00:00 to 23:59:59.

Table 7-27 Possible reasons why LTE load-based deep dormancy or multi-frequency flexible dormancy does not take effect

Cause	Description
CELL_IN_BS_NOT_BOUND_TO_BBP	None
SWITCH_OFF	None
RF_GROUP_CFG_COND_NOT_MET	The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD . Table 7-21 lists the possible reasons.
CELL_UNAVAILABLE	The cell is blocked or deactivated, fails to be set up, or is unavailable due to other reasons.
IN_MAINTENANCE_TEST_PROTECTION	- Service-related maintenance and test operations are being performed. - Protection will be canceled within 30 minutes after service-related maintenance and test operations are completed. Dormancy will be allowed after the protection is canceled.
RELIABILITY_NOT_MET	Alarms that affect the dormancy function exist, such as the High-Risk Function Enable alarm.
NOT_IN_SHUTDOWN_TIME	The current time falls outside the specified energy-saving effective period.
IN_PENALTY_STATE	- A cell enters the penalty state after exiting dormancy within the specified effective period. It can re-enter dormancy once the penalty state ends. The penalty time is specified by the CellPwrSavingPolicy.PenaltyTime parameter. - A cell enters the penalty state when some UEs fail to be transferred out of the cell. The penalty time is specified by the CellPwrSavingPolicy.PenaltyTime parameter.
HIGH_PRIORITY_FEATURE_TAKE_EFFECT	None

Cause	Description
NEIGHBORING_CELL_NOT_CONFIG	- No inter-frequency neighboring cells are configured for the current capacity-layer cell. - The current capacity-layer cell has been configured with inter-frequency neighboring cells, but the number of neighboring cells with the POWER_SAVING_OVERLAP_IND option selected is 0. - In multi-operator sharing scenarios, the operators of the local cell must be a subset of the operators of its co-coverage neighboring basic cells. If optimization for cell PLMN reservation scenarios is enabled by selecting the CELL_PLMN_RSVD_OPT_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter: - If the operators of the local cell are non-reserved operators, the operators must be a subset of non-reserved operators of its co-coverage neighboring basic cells. - If the operators of the local cell are reserved operators, the operators must be a subset of non-reserved operators or reserved operators of its co-coverage neighboring basic cells.
X2_INTERFACE_UNAVAILABLE	The X2 interface is abnormal when the capacity-layer and basic-layer cells are provided by different base stations.
NEIGHBORING_CELL_UNAVAILABLE	- A neighboring cell for which POWER_SAVING_OVERLAP_IND is set to ON is unavailable. - LTE sector split cells are configured as basic-layer cells, and the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter is deselected.
TRIG_JUDGE_PERIOD_NOT_MET	The current time falls within the specified effective period, but the cell does not meet the dormancy entry conditions throughout the period specified by the CellPwrSavingPolicy.TriggerJudgePeriod parameter.
NOT_IN_UE_NUM_THRESHOLD	None
NOT_IN_DL_PRB_THRESHOLD	None
NOT_IN_UL_PRB_THRESHOLD	None

Cause	Description
MULTI_CARR_NOT_COORDINATED	- After the multi-RRU cell dormancy compatibility switch is turned on (by selecting the MULTI_RRU_CELL_DORM_COMPAT_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter), not all RF modules serving the same cell as the RF module meet the dormancy conditions. - All cells in a sector split group must be enabled with load-based dormancy and meet the trigger conditions.
RF_DORM_BASE_CAP_CONFIG_NOT_MEET	The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD . Table 7-21 lists the possible reasons.
OTHERS	- LTE cells meet the dormancy entry conditions but the local cell does not receive an instruction to enter dormancy. The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD. Table 7-21 lists the possible reasons. - Internal software reasons cause the function not to take effect.

Table 7-28 Possible reasons why LTE load-based precise dormancy does not take effect

Cause	Description
CELL_IN_BS_NOT_BOUND_TO_BBP	None
SWITCH_OFF	None
RF_GROUP_CONFIG_CONDITION_NOT_MET	The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD . Table 7-21 lists the possible reasons.
CELL_UNAVAILABLE	The cell is blocked or deactivated, fails to be set up, or is unavailable due to other reasons.
IN_MAINTENANCE_TEST_PROTECTION	- Service-related maintenance and test operations are being performed. - Protection will be canceled within 30 minutes after service-related maintenance and test operations are completed. Dormancy will be allowed after the protection is canceled.

Cause	Description
RELIABILITY_NOT_MET	Alarms that affect the dormancy function exist, such as the High-Risk Function Enable alarm.
NOT_IN_SHUTDOWN_TIME	The current time falls outside the specified energy-saving effective period.
IN_PENALTY_STATE	- A cell enters the penalty state after exiting dormancy within the specified effective period. It can re-enter dormancy once the penalty state ends. The penalty time is specified by the CellPwrSavingPolicy.PenaltyTime parameter. - A cell enters the penalty state when some UEs fail to be transferred out of the cell. The penalty time is specified by the CellPwrSavingPolicy.PenaltyTime parameter.
HIGH_PRIORITY_FEATURE_TAKEN_EFFECT	None
NEIGHBORING_CELL_NOT_CONFIG	- No inter-frequency neighboring cells are configured or obtained for the current capacity-layer cell. - The LTE_DORMANCY_BY_LOAD_PE_SW option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter is deselected for all inter-frequency neighboring cells. - The CellPwrSavingAlgo.DormancyCellType parameter is not set to BASIC_CELL. - The EutranInterFreqNCell.OverlapRate parameter is set to a value less than 20 for the inter-frequency neighboring cell. - In multi-operator sharing scenarios, the operators of the local cell must be a subset of the operators of its co-coverage neighboring basic cells. If optimization for cell PLMN reservation scenarios is enabled by selecting the CELL_PLMN_RSVD_OPT_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter: - If the operators of the local cell are non-reserved operators, the operators must be a subset of non-reserved operators of its co-coverage neighboring basic cells. - If the operators of the local cell are reserved operators, the operators must be a subset of non-reserved operators or reserved operators of its co-coverage neighboring basic cells.

Cause	Description
NEIGHBORING_CELL_UNAVAILABLE	- There are basic-layer and capacity-layer cells served by the same RF module. - The basic-layer cells are unavailable. - LTE sector split cells are configured as basic-layer cells, and the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter is deselected.
TRIG_JUDGE_PERIOD_NOT_MET	The current time falls within the specified effective period, but the cell does not meet the dormancy entry conditions throughout the period specified by the CellPwrSavingPolicy.TriggerJudgePeriod parameter.
NOT_IN_UE_NUM_THRESHOLD	None
NOT_IN_DL_PRB_THRESHOLD	None
NOT_IN_UL_PRB_THRESHOLD	None
BASIC_CELL_PRB_USAGE_EXCEED	The uplink or downlink PRB usage of a basic-layer cell exceeds the congestion threshold.
MULTI_CARR_NOT_COORDINATED	After the multi-RRU cell dormancy compatibility switch is turned on (by selecting the MULTI_RRU_CELL_DORM_COMPAT_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter), not all RF modules serving the same cell as the RF module meet the dormancy conditions.
RF_DORM_BASE_CAP_COVD_NOT_MEET	The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD . Table 7-21 lists the possible reasons.
OTHERS	- LTE cells meet the dormancy entry conditions but the local cell does not receive an instruction to enter dormancy. The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD. Table 7-21 lists the possible reasons. - Internal software reasons cause the function not to take effect.

- In the case of LTE load-based RF module deep dormancy, use the following methods to query the cell's operator reservation status:
 - Run the **DSP EUTRANEXTERNALCELL** command to query the external E-UTRAN cell's operator reservation status.
 - If the value of **Cell Reserved for Operator Use** is **NOT_RESERVED**, the cell is not reserved for the operator.
 - If the value of **Cell Reserved for Operator Use** is **RESERVED**, the cell is reserved for the operator.
 - Run the **DSP CELLCNOPERATORSTATUS** command to query the intra-eNodeB cell's operator reservation status.
 - If the value of **Cell Operator Reserved State** is **NULL**, the cell is not activated or the PLMN ID of the operator is not on the list broadcast in SIB1 of the cell.
 - If the value of **Cell Operator Reserved State** is **Not Reserved**, the cell is not reserved for the operator.
 - If the value of **Cell Operator Reserved State** is neither **NULL** nor **Not Reserved**, the cell is reserved for the operator.

7.2.4.3 Network Monitoring

On the live network, the one-week energy consumption data obtained before and after the energy saving feature is enabled can be compared to evaluate the energy saving gains. It is expected that the energy consumption decreases significantly after this feature is enabled. The specific data and observation methods are as follows:

- Energy consumption of the RF module. Check the **VS.EnergyCons.BTSBoard (LTE eNodeB, 5G gNodeB)** counter on the MAE-Access.
- Unavailability duration of the NR cell in the energy saving state. Check the **N.Cell.Unavail.Dur.EnergySaving** counter on the MAE-Access.
- Duration of the NR cell in the dormancy state. Check the **N.Cell.Unavail.Dur.RFModuleDeepDormancy** counter on the MAE-Access.
- Unavailability duration of the LTE cell in the energy saving state. Check the **L.Cell.Unavail.Dur.EnergySaving** counter on the MAE-Access.
- Duration of the LTE cell in the dormancy state. Check the **L.Cell.Unavail.Dur.RFModuleDeepDormancy** counter on the MAE-Access.
- Duration of the UMTS cell in the dormancy state. Check the **VS.Cell.UnavailTime.RFModuleDormancy** counter on the MAE-Access.
- Duration of the RF module in the deep dormancy state. Check the **VS.RADIOEQM.DeepDormancy.Dur (5G gNodeB, LTE eNodeB)** counter on the MAE-Access.
- Check the **VS.RADIOEQM.MultiFreqFlexDormancy.Dur (5G gNodeB, LTE eNodeB)** counter (which indicates the RF Unit Multi-Freq Flexible Dormancy Duration) on the MAE-Access to observe the duration in which an RF module is in the multi-frequency flexible dormancy state.

 NOTE

- There is delay in making decisions on whether to enter or exit the energy saving state. Therefore, the duration measured by the performance counters has an approximately 1-minute deviation from the actual duration.
- The sampling period is not synchronous with the reporting period, so a deviation of one sampling period may occur.
- If multiple RATs share an RF module through SDR, it is recommended that the power consumption be checked using module-level counters.
- For UMTS RF module deep dormancy, the RNC and NodeB must cooperate to measure the **VS.Cell.UnavailTime.RFModuleDormancy** counter. If the RNC becomes disconnected from the NodeB, the measured duration may deviate from the actual value by approximately 1 to 5 minutes.

7.3 RF Module Fast Deep Dormancy

7.3.1 Principles

It takes minutes for an RF module in traditional deep dormancy to wake up, making it more suitable for activation at night. To enhance energy savings during the daytime, the RF module fast deep dormancy function, which enables rapid wakeup from deep dormancy, has been proposed.

This function enables RF modules to enter dormancy to reduce power consumption when the network is under light or no load, while maintaining RF module reliability. When the network load is high, cells can be quickly woken without reestablishment, minimizing impact on user experience while saving energy.

7.1.2 Differences in Dormancy Triggering Factors lists the sub-functions of RF module fast deep dormancy by control approach.

In addition to the preceding sub-functions, RF module fast deep dormancy provides the following functions:

- Dormancy pending and pending cancellation
If operations on the base station (such as upgrades) require all RF modules to remain active, dormancy pending can be enabled. For details, see **7.10 Dormancy Pending and Pending Cancellation**.
- Reliability protection during dormancy
The reliability protection involves license reliability protection, configuration reliability protection, reliability protection against faults, and O&M reliability protection during dormancy. For details, see **7.12 Reliability Protection During Fast Deep Dormancy**.

7.3.1.1 NR Load-based RF Module Fast Deep Dormancy

In addition to providing greater network capacity, NR multi-carrier co-coverage networking delivers higher data throughput on a heavily-loaded network. However, it wastes base station energy when the network is under light or no load. To address this, NR load-based RF module fast deep dormancy is introduced. The RF module shutdown measures are optimized to allow cells to exit dormancy without reestablishment, enabling faster wakeup.

NR load-based RF module fast deep dormancy enables the gNodeB to transfer UEs from capacity-layer cells to basic-layer cells and instruct the RF modules serving the capacity-layer cells to enter fast deep dormancy within a specified period when the network is under light or no load. When the load of the basic-layer cells increases to a specified threshold, the dormant RF modules are automatically woken and the capacity-layer cells are restored.

This function requires the following switches to be configured:

- RF module-level switch
FAST_DEEP_DORM_SW of the **RRUEXTINFO.RRUDOROPTFEATSW (5G gNodeB, LTE eNodeB)** parameter is selected.
- NR cell-level switch
NR_FAST_DORMANCY_BY_LOAD_SW of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter is selected.

The effective period and threshold parameters for NR load-based RF module fast deep dormancy can be uniquely determined by using the setting combinations of the **NRDUCellPowerSaving.PowerSavingPolicyIndex** and **NRDUCellPowerSaving.PowerSavingType** parameters in the **NRDUCellPowerSaving** MOs on the NR side. Where,

- The **NRDUCellPowerSaving.PowerSavingPolicyIndex** parameter specifies an energy saving policy index. It uniquely identifies an energy saving policy within the local cell. With the index specified, the corresponding policy can be added, modified, queried, or removed at the local cell level.
- The **NRDUCellPowerSaving.PowerSavingType** parameter specifies the energy saving type indicated by an energy saving policy index. NR load-based RF module fast deep dormancy requires this parameter to be set to **NR_FAST_DORMANCY_BY_LOAD**. Then under the corresponding energy saving policy index, the local cell performs load-based fast deep dormancy during the specified period when certain load conditions are met.

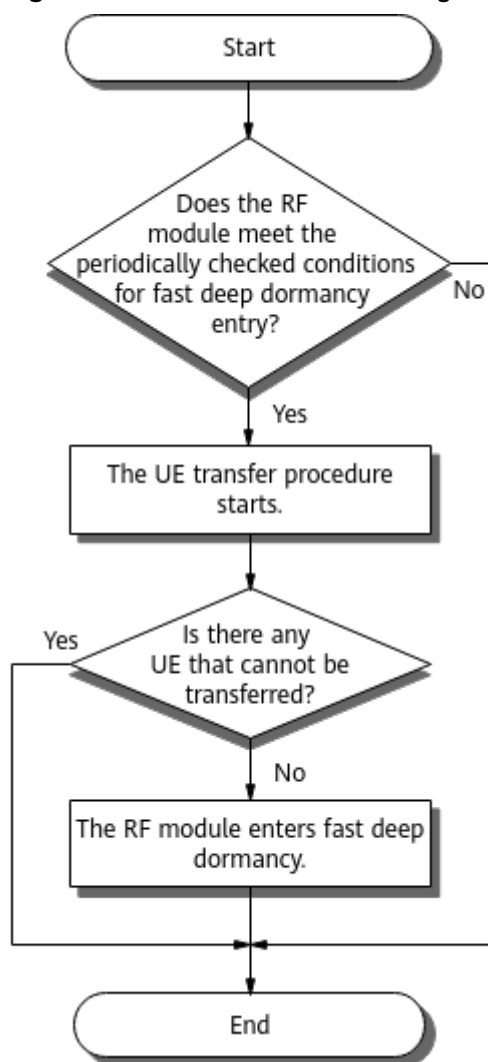
Related Concepts

For the concepts of basic-layer carriers, capacity-layer carriers, basic-layer cells, and capacity-layer cells, refer to [Related Concepts](#).

Entering Fast Deep Dormancy

Figure 7-14 shows how the RF module serving a capacity-layer cell enters fast deep dormancy after NR load-based RF module fast deep dormancy is enabled.

Figure 7-14 Procedure for entering NR load-based RF module fast deep dormancy



1. The base station determines whether the entry conditions are met.
The base station periodically checks whether the dormancy entry conditions are met. The evaluation period granularity is specified by the **NRDUCellPowerSaving.EvalPeriodGranularity** parameter.

- If this parameter is set to **MINUTE_LEVEL**, the evaluation period is expressed in minutes. In this case, the period is specified by the **NRDUCellPowerSaving.ShutdownTrigJudgePeriod** parameter.
- If this parameter is set to **SECOND_LEVEL**, the evaluation period is expressed in seconds. In this case, the period is specified by the **NRDUCellPowerSaving.SecondLevelTrigEvalPrd** parameter.

A capacity-layer cell must meet all of the following conditions:

- The current time falls within the effective energy saving period determined by the **NRDUCellPowerSaving.StartTime** and **NRDUCellPowerSaving.StopTime** parameters.
- The capacity-layer cell is not in an energy saving penalty state.

The basic-layer cells associated with the capacity-layer cell must meet one of the following conditions:

- All basic-layer carriers are available.
- Some basic-layer carriers are available, while others, functioning as capacity-layer carriers of other basic-layer carriers, are in the energy saving state due to NR intelligent carrier shutdown, LTE and NR intelligent carrier shutdown, NR load-based RF module dormancy, or LNR load-based RF module dormancy.

The capacity-layer cell and its associated basic-layer cells must meet all the following conditions:

- The total downlink PRB usage of the capacity-layer and basic-layer carriers is less than or equal to the downlink PRB usage threshold specified by the **NRDUCellPowerSaving.DlPrbThld** parameter. Where,
$$\text{Total downlink PRB usage of the capacity-layer and basic-layer carriers} = (\text{Number of PRBs used by PDSCH DRBs on the capacity-layer carrier} + \text{Number of downlink PRBs used on the basic-layer carrier}) / \text{Number of available downlink PRBs on the basic-layer carrier}$$

If multiple basic-layer carriers are associated with one capacity-layer carrier, the sum of PRBs of these basic-layer carriers is used in the preceding formula.
- The total uplink PRB usage of the capacity-layer and basic-layer carriers is less than or equal to the uplink PRB usage threshold specified by the **NRDUCellPowerSaving.UlPrbThld** parameter. Where,
$$\text{Total uplink PRB usage of the capacity-layer and basic-layer carriers} = (\text{Number of uplink PRBs used on the capacity-layer carrier} + \text{Number of uplink PRBs used on the basic-layer carriers}) / \text{Number of available uplink PRBs on the basic-layer carriers}$$
- The total number of RRC_CONNECTED UEs on the capacity-layer carrier and a basic-layer carrier is less than or equal to the UE number threshold specified by the **NRDUCellPowerSaving.UserNumThld** parameter.

If multiple basic-layer carriers are associated with one capacity-layer carrier, each pair of the basic-layer and capacity-layer carriers must meet this condition.
- In multi-operator sharing scenarios, operators are divided into normal operators and reserved operators (with **NRExternalNCellPlmn.PlmnReservedFlag** set to **TRUE**) for the local cell and its co-coverage neighboring basic cells.
 - If the operators of the local cell are normal operators, the operators must be a subset of normal operators of its co-coverage neighboring basic cells.
 - If the operators of the local cell are reserved operators, the operators must be a subset of normal or reserved operators of its co-coverage neighboring basic cells.
- In SUL co-deployment scenarios, the total number of SUL UEs in the capacity-layer and basic-layer cells is less than or equal to the value of the **NRDUCellPowerSaving.SulUserNumThld** parameter. If the capacity-layer cell is associated with multiple basic-layer cells, each pair of the basic-layer and capacity-layer cells must meet this condition.
- The Xn interface is normal in multi-carrier inter-base-station scenarios.

 NOTE

When evaluating the total uplink or downlink PRB usage of capacity-layer and basic-layer cells against the entry threshold, the base station performs conversion on the cell bandwidth, subcarrier spacing, and slot configuration.

2. A UE transfer procedure starts. The gNodeB sends system information to prohibit UE access and handovers to the capacity-layer cell and starts the UE transfer procedure with a specified duration.

In SA networking, UEs are transferred from the local cell to basic-layer neighboring cells as follows:

- Non-CA UEs: UEs not performing voice services are transferred to neighboring cells through inter-frequency handovers first. If the handovers fail, the UEs are transferred to inter-frequency neighboring cells through blind redirections. UEs performing voice services are handed over to inter-frequency neighboring cells.
- CA UEs: If the local cell serves as an SCell of a CA UE, the SCell is removed. If the local cell serves as the PCell of a CA UE, the UE is transferred to a neighboring cell in the same way as a non-CA UE.

If UEs cannot be transferred out of the capacity-layer cell within 30 seconds after the UE transfer procedure starts (for example, due to handover failures or unsupported operating bands of the basic-layer cells), the procedure stops and cannot be re-triggered for the capacity-layer cell within 30 minutes.

In NSA networking, UE connections on the NR side are released as follows:

- If energy saving takes effect in the PSCell of a UE, all connections on the NR side are released.
- If energy saving takes effect in an SCell of a UE, its connection to this SCell on the NR side is released but its other connections on the NR side are retained.

3. The RF module enters fast deep dormancy.

Once all UEs are transferred out of the capacity-layer cell, no RRC_CONNECTED UEs remain, and the cell and its serving RF module enter fast deep dormancy. If the RF module serves multiple NR cells, it enters fast deep dormancy only when load-based fast deep dormancy is triggered for all these cells. If the RF module also serves LTE cells, RF module fast deep dormancy must also be enabled on the LTE side.

 NOTE

When NR load-based RF module fast deep dormancy and LNR load-based RF module fast deep dormancy are both enabled, cells enter dormancy only if the entry conditions of both functions are met.

Exiting Fast Deep Dormancy

The base station periodically checks whether the dormancy exit conditions are met. The evaluation period granularity is specified by the **NRDUCellPowerSaving.EvalPeriodGranularity** parameter.

- If this parameter is set to **MINUTE_LEVEL**, the evaluation period is expressed in minutes. In this case, the period is specified by the **NRDUCellPowerSaving.ExitEvaluationPeriod** parameter.

- If this parameter is set to **SECOND_LEVEL**, the evaluation period is expressed in seconds. In this case, the period is specified by the **NRDUCellPowerSaving.SecondLevelExitEvalPrd** parameter.

A capacity-layer cell and its serving RF modules automatically exit dormancy when any of the following conditions is met:

- The current time falls outside the effective energy saving period determined by the **NRDUCellPowerSaving.StartTime** and **NRDUCellPowerSaving.StopTime** parameters.
- The total downlink PRB usage of basic-layer carriers is greater than the sum of the **NRDUCellPowerSaving.DlPrbThld** and **NRDUCellPowerSaving.DlPrbOffset** parameter values.
- The total uplink PRB usage of basic-layer carriers is greater than the sum of the **NRDUCellPowerSaving.UlPrbThld** and **NRDUCellPowerSaving.NCellUlPrbOffset** parameter values.
- The number of RRC_CONNECTED UEs on any basic-layer carrier is greater than the sum of the **NRDUCellPowerSaving.UserNumThld** and **NRDUCellPowerSaving.UserNumOffset** parameter values.
- A basic-layer neighboring cell is added for the capacity-layer cell.
- In multi-operator sharing scenarios:
 - The PLMN IDs of the normal operators of the capacity-layer cell are not a subset of the PLMN IDs of normal operators of basic-layer cells.
 - The PLMN IDs of the reserved operators of the capacity-layer cell are not a subset of the PLMN IDs of normal or reserved operators of basic-layer cells.
- In SUL co-deployment scenarios, the number of SUL UEs in a basic-layer cell is greater than the sum of the **NRDUCellPowerSaving.SulUserNumThld** and **NRDUCellPowerSaving.SulUserNumOffset** parameter values. If the capacity-layer cell is associated with multiple basic-layer cells, the number of SUL UEs in one of the basic-layer cells must meet the condition.
- The Xn interface is abnormal when basic-layer carriers are provided by different base stations.
- Any of the basic-layer carriers becomes unavailable for a reason other than NR intelligent carrier shutdown, LTE and NR intelligent carrier shutdown, and NR load-based RF module dormancy.

Upon the exit from fast deep dormancy:

- The capacity-layer cell cannot re-enter fast deep dormancy within the penalty period specified by the **NRDUCellPowerSaving.PwrSavingPenaltyTime** parameter if it exited dormancy due to uplink/downlink PRB usage or UE number exceptions or unavailability of the basic-layer cell, or Xn interface exceptions.
- A penalty is not imposed if the capacity-layer cell exited fast deep dormancy because the function switch was turned off, the effective period elapsed, or the base station software was upgraded.

NR Load-based RF Module Fast Deep Dormancy Assurance

To reduce the impact of NR load-based RF module fast deep dormancy on user-perceived rates, RF module fast deep dormancy assurance is introduced. This

function includes two sub-functions: NR load-based RF module fast deep dormancy service assurance and NR load-based RF module fast deep dormancy load assurance. After NR load-based RF module fast deep dormancy is enabled, the following two sub-functions can be enabled.

- NR load-based RF module fast deep dormancy service assurance (by selecting the **NR_FAST_DORM_SERVICE_ASSUR_SW** option of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter for capacity-layer cells and the **PWR_SAVE_SERVICE_ASSUR_SW** option of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter for basic-layer cells). Once enabled, the cell determines whether to exit dormancy for user experience assurance at an interval of 5s as follows:
 - If the downlink PRB usage of each basic-layer cell is less than the PRB usage threshold for fast evaluation on carrier shutdown (specified by the **NRDUCellMobility.CarrShutdFastEvalPrbThld** parameter) and any basic-layer cell serves UEs with heavy downlink traffic, the capacity-layer cell exits fast deep dormancy.
 - If the downlink PRB usage of any basic-layer cell is greater than or equal to the PRB usage threshold for fast evaluation on carrier shutdown (specified by the **NRDUCellMobility.CarrShutdFastEvalPrbThld** parameter), the capacity-layer cell exits fast deep dormancy based on the setting of the **NRDUCellPowerSaving.PwrSaveServiceAssurLevel** parameter. A larger value of this parameter results in a smaller impact on user experience and fewer energy saving gains.
- NR load-based RF module fast deep dormancy load assurance (by selecting the **NR_FAST_DORM_LOAD_ASSUR_SW** option of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter for capacity-layer cells). Once enabled, the downlink PRB usage of each basic-layer cell is evaluated per second. If the downlink PRB usage of any basic-layer cell is greater than the threshold specified by the **NRDUCellMobility.CarrShdnEventEvalPrbThld** parameter in the current period but does not exceed this threshold in the previous period, the capacity-layer cell exits fast deep dormancy. NR load-based RF module fast deep dormancy load assurance takes effect only when the **NRDUCellMobility.CarrShdnEventEvalPrbThld** parameter is set to a value other than 255. A smaller value of this parameter results in a smaller impact on user experience and fewer energy saving gains.

After the capacity-layer cell exits fast deep dormancy, a procedure is immediately triggered for each UE with heavy downlink traffic in the basic-layer cells to evaluate whether a frequency-priority-based handover can be performed, as long as this type of handover is enabled in the basic-layer cells. This enables such UEs to be transferred back to the capacity-layer cell promptly. The time-to-trigger in this procedure is specified by the **NRCCellMobilityConfig.CarrShutUserMeasTimeToTrig** parameter. Frequency-priority-based handover will not be triggered in a basic-layer cell with this type of handover disabled.

For details about frequency-priority-based handover, see the section on frequency-priority-based inter-frequency PSCell change in *NSA Mobility Management* and the section on frequency-priority-based inter-frequency handover in *SA Mobility Management in Connected Mode* in *5G RAN Feature Documentation*.

7.3.1.2 LNR Load-based RF Module Fast Deep Dormancy

NR cells in operation continue to consume energy even under light or no traffic in areas covered by both LTE and NR, resulting in unnecessary base station energy consumption. To address this, LNR load-based RF module fast deep dormancy is introduced. By optimizing RF module shutdown measures, cells can exit dormancy without reestablishment, enabling faster wakeup.

LNR load-based RF module fast deep dormancy enables the gNodeB to transfer UEs from NR cells to LTE cells and instruct the RF modules serving the NR cells to enter fast deep dormancy within a specified period when the network is under light or no load. When the load of the LTE cells increases to a specified threshold, the dormant RF modules are automatically woken and the NR cells are restored.

This function requires the following switches to be configured:

- RF module-level switch
FAST_DEEP_DORM_SW of the **RRUEXTINFO.RRUDOROPTFEATSW (5G gNodeB, LTE eNodeB)** parameter is selected.
- LTE cell-level switch
LNR_LOAD_BASED_DORMANCY_SW of the **CellPwrSavingAlgo.PwrSavingAlgoSwitch** parameter is selected.
- NR cell-level switch
LNR_FAST_DORMANCY_BY_LOAD_SW of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter is selected.

The effective period for LNR load-based RF module fast deep dormancy can be uniquely determined by using the setting combinations of the **NRDUCellPowerSaving.PowerSavingPolicyIndex** and **NRDUCellPowerSaving.PowerSavingType** parameters in the **NRDUCellPowerSaving** MOs on the NR side. Where,

- The **NRDUCellPowerSaving.PowerSavingPolicyIndex** parameter specifies an energy saving policy index. It uniquely identifies an energy saving policy within the local cell. With the index specified, the corresponding policy can be added, modified, queried, or removed at the local cell level.
- The **NRDUCellPowerSaving.PowerSavingType** parameter specifies the energy saving type indicated by an energy saving policy index. LNR load-based RF module fast deep dormancy requires that this parameter be set to **LNR_FAST_DORMANCY_BY_LOAD**. Then under the corresponding energy saving policy index, the local cell performs LNR load-based RF module fast deep dormancy during the specified period when certain load conditions are met.

Identifying Overlapping Neighboring Cells

The mechanism of identifying overlapping neighboring cells for LNR load-based RF module fast deep dormancy is the same as that for LTE and NR intelligent carrier shutdown. For details, see [6.1.1.1 Identifying Overlapping Neighboring Cells](#).

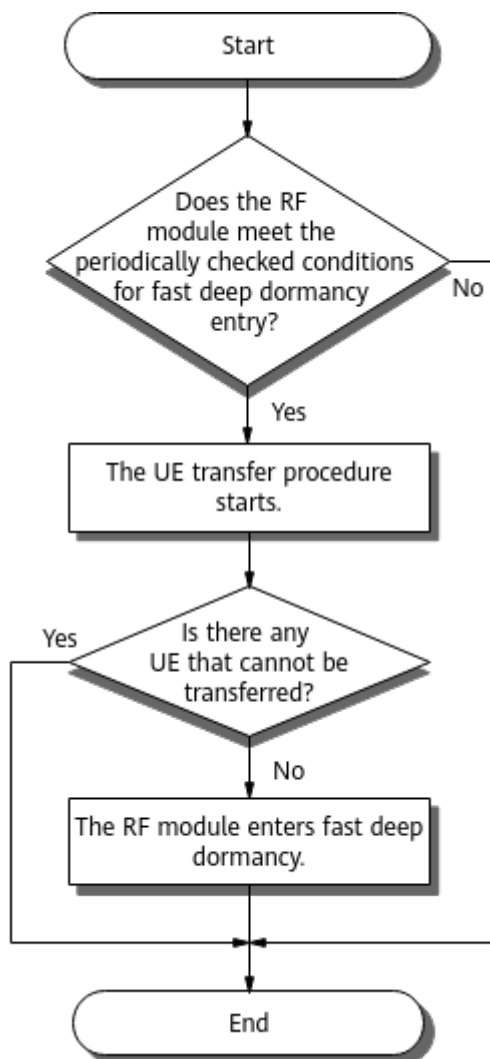
When LNR load-based RF module fast deep dormancy is enabled, NR cells act as capacity-layer cells and LTE cells act as the corresponding basic-layer cells. After an NR cell enters the energy saving state, its overlapping neighboring LTE cells

provide coverage. In this case, the capacity-layer and basic-layer cells cannot share the same RF module.

Entering Fast Deep Dormancy

Figure 7-15 shows how the RF module serving an NR cell enters fast deep dormancy after LNR load-based RF module fast deep dormancy is enabled.

Figure 7-15 Procedure for entering LNR load-based RF module fast deep dormancy



1. The base station determines whether the entry conditions are met.
The current time falls within the effective energy saving period determined by the **NRDUCellPowerSaving.StartTime** and **NRDUCellPowerSaving.StopTime** parameters, and the NR cell is not in an energy saving penalty state.
The base station periodically checks whether the dormancy entry conditions are met. The evaluation period granularity is specified by the **NRDUCellPowerSaving.EvalPeriodGranularity** parameter.

- If this parameter is set to **MINUTE_LEVEL**, the evaluation period is expressed in minutes. In this case, the period is specified by the **NRDUCellPowerSaving.ShutdownTrigJudgePeriod** parameter.
- If this parameter is set to **SECOND_LEVEL**, the evaluation period is expressed in seconds. In this case, the period is specified by the **NRDUCellPowerSaving.SecondLevelTrigEvalPrd** parameter.

The NR cell must meet all of the following conditions:

- Average number of active UEs in the NR cell $\leq$ UE number threshold (specified by the **NRDUCellPowerSaving.UserNumThld** parameter)
- Downlink PRB usage of the NR cell $\leq$ Downlink PRB usage threshold (specified by the **NRDUCellPowerSaving.DlPrbThld** parameter)
- Uplink PRB usage of the NR cell $\leq$ Uplink PRB usage threshold (specified by the **NRDUCellPowerSaving.UlPrbThld** parameter)

 NOTE

- If the NR cell is a DSS cell, the PRB usage equals the number of used RBs divided by the number of available RBs.
- Active UEs are those in RRC_CONNECTED mode in an NR cell.
- Downlink PRB usage of an NR cell = Average number of PRBs used by PDSCH DRBs/Average number of available downlink PRBs
- Uplink PRB usage of an NR cell = Average number of PRBs used by PUSCH DRBs/Average number of available uplink PRBs
- In SUL co-deployment scenarios, the total number of SUL UEs in the NR cell is less than or equal to the value of the **NRDUCellPowerSaving.SulUserNumThld** parameter.
- In multi-operator sharing scenarios, operators are divided into normal operators and reserved operators for the NR cell and neighboring LTE cells.
 - The reserved operator of the NR cell is an operator with the **NRExternalNCellPlmn.PlmnReservedFlag** parameter set to **TRUE**.
 - The reserved operator of the LTE cell is an operator with the **CellOp.CellReservedForOp** parameter set to **CELL_RESERVED_FOR_OP** or **CELL_DYNAMIC_RESERVED_FOR_OP**.
- In multi-operator scenarios, primary operator recognition for LNR load-based dormancy is supported. This function is controlled by the **LNR_LOAD_DORM_PRIM_OP_RECOG_SW** option of the **gNBpwrSavingAlgo.PwrSavingAlgoSwitch** parameter.
 - If this option is deselected, the operator requirements are as follows:
 - The PLMN IDs of the normal operators of an NR cell must be a subset of the PLMN IDs of normal operators of neighboring LTE cells.
 - The PLMN IDs of the reserved operators of an NR cell must be a subset of the PLMN IDs of normal or reserved operators of neighboring LTE cells.
 - If this option is selected, the operator requirements are as follows:

- The PLMN ID of the primary operator of an NR cell served by normal operators must be obtained from the normal operators of neighboring LTE cells.
- The PLMN ID of the primary operator of an NR cell served by reserved operators must be obtained from the normal or reserved operators of neighboring LTE cells.
- The eNodeB sends the load information about the neighboring LTE cells of the NR cell to the gNodeB. The gNodeB requires that the carriers of neighboring cells be available.

If the **INTER_RAT_GRAD_PWR_SAVING_SW** option of the **gNBpwrSavingAlgo.PwrSavingAlgoSwitch** parameter is selected, some neighboring LTE cells can act as capacity-layer cells of other basic-layer carriers and enter the energy saving state due to functions such as intelligent power-off of carriers in the same coverage, Cell Switch-off Based on Automatic Co-coverage Identification, multi-carrier coordinated energy saving, GUL joint carrier shutdown, dynamic carrier shutdown, LTE load-based deep dormancy, or LTE load-based fast deep dormancy.

The gNodeB checks whether a proportion of available neighboring cells (the proportion is specified by the **NRDUCellPowerSaving.EutranCarrShutdownRatio** parameter) meet the following conditions:

- Downlink PRB usage of the neighboring LTE cell < E-UTRAN downlink PRB threshold (specified by the **NRDUCellPowerSaving.EutranDlPrbThld** parameter)
- Uplink PRB usage of the neighboring LTE cell < Neighboring cell uplink PRB usage threshold (specified by the **NRDUCellPowerSaving.NCellUlPrbThld** parameter)
- Number of NR UEs in the neighboring LTE cell < E-UTRAN NR user number threshold (specified by the **NRDUCellPowerSaving.EutranNrUserNumThld** parameter)
- Downlink PRB usage of NR UEs in the neighboring LTE cell < E-UTRAN NR user downlink PRB threshold (specified by the **NRDUCellPowerSaving.EutranNrUserDlPrbThld** parameter)
- Uplink PRB usage of NR UEs in the neighboring LTE cell < E-UTRAN NR user uplink PRB threshold (specified by the **NRDUCellPowerSaving.EutranNrUserUlPrbThld** parameter)

 NOTE

Downlink PRB usage of an LTE cell = Average number of used downlink PRBs/Number of available downlink PRBs. The number of RBs for common channels is included in the calculation of downlink PRB usage.

Uplink PRB usage of an LTE cell = Average number of used uplink PRBs/Number of available uplink PRBs

2. A UE transfer procedure starts. The gNodeB sends system information to prohibit UE access and handovers to the NR cell and starts the UE transfer procedure with a specified duration.

In SA networking, UEs are transferred from the local cell to basic-layer neighboring cells as follows:

- Non-CA UEs: UEs not performing voice services are handed over to inter-RAT neighboring cells. If the handovers fail, they are redirected to inter-

RAT neighboring cells through blind redirection. UEs performing voice services are handed over to inter-RAT neighboring cells.

- CA UEs: If the local cell serves as an SCell of a CA UE, the SCell is removed. If the local cell serves as the PCell of a CA UE, the UE is transferred to a neighboring cell in the same way as a non-CA UE.

If UEs cannot be transferred out of the NR cell within 30s after the UE transfer procedure starts (for example, due to handover failures or unsupported operating bands of the basic-layer cells), the procedure stops and cannot be re-triggered for the NR cell within the period specified by the **NRDUCellPowerSaving.PwrSavingPenaltyTime** parameter.

In NSA networking, UE connections on the NR side are released as follows:

- If energy saving takes effect in the PSCell of a UE, all connections on the NR side are released.
- If energy saving takes effect in an SCell of a UE, its connection to this SCell on the NR side is released but its other connections on the NR side are retained.

3. The RF module enters fast deep dormancy.

Once all UEs are transferred out of the NR cell, no RRC_CONNECTED UEs remain, and the cell and its serving RF modules enter fast deep dormancy. If the RF module serves multiple NR cells, it enters fast deep dormancy only when load-based fast deep dormancy is triggered for all these cells. If the RF module also serves LTE cells, RF module fast deep dormancy must also be enabled on the LTE side.

 NOTE

When NR load-based RF module fast deep dormancy and LNR load-based RF module fast deep dormancy are both enabled, cells enter dormancy only if the entry conditions of both functions are met.

Exiting Fast Deep Dormancy

The base station periodically checks whether the dormancy exit conditions are met. The evaluation period granularity is specified by the **NRDUCellPowerSaving.EvalPeriodGranularity** parameter.

- If this parameter is set to **MINUTE_LEVEL**, the evaluation period is expressed in minutes. In this case, the period is specified by the **NRDUCellPowerSaving.ExitEvaluationPeriod** parameter.
- If this parameter is set to **SECOND_LEVEL**, the evaluation period is expressed in seconds. In this case, the period is specified by the **NRDUCellPowerSaving.SecondLevelExitEvalPrd** parameter.

The dormancy state is exited and the NR carrier is activated when any of the following conditions is met:

- The current time falls outside the shutdown period.
- Proportion of neighboring LTE cells that meet one of the following conditions $\geq$ E-UTRAN carrier shutdown ratio (specified by the **NRDUCellPowerSaving.EutranCarrShutdownRatio** parameter) – E-UTRAN carrier shutdown exit offset (specified by the **NRDUCellPowerSaving.EutranCarrShutdownExitOfs** parameter)

- Downlink PRB usage of the neighboring LTE cell $\geq$ E-UTRAN downlink PRB threshold (specified by the **NRDUCellPowerSaving.EutranDlPrbThld** parameter) + E-UTRAN downlink PRB offset (specified by the **NRDUCellPowerSaving.EutranDlPrbOffset** parameter)
- Uplink PRB usage of the neighboring LTE cell $\geq$ Neighboring cell uplink PRB threshold (specified by the **NRDUCellPowerSaving.NCellUlPrbThld** parameter) + Neighboring cell uplink PRB offset (specified by the **NRDUCellPowerSaving.NCellUlPrbOffset** parameter)
- Number of NR UEs in the neighboring LTE cell $\geq$ E-UTRAN NR user number threshold (specified by the **NRDUCellPowerSaving.EutranNrUserNumThld** parameter) + E-UTRAN NR user number threshold offset (specified by the **NRDUCellPowerSaving.EutranNrUserNumThldOfs** parameter)
- Downlink PRB usage of NR UEs in the neighboring LTE cell $\geq$ E-UTRAN NR user downlink PRB threshold (specified by the **NRDUCellPowerSaving.EutranNrUserDlPrbThld** parameter) + E-UTRAN NR user downlink PRB threshold offset (specified by the **NRDUCellPowerSaving.EutranNrUserDlPrbThldOfs** parameter)
- Uplink PRB usage of NR UEs in the neighboring LTE cell $\geq$ E-UTRAN NR user uplink PRB threshold (specified by the **NRDUCellPowerSaving.EutranNrUserUlPrbThld** parameter) + E-UTRAN NR user uplink PRB threshold offset (specified by the **NRDUCellPowerSaving.EutranNrUserUlPrbThldOfs** parameter)
- The PLMN IDs of the normal operators of the NR cell are not a subset of the PLMN IDs of normal operators of neighboring LTE cells.
- The PLMN IDs of the reserved operators of the NR cell are not a subset of the PLMN IDs of normal or reserved operators of neighboring LTE cells.
- When a neighboring LTE cell functions as a capacity-layer cell of other basic-layer carriers and enters the energy saving state, whether it is considered as a neighboring cell that meets the exit conditions depends on the setting of the **INTER_RAT_GRAD_PWR_SAVING_SW** option of the **gNBPwrSavingAlgo.PwrSavingAlgoSwitch** parameter.
 - If this option is deselected, this LTE cell is considered as a cell that meets the exit conditions.
 - If this option is selected, this LTE cell is not considered as a cell that meets the exit conditions and does not trigger exit from fast deep dormancy.

Upon the exit from fast deep dormancy:

- The NR cell cannot re-enter fast deep dormancy within the penalty duration specified by the **NRDUCellPowerSaving.PwrSavingPenaltyTime** parameter if it exited dormancy due to uplink/downlink PRB usage or UE number exceptions, or unavailability of the LTE cell.
- A penalty is not imposed if the NR cell exited fast deep dormancy because the function switch was turned off, the effective period elapsed, or the base station software was upgraded.

 NOTE

- NR UEs in neighboring LTE cells of the NR cell are counted based on the UE capability (for example, whether a UE supports SA or NSA).
- In CA scenarios, only NR UEs on the primary component carrier (PCC) are counted.

7.3.1.3 LTE Load-based RF Module Fast Deep Dormancy

In addition to providing greater network capacity, LTE multi-carrier co-coverage networking delivers higher data throughput on a heavily-loaded network. However, it wastes base station energy when the network is under light or no load. To address this, LTE load-based RF module fast deep dormancy is introduced. By optimizing RF module shutdown measures, cells can exit dormancy without reestablishment, enabling faster wakeup.

LTE load-based RF module fast deep dormancy enables the eNodeB to transfer UEs from capacity-layer cells to basic-layer cells and instruct the RF modules serving the capacity-layer cells to enter fast deep dormancy within a specified period when the network is under light or no load. When the load of the basic-layer cells increases to a specified threshold, the RF modules in dormancy are automatically woken up and the capacity-layer cells are restored.

This function requires the following switches to be configured:

- RF module-level switch
Select **FAST_DEEP_DORM_SW** of the **RRUEXTINFO.RRUDOROPTFEATSW (5G gNodeB, LTE eNodeB)** parameter.
- LTE cell-level switch
Select **LTE_FAST_DORMANCY_BY_LOAD_SW** of the **CellPwrSavingAlgo.PwrSavingAlgoSwitch** parameter.

The effective period and threshold parameters for LTE load-based RF module fast deep dormancy can be uniquely determined by using the combinations of the **CellPwrSavingPolicy.PolicyIndex** and **CellPwrSavingPolicy.PowerSavingType** parameters in the **CellPwrSavingPolicy** MO on the LTE side. Where,

- The **CellPwrSavingPolicy.PolicyIndex** parameter specifies an energy saving policy index. It uniquely identifies an energy saving policy within the local cell. With the index specified, the corresponding policy can be added, modified, queried, or removed at the local cell level.
- The **CellPwrSavingPolicy.PowerSavingType** parameter specifies the energy saving type indicated by an energy saving policy index. LTE load-based RF module fast deep dormancy requires this parameter to be set to **LTE_FAST_DORMANCY_BY_LOAD**. Then under the corresponding energy saving policy index, the local cell performs load-based fast deep dormancy during the specified period when certain load conditions are met.

Related Concepts

- Basic-layer carriers provide basic coverage for the network. RF modules supporting these carriers do not enter dormancy. Cells served by basic-layer carriers are referred to as basic-layer cells.
- Capacity-layer carriers handle high traffic volumes during peak hours. RF modules supporting these carriers may enter dormancy. Cells served by capacity-layer carriers are referred to as capacity-layer cells.

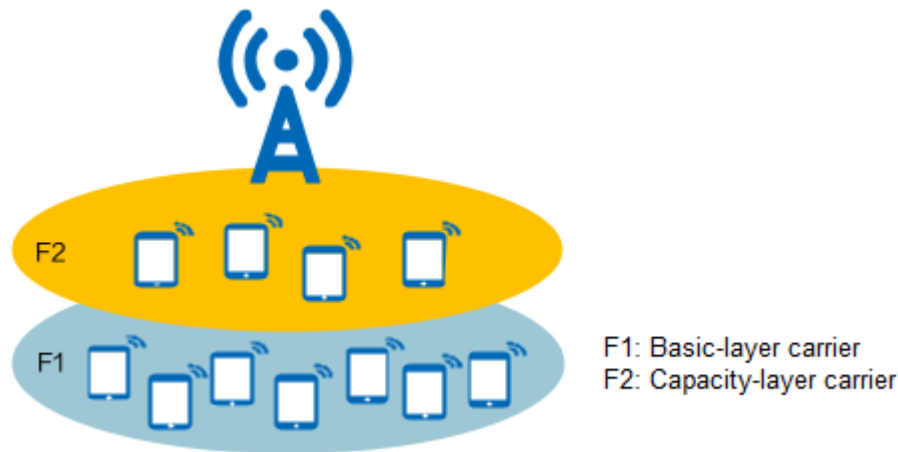
 NOTE

Load-based RF module fast deep dormancy is enabled for capacity-layer cells and their associated RF modules. A capacity-layer cell can be associated with one or more basic-layer cells. Basic-layer cells and capacity-layer cells must be configured as inter-frequency neighbors of each other.

Basic-layer and capacity-layer cells are relative concepts. When configuring neighbor relationships, it is necessary to identify neighboring cells using the **POWER_SAVING_OVERLAP_IND** option of the **EutranInterFreqNCell.OverlapIndicatorExtension** parameter.

- If this option is deselected, no energy-saving relationship exists between the neighboring cell and the local cell, and the neighboring cell is not considered for load-based fast deep dormancy.
- If this option is selected, the neighboring cell is treated as a basic-layer cell, the local cell as a capacity-layer cell, and these cells overlap with each other in coverage.

Figure 7-16 Basic-layer and capacity-layer cells



In [Figure 7-16](#), for example, the capacity-layer cell on F2 is listed in the neighboring relation table (NRT) of the basic-layer cell on F1, with the **POWER_SAVING_OVERLAP_IND** option of the **EutranInterFreqNCell.OverlapIndicatorExtension** parameter selected.

In LTE multi-carrier networking, it is recommended that capacity-layer and basic-layer cells be configured as described in [Table 7-29](#).

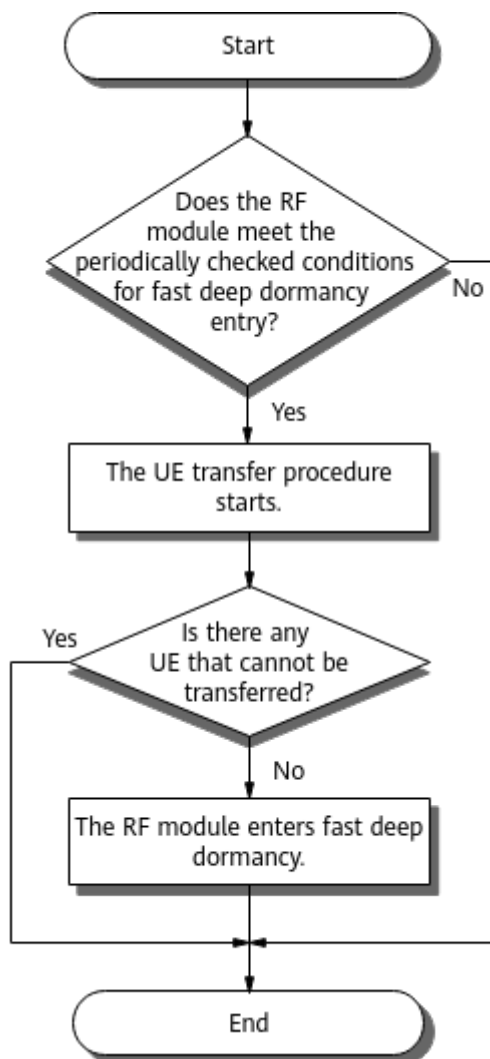
Table 7-29 Suggestions for configuring capacity-layer and basic-layer cells

Networking	Capacity-Layer Cell	Basic-Layer Cell
LTE TDD and LTE FDD coordinated networking	LTE TDD cells	LTE FDD cells
LTE TDD or LTE FDD intra-band multi-carrier networking	Cells on a higher frequency	Cells on a lower frequency

Entering Fast Deep Dormancy

Figure 7-17 shows how the RF module serving a capacity-layer cell enters fast deep dormancy after LTE load-based RF module fast deep dormancy is enabled.

Figure 7-17 Procedure for entering LTE load-based RF module fast deep dormancy



1. The base station determines whether the entry conditions are met.
The base station periodically checks whether the fast deep dormancy entry conditions are met within an evaluation period specified by the **CellPwrSavingPolicy.TriggerJudgePeriod** parameter.
 - A capacity-layer cell must meet all of the following conditions:
 - The current time falls within the effective energy saving period, as determined by the **CellPwrSavingPolicy.StartTime** and **CellPwrSavingPolicy.StopTime** parameters.
 - The capacity-layer cell is not in an energy saving penalty state.
 - The basic-layer cells associated with the capacity-layer cell must meet one of the following conditions:

- All basic-layer cells are available.
- Some basic-layer cells are available, while others are in the energy saving state. For example, these cells enter the energy saving state when functioning as capacity-layer cells of other basic-layer cells. (Whether this condition applies depends on the setting of the **INTRA_RAT_GRAD_PWR_SAVING_SW** option of the **EnodebPwrSavingAlgo.PwrSavingAlgoSwitch** parameter.)
- The capacity-layer cell and its associated basic-layer cells must meet all the following conditions:
 - The number of RRC_CONNECTED UEs in the capacity-layer cell is less than or equal to the UE number threshold specified by the **CellPwrSavingPolicy.UENumThld** parameter.
 - In multi-operator sharing scenarios, the PLMN IDs of operators serving the capacity-layer cell are included in the PLMN ID set of operators of at least one basic-layer cell.
 - In multi-operator sharing scenarios, the operators of the local cell must be a subset of the operators of co-coverage neighboring basic cells. If optimization for cell PLMN reservation scenarios is enabled by selecting the **CELL_PLMN_RSVD_OPT_SW** option of the **EnodebPwrSavingAlgo.PwrSavingAlgoSwitch** parameter:
 - If the operators of the local cell are non-reserved operators, the operators must be a subset of non-reserved operators of its co-coverage neighboring basic cells.
 - If the operators of the local cell are reserved operators, the operators must be a subset of non-reserved operators or reserved operators of its co-coverage neighboring basic cells.

The operator status of the local cell and intra-base-station co-coverage neighboring basic cells is obtained as follows:

- Using the effective operator status of the cells if the **RSVD_FOR_OP_X2_BROADCAST_SW** option of the **eNodeBOamParam.ConfigAdaptOptSwitch** parameter is selected
- Using the value of the **CellOp.CellReservedForOp** parameter if the **RSVD_FOR_OP_X2_BROADCAST_SW** option of the **eNodeBOamParam.ConfigAdaptOptSwitch** parameter is deselected

The operator status of inter-base-station co-coverage neighboring basic cells is obtained as follows:

- Using the **PLMN_RESERVED_FLAG** option of the **EutranExternalCellPlmn.AggregationAttribute** parameter if the **RSVD_FOR_OP_X2_BROADCAST_SW** option of the **eNodeBOamParam.ConfigAdaptOptSwitch** parameter is deselected
- Using the following methods if the **RSVD_FOR_OP_X2_BROADCAST_SW** option of the **eNodeBOamParam.ConfigAdaptOptSwitch** parameter is selected:

Method 1: Using the **PLMN_RESERVED_FLAG** option of the **EutranExternalCellPlmn.AggregationAttribute** parameter

Method 2: The operator status of co-coverage neighboring basic cells is automatically obtained through broadcast over the X2 interface when neighbor relationships are properly configured and the X2 interface between base stations is working properly.

If both of the preceding methods are available, the operator status obtained through these methods must meet the triggering conditions.

- The X2 interface is normal when the capacity-layer cell and basic-layer cells are provided by different base stations.
- For sector split cells (excluding FDD Dynamic Massive Beam cells in massive MIMO scenarios), a capacity-layer cell and its associated basic-layer cell must meet the following conditions:
 - Uplink PRB usage of the local cell + Uplink PRB usage of co-coverage neighboring basic cells + Uplink PRB usage of other cells in the same sector split group < Uplink PRB threshold (**CellPwrSavingPolicy.UlPrbThld**)
 - Downlink PRB usage of the local cell + Downlink PRB usage of co-coverage neighboring basic cells + Downlink PRB usage of other cells in the same sector split group < Downlink PRB threshold (**CellPwrSavingPolicy.DlPrbThld**)
 - Uplink PRB usage of the local cell + Uplink PRB usage of a co-coverage neighboring basic cell + Uplink PRB usage of other cells in the same sector split group < Threshold for a co-coverage neighboring basic cell to start load adjustment for inter-frequency load balancing (**CellMLB.InterFreqMlbThd** plus **CellMLB.LoadOffset**)
 - Downlink PRB usage of the local cell + Downlink PRB usage of a co-coverage neighboring basic cell + Downlink PRB usage of other cells in the same sector split group < Threshold for a co-coverage neighboring basic cell to start load adjustment for inter-frequency load balancing (**CellMLB.InterFreqMlbThd** plus **CellMLB.LoadOffset**)

All cells in a sector split group are enabled with load-based dormancy and meet the triggering conditions.

2. A UE transfer procedure starts.

- The capacity-layer cell prohibits the access of new UEs and incoming handovers.

The eNodeB broadcasts the barred state of the capacity-layer cell in SIB1. In addition, the eNodeB sends an eNodeB Configuration Update message over the X2 interface to each neighboring eNodeB. The message contains the *Deactivation Indication* IE, indicating that the local cell is unavailable due to energy saving.

- The eNodeB attempts to hand over all RRC_CONNECTED UEs from the capacity-layer cell to the cells on the operating frequencies of the co-coverage neighboring basic cells within 60s.

The eNodeB delivers A4 measurement configurations for the frequencies of co-coverage basic cells to RRC_CONNECTED UEs. The RSRP threshold for event A4 is specified by the **InterFreqHoGroup.InterFreqHoA4ThdRsrp** parameter. After receiving A4 measurement reports from the UEs, the eNodeB initiates inter-frequency handovers to the cells on those frequencies.

During the handover attempt, the eNodeB checks the number of RRC_CONNECTED UEs in the capacity-layer cell every 5s.

- If the number of RRC_CONNECTED UEs is 0, the capacity-layer cell enters dormancy.
- If the number of RRC_CONNECTED UEs is not 0, the eNodeB checks after 60s whether the number of RRC_CONNECTED UEs is less than or equal to the value of the **CellPwrSavingPolicy.ForcibleTriggerUeNumThld** parameter and whether no UEs are performing QCI-1 services in the capacity-layer cell. If these conditions are met, the capacity-layer cell enters dormancy. If these conditions are not met, the eNodeB cancels the barred state of the capacity-layer cell and notifies all neighboring eNodeBs over the X2 interfaces that the capacity-layer cell has exited the energy saving state. It does not recheck the entry conditions until the penalty period specified by the **CellPwrSavingPolicy.PenaltyTime** parameter has elapsed.

3. The RF module enters fast deep dormancy.

Once all UEs are transferred out of the capacity-layer cell, no RRC_CONNECTED UEs remain, and the cell and its associated RF module enter fast deep dormancy. If the RF module serves multiple LTE cells, it enters deep dormancy only when load-based fast deep dormancy is triggered for all these cells. If the RF module also serves NR cells, RF module fast deep dormancy must also be enabled on the NR side.

If intra-RAT level-by-level energy saving is enabled by selecting the **INTRA_RAT_GRAD_PWR_SAVING_SW** option of the **EnodebPwrSavingAlgo.PwrSavingAlgoSwitch** parameter, some basic-layer cells can act as capacity-layer cells of other basic-layer carriers and enter the energy saving state due to functions such as intelligent power-off of carriers in the same coverage, Cell Switch-off Based on Automatic Co-coverage Identification, multi-carrier coordinated energy saving, GUL joint carrier shutdown, dynamic carrier shutdown, or LTE load-based fast deep dormancy.

Exiting Fast Deep Dormancy

The base station periodically checks whether the dormancy exit conditions are met. The exit evaluation period is specified by the **CellPwrSavingPolicy.ExitJudgePeriod** parameter.

The base station continuously checks whether the dormancy exit conditions are met at intervals specified by the **CellPwrSavingPolicy.ExitJudgePeriod** parameter. The cell can exit the energy saving state only when the conditions are met at each evaluation occasion. The value of this parameter ranges from 1 to 120 (unit: 5s), meaning the actual value is an integer multiple of 5s.

The **EnodebPwrSavingAlgo.InterEnbInfoExchangeMode** parameter can be used to specify the mode of information exchange between base stations. This enables

base stations to obtain the latest load information more quickly and evaluate whether the dormancy exit conditions are met.

- If this parameter is set to **SLOW_EXCHANGE_MODE**, the slow exchange mode is used, meaning information is exchanged at long intervals.
- If this parameter is set to **FAST_EXCHANGE_MODE**, the fast exchange mode is used, meaning information is exchanged at a short interval.

A capacity-layer cell and its associated RF module automatically exit fast deep dormancy when any of the following conditions is met:

- The current time falls outside the effective energy saving period, as determined by the **CellPwrSavingPolicy.StartTime** and **CellPwrSavingPolicy.StopTime** parameters.
- The total downlink PRB usage of basic-layer cells is greater than the sum of the **CellPwrSavingPolicy.DlPrbThld** and **CellPwrSavingPolicy.DlPrbOffset** parameter values.
- The total uplink PRB usage of basic-layer cells is greater than the sum of the **CellPwrSavingPolicy.UlPrbThld** and **CellPwrSavingPolicy.UlPrbOffset** parameter values.
- A basic-layer cell is added for the capacity-layer cell.
- In multi-operator sharing scenarios, the PLMN IDs of operators serving the capacity-layer cell are not included in the PLMN ID set of operators serving any basic-layer cell.
- In multi-operator sharing scenarios, the operators of the local cell are not a subset of the operators of its co-coverage neighboring basic cells. If optimization for cell PLMN reservation scenarios is enabled by selecting the **CELL_PLMN_RSVD_OPT_SW** option of the **ENodebPwrSavingAlgo.PwrSavingAlgoSwitch** parameter:
 - If the operators of the local cell are non-reserved operators, the operators are not a subset of non-reserved operators of its co-coverage neighboring basic cells.
 - If the operators of the local cell are reserved operators, the operators are not a subset of non-reserved operators or reserved operators of its co-coverage neighboring basic cells.

The operator status of the local cell and intra-base-station co-coverage neighboring basic cells is obtained as follows:

- Using the effective operator status of the cells if the **RSVD_FOR_OP_X2_BROADCAST_SW** option of the **eNodeBOamParam.ConfigAdaptOptSwitch** parameter is selected
- Using the value of the **CellOp.CellReservedForOp** parameter if the **RSVD_FOR_OP_X2_BROADCAST_SW** option of the **eNodeBOamParam.ConfigAdaptOptSwitch** parameter is deselected

The operator status of inter-base-station co-coverage neighboring basic cells is obtained as follows:

- Using the **PLMN_RESERVED_FLAG** option of the **EutranExternalCellPlmn.AggregationAttribute** parameter if the **RSVD_FOR_OP_X2_BROADCAST_SW** option of the **eNodeBOamParam.ConfigAdaptOptSwitch** parameter is deselected

- Using the following methods if the **RSVD_FOR_OP_X2_BROADCAST_SW** option of the **eNodeBOamParam.ConfigAdaptOptSwitch** parameter is selected:
 - Method 1: Using the **PLMN_RESERVED_FLAG** option of the **EutranExternalCellPlmn.AggregationAttribute** parameter
 - Method 2: The operator status of co-coverage neighboring basic cells is automatically obtained through broadcast over the X2 interface when neighbor relationships are properly configured and the X2 interface between base stations is working properly.
 - If both of the preceding methods are available, the operator status obtained through these methods must meet the triggering conditions.
- The X2 interface is abnormal when the capacity-layer cell and basic-layer cells are provided by different base stations.
- Any basic-layer cell becomes unavailable due to other causes, such as cell faults or deactivation. This condition does not apply when the **INTRA_RAT_GRAD_PWR_SAVING_SW** option of the **EnodebPwrSavingAlgo.PwrSavingAlgoSwitch** parameter is selected and basic-layer cells become unavailable due to carrier shutdown for energy saving or RF module fast deep dormancy.

Upon the exit from fast deep dormancy:

- The capacity-layer cell cannot re-enter fast deep dormancy within the penalty period specified by the **CellPwrSavingPolicy.PenaltyTime** parameter if it exited dormancy due to uplink/downlink PRB usage exceptions or unavailability of the basic-layer cell, or X2 interface exceptions.
- The penalty mechanism is not triggered for the capacity-layer cell if:
 - Fast deep dormancy is exited because the function switch is turned off or the energy saving period has elapsed.
 - Fast deep dormancy is pending due to a base station software upgrade or configuration.

7.3.2 Network Analysis

7.3.2.1 Benefits

It is recommended that RF module fast deep dormancy be enabled during fixed daily periods with light or no load to reduce base station power consumption. To ensure that the actual operating duration of an RF module meets lifecycle management requirements in typical scenarios, the module can enter dormancy when its reliability requirements (such as temperature and humidity) are met. If the RF module is reset during the dormancy period for purposes such as upgrade, the module and its cells will return to the normal state, which leads to reduced energy saving gains.

When the network is under heavy load, the RF module and its served cells can be quickly woken. This mitigates the impacts on user experience while achieving energy saving effects.

When load-based RF module fast deep dormancy is enabled and scheduled energy saving functions (such as timing carrier shutdown, scheduled RF module deep

dormancy, and scheduled RF module super dormancy) are enabled on neighboring base stations or the neighboring base stations or RF modules are reset or powered off, the load of capacity-layer or basic-layer cells served by the RF modules may increase above the preset threshold. As a result, the RF modules may exit or fail to enter dormancy, reducing energy saving gains.

7.3.2.2 Impacts

The impacts between this function and other functions are described in [7.3.3.2.3 Function Impacts](#).

For NR load-based RF module fast deep dormancy, LNR load-based RF module fast deep dormancy, and LTE load-based RF module fast deep dormancy:

- After the RF module enters fast deep dormancy, RAT introduction to the base station and Fast Fourier Transform (FFT) frequency scanning are not supported. Related RF module hardware status checks cannot be started using MML commands such as **STR RRUCHK**, **STR HWTST**, and **STR HWOLTST**.
- Dormancy pending is triggered in scenarios such as execution of the **PND DORMANCY** command or software download and activation. If the main control board is powered off before the pending duration elapses, dormancy pending resumes after the board is powered on and continues until the pending duration elapses.
- If a switchover occurs between the active and standby main control boards, the pending duration needs to be reconfigured.
- By default, the base station triggers a 30-minute pending period after startup. During this period, dormancy does not take effect.
- If dormancy entry conditions are met during a base station version upgrade, the RF module will enter dormancy only after the upgrade is complete. If an upgrade is performed while the RF module is in fast deep dormancy, fast deep dormancy pending will be triggered. Assume that the main control board on one side requires the RF module to be upgraded while the main control board on the other side requires the RF module to exit dormancy, when the RF module is shared between the two sides of a separate-MPT base station using SDR technology. In this case, cells on the side where the RF module is required to exit dormancy may fail to be reestablished, and alarms related to these cells are reported. These cells become normal after the upgrade is complete.
- Assume that both the local and neighboring base stations are in fast deep dormancy, and an upgrade is performed on the local base station but not on the neighboring ones. After the upgrade, the local base station returns to the normal state, and cell edge users (CEUs) in weak coverage areas may access the available cells served by the local base station. This affects KPIs related to the aspects such as access and service drop of the local base station. It is recommended that the **PND DORMANCY** command be executed on both the base station to be upgraded and its neighboring base stations within the upgrade time window.
- If the dormancy type is changed while dormancy is in effect, the current dormancy does not exit.
- After RF module dormancy takes effect, if the configured end time is changed to a later value, the modification applies from the next dormancy cycle. If the configured end time is changed to an earlier value, the ongoing dormancy ends at the updated end time.

- RF module fast deep dormancy triggers handovers or redirections for active UEs, increasing the numbers of handover requests and redirections from the cell.
- In areas served by an RF module in fast deep dormancy, signal quality deteriorates because the module does not transmit signals. As a result, UEs may experience service drops or access failures.
- If a batch processing task on the OSS contains an eNodeBFunction or gNodeBFunction MML command that affects cell setup, or a Node MML command, and an NE of the base station is in fast deep dormancy, the OSS can execute the related configurations only after the base station exits fast deep dormancy. This results in increased waiting time before configuration execution. On average, the base station exits fast deep dormancy in less than 10 seconds.
- If a power-related command is executed on a base station with an NE in fast deep dormancy, the base station first exits fast deep dormancy and then executes the command. On average, the base station exits fast deep dormancy in less than 10 seconds.
- Before an RF module enters fast deep dormancy, new UEs are prohibited from accessing the NR cells of the module during NR carrier shutdown. If UEs attempt to be handed over to the NR cells, the values of the counters related to handover preparation failures (such as **N.HO.IntraFreq.FailOut.PrepFailure**) slightly increase. Such counters are not used for calculating Huawei-defined KPIs outgoing intra- and inter-frequency handover success rates, but may be used for calculating user-defined KPIs. When dormancy is set to take effect on the hour, no UEs can be handed over to the deactivated NR cell. Therefore, even slight increases in the counter values may significantly affect the user-defined outgoing handover success rates. If fast deep dormancy is scheduled to take effect on the hour, you are advised to set the dormancy start time 5 minutes earlier, and set the lengths of penalty timers for handover failures to 30s. After penalty timers for handover failures are set, UEs may experience service drops due to delayed handovers. The penalty timers for handover failures are specified by the **gNBMobilityCommParam.NecHoPrepFailPunishTimer** and **gNBMobilityCommParam.OptHoPrepFailPunishTimer** parameters.
- Assuming that the neighboring base stations are in the dormancy state, CEUs in weak coverage areas may access the cells served by the local base station. This affects KPIs related to the aspects such as access and service drop of the local base station. For example, the value of **N.NsaDc.DRB.AbnormRel** may increase.

In addition to the preceding impacts, some functions have their respective network impacts:

For NR load-based RF module fast deep dormancy:

- When NR load-based RF module fast deep dormancy takes effect, counters related to inter-frequency handovers and redirections are updated.
- If a UE attempts to hand over to a capacity-layer cell that is about to enter fast deep dormancy during UE transfer (lasting up to 30s), the source cell does not initiate handover execution. In this case, the values of the counters related to outgoing and incoming handover attempts increase in the source and capacity-layer cells, respectively, while the values of the counters related to handover executions and successful handovers do not increase. As Huawei-

defined KPIs **Inter-Frequency Handover Out Success Rate (CU)** and **Intra-RAT Handover In Success Rate (CU)** are calculated using the numbers of outgoing and incoming handover executions as their denominators, these KPIs are not affected in this scenario. However, if the user-defined KPIs are calculated using the numbers of outgoing and incoming handover attempts as their denominators, these KPIs may be slightly affected in this scenario.

For LNR load-based RF module fast deep dormancy:

- When LNR load-based RF module fast deep dormancy takes effect, performance counters on the LTE and NR sides are affected. The affected performance counters are the same as those described in [6.1.2.2 Impacts](#) for LTE and NR intelligent carrier shutdown.
- In inter-base-station SA networking, X2 link setup may be triggered for energy saving purposes. This occupies the specifications of EN-DC X2 links and affects the experience of NSA UEs. The probability of ALM-29225 Maximum Number of X2 Interfaces Reached being reported also increases. Therefore, this function is not recommended when the X2 specifications are less than required.
- In inter-base-station SA networking, if the **LNR_INTER_BS_PWR_SAVE_SW** option of the **ENodebPwrSavingAlgo.PwrSavingAlgoSwitch** parameter or the **LNR_SA_GNB_X2_SETUP_SW** option of the **eNodeBOamParam.InterfaceSonConfigSwitch** parameter is selected but the **X2_SETUP_FOR_SA_SW** option of the **gNBInterfaceParam.X2ConfigSw** parameter is deselected on the gNodeB side, ALM-29204 X2 Interface Fault and ALM-25888 SCTP Link Fault may be reported.
- In inter-base-station SA networking, the configurations of the neighboring NR cell with overlapping coverage areas on the LTE side must be the same as those on the NR side, such as the cell ID, base station ID, and operator information. Otherwise, the SA X2 link may fail to be set up.

For LTE load-based RF module fast deep dormancy:

- When LTE load-based RF module fast deep dormancy is triggered, inter-frequency handovers are performed for UEs in the capacity-layer cell, increasing the values of counters listed in [Table 7-30](#) and [Table 7-31](#). In addition, to promptly transfer UEs out of capacity-layer cells, the eNodeB prohibits UEs from accessing the cells and broadcasts the barred state of the cells in SIB1. However, a small number of UEs may still send RRC connection setup requests to access the cells, causing an increase in the **L.RRC.SetupFail.Rej** counter value and a decrease in the **RRC Setup Success Rate** value. The number of such access failures is indicated by the **L.RRC.SetupFail.ResFail.CellBarred** counter.

Table 7-30 Performance counters of capacity-layer cells

Networking	Counter Description	Counter Name
Macro networking	Number of intra-eNodeB inter-frequency outgoing handover attempts in a cell	L.HHO.IntraeNB.InterFreq.PrepAttOut

Networking	Counter Description	Counter Name
	Number of intra-eNodeB inter-frequency outgoing handover executions in a cell	L.HHO.IntraeNB.InterFreq.ExecAttOut
	Number of successful intra-eNodeB inter-frequency outgoing handovers in a cell	L.HHO.IntraeNB.InterFreq.ExecSuccOut
Inter-eNodeB networking	Number of inter-eNodeB inter-frequency outgoing handover attempts in a cell	L.HHO.IntereNB.InterFreq.PrepAttOut
	Number of inter-eNodeB inter-frequency outgoing handover executions in a cell	L.HHO.IntereNB.InterFreq.ExecAttOut
	Number of successful inter-eNodeB inter-frequency outgoing handovers in a cell	L.HHO.IntereNB.InterFreq.ExecSuccOut

Table 7-31 Performance counters of basic-layer cells

Networking	Counter Description	Counter Name
Macro networking	Number of intra-eNodeB intra-duplex-mode incoming handover attempts in a cell	L.HHO.IntraeNB.PrepAttIn
	Number of intra-eNodeB intra-duplex-mode incoming handover executions in a cell	L.HHO.IntraeNB.ExecAttIn
	Number of successful intra-eNodeB intra-duplex-mode incoming handovers in a cell	L.HHO.IntraeNB.ExecSuccIn
Inter-eNodeB networking	Number of inter-eNodeB intra-duplex-mode incoming handover attempts in a cell	L.HHO.IntereNB.PrepAttIn

Networking	Counter Description	Counter Name
	Number of inter-eNodeB intra-duplex-mode incoming handover executions in a cell	L.HHO.IntereNB.ExecAttIn
	Number of successful inter-eNodeB intra-duplex-mode incoming handovers in a cell	L.HHO.IntereNB.ExecSucIn

7.3.3 Requirements

7.3.3.1 Licenses

RAT	Feature ID	Feature Name	Model	NE	Sales Unit
NR	MRFD-2 21261	RF Module Fast Deep Dormancy (NR)	NR0SRF MFDD10	gNodeB	per RU
LTE FDD	MRFD-2 21221	RF Module Fast Deep Dormancy (LTE FDD)	LT1SRFM FDD00	eNodeB	per RU
LTE TDD	MRFD-2 21231	RF Module Fast Deep Dormancy (LTE TDD)	LT1RFM FDD00	eNodeB	per RU

CAUTION

- For an RF module shared by multiple RATs using SDR technology, it can enter fast deep dormancy only when the licenses for all RATs of the RF module are acquired.
- It is recommended that the RF unit working mode (specified by the **RRU.RS (LTE eNodeB, 5G gNodeB)** parameter) be set to a single RAT for single-mode RF modules. If this parameter is set to a value indicating multiple RATs for an RF module and RAT introduction has been performed on the base station, the RF module can enter dormancy only when the licenses for the newly introduced RATs are loaded.

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

7.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.3.3.2.1 Prerequisite Functions

None

7.3.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	The time period configured for timing carrier shutdown cannot overlap with the period configured for NR or LNR load-based RF module fast deep dormancy, which is a sub-function of RF module deep dormancy of the local cell. You are advised to preferentially plan the periods during which timing carrier shutdown takes effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Deployment Position	NRDUCell.DeployPosition set to DEPLOY_IAB	None	If this parameter is set to DEPLOY_IAB , NR or LNR load-based RF module fast deep dormancy cannot be enabled.
NR FDD	MBS broadcast	MBS_BROADCAST_SWITCH option of the NRCellFeatureSwitch.MbsSwitch parameter	None	When the MBS broadcast switch is on, NR or LNR load-based RF module fast deep dormancy cannot be enabled.
LTE FDD LTE TDD	Flag of Multi-RRU Cell	Cell.MultiRruCellFlag set to BOOLEAN_TRUE	None	LTE load-based RF module fast deep dormancy is incompatible with multi-RRU cells.
NR FDD Low-frequency NR TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC in 5G RAN Feature Documentation</i>	NR or LNR load-based RF module fast deep dormancy is not supported in low latency and high reliability scenarios.
Low-frequency NR TDD	Time-domain reservation mode	NRDUCellTimeIntrf.TimeDomainRsvMode	None	RF module fast deep dormancy cannot be enabled when the NRDUCellTimeIntrf.TimeDomainRsvMode parameter is set to a value other than DEFAULT .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Dual-BBU backup in dual-network redundancy networking	DUAL_BBU_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	Dual-BBU backup in dual-network redundancy networking is incompatible with NR or LNR load-based RF module fast deep dormancy.
Low-frequency NR TDD	Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios	DUAL_PLANE_TOB_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	<i>Base Station Equipment Reliability</i>	"Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "NR or LNR load-based RF module fast deep dormancy" cannot be both enabled.
NR FDD Low-frequency NR TDD	Primary/Backup function in intra-frequency dual-network redundancy networking	PRIMARY_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	Primary/Backup function in intra-frequency dual-network redundancy networking is incompatible with NR or LNR load-based RF module fast deep dormancy.
NR FDD	Dual-plane function in intra-frequency dual-network redundancy networking	DUAL_PLANE_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	"Dual-plane function in intra-frequency dual-network redundancy networking" and "NR or LNR load-based RF module fast deep dormancy" cannot be both enabled.

7.3.3.2.3 Function Impacts

When RF module fast deep dormancy and another energy saving function related to RF modules (such as symbol power saving, channel shutdown, or carrier shutdown) are both enabled, RF module fast deep dormancy preferentially takes effect.

When RF module fast deep dormancy and RF module deep dormancy (or RF module super dormancy) are both enabled, RF module deep dormancy (or RF module super dormancy) preferentially takes effect.

Cells do not provide services after entering the dormancy state, and related features do not take effect in these cells.

The following tables list other function impacts.

- **Table 7-32** lists the function impacts of NR load-based RF module fast deep dormancy.
- In addition to the function impacts of NR load-based RF module fast deep dormancy (except the impact relationships with intra-RAT ANR) listed in **Table 7-32**, LNR load-based RF module fast deep dormancy has the function impacts listed in **Table 7-33**.
- **Table 7-34** lists the function impacts of LTE load-based RF module fast deep dormancy.
- **Table 7-32 Function impacts of NR load-based RF module fast deep dormancy**

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Downlink intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation in 5G RAN Feature Documentation</i>	After RF module fast deep dormancy takes effect, the capacity-layer cells that have been shut down cannot be configured as SCells for UEs.
NR FDD Low-frequency NR TDD	Downlink intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation in 5G RAN Feature Documentation</i>	After RF module fast deep dormancy takes effect, the capacity-layer cells that have been shut down cannot be configured as SCells for UEs.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Inter-gNodeB CA	INTER_GNODEB_CA_SWITCH option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>CA Experience Improvement in 5G RAN Feature Documentation</i>	After RF module fast deep dormancy takes effect, the capacity-layer cells that have been shut down cannot be configured as SCells for UEs.
NR FDD Low-frequency NR TDD	Uplink symbol power saving	UL_SYMBOL_SHUTDOWN_SWITCH option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Uplink symbol power saving does not take effect when RF module fast deep dormancy is in effect.
NR FDD Low-frequency NR TDD	Symbol power saving	NRDUCellAlgoSwitch.SymbolShutdownSwitch	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over symbol power saving when both functions are enabled. Symbol power saving does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Deep symbol power saving	DEEP_SYMBOL_SHUTDOWN_SWITCH option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over deep symbol power saving when both functions are enabled. Deep symbol power saving does not take effect when RF module fast deep dormancy is in effect.
NR FDD Low-frequency NR TDD	Power saving scheduling	POWER_SAVING_SCHEDULE_SWITCH option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over power saving scheduling when both functions are enabled. Power saving scheduling does not take effect when RF module fast deep dormancy is in effect.
Low-frequency NR TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SWITCH option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over RF channel intelligent shutdown when both functions are enabled. RF channel intelligent shutdown does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over energy saving based on flexible frequency-domain scheduling when both functions are enabled. Energy saving based on flexible frequency-domain scheduling does not take effect when RF module fast deep dormancy is in effect.
NR FDD Low-frequency NR TDD	TTI-level channel shutdown	TTI_RF_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over TTI-level channel shutdown when both functions are enabled. TTI-level channel shutdown does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Low-power RF channel dynamic shutdown	LOW_POWER_RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over low-power RF channel dynamic shutdown when both functions are enabled. Low-power RF channel dynamic shutdown does not take effect when RF module fast deep dormancy is in effect.
NR FDD Low-frequency NR TDD	Intelligent power control energy saving	SMART_POWER_CTRL_ENERGY_SAVE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over intelligent power control energy saving when both functions are enabled. Intelligent power control energy saving does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Dynamic RF channel shutdown	RF_CHN_DYN_SHUTDOWN_SW option of the NRDUCellParameters.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over dynamic RF channel shutdown when both functions are enabled. Dynamic RF channel shutdown does not take effect when RF module fast deep dormancy is in effect.
NR FDD	Uplink TTI-level RF channel dynamic muting	UL_TTI_CHANNEL_DYN_MUTING_SW option of the NRDUCellParameters.PowerSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over uplink TTI-level RF channel dynamic muting when both functions are enabled. Uplink TTI-level RF channel dynamic muting does not take effect when RF module fast deep dormancy is in effect.
NR FDD Low-frequency NR TDD	Initial transmission power reduction	INIT_TRANSMISSION_POWER_REDUCE_SW option of the NRDUCellParameters.PowerSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over initial transmission power reduction when both functions are enabled. Initial transmission power reduction does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Retransmission power reduction	RETRANS_POWER_REDUCTION_SW option of the NRDUCellPowerSavingAlgorithmSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over retransmission power reduction when both functions are enabled. Retransmission power reduction does not take effect when RF module fast deep dormancy is in effect.
NR FDD Low-frequency NR TDD	Intelligent carrier shutdown in precise mode	INTRANB_MULTICARRIER_SHUTDOWN_SW or INTERNANB_MULTICARRIER_SHUTDOWN_SW option of the NRDUCellPowerSavingAlgorithmSwitch parameter MULTICARRIER_PRECISE_SHUTDOWN_SW option of the NRDUCellPowerSavingAlgorithmSwitch parameter selected	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over intelligent carrier shutdown when both functions are enabled. Intelligent carrier shutdown does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Intelligent carrier shutdown in non-precise mode	• INTRA_GNB_MULTI_CARRIER_SHUTDOWN_SW or INTER_GNB_MULTI_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter • MULTI_CARRIER_PRECISE_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter deselected	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over intelligent carrier shutdown when both functions are enabled. Intelligent carrier shutdown does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Cell load simulation	STRNRDUCELLSIMULATELOAD	None	• If the SIMU_LOAD_DORMANCY_IMP_SW option of the gNBWPwrSavingsAlgo.PwrSavingsAlgoSwitch parameter is selected, and cell load simulation takes effect, the RF module serving the local cell is not allowed to enter the dormancy state, but the RF modules serving the cells not enabled with cell load simulation are allowed to enter the dormancy state. • If the SIMU_LOAD_DORMANCY_IMP_SW option of the gNBWPwrSavingsAlgo.PwrSavingsAlgoSwitch parameter is deselected, and cell load simulation takes effect, all RF modules of the base station are not allowed

RAT	Function Name	Function Switch	Reference	Description
				to enter the dormancy state.
NR FDD Low-frequency NR TDD	Automatic removal of neighboring cells	NR_NR_ANR_DEL_SW option of the NRCellAlgoSwitch.AnrSwitch parameter	<i>ANR in 5G RAN Feature Documentation</i>	When automatic removal of neighboring cells in ANR takes effect and the neighbor relationships with the basic-layer cells are automatically removed for capacity-layer cells, RF module fast deep dormancy does not take effect.
NR FDD Low-frequency NR TDD	CMAS	gNBsibConfig.SibType set to SIB_TYPE_8	<i>CMAS in 5G RAN Feature Documentation</i>	When RF module fast deep dormancy is in effect, the corresponding NR cell is blocked and CMAS warning notifications cannot be properly sent in this cell. However, UEs can still receive CMAS warning notifications from other carriers.
Low-frequency NR TDD	Super uplink with SUL and NR FDD co-deployment	FLEX_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter	<i>Super Uplink in 5G RAN Feature Documentation</i>	After RF module fast deep dormancy takes effect, SUL scheduling cannot be performed for super uplink UEs.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Basic VoNR functions	VONR_SW option of the NRCellAlgoSwitch.VonrSwitch parameter	<i>VoNR in 5G RAN Feature Documentation</i>	RF module fast deep dormancy does not take effect if any VoNR UE fails to be handed over from a capacity-layer cell during UE transfer.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Virtual Grid-based Multi-Frequency Coordination	• RSRP prediction model – VG_MODEL_ALLOW_BUILDFLAG option of the NRCellFreqRelationAggregationAttribute parameter – NR_IF_HO_MEAS_QTY_PRED_FUN_SW option of the NRCellSmartMultiCarr.NrMultiCarrierAlgoSwitch parameter • Downlink intra-frequency spectral efficiency prediction model	<i>Virtual Grid-based Multi-Frequency Coordination in 5G RAN Feature Documentation</i>	After RF module fast deep dormancy takes effect, the building of virtual grid models will be suspended and existing virtual grid models will go offline and cannot be used.

RAT	Function Name	Function Switch	Reference	Description
		- DL_IN TRAF_SE_MODEL_BUILT_ALLOW option of the NRCellSmartMultiCarr. <i>ModelBuildingSwitch</i> parameter - DL_IN TRAF_REQ_SELECT_ENABLED option of the NRCellSmartMultiCarr. <i>NrMultiCarrierAlgorithmSwitch</i> parameter • Intra-frequency SRS beam prediction model INTRAF_SRS_MODEL_BUILT_ALLOW option of the NRCellSmartMultiCarr. <i>IntraFrequencySRSBeamPredictionModelSwitch</i> parameter		

RAT	Function Name	Function Switch	Reference	Description
		LD_ALLO W option of the NRCells martMul tiCarr.M odelBuil dingSwit ch parameter		
NR FDD Low-frequency NR TDD	Channel calibration	Macro base station: None LampSite base station: INTER_RRU _CHN_CALI B_SW option of the NRDUCellAl goSwitch.C hnCalibSwi tch parameter	None	If channel calibration is being performed when RF module fast deep dormancy takes effect, channel calibration fails and will not be started again. You can run the DSP NRDUCELLCHN CALIB command to check whether channel calibration is successful.
NR FDD Low-frequency NR TDD	RB allocation for downlink experience optimization	NRDUCellD lSchRes.DIE xpOptRblnc easePct	<i>Downlink Scheduling in 5G RAN Feature Documentation</i>	When both RB allocation for downlink experience optimization and RF module fast deep dormancy take effect, an exit from RF module fast deep dormancy may be performed earlier than expected.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Cell-level MRO	MRO_SCENARIO_IDENTITY_SWITCH option of the gNB-MRO-ScenarioIdentitySwitch parameter and INTRA_FREQ_MRO_SWITCH option of the NRCell-MRO-MroSwitch parameter	<i>MRO in 5G RAN Feature Documentation</i>	When both RF module fast deep dormancy and intra-frequency MRO are enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, MRO may take effect later than expected.
NR FDD Low-frequency NR TDD	UE-level MRO	UE_MRO_SWITCH option of the NRCell-MRO-MroSwitch parameter	<i>MRO in 5G RAN Feature Documentation</i>	When both RF module fast deep dormancy and UE-level MRO are enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, the penalty mechanism accuracy of UE-level MRO may be affected.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SWITCH option of the NRDUCellAIgoSwitch.DynamicPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation in 5G RAN Feature Documentation</i>	A carrier exits a power sharing group once RF module fast deep dormancy takes effect for this carrier. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control messages changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	NR multi-band power adaptive scheduling	NR_MULTI_BAND_PWR_ADAPT_SCH_SW option of the NRDUCellAIgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation in 5G RAN Feature Documentation</i>	A carrier exits a power sharing group once RF module fast deep dormancy takes effect for this carrier. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control messages changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.
NR FDD Low-frequency NR TDD	Inter-RAT redirection back to the HPLMN based on roaming UE identification	VOICE_CAPB_MOB_TO_HPLMN_SW option of the gNBOperator.HplmnAIgoSwitch parameter	<i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i>	When RF module fast deep dormancy is enabled, only blind redirection can be performed to transfer VoNR-incapable roaming UEs from the visited 5G network to the home 5G network.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Clock out-of-synchronization detection	CLK_DETECTION_SW option of the gNodeBParameter.ClkOutofsyncDetection parameter	<i>Synchronization</i>	If clock out-of-synchronization occurs during the dormancy period, this issue may fail to be detected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Ultra-long holdover for sites with abnormal clock sources	CLK_OTASY NC_PROC_SW option of the TASM.MSCLKDIAGPROCESSSW parameter	<i>Synchronization</i>	Assume that the ultra-long hold switch is turned on but the ultra-long hold enhancement switch is turned off. If sequence detection is performed on a to-be-detected base station during the period when RF module fast deep dormancy takes effect for all cells of the base station, the sequence detection fails. In this case, the base station cannot enter the ultra-long holdover state even if its clock is faulty, or it exits the ultra-long holdover state as sequence detection cannot be performed for a long time due to energy saving. If the ultra-long hold enhancement switch is turned on, RF module fast deep dormancy does not take effect for the base station in the ultra-long holdover state and its reference

RAT	Function Name	Function Switch	Reference	Description
				neighboring base stations.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Voice-capability-based inter-frequency handover back to the HPLMN	VOICE_CAP B_MOB_TO_ HPLMN_SW option of the gNBOperat or.HplmnAl goSwitch parameter	<i>SA Mobility Managemen t in Connected Mode in 5G RAN Feature Documentat ion</i>	Assume that voice-capability-based inter-frequency handover back to the HPLMN is enabled on the gNodeB for the home operator of a VoNR-incapable roaming UE, the INTRA_GNB_MULTICAST_CARR_SWITCH or INTER_GNB_MULTICAST_CARR_SWITCH option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter is selected for the cell serving the UE, handover evaluation for RF module fast deep dormancy is ongoing, and the related timer has not expired. When the gNBFreqPriorityGroup.RoamingUeHplmnInterFHoflag parameter is set to TRUE for the target cell's frequency in the frequency list group associated with the roaming UE, the following mechanism is applied:

RAT	Function Name	Function Switch	Reference	Description
				- If the roaming UE supports handovers, the gNodeB instructs the roaming UE to immediately perform a handover to the target cell. - If the roaming UE does not support handovers, inter-frequency handover for RF module fast deep dormancy is not performed for the roaming UE. • When the gNBFreqPriorityGroup.RoamingUeHplmnInterFHoFlag parameter is set to FALSE for the target cell's frequency in the frequency list group associated with the

RAT	Function Name	Function Switch	Reference	Description
				roaming UE, inter-frequency handover for RF module fast deep dormancy is not performed for the roaming UE.

RAT	Function Name	Function Switch	Reference	Description
NR FDD	PIM interference cancellation (FDD) (trial)	PIM_INTERF_CANCELLATION_SWITCH option of the NRDUCellPimIntrfAvd.PimAlgoSwitch parameter	<i>Interference Avoidance in 5G RAN Feature Documentation</i>	After RF module fast deep dormancy takes effect in an interfering or victim cell, the gNodeB evaluates whether to continue PIM interference cancellation in each cell. • If PIM interference persists after RF module fast deep dormancy takes effect, PIM interference cancellation continues. • If PIM interference disappears after RF module fast deep dormancy takes effect, PIM interference cancellation is stopped to reduce resource consumption caused by periodic PIM interference characteristic training and interference characteristic extraction.

RAT	Function Name	Function Switch	Reference	Description
NR FDD	RF PIM interference cancellation	RF_PIM_CANN_SW option of the RRUEXTINFO.RruOptInterfSw parameter	<i>Interference Avoidance in 5G RAN Feature Documentation</i>	For RF PIM interference cancellation, after RF module fast deep dormancy takes effect in an interfering or victim cell, PIM interference characteristics change. In this case, the PIM interference cancellation process is restarted, causing fluctuations in PIM interference cancellation performance.
NR FDD	Intelligent PIM interference avoidance	PIM_INTERF_SMART_AVOID_SW option of the NRDUCellPimIntrfAvoid.PimAlgoSwitch parameter	<i>Interference Avoidance in 5G RAN Feature Documentation</i>	RF module fast deep dormancy is terminated during PIM interference detection.

RAT	Function Name	Function Switch	Reference	Description
NR FDD	Time-frequency two-dimensional unified scheduling optimization	TIME_FREQ_2D_UNIF_OPT_SCH_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgorithmExtSwitch parameter	<i>Multi-Frequency Smart Aggregation in 5G RAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over time-frequency two-dimensional unified scheduling optimization when both functions are enabled. Time-frequency two-dimensional unified scheduling optimization does not take effect when RF module fast deep dormancy is in effect.
Low-frequency NR TDD	CA Performance Boost Based on Multi-dimensional Optimization	CA_PRFM_BOOST_BSD_ON_MD_OPT_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgorithmExtSwitch parameter	<i>Multi-Frequency Smart Aggregation in 5G RAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over CA Performance Boost Based on Multi-dimensional Optimization when both functions are enabled. CA Performance Boost Based on Multi-dimensional Optimization does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Co-MM	INTRA_GNB_SU_COMM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD) in 5G RAN Feature Documentation</i>	When RF module fast deep dormancy takes effect, the probability that Co-MM takes effect decreases.
Low-frequency NR TDD	LTE and NR multi-band power adaptive scheduling	• LTE: LNR_MULTIBAND_POWER_ADPT_SWITCH option of the CellDynaPowerSharing.DynaPowerSharingSwitch parameter • NR: LNR_MULTIBAND_POWER_ADPT_SWITCH option of the NRDUCellAlgoSwitch.DynaPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	A carrier exits a power adaptive scheduling group once RF module fast deep dormancy takes effect for this carrier. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control messages changes due to system parameter adjustment. This causes changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference	Description
NR FDD	LTE FDD and NR Flash Dynamic Power Sharing	LTE_NR_DYNAMIC_POWER_SHARING_SWITCH option of the NRDUCellAlgorithmSwitch.DynamicPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	RF module fast deep dormancy takes precedence over LTE FDD and NR Flash Dynamic Power Sharing when both functions are enabled. LTE FDD and NR Flash Dynamic Power Sharing does not take effect when RF module fast deep dormancy is in effect.
Low-frequency NR TDD	LTE TDD and NR Flash Dynamic Power Sharing	LTE_NR_DYNAMIC_POWER_SHARING_SWITCH option of the NRDUCellAlgorithmSwitch.DynamicPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	RF module fast deep dormancy takes precedence over LTE TDD and NR Flash Dynamic Power Sharing when both functions are enabled. LTE TDD and NR Flash Dynamic Power Sharing does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	L/NR Inter-band Deep Coordination	LNR_INTER_BAND_DEEP_COORD_SW option of the NRDUCellResShrCfg.DynPwrShrAlgorithmSwitch parameter	<i>LTE and NR Power Allocation</i>	RF module fast deep dormancy takes precedence over L/NR Inter-band Deep Coordination when both functions are enabled. L/NR Inter-band Deep Coordination does not take effect when RF module fast deep dormancy is in effect.
NR FDD	LTE-NR multi-dimensional scheduling optimization	LNR_MULTID_OPT_SCHE_SW option of the NRDUCellGigaBand.GigaBandCellAlgorithmSw parameter	<i>LTE FDD and NR Inter-RAT Unified Scheduling</i>	LTE-NR multi-dimensional scheduling optimization does not take effect for cells in the RF module fast deep dormancy state.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Speed turbo UE identification	gNodeBParam.StUelidentSw	None	When any of the following configuration operations is performed and the configured function takes effect on the base station, the served cells already in the dormancy state remain in this state but other served cells will not enter this state, resulting in a decrease in the energy saving gains: - Any option of the gNodeBParam.StUelidentSw parameter is selected. - The gNBAsAppParam.AppDomainName parameter is set. - The gNBAsAppParam.AppIpv4 parameter is set. - The gNBAsAppParam.AppIpv6 parameter is set. The following methods can be used to address the preceding issue:
NR FDD Low-frequency NR TDD	gNodeB acceleration service application	gNBAsAppParam.AppDomainName gNBAsAppParam.AppIpv4 gNBAsAppParam.AppIpv6	None	

RAT	Function Name	Function Switch	Reference	Description
				• Select the SPEED_TURBO_PWR_SAVING_SW option of the gNBWPwrSavingsAlgo.PwrSavingAlgoSwitch parameter. Dormancy can take effect for cells served by the base station in some of the preceding scenarios within the period specified by the gNBWPwrSavingsAlgo.SpeedTurboPwrSavingsStartTime and gNBWPwrSavingsAlgo.SpeedTurboPwrSavingsEndTime parameters, increasing the energy saving gains. • Select the STATIC_ST_PWR_SAVING_SW option of the gNBWPwrSavingsAlgo.PwrSavingAlgoSwitch parameter. Dormancy can take effect for cells served by the base station in some of the preceding

RAT	Function Name	Function Switch	Reference	Description
				scenarios within the period specified by the gNB PwrSavin gAlgo.StaticS TPwrSavStartTime and gNB PwrSavin gAlgo.StaticS TPwrSavEndTime parameters, increasing the energy saving gains.
NR FDD Low-frequency NR TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag set to HIGH_SPEED	<i>High Speed Mobility in 5G RAN Feature Documentation</i>	When the high-speed cell function in high speed specified policy management is enabled, you are advised not to enable NR load-based RF module fast deep dormancy or LNR load-based RF module fast deep dormancy.

In addition to the function impacts of NR load-based RF module fast deep dormancy (except the impact relationship with intra-RAT ANR), LNR load-based RF module fast deep dormancy has the following function impacts.

Table 7-33 Function impacts of LNR load-based RF module fast deep dormancy

RA T	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Work mode	Cell.WorkMode	None	- An LTE FDD cell with the Cell.WorkMode parameter set to DL_ONLY or LAA cannot be configured as an overlapping neighboring cell of an NR cell. - An LTE TDD cell with the Cell.WorkMode parameter set to LAA cannot be configured as an overlapping neighboring cell of an NR cell.
LTE FDD LTE TDD	Cell System Inform ation Switch	CellSiMap.SiSwitch	None	A cell whose ForbidCellSiSwitch option of the CellSiMap.SiSwitch parameter is selected cannot be configured as a basic-layer cell.
LTE FDD LTE TDD	eMTC introduction	EMTC_SWITCH option of the CellEmtcAlgo.EmtcAlgoSwitch parameter	<i>eMTC in eRAN Feature Documentation</i>	An eMTC-only cell (specified by the EMTC_SWITCH and EMTC_ONLY_CELL_SWITCH options of the CellEmtcAlgo.EmtcAlgoSwitch parameter) cannot be configured as a basic-layer cell.
LTE FDD	Static Multiple Beam	SectorSplitGroup.BeamTemplateId set to a value greater than 1	<i>Massive MIMO Basics (FDD) in eRAN Feature Documentation</i>	For LTE FDD, if a Static Multiple Beam cell is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in NR cells, the user-perceived rate may decrease in the split cell that serves as a basic-layer cell.

RA T	Function Name	Function Switch	Reference	Description
LTE TDD	Intelligent Multi- Beam of 8T8R	None	<i>Intelligent Multi-Beam of 8T8R (TDD) in eRAN Feature Documentation</i>	For LTE TDD, if a cell enabled with Intelligent Multi-Beam of 8T8R is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in NR cells, the user-perceived rate may decrease in the split cell that serves as a basic-layer cell.
LTE FDD	Static Multiple Beam	None	<i>Smart 8T8R (FDD) in eRAN Feature Documentation</i>	For LTE FDD, if a Static Multiple Beam cell is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in NR cells, the user-perceived rate may decrease in the split cell that serves as a basic-layer cell.
LTE TDD	Intra- frequency split	None	<i>Soft Split Resource Duplex (TDD) in eRAN Feature Documentation</i>	For LTE TDD, if an intra-frequency split cell is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in NR cells, the user-perceived rate may decrease in the split cell that serves as a basic-layer cell.

RA T	Function Name	Function Switch	Reference	Description
LTE FDD	RuralLink antenna solution (LTE)	None	<i>RuralLink Solution</i>	For LTE FDD, if a split cell established through the RuralLink antenna solution (LTE) is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in NR cells, the user-perceived rate may decrease in the split cell that serves as a basic-layer cell.
LTE FDD	LumiLink solution (LTE)	• SectorSplitGroup.BeamTemplateId set to 0 • SectorSplitGroup.BeamDeploymentMode set to MODE_5 • SectorSplitCell.CellBeamIndex set to 0, 1, or 2	<i>LumiLink Solution</i>	For LTE FDD, if a split cell established through the LumiLink solution (LTE) is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in NR cells, the user-perceived rate may decrease in the split cell that serves as a basic-layer cell.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	RAN sharing	RAN sharing with common carrier: ENodeBSharingMode.ENodeBSharingMode set to SHARED_FREQ RAN sharing with dedicated carrier: ENodeBSharingMode.ENodeBSharingMode set to SEPARATED_FREQ Hybrid RAN sharing: ENodeBSharingMode.ENodeBSharingMode set to HYBRID_SHARED	<i>RAN Sharing in eRAN Feature Documentation</i>	In RAN sharing scenarios, the PLMN IDs of a capacity-layer cell must be a subset of the PLMN IDs of its co-coverage basic-layer cells. Otherwise, the capacity-layer cell cannot enter the dormancy state.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	NSA anchor carrier shutdown	NSA_ANCHOR_CARRIER_SHUTDOWN_SW option of the CellShutdown.CarrierShutdownEnhancedSw parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	In NSA or NSA and SA hybrid networking, it is recommended that the INTER_RAT_GRAD_PWR_SAVING_SW option of the gNBpwrSavingAlgo.PwrSavingAlgoSwitch parameter be selected when NSA anchor carrier shutdown is enabled for the LTE basic-layer cell and NSA anchor cells and LNR load-based RF module fast deep dormancy is enabled for NR capacity-layer cells and NSA anchor cells. • If NSA anchor carrier shutdown takes effect first, carriers are shut down in NSA anchor cells. If NR capacity-layer cells do not have other neighboring LTE cells, LNR load-based RF module fast deep dormancy does not take effect. • If LNR load-based RF module fast deep dormancy takes effect first, this function can take effect in NR capacity-layer cells.

Table 7-34 Function impacts of LTE load-based RF module fast deep dormancy

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Channel calibration	None	None	If channel calibration is being performed while PAs are shut down after RF module fast deep dormancy takes effect, channel calibration fails. Channel calibration is not started during RF module fast deep dormancy. You can run the DSP CELLCALIBRATION command to check whether channel calibration is successful.
LTE FDD LTE TDD	RAN sharing	RAN sharing with common carrier: ENodeBSharingMode.ENABLEBSharingMode set to SHARED_FREQ RAN sharing with dedicated carrier: ENodeBSharingMode.ENABLEBSharingMode set to SEPARATED_FREQ Hybrid RAN sharing: ENodeBSharingMode.ENABLEBSharingMode set to HYBRID_SHARED	<i>RAN Sharing in eRAN Feature Documentation</i>	In RAN sharing scenarios, the PLMN IDs of a capacity-layer cell must be a subset of the PLMN IDs of its co-coverage neighboring basic cells. Otherwise, the capacity-layer cell cannot be shut down.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	RF channel intelligent shutdown	EnodebMpruEs.MpruDormancyDLEarfcn	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over RF channel intelligent shutdown when both functions are enabled. RF channel intelligent shutdown does not take effect when RF module fast deep dormancy is in effect.
LTE TDD	Low-power RF channel dynamic shutdown	LOW_POWER_RF_DYN_MUTING_SW option of the CellRfChnDynamicMuting.RfChnDynMutin gAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over low-power RF channel dynamic shutdown when both functions are enabled. Low-power RF channel dynamic shutdown does not take effect when RF module fast deep dormancy is in effect.
LTE FDD	RF channel dynamic muting in TM4	TM4_TTI_CHN_DYN_MUTING_SW option of the CellRfChnDynamicMuting.RfChnDynMutin gAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over RF channel dynamic muting in TM4 when both functions are enabled. RF channel dynamic muting in TM4 does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	RF channel dynamic muting in TM9	TM9_TTI_CHN_DYN_MUTING_SW option of the CellRfChnDynamicMuting.RfChannelDynamicMutingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over RF channel dynamic muting in TM9 when both functions are enabled. RF channel dynamic muting in TM9 does not take effect when RF module fast deep dormancy is in effect.
LTE FDD	Uplink TTI-level RF channel dynamic muting	UL_TTI_CHN_DYN_MUTING_SW option of the CellRfChnDynamicMuting.RfChannelDynamicMutingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over uplink TTI-level RF channel dynamic muting when both functions are enabled. Uplink TTI-level RF channel dynamic muting does not take effect when RF module fast deep dormancy is in effect.
LTE FDD	Port sparseness	PORT_SPARSENESS_SW option of the CellPwrSavingAlgo.PowerSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over port sparseness when both functions are enabled.
LTE FDD LTE TDD	CRS dynamic muting	CRS_DYNAMIC_MUTING_SW option of the SymbolPowerSaving.SymbolShutdownEnhancedSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over CRS dynamic muting when both functions are enabled. CRS dynamic muting does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Dynamic voltage adjustment	CellAlgoSwitch.DynAdjVoltageSwitch	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over dynamic voltage adjustment when both functions are enabled. Dynamic voltage adjustment does not take effect when fast deep dormancy is in effect.
LTE FDD LTE TDD	Automatic cell shutdown	CellAutoShutdown.CellAutoShutdownSwitch	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over automatic cell shutdown when both functions are enabled. Automatic cell shutdown does not take effect when RF module fast deep dormancy is in effect.
LTE FDD LTE TDD	Intelligent power-off of carriers in the same coverage	CellShutdown.CellShutdownSwitch set to ON	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over intelligent power-off of carriers in the same coverage when both functions are enabled. Intelligent power-off of carriers in the same coverage does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Cell Switch-off Based on Automatic Co-coverage Identification	CellShutdown n.CellShutdownSwitch set to ON_CO_COV_AUTO_IDENT	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over Cell Switch-off Based on Automatic Co-coverage Identification when both functions are enabled. Cell Switch-off Based on Automatic Co-coverage Identification does not take effect when RF module fast deep dormancy is in effect.
LTE FDD LTE TDD	Multi-carrier coordinated energy saving	CellShutdown n.CellShutdownSwitch set to ON_MULTICARRIER_HIER_SHUTDOWN , ON_MULTICARRIER_HIER_MANUAL_SHUTDOWN , or ON_CARRIER_PRECISE_SHUTDOWN	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over multi-carrier coordinated energy saving when both functions are enabled. Multi-carrier coordinated energy saving does not take effect when RF module fast deep dormancy is in effect.
LTE FDD LTE TDD	Dynamic carrier shutdown	CARRIER_DYNAMIC_SHUTDOWN_SWITCH option of the CellShutdown n.CarrierShutdownEnhancedSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over dynamic carrier shutdown when both functions are enabled. Dynamic carrier shutdown does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Intelligent power-off of carriers in the same coverage as UMTS networks	InterRatCells hutdown.For ceShutdownSwitch	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over intelligent power-off of carriers in the same coverage as UMTS networks when both functions are enabled. Intelligent power-off of carriers in the same coverage as UMTS networks does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Intelligent power control energy saving	- Single-cell intelligent power control energy saving: INTEL_INTRA_CELL_POWER_SAVING option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch - Multi-cell coordinated intelligent power control energy saving: INTEL_INTRA_CELL_POWER_SAVING and INTEL_MULTICELL_COOR_PC_SWITCH options of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over intelligent power control energy saving when both functions are enabled. Intelligent power control energy saving does not take effect when RF module fast deep dormancy is in effect.
LTE FDD LTE TDD	Deep symbol power saving	DEEP_SYMBOL_SHUTDOWN_SWITCH option of the SymbolPwrSaving.SymbolShutdownEnhancedSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	Deep symbol power saving does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Dynamic carrier shutdown phase 2	CellPwrSavin gAlgo.Carrier DynShutdown nPh2Sw set to CAPACITY_CEL LL	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over dynamic carrier shutdown phase 2 when both functions are enabled. Dynamic carrier shutdown phase 2 does not take effect when RF module fast deep dormancy is in effect.
LTE FDD LTE TDD	Low power consumption mode	CellLowPower.LowPwrSwitch	<i>Green Site</i>	RF module fast deep dormancy takes precedence over low power consumption mode when both functions are enabled. Low power consumption mode does not take effect when RF module fast deep dormancy is in effect.
LTE FDD LTE TDD	SOC level-based low power consumption mode	IPWRM.SOC CLVLADJS W SOCLVL_B ASED_LO W_PWR_S W option of the CellLowPower.LowPwrConsum eModeExt Sw parameter	<i>Green Site</i>	RF module fast deep dormancy takes precedence over SOC level-based low power consumption mode when both functions are enabled. SOC level-based low power consumption mode does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Dynamic Power Sharing Between LTE Carriers	LTE_DYN_POWER_SHARING_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter	<i>Dynamic Power Sharing Between LTE Carriers in eRAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over Dynamic Power Sharing Between LTE Carriers when both functions are enabled. Dynamic Power Sharing Between LTE Carriers does not take effect when RF module fast deep dormancy is in effect.
LTE FDD	Extreme power sharing	LTE_EXTREME_POWER_SHARING_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter	<i>Dynamic Power Sharing Between LTE Carriers in eRAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over extreme power sharing when both functions are enabled. Extreme power sharing does not take effect when RF module fast deep dormancy is in effect.
LTE FDD LTE TDD	Smart carrier selection based on virtual grids	SMART_CARRIER_SELECTION_SW option of the MultiCarrierUnifiedSch.MultiCarrierUnifiedSchSw parameter	<i>Multi-carrier Unified Scheduling in eRAN Feature Documentation</i>	The progress of virtual grid model building will be affected after RF module fast deep dormancy takes effect.
LTE FDD	Optimization-based scheduling	TIME_FREQ_2D_UNIF_OPT_SCH_SW and TIME_FREQ_UNIF_OPT_SCH_ENH_SW options of the CellMultiCarrierUnifiedSch.CellMultiCarrierUnifiedSchSw parameter	<i>Multi-carrier Unified Scheduling in eRAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over optimization-based scheduling when both functions are enabled. Optimization-based scheduling does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Optimization-based carrier selection	OPT_MODEL_BASED_CARR_SEL_SW option of the MultiCarrUnifiedSch.MultiCarrierUnifiedSchEnSw parameter	<i>Multi-carrier Unified Scheduling</i> in <i>eRAN Feature Documentation</i>	When the optimization-based carrier selection function is enabled, the base station adjusts UE distribution across cells working in different bands, which leads to changes in the traffic volumes in cells in the dormancy state. When the traffic volume in a cell in the dormancy state decreases, the dormancy duration may increase accordingly. As a result, the average downlink UE throughput may decrease.
LTE FDD LTE TDD	eMTC introduction	EMTC_SWITCH option of the CellEmtcAlgo.EmtcAlgorithmSwitch parameter	<i>eMTC</i> in <i>eRAN Feature Documentation</i>	An eMTC-only cell (specified by the EMTC_SWITCH and EMTC_ONLY_CELL_SWITCH options of the CellEmtcAlgo.EmtcAlgorithmSwitch parameter) cannot be configured as a basic-layer cell.
LTE FDD LTE TDD	Carrier aggregation	None	<i>Carrier Aggregation</i> in <i>eRAN Feature Documentation</i>	Cells in the RF module fast deep dormancy state do not participate in CA.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	PIM Interference Cancellation (FDD) (Trial)	PIM_INTERF_CANCELLATION_SW option of the PimInterfSmartAvoid.PimAlgoSwitch parameter	<i>Interference Detection and Suppression in eRAN Feature Documentation</i>	After RF module fast deep dormancy takes effect in an interfering or victim cell, the eNodeB evaluates whether to continue PIM interference cancellation in each cell. If PIM interference persists after RF module fast deep dormancy takes effect, PIM interference cancellation continues. If PIM interference disappears after RF module fast deep dormancy takes effect, PIM interference cancellation is stopped to reduce resource consumption caused by periodic PIM interference characteristic training and interference characteristic extraction.
LTE FDD	RF PIM interference cancellation	RF_PIM_CANCELLATION_SW option of the RRUEXTINFO.RruOptlFeatureSw parameter	<i>Interference Detection and Suppression in eRAN Feature Documentation</i>	After RF module fast deep dormancy takes effect in an interfering or victim cell, PIM interference characteristics change. In this case, the PIM interference cancellation process is restarted, causing fluctuations in PIM interference cancellation performance.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Intelligent PIM interference avoidance	PIM_INTERF_SMART_AVOID_SW option of the PimInterfSmartAvoid.PimAlgoSwitch parameter	<i>Interference Detection and Suppression in eRAN Feature Documentation</i>	RF module fast deep dormancy is terminated during PIM interference detection.
LTE TDD	Clock synchronization detection (TDD)	ClkDetect.ClkAsyncDetectSwitch	<i>Synchronization</i>	If clock out-of-synchronization occurs during the dormancy period, this issue may fail to be detected.
LTE TDD	IMB beam quantity adaptation	IMB_BEAM_NUM_SWITCH_OVER_SW option of the BeamNumAdjAlgo.BeamNumSwitchoverSwitch parameter	<i>Intelligent Multi-Beam of 8T8R (TDD) in eRAN Feature Documentation</i>	A TDD cell with IMB beam quantity adaptation in Intelligent Multi-Beam of 8T8R enabled cannot be configured as a basic-layer cell.
LTE FDD LTE TDD	Cell System Information Switch	CellSiMap.SiSwitch	None	A cell whose ForbidCellSiSwitch option of the CellSiMap.SiSwitch parameter is selected cannot be configured as a basic-layer cell.
LTE FDD LTE TDD	Work mode	Cell.WorkMode	None	An FDD cell with the Cell.WorkMode parameter set to DL_ONLY or LAA cannot be configured as a basic-layer cell. A TDD cell with the Cell.WorkMode parameter set to LAA cannot be configured as a basic-layer cell.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over energy saving based on flexible frequency-domain scheduling when both functions are enabled. Energy saving based on flexible frequency-domain scheduling does not take effect when RF module fast deep dormancy is in effect.
LTE FDD	Beam number adjustment	BEAM_NUM_ADJ_SWITCH option of the SectorSplitGroup.SectorSplitSwitch parameter	<i>Massive MIMO Enhancements (FDD) in eRAN Feature Documentation</i>	FDD cells with beam number adjustment enabled cannot be configured as basic-layer cells.
LTE FDD	Massive MIMO downlink coordinated power control	MM_DL_COORD_PC_SW option of the SectorSplitGroup.IntelligentBeamSwitch parameter	<i>Massive MIMO Enhancements (FDD) in eRAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over massive MIMO downlink coordinated power control when both functions are enabled. Massive MIMO downlink coordinated power control does not take effect when RF module fast deep dormancy is in effect.
LTE FDD	Inter-carrier power sharing	INTER_CARRIER_PWR_SHR_SWITCH option of the SectorSplitGroup.IntelligentBeamSwitch parameter	<i>Massive MIMO Enhancements (FDD) in eRAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over inter-carrier power sharing when both functions are enabled. Inter-carrier power sharing does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	LTE FDD and NR Flash Dynamic Power Sharing	LTE_NR_DYN_POWER_SHARING_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	RF module fast deep dormancy takes precedence over LTE FDD and NR Flash Dynamic Power Sharing when both functions are enabled. LTE FDD and NR Flash Dynamic Power Sharing does not take effect when RF module fast deep dormancy is in effect.
LTE TDD	LTE TDD and NR Flash Dynamic Power Sharing	LTE_NR_DYN_POWER_SHARING_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	RF module fast deep dormancy takes precedence over LTE TDD and NR Flash Dynamic Power Sharing when both functions are enabled. LTE TDD and NR Flash Dynamic Power Sharing does not take effect when RF module fast deep dormancy is in effect.
LTE TDD	L/NR Inter-band Deep Coordination	LNR_INTER_BAND_DEEP_COORD_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	RF module fast deep dormancy takes precedence over L/NR Inter-band Deep Coordination when both functions are enabled. L/NR Inter-band Deep Coordination does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE TDD	LTE and NR multi-band power adaptive scheduling	• LTE: LNR_MULT I_BAND_P WR_ADPT _SCH_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter • NR: LNR_MULT I_BAND_P WR_ADPT _SCH_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	RF module fast deep dormancy takes precedence over LTE and NR multi-band power adaptive scheduling when both functions are enabled. LTE and NR multi-band power adaptive scheduling does not take effect when RF module fast deep dormancy is in effect.
LTE FDD	DSS and NR Flexible Refarming	DSS_FLEX_RE FARM_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>	DSS and NR Flexible Refarming cannot take effect in spectrum sharing cells that are configured as basic-layer cells for RF module fast deep dormancy when both functions are enabled.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	LTE-NR multi-dimensional scheduling optimization	LNR_MULTI_D_OPT_SCH_SW option of the GigaBandCell . GigaBandCellAlgoSw parameter	<i>LTE FDD and NR Inter-RAT Unified Scheduling</i>	RF module fast deep dormancy takes precedence over LTE-NR multi-dimensional scheduling optimization when both functions are enabled. LTE-NR multi-dimensional scheduling optimization does not take effect when RF module fast deep dormancy is in effect.
LTE FDD LTE TDD	Specified service carrier	Cell.SpecifiedCellFlag	<i>WBB in eRAN Feature Documentation</i>	WBB and non-emergency-call MBB UEs can only be handed over to the corresponding specified-service cells. RF module fast deep dormancy enables UEs to be transferred out of a capacity-layer cell when the trigger conditions are met. If the number of UEs in the cell after the transfer is greater than the value of the CellPwrSavingPolicy.ForcibleTriggerUeNumThld parameter, RF module fast deep dormancy cannot take effect; if the number of UEs in the cell after the transfer is less than or equal to the value of this parameter, RF module fast deep dormancy can take effect and the remaining UEs in the cell are directly released.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	LNR CRS muting	- On the LTE side: LNR_CRS_MUTE_SW option of the CellAlgoExtSwitch.LnrZeroBufferzoneSw parameter - On the NR side: LNR_CRS_INTRF_REDUCE_SW option of the NRDUCellResShrCfg.LnrZeroBufferZoneSw parameter	<i>LTE and NR Zero Bufferzone</i>	RF module fast deep dormancy takes precedence over LNR CRS muting when both functions are enabled. LNR CRS muting does not take effect when RF module fast deep dormancy is in effect.
LTE FDD	Initial transmission power reduction	INIT_TRANSPower_REDUCE_SW option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over initial transmission power reduction when both functions are enabled.
LTE FDD LTE TDD	Retransmission power reduction	RETRANS_Power_REDUCE_SW option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over retransmission power reduction when both functions are enabled.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	CMAS	CellAlgoSwitch.CmasSwitch	CMAS in eRAN Feature Documentation	The following applies when cells enter RF module fast deep dormancy: • With PWS message-triggered exit of cells from energy saving disabled, CMAS warning notifications cannot be properly sent in LTE cells in the dormancy state. However, UEs can still receive CMAS warning notifications from other cells. • With PWS message-triggered exit of cells from energy saving enabled, cells exit the dormancy state upon receipt of a PWS message.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	ETWS	None	<i>ETWS in eRAN Feature Documentation</i>	The following applies when cells enter RF module fast deep dormancy: • With PWS message-triggered exit of cells from energy saving disabled, ETWS warning notifications cannot be sent in LTE cells in the dormancy state. However, UEs can still receive ETWS warning notifications from other cells. • With PWS message-triggered exit of cells from energy saving enabled, cells exit the dormancy state upon receipt of a PWS message.
LTE FDD	GSM and LTE Buffer Zone Optimization	CellAlgoSwitch.AvoidInterfSwitch	<i>GSM and LTE Buffer Zone Optimization</i>	Cells in the RF module fast deep dormancy state do not participate in GSM and LTE Buffer Zone Optimization.
LTE FDD	GSM and LTE Spectrum Concurrency	SpectrumCloud.SpectrumCloudSwitch set to GL_SPECTRUM_CONCURRENCY	<i>GSM and LTE Spectrum Concurrency</i>	Cells in the RF module fast deep dormancy state do not participate in GSM and LTE Spectrum Concurrency.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	GSM and LTE Spectrum Concurrency Phase 2	GL_SPCT_CONCURRENCY_PH2_SW option of the SpectrumCloudEnhSwitch parameter	<i>GSM and LTE Spectrum Concurrency</i>	Cells in the RF module fast deep dormancy state do not participate in GSM and LTE Spectrum Concurrency Phase 2.
LTE FDD	UMTS and LTE Spectrum Sharing Based on DC-HSDPA (LTE FDD)	SpectrumCloud.SpectrumCloudSwitch set to DC_HSDPA_BASED_UL_SPECTRUM_SHR	<i>UMTS and LTE Spectrum Sharing Based on DC-HSDPA</i>	RF module fast deep dormancy cannot take effect in cells where UMTS and LTE Spectrum Sharing Based on DC-HSDPA is enabled.
LTE FDD	UMTS and LTE Spectrum Sharing (LTE FDD)	SpectrumCloud.SpectrumCloudSwitch set to UL_SPECTRUM_SHARING	<i>UMTS and LTE Spectrum Sharing</i>	UMTS and LTE Spectrum Sharing does not take effect when RF module fast deep dormancy is in effect.
LTE FDD	UMTS and LTE Spectrum Sharing Phase 2 (LTE FDD)	UL_SPECTRUM_SHR_PH2_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter	<i>UMTS and LTE Spectrum Sharing</i>	UMTS and LTE Spectrum Sharing Phase 2 does not take effect when RF module fast deep dormancy is in effect.
LTE FDD	UMTS and LTE Zero Bufferzone (LTE FDD)	UMTS_LTE_ZERO_BUFFER_ZONE_SW option of the ULZeroBufferZone.ZeroBufferZoneSwitch parameter	<i>UMTS and LTE Zero Bufferzone</i>	UMTS and LTE Zero Bufferzone does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Cell RX channel interference and noise query	STR CELLRXINTRF NOIQUERY	<i>MIMO in eRAN Feature Documentation</i>	If LTE load-based RF module fast deep dormancy is in effect and reporting of invalid uplink interference counter values is enabled (controlled by the UL_IF_INVALID_IN_CARROFF_SW option of the EnodebCounterParaGrp.EnodebCounterAlgoSwitch parameter), then RX channel interference and noise query for the cell will not produce any results.

7.3.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

Boards

- Main control boards: Only the UMPTe, UMPTg, UMPTga, UMPTha, and UMPThb series boards support this function.
- Baseband processing units: Only the UBBPg and UBBPh series boards support this function. The UBBPg1 and UBBPg1a do not support this function.

RF Modules

The AAU5685md is compatible with RF module fast deep dormancy.

7.3.3.4 Networking

- An RF module shared by multiple RATs can enter dormancy only when the dormancy conditions of all involved RATs are met.
- In multi-operator scenarios, the procedure of entering this function for both primary and secondary operators is determined by the networking of the primary operator.

- In a separate-MPT base station, if two main control boards are configured with different dormancy types, the expected dormancy type does not take effect on the RF module shared by two sides of the base station. The same dormancy type must be configured for the two main control boards.
- Requirements of NR load-based RF module fast deep dormancy:
In NSA networking, if the capacity layer and basic layer are deployed at different base stations, information is transmitted over the Xn interface. If no Xn interface is deployed between the gNodeBs enabled with this function, IP addresses need to be allocated for Xn setup. Otherwise, NR load-based RF module fast deep dormancy cannot take effect in inter-base-station scenarios.
- Requirements of LNR load-based RF module fast deep dormancy:
 - X2 links need to be set up in inter-base-station SA networking. X2 links can be set up automatically or manually. Automatic setup is recommended. For details about X2 configurations, see *X2 Self-Management in SA Networking*.
 - In NSA networking, the NSA DC capability must be enabled for all basic- and capacity-layer cells. To do so, select the **NSA_DC_CAPABILITY_SWITCH** option of the **NsaDcMgmtConfig.NsaDcAlgoSwitch** parameter on the eNodeB and turn on the switch controlled by the **NRCCellAlgoSwitch.NsaDcSwitch** parameter on the gNodeB.
 - In NSA and SA hybrid networking, the configuration and activation procedures of this function are the same as those in NSA networking. The load information is transmitted between the eNodeB and gNodeB over the X2 interface.
- When LNR or NR load-based fast deep dormancy is enabled and the NR capacity-layer cell is MOCN-enabled, ensure that the operator group configured for the basic-layer cells (co-coverage neighboring LTE or NR cells) contains the operator group of the capacity-layer cell. MOCN refers to multi-operator core network.

7.3.3.5 Cells

- NR cells:
 - **NRDUCell.NrDuCellNetworkingMode** is set to **NORMAL_CELL** and **NRDUCellTimeIntrf.TimeDomainRsvMode** is set to **DEFAULT**.
 - **NRDUCellTrp.TxRxMode** is set to **32T32R** or **64T64R**.
 - SDL cells (with **NRDUCell.DuplexMode** set to **CELL_SDL**) do not support LNR load-based RF module fast deep dormancy.
 - SUL cells (with **NRDUCell.DuplexMode** set to **CELL_SUL**) do not support NR or LNR load-based RF module fast deep dormancy.
 - SDL cells cannot serve as basic-layer cells. If an SDL cell is configured as a basic-layer cell, dormancy-related functions cannot take effect in the corresponding capacity-layer cells.
- LTE cells:
 - NB-IoT cells are incompatible with this function.
 - For FDD cells, **CELL.TxRxMode (LTE eNodeB, 5G gNodeB)** is set to **32T32R**.

- For TDD cells, **CELL.TxRxMode (LTE eNodeB, 5G gNodeB)** is set to **32T32R** or **64T64R**.
- RF module fast deep dormancy cannot take effect in cells where UMTS and LTE Spectrum Sharing Based on DC-HSDPA is enabled.

7.3.3.6 Others

The working mode of RF modules (specified by **RRU.RS (5G gNodeB, LTE eNodeB)**) must be set to **NO**, **TDL** or **LO**, or any combination of these modes.

7.3.4 Operation and Maintenance

7.3.4.1 Data Configuration

7.3.4.1.1 Data Preparation

Table 7-35 describes the parameter shared by all sub-functions. The parameters that need to be separately set for each sub-function are as follows:

- **Table 7-36** and **Table 7-37** describe the parameters used for NR load-based RF module fast deep dormancy.
- **Table 7-36**, **Table 7-38**, and **Table 7-39** describe the parameters used for LNR load-based RF module fast deep dormancy.
- **Table 7-40** describes the parameters used for LTE load-based RF module fast deep dormancy.

Table 7-35 Parameter shared by all sub-functions

Parameter Name	Parameter ID	Setting Notes
Dormancy Switch	RRUEXTINFO.RRUDOROPTFEATSW (5G gNodeB, LTE eNodeB)	To enable RF module fast deep dormancy, select the FAST_DEEP_DORM_SW option.

Parameter Name	Parameter ID	Setting Notes
Equipment Dormancy Optional Feature Switch	EQUIPMENT.EQMDO ROPTFEATSW (5G gNodeB, LTE eNodeB)	Deselect the GU_DC_SW option, which is the default setting. If the option is deselected and a configuration command is executed to trigger dormancy pending or dormancy pending is enabled on the local end that works in UMTS, then dormancy pending will take effect on the entire base station. In this case, the dormancy duration of the entire base station is shortened. This option must also be deselected for the NE to which the RF module belongs if the RRU.RS parameter is configured to include GSM or UMTS and the RRU.DORMANCYSW parameter is set to ON , ON_SUPER , or ON_DEEP_SUPER .

Table 7-36 Parameters used for NR or LNR load-based RF module fast deep dormancy

Parameter Name	Parameter ID	Setting Notes
Power Saving Policy Index	NRDUCellPowerSav- ing.PowerSavingPoli- cyIndex	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Start Time	NRDUCellPowerSaving.StartTime	Set this parameter based on the network plan and the following requirements: - For the same policy of the same energy saving type, the interval between the stop time and the start time must be longer than 20 minutes. If the stop time is earlier than or equal to the start time, the stop time is interpreted as occurring on the next day. For example: If the start time is 8:00 a.m. and the stop time is 7:00 a.m., the power saving function is enabled from 8:00 a.m. to 7:00 a.m. on the next day. - For different policies of the same energy saving type, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving type needs to be enabled after policy A of this type has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A. - The period for RF module fast deep dormancy must not overlap with the period specified when the NRDUCellPowerSaving.PowerSavingType parameter is set to TIMING_CARRIER_SHUTDOWN for the local cell.
Stop Time	NRDUCellPowerSaving.StopTime	
Downlink PRB Threshold	NRDUCellPowerSaving.DlPrbThld	Use the recommended value.
Uplink PRB Threshold	NRDUCellPowerSaving.UlPrbThld	Use the recommended value.
Neighboring Cell Uplink PRB Offset	NRDUCellPowerSaving.NCellUlPrbOffset	Use the recommended value.
User Number Offset	NRDUCellPowerSaving.UserNumOffset	Use the recommended value.

Parameter Name	Parameter ID	Setting Notes
UE Number Threshold	NRDUCellPowerSaving.UserNumThld	Use the recommended value. The maximum value is 150 .
SUL User Number Threshold	NRDUCellPowerSaving.SulUserNumThld	You are advised to set this parameter to its recommended value. - This parameter is invalid when it is set to 65535. - If the FLEX_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter is deselected for a cell and the cell has an NRDUCellPowerSaving MO with power saving type set to NR or LNR load-based RF module dormancy, the corresponding NRDUCellPowerSaving.SulUserNumThld parameter must be set to 65535.
Evaluation Period Granularity	NRDUCellPowerSaving.EvalPeriodGranularity	Use the recommended value.
Triggering Evaluation Period	NRDUCellPowerSaving.ShutdownTrigJudgePeriod	Use the recommended value.
Exit Evaluation Period	NRDUCellPowerSaving.ExitEvaluationPeriod	Use the recommended value.

Parameter Name	Parameter ID	Setting Notes
Second-Level Trigger Evaluation Period	NRDUCellPowerSaving.SecondLevelTrigEvalPrd	Use the recommended value. - This parameter can be configured and take effect only when the NRDUCellPowerSaving.EvalPeriodGranularity parameter is set to SECOND_LEVEL. - When the NRDUCellPowerSaving.PowerSavingType parameter is set to INTRA_NR_DORMANCY_BY_LOAD and the NRDUCellPowerSaving.EvalPeriodGranularity parameter is set to SECOND_LEVEL, the NRDUCellPowerSaving.SecondLevelTrigEvalPrd parameter must be set to a value greater than or equal to 6. - The smaller the value of this parameter, the higher the probability of triggering power saving. The larger the value of this parameter, the lower the probability.
Second-Level Exit Evaluation Period	NRDUCellPowerSaving.SecondLevelExitEvalPrd	Use the recommended value. - This parameter can be configured and take effect only when the NRDUCellPowerSaving.EvalPeriodGranularity parameter is set to SECOND_LEVEL. - The smaller the value of this parameter, the higher the probability of triggering exit from power saving and the smaller the power saving gains. The larger the value of this parameter, the lower the probability.
Power Saving Penalty Duration	NRDUCellPowerSaving.PwrSavingPenaltyTime	Use the recommended value.

Parameter Name	Parameter ID	Setting Notes
Power Saving Cell Flag	NRCellRelation.PowerSavingCellFlag	This parameter specifies the flag of a neighboring cell for load-based fast deep dormancy. Set this parameter based on the network plan. When NRDUCellPowerSaving.PowerSavingType is set to NR_FAST_DORMANCY_BY_LOAD or LNR_FAST_DORMANCY_BY_LOAD and NRDUCellPowerSaving.EvalPeriodGranularity is set to SECOND_LEVEL for a cell, the numbers of corresponding NRCellRelation MOs with NRCellRelation.PowerSavingCellFlag set to COVERAGE and CAPACITY must each be less than or equal to 16.
No Removal Flag	NRCellRelation.NoRmvFlag	When automatic removal of neighboring cells in ANR is enabled (with the NR_NR_ANR_DEL_SW option of the NRCellAlgoSwitch.AnrSwitch parameter selected), it is recommended that the NRCellRelation.NoRmvFlag parameter be set to FORBID_ANR_RMV for the mutual neighbor relationships between the capacity-layer and basic-layer cells.
PLMN Reserved Flag	NRExternalNCellPlmn.PlmnReservedFlag	In multi-operator sharing scenarios, set this parameter for capacity-layer cells based on the network plan.
Power Saving Algorithm Switch	gNBpwrSavingAlgo.PwrSavingAlgoSwitch	Assume that the cell load simulation function is enabled. To forbid the RF module serving the local cell to enter dormancy and allow the RF modules serving the cells not enabled with cell load simulation to enter dormancy, select the SIMU_LOAD_DORMANCY_IMP_SW option.

Table 7-37 Parameters used for NR load-based RF module fast deep dormancy

Parameter Name	Parameter ID	Setting Notes
Power Saving Algorithm Switch	NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch	- To enable NR load-based RF module fast deep dormancy, select the NR_FAST_DORMANCY_BY_LOAD_SW option. - To enable NR load-based RF module fast deep dormancy service assurance, select the options for capacity-layer and basic-layer cells as follows: - Capacity-layer cell: NR_FAST_DORM_LOAD_ASSUR_SW - Basic-layer cell: PWR_SAVE_SERVICE_ASSUR_SW When NR load-based RF module fast deep dormancy takes effect, it can exit quickly if UEs with heavy downlink traffic are present. - To enable NR load-based RF module fast deep dormancy load assurance, select the NR_FAST_DORM_LOAD_ASSUR_SW option for capacity-layer cells. The NR_FAST_DORMANCY_BY_LOAD_SW option must also be selected.
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	Set this parameter to NR_FAST_DORMANCY_BY_LOAD .
Downlink PRB Offset	NRDUCellPowerSaving.DlPrbOffset	Use the recommended value. The sum of NRDUCellPowerSaving.DlPrbThld and NRDUCellPowerSaving.DlPrbOffset cannot exceed 100.

Parameter Name	Parameter ID	Setting Notes
SUL User Number Offset	NRDUCellPowerSaving.SulUserNumOffset	You are advised to set this parameter to its recommended value. If NRDUCellPowerSaving.SulUserNumThld is not set to 65535, the sum of NRDUCellPowerSaving.SulUserNumThld and NRDUCellPowerSaving.SulUserNumOffset must be less than 65535, and NRDUCellPowerSaving.SulUserNumThld must be less than NRDUCellPowerSaving.UserNumThld.
Power Save Service Assurance Level	NRDUCellPowerSaving.PwrSaveServiceAssurLevel	To use NR load-based RF module fast deep dormancy service assurance: - The value LEVEL4 is recommended when the impact of NR load-based RF module fast deep dormancy on service experience needs to be minimized. - The value LEVEL0 is recommended when the gains of NR load-based RF module fast deep dormancy need to be maximized. A larger value of this parameter results in a smaller impact on user experience, a shorter fast-deep-dormancy duration, and lower energy saving gains.
Carrier Shutdown Fast Eval PRB Threshold	NRDUCellMobility.CarrrShutdFastEvalPrbThld	To use NR load-based RF module fast deep dormancy service assurance, set this parameter to the recommended value. Set this parameter for both capacity-layer and basic-layer cells.

Parameter Name	Parameter ID	Setting Notes
Fast Carrier Shutdown Event Evaluation PRB Thld	NRDUCellMobility.Ca rrShdnEventEvalPrbThld	To use NR load-based RF module fast deep dormancy load assurance, set this parameter to the sum of NRDUCellPowerSaving.DlPrbThld and NRDUCellPowerSaving.DlPrbOffset . An improper setting can easily prevent quick exit from fast deep dormancy or cause frequent entries into and exits from fast deep dormancy. This parameter needs to be set only for basic-layer cells.
Necessary HO Prepare Fail Punish Timer	gNBMobilityComm-Param.NecHoPrepFailPunishTimer	The value 30 is recommended.
Optimized HO Prepare Fail Punish Timer	gNBMobilityComm-Param.OptHoPrepFailPunishTimer	It is recommended that this parameter be set to 30 in scheduled RF module deep dormancy scenarios.

Table 7-38 LTE parameters used for LNR load-based RF module fast deep dormancy

Parameter Name	Parameter ID	Setting Notes
Power Saving Algorithm Switch	CellPwrSavingAlgo.PwrSavingAlgoSwitch	To enable LNR load-based RF module fast deep dormancy, select the LNR_LOAD_BASED_DORMANCY_SW option.
Aggregation Attribute	NrNRRelationship.AggregationAttribute	Set the CARRIER_SHUTDOWN_OVERLAP_FLAG option of this parameter to the recommended value.
Power Saving Algorithm Switch	EnodebPwrSavingAlgo.PwrSavingAlgoSwitch	To enable LTE sector split cells to serve as basic-layer cells for load-based dormancy, select the SPLIT_CELL_RF_DORMANCY_SW option.

Parameter Name	Parameter ID	Setting Notes
Interface SON Configuration Switch	eNodeBOamParam.InterfaceSonConfigSwitch	To enable the eNodeB to set up X2 links with gNodeBs that do not have NSA cells, select the LNR_SA_GNB_X2_SETUP_SW option.
Power Saving Algorithm Switch	EnodebPwrSavingAlgo.PwrSavingAlgoSwitch	To enable inter-base-station power saving between LTE and SA NR cells in LTE and NR scenarios, select the LNR_INTER_BS_PWR_SAVE_SW option. The LNR_SA_GNB_X2_SETUP_SW option of the eNodeBOamParam.InterfaceSonConfigSwitch parameter must also be selected.

Table 7-39 NR parameters used for LNR load-based RF module fast deep dormancy

Parameter Name	Parameter ID	Setting Notes
Power Saving Switch	NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch	To enable LNR load-based RF module fast deep dormancy, select the LNR_FAST_DORMANCY_BY_LOAD_SW option.
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	Set this parameter to LNR_FAST_DORMANCY_BY_LOAD .
E-UTRAN Downlink PRB Threshold	NRDUCellPowerSaving.EutranDlPrbThld	Set this parameter based on the network plan.
Neighboring Cell Uplink PRB Threshold	NRDUCellPowerSaving.NCellUlPrbThld	Set this parameter based on the network plan. The sum of NRDUCellPowerSaving.NCellUlPrbThld and NRDUCellPowerSaving.NCellUlPrbOffset must be less than or equal to 100.
E-UTRAN Downlink PRB Offset	NRDUCellPowerSaving.EutranDlPrbOffset	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
E-UTRAN Carrier Shutdown Ratio	NRDUCellPowerSav- ing.EutranCarrShutdown- Ratio	Set this parameter based on the network plan.
E-UTRAN Carrier Shutdown Offset	NRDUCellPowerSav- ing.EutranCarrShutdown- nExitOfs	To ensure that the exit conditions for energy saving are not triggered when the entry conditions are met, the following conditions must be met simultaneously: • NRDUCellPowerSav- ing.EutranCarrShutdown- Ratio > NRDUCellPowerSav- ing.EutranCarrShutdown- nExitOfs • (NRDUCellPowerSav- ing.EutranCarrShutdown- Ratio – NRDUCellPowerSav- ing.EutranCarrShutdown- nExitOfs) > (100 – NRDUCellPowerSav- ing.EutranCarrShutdown- Ratio)
E-UTRAN NR User Number Threshold	NRDUCellPowerSav- ing.EutranNrUserNumThld	Set this parameter based on the network plan. If this parameter is set to 65535, this parameter and the NRDUCellPowerSav- ing.EutranNrUserNumThldOfs parameter are invalid.
E-UTRAN NR User Number Threshold Offset	NRDUCellPowerSav- ing.EutranNrUserNumThldOfs	Set this parameter based on the network plan.
E-UTRAN NR User Downlink PRB Threshold	NRDUCellPowerSav- ing.EutranNrUserDlPrbThld	Set this parameter based on the network plan.
E-UTRAN NR User Uplink PRB Threshold	NRDUCellPowerSav- ing.EutranNrUserUlPrbThld	Set this parameter based on the network plan.
E-UTRAN NR User Downlink PRB Threshold Offset	NRDUCellPowerSav- ing.EutranNrUserDlPrbThldOfs	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
E-UTRAN NR User Uplink PRB Threshold Offset	NRDUCellPowerSaving.EutranNrUserULPrbThldOfs	Set this parameter based on the network plan.
Power Saving Overlapping Indicator	NRCellEutraNRelation.PowerSavingOverlapInd	Configure the LNR_CARR_SHUTDOWN_OVERLAP_IND option based on the network plan. This parameter is mandatory only in SA networking.
Power Saving Algorithm Switch	gNBPwrSavingAlgo.PwrSavingAlgoSwitch	To enable inter-RAT level-by-level energy saving, select the INTER_RAT_GRAD_PWR_SAVING_SW option.
Power Saving Algorithm Switch	gNBPwrSavingAlgo.PwrSavingAlgoSwitch	When the LNR_LOAD_DORM_PRIM_OP_RECOG_SW option is selected, primary operator recognition for LNR load-based dormancy in MOCN scenarios is enabled. Only neighboring LTE cells served by the same operator as the primary operator of an NR cell can be configured as basic-layer cells for LNR load-based dormancy in the NR cell.
X2 Config Switch	gNBInterfaceParam.X2ConfigSw	To allow X2 link setup in SA scenarios, select the X2_SETUP_FOR_SA_SW option.

Parameter Name	Parameter ID	Setting Notes
Power Saving Algorithm Switch	gNBPwrSavingAlgo.PwrSavingAlgoSwitch	- To enable energy saving for SA NR cells in LTE-NR inter-base-station energy saving scenarios, select the SA_LNR_INTER_BS_PWR_SAVE_SW option. The X2_SETUP_FOR_SA_SW option of the gNBInterfaceParam.X2ConfigSw parameter must also be selected. - When the SA_LNR_INTER_BS_PWR_SAVE_SW option is deselected, the NRCellEutraNRelation.EnodebId parameter must have the same value in all entries where the LNR_CARR_SHUTDOWN_OVERLAP_IND option of the NRCellEutraNRelation.PowerSavingOverlapInd parameter is set to Yes.

Table 7-40 Parameters used for LTE load-based RF module fast deep dormancy

Parameter Name	Parameter ID	Setting Notes
Power Saving Algorithm Switch	CellPwrSavingAlgo.PwrSavingAlgoSwitch	To enable LTE load-based RF module fast deep dormancy, select the LTE_FAST_DORMANCY_BY_LOAD_SW option.
Power Saving Type	CellPwrSavingPolicy.PowerSavingType	Set this parameter to LTE_FAST_DORMANCY_BY_LOAD .
Policy Index	CellPwrSavingPolicy.PolicyIndex	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Start Time	CellPwrSavingPolicy.StartTime	Set these parameters based on the network plan and the following requirements: - For the same policy of the same energy saving type, the interval between the stop time and the start time must be longer than 20 minutes. If the end time is earlier than the start time, the end time is interpreted as occurring on the next day. For example: If the start time is 8:00 a.m. and the stop time is 7:00 a.m., the power saving function is enabled from 8:00 a.m. to 7:00 a.m. on the next day. - For different policies of the same energy saving type, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving type needs to be enabled after policy A of this type has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A.
Stop Time	CellPwrSavingPolicy.StopTime	
Trigger Judge Period	CellPwrSavingPolicy.TriggerJudgePeriod	Use the recommended value.
Downlink PRB Threshold	CellPwrSavingPolicy.DlPrbThld	Use the recommended value. CellPwrSavingPolicy.DlPrbThld + CellPwrSavingPolicy.DlPrbOffset ≤ 100.
Downlink PRB Offset	CellPwrSavingPolicy.DlPrbOffset	

Parameter Name	Parameter ID	Setting Notes
Uplink PRB Threshold	CellPwrSavingPolicy.UlPrbThld	Use the recommended value. The sum of CellPwrSavingPolicy.UlPrbThld and CellPwrSavingPolicy.UlPrbOffset must be less than or equal to 100.
Uplink PRB Offset	CellPwrSavingPolicy.UlPrbOffset	
UE Number Threshold	CellPwrSavingPolicy.UENumThld	Use the recommended value.
Forcible Trigger UE Number Threshold	CellPwrSavingPolicy.ForcibleTriggerUeNumThld	Use the recommended value.
Exit Judge Period	CellPwrSavingPolicy.ExitJudgePeriod	Use the recommended value. The smaller the value of this parameter, the higher the probability of triggering exit from power saving and the smaller the power saving gains. The larger the value of this parameter, the lower the probability.
Penalty Time	CellPwrSavingPolicy.PenaltyTime	Use the recommended value.
Inter-Frequency Mobility Load Balancing Threshold	CellMLB.InterFreqMlbThd	You are advised to set this parameter to its recommended value.
Load Offset	CellMLB.LoadOffset	You are advised to set this parameter to its recommended value.
Overlap Indicator Extension	EutranInterFreqN-Cell.OverlapIndicatorExtension	Set this parameter based on the network plan. If this parameter is set to POWER_SAVING_OVERLAP_IND , the neighboring cell has overlapping coverage with the serving cell.
Power Saving Algorithm Switch	EnodebPwrSavingAlgo.PwrSavingAlgoSwitch	Select the INTRA_RAT_GRAD_PWR_SAVING_SW option of this parameter if intra-RAT level-by-level energy saving is required.

Parameter Name	Parameter ID	Setting Notes
Inter-eNodeB Information Exchange Mode	EnodebPwrSavingAlgo.InterEnbInfoExchange-Mode	Use the recommended value. - If this parameter is set to SLOW_EXCHANGE_MODE, the slow exchange mode is used, meaning that information is exchanged at long intervals. - If this parameter is set to FAST_EXCHANGE_MODE, the fast exchange mode is used, meaning that information is exchanged at short intervals. - If Exit Judge Period is set to a value less than 12, you are advised to set this parameter to FAST_EXCHANGE_MODE.
Power Saving Algorithm Switch	EnodebPwrSavingAlgo.PwrSavingAlgoSwitch	To enable optimization for cell PLMN reservation scenarios in multi-operator sharing scenarios, select the CELL_PLMN_RSVD_OPT_SW option.
Aggregation Attribute	EutranExternalCellPlmn.AggregationAttribute	In multi-operator sharing scenarios, set this parameter based on the network plan. To set the operator as a reserved operator, select the PLMN_RESERVED_FLAG option of this parameter.
Configuration Adaptation Opt Sw	eNodeBOamParam.ConfigAdaptOptSwitch	In multi-operator sharing scenarios, set this parameter based on the network plan. To enable the cell reservation status to be automatically broadcast to neighboring base stations through X2 signaling messages, select the RSVD_FOR_OP_X2_BROADCAST_SW option of this parameter.

Parameter Name	Parameter ID	Setting Notes
Power Saving Algorithm Switch	EnodebPwrSavingAlgo.PwrSavingAlgoSwitch	To enable LTE sector split cells to serve as basic-layer cells for load-based dormancy, select the SPLIT_CELL_RF_DORMANCY_SW option.

7.3.4.1.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [7.3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

For details about configuring intra-base-station neighboring cells, see *SA Mobility Management in Connected Mode* in *5G RAN Feature Documentation*.

Setting parameters shared by sub-functions

```
//Turning on the fast deep dormancy switch
ADD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=FAST_DEEP_DORM_SW-1;
```

Activating NR load-based RF module fast deep dormancy

Assume that a cell with NrCellId 0 is a capacity-layer cell and another cell with NrCellId 10 is a basic-layer cell.

```
//Turning on the switch on the NR DU cell side
MOD NRDUCELLPWRSALGO: NrDuCellId=0,
PowerSavingAlgoSwitch=NR_FAST_DORMANCY_BY_LOAD_SW-1;
//Configuring an energy saving policy
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=10,
PowerSavingType=NR_FAST_DORMANCY_BY_LOAD, StartTime=00&00&00, StopTime=06&00&00,
DlPrbThld=25, DlPrbOffset=25, UserNumThld=20, UserNumOffset=20,
EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=1, ExitEvaluationPeriod=1,
PwrSavingPenaltyTime=30, UlPrbThld=25, NCellUlPrbOffset=25;
//Setting Power Saving Cell Flag, and (recommended) setting No Removal Flag to FORBID_ANR_RMV for
the bidirectional neighbor relationships between capacity-layer and basic-layer cells when automatic
removal of neighboring cells in ANR takes effect
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=0, CellId=10,
PowerSavingCellFlag=COVERAGE, NoRmvFlag=FORBID_ANR_RMV;
MOD NRCELLRELATION: NrCellId=10, Mcc="302", Mnc="220", gNBId=0, CellId=0,
PowerSavingCellFlag=CAPACITY, NoRmvFlag=FORBID_ANR_RMV;
```

(Optional) Multi-operator sharing scenarios

```
//Setting the PlmnReservedFlag parameter in the NR external-cell PLMN list for the capacity cell  
ADD NREXTERNALNCELLPLMN: Mcc="302", Mnc="220", gNBId=0, CellId=10, SharedMcc=xx, SharedMnc=xx,  
PlmnReservedFlag=FALSE;
```

(Optional) In SUL co-deployment scenarios

```
//Modifying the energy saving policy  
MOD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=10,  
PowerSavingType=INTRA_NR_DORMANCY_BY_LOAD, SulUserNumThld=4, SulUserNumOffset=4;
```

(Optional) Enabling NR load-based RF module fast deep dormancy service assurance

```
//Enabling service assurance  
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=0,  
PowerSavingAlgoSwitch=NR_FAST_DORM_LOAD_ASSUR_SW-1;  
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=10,  
PowerSavingAlgoSwitch=PWR_SAVE_SERVICE_ASSUR_SW-1;  
//Configuring the power saving service assurance level  
MOD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0,  
PowerSavingType=NR_FAST_DORMANCY_BY_LOAD, PwrSaveServiceAssurLevel=LEVEL4;  
//Configuring the carrier shutdown fast evaluation PRB threshold  
MOD NRDUCELLMOBILITY: NrDuCellId=0, CarrShutdFastEvalPrbThld=1;  
MOD NRDUCELLMOBILITY: NrDuCellId=10, CarrShutdFastEvalPrbThld=1;
```

(Optional) Enabling NR load-based RF module fast deep dormancy load assurance

```
//Enabling load assurance  
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=0,  
PowerSavingAlgoSwitch=NR_FAST_DORM_LOAD_ASSUR_SW-1;  
//Configuring the CarrShdnEventEvalPrbThld parameter  
MOD NRDUCELLMOBILITY: NrDuCellId=10, CarrShdnEventEvalPrbThld=50;
```

Activating LNR load-based RF module fast deep dormancy

```
//LTE side:  
//Turning on the LNR load-based RF module fast deep dormancy switch  
MOD CELLPWRSAVINGALGO: LocalCellId=0, PwrSavingAlgoSwitch=LNR_LOAD_BASED_DORMANCY_SW-1;  
//(Optional) Turning on the LTE-NR SA gNodeB X2 setup switch in inter-base-station SA networking  
MOD ENODEBOAMPARAM: InterfaceSonConfigSwitch=LNR_SA_GNB_X2_SETUP_SW-1;  
//(Optional) Turning on LTE-NR SA inter-base-station energy saving switch in inter-base-station SA  
networking  
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=LNR_INTER_BS_PWR_SAVE_SW-1;  
//(Optional, only in NSA networking, NSA and SA hybrid networking, or inter-base-station SA networking)  
Manually configuring all the neighbor relationships between LTE and NR cells with overlapping coverage  
areas with the parameters set as follows: MCC is 460, MNC is 01, and gNodeB ID is 441233  
MOD NRNRELATIONSHIP: LocalCellId=0, Mcc="460", Mnc="01", GnodebId=441233, CellId=1,  
AggregationAttribute=CARRIER_SHUTDOWN_OVERLAP_FLAG-1;  
//(Optional) Turning on the RF module dormancy switch for the LTE sector split cell that serves as the basic-  
layer cell for load-based dormancy  
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=SPLIT_CELL_RF_DORMANCY_SW-1;  
//NR side:  
//Turning on the LNR load-based RF module fast deep dormancy switch  
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=1,  
PowerSavingAlgoSwitch=LNR_FAST_DORMANCY_BY_LOAD_SW-1;  
//(Optional, required only in SA networking) Configuring all neighbor relationships in the overlapping  
coverage area with the parameters set as follows: MCC is 460, MNC is 01, and eNodeB ID is 441282  
MOD NRCELLEUTRANRELATION: NrCellId=1, Mcc="460", Mnc="01", EnodebId=441282, CellId=0,  
PowerSavingOverlapInd=LNR_CARR_SHUTDOWN_OVERLAP_IND-1;  
//Configuring an energy saving policy  
ADD NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=0, StartTime=00&00&00,  
StopTime=06&00&00, PowerSavingType=LNR_FAST_DORMANCY_BY_LOAD, DIPrbThld=5, UIPrbThld=5,  
UserNumThld=20, PwrSavingPenaltyTime=30, EutranDIPrbThld=30, EutranDIPrbOffset=30,  
NCellUIPrbThld=30, NCellUIPrbOffset=30, EutranCarrShutdownRatio=90, EutranCarrShutdownExitOfs=70,  
EutranNrUserNumThld=20, EutranNrUserNumThldOfs=20, EutranNrUserDIPrbThld=15,  
EutranNrUserDIPrbThldOfs=15, EutranNrUserUIPrbThld=15, EutranNrUserUIPrbThldOfs=15,  
EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=1, ExitEvaluationPeriod=1;  
//(Optional, in SUL co-deployment scenarios) Modifying the energy saving policy  
MOD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0,  
PowerSavingType=LNR_FAST_DORMANCY_BY_LOAD, SulUserNumThld=4;
```

```
//Optional Turning on the inter-RAT level-by-level energy saving switch
MOD GNBPWRSAVINGALGO: PwrSavingAlgoSwitch=INTER_RAT_GRAD_PWR_SAVING_SW-1;
//Optional Turning on the switch for primary operator recognition for LNR load-based dormancy
MOD GNBPWRSAVINGALGO: PwrSavingAlgoSwitch=LNR_LOAD_DORM_PRIM_OP_RECOG_SW-1;
//Optional Turning on the SA X2 setup switch in inter-base-station SA networking
MOD GNBINTERFACEPARAM: X2ConfigSw=X2_SETUP_FOR_SA_SW-1;
//Optional Turning on LTE-NR SA inter-base-station energy saving switch in inter-base-station SA
networking
MOD GNBPWRSAVINGALGO: PwrSavingAlgoSwitch=SA_LNR_INTER_BS_PWR_SAVE_SW-1;
```

Activating LTE load-based RF module fast deep dormancy

```
//Turning on the switch on the LTE cell side
MOD CELLPWRSAVINGALGO: LocalCellId=0, PwrSavingAlgoSwitch=LTE_FAST_DORMANCY_BY_LOAD_SW-1;
//Adding an energy saving policy
ADD CELLPWRSAVINGPOLICY: LocalCellId=0, PolicyIndex=0,
PowerSavingType=LTE_FAST_DORMANCY_BY_LOAD, StartTime=00&00&00, StopTime=06&00&00,
DlPrbThld=20, DlPrbOffset=20, UlPrbThld=20, UlPrbOffset=20, UENumThld=20, ExitJudgePeriod=1;
//Configuring the overlap indicator extension for an inter-frequency neighboring cell
MOD EUTRANINTERFREQNCELL: LocalCellId=0, Mcc="460", Mnc="00", eNodeBId=720800, CellId=1,
OverlapIndicatorExtension=POWER_SAVING_OVERLAP_IND-1;
//Optional Turning on the intra-RAT level-by-level energy saving switch
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=INTRA_RAT_GRAD_PWR_SAVING_SW-1;
//Optional Setting the inter-eNodeB information exchange mode
MOD ENODEBPWRSAVINGALGO: InterEnbInfoExchangeMode=FAST_EXCHANGE_MODE;

//Optional Turning on the RF module dormancy switch for the LTE sector split cell that serves as the basic-
layer cell for load-based dormancy
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=SPLIT_CELL_RF_DORMANCY_SW-1;
//Optional Setting the inter-frequency MLB threshold and load offset
MOD CELLMLB: LocalCellId=0, InterFreqMlbThld=40, LoadOffset=8;
```

(Optional) Multi-operator sharing scenarios

```
//Enabling optimization for cell PLMN reservation scenarios
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=CELL_PLMN_RSVD_OPT_SW-1;
//Setting PLMN_RESERVED_FLAG for a record in the external-cell PLMN list based on the network plan
ADD EUTRANEXTERNALCELLPLMN: Mcc="460", Mnc="11", eNodeBId=222, CellId=1, ShareMcc="460",
ShareMnc="20", AggregationAttribute=PLMN_RESERVED_FLAG-1;
//Turning on the switch for broadcast of operator-specific cell reservation status over the X2 interface based
on the network plan
MOD ENODEBOAMPARAM: ConfigAdaptOptSwitch=RSVD_FOR_OP_X2_BROADCAST_SW-1;
```

(Optional) Enabling or canceling fast deep dormancy pending (applicable to all the preceding energy saving functions)

```
//Enabling fast deep dormancy pending
PND DORMANCY: OPERATIONTYPE=PENDING, PENDINGDURATION=30;
```

Optimization Command Examples

```
//NR side:
//Changing the lengths of the penalty timers applied to resource-related handovers to 30s when the
handover success rate decreases due to NR load-based RF module fast deep dormancy
MOD GNBMOBILITYCOMMPARAM: NecHoPrepFailPunishTimer=30, OptHoPrepFailPunishTimer=30;
//Turning on the simulated load dormancy impact switch
MOD GNBPWRSAVINGALGO: PwrSavingAlgoSwitch=SIMU_LOAD_DORMANCY_IMP_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

Common parameter used for deactivation

```
//Turning off the fast deep dormancy switch
MOD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=FAST_DEEP_DORM_SW-0;
```

Deactivating NR load-based RF module fast deep dormancy

```
//Turning off the switch for NR load-based fast deep dormancy on the NR DU cell side  
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=0,  
PowerSavingAlgoSwitch=NR_FAST_DORMANCY_BY_LOAD_SW-0;
```

Deactivating LNR load-based RF module fast deep dormancy

```
//LTE side:  
//(Optional) Turning off the LTE-NR SA inter-base-station energy saving switch in inter-base-station SA  
networking  
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=LNR_INTER_BS_PWR_SAVE_SW-0;  
//(Optional) Turning off the LTE-NR SA gNodeB X2 setup switch in inter-base-station SA networking  
MOD ENODEBOAMPARAM: InterfaceSonConfigSwitch=LNR_SA_GNB_X2_SETUP_SW-0;  
//Turning off the LNR load-based RF module fast deep dormancy switch  
MOD CELLPWRSAVINGALGO: LocalCellId=0, PwrSavingAlgoSwitch=LNR_LOAD_BASED_DORMANCY_SW-0;  
//NR side:  
//(Optional) Turning off the switch for primary operator recognition for LNR load-based dormancy  
MOD GNBWPWRSAVINGALGO: PwrSavingAlgoSwitch=LNR_LOAD_DORM_PRIM_OP_RECOG_SW-0;  
//(Optional) Turning off the LTE-NR SA inter-base-station energy saving switch in inter-base-station SA  
networking  
MOD GNBWPWRSAVINGALGO: PwrSavingAlgoSwitch=SA_LNR_INTER_BS_PWR_SAVE_SW-0;  
//(Optional) Turning off the SA X2 setup switch in inter-base-station SA networking  
MOD GNBINTERFACEPARAM: X2ConfigSw=X2_SETUP_FOR_SA_SW-0;  
//Turning off the LNR load-based RF module fast deep dormancy switch  
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=1,  
PowerSavingAlgoSwitch=LNR_FAST_DORMANCY_BY_LOAD_SW-0;  
//(Optional, required only in SA networking) Removing the overlapping relationship (between the local cell  
and the neighboring LTE cell) for energy saving  
MOD NRCELLEUTRANRELATION: NrCellId=1, Mcc="460", Mnc="01", EnodebId=441282, CellId=0,  
PowerSavingOverlapInd=LNR_CARR_SHUTDOWN_OVERLAP_IND-0;  
//Removing energy saving configurations  
RMV NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=0;
```

Deactivating LTE load-based RF module fast deep dormancy

```
//Turning off the switch on the LTE cell side  
MOD CELLPWRSAVINGALGO: LocalCellId=0, PwrSavingAlgoSwitch=LTE_FAST_DORMANCY_BY_LOAD_SW-0;  
//(Optional) Turning off the intra-RAT level-by-level energy saving switch  
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=INTRA_RAT_GRAD_PWR_SAVING_SW-0;
```

(Optional) Multi-operator sharing scenarios

```
//Turning off the switch for broadcast of operator-specific cell reservation status over the X2 interface based  
on the network plan  
MOD ENODEBOAMPARAM: ConfigAdaptOptSwitch=RSVD_FOR_OP_X2_BROADCAST_SW-0;  
//Disabling optimization for cell PLMN reservation scenarios  
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=CELL_PLMN_RSVD_OPT_SW-0;
```

7.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.3.4.2 Activation Verification

The activation of this function can be verified during the specified dormancy period using the following common methods:

- You can run the **DSP BRDDORMANCY** command to observe the current or latest RF module dormancy status. The values displayed in the query results

are periodically refreshed and may not always reflect the most recent updates.

- Run the **DSP BRDDORMANCY** command with the **QUERYTYPE** parameter set to **BOARD** and the **QUERYMODE** parameter set to **CURRENT**.
 - If the value of **Board Dormancy State** is **Fast Deep Dormancy** in the command output, RF module fast deep dormancy has taken effect.
 - If the value of **Board Dormancy State** is **Not in the Dormancy State** in the command output, the possible reasons can be identified based on the value of **Reason for Not Entering Fast Deep Dormancy**, as listed in [Table 7-41](#).

Table 7-41 Possible reasons why board dormancy does not take effect

Cause	Description
Maintenance Link Abnormal	The RF module is disconnected, and the BBU cannot obtain the dormancy state of the RF module.
Board Unavailable	The link is normal, and the RF module is starting up.
Dormancy Capability Not Supported by the Hardware	The RF module does not support fast deep dormancy.
Configuration Conditions Not Met	• The fast deep dormancy switch is turned off. • The working RAT does not match any available application. • The RF module is powered by the iSDU, and the customized timing power-off switch is turned on for the power supply port on the iSDU (by setting the PDPATTR.CTSSW (LTE eNodeB, 5G gNodeB) parameter to ON). • The RF module is configured as the iSDU manager, and the customized timing power-off switch is turned on for any port on the iSDU (by setting the PDPATTR.CTSSW (LTE eNodeB, 5G gNodeB) parameter to ON).
License Insufficiency	The fast deep dormancy license items of an RAT are insufficient.

Cause	Description
Upgrading	- The base station is being upgraded. - The base station is in the protection period for a successful upgrade. The protection period lasts less than 15 minutes.
Base Station in Dormancy Suspension	- Dormancy pending is triggered due to the execution of the PND DORMANCY command. - Dormancy pending is triggered due to base station reset or configuration verification. - Dormancy pending is triggered due to reliability protection.
Protection Due to Ongoing Device Maintenance and Test	- Device-related maintenance and test operations are being performed. - Protection will be canceled within 30 minutes after device-related maintenance and test operations are completed. Dormancy will be allowed after the protection is canceled.
Reliability Requirements Not Met	- The reliability requirements, such as those for solder joints or temperature difference, are not met. - The current RF module is faulty and does not meet the dormancy entry conditions. - In separate-MPT scenarios, the peer NE is stable for less than 5 minutes.
Service Conditions Not Met	Query the reasons why dormancy does not take effect for all RATs of the RF module. LTE side: For load-based fast deep dormancy, run the DSP CELLPWRSAVINGPOLICY command to query the reason why the cell does not meet the dormancy entry conditions. For details, see Table 7-45. NR side: For load-based fast deep dormancy, run the DSP NRDUCELLPOWERSAVING command to query the reason why the cell does not meet the dormancy entry conditions. For details, see Table 7-42 or Table 7-44.
Requirements Not Met by the Peer NE of the Separate-MPT Base Station	In separate-MPT scenarios, the local end meets the entry conditions, but the peer end does not. The reasons why dormancy does not take effect at the peer end can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD. Table 7-41 lists the possible reasons.

Cause	Description
Impacted Features Exist	- RF module super dormancy, RF module timing dormancy, or RF module iSDU power-off dormancy is in effect. - RF module super dormancy, RF module timing dormancy, or RF module iSDU power-off dormancy is taking effect. - RF module super dormancy, RF module timing dormancy, or RF module iSDU power-off dormancy is being exited.
Process Being Protected	- The RF module link recovery will experience a delay of approximately 5 minutes. Once the link status stabilizes, the RF module enters dormancy. - The RF module is entering the dormancy state.
Carriers Existed on the RF Module	None
Other Reason	- A plug-and-play module is used, which does not meet the function entry conditions. - The internal software is faulty, or the evaluation period defined in the software is not reached.

- Query the cell dormancy status using the following methods if the outputs of the preceding commands indicate that RF module fast deep dormancy does not take effect:
 - LTE side: Run the **DSP CELL** command. If the value of the **Cell power save state** parameter is **FAST_DEEP_DORMANCY** in the command output, the cell has entered dormancy.
 - NR side: Run the **DSP NRDUCELL** command. If the **Power Saving State** parameter has a value of **Dormancy of All RF Modules:Enabled** in the command output, the cell has entered dormancy.
 - If **Extension Power Saving Status** has a value of **NR Load-based Fast Dormancy:Enabled**, NR load-based RF module fast dormancy has taken effect in the cell.
 - If the **Extension Power Saving Status** parameter has a value of **LNR Load-based Fast Dormancy:Enabled**, LNR load-based RF module fast dormancy has taken effect in the cell.

In addition to the preceding observation methods, load-based RF module fast deep dormancy can be monitored using the **DSP NRDUCELLPOWERSAVING** and **DSP CELLPWRSAVINGPOLICY** commands. The values displayed in the query results are periodically refreshed and may not always reflect the most recent updates.

- You can run the **DSP NRDUCELLPOWERSAVING** command to query the status of NR load-based RF module fast deep dormancy and the reason why it does not take effect.
 - If the value of **Power Saving Type** is **NR Load-based Fast Dormancy** and the value of **Current State** is **Enabled**, dormancy has taken effect.
 - If the value of **Power Saving Type** is **NR Load-based Fast Dormancy** and the value of **Current State** is **Disabled**, check **Current Disabled Cause** to determine the reason why this function does not take effect. [Table 7-42](#) lists the possible reasons.
 - If the value of **Power Saving Type** is **NR Load-based Fast Dormancy**, check **Latest Exit Cause** and **Latest Disabled Cause** to determine the reason why this function does not take effect within the specified period. [Table 7-42](#) lists the possible reasons.

[Table 7-43](#) describes the meanings of **Latest Exit Cause**, **Latest Disabled Cause**, **Latest Exit Time**, and **Latest Disabled Time**.

Latest Exit Cause and **Latest Disabled Cause** indicate the reasons for the most recent exit and failure of the function. These reasons can be queried after the effective period has ended. For example, if the effective period is from 00:00:00 to 06:00:00, **Latest Exit Cause** and **Latest Disabled Cause** record the reasons for the most recent exit and failure of the function from 00:00:00 to 06:00:00. These reasons can be queried by an operation performed within the time from 06:00:00 to 23:59:59.

Table 7-42 Possible reasons why NR load-based RF module fast deep dormancy does not take effect

Cause	Description
Switch Turned Off	None
RF Group Configuration Condition Not Met	The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD . Table 7-41 lists the possible reasons.
Cell Unavailable	The cell is blocked or deactivated, fails to be set up, or is unavailable due to other reasons.
Protection Due to Ongoing Maintenance and Test	• Service-related maintenance and test operations are being performed. • Protection will be canceled within 30 minutes after service-related maintenance and test operations are completed. Dormancy will be allowed after the protection is canceled.
Triggering Time Not Reached	The current time falls outside the specified energy-saving effective period.

Cause	Description
Ultra-Long Holdover in Progress	None
Impacted Feature Taken Effect	None
Neighboring Cell Not Configured	- No inter-frequency neighboring cells are configured for the serving cell. - Inter-frequency neighboring cells are configured for the serving cell, but there is no neighboring cell for which Power Saving Cell Flag is set to COVERAGE. - In multi-operator sharing scenarios: - The PLMN IDs of the normal operators of the capacity-layer cell are not a subset of the PLMN IDs of normal operators of basic-layer cells. - The PLMN IDs of the reserved operators of the capacity-layer cell are not a subset of the PLMN IDs of normal or reserved operators of basic-layer cells.
In Penalty State	- A cell enters the penalty state if it exits fast deep dormancy within the specified effective period. After the penalty ends, it can re-enter fast deep dormancy. The penalty duration is specified by the NRDUCellPowerSaving.PwrSavingPenaltyTime parameter. - A cell enters the penalty state when some UEs failed to be transferred out of the cell. The penalty lasts 30 minutes. The causes of UE transfer failures include UE capability mismatch and missing configuration of neighboring frequency relationships.
Xn Interface Unavailable	The Xn interface is abnormal in multi-carrier inter-base-station scenarios.

Cause	Description
Neighboring Cell Unavailable	- A neighboring cell for which Power Saving Cell Flag is set to COVERAGE is unavailable. - A neighboring cell for which Power Saving Cell Flag is set to COVERAGE is a hyper cell or combined cell, and some TRPs of this cell are unavailable. - A neighboring cell for which Power Saving Cell Flag is set to COVERAGE is an NSA cell. - When the NR_FAST_DORM_SERVICE_ASSUR_SW option of the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch is selected for a capacity-layer cell, the PWR_SAVE_SERVICE_ASSUR_SW option of the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter is deselected for the corresponding basic-layer cell.
Downlink PRB Threshold Not Reached	For NR load-based RF module fast deep dormancy, the total downlink PRB usage of the capacity-layer and basic-layer cells does not reach the downlink PRB usage threshold for triggering this function.
Uplink PRB Threshold Not Reached	For NR load-based RF module fast deep dormancy, the total uplink PRB usage of the capacity-layer and basic-layer cells does not reach the uplink PRB usage threshold for triggering this function.
UE Number Threshold Not Reached	None
Service Assurance in Progress	None
Triggering Decision Period Not Met	For NR load-based RF module fast deep dormancy, the current time falls within the specified effective period, but the cell does not meet the dormancy entry conditions throughout the period specified by the NRDUCellPowerSaving.ShutdownTrigJudgePeriod parameter.
Multi-Carrier Not Coordinated	For activated cells sharing the same RF module, not all the intra-RAT carriers meet the conditions for entering load-based RF module fast deep dormancy.
RF Dormancy Basic Capb Condition Not Met	The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD . Table 7-41 lists the possible reasons.

Cause	Description
Others	- The local NR cell meets the dormancy entry conditions but does not receive an instruction to enter dormancy. The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD. Table 7-41 lists the possible reasons. If the DSP BRDDORMANCY command output displays Service Conditions Not Met as the reason dormancy does not take effect, the possible cause is that a cell of another RAT sharing the same RF module is not configured with fast deep dormancy, or the cell is configured with fast deep dormancy but the service conditions are not met. - In the current period, the conditions for this function to take effect are met, but the conditions for LNR load-based RF module fast deep dormancy are not met. - Internal software reasons cause the function not to take effect.

Table 7-43 Time and causes for the most recent exit and failure of a function

Command Output	Description
Latest Exit Cause	Indicates the cause of the most recent exit of a function during its effective period.
Latest Disabled Cause	Indicates the cause of the most recent failure of a function to take effect during its effective period.
Latest Exit Time	Indicates the recorded time of the latest exit of a function during its effective period.
Latest Disabled Time	Indicates the recorded time of the latest failure for a function to take effect during its effective period.

- You can also run the **DSP NRDUCELLPOWERSAVING** command to query the status of LNR load-based RF module fast dormancy and the reason why it does not take effect. The values displayed in the query results are periodically refreshed and may not always reflect the most recent updates.
 - If the value of the **Power Saving Type** parameter is **LNR Load-based Fast Dormancy** and the value of the **Current State** parameter is **Enabled**, this function has taken effect.

- If the value of **Power Saving Type** is **LNR Load-based Fast Dormancy** and the value of **Current State** is **Disabled**, check **Current Disabled Cause** to determine the reason why this function does not take effect. **Table 7-44** lists the possible reasons.
- If the value of **Power Saving Type** is **LNR Load-based Fast Dormancy**, check **Latest Exit Cause** and **Latest Disabled Cause** to determine the reason why this function does not take effect within the specified period. **Table 7-44** lists the possible reasons.

Table 7-43 describes the meanings of **Latest Exit Cause**, **Latest Disabled Cause**, **Latest Exit Time**, and **Latest Disabled Time**.

Latest Exit Cause and **Latest Disabled Cause** indicate the reasons for the most recent exit and failure of the function. These reasons can be queried after the effective period has ended. For example, if the effective period is from 00:00:00 to 06:00:00, **Latest Exit Cause** and **Latest Disabled Cause** record the reasons for the most recent exit and failure of the function from 00:00:00 to 06:00:00. These reasons can be queried by an operation performed within the time from 06:00:00 to 23:59:59.

Table 7-44 Possible reasons why LNR load-based RF module fast deep dormancy does not take effect

Cause	Description
Switch Turned Off	The function switch for the NR cell is turned off.
RF Group Configuration Condition Not Met	The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD . Table 7-41 lists the possible reasons.
Cell Unavailable	The NR cell is blocked or deactivated, fails to be set up, or is unavailable due to other reasons.
Protection Due to Ongoing Maintenance and Test	• Service-related maintenance and test operations are being performed. • Protection will be canceled within 30 minutes after service-related maintenance and test operations are completed. Dormancy will be allowed after the protection is canceled.
Triggering Time Not Reached	The current time falls outside the specified effective period for the function.
Ultra-Long Holdover in Progress	None
Impacted Feature Taken Effect	None

Cause	Description
Neighboring Cell Not Configured	• In SA networking, the neighbor relationships with LTE cells are not configured for the NR cell, the overlap indicator is not configured for the neighboring LTE cells of the NR cell, or the NR cell fails to obtain information about its neighboring LTE cells. • In SA networking: – The PLMN IDs of the normal operators of the NR cell are not a subset of the PLMN IDs of normal operators of neighboring LTE cells. – The PLMN IDs of the reserved operators of the NR cell are not a subset of the PLMN IDs of normal or reserved operators of neighboring LTE cells.
Obtaining NCell Configurations	• In NSA networking or NSA and SA hybrid networking, the neighbor relationships with LTE cells are not configured for the NR cell, the overlap indicator is not configured for the neighboring LTE cells of the NR cell, the X2 link is faulty, or the NR cell fails to obtain information about its neighboring LTE cells. • In SA networking, the system is subscribing to neighboring LTE cell information. • In NSA networking or NSA and SA hybrid networking: – The PLMN IDs of the normal operators of the NR cell are not a subset of the PLMN IDs of normal operators of neighboring LTE cells. – The PLMN IDs of the reserved operators of the NR cell are not a subset of the PLMN IDs of normal or reserved operators of neighboring LTE cells.
In Penalty State	• A cell enters the penalty state after exiting fast deep dormancy within the specified effective period. It can re-enter dormancy once the penalty state ends. The penalty duration is specified by the NRDUCellPowerSaving.PwrSavingPenaltyTime parameter. • A cell enters the penalty state when some UEs failed to be transferred out of the cell. The penalty lasts 30 minutes.

Cause	Description
Neighboring Cell Unavailable	- The function switches of the neighboring LTE cells are not turned on, the licenses for the neighboring LTE cells are insufficient, the neighboring LTE cells are in barred state or not activated, no neighboring LTE cell is an anchor cell, or the LTE cells are unavailable due to other reasons. - If NSA anchor carrier shutdown is enabled in neighboring LTE cells, it is recommended that the INTER_RAT_GRAD_PWR_SAVING_SW option of the gNBpwrSavingAlgo.PwrSavingAlgoSwitch parameter be selected for the NR cell. This will prevent the NR cell from exiting LNR load-based RF module fast deep dormancy in this case. For details about NSA anchor carrier shutdown, see <i>Energy Conservation and Emission Reduction</i> in <i>eRAN Feature Documentation</i>. - LTE sector split cells are configured as basic-layer cells, and the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter is deselected.
UE Number Threshold Not Reached	- The downlink PRB usage in the NR cell has reached the trigger threshold, but the number of UEs remains below the trigger threshold. - The downlink PRB usage and the number of UEs in the NR cell have not reached their respective trigger thresholds.
Downlink PRB Threshold Not Reached	The number of UEs in the NR cell has reached the trigger threshold, but the downlink PRB usage remains below the trigger threshold.
Uplink PRB Threshold Not Reached	The number of UEs in the NR cell has reached the trigger threshold, but the uplink PRB usage remains below the trigger threshold.
Neighboring Cell Downlink PRB Exceeded Threshold	The current downlink PRB usage of neighboring cells exceeds the configured downlink PRB usage threshold. That is, the proportion of neighboring LTE cells meeting the PRB usage threshold does not meet the requirement specified by the NRDUCellPowerSaving.EutranCarrShutdownRatio parameter.

Cause	Description
Neighboring Cell Uplink PRB Exceeded Threshold	The current uplink PRB usage of neighboring cells exceeds the configured uplink PRB usage threshold. That is, the proportion of neighboring LTE cells meeting the PRB usage threshold does not meet the requirement specified by the NRDUCellPowerSaving.EutranCarrShutdownRatio parameter.
NR UEs in NCell Exceeding Threshold	The number of NR UEs in neighboring LTE cells of an NR cell exceeds the threshold. That is, the proportion of neighboring LTE cells meeting the NR UE number threshold does not meet the requirement specified by the NRDUCellPowerSaving.EutranCarrShutdownRatio parameter.
Neighboring Cell NR UE Downlink PRB Exceeded Threshold	The current downlink PRB usage of NR UEs exceeds the configured downlink PRB threshold. That is, the proportion of neighboring LTE cells meeting the NR UE downlink PRB usage threshold does not meet the requirement specified by the NRDUCellPowerSaving.EutranCarrShutdownRatio parameter.
Neighboring Cell NR UE Uplink PRB Exceeded Threshold	The current uplink PRB usage of NR UEs exceeds the configured uplink PRB usage threshold. That is, the proportion of neighboring LTE cells meeting the NR UE uplink PRB usage threshold does not meet the requirement specified by the NRDUCellPowerSaving.EutranCarrShutdownRatio parameter.
Triggering Decision Period Not Met	The current time falls within the check period specified by the NRDUCellPowerSaving.ShutdownTrigJudgePeriod parameter, but the thresholds for function triggering have not been met.
UE Being Transferred or Transfer Failed	None
Multi-Carrier Not Coordinated	For activated cells sharing the same RF module, not all the intra-RAT carriers meet the conditions for entering LNR load-based RF module fast deep dormancy.
RF Dormancy Basic Capb Condition Not Met	The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD . Table 7-41 lists the possible reasons.

Cause	Description
Other Reason	- In the current period, the conditions for this function to take effect are met, but the conditions for NR load-based RF module fast deep dormancy are not met. - The current time falls within the effective period specified for LNR load-based RF module fast deep dormancy, but the gNodeB has not obtained data such as the PRB usage and UE number. - The local NR cell meets the dormancy entry conditions but does not receive an instruction to enter dormancy. The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD. Table 7-41 lists the possible reasons. If the DSP BRDDORMANCY command output displays Service Conditions Not Met as the reason dormancy does not take effect, the possible cause is that a cell of another RAT sharing the same RF module is not configured with fast deep dormancy, or the cell is configured with fast deep dormancy but the service conditions are not met. - Other reasons, such as internal software issues, cause the function not to take effect. - The configurations are incorrect. For example, in multi-operator sharing scenarios, the neighbor relationships with LTE cells served by the LTE primary operators are not configured for the NR cell.

- You can also run the **DSP CELLPWRSAVINGPOLICY** command to query the status of LTE load-based fast deep dormancy and the reason why it does not take effect. The values displayed in the query results are periodically refreshed and may not always reflect the most recent updates.
 - If the value of the **Power Saving Type** parameter is **Intra-LTE Load-based Fast Dormancy** and the value of the **Current State** parameter is **Enabled**, this function has taken effect.
 - If the value of **Power Saving Type** is **Intra-LTE Load-based Fast Dormancy** and **Current State** has any other value, check [Table 7-45](#) for possible reasons why the function does not take effect.
 - If the value of **Power Saving Type** is **Intra-LTE Load-based Fast Dormancy**, check **Latest Exit Cause** and **Latest Disabled Cause** to determine the reason why this function does not take effect within the specified period. [Table 7-45](#) lists the possible reasons.

[Table 7-43](#) describes the meanings of **Latest Exit Cause**, **Latest Disabled Cause**, **Latest Exit Time**, and **Latest Disabled Time**.

Latest Exit Cause and **Latest Disabled Cause** indicate the reasons for the most recent exit and failure of the function. These reasons can be queried after the effective period has ended. For example, if the effective

period is from 00:00:00 to 06:00:00, **Latest Exit Cause** and **Latest Disabled Cause** record the reasons for the most recent exit and failure of the function from 00:00:00 to 06:00:00. These reasons can be queried by an operation performed within the time from 06:00:00 to 23:59:59.

Table 7-45 Possible reasons why LTE load-based fast deep dormancy does not take effect

Cause	Description
SWITCH_OFF	None
RF_GROUP_CFG_COND_NOT_MET	The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD . Table 7-41 lists the possible reasons.
CELL_UNAVAILABLE	The cell is blocked or deactivated, fails to be set up, or is unavailable due to other reasons.
BBP_CAPB_NOT_SUPPORTED	None
IN_MAINTENANCE_TEST_PROTECTION	- Service-related maintenance and test operations are being performed. - Protection will be canceled within 30 minutes after service-related maintenance and test operations are completed. Dormancy will be allowed after the protection is canceled.
NOT_IN_SHUTDOWN_TIME	The current time falls outside the specified energy-saving effective period.
IN_PENALTY_STATE	- A cell enters the penalty state after exiting dormancy within the specified effective period. It can re-enter dormancy once the penalty state ends. The penalty time is specified by the CellPwrSavingPolicy.PenaltyTime parameter. - A cell enters the penalty state when some UEs fail to be transferred out of the cell. The penalty time is specified by the CellPwrSavingPolicy.PenaltyTime parameter.
HIGH_PRIORITY_TAKE_EFFECT	None

Cause	Description
NEIGHBORING_CELL_NOT_CONFIG	- No inter-frequency neighboring cells are configured for the current capacity-layer cell. - The current capacity-layer cell has been configured with inter-frequency neighboring cells, but the number of neighboring cells with the POWER_SAVING_OVERLAP_IND option selected is 0. - In multi-operator sharing scenarios, the operators of the local cell must be a subset of the operators of its co-coverage neighboring basic cells. If optimization for cell PLMN reservation scenarios is enabled by selecting the CELL_PLMN_RSVD_OPT_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter: - If the operators of the local cell are non-reserved operators, the operators must be a subset of non-reserved operators of its co-coverage neighboring basic cells. - If the operators of the local cell are reserved operators, the operators must be a subset of non-reserved operators or reserved operators of its co-coverage neighboring basic cells.
X2_INTERFACE_UNAVAILABLE	The X2 interface is abnormal when the capacity-layer and basic-layer cells are provided by different base stations.
NEIGHBORING_CELL_UNAVAILABLE	- A neighboring cell for which POWER_SAVING_OVERLAP_IND is set to ON is unavailable. - LTE sector split cells are configured as basic-layer cells, and the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter is deselected.
TRIG_JUDGE_PERIOD_NOT_MET	The current time falls within the specified effective period, but the cell does not meet the dormancy entry conditions throughout the period specified by the CellPwrSavingPolicy.TriggerJudgePeriod parameter.
NOT_IN_UE_NUM_THRESHOLD	None
NOT_IN_DL_PRB_THRESHOLD	None
NOT_IN_UL_PRB_THRESHOLD	None

Cause	Description
MULTI_CARR_NOT_COORDINATED	For activated cells sharing the same RF module, not all the intra-RAT carriers meet the conditions for entering load-based RF module fast deep dormancy.
RF_DORM_BASE_CAP_COIN_NOT_MET	The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD . Table 7-41 lists the possible reasons.
OTHERS	- The local LTE cell meets the dormancy entry conditions but does not receive an instruction to enter dormancy. The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD. Table 7-41 lists the possible reasons. If the DSP BRDDORMANCY command output displays Service Conditions Not Met as the reason dormancy does not take effect, the possible cause is that a cell of another RAT sharing the same RF module is not configured with fast deep dormancy, or the cell is configured with fast deep dormancy but the service conditions are not met. - Internal software reasons cause the function not to take effect.

- In the case of LTE load-based RF module fast deep dormancy, use the following methods to query the cell's operator reservation status:
 - Run the **DSP EUTRANEXTERNALCELL** command to query the external E-UTRAN cell's operator reservation status.
 - If the value of **Cell Reserved for Operator Use** is **NOT_RESERVED**, the cell is not reserved for the operator.
 - If the value of **Cell Reserved for Operator Use** is **RESERVED**, the cell is reserved for the operator.
 - Run the **DSP CELLCNOPERATORSTATUS** command to query the intra-eNodeB cell's operator reservation status.
 - If the value of **Cell Operator Reserved State** is **NULL**, the cell is not activated or the PLMN ID of the operator is not on the list broadcast in SIB1 of the cell.
 - If the value of **Cell Operator Reserved State** is **Not Reserved**, the cell is not reserved for the operator.
 - If the value of **Cell Operator Reserved State** is neither **NULL** nor **Not Reserved**, the cell is reserved for the operator.

7.3.4.3 Network Monitoring

On the live network, the one-week energy consumption data obtained before and after the energy saving feature is enabled can be compared to evaluate the energy

saving gains. It is expected that the energy consumption decreases significantly after this feature is enabled. The specific data and observation methods are as follows:

- Energy consumption of the RF module. Check the **VS.EnergyCons.BTSBoard (LTE eNodeB, 5G gNodeB)** counter on the MAE-Access.
- Unavailability duration of the NR cell in the energy saving state. Check the **N.Cell.Unavail.Dur.EnergySaving** counter on the MAE-Access.
- Duration of the NR cell in the fast deep dormancy state. Check the **N.Cell.Unavail.Dur.RFModuleFastDormancy** counter on the MAE-Access.
- Unavailability duration of the LTE cell in the energy saving state. Check the **L.Cell.Unavail.Dur.EnergySaving** counter on the MAE-Access.
- Duration of the LTE cell in the fast deep dormancy state. Check the **L.Cell.Unavail.Dur.RFModuleFastDormancy** counter on the MAE-Access.
- Duration of the RF module in the fast deep dormancy state. Check the **VS.RADIOEQM.FastDeepDormancy.Dur (LTE eNodeB, 5G gNodeB)** counter on the MAE-Access.

 NOTE

- There is delay in determining whether to enter or exit the energy saving state. As a result, the duration measured by the performance counters may deviate from the actual duration by approximately 1 minute.
- The sampling period is not synchronous with the reporting period, so a deviation of one sampling period may occur.
- If multiple RATs share an RF module through SDR, it is recommended that the power consumption be checked using module-level counters.

7.4 RF Module Super Dormancy

7.4.1 Principles

The RF module super dormancy function allows RF modules to enter the dormancy state when the network is under light or no load. This reduces power consumption without affecting the reliability of RF modules.

[7.1.2 Differences in Dormancy Triggering Factors](#) lists the sub-functions of RF module super dormancy by control approach.

For high frequency NR TDD, only scheduled RF module super dormancy and NR load-based RF module super dormancy are supported.

RF module super dormancy also provides the following functions:

- Dormancy pending and pending cancellation
If operations on the base station (such as upgrades) require all RF modules to remain active, dormancy pending can be enabled. For details, see [7.10 Dormancy Pending and Pending Cancellation](#).
- Reliability protection during dormancy
The reliability protection involves license reliability protection, configuration reliability protection, reliability protection against faults, and O&M reliability

protection during dormancy. For details, see [7.11 Reliability Protection for Deep Dormancy and Super Dormancy](#).

- Multi-RRU cell dormancy compatibility

For LTE, this function is controlled by the **MULTI_RRU_CELL_DORM_COMPAT_SW** option of the **ENodebPwrSavingAlgo.PwrSavingAlgoSwitch** parameter. When this option is selected, RF module super dormancy can take effect in multi-RRU cells, such as combined-RRU cells, combined cells, and SFN cells.

This function requires that energy saving features take effect and terminate for involved RRUs simultaneously since these RRUs provide services as a whole. Therefore, it is recommended that RF modules be configured with the same dormancy function and dormancy period.

- Wake-up algorithm optimization for RF module super dormancy

Wake-up algorithm optimization for RF module super dormancy is controlled by the **RF_SDWAO_SW** option of the **EQUIPMENT.EQMDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter. If this option is selected, the base station optimizes the algorithm for RF module wake-up. When an RF module exits the dormancy state, services can be restored without restarting the RF module, thereby shortening the service restoration duration of the RF module. This function must be configured under the guidance of Huawei engineers.

7.4.1.1 Scheduled RF Module Super Dormancy

The scheduled RF module super dormancy function allows RF modules to enter the dormancy state within specified periods when the network is under light or no load. This reduces power consumption without affecting the reliability of RF modules.

This function is controlled by setting the **RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB)** parameter to **ON_SUPER** or **ON_DEEP_SUPER**, specifying the dormancy period by the **RRU.DORMANCYSTARTTIME (LTE eNodeB, 5G gNodeB)** and **RRU.DORMANCYSTOPTIME (LTE eNodeB, 5G gNodeB)** parameters, and setting the **RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter to **ANYTIME_WAKEUP_SW**. The value **ON_DEEP_SUPER** of **RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB)** indicates that RF module deep dormancy and RF module super dormancy can be both enabled. For details about RF module deep dormancy, see [7.2 RF Module Deep Dormancy](#).

When the **RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB)** parameter is set to **ON_DEEP_SUPER** within the configured dormancy period, and both the conditions for entering RF module deep dormancy and those for entering RF module super dormancy are met:

- RF module super dormancy takes effect if the **ANYTIME_WAKEUP_SW** option of the **RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter is selected.
- If the **ANYTIME_WAKEUP_SW** option of the **RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter is not selected:
 - If the **ANYTIME_WAKEUP_SW** option of the **RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter is selected within 30s, RF module super dormancy takes effect.

- If the **ANYTIME_WAKEUP_SW** option of the **RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter is not selected within 30s, RF module deep dormancy takes effect.
- For a combined-RRU cell or multi-TRP NR cell (such as an SFN cell, hyper cell, or combined cell):
All RF modules serving the cell need to be configured with the same dormancy period. Otherwise, RF modules enter the super dormancy state only when the current time is within the overlapping range of all dormancy periods configured for the RF modules and other dormancy conditions are met.
- In co-RF module scenarios involving GSM, UMTS, LTE, or NR, RAT-specific deep dormancy can be enabled by turning on **RRU.CUSTOMDORMANCYSTDSW (LTE eNodeB, 5G gNodeB)** and specifying **RRU.DORMANCYSTD (LTE eNodeB, 5G gNodeB)**. Once RAT-specific dormancy is enabled, only the cells of the specified RAT can enter dormancy. If RAT-specific dormancy is not enabled, RF modules can enter dormancy only when dormancy conditions are met across all configured RATs.
- For UMTS inter-frequency handovers, the **HO_INTER_FREQ_HARD_HO_SWITCH** option of the **UCORRMALGOSWITCH.HoSwitch** parameter must be selected.
- For inter-RAT handovers involving UMTS, the **HO_INTER_RAT_CS_OUT_SWITCH** and **HO_INTER_RAT_PS_OUT_SWITCH** options of the **UCORRMALGOSWITCH.HoSwitch** parameter must be selected.
- For dormancy involving GSM or UMTS, the **GU_DC_SW** option of the base-station-specific **EQUIPMENT.EQMDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter must be deselected. If the option is deselected and a configuration command is executed to trigger dormancy pending or dormancy pending is enabled on the local end that works in GSM or UMTS, then dormancy pending will take effect on the entire base station.

 CAUTION

- You are advised not to perform base station reconfiguration (for example, running the **MOD RETPORT** command to modify RET port settings), install or activate a license, or download software during the specified dormancy period. If any of these operations is required, run the **PND DORMANCY** command and wait 10 minutes before performing the operation.
 - During RF module dormancy, cell-related online detections (such as inter-cell cable sequence detection) and RF module-related detections (such as VSWR test, cross feeder connection detection, and intermodulation interference detection) cannot be performed. If any of these detections is required, run the **PND DORMANCY** command and wait 10 minutes before performing the detection.
-

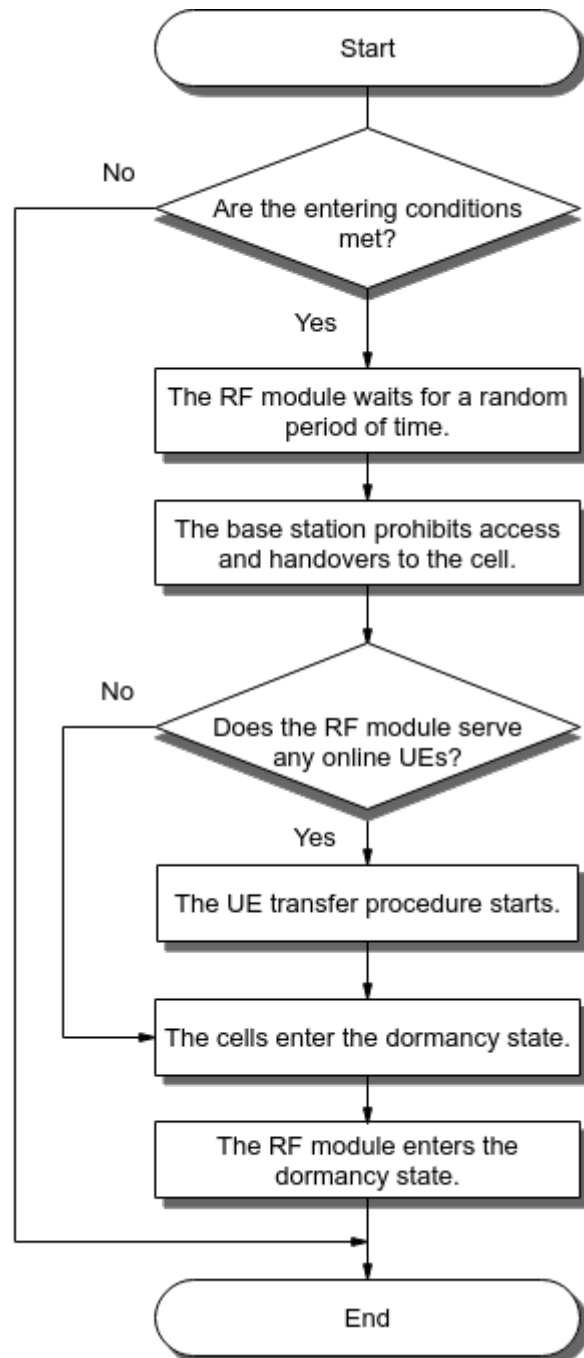
Entering the Super Dormancy State

The prerequisites for this function to take effect are as follows:

- The RF module works in GSM, UMTS, LTE, or NR or is shared by multiple RATs including any of the preceding RATs.
- In the case of UMTS, LTE, or NR, all activated cells served by the base station must be bound to specified BBPs. For UMTS, each piece of uplink and downlink baseband equipment can contain only one baseband processing unit.

Figure 7-18 shows the procedure for entering the super dormancy state.

Figure 7-18 Procedure for entering the super dormancy state



1. The base station evaluates whether entering conditions are met.
The base station checks whether the current time falls within the effective period.
 - If the **RRU.PWRSVGPOLICYID (LTE eNodeB, 5G gNodeB)** parameter is set to 0, the effective period is determined by the **RRU.DORMANCYSTARTTIME (LTE eNodeB, 5G gNodeB)** and **RRU.DORMANCYSTOPTIME (LTE eNodeB, 5G gNodeB)** parameters.
 - If the **RRU.PWRSVGPOLICYID (LTE eNodeB, 5G gNodeB)** parameter is set to a non-zero value, the effective period is determined by the **BRDPWRSVGPOLICYTIME.STARTTIME (5G gNodeB, LTE eNodeB)** and **BRDPWRSVGPOLICYTIME.STOPTIME (LTE eNodeB, 5G gNodeB)** parameters of the policy whose **BRDPWRSVGPOLICYTIME.POLICYID (LTE eNodeB, 5G gNodeB)** is also set to this non-zero value.
2. The system waits for a random duration.
The system waits for a random period of time specified by the **EQUIPMENT.MAXRANDORMANCYWAITTIME (LTE eNodeB, 5G gNodeB)** parameter and then enters the dormancy state. This prevents a large number of sites from entering the super dormancy state at the same time and the consequent impact on signaling of the core network.
3. The base station prohibits access and handovers to the cell.
4. The base station checks whether the RF module is serving any online UEs.
 - If all cells for all RATs of the RF module are not serving any online UEs, the cells enter the dormancy state. The RATs of the RF module are indicated by the **RRU.RS** parameter.
 - If there are online UEs served by the RF module, the base station waits until the specified waiting time elapses. If the waiting time elapses, the process proceeds to 5.
 - If the **RRU.PWRSVGPOLICYID (LTE eNodeB, 5G gNodeB)** parameter is set to 0, the waiting time is specified by the **RRU.DORMANCYTIMER (LTE eNodeB, 5G gNodeB)** parameter.
 - If the **RRU.PWRSVGPOLICYID (LTE eNodeB, 5G gNodeB)** parameter is set to a non-zero value, the waiting time is specified by the **BRDPWRSVGPOLICY.DORMANCYTIMER (LTE eNodeB, 5G gNodeB)** parameter of the policy whose **BRDPWRSVGPOLICYTIME.POLICYID (LTE eNodeB, 5G gNodeB)** is also set to this non-zero value.
5. A UE transfer procedure starts.
 - If there are online UEs in LTE cells served by the RF module, all UEs in these cells need to be handed over to co-coverage neighboring basic-layer cells. The RATs of the RF module are indicated by the **RRU.RS (LTE eNodeB, 5G gNodeB)** parameter. The handover procedure is described as follows:
The eNodeB delivers event A4 measurement configurations about frequencies of the co-coverage neighboring basic-layer cells to UEs in connected mode. The RSRP threshold for event A4 is specified by the **InterFreqHoGroup.InterFreqHoA4ThdRsrp** parameter. After receiving event A4 measurement reports from UEs, the eNodeB initiates inter-frequency handovers to the cells served by the frequencies of basic-layer cells for the UEs.

- If there are online UEs in NR cells served by the RF module, a scheduled UE transfer procedure is started. The RATs of the RF module can be queried using the **RRU.RS (LTE eNodeB, 5G gNodeB)** parameter.
 - In SA networking, UEs are transferred from the local cell to neighboring cells as follows:
 - Non-CA UEs: UEs without voice services are transferred to inter-frequency or inter-RAT neighboring cells through blind redirections. Voice service UEs are preferentially handed over to inter-frequency or inter-RAT neighboring cells. If the handovers fail, inter-frequency or inter-RAT redirections are performed.
 - CA UEs: If the cell is an SCell of a CA UE, the SCell is removed. If the cell is the PCell of a CA UE, the CA UE is transferred to a neighboring cell in the same way as non-CA UEs.
 - In NSA networking, UE connections on the NR side are released as follows:
 - If energy saving takes effect in the PCell of a UE, all connections on the NR side are released.
 - If energy saving takes effect in an SCell of a UE, its connection to this SCell on the NR side is released but its other connections on the NR side are retained.
- If there are online UEs in UMTS cells served by the RF module, a scheduled UE transfer procedure is started. The RATs of the RF module are indicated by the **RRU.RS** parameter.
 - CELL_DCH UEs are preferentially handed over to inter-band or inter-RAT neighboring cells based on measurement results, and are redirected if the handovers fail.
 - The RNC periodically attempts to perform inter-frequency or inter-RAT handovers for CELL_DCH UEs.

Inter-frequency handovers: The RNC delivers load-based inter-frequency measurement configurations in which the measurement frequencies are those of inter-band neighboring cells. After receiving a measurement report from the UE, the RNC hands over the UE to an inter-band neighboring cell with signal quality higher than the RSCP (**UCCELLMCLDR.TargetFreqThdRscp**) and Ec/No (**UCCELLMCLDR.TargetFreqThdEcNo**) thresholds of the local cell.

Inter-RAT handovers: The RNC delivers load-based inter-RAT measurement configurations and hands over a UE to the inter-RAT neighboring cell after receiving a measurement report from the UE.
 - CELL_FACH UEs are redirected.
 - By default, cell broadcast service (CBS) takes precedence over dormancy. Therefore, when CBS is enabled in a cell for which the **URNCCCELLDORMANCYPARA.CellIgnoreCbsSwitch** parameter is set to **OFF** (default value), the cell cannot enter dormancy. If dormancy is required in scenarios where CBS is enabled, this parameter must be set to **ON**. Once it is set to **ON**:

- If no CBS message is sent and the current time falls outside the CBS protection period (specified by the **URNCELLDORMANCYPARA.CbsServiceHoldDuration** parameter), the cell can enter dormancy.
- If CBS messages are sent, the cell cannot enter dormancy.

 NOTE

After receiving a CBS message, the RNC immediately wakes the dormant cell and processes the CBS message once the cell is successfully set up. Because the cell requires time to wake, CBS messages may not be processed promptly. If CBS has a higher priority, it is recommended that **URNCELLDORMANCYPARA.CellIgnoreCbsSwitch** be set to its default value **OFF**. For details about CBS, see *CBS* in *RAN Feature Documentation*.

- If the local cell and its inter-band neighboring cells belong to different operators, **URNCELLDORMANCYPARA.CellInterPlmnHoSwitch** must be set to **ON** and **UOPERATORSHARINGMODE.InterPlmnHoAllowedIntraRat** must be set to **YES** to enable UE transfer.
- If the local cell and its inter-RAT neighboring cells belong to different operators, **UOPERATORSHARINGMODE.InterPlmnHoAllowedInterRat** must be set to **YES** to enable UE transfer.
- Emergency UEs (if any) are not forcibly released, and the cells enter dormancy only after these UEs are released.
- If there are online UEs in GSM cells served by the RF module, a scheduled UE transfer procedure is started. The RATs of the RF module can be queried using the **RRU.RS (5G gNodeB, LTE eNodeB)** parameter.
 - Graceful transfer is preferentially performed to transfer UEs to co-coverage cells or neighboring cells.
 - If some UEs fail to be transferred, they will be forcibly transferred. Calls that cannot be handed over are directly released. However, emergency UEs (if any) are not forcibly released and the cells can enter the dormancy state only after these UEs are normally released.
 - All UEs are transferred or released.
 - By default, CBC services take precedence over dormancy. Therefore, when CBC is enabled in a cell for which the **BSEXSOFTPARA.IgnoreCbcSwitch** parameter is set to **OFF** (default value), the cell cannot enter the dormancy state. If dormancy is required in scenarios where CBC is enabled, this parameter must be set to **ON**.
 - If no CBC message is sent and the current time falls outside the CBC protection period (specified by **BSEXSOFTPARA.CbcServiceHoldDuration**), the cell can enter dormancy.
 - If CBC messages are sent, the cell cannot enter dormancy.

 CAUTION

After receiving a CBC message, the BSC immediately wakes up the dormant cell and processes the CBC message once the cell is successfully set up. Since the cell needs time to wake up, CBC messages may not be processed promptly. If CBC has a higher priority, it is recommended that **BSCXSOFTPARA.IgnoreCbcSwitch** be set to its default value **OFF**. For details about CBC, see *Cell Broadcast* in *GBSS Feature Documentation*.

6. The cells enter the dormancy state.
7. The RF module enters the dormancy state.

Exiting the Super Dormancy State

- When the specified dormancy stop time arrives, the RF module exits the dormancy state. The RF module restarts when exiting the deep dormancy state. After the restart, it works properly.
- After RF module dormancy is disabled, the RF module starts to exit the dormancy state.

7.4.1.2 NR Load-based RF Module Super Dormancy

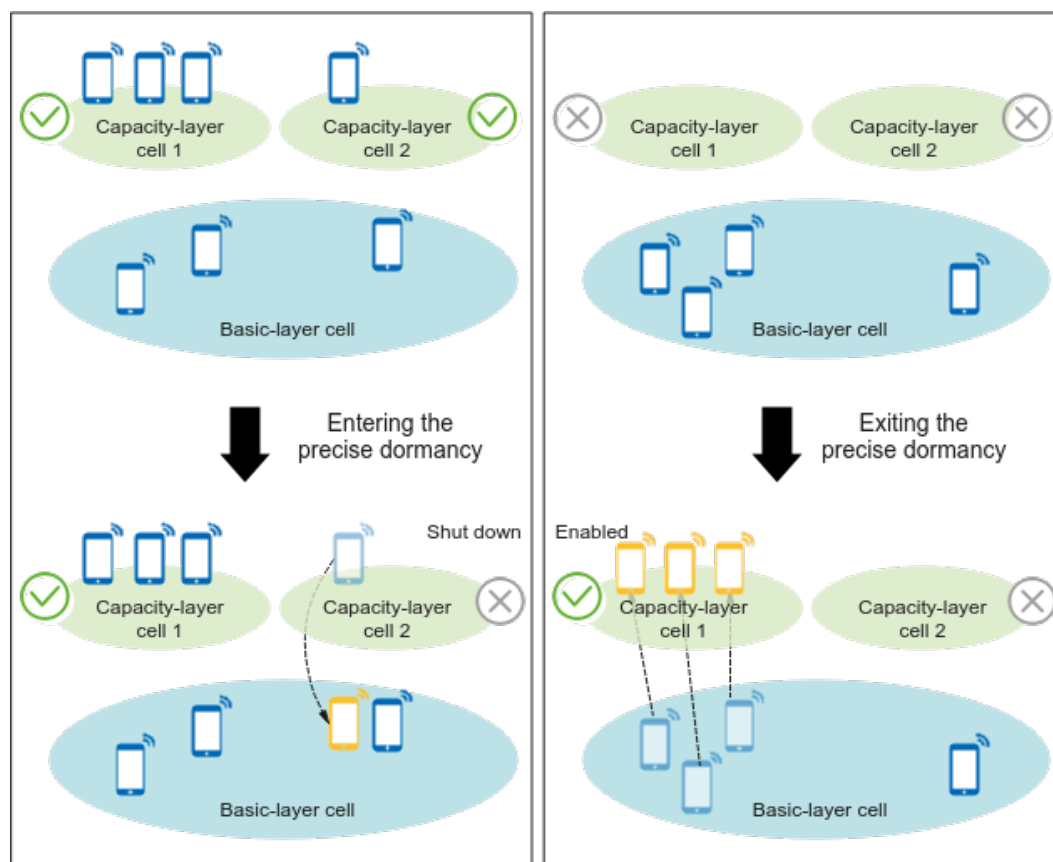
Multi-carrier co-coverage NR networking provides a larger network capacity and, upon a heavy traffic load, a higher data throughput than single-carrier NR networking. However, such networking wastes base station energy when the network is under light or no load. NR load-based RF module super dormancy is introduced to reduce the waste.

Load-based RF module super dormancy enables the gNodeB to transfer UEs from capacity-layer cells to basic-layer cells and instruct the RF module serving the capacity-layer cells to enter the super dormancy state within a specified period when the network is under light or no load. When the load of the basic-layer cells increases to a specified threshold, the RF modules in dormancy are automatically woken up and the capacity-layer cells are restored.

NR load-based RF module super dormancy can be classified into non-precise dormancy mode and precise dormancy mode. For high-frequency NR TDD, only the non-precise dormancy mode is supported in NR load-based RF module super dormancy.

In scenarios where basic-layer cells cover large areas, capacity-layer cells cover small areas, one basic-layer cell corresponds to multiple capacity-layer cells, and UEs are unevenly distributed, the precise dormancy mode enables the base station to accurately identify which cells can enter or exit the dormancy state based on UE distribution in the basic-layer and capacity-layer cells. This maximizes the probability of cells entering the dormancy state and minimizes the probability of cells exiting this state, extending the dormancy duration, as shown in [Figure 7-19](#).

Figure 7-19 Working principles for precise dormancy mode



This function requires the following switches to be configured:

- RF module-level switches

The **RRU.DORMANCYSW (5G gNodeB, LTE eNodeB)** parameter is set to **ON_SUPER** or **ON_DEEP_SUPER**, the **RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter is set to **ANYTIME_WAKEUP_SW**, and either of the following conditions is met:

- The **RRU.CUSTOMDORMANCYSTDSW (5G gNodeB, LTE eNodeB)** parameter is set to **OFF**.
- The **RRU.CUSTOMDORMANCYSTDSW (5G gNodeB, LTE eNodeB)** parameter is set to **ON**, and the **NR** option of the **RRU.DORMANCYSTD (5G gNodeB, LTE eNodeB)** parameter is selected.

The value **ON_DEEP_SUPER** of **RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB)** indicates that RF module deep dormancy and RF module super dormancy can be both enabled. For details about RF module deep dormancy, see [7.2 RF Module Deep Dormancy](#).

When the **RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB)** parameter is set to **ON_DEEP_SUPER** within the configured dormancy period, and both the conditions for entering RF module deep dormancy and those for entering RF module super dormancy are met:

- RF module super dormancy takes effect if the **ANYTIME_WAKEUP_SW** option of **RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** has been selected.

- If the **ANYTIME_WAKEUP_SW** option of **RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** is not selected:
 - If the **ANYTIME_WAKEUP_SW** option of the **RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter is selected within 30s, RF module super dormancy takes effect.
 - If the **ANYTIME_WAKEUP_SW** option of the **RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter is not selected within 30s, RF module deep dormancy takes effect.
- NR cell-level switch
 - In non-precise dormancy mode, the **INTRA_NR_DORMANCY_BY_LOAD_SW** option of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter is selected.
 - In precise dormancy mode, both the **INTRA_NR_DORMANCY_BY_LOAD_SW** option of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter and the **PRECISE_DORMANCY_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter are selected.

The effective period and threshold parameters for NR load-based RF module super dormancy can be uniquely determined by using the combinations of the **NRDUCellPowerSaving.PowerSavingPolicyIndex** and **NRDUCellPowerSaving.PowerSavingType** parameters in the **NRDUCellPowerSaving** MO on the NR side. Specifically:

- The **NRDUCellPowerSaving.PowerSavingPolicyIndex** parameter specifies an energy-saving policy index. It uniquely identifies an energy-saving policy within the local cell. With the index specified, the corresponding policy can be added, modified, queried, or removed from the local cell.
- The **NRDUCellPowerSaving.PowerSavingType** parameter specifies the energy saving type indicated by an energy-saving policy index. NR load-based RF module super dormancy requires that this parameter be set to **INTRA_NR_DORMANCY_BY_LOAD**, meaning that under the corresponding energy-saving policy, the local cell performs load-based dormancy in the specified period under certain load conditions.

Related Concepts

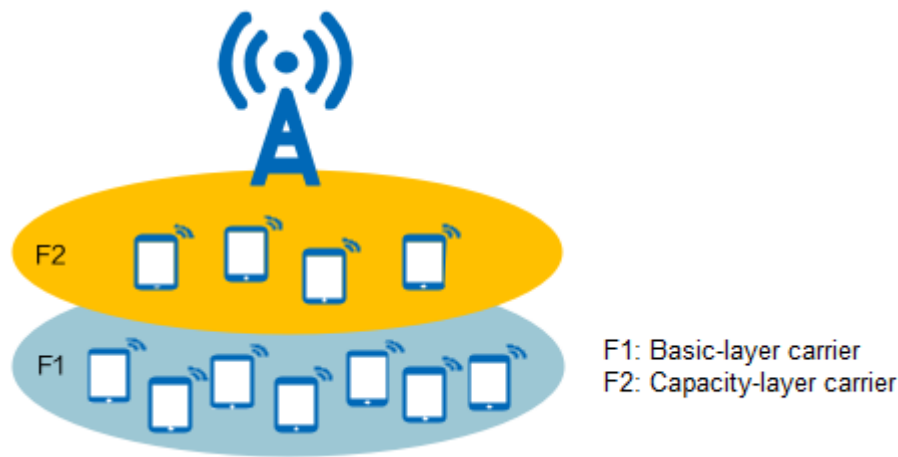
- Basic-layer carriers provide basic coverage for the network. RF modules providing such carriers will not enter the dormancy state. The cells served by basic-layer carriers are referred to as basic-layer cells.
- Capacity-layer carriers support large traffic volumes during peak hours. RF modules providing such carriers can enter the dormancy state. The cells served by capacity-layer carriers are referred to as capacity-layer cells.

 NOTE

Load-based RF module super dormancy is enabled for capacity-layer cells and corresponding RF modules. One capacity-layer cell corresponds to one or more basic-layer cells. Basic-layer cells and capacity-layer cells must be inter-frequency neighboring cells of each other.

- Basic-layer and capacity-layer cells are relative concepts identified by the **NRCellRelation.PowerSavingCellFlag** parameter during neighbor relationship configuration.
- If this parameter is set to **COVERAGE**, the neighboring cell is a basic-layer cell and the serving cell is a capacity-layer cell.
 - If this parameter is set to **CAPACITY**, the neighboring cell is a capacity-layer cell and the serving cell is a basic-layer cell.
 - If this parameter is set to **NOT_CONFIG**, there is no energy-saving relationship between the neighboring cell and the serving cell. The neighboring cell is not considered in load-based RF module dormancy.

Figure 7-20 Basic-layer and capacity-layer cells



For example, the following table describes how to set the Power Saving Cell Flag for cells in the preceding figure.

Table 7-46 Configuring the Power Saving Cell Flag
(NRCellRelation.PowerSavingCellFlag)

Value of NRCellRelation.PowerSavingCellFlag	NRCellRelation.NrCellId	NRCellRelation.CellId
COVERAGE	ID of the cell operating on F2	ID of the cell operating on F1
CAPACITY	ID of the cell operating on F1	ID of the cell operating on F2

In NR multi-carrier networking scenarios, it is recommended that capacity-layer and basic-layer cells be configured as described in [Table 7-47](#).

Table 7-47 Suggestions on designation of capacity-layer and basic-layer cells

Networking	Capacity-Layer Cell	Basic-Layer Cell
NR TDD (low-frequency) and NR FDD coordinated networking	NR TDD (low-frequency) cells	NR FDD cells
NR TDD (low-frequency) sub-6 GHz and NR TDD (low-frequency) sub-3 GHz coordinated networking	NR TDD (low-frequency) sub-6 GHz cells	NR TDD (low-frequency) sub-3 GHz cells
Inter-band NR FDD coordinated networking	Cells on a higher frequency	Cells on a lower frequency
Intra-band multi-carrier networking	Cells on a higher frequency	Cells on a lower frequency
NR TDD (high-frequency) and NR TDD (low-frequency) coordinated networking	NR TDD (high-frequency) cells	NR TDD (low-frequency) cells
NR TDD (high-frequency) and NR FDD (low-frequency) coordinated networking	NR TDD (high-frequency) cells	NR FDD (low-frequency) cells

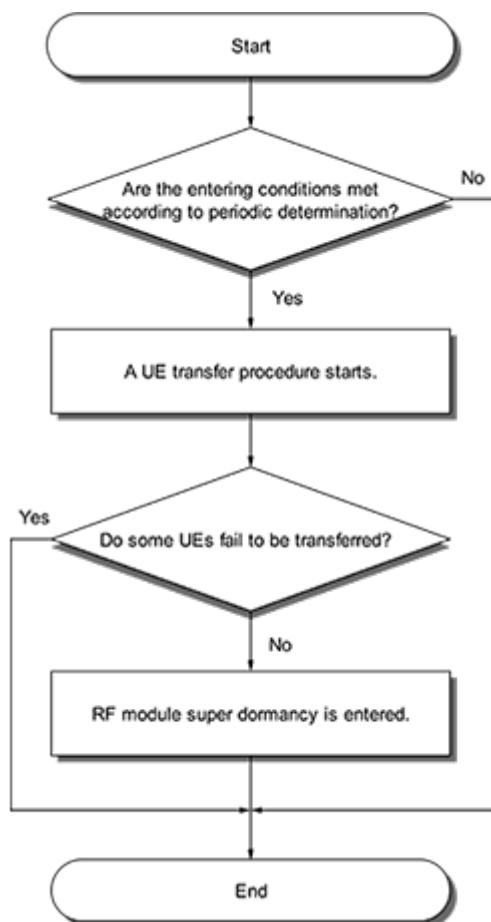
NOTE

The automatic identification of co-coverage neighbor relationships (with basic-layer cells) and the configuration of the power saving cell flag are implemented on the MAE-Optimization. For details, see *MAE-Optimization* in *MAE Product Documentation (EulerOS, TaiShan)*.

Entering the Super Dormancy State

[Figure 7-21](#) shows how the RF module serving a capacity-layer cell enters the super dormancy state after NR load-based RF module super dormancy is enabled.

Figure 7-21 Procedure for entering NR load-based RF module super dormancy



1. The base station determines whether entering conditions are met.
The base station periodically checks whether the conditions for entering the dormancy state are met. The granularity of the evaluation period is specified by the **NRDUCellPowerSaving.EvalPeriodGranularity** parameter.
 - If this parameter is set to **MINUTE_LEVEL**, the evaluation period is expressed in minutes. In this case, the period is specified by the **NRDUCellPowerSaving.ShutdownTrigJudgePeriod** parameter. For high-frequency TDD, the **NRDUCellPowerSaving.EvalPeriodGranularity** parameter must be set to **MINUTE_LEVEL**.
 - If this parameter is set to **SECOND_LEVEL**, the evaluation period is expressed in seconds. In this case, the period is specified by the **NRDUCellPowerSaving.SecondLevelTrigEvalPrd** parameter.
 - A capacity-layer cell must meet all of the following conditions:
 - The current time is within the effective period for this function, which is determined by the **NRDUCellPowerSaving.StartTime** and **NRDUCellPowerSaving.StopTime** parameters.
 - The capacity-layer cell is not in an energy saving penalty state.
 - The basic-layer cells corresponding to the capacity-layer cell must meet one of the following conditions:

- All basic-layer carriers are available.
- Some basic-layer carriers are available, and some basic-layer carriers, functioning as capacity-layer carriers of other basic-layer carriers, are in the energy saving state, due to NR intelligent carrier shutdown, LTE and NR intelligent carrier shutdown, NR load-based RF module dormancy, or LNR load-based RF module dormancy.
- The capacity-layer cell and the corresponding basic-layer cells must meet all the following conditions:
 - The total downlink PRB usage of the capacity-layer carrier and basic-layer carriers is less than or equal to the downlink PRB usage threshold specified by the **NRDUCellPowerSaving.DlPrbThld** parameter. Specifically:

NR TDD (low frequency) and NR FDD in non-precise dormancy: Total downlink PRB usage of the capacity-layer carrier and basic-layer carriers = (Number of PRBs used by PDSCH DRBs of the capacity-layer carrier + Number of downlink PRBs used by the basic-layer carriers)/Number of available downlink PRBs of the basic-layer carriers

For NR TDD (high frequency), total downlink PRB usage of the capacity-layer carrier and basic-layer carriers = ((Number of PRBs used by PDSCH DRBs of the high-frequency sector x (Subcarrier spacing in the high-frequency sector/Subcarrier spacing in the basic-layer)) + Number of downlink PRBs used by the basic-layer carriers)/Number of available downlink PRBs of the basic-layer carriers

If multiple basic-layer carriers correspond to one capacity-layer carrier, the sum of PRBs of these basic-layer carriers is used in the preceding formula.

In precise dormancy mode: Total downlink PRB usage of the capacity-layer carrier and basic-layer carriers = (Number of PRBs used by PDSCH DRBs of the capacity-layer carrier + Overlap rate of each inter-frequency neighboring cell x Number of used downlink PRBs of each basic-layer carrier) / (Number of PRBs available for PDSCH DRBs of the capacity-layer carrier + Overlap rate of each inter-frequency neighboring cell x Number of available downlink PRBs of each basic-layer carrier)

 NOTE

In precise dormancy mode, the overlap rate of each inter-frequency neighboring cell is specified by the **NRCellRelation.OverlapRate** parameter.

- The total uplink PRB usage of the capacity-layer carrier and basic-layer carriers is less than or equal to the uplink PRB usage threshold specified by the **NRDUCellPowerSaving.UlPrbThld** parameter. Specifically:

For NR TDD (low frequency) and NR FDD, total uplink PRB usage of the capacity-layer carrier and basic-layer carriers = (Number of uplink PRBs used by the capacity-layer carrier + Number of uplink PRBs used by the basic-layer carriers)/Number of available uplink PRBs of the basic-layer carriers

For NR TDD (high frequency), total uplink PRB usage of the capacity-layer carrier and basic-layer carriers = ((Number of uplink PRBs used in the high-frequency sector x (Subcarrier spacing in the high-frequency sector/Subcarrier spacing in the basic-layer)) + Number of uplink PRBs used by the basic-layer carriers)/Number of available uplink PRBs of the basic-layer carriers

In precise dormancy mode, the total uplink PRB usage of the capacity-layer carrier and basic-layer carriers is equal to (Number of uplink PRBs used by the capacity-layer carrier + Overlap rate of each inter-frequency neighboring cell x Number of uplink PRBs used by each basic-layer carrier)/(Number of available uplink PRBs of the capacity-layer carrier + Overlap rate of each inter-frequency neighboring cell x Number of available uplink PRBs of each basic-layer carrier).

- In precise dormancy mode, the following conditions must also be met:
 - Uplink PRB usage of each basic-layer carrier corresponding to the capacity-layer carrier <
NRDUCellPowerSaving.ProtectUlPrbThld – NRDUCellPowerSaving.ProtectUlPrbOffset
Uplink PRB usage of each basic-layer carrier corresponding to the capacity-layer carrier = (Number of used uplink PRBs on the basic-layer carrier + Overlap rate x Number of used uplink PRBs of the capacity-layer carrier)/Total number of available uplink PRBs on the basic-layer carrier
 - Downlink PRB usage of each basic-layer carrier corresponding to the capacity-layer carrier <
NRDUCellPowerSaving.ProtectDlPrbThld – NRDUCellPowerSaving.ProtectDlPrbOffset
Downlink PRB usage of each basic-layer carrier corresponding to the capacity-layer carrier = (Number of used downlink PRBs on the basic-layer carrier + Overlap rate x Number of PRBs used by PDSCH DRBs of the capacity-layer carrier)/Total number of available downlink PRBs on the basic-layer carrier
- The overlap rate is the proportion of overlapping area in a capacity-layer cell to a basic-layer cell, which is specified by the **NRCCellRelation.OverlapRate** parameter at the basic-layer cell.
- The total number of RRC_CONNECTED UEs on the capacity-layer carrier and a basic-layer carrier is less than or equal to the UE number threshold specified by the **NRDUCellPowerSaving.UserNumThld** parameter.
If multiple basic-layer carriers correspond to one capacity-layer carrier, each combination of the basic-layer and capacity-layer carriers must meet this condition.
For NR TDD (high frequency), the capacity layer carrier indicates all cells in a high-frequency sector.
- In multi-operator sharing scenarios, the local cell and co-coverage neighboring basic cells are classified based on normal operators and

reserved operators (whose **NRExternalNCellPlmn.PlmnReservedFlag** is set to **TRUE**).

- If the operators of the local cell are normal operators, the operators must be a subset of normal operators of the co-coverage neighboring basic cells.
- If the operators of the local cell are reserved operators, the operators must be a subset of normal or reserved operators of the co-coverage neighboring basic cells.
- In SUL co-deployment scenarios, the total number of SUL UEs in the capacity-layer and basic-layer cells is less than or equal to the value of the **NRDUCellPowerSaving.SulUserNumThld** parameter. If the capacity-layer cell is associated with multiple basic-layer cells, each pair of the basic-layer and capacity-layer cells must meet this condition.
- (When the carriers are provided by different base stations) The Xn interface is in the normal state.

 NOTE

The same dormancy period is configured for the RF modules bound to a combined-RRU cell or multi-TRP cell (such as an SFN cell, hyper cell, or combined cell).

If the capacity-layer cell is a hyper cell or combined cell:

- The total downlink or uplink PRB usage of the capacity-layer carrier and basic-layer carriers is calculated based on the maximum number of used PRBs among all TRPs. When load-based dormancy takes effect, all TRPs and their carriers enter the dormancy state.
- It is recommended that basic-layer cells also be hyper cells or combined cells. It is not recommended that common cells be configured as basic-layer cells.

When comparing the total uplink or downlink PRB usage of the capacity-layer cell and basic-layer cell with the entering threshold, the base station performs conversion on the cell bandwidth, subcarrier spacing, and uplink-downlink subframe configuration.

2. A UE transfer procedure starts. The gNodeB sends system information to prohibit UE access and handovers to the NR cell and starts the UE transfer procedure with a specified duration.
 - In SA networking, UEs are transferred from the local cell to basic-layer neighboring cells as follows:
 - Non-CA UEs: UEs not performing voice services are transferred to neighboring cells through inter-frequency handovers. If the handovers fail, the UEs are transferred to inter-frequency neighboring cells through blind redirections. Voice service UEs are handed over to inter-frequency neighboring cells.
 - CA UEs: If the cell is an SCell of a CA UE, the SCell is removed. If the cell is the PCell of a CA UE, the CA UE is transferred to a neighboring cell in the same way as non-CA UEs.

If UEs cannot be transferred out of the capacity-layer cell 30 seconds after the UE transfer procedure starts (for example, due to handover failures or unsupported operating bands of the basic-layer cells), the procedure stops and cannot be re-triggered for the capacity-layer cell within 30 minutes.

- In NSA networking, UE connections on the NR side are released as follows:
 - If energy saving takes effect in the PSCell of a UE, all connections on the NR side are released.
 - If energy saving takes effect in an SCell of a UE, its connection to this SCell on the NR side is released but its other connections on the NR side are retained.
- 3. The RF module enters the super dormancy state.

Once all UEs are transferred out of the capacity-layer cell, no RRC_CONNECTED UEs remain in the cell, and the cell and the corresponding RF module enter the super dormancy state. If an RF module serves multiple NR cells, it enters the super dormancy state only when load-based RF module dormancy is triggered for all these NR cells. If an RF module also serves LTE cells, RF module super dormancy must also be enabled on the LTE side.

Exiting the Super Dormancy State

The base station periodically checks whether the conditions for exiting the dormancy state are met. The granularity of the evaluation period is specified by the **NRDUCellPowerSaving.EvalPeriodGranularity** parameter.

- If this parameter is set to **MINUTE_LEVEL**, the evaluation period is expressed in minutes. In this case, the period is specified by the **NRDUCellPowerSaving.ExitEvaluationPeriod** parameter. For high-frequency TDD, the **NRDUCellPowerSaving.EvalPeriodGranularity** parameter must be set to **MINUTE_LEVEL**.
- If this parameter is set to **SECOND_LEVEL**, the evaluation period is expressed in seconds. In this case, the period is specified by the **NRDUCellPowerSaving.SecondLevelExitEvalPrd** parameter.

A capacity-layer cell and the corresponding RF module automatically exit the dormancy state when any of the following conditions is met:

- The current time is not within the effective periods (determined by **NRDUCellPowerSaving.StartTime** and **NRDUCellPowerSaving.StopTime**) for this function.
- The total downlink PRB usage of basic-layer carriers is greater than **NRDUCellPowerSaving.DlPrbThld** plus **NRDUCellPowerSaving.DlPrbOffset**. Additionally for high-frequency NR TDD, the number of high-frequency UEs on the basic-layer carrier is greater than **NRDUCellPowerSaving.MmwaveUserNumThld**.

NOTE

- If the sum of **NRDUCellPowerSaving.DlPrbThld** and **NRDUCellPowerSaving.DlPrbOffset** is greater than 100, capacity-layer cells cannot exit the super dormancy state based on the PRB usage.
- The total uplink PRB usage of basic-layer carriers is greater than **NRDUCellPowerSaving.UlPrbThld** plus **NRDUCellPowerSaving.NCellUlPrbOffset**. Additionally for high-frequency NR TDD, the number of high-frequency UEs on the basic-layer carrier is greater than **NRDUCellPowerSaving.MmwaveUserNumThld**.

- In precise dormancy mode, the uplink PRB usage of any basic-layer carrier corresponding to the capacity-layer carrier is greater than or equal to the value of the **NRDUCellPowerSaving.ProtectUlPrbThld** parameter.
Uplink PRB usage of a basic-layer carrier corresponding to the capacity-layer carrier = Number of used uplink PRBs on the basic-layer carrier/Total number of available uplink PRBs on the basic-layer carrier
- In precise dormancy mode, the downlink PRB usage of any basic-layer carrier corresponding to the capacity-layer carrier is greater than or equal to the value of the **NRDUCellPowerSaving.ProtectDlPrbThld** parameter.
Downlink PRB usage of a basic-layer carrier corresponding to the capacity-layer carrier = Number of used downlink PRBs on the basic-layer carrier/Total number of available downlink PRBs on the basic-layer carrier
- The number of UEs in connected mode served by the basic-layer carriers is greater than **NRDUCellPowerSaving.UserNumThld** plus **NRDUCellPowerSaving.UserNumOffset**. Additionally for high-frequency NR TDD, the number of high-frequency UEs on the basic-layer carriers is greater than **NRDUCellPowerSaving.MmwaveUserNumThld**.

 NOTE

- If the base station software version of a low-frequency basic-layer cell is V100R020C00 or earlier, the number of high-frequency-capable UEs cannot be reported.
- If the base station software versions of all low-frequency basic-layer cells are V100R020C00 or earlier, the high-frequency capacity-layer cell does not need to check the high-frequency UEs in the basic-layer cells. In this scenario, the cell can exit the dormancy state if the PRB usage or the number of UEs in the basic-layer cell is greater than the threshold.
- If the base station software version of some low-frequency basic-layer cells is V100R020C00 or earlier:
 - For a low-frequency basic-layer cell that can properly report the number of high-frequency-capable UEs, the high-frequency capacity-layer cell needs to check the number of high-frequency UEs in this low-frequency basic-layer cell. In this scenario, the capacity-layer cell can exit the dormancy state if the number of high-frequency-capable UEs in the low-frequency basic-layer cell meets the configured threshold and the PRB usage or the number of UEs in connected mode in the basic-layer cell meets the exit conditions.
 - For low-frequency basic-layer cells that do not support reporting the number of high-frequency UEs, the number of high-frequency UEs is 0 by default. In this scenario, the capacity-layer cell cannot exit the dormancy state, even if the actual number of high-frequency UEs in the low-frequency basic-layer cell meets the configured threshold and the PRB usage or the number of UEs in connected mode meets the exit conditions.
- A basic-layer neighboring cell is added for the capacity-layer cell.
- In multi-operator sharing scenarios:
 - The PLMN IDs of the normal operators of the capacity-layer cell are not a subset of the PLMN IDs of normal operators of basic-layer cells.
 - The PLMN IDs of the reserved operators of the capacity-layer cell are not a subset of the PLMN IDs of normal or reserved operators of basic-layer cells.
- In SUL co-deployment scenarios, the number of SUL UEs in a basic-layer cell is greater than the sum of **NRDUCellPowerSaving.SulUserNumThld** and **NRDUCellPowerSaving.SulUserNumOffset**. If the capacity-layer cell is

associated with multiple basic-layer cells, the number of SUL UEs in one of the basic-layer cells must meet the condition.

- (When the basic-layer carriers are provided by different base stations) The Xn interface is abnormal.
- Any of the basic-layer carriers becomes unavailable for a reason other than NR intelligent carrier shutdown, LTE and NR intelligent carrier shutdown, and NR load-based dormancy.

Upon the exit from super dormancy:

- The capacity-layer cell cannot enter the dormancy state again within the penalty duration specified by the **NRDUCellPowerSaving.PwrSavingPenaltyTime** parameter if it exited this state because the uplink or downlink PRB usage, number of UEs, and availability of the basic-layer cells is abnormal or because the Xn interface is abnormal.
- A penalty is not imposed if the capacity-layer cell exited the dormancy state because the function switch was turned off, the effective period elapsed, or due to RRU carbonization detection.

7.4.1.3 NR Local-Cell-Load-based RF Module Super Dormancy

NR local-cell-load-based RF module super dormancy allows RF modules to enter dormancy during specified periods based on the local cell load when the network is under light or no load, thereby reducing energy consumption without compromising RF module reliability.

This function is controlled by setting the **RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB)** parameter to **ON_SUPER** or **ON_DEEP_SUPER**, specifying the dormancy period by the **RRU.DORMANCYSTARTTIME (LTE eNodeB, 5G gNodeB)** and **RRU.DORMANCYSTOPTIME (LTE eNodeB, 5G gNodeB)** parameters, and setting the **RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter to **ANYTIME_WAKEUP_SW**. The value **ON_DEEP_SUPER** of **RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB)** indicates that RF module deep dormancy and RF module super dormancy can be both enabled. For details about RF module deep dormancy, see [7.2 RF Module Deep Dormancy](#).

When the **RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB)** parameter is set to **ON_DEEP_SUPER** within the configured dormancy period, and both the conditions for entering RF module deep dormancy and those for entering RF module super dormancy are met:

- RF module super dormancy takes effect if the **ANYTIME_WAKEUP_SW** option of the **RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter is selected.
- If the **ANYTIME_WAKEUP_SW** option of the **RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter is not selected:
 - If the **ANYTIME_WAKEUP_SW** option of the **RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter is selected within 30s, RF module super dormancy takes effect.
 - If the **ANYTIME_WAKEUP_SW** option of the **RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)**

parameter is not selected within 30s, RF module deep dormancy takes effect.

- For a combined-RRU cell or multi-TRP NR cell (such as an SFN cell, hyper cell, or combined cell):
All RF modules serving the cell need to be configured with the same dormancy period. Otherwise, RF modules will enter super dormancy only when the current time falls within the overlap of all configured dormancy periods and other dormancy conditions are met.
- In co-MPT scenarios involving NR, RAT-specific dormancy needs to be enabled by turning on **RRU.CUSTOMDORMANCYSTDSW (LTE eNodeB, 5G gNodeB)** and specifying **RRU.DORMANCYSTD (LTE eNodeB, 5G gNodeB)**. Once RAT-specific dormancy is enabled, only the cells of the specified RAT can enter dormancy.

 CAUTION

- You are advised not to perform base station reconfiguration (for example, running the **MOD RETPORT** command to modify RET port settings), install or activate a license, or download software during the specified dormancy period. If any of these operations is required, run the **PND DORMANCY** command and wait 10 minutes before performing the operation.
 - During RF module dormancy, cell-related online detections (such as inter-cell cable sequence detection) and RF module-related detections (such as VSWR test, cross feeder connection detection, and intermodulation interference detection) cannot be performed. If any of these detections is required, run the **PND DORMANCY** command and wait 10 minutes before performing the detection.
-

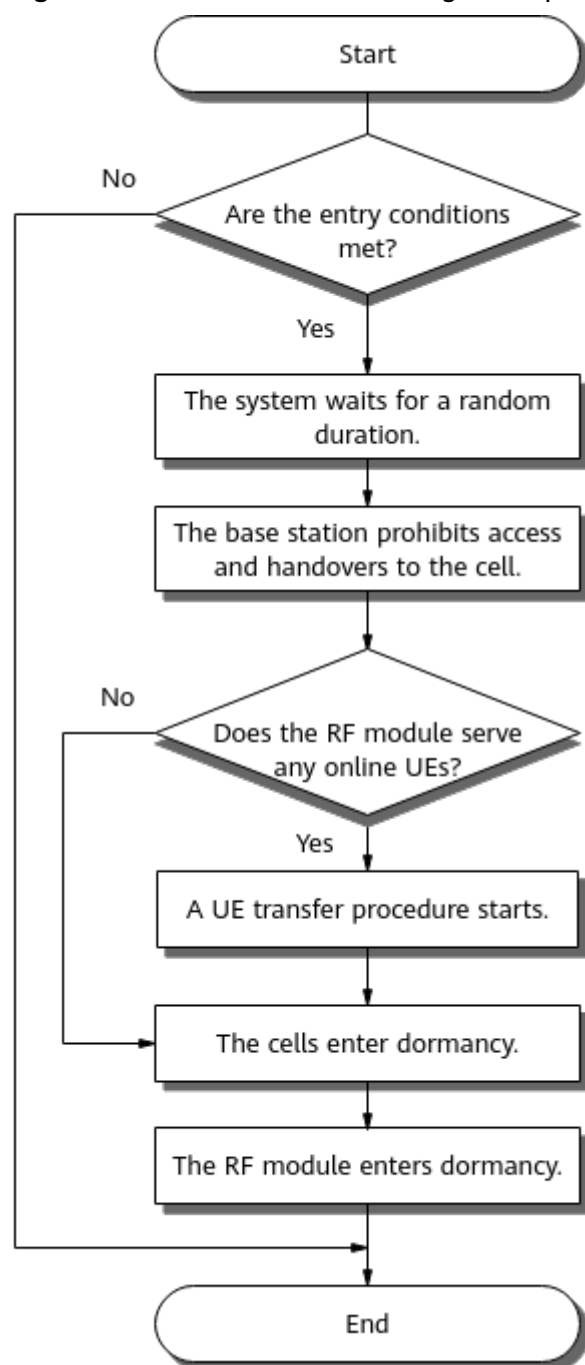
Entering the Super Dormancy State

The prerequisites for this function to take effect are as follows:

- The RF module works in NR or is shared by multiple RATs including NR.
- All activated cells served by the base station must be bound to specified BBPs. For UMTS, each piece of uplink and downlink baseband equipment can contain only one baseband processing unit.

Figure 7-22 shows the procedure for entering the super dormancy state.

Figure 7-22 Procedure for entering the super dormancy state



1. The base station determines whether the entry conditions are met.
The base station checks whether the current time falls within the effective period.
 - If the **RRU.PWRSVGPOLICYID (LTE eNodeB, 5G gNodeB)** parameter is set to **0**, the effective period is determined by the **RRU.DORMANCYSTARTTIME (LTE eNodeB, 5G gNodeB)** and **RRU.DORMANCYSTOPTIME (LTE eNodeB, 5G gNodeB)** parameters.
 - If the **RRU.PWRSVGPOLICYID (LTE eNodeB, 5G gNodeB)** parameter is set to a non-zero value, the effective period is determined by the

BRDPWRSVGPOLICYTIME.STARTTIME (5G gNodeB, LTE eNodeB) and **BRDPWRSVGPOLICYTIME.STOPTIME** (LTE eNodeB, 5G gNodeB) parameters of the policy whose **BRDPWRSVGPOLICYTIME.POLICYID** (LTE eNodeB, 5G gNodeB) is also set to this non-zero value.

If NR is included in the RATs indicated by the **RRU.RS** parameter for the RF module and optimized evaluation of NR RF module dormancy is enabled (by selecting the **NR_DORMANCY_JUDGE_THLD_OPT_SW** option of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter), the base station periodically evaluates whether the local cell can enter scheduled RF module dormancy based on the cell load.

- An energy saving policy is configured for the local cell. The **NRDUCellPowerSaving.PowerSavingPolicyIndex** parameter specifies an energy saving policy index. It uniquely identifies an energy saving policy within the local cell. The **NRDUCellPowerSaving.PowerSavingType** parameter must be set to **NR_DORMANCY_JUDGE_THLD_OPT**. The **NRDUCellPowerSaving.EvalPeriodGranularity** parameter must be set to **MINUTE_LEVEL**.
- An NR cell cannot enter dormancy unless both of the following conditions are met:
 - The current time falls within the effective energy saving period determined by the **NRDUCellPowerSaving.StartTime** and **NRDUCellPowerSaving.StopTime** parameters.
 - The NR cell is not in an energy saving penalty state.
- Regarding cell load, an NR cell cannot enter dormancy unless all of the following conditions are met:
 - Downlink PRB usage of the cell $\leq$ **NRDUCellPowerSaving.DlPrbThld**.
Downlink PRB usage of the cell = $(N.PR.B.DL.DrbUsed.Avg - N.PR.B.DL.ExtendUsed.Avg) / N.PR.B.DL.Actual.Avail.Avg \times 100\%$
 - Uplink PRB usage of the cell $\leq$ **NRDUCellPowerSaving.UlPrbThld**.
Uplink PRB usage of the cell = $N.PR.B.PUSCH.Used.Avg / N.PR.B.UL.PUSCH.Actual.Avail.Avg \times 100\%$
 - Number of RRC_CONNECTED UEs in the cell $\leq$ **NRDUCellPowerSaving.UserNumThld**
 - The number of RRC_CONNECTED UEs in an SUL cell with co-deployment is the sum of the **N.User.RRConn.Avg** and **N.User.SUL.UL.Avg** values.
 - The number of RRC_CONNECTED UEs in any other cell is indicated by **N.User.RRConn.Avg**.
 - In SUL co-deployment scenarios, the total number of SUL UEs in the local cell is less than or equal to the value of the **NRDUCellPowerSaving.SulUserNumThld** parameter.
 - For NR TDD, when the **ELIM_INCR_PWR_SAVING_JUDGE_SW** option of the **gNBPwrSavingAlgo.PwrSavingAlgoSwitch** parameter is selected, the number of UEs in the cell excludes the number of iNCR modules. For details about iNCR, see *LampSite iNCR Product Documentation*.

2. The system waits for a random duration.
The system waits for a random duration determined by the **EQUIPMENT.MAXRAMDORMANCYWAITTIME (LTE eNodeB, 5G gNodeB)** parameter before entering dormancy. This prevents a large number of sites from entering super dormancy simultaneously and avoids the resulting impact on core network signaling.
3. The base station prohibits access and handovers to the cell.
4. The base station checks whether the RF module is serving any online UEs.
 - If the cells for all RATs of the RF module are not serving any online UEs, the cells enter dormancy. The RATs of the RF module are indicated by the **RRU.RS (LTE eNodeB, 5G gNodeB)** parameter.
 - If online UEs are being served by the RF module, the base station waits until the specified waiting time elapses. If the waiting time elapses, go to 5.
 - If the **RRU.PWRSVGPOLICYID (LTE eNodeB, 5G gNodeB)** parameter is set to 0, the waiting time is specified by the **RRU.DORMANCYTIMER (LTE eNodeB, 5G gNodeB)** parameter.
 - If the **RRU.PWRSVGPOLICYID (LTE eNodeB, 5G gNodeB)** parameter is set to a non-zero value, the waiting time is specified by the **BRDPWRSVGPOLICY.DORMANCYTIMER (LTE eNodeB, 5G gNodeB)** parameter of the policy whose **BRDPWRSVGPOLICYTIME.POLICYID (LTE eNodeB, 5G gNodeB)** is also set to this non-zero value.
5. A UE transfer procedure starts.
If there are online UEs in NR cells served by the RF module, a scheduled UE transfer procedure is started. The RATs of the RF module can be queried using the **RRU.RS (LTE eNodeB, 5G gNodeB)** parameter.
 - In SA networking, UEs are transferred from the local cell to neighboring cells as follows:
 - Non-CA UEs: UEs without voice services are transferred to inter-frequency or inter-RAT neighboring cells through blind redirections. Voice service UEs are preferentially handed over to inter-frequency or inter-RAT neighboring cells. If the handovers fail, inter-frequency or inter-RAT redirections are performed.
 - CA UEs: If the local cell serves as an SCell of a CA UE, the SCell is removed. If the local cell serves as the PCell of a CA UE, the CA UE is transferred to a neighboring cell in the same way as a non-CA UE.
 - In NSA networking, UE connections on the NR side are released as follows:
 - If energy saving takes effect in the PSCell of a UE, all connections on the NR side are released.
 - If energy saving takes effect in an SCell of a UE, its connection to this SCell on the NR side is released but its other connections on the NR side are retained.
6. The cells enter dormancy.

7. The RF module enters dormancy.

Exiting the Super Dormancy State

- When the specified dormancy stop time arrives, the RF module starts to exit dormancy. During the exit, the RF module restarts to return to the normal state.
- After RF module dormancy is disabled, the RF module starts to exit dormancy.

7.4.1.4 LNR Load-based RF Module Super Dormancy

NR cells in operation still consume energy when there is light or even no traffic in the area covered by both LTE and NR networks, wasting base station energy. Therefore, LNR load-based RF module super dormancy is introduced.

Load-based RF module super dormancy enables the gNodeB to transfer UEs from NR cells to LTE cells and instruct the RF module serving the NR cells to enter super dormancy within a specified period when the network is under light or no load. When the load of LTE cells increases and reaches a specified threshold, the RF module in the dormancy state is automatically woken up and the NR cells are restored.

This function requires the following switches to be configured:

- RF module-level switches

The **RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB)** parameter is set to **ON_SUPER** or **ON_DEEP_SUPER**, the **RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter is set to **ANYTIME_WAKEUP_SW**, and either of the following conditions is met:

- The **RRU.CUSTOMDORMANCYSTDSW** parameter is set to **OFF**.
- The **RRU.CUSTOMDORMANCYSTDSW** parameter is set to **ON**, and the **NR** option of the **RRU.DORMANCYSTD** parameter is selected.

The value **ON_DEEP_SUPER** of **RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB)** indicates that RF module deep dormancy and RF module super dormancy can be both enabled. For details about RF module deep dormancy, see [7.2 RF Module Deep Dormancy](#).

When the **RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB)** parameter is set to **ON_DEEP_SUPER** within the configured dormancy period, and both the conditions for entering RF module deep dormancy and those for entering RF module super dormancy are met:

- RF module super dormancy takes effect if the **ANYTIME_WAKEUP_SW** option of **RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** is selected.
- If the **ANYTIME_WAKEUP_SW** option of the **RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter is not selected:
 - If the **ANYTIME_WAKEUP_SW** option of the **RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter is selected within 30s, RF module super dormancy takes effect.

The base station periodically checks whether the conditions for entering the dormancy state are met. The granularity of the evaluation period is specified by the **NRDUCellPowerSaving.EvalPeriodGranularity** parameter.

- If this parameter is set to **MINUTE_LEVEL**, the evaluation period is expressed in minutes. In this case, the period is specified by the **NRDUCellPowerSaving.ShutdownTrigJudgePeriod** parameter.
- If this parameter is set to **SECOND_LEVEL**, the evaluation period is expressed in seconds. In this case, the period is specified by the **NRDUCellPowerSaving.SecondLevelTrigEvalPrd** parameter.
- The NR cell must meet all of the following conditions:
 - Average number of active UEs in the NR cell $\leq$ UE number threshold (specified by the **NRDUCellPowerSaving.UserNumThld** parameter)
 - Downlink PRB usage of the NR cell $\leq$ Downlink PRB usage threshold (specified by the **NRDUCellPowerSaving.DlPrbThld** parameter)
 - Uplink PRB usage of the NR cell $\leq$ Uplink PRB usage threshold (specified by the **NRDUCellPowerSaving.UlPrbThld** parameter)

 NOTE

- If the NR cell is a hyper cell or combined cell, the TRP with the highest PRB usage of the NR cell is calculated.
 - If the NR cell is a DSS cell, the PRB usage equals the number of used RBs divided by the number of available RBs.
 - Active UEs are UEs in RRC_CONNECTED mode in an NR cell.
 - Downlink PRB usage of an NR cell = Average number of PRBs used by downlink PDSCH DRB services/Average number of available PRBs in the downlink
 - Uplink PRB usage of an NR cell = Average number of PRBs used by uplink PUSCH DRB services/Average number of available PRBs in the uplink
- In SUL co-deployment scenarios, the total number of SUL UEs in the NR cell is less than or equal to the value of the **NRDUCellPowerSaving.SulUserNumThld** parameter.
 - In multi-operator sharing scenarios, operators are divided into normal operators and reserved operators for the NR cell and neighboring LTE cells.
 - For an NR cell, an operator is reserved if the **NRExternalNCellPlmn.PlmnReservedFlag** parameter is set to **TRUE**.
 - For an LTE cell, an operator is reserved if the **CELLOP.CellReservedForOp** parameter is set to **CELL_RESERVED_FOR_OP** or **CELL_DYNAMIC_RESERVED_FOR_OP**.
 - In multi-operator scenarios, primary operator recognition for LNR load-based dormancy is supported. This function is controlled by the **LNR_LOAD_DORM_PRIM_OP_RECOG_SW** option of the **gNBpwrSavingAlgo.PwrSavingAlgoSwitch** parameter.
 - If this option is deselected, the requirements for the operator are as follows:

- The PLMN IDs of the normal operators of the NR cell must be a subset of the PLMN IDs of normal operators of neighboring LTE cells.
 - The PLMN IDs of the reserved operators of the NR cell must be a subset of the PLMN IDs of normal operators or the PLMN IDs of reserved operators of neighboring LTE cells.
- If this option is selected, the requirements for the operator are as follows:
 - If an NR cell is served by normal operators, the PLMN ID of the primary operator of the NR cell is the same as that of the neighboring LTE cells served by normal operators.
 - If an NR cell is served by reserved operators, the PLMN ID of the primary operator of the NR cell is the same as that of the neighboring LTE cells served by normal or reserved operators.
- The eNodeB sends the following load information about the neighboring LTE cells of the NR cell to the gNodeB. The gNodeB requires that the carriers of neighboring cells be available. The gNodeB checks whether a proportion of available neighboring cells (the proportion is specified by the **NRDUCellPowerSaving.EutranCarrShutdownRatio** parameter) meet the following conditions:
 - Downlink PRB usage of the neighboring LTE cell < E-UTRAN downlink PRB threshold (specified by the **NRDUCellPowerSaving.EutranDlPrbThld** parameter)
 - Uplink PRB usage of the neighboring LTE cell < Neighboring-cell uplink PRB usage threshold (specified by the **NRDUCellPowerSaving.NCellUlPrbThld** parameter)
 - Number of NR UEs in the neighboring LTE cell < E-UTRAN NR user number threshold (specified by the **NRDUCellPowerSaving.EutranNrUserNumThld** parameter)
 - Downlink PRB usage of NR UEs in the neighboring LTE cell < E-UTRAN NR user downlink PRB threshold (specified by the **NRDUCellPowerSaving.EutranNrUserDlPrbThld** parameter)
 - Uplink PRB usage of NR UEs in the neighboring LTE cell < E-UTRAN NR user uplink PRB threshold (specified by the **NRDUCellPowerSaving.EutranNrUserUlPrbThld** parameter)

 NOTE

Downlink PRB usage of an LTE cell = Average number of used downlink PRBs/Number of available downlink PRBs. The number of RBs for common channels is included in the calculation of downlink PRB usage.

Uplink PRB usage of an LTE cell = Average number of used uplink PRBs/Number of available uplink PRBs

2. A UE transfer procedure starts. The gNodeB sends system information to prohibit UE access and handovers to the NR cell and starts the UE transfer procedure with a specified duration.
 - In SA networking, UEs are transferred from the local cell to basic-layer neighboring cells as follows:

- Non-CA UEs: UEs not performing voice services are transferred to neighboring cells through inter-RAT handovers. If the handovers fail, the UEs are transferred to inter-RAT neighboring cells through blind redirections. Voice service UEs are handed over to inter-RAT neighboring cells.
- CA UEs: If the cell is an SCell of a CA UE, the SCell is removed. If the cell is the PCell of a CA UE, the CA UE is transferred to a neighboring cell in the same way as non-CA UEs.

If UEs cannot be transferred out of the NR cell 30s after the UE transfer procedure starts (for example, they do not support the bands of the basic-layer cells or the handover fails), the procedure stops and cannot be triggered for the NR cell again within a certain period of time specified by the **NRDUCellPowerSaving.PwrSavingPenaltyTime** parameter.

- In NSA networking, UE connections on the NR side are released as follows:
 - If energy saving takes effect in the PSCell of a UE, all connections on the NR side are released.
 - If energy saving takes effect in an SCell of a UE, its connection to this SCell on the NR side is released but its other connections on the NR side are retained.

3. The RF module enters the super dormancy state.

Once all UEs are transferred out of the NR cell, no RRC_CONNECTED UEs remain in the cell, and the cell and the corresponding RF module enter the super dormancy state. If an RF module serves multiple NR cells, it enters the super dormancy state only when load-based RF module dormancy is triggered for all these NR cells. If an RF module also serves LTE cells, RF module super dormancy must also be enabled on the LTE side.

If the **INTER_RAT_GRAD_PWR_SAVING_SW** option of the **gNB PwrSavingAlgo.PwrSavingAlgoSwitch** parameter is selected, some neighboring LTE cells can be used as capacity-layer cells of other basic-layer cells and enter the energy saving state due to functions such as intelligent power-off of carriers in the same coverage, Cell Switch-off Based on Automatic Co-coverage Identification, multi-carrier coordinated energy saving, GUL joint carrier shutdown, LTE load-based RF module dormancy, or dynamic carrier shutdown.

Exiting the Super Dormancy State

The base station periodically checks whether the conditions for exiting the dormancy state are met. The granularity of the evaluation period is specified by the **NRDUCellPowerSaving.EvalPeriodGranularity** parameter.

- If this parameter is set to **MINUTE_LEVEL**, the evaluation period is expressed in minutes. In this case, the period is specified by the **NRDUCellPowerSaving.ExitEvaluationPeriod** parameter.
- If this parameter is set to **SECOND_LEVEL**, the evaluation period is expressed in seconds. In this case, the period is specified by the **NRDUCellPowerSaving.SecondLevelExitEvalPrd** parameter.

The dormancy state will be exited and the NR carrier will be enabled if any of the following conditions is met:

- The current time is not within the shutdown period.
- Proportion of neighboring LTE cells that meet one of the following conditions $\geq \text{NRDUCellPowerSaving.EutranCarrShutdownRatio} - \text{NRDUCellPowerSaving.EutranCarrShutdownExitOfs}$
 - Downlink PRB usage of the neighboring LTE cell $\geq \text{NRDUCellPowerSaving.EutranDlPrbThld} + \text{NRDUCellPowerSaving.EutranDlPrbOffset}$
 - Uplink PRB usage of the neighboring LTE cell $\geq$ Neighboring cell uplink PRB threshold (specified by the **NRDUCellPowerSaving.NCellUlPrbThld** parameter) + Neighboring cell uplink PRB offset (specified by the **NRDUCellPowerSaving.NCellUlPrbOffset** parameter)
 - Number of NR UEs in the neighboring LTE cell $\geq \text{NRDUCellPowerSaving.EutranNrUserNumThld} + \text{NRDUCellPowerSaving.EutranNrUserNumThldOfs}$
 - Downlink PRB usage of NR UEs in the neighboring LTE cell $\geq \text{NRDUCellPowerSaving.EutranNrUserDlPrbThld} + \text{NRDUCellPowerSaving.EutranNrUserDlPrbThldOfs}$
 - Uplink PRB usage of NR UEs in the neighboring LTE cell $\geq \text{NRDUCellPowerSaving.EutranNrUserUlPrbThld} + \text{NRDUCellPowerSaving.EutranNrUserUlPrbThldOfs}$
 - The PLMN IDs of the normal operators of the NR cell are not a subset of the PLMN IDs of normal operators of neighboring LTE cells.
 - The PLMN IDs of the reserved operators of the NR cell are not a subset of the PLMN IDs of normal operators or the PLMN IDs of reserved operators of neighboring LTE cells.

When a neighboring LTE cell, functioning as a capacity-layer cell of other basic-layer cells, is in the energy saving state, whether this cell is considered as a neighboring cell meeting the exit conditions depends on the setting of the **INTER_RAT_GRAD_PWR_SAVING_SW** option of the **gNBWPwrSavingAlgo.PwrSavingAlgoSwitch** parameter.

- If this option is deselected, this LTE cell is considered as a cell meeting the exit conditions.
- If this option is selected, this LTE cell is not considered as a cell meeting the exit conditions and will not trigger the exit from the dormancy state.

Upon the exit from dormancy:

- The NR cell cannot enter the dormancy state again within the penalty duration specified by the **NRDUCellPowerSaving.PwrSavingPenaltyTime** parameter if it exited this state because the uplink or downlink PRB usage, number of UEs, or availability of the LTE cell is abnormal.
- A penalty is not imposed if the NR cell exited the dormancy state because the function switch was turned off, the effective period elapsed, or the base station software was upgraded.

NOTE

- The number of NR UEs in neighboring LTE cells of the NR cell is counted based on the UE capability (that is, whether a UE supports SA or NSA).
- In CA scenarios, only NR UEs on the primary component carrier (PCC) are counted.

7.4.1.5 LTE Load-based RF Module Super Dormancy

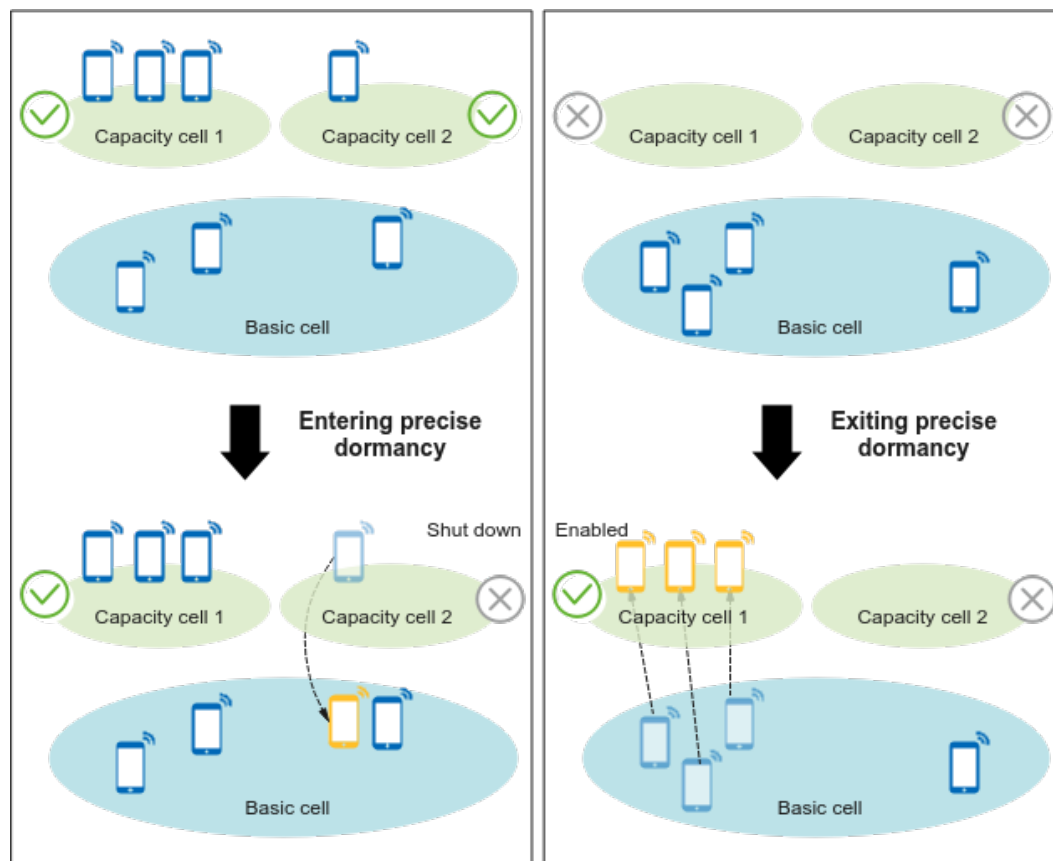
Multi-carrier co-coverage LTE networking provides a larger network capacity and, upon a heavy traffic load, a higher data throughput than single-carrier LTE networking. However, such networking wastes base station energy when the network is under light or no load. LTE load-based RF module super dormancy is introduced to reduce the waste.

Load-based RF module super dormancy enables the eNodeB to transfer UEs from capacity cells to basic cells and instruct the RF module serving the capacity cells to enter the super dormancy state within a specified period when the network is under light or no load. When the load of basic cells increases and reaches a specified threshold, the RF module in the dormancy state is automatically woken up and the capacity cells are restored.

LTE load-based RF module super dormancy can be classified into non-precise dormancy mode and precise dormancy mode.

In scenarios where basic cells cover large areas, capacity cells cover small areas, and UEs are unevenly distributed, the precise dormancy mode enables the base station to precisely identify cells that can enter or exit the dormancy state based on the UE distribution in the basic and capacity cells. This maximizes the probability of cells entering the dormancy state and minimizes the probability of cells exiting this state, prolonging the dormancy duration, as shown in [Figure 7-24](#).

Figure 7-24 Working principles for precise dormancy mode



This function requires the following switches to be configured:

- RF module-level switches:

The **RRU.DORMANCYSW (5G gNodeB, LTE eNodeB)** parameter is set to **ON_SUPER** or **ON_DEEP_SUPER**, the **RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter is set to **ANYTIME_WAKEUP_SW**, and either of the following conditions is met:

- The **RRU.CUSTOMDORMANCYSTDSW** parameter is set to **OFF**.
- The **RRU.CUSTOMDORMANCYSTDSW** parameter is set to **ON**, and the **LTE_TDD** or **LTE_FDD** option of the **RRU.DORMANCYSTD** parameter is selected.

The value **ON_DEEP_SUPER** of **RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB)** indicates that RF module deep dormancy and RF module super dormancy can be both enabled. For details about RF module deep dormancy, see [7.2 RF Module Deep Dormancy](#).

When the **RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB)** parameter is set to **ON_DEEP_SUPER** within the configured dormancy period, and both the conditions for entering RF module deep dormancy and RF module super dormancy are met:

- RF module super dormancy takes effect if the **ANYTIME_WAKEUP_SW** option of **RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** is selected.
- If the **ANYTIME_WAKEUP_SW** option of the **RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter is not selected:
 - If the **ANYTIME_WAKEUP_SW** option of the **RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter is selected within 30s, RF module super dormancy takes effect.
 - If the **ANYTIME_WAKEUP_SW** option of the **RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter is not selected within 30s, RF module deep dormancy takes effect.

- LTE cell-level switch:

- In non-precise dormancy mode, the **INTRA_LTE_DORMANCY_BY_LOAD_SW** option of the **CellPwrSavingAlgo.PwrSavingAlgoSwitch** parameter is selected.
- In precise dormancy mode, the **LTE_DORMANCY_BY_LOAD_PE_SW** option of the **CellPwrSavingAlgo.PwrSavingAlgoSwitch** parameter is selected.

The effective period and threshold parameters for LTE load-based RF module dormancy can be uniquely determined by using the combinations of the **CellPwrSavingPolicy.PolicyIndex** and **CellPwrSavingPolicy.PowerSavingType** parameters in the **CellPwrSavingPolicy** MO on the LTE side.

- The **CellPwrSavingPolicy.PolicyIndex** parameter specifies an energy-saving policy index. It uniquely identifies an energy-saving policy within the local cell. With the index specified, the corresponding policy can be added, modified, queried, or removed from the local cell.

- The **CellPwrSavingPolicy.PowerSavingType** parameter specifies the energy saving type indicated by an energy-saving policy index. Non-precise LTE load-based dormancy requires that this parameter be set to **INTRA_LTE_DORMANCY_BY_LOAD**, and LTE load-based precise dormancy requires that this parameter be set to **LTE_PRECISE_DORMANCY_BY_LOAD**, meaning that under the corresponding energy-saving policy, the local cell performs load-based dormancy in the specified effective period under certain load conditions.

Related Concepts

- Basic-layer carriers provide basic coverage for the network. RF modules providing such carriers will not enter the dormancy state. The cells served by basic-layer carriers are referred to as basic cells.
- Capacity-layer carriers support large traffic volumes during peak hours. RF modules providing such carriers can enter the dormancy state. The cells served by capacity-layer carriers are referred to as capacity cells.

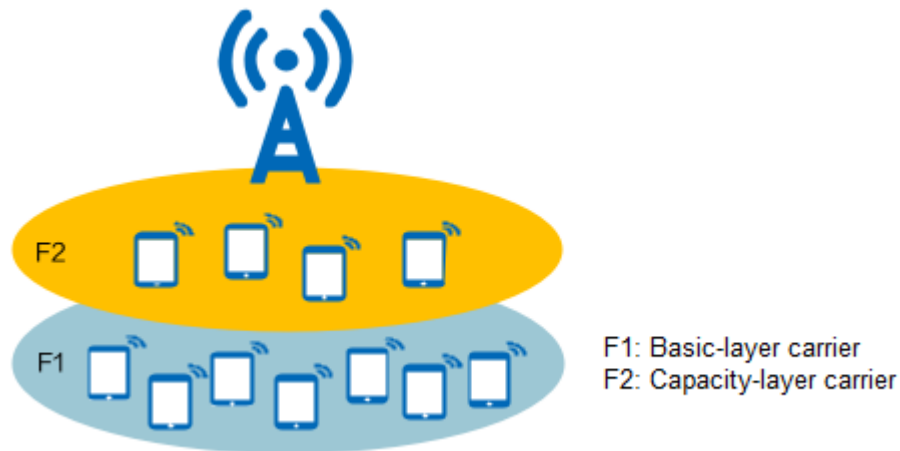
NOTE

Load-based RF module super dormancy is enabled for capacity cells and corresponding RF modules. One capacity cell corresponds to one or more basic cells. Basic cells and capacity cells must be inter-frequency neighboring cells of each other.

Basic-layer and capacity cells are relative concepts identified by the **POWER_SAVING_OVERLAP_IND** option of the **EutranInterFreqNCell.OverlapIndicatorExtension** parameter during neighbor relationship configuration.

- If this option is deselected, there is no energy-saving relationship between the neighboring cell and the serving cell. The neighboring cell is not considered in load-based RF module dormancy.
- If this option is selected, the neighboring cell is a basic cell, the serving cell is a capacity cell, and the cells have overlapping coverage.

Figure 7-25 Basic cells and capacity cells



For example, on the capacity cell (on F2 in [Figure 7-25](#)) side, the **POWER_SAVING_OVERLAP_IND** option of the

EutranInterFreqNCell.OverlapIndicatorExtension parameter is selected for the basic cell on F1, which is a neighboring cell.

In precise dormancy mode, independent configurations are used to identify capacity cells and basic cells.

- When the **CellPwrSavingAlgo.DormancyCellType** parameter is set to **BASIC_CELL** for a cell, the cell serves as a basic cell for load-based precise dormancy.
- When the **CellPwrSavingAlgo.DormancyCellType** parameter is set to **CAPACITY_CELL** for a cell, the cell serves as a capacity cell for load-based precise dormancy.

In LTE multi-carrier networking scenarios, it is recommended that capacity and basic cells be configured as described in [Table 7-48](#).

Table 7-48 Suggestions on designation of capacity and basic cells

Networking	Capacity Cell	Basic Cell
LTE TDD and LTE FDD coordinated networking	LTE TDD cells	LTE FDD cells
LTE TDD or LTE FDD intra-band multi-carrier networking	Cells on a higher frequency	Cells on a lower frequency

Entering the Super Dormancy State

[Figure 7-26](#) shows how the RF module serving a capacity cell enters the dormancy state after LTE load-based RF module dormancy is enabled.

Figure 7-26 Procedure for entering LTE load-based RF module super dormancy



1. The base station determines whether entering conditions are met.
The base station periodically checks whether the conditions for entering the dormancy state are met within an evaluation period specified by the **CellPwrSavingPolicy.TriggerJudgePeriod** parameter.
 - A capacity cell must meet all of the following conditions:
 - The current time is within the effective period for this function, which is determined by the **CellPwrSavingPolicy.StartTime** and **CellPwrSavingPolicy.StopTime** parameters.
 - The capacity cell is not in any energy saving penalty state.
 - The basic cells corresponding to the capacity cell must meet one of the following conditions:

- All basic cells are available.
- Some basic cells are available and some basic cells are in the energy saving state. For example, these cells enter the energy saving state when functioning as capacity cells of other basic cells. (Whether this condition applies depends on the setting of the **INTRA_RAT_GRAD_PWR_SAVING_SW** option of the **EnodebPwrSavingAlgo.PwrSavingAlgoSwitch** parameter.)
- For common cells (including multi-RRU cells, FDD Dynamic Massive Beam cells (8T3S scenarios), and TDD cells split through soft split resource duplex (inter-frequency split)), and FDD Dynamic Massive Beam cells in massive MIMO scenarios, a capacity cell and its corresponding basic cell must both meet the following conditions:
 - In non-precise dormancy mode, the following conditions must also be met:
 - The total downlink PRB usage of the capacity and basic cells is less than or equal to the downlink PRB usage threshold specified by the **CellPwrSavingPolicy.DlPrbThld** parameter, where:
 Total downlink PRB usage of the capacity and basic cells = $[(\text{Downlink PRB usage of the capacity cell} \times \text{System bandwidth of the capacity cell} \times \text{Number of downlink subframes per unit time}) + \sum_{\text{basic cell}} (\text{Downlink PRB usage of the basic cell} \times \text{System bandwidth of the basic cell} \times \text{Number of downlink subframes per unit time})] / \sum_{\text{basic cell}} (\text{System bandwidth of the basic cell} \times \text{Number of downlink subframes per unit time})$
 - The total uplink PRB usage of the capacity and basic cells is less than or equal to the uplink PRB usage threshold specified by the **CellPwrSavingPolicy.UlPrbThld** parameter, where:
 Total uplink PRB usage of the capacity and basic cells = $[(\text{Uplink PRB usage of the capacity cell} \times \text{System bandwidth of the capacity cell} \times \text{Number of uplink subframes per unit time}) + \sum_{\text{basic cell}} (\text{Uplink PRB usage of the basic cell} \times \text{System bandwidth of the basic cell} \times \text{Number of uplink subframes per unit time})] / \sum_{\text{basic cell}} (\text{System bandwidth of the basic cell} \times \text{Number of uplink subframes per unit time})$
 - In precise dormancy mode, the following conditions must also be met:
 - Uplink PRB usage in the co-coverage area < Maximum uplink PRB usage in the co-coverage area after the local cell is shut down – Uplink PRB offset – 5%
 - Downlink PRB usage in the co-coverage area < Maximum downlink PRB usage in the co-coverage area after the local cell is shut down – Downlink PRB offset – 5%
 - Uplink PRB usage of each basic-layer carrier corresponding to the capacity-layer carrier < **CellPwrSavingPolicy.ProtectUlPrbThld** – **CellPwrSavingPolicy.ProtectUlPrbOffset**

Uplink PRB usage of each basic-layer carrier corresponding to the capacity-layer carrier = Uplink PRB usage of the basic cell + Overlap rate x Uplink PRB usage of the capacity cell

- Downlink PRB usage of each basic-layer carrier corresponding to the capacity-layer carrier <

CellPwrSavingPolicy.ProtectDlPrbThld - CellPwrSavingPolicy.ProtectDlPrbOffset

Downlink PRB usage of each basic-layer carrier corresponding to the capacity-layer carrier = Downlink PRB usage of the basic cell + Overlap rate x Downlink PRB usage of the capacity cell

The overlap rate is the proportion of overlapping area in a capacity cell to a basic cell, which is specified by the **EutranInterFreqNCell.OverlapRate** parameter at the basic cell side.

NOTE

Downlink PRB usage of an LTE cell = Average number of used downlink PRBs/ Number of available downlink PRBs. The number of RBs for common channels is included in the calculation of downlink PRB usage.

Uplink PRB usage of an LTE cell = Average number of used uplink PRBs/Number of available uplink PRBs

In precise dormancy mode:

- Total downlink PRB usage of the capacity cell and basic cells = [(Downlink PRB usage of the capacity cell x System bandwidth of the capacity cell x Number of downlink subframes per unit time) + $\sum_{\text{basic cell}} (\text{Overlap rate} \times \text{Downlink PRB usage of the basic cell} \times \text{System bandwidth of the basic cell} \times \text{Number of downlink subframes per unit time})$]/[System bandwidth of the capacity cell x Number of downlink subframes per unit time + $\sum_{\text{basic cell}} (\text{Overlap rate} \times \text{System bandwidth of the basic cell} \times \text{Number of downlink subframes per unit time})$]
- Total uplink PRB usage of the capacity cell and basic cells = [(Uplink PRB usage of the capacity cell x System bandwidth of the capacity cell x Number of uplink subframes per unit time) + $\sum_{\text{basic cell}} (\text{Overlap rate} \times \text{Uplink PRB usage of the basic cell} \times \text{System bandwidth of the basic cell} \times \text{Number of uplink subframes per unit time})$]/[System bandwidth of the capacity cell x Number of uplink frames per unit time + $\sum_{\text{basic cell}} (\text{Overlap rate} \times \text{System bandwidth of the basic cell} \times \text{Number of uplink frames per unit time})$]

The overlap rate between capacity cells and basic cells for precise dormancy is specified by the **EutranInterFreqNCell.OverlapRate** parameter.

- The number of RRC_CONNECTED UEs in the capacity cell is less than or equal to the UE number threshold specified by the **CellPwrSavingPolicy.UENumThld** parameter.
- In multi-operator sharing scenarios, the PLMN IDs of the operators serving the capacity cell are included in the PLMN ID set of operators of at least one basic cell.
- In multi-operator sharing scenarios, the operators of the local cell must be a subset of the operators of co-coverage neighboring basic cells. If optimization for cell PLMN reservation scenarios is enabled by selecting the **CELL_PLMN_RSVD_OPT_SW** option of the **EnodebPwrSavingAlgo.PwrSavingAlgoSwitch** parameter:

- If the operators of the local cell are non-reserved operators, the operators must be a subset of non-reserved operators of the co-coverage neighboring basic cells.
- If the operators of the local cell are reserved operators, the operators must be a subset of non-reserved operators or reserved operators of the co-coverage neighboring basic cells.

The operator status of the local cell and intra-base-station co-coverage neighboring basic cells is obtained as follows:

- Using the effective operator status of the cells if the **RSVD_FOR_OP_X2_BROADCAST_SW** option of the **eNodeBOamParam.ConfigAdaptOptSwitch** parameter is selected
- Using the value of the **CellOp.CellReservedForOp** parameter if the **RSVD_FOR_OP_X2_BROADCAST_SW** option of the **eNodeBOamParam.ConfigAdaptOptSwitch** parameter is deselected

The operator status of inter-base-station co-coverage neighboring basic cells is obtained as follows:

- Configuring through the **PLMN_RESERVED_FLAG** option of the **EutranExternalCellPlmn.AggregationAttribute** parameter if the **RSVD_FOR_OP_X2_BROADCAST_SW** option of the **eNodeBOamParam.ConfigAdaptOptSwitch** parameter is deselected
- Using the following methods if the **RSVD_FOR_OP_X2_BROADCAST_SW** option of the **eNodeBOamParam.ConfigAdaptOptSwitch** parameter is selected:

Method 1: Using the **PLMN_RESERVED_FLAG** option of the **EutranExternalCellPlmn.AggregationAttribute** parameter

Method 2: The operator status of co-coverage neighboring basic cells is automatically obtained through broadcast over the X2 interface when neighbor relationships are properly configured and the X2 interface between base stations is working properly.

If both of the preceding methods are available, the operator status obtained through these methods must meet the triggering conditions.

- (When the capacity and basic cells are provided by different base stations) The X2 interface is normal.
- For sector split cells (excluding FDD Dynamic Massive Beam cells in massive MIMO scenarios), a capacity cell and its corresponding basic cell must meet the following conditions:
 - Uplink PRB usage of the local cell + Uplink PRB usage of co-coverage neighboring basic cells + Uplink PRB usage of other cells in the same sector split group < Uplink PRB threshold for starting intelligent power-off of carriers in the same coverage
 - Downlink PRB usage of the local cell + Downlink PRB usage of co-coverage neighboring basic cells + Downlink PRB usage of other cells

in the same sector split group < Downlink PRB threshold for starting intelligent power-off of carriers in the same coverage

- Uplink PRB usage of the local cell + Uplink PRB usage of a co-coverage neighboring basic cell + Uplink PRB usage of other cells in the same sector split group < Threshold for a co-coverage neighboring basic cell to start load adjustment of inter-frequency load balancing (**CellMLB.InterFreqMlbThd** plus **CellMLB.LoadOffset**)
- Downlink PRB usage of the local cell + Downlink PRB usage of a co-coverage neighboring basic cell + Downlink PRB usage of other cells in the same sector split group < Threshold for a co-coverage neighboring basic cell to start load adjustment of inter-frequency load balancing (**CellMLB.InterFreqMlbThd** plus **CellMLB.LoadOffset**)

All cells in a sector split group are enabled with load-based dormancy and meet the triggering conditions.

 NOTE

Sector split cells cannot be configured as capacity cells for LTE load-based RF module dormancy in precise dormancy mode.

2. A UE transfer procedure starts.

- The capacity cell prohibits the access of new UEs and incoming handovers.

The eNodeB broadcasts the barred state of the capacity cell through SIB1. In addition, the eNodeB sends an eNodeB Configuration Update message over the X2 interface to each neighboring eNodeB. The message contains the *Deactivation Indication* IE, indicating that the local cell is unavailable due to energy saving.

- The eNodeB attempts to hand over all RRC_CONNECTED UEs from the capacity cell to the cells on the operating frequencies of the co-coverage neighboring basic cells within 60s.

The eNodeB delivers event A4 measurement configurations about frequencies of the co-coverage basic cells to UEs in connected mode. The RSRP threshold for event A4 is specified by the **InterFreqHoGroup.InterFreqHoA4ThdRsrp** parameter. After receiving event A4 measurement reports from UEs, the eNodeB initiates inter-frequency handovers to the cells in the frequencies of the basic layers for UEs.

During the handover attempt, the eNodeB checks the number of RRC_CONNECTED UEs in the capacity cell every 5s.

- If the number of RRC_CONNECTED UEs is 0, the capacity cell enters the dormancy state.
- If the number of RRC_CONNECTED UEs is not 0, the eNodeB checks whether the number of RRC_CONNECTED UEs is less than or equal to the value of the **CellPwrSavingPolicy.ForcibleTriggerUeNumThld** parameter and whether there are no UEs performing QCI-1 services in the capacity cell after 60s elapses. If so, the capacity cell enters the dormancy

state. If not, the eNodeB cancels the barred state of the capacity cell and notifies all neighboring eNodeBs over the X2 interfaces that the capacity cell has exited the energy saving state. It does not check the entering conditions again until the penalty period specified by the **CellPwrSavingPolicy.PenaltyTime** parameter elapses.

3. The RF module enters the super dormancy state.

Once all UEs are transferred out of the capacity cell, no RRC_CONNECTED UEs remain in the cell, and the cell and the corresponding RF module enter the super dormancy state. If an RF module serves multiple LTE cells, it enters the super dormancy state only when load-based RF module dormancy is triggered for all these LTE cells. If an RF module also serves NR cells, RF module super dormancy must also be enabled on the NR side.

If intra-RAT level-by-level energy saving is enabled by selecting the **INTRA_RAT_GRAD_PWR_SAVING_SW** option of the **EnodebPwrSavingAlgo.PwrSavingAlgoSwitch** parameter, some basic cells can be used as capacity cells of other basic cells and enter the energy saving state due to functions such as intelligent power-off of carriers in the same coverage, Cell Switch-off Based on Automatic Co-coverage Identification, multi-carrier coordinated energy saving, GUL joint carrier shutdown, LTE load-based RF module dormancy, or dynamic carrier shutdown.

Exiting the Super Dormancy State

The base station periodically checks whether the conditions for exiting the dormancy state are met. The exit evaluation period is specified by the **CellPwrSavingPolicy.ExitJudgePeriod** parameter. Within an exit evaluation period, the base station evaluates whether the conditions are met at an interval specified by the **CellPwrSavingPolicy.EvalInterval** parameter. The cell can exit the power saving state only when the conditions are met at each evaluation occasion within the period.

- If the **CellPwrSavingPolicy.EvalInterval** parameter is set to **EVAL_INTERVAL_60S**, the exit evaluation is performed at an interval of 60s in the exit evaluation period. In this case, the **CellPwrSavingPolicy.ExitJudgePeriod** parameter must be set to an integer multiple of 12 (unit: 5s), that is, the exit evaluation period must be an integer multiple of 1 minute.
- If the **CellPwrSavingPolicy.EvalInterval** parameter is set to **EVAL_INTERVAL_5S**, the exit evaluation is performed at an interval of 5s in the exit evaluation period. In this case, the **CellPwrSavingPolicy.ExitJudgePeriod** parameter must be set to a value within the range of 1 to 120 (unit: 5s), that is, the exit evaluation period must be an integer multiple of 5s.

To support second-level evaluation periods, the **EnodebPwrSavingAlgo.InterEnbInfoExchangeMode** parameter has been added to specify how frequently information is exchanged between base stations. This enables base stations to obtain the latest load information more quickly, allowing them to evaluate whether the conditions for exiting the dormancy state are met.

- If this parameter is set to **SLOW_EXCHANGE_MODE**, the slow exchange mode is used, that is, information is exchanged at a long interval.
- If this parameter is set to **FAST_EXCHANGE_MODE**, the fast exchange mode is used, that is, information is exchanged at a short interval.

A capacity cell and the corresponding RF module automatically exit the dormancy state when any of the following conditions is met:

- The current time is not within the effective period for this function, which is determined by the **CellPwrSavingPolicy.StartTime** and **CellPwrSavingPolicy.StopTime** parameters.
- The total downlink PRB usage of basic cells is greater than the sum of the **CellPwrSavingPolicy.DlPrbThld** and **CellPwrSavingPolicy.DlPrbOffset** parameter values.
- The total uplink PRB usage of basic cells is greater than the sum of the **CellPwrSavingPolicy.UlPrbThld** and **CellPwrSavingPolicy.UlPrbOffset** parameter values.
- In precise dormancy mode, uplink PRB usage in the co-coverage area $\geq$ Maximum uplink PRB usage in the co-coverage area after the local cell is shut down – 5%
- In precise dormancy mode, downlink PRB usage in the co-coverage area $\geq$ Maximum downlink PRB usage in the co-coverage area after the local cell is shut down – 5%
- In precise dormancy mode, the uplink PRB usage of any basic-layer carrier corresponding to the capacity-layer carrier is greater than or equal to the value of the **CellPwrSavingPolicy.ProtectUlPrbThld** parameter.

Uplink PRB usage of a basic-layer carrier corresponding to the capacity-layer carrier = Number of used uplink PRBs in the basic cell/Total number of available uplink PRBs in the basic cell

- In precise dormancy mode, the downlink PRB usage of any basic-layer carrier corresponding to the capacity-layer carrier is greater than or equal to the value of the **CellPwrSavingPolicy.ProtectDlPrbThld** parameter.

Downlink PRB usage of a basic-layer carrier corresponding to the capacity-layer carrier = Number of used downlink PRBs in the basic cell/Total number of available downlink PRBs in the basic cell

- A basic cell is added for the capacity cell.
- In multi-operator sharing scenarios, the PLMN IDs of the operators serving the capacity cell are not included in the PLMN ID set of operators serving any basic cell.
- In multi-operator sharing scenarios, the operators of the local cell are not a subset of the operators of co-coverage neighboring basic cells. If optimization for cell PLMN reservation scenarios is enabled by selecting the **CELL_PLMN_RSVD_OPT_SW** option of the **ENodebPwrSavingAlgo.PwrSavingAlgoSwitch** parameter:

- If the operators of the local cell are non-reserved operators, the operators are not a subset of non-reserved operators of the co-coverage neighboring basic cells.
- If the operators of the local cell are reserved operators, the operators are not a subset of non-reserved operators or reserved operators of the co-coverage neighboring basic cells.

The operator status of the local cell and intra-base-station co-coverage neighboring basic cells is obtained as follows:

- Using the effective operator status of the cells if the **RSVD_FOR_OP_X2_BROADCAST_SW** option of the **eNodeBOamParam.ConfigAdaptOptSwitch** parameter is selected
- Using the value of the **CellOp.CellReservedForOp** parameter if the **RSVD_FOR_OP_X2_BROADCAST_SW** option of the **eNodeBOamParam.ConfigAdaptOptSwitch** parameter is deselected

The operator status of inter-base-station co-coverage neighboring basic cells is obtained as follows:

- Configuring through the **PLMN_RESERVED_FLAG** option of the **EutranExternalCellPlmn.AggregationAttribute** parameter if the **RSVD_FOR_OP_X2_BROADCAST_SW** option of the **eNodeBOamParam.ConfigAdaptOptSwitch** parameter is deselected
- Using the following methods if the **RSVD_FOR_OP_X2_BROADCAST_SW** option of the **eNodeBOamParam.ConfigAdaptOptSwitch** parameter is selected:

Method 1: Using the **PLMN_RESERVED_FLAG** option of the **EutranExternalCellPlmn.AggregationAttribute** parameter

Method 2: The operator status of co-coverage neighboring basic cells is automatically obtained through broadcast over the X2 interface when neighbor relationships are properly configured and the X2 interface between base stations is working properly.

If both of the preceding methods are available, the operator status obtained through these methods must meet the triggering conditions.

- (When the capacity and basic cells are provided by different base stations) The X2 interface is abnormal.
- Any basic cell becomes unavailable due to other reasons, such as cell faults and cell deactivation.
- When the **INTRA_RAT_GRAD_PWR_SAVING_SW** option of the **EnodebPwrSavingAlgo.PwrSavingAlgoSwitch** parameter is deselected, the cells that serve as the capacity cells of other basic cells and enter the energy saving state are considered as cells meeting the exit conditions.
- When the **INTRA_RAT_GRAD_PWR_SAVING_SW** option of the **EnodebPwrSavingAlgo.PwrSavingAlgoSwitch** parameter is selected, the cells that serve as the capacity cells of other basic cells and enter the energy saving state are not considered as cells meeting the exit conditions and will not trigger the exit from the dormancy state.

Upon the exit from dormancy:

- The capacity cell cannot enter the dormancy state again within the penalty period specified by the **CellPwrSavingPolicy.PenaltyTime** parameter if it exited this state because of the uplink or downlink PRB usage, unavailability of a basic cell, or abnormal X2 interface.
- The penalty mechanism will not be triggered for the capacity cell in the following scenarios:
 - Dormancy is exited because the function switch is turned off or the energy saving period has elapsed.
 - Dormancy is pending due to base station software upgrade or base station configuration.

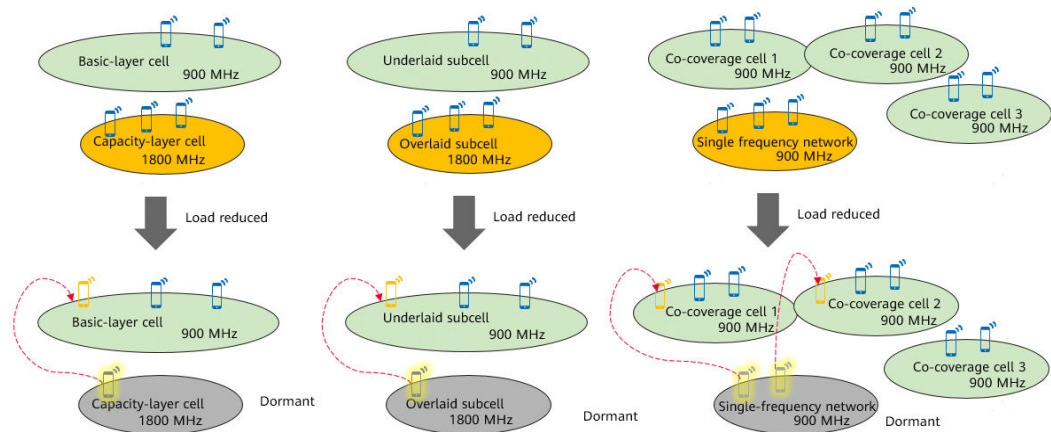
7.4.1.6 GSM Load-based RF Module Super Dormancy

On GSM dual-band networks, low-band basic-layer cells are used to provide wide network coverage, and high-band capacity-layer cells are used to absorb traffic. In low-load scenarios, capacity-layer cells still consume much power, resulting in a waste of energy. Therefore, GSM load-based RF module super dormancy is required. This function enables UEs to be transferred from capacity-layer cells to basic-layer cells and allows the RF module serving the capacity-layer cells to enter the super dormancy state within a specified period when the network is under light or no load. When the load of the basic-layer cells increases to a specified threshold, the RF modules in dormancy are automatically woken up and the capacity-layer cells are restored, achieving a balance between services and energy saving.

In GSM concentric cell scenarios, the underlaid and overlaid subcells function as basic-layer and capacity-layer cells, respectively. When the load of the overlaid subcell is low, traffic needs to be transferred to the underlaid subcell, and the overlaid subcell can then enter the dormancy state. When the load of the underlaid subcell is high, the overlaid subcell is woken up to carry services.

In GSM single frequency networks (SFNs), physical co-coverage cells do not exist. However, some neighboring cells of a single-frequency cell can be considered as co-coverage cells of this single-frequency cell based on signal levels measured by these neighboring cells. These co-coverage cells can be configured as basic-layer cells. When the load is low, the traffic of the single-frequency cell can be transferred to its co-coverage cells and then the single-frequency cell can enter the dormancy state. When the load of the co-coverage cells is high, the single-frequency cell can be woken up to carry some traffic.

Figure 7-27 Schematic diagram of GSM load-based RF module super dormancy



This function requires the following switches to be configured:

- RF module-level switches

The **RRU.DORMANCYSW (5G gNodeB, LTE eNodeB)** parameter is set to **ON_SUPER**, the **RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter is set to **ANYTIME_WAKEUP_SW**, and either of the following conditions is met:

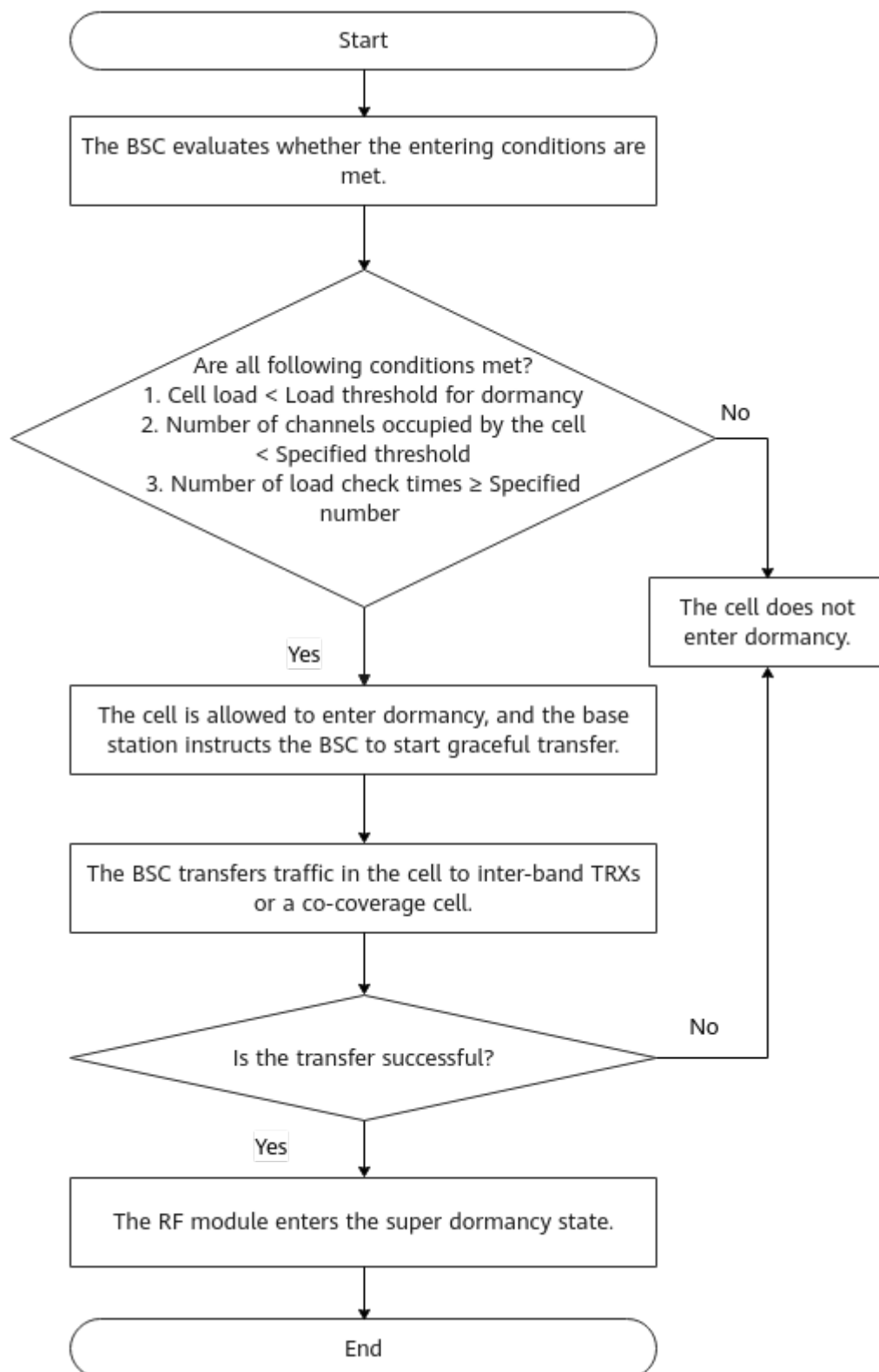
- The **RRU.CUSTOMDORMANCYSTDSW (5G gNodeB, LTE eNodeB)** parameter is set to **OFF**.

- The **RRU.CUSTOMDORMANCYSTDSW (5G gNodeB, LTE eNodeB)** parameter is set to **ON**, and the **GSM** option of the **RRU.DORMANCYSTD (5G gNodeB, LTE eNodeB)** parameter is selected.
- Switch on the BSC side
 - If the **GCELLOTHPARA.DormancyEnable** parameter is set to **ENABLE**, load-based RF module super dormancy is enabled on the BSC side.
 - For a non-concentric cell, at least one co-coverage cell must be configured (co-coverage cells are specified by parameters **GCELLOTHPARA.CoCoverageCellId1** to **GCELLOTHPARA.CoCoverageCellId4**).
 - On the BSC side, **GCELLDORMANCY** MOs are used to set the effective periods of GSM load-based RF module super dormancy and other dormancy algorithm parameters. An effective period and other corresponding dormancy algorithm parameters are uniquely determined by the **GCELLDORMANCY.DormancyPolicyIndex** parameter, which uniquely identifies a dormancy policy for a cell. With the index specified, the corresponding dormancy policy can be added, modified, queried, or removed for a cell.
- For dormancy involving GSM, the **GU_DC_SW** option of the base-station-specific **EQUIPMENT.EQMDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter must be deselected. If the option is deselected and a configuration command is executed to trigger dormancy pending or dormancy pending is enabled on the local end that works in GSM, then dormancy pending will take effect on the entire base station.

Entering the Super Dormancy State

Figure 7-28 shows how the RF module serving a capacity-layer cell enters dormancy after GSM load-based RF module dormancy is enabled.

Figure 7-28 Procedure for entering GSM load-based RF module super dormancy



1. The BSC evaluates whether the entering conditions are met.

The current time is within an effective period (determined by **GCELLDORMANCY.DormancyCellStrTime** and **GCELLDORMANCY.DormancyCellStpTime**) for this function.

- If the cell is a common cell, co-coverage cells must be configured for the cell. The BSC checks the working status and load of the cell (expected to enter dormancy) and its co-coverage cells every minute for N (specified by the **GCELLDORMANCY.DormancyCellLoadStatTm** parameter) times. The following conditions must be met in each check:
 - The load of each co-coverage cell is less than the value of the **GCELLDORMANCY.DormancyLoadThrD** parameter. When calculating the load of a co-coverage cell, the BSC determines whether to include dynamic PDCHs in the load statistics based on the setting of the **GCELLCHMGAD.LOADSTATYPE** parameter.
 - The number of channels (TCHs and PDCHs) occupied by the cell expected to enter dormancy is less than the value of the **GCELLDORMANCY.DormancyCellChanNum** parameter.
- If the cell is a concentric cell, the underlaid subcell serves as the co-coverage cell of the overlaid subcell. The BSC checks the load of the overlaid and underlaid subcells every minute for N (specified by the **GCELLDORMANCY.DormancyCellLoadStatTm** parameter) times. The following conditions must be met in each check:
 - The load of the underlaid subcell, indicated by Number of calls in the underlaid subcell/Total number of channels occupied by the underlaid subcell, is less than the value of the **GCELLDORMANCY.DormancyLoadThrD** parameter.
 - The overall load of the overlaid and underlaid subcells, indicated by (Number of calls in the overlaid subcell + Number of calls in the underlaid subcell)/Total number of channels occupied by the underlaid subcell, is less than the value of the **GCELLDORMANCY.WakeUpLoadThrD** parameter. This ensures that calls in the overlaid subcell can be handed over to the underlaid subcell.
 - The number of active channels (TCHs and PDCHs) in the overlaid subcell is less than the value of the **GCELLDORMANCY.DormancyCellChanNum** parameter.

By default, CBC services take precedence over dormancy. Therefore, when CBC is enabled in a cell for which the **BSCXSOFTPARA.IgnoreCbcSwitch** parameter is set to **OFF** (default value), the cell cannot enter the dormancy state. If dormancy is required in scenarios where CBC is enabled, this parameter must be set to **ON**. In this situation:

- If no CBC message is sent and the current time falls outside the CBC protection period (specified by **BSCXSOFTPARA.CbcServiceHoldDuration**), the cell can enter dormancy.
- If CBC messages are sent, the cell cannot enter dormancy.

 NOTE

After receiving a CBC message, the BSC immediately wakes up the dormant cell and processes the CBC message once the cell is successfully set up. Since the cell needs time to wake up, CBC messages may not be processed promptly. If CBC has a higher priority, it is recommended that **BSCEXSOFTPARA.IgnoreCbcSwitch** be set to its default value **OFF**. For details about CBC, see *Cell Broadcast* in *GBSS Feature Documentation*.

2. The base station instructs the BSC to start a graceful transfer procedure.
 - In the case of a common cell, as all its TRXs are expected to enter the dormancy state, calls in the cell are forcibly handed over to its co-coverage cell and PS services are directly released.
 - In the case of a concentric cell, as only some of its TRXs are expected to enter the dormancy state, calls on the TRXs to enter dormancy are forcibly handed over to inter-band TRXs and PS services are directly released. When a new call (CS or PS call) is admitted, the call can be admitted only to TRXs that do not enter the dormancy state.
 - During the graceful transfer, if the forced handover fails, the dormancy procedure is terminated. The cell is not allowed to enter the dormancy state again within the period specified by the **GCELLDORMANCY.DormancyPenaltyTime** parameter. After the penalty period ends, the system checks the trigger conditions again.

3. The RF module enters the super dormancy state.

After the traffic is successfully transferred, the cell and its associated RF module enter the super dormancy state. At the same time, the system starts to check the dormancy exit conditions.

Exiting the Super Dormancy State

1. The BSC determines whether the dormancy exit conditions are met.

A cell and its associated RF module automatically exit dormancy when any of the following conditions is met:

 - The current time is not within the effective periods (determined by **GCELLDORMANCY.DormancyCellStrTime** and **GCELLDORMANCY.DormancyCellStpTime**) for this function.
 - If the cell in dormancy is a common cell, the BSC checks the load of all its co-coverage cells every minute for *N* (specified by the **GCELLDORMANCY.DormancyCellLoadStatTm** parameter) times. The exit condition is that the load of any of its co-coverage cells is greater than or equal to **GCELLDORMANCY.WakeUpLoadThrD** in each check.
 - If the cell in dormancy is a concentric cell, the BSC checks the load of the underlaid subcell every minute for *N* (specified by the **GCELLDORMANCY.DormancyCellLoadStatTm** parameter) times. The exit condition is that the load of the underlaid subcell is greater than or equal to **GCELLDORMANCY.WakeUpLoadThrD** in each check.
 - The cell receives a CBC message from the CBC server.
2. The BSC instructs the base station to start the dormancy exit process.

The BSC sends a message to the base station, instructing the base station to get the cell to exit the dormancy state. The base station then sets up carriers.

If the dormancy exit is triggered by a CBC message, the cell is not allowed to enter dormancy during the CBS hold duration (specified by **BSCEXSOFTPARA.CbcServiceHoldDuration**), which starts upon the exit.

7.4.1.7 UMTS Load-based RF Module Super Dormancy

Multi-carrier co-coverage UMTS networking provides a larger network capacity and, upon a heavy traffic load, a higher data throughput than single-carrier UMTS networking. However, such networking wastes base station energy when the network is under light or no load. To address this, UMTS load-based RF module super dormancy is introduced.

Load-based RF module super dormancy enables the RNC to transfer UEs from capacity-layer cells to basic-layer cells and instruct the RF module serving the capacity-layer cells to enter the super dormancy state within a specified period when the network is under light or no load. When the load of the basic-layer cells increases to a specified threshold, the RF modules in dormancy are automatically woken up and the capacity-layer cells are restored.

This function requires the following switches to be configured:

- RF module-level switches

The **RRU.DORMANCYSW (5G gNodeB, LTE eNodeB)** parameter is set to **ON_SUPER** or **ON_DEEP_SUPER**, the **RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter is set to **ANYTIME_WAKEUP_SW**, and either of the following conditions is met:

- The **RRU.CUSTOMDORMANCYSTDSW (5G gNodeB, LTE eNodeB)** parameter is set to **OFF**.
- The **RRU.CUSTOMDORMANCYSTDSW (5G gNodeB, LTE eNodeB)** parameter is set to **ON**, and the **UMTS** option of the **RRU.DORMANCYSTD (5G gNodeB, LTE eNodeB)** parameter is selected.

The value **ON_DEEP_SUPER** of **RRU.DORMANCYSW (5G gNodeB, LTE eNodeB)** indicates that RF module deep dormancy and RF module super dormancy can be both enabled. For details about RF module deep dormancy, see [7.2 RF Module Deep Dormancy](#).

When the **RRU.DORMANCYSW (5G gNodeB, LTE eNodeB)** parameter is set to **ON_DEEP_SUPER** within the configured dormancy period, and both the conditions for entering RF module deep dormancy and those for entering RF module super dormancy are met:

- RF module super dormancy takes effect if the **ANYTIME_WAKEUP_SW** option of the **RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter is selected.
- If the **ANYTIME_WAKEUP_SW** option of the **RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter is not selected:
 - If the **ANYTIME_WAKEUP_SW** option of the **RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter is selected within 30s, RF module super dormancy takes effect.
 - If the **ANYTIME_WAKEUP_SW** option of the **RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)**

parameter is not selected within 30s, RF module deep dormancy takes effect.

- UMTS cell-level switch
 - The **DORMANCY_BY_LOAD_SW** option of the **ULOCCELLALGPARA.DORMANCYALGOSW** parameter is selected.
 - The effective period and threshold parameters for UMTS load-based RF module super dormancy can be uniquely determined by using the combinations of the **UCELLDORMANCYPOLICY.PolicyIndex** and **UCELLDORMANCYPOLICY.DormancyType** parameters in the **UCELLDORMANCYPOLICY** MO on the RNC side. Specifically:
 - The **UCELLDORMANCYPOLICY.PolicyIndex** parameter specifies an energy saving policy index. It uniquely identifies an energy saving policy within the local cell. With the index specified, the corresponding policy can be added, modified, queried, or removed from the local cell.
 - The **UCELLDORMANCYPOLICY.DormancyType** parameter specifies the energy saving type corresponding to an energy saving policy index. UMTS load-based dormancy requires this parameter to be set to **INTRA_DORMANCY_BY_LOAD**. Then under the corresponding energy saving policy index, the local cell performs load-based dormancy in the preset duration based on the load of the local cell and that of inter-band neighboring cells.
- RNC-level switch

The **HO_INTER_FREQ_HARD_HO_SWITCH** option of the **UCORRMALGOSWITCH.HoSwitch** parameter is selected to ensure that inter-frequency handovers can be performed.
- For dormancy involving UMTS, the **GU_DC_SW** option of the base-station-specific **EQUIPMENT.EQMDOROPTFEATSW** parameter must be deselected. If the option is deselected and a configuration command is executed to trigger dormancy pending or dormancy pending is enabled on the local end that works in UMTS, then dormancy pending will take effect on the entire base station.

Related Concepts

For the concepts of basic-layer carriers, capacity-layer carriers, basic-layer cells, and capacity-layer cells, refer to [Related Concepts](#).

Entering the Super Dormancy State

[Figure 7-29](#) shows how the RF module serving a capacity-layer cell enters dormancy after UMTS load-based RF module super dormancy is enabled.

Figure 7-29 Procedure for entering UMTS load-based RF module super dormancy



1. The base station evaluates whether entering conditions are met.
The RNC periodically checks whether the dormancy entry conditions are met within an evaluation period specified by the **UCELLDORMANCYPOLICY.TriggerJudgePeriod** parameter.
 - A capacity-layer cell must meet all of the following conditions:
 - The current time falls within the effective energy saving period, which is determined by the **UCELLDORMANCYPOLICY.StartTime** and **UCELLDORMANCYPOLICY.StopTime** parameters.
 - The capacity-layer cell is not in an energy saving penalty state.
 - There are no UEs performing emergency calls in the capacity-layer cell.
 - By default, cell broadcast service (CBS) takes precedence over dormancy. Therefore, when CBS is enabled in a cell for which the **URNCELLDORMANCYPARA.CellIgnoreCbsSwitch** parameter is set to **OFF** (default value), the cell cannot enter dormancy. If dormancy is required in scenarios where CBS is enabled, this parameter must be set to **ON**. Once it is set to **ON**:
 - If no CBS message is sent and the current time falls outside the CBS protection period (specified by the **URNCELLDORMANCYPARA.CbsServiceHoldDuration** parameter), the cell can enter dormancy.
 - If CBS messages are sent, the cell cannot enter dormancy.

 NOTE

After receiving a CBS message, the RNC immediately wakes the dormant cell and processes the CBS message once the cell is successfully set up. Because the cell requires time to wake, CBS messages may not be processed promptly. If CBS has a higher priority, it is recommended that **URNCELLDORMANCYPARA.CellIgnoreCbsSwitch** be set to its default value **OFF**. For details about CBS, see *CBS in RAN Feature Documentation*.

- The number of HSDPA UEs is less than or equal to the value of **UCELLDORMANCYPOLICY.HsdpaUserNumThd**.
- The number of HSUPA UEs is less than or equal to the value of **UCELLDORMANCYPOLICY.HsupaUserNumThd**.
- In multi-carrier HSDPA cell scenarios, the number of multi-carrier HSDPA UEs served by a multi-carrier HSDPA cell is less than or equal to the value of **UCELLDORMANCYPOLICY.McHsdpaUserNumThd**.
- When **UCELLDORMANCYPOLICY.DormancyCchUserSwitch** is set to **ON**, no UEs are in the CELL_FACH state, and the number of UEs in the CELL_DCH state is less than or equal to the value of **UCELLDORMANCYPOLICY.TotalUserNumThd**.
- When **UCELLDORMANCYPOLICY.DormancyCchUserSwitch** is set to **OFF**, the total number of UEs in the CELL_DCH and CELL_FACH states is less than or equal to the value of **UCELLDORMANCYPOLICY.TotalUserNumThd**.

- The basic-layer cells associated with a capacity-layer cell must meet all the following conditions:
 - None of the basic-layer cells is in the LDR or OLC state.
 - All basic-layer cells are free of faults (such as FACH faults) and not manually deactivated.
 - At least one basic-layer cell meets the following load margin conditions:
 - The uplink power load of the neighboring cell is less than the value of **UCELLDM.UllDrTrigThd** minus the value of **UCELLDORMANCYPOLICY.NCellLdrRemainThd**.
 - The downlink power load of the neighboring cell is less than the value of **UCELLDM.DllDrTrigThd** minus the value of **UCELLDORMANCYPOLICY.NCellLdrRemainThd**.
 - The value of **UCELLDM.UllDrTrigThd** is greater than the value of **UCELLDORMANCYPOLICY.NCellLdrRemainThd**.
 - The value of **UCELLDM.DllDrTrigThd** is greater than the value of **UCELLDORMANCYPOLICY.NCellLdrRemainThd**.

 NOTE

The uplink and downlink power loads of a cell are related to the cell's admission control algorithm. For details, see *Call Admission Control in RAN Feature Documentation*.

2. A UE transfer procedure starts.

- The capacity-layer cell prohibits the access of new UEs and incoming handovers.

The RNC broadcasts the barred state of the capacity-layer cell in SIB3.

- The RNC attempts to hand over all UEs in the CELL_DCH and CELL_FACH states from the capacity-layer cell to its co-coverage basic-layer cells within 60s.

The RNC selects basic-layer cells that meet the load margin conditions as the candidate set for handover.

- For CELL_DCH UEs, the RNC delivers load-based inter-frequency measurement configurations in which the measurement frequencies are those of inter-band neighboring cells. After receiving a measurement report from a UE, the RNC hands over the UE to an inter-band neighboring cell with signal quality higher than the RSCP (**UCELLMCLDR.TargetFreqThdRscp**) and Ec/No (**UCELLMCLDR.TargetFreqThdEcNo**) thresholds of the local cell.
 - For UEs not performing CS services, measurement-based handovers to inter-band neighboring cells are executed, and if the handovers fail, redirections to inter-band neighboring cells are executed.
 - For UEs performing CS services, measurement-based handovers to inter-band neighboring cells are executed, and if the handovers fail, measurement-based handovers to inter-band neighboring cells are executed again.

- For UEs in the CELL_FACH state, the RNC initiates redirections to the basic-layer cells.

During inter-frequency handover and redirection, the RNC periodically checks the number of UEs in the CELL_DCH and CELL_FACH states in the capacity-layer cell.

- If the number of UEs is 0, the capacity-layer cell enters dormancy.
- If the number of UEs is not 0, the RNC checks after 60s whether the number of UEs in the capacity-layer cell is less than or equal to the value of the **UCELLDORMANCYPOLICY.ForcibleTriggerUeNumThld** parameter. If the number of UEs is less than or equal to this value, the capacity-layer cell enters dormancy. Otherwise, the RNC cancels the barred state of the capacity-layer cell, and the cell enters the energy saving penalty state and exits the energy saving state. It does not recheck the entry conditions until the penalty period specified by the **UCELLDORMANCYPOLICY.PenaltyTime** parameter has elapsed.

3. The RF module enters the super dormancy state.

Once all UEs are transferred out of the capacity-layer cell, the cell and its associated RF module enter the super dormancy state. At the same time, the system starts to check the dormancy exit conditions.

Exiting the Super Dormancy State

The base station periodically checks whether the dormancy exit conditions are met. The exit evaluation period is specified by the **UCELLDORMANCYPOLICY.ExitJudgePeriod** parameter.

A capacity-layer cell and its associated RF module automatically exit dormancy when any of the following conditions is met:

- The current time falls outside the effective energy saving period, which is determined by the **UCELLDORMANCYPOLICY.StartTime** and **UCELLDORMANCYPOLICY.StopTime** parameters.
- The capacity-layer cell receives a CBS message from the CBS server.
- No associated basic-layer cell is operating in a different frequency band.
- Any associated basic-layer cell meets either of the following conditions:
 - The basic-layer cell is in the LDR or OLC state.
 - The basic-layer cell is faulty or manually deactivated.

Upon the exit from dormancy:

- The capacity-layer cell wakes up and can provide services properly.
- If the wakeup is triggered by a CBS message, the cell is not allowed to enter dormancy during the CBS hold duration (specified by **URNCCCELLDORMANCYPARA.CbsServiceHoldDuration**).

7.4.1.8 GU Load-based RF Module Super Dormancy

UMTS cells in operation still consume energy when there is light or even no traffic in the area covered by both GSM and UMTS networks, wasting base station energy. To address this, GU load-based RF module super dormancy is introduced.

Load-based RF module super dormancy enables the RNC to transfer UEs from UMTS cells to GSM cells and instruct the RF modules serving the UMTS cells to enter super dormancy within a specified period when the network is under light or no load. When the load of GSM cells increases to a specified threshold, the RF modules in dormancy are automatically woken up and the UMTS cells are restored.

This function requires the following switches to be configured:

- RF module-level switches

The **RRU.DORMANCYSW** parameter is set to **ON_SUPER** or **ON_DEEP_SUPER**, the **RRUEXTINFO.RRUDOROPTFEATSW** parameter is set to **ANYTIME_WAKEUP_SW**, and either of the following conditions is met:

- The **RRU.CUSTOMDORMANCYSTDSW** parameter is set to **OFF**.
- The **RRU.CUSTOMDORMANCYSTDSW** parameter is set to **ON**, and the **UMTS** option of the **RRU.DORMANCYSTD** parameter is selected.

The value **ON_DEEP_SUPER** of **RRU.DORMANCYSW** indicates that RF module deep dormancy and RF module super dormancy can be both enabled. For details about RF module deep dormancy, see [7.2 RF Module Deep Dormancy](#).

When the **RRU.DORMANCYSW** parameter is set to **ON_DEEP_SUPER** within the configured dormancy period, and both the conditions for entering RF module deep dormancy and those for entering RF module super dormancy are met:

- RF module super dormancy takes effect if the **ANYTIME_WAKEUP_SW** option of the **RRUEXTINFO.RRUDOROPTFEATSW** parameter is selected.
- If the **ANYTIME_WAKEUP_SW** option of the **RRUEXTINFO.RRUDOROPTFEATSW** parameter is not selected:
 - If the **ANYTIME_WAKEUP_SW** option of the **RRUEXTINFO.RRUDOROPTFEATSW** parameter is selected within 30s, RF module super dormancy takes effect.
 - If the **ANYTIME_WAKEUP_SW** option of the **RRUEXTINFO.RRUDOROPTFEATSW** parameter is not selected within 30s, RF module deep dormancy takes effect.

- UMTS cell-level switch

- The **DORMANCY_BY_LOAD_SW** option of the **ULOCCELLALGPARA.DORMANCYALGOSW** parameter is selected.
- The effective period and threshold parameters for UMTS load-based RF module super dormancy can be uniquely determined by using the combinations of the **UCELLDORMANCYPOLICY.PolicyIndex** and **UCELLDORMANCYPOLICY.DormancyType** parameters in the **UCELLDORMANCYPOLICY** MO on the RNC side. Specifically:
 - The **UCELLDORMANCYPOLICY.PolicyIndex** parameter specifies an energy saving policy index. It uniquely identifies an energy saving policy within the local cell. With the index specified, the corresponding policy can be added, modified, queried, or removed at the local cell level.

- The **UCELLDORMANCYPOLICY.DormancyType** parameter specifies the energy saving type corresponding to an energy saving policy index. GU load-based RF module dormancy requires that this parameter be set to **INTER_GU_DORMANCY_BY_LOAD**, meaning that under the corresponding energy saving policy, the local cell performs load-based dormancy in the specified effective period under certain load conditions.
- RNC-level switch
The **HO_INTER_RAT_CS_OUT_SWITCH** and **HO_INTER_RAT_PS_OUT_SWITCH** options of the **UCORRMALGOSWITCH.HoSwitch** parameter are selected to ensure that inter-RAT handovers can be performed.
- BSC-level switch
 - The **GCELLOTHEXT.IURGINFCTRL** parameter is set to **YES** to ensure information exchange on the Iur-g interface, when the BSC6910 is used on the GSM side. This parameter does not need to be set when the BSC6960 is used on the GSM side.
 - The **GCELLSOFT.IRATShutdownSwitch** parameter is set to **ON** to ensure inter-RAT energy saving information exchange.
- For dormancy involving UMTS, the **GU_DC_SW** option of the base-station-specific **EQUIPMENT.EQMDOROPTFEATSW** parameter must be deselected. If the option is deselected and a configuration command is executed to trigger dormancy pending or dormancy pending is enabled on the local end that works in UMTS, then dormancy pending will take effect on the entire base station.

Related Concepts

For the concepts of basic-layer carriers, capacity-layer carriers, basic-layer cells, and capacity-layer cells, refer to [Related Concepts](#). In this scenario, the basic-layer and capacity-layer cells must be inter-RAT neighboring cells. In this situation:

- The **U2GNCELL.NIRATOverLap** parameter must be set to **TRUE** for co-coverage neighboring GSM cells (basic-layer cells) of the UMTS cell that acts as the capacity-layer cell. When the UMTS cell enters the energy saving state, neighboring GSM cells with this parameter set to **TRUE** provide coverage.
- If the local cell and its inter-RAT neighboring cells belong to different operators, the **UOPERATORSHARINGMODE.InterPlmnHoAllowedInterRat** parameter must be set to **YES** to enable UE transfer.

Entering the Super Dormancy State

[Figure 7-30](#) shows how the RF module serving a capacity-layer cell enters dormancy after GU load-based RF module dormancy is enabled.

Figure 7-30 Procedure for entering GU load-based RF module super dormancy



1. The RNC determines whether the entry conditions are met.

The RNC periodically checks whether the dormancy entry conditions are met within an evaluation period specified by the **UCELLDORMANCYPOLICY.TriggerJudgePeriod** parameter. The information about the basic-layer cells is obtained from inter-RAT common measurements over the Iur-g interface (**UMBSCCRRM.IRATCommonMeasSwitch** must be set to **ON**).

- A capacity-layer cell must meet all of the following conditions:
 - The current time falls within the effective energy saving period determined by the **UCELLDORMANCYPOLICY.StartTime** and **UCELLDORMANCYPOLICY.StopTime** parameters.
 - The capacity-layer cell is not in an energy saving penalty state.
 - No UEs performing emergency calls exist in the capacity-layer cell.
 - By default, cell broadcast service (CBS) takes precedence over dormancy. Therefore, when CBS is enabled in a cell for which the **URNCCELLDORMANCYPARA.CellIgnoreCbsSwitch** parameter is set to **OFF** (default value), the cell cannot enter dormancy. If dormancy is required in scenarios where CBS is enabled, this parameter must be set to **ON**. Once it is set to **ON**:
 - If no CBS message is sent and the current time falls outside the CBS protection period (specified by the **URNCCELLDORMANCYPARA.CbsServiceHoldDuration** parameter), the cell can enter dormancy.
 - If CBS messages are sent, the cell cannot enter dormancy.

 NOTE

After receiving a CBS message, the RNC immediately wakes the dormant cell and processes the CBS message once the cell is successfully set up. Because the cell requires time to wake, CBS messages may not be processed promptly. If CBS has a higher priority, it is recommended that **URNCCELLDORMANCYPARA.CellIgnoreCbsSwitch** be set to its default value **OFF**. For details about CBS, see *CBS* in *RAN Feature Documentation*.

- The total downlink load of the capacity-layer cell is less than the value of **UCELLDORMANCYPOLICY.IRATShutdownDLLoadThd**.
 - The total uplink load of the capacity-layer cell is less than the value of **UCELLDORMANCYPOLICY.IRATShutdownULLoadThd**.
 - When **UCELLDORMANCYPOLICY.DormancyCchUserSwitch** is set to **ON**, no UEs are in the CELL_FACH state, and the number of UEs in the CELL_DCH state is less than or equal to the value of **UCELLDORMANCYPOLICY.TotalUserNumThd**.
 - When **UCELLDORMANCYPOLICY.DormancyCchUserSwitch** is set to **OFF**, the total number of UEs in the CELL_DCH and CELL_FACH states is less than or equal to the value of **UCELLDORMANCYPOLICY.TotalUserNumThd**.
- Any basic-layer GSM cell associated with a capacity-layer cell must meet all the following conditions:

- The timeslot usage in the GSM cell is less than the value of **UCELLDORMANCYPOLICY.IRATshutdownGSMLoadThd**.
 - The GSM cell supports PS services.
 - The GSM cell supports inter-RAT power saving (with **GCELLSOFT.IRATShutdownSwitch** set to **ON**).
2. A UE transfer procedure starts.
- The capacity-layer cell prohibits the access of new UEs and incoming handovers.
The RNC broadcasts the barred state of the capacity-layer cell in SIB3.
 - The RNC attempts to hand over all UEs in the CELL_DCH and CELL_FACH states from the capacity-layer cell to its co-coverage basic-layer GSM cells within 60s.
The RNC selects qualified basic-layer GSM cells as the candidate set for handover.
- For CELL_DCH UEs, the RNC delivers load-based inter-RAT measurement configurations in which the measurement frequencies are those of basic-layer cells. After receiving a measurement report from a UE, the RNC hands over the UE to the corresponding basic-layer cell.
 - For UEs not performing CS services, measurement-based handovers to inter-RAT neighboring cells are executed, and if the handovers fail, redirections to inter-RAT neighboring cells are executed.
 - For UEs performing CS services, measurement-based handovers to inter-RAT neighboring cells are executed, and if the handovers fail, measurement-based handovers to inter-RAT neighboring cells are executed again.
 - For UEs in the CELL_FACH state, the RNC initiates redirections to the basic-layer cells.
- During inter-RAT handover and redirection, the RNC periodically checks the number of UEs in the CELL_DCH and CELL_FACH states in the capacity-layer cell.
- If the number of UEs is 0, the capacity-layer cell enters dormancy.
 - If the number of UEs is not 0, the RNC checks after 60s whether the number of UEs in the capacity-layer cell is less than or equal to the value of the **UCELLDORMANCYPOLICY.ForcibleTriggerUeNumThld** parameter. If the number of UEs is less than or equal to this value, the capacity-layer cell enters dormancy. Otherwise, the RNC cancels the barred state of the capacity-layer cell, and the cell enters the energy saving penalty state and exits the energy saving state. It does not recheck the entry conditions until the penalty period specified by the **UCELLDORMANCYPOLICY.PenaltyTime** parameter has elapsed.
3. The RF module enters the super dormancy state.

Once all UEs are transferred out of the capacity-layer cell, the cell and its associated RF module enter the super dormancy state. At the same time, the system starts to check the dormancy exit conditions.

Exiting the Super Dormancy State

The base station periodically checks whether the dormancy exit conditions are met. The exit evaluation period is specified by the **UCELLDORMANCYPOLICY.ExitJudgePeriod** parameter.

A capacity-layer cell and its associated RF module automatically exit dormancy when any of the following conditions is met:

- The current time falls outside the effective energy saving period, which is determined by the **UCELLDORMANCYPOLICY.StartTime** and **UCELLDORMANCYPOLICY.StopTime** parameters.
- There is no associated basic-layer GSM cell.
- The capacity-layer cell receives a CBS message from the CBS server.
- Any associated basic-layer cell meets either of the following conditions:
 - The timeslot usage of the GSM cell is greater than or equal to the sum of **UCELLDORMANCYPOLICY.IRATshutdownGSMLoadThd** and **UCELLDORMANCYPOLICY.IRATshutdownGSMLoadHyst**.
 - The GSM cell does not support inter-RAT power saving (with **GCELLSOFT.IRATShutdownSwitch** set to **OFF**).
 - The load information of the GSM cell is unavailable. For example, the GSM cell is deactivated, deleted, or faulty, or the Iur-g transmission is faulty.

Upon the exit from dormancy:

- The capacity-layer cell wakes up and can provide services properly.
- If the wakeup is triggered by a CBS message, the cell is not allowed to enter dormancy during the CBS hold duration (specified by **URNCCCELLDORMANCYPARA.CbsServiceHoldDuration**).

NOTE

When both UMTS load-based RF module super dormancy and GU load-based RF module super dormancy are enabled, capacity-layer cells can enter dormancy during the overlap of their respective dormancy periods only if the dormancy conditions of both functions are met.

7.4.2 Network Analysis

7.4.2.1 Benefits

It is recommended that RF module super dormancy be enabled during a fixed time period with light or no load every day to reduce the power consumption of the base station. To ensure that the actual operating duration of an RF module meets lifecycle management requirements in typical scenarios, the module can enter dormancy when its reliability requirements (such as temperature and humidity) are met. If the RF module is reset during the dormancy period for

purposes such as upgrade, the module and its cells will return to the normal state, which leads to reduced energy saving gains.

In the case of LTE and NR, it is recommended that the same dormancy period be configured for the RF modules bound to a combined-RRU cell or multi-TRP cell (such as an SFN cell, hyper cell, or combined cell) to improve energy saving gains.

In the case of UMTS, it is recommended that the same dormancy period be configured for the RF modules bound to a combined-RRU cell or multi-RRU cell (such as a distributed cell, 0.5/0.5 cell, or cell with "independent demodulation of signals from multiple RRUs in one cell" enabled) to improve energy saving gains.

In the case of GSM, it is recommended that the same dormancy period be configured for the RF modules bound to a multi-RRU cell (such as a multi-site cell) to improve energy saving gains.

It is not recommended that the dormancy period be frequently changed or the dormancy feature be frequently enabled or disabled every day. Otherwise, the dormancy period starts about 10 minutes later than it started on the previous day, consequently decreasing energy saving gains.

In a separate-MPT base station where the local RAT is limited to GSM or UMTS, the peer RAT includes LTE or NR, and RF modules are configured on both sides with the working RAT only including LTE or NR, RF module super dormancy can be enabled to increase energy savings.

RF modules with load-based RF module super dormancy enabled may exit or fail to enter dormancy when scheduled energy saving functions (such as timing carrier shutdown, scheduled RF module deep dormancy, and scheduled RF module super dormancy) are enabled on neighboring base stations or the neighboring base stations or RF modules are reset or powered off. This is because such situations of the neighboring base stations or RF modules may cause the load of capacity-layer or basic-layer cells served by the RF modules with load-based RF module super dormancy enabled to increase to above the preset threshold. If this is the case, energy saving gains decrease.

7.4.2.2 Impacts

The impacts between this function and other functions are described in [7.4.3.2.3 Function Impacts](#).

Impacts related to scheduled RF module super dormancy, NR load-based RF module super dormancy, LNR load-based RF module super dormancy, LTE load-based RF module super dormancy, GSM load-based RF module super dormancy, UMTS load-based RF module super dormancy, and GU load-based RF module super dormancy:

- Assume that RF module super dormancy has taken effect. If the end time is changed to a later time, the current RF module dormancy will be exited at the original end time. After the exit, dormancy will be triggered again if the conditions are met. Otherwise, dormancy will be triggered until the next day when the conditions are met (in which case, dormancy takes effect in the new entire effective period). If the end time is changed to an earlier time, the current RF module dormancy will be exited in advance.
- RF module dormancy triggers handovers or redirections for active UEs, increasing the numbers of handovers and redirections from the cell.

- The signal strength in an area served by an RF module in the dormancy state decreases because the RF module does not transmit signals. UEs may experience service drops and access failures.
- To ensure that an RF module meets reliability requirements (such as temperature and humidity differences), it automatically exits the dormancy state after being in dormancy for a consecutive 23.5 hours. After the dormancy conditions are met, the dormancy state can be entered again.
- In RRU cascading scenarios, RRUs support the RF module super dormancy function, but the function cannot be configured together with RF module deep dormancy.
- In RRU cascading scenarios, the prerequisite for super dormancy to take effect on the current-level RRU is that its lower-level RRUs are already in the super dormancy state.
- If RRUs are cascaded, RRUs exit the super dormancy state level by level, with the RRUs at the chain tail being the last. In this case, the RRUs may need more time to exit the dormancy state.
- When an RF module prepares to enter the super dormancy state, the maintenance link between the BBU and the RF module is disconnected for a short period of time. During this period, the response to configuration or maintenance MML commands (such as **DSP RRU**) related to the RF module may time out.
- After the BBU instructs the RF module to enter the dormancy state, the RF module components enter the dormancy state one after another. During this period, the RF module is still in the process of entering dormancy and wake-up is not allowed. It can be woken up only after its status becomes stable.
- In RRU cascading scenarios, if RRUs at level n enter the super dormancy state and reset of RF modules at level $(n-1)$ are triggered, a link alarm is reported after level $(n-1)$ RF modules start up. The alarm is automatically cleared after level n RF modules exit the super dormancy state.
- In RRU cascading scenarios, if the CPRI line rate for the RRU chain or ring is set to auto-negotiation (with the **RRUCHAIN.CR (LTE eNodeB, 5G gNodeB)** parameter set to **AUTO**), and the maximum rates supported by cascaded RRUs differ, the RRUs on the chain or ring will not enter the dormancy state. The maximum rates supported by cascaded RRUs on a chain or ring can be queried by running the **DSP CPRILBR** command.
- If the RF module detects that the optical module has reliability risks when entering super dormancy, super dormancy will not take effect. For example, RX_LOS error is reported.
- If an ALD-related alarm is reported before the RF module enters super dormancy, super dormancy does not take effect.
- Optical modules are consumables. If the reliability of existing optical modules deteriorates, connection exceptions may occur.
- If an optical module does not have a BOM code, it is identified as a non-standard optical module, and super dormancy will not take effect by default. If the label indicates a standard optical module, the **RF_SDC_SW** option of the **EQUIPMENT.EQMDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter needs to be selected for super dormancy to take effect.
- If there is a configured optical module with unknown information on the BBP or RF module, super dormancy will not take effect.

- If there is an unconfigured optical module with unknown information on the BBP or RF module, super dormancy will not take effect.
- ALDs with non-Huawei vendor code are identified as non-standard ALDs, for which super dormancy does not take effect by default. To enable super dormancy in this scenario, select the **RF_ALD_SW** option of the **EQUIPMENT.EQMDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter.
- After the RF module enters the dormancy state or is in the process of exiting this state, the following alarms caused by the RF module super dormancy are not reported:
 - Node alarm
 - ALM-26235 RF Unit Maintenance Link Failure
 - NR alarms
 - ALM-29840 gNodeB Out of Service
 - ALM-29841 NR Cell Unavailable
 - ALM-29844 NR DU Cell Unavailable
 - ALM-29870 NR DU Cell TRP Unavailable
 - ALM-29871 NR DU Cell TRP Capability Degraded
 - LTE alarms
 - ALM-29240 Cell Unavailable
 - ALM-29243 Cell Capability Degraded
 - ALM-29249 eNodeB Out of Service
 - UMTS alarms
 - ALM-28203 Local Cell Unusable
 - ALM-28206 Local Cell Capability Decline
 - ALM-22202 UMTS Cell Unavailable (RNC)
 - ALM-22206 UMTS Cell Setup Failed (RNC)
 - ALM-22214 NodeB Unavailable (RNC)
 - ALM-22215 UMTS Cell Max DL Power Mismatch (RNC)
 - ALM-22203 UMTS Cell Common Channel Faulty (RNC)
 - ALM-22207 UMTS Cell Reconfiguration Failed (RNC)
 - ALM-22208 UMTS Cell Common Channel Setup Failed (RNC)
 - ALM-22209 UMTS Cell Common Channel Reconfiguration Failed (RNC)
 - ALM-22217 UMTS Cell HSDPA Function Fault (RNC)

- ALM-22218 UMTS Cell HSUPA Function Fault (RNC)
- ALM-22219 UMTS Cell EFACH Function Fault (RNC)
- ALM-22220 UMTS Cell MIMO Function Faulty (RNC)
- ALM-22222 UMTS Cell ERACH Function Fault (RNC)
- ALM-22223 UMTS Cell EPCH Function Fault (RNC)
- ALM-22237 UMTS Cell DC-HSUPA Function Fault (RNC)
- ALM-22241 UMTS Cell MC-HSDPA Function Fault (RNC)
- GSM alarms
 - ALM-28006 Radio Signaling Link Disconnected
 - ALM-28008 Radio Link Failure
 - ALM-28009 Carrier No Traffic
 - ALM-28010 Carrier Configuration Error
 - ALM-28016 GSM Local Cell Unusable
 - ALM-28017 GSM Local Cell Capability Decline
 - ALM-21801 GSM Cell out of Service (BSC)

An RF module is restarted to restore its services during the dormancy exit process. After the restart, RF module startup events (for purposes such as power-off reset and self-reset) are reported normally.

- During dormancy, the maintenance link between the BBU and the RF module is disconnected. If the RF module is powered off unexpectedly, the generated alarms cannot be reported. After exiting the dormancy state, no power failure alarm (such as ALM-26540 RF Unit AC Input Power Failure or ALM-26564 RF Unit DC Input Power Failure) is reported.
- During dormancy, the maintenance link between the BBU and the RF module is disconnected. The BBU cannot detect the unexpected power-off of the RF module, and therefore, RF module does not enter the dormancy state again after being reset. If unexpected power-off occurs during the configured dormancy period, you are advised not to enable super dormancy.
- In the dormancy state, the status of the RUN indicator cannot be controlled through configuration.
- The **DOR_LM_SW** option of the **EQUIPMENT.EQMDOROPTFEATSW** parameter specifies whether to enable functions related to RF module dormancy when an NE is connected to the NetEco or RPCU. When this option is deselected and an NE is connected to the NetEco or RPCU, dormancy-related functions cannot take effect. When this option is selected and an NE is connected to the NetEco, dormancy-related functions can take effect. Once an RF module enters the dormancy state, the maintenance link between the NetEco and the power supply that transmits signals through the RF module will be disconnected. The power supply status (for example, backup power saving) can be updated only after the RF module exits the dormancy state.

When this option is selected and an NE is connected to the RPCU, dormancy-related functions can take effect. However, the upper-level board connected to the RPCU cannot enter the dormancy state. (RAT-specific dormancy can take effect on the upper-level RF module connected to the RPCU.)

- If the upper-level RF module connected to the RPCU differs from the RF module supplying power to the RPCU, only the latter RF module can enter the dormancy state. When it happens, the power supply link of the RPCU is interrupted, ALM-26918 CXU Maintenance Link Failure is reported for the RPCU, ALM-26504 RF Unit CPRI Interface Error is reported for the upper-level RF module connected to the RPCU, thereby affecting RPCU-related functions.
- For details about the impact of configuration modification during dormancy, see [7.11.2 Configuration Reliability](#).
 - Base station reconfiguration during the specified dormancy period is not recommended. If reconfiguration is required, the dormancy function must be disabled first. Cell reconfigurations performed when the RF module is in the dormancy state take effect only after the RF module exits the dormancy state. If new configurations do not meet the conditions for the RF module to enter the dormancy state, the RF module exits the state.
 - Assume that the batch processing tasks of the OSS contain GBTSFunction, NodeBFunction, gNodeBFunction, or eNodeBFunction MML commands that affect cell setup or Node MML commands, and an NE of the base station is in the dormancy state. The OSS executes related configurations after the base station exits the dormancy state. This increases the waiting time before configuration execution. It takes less than 10 minutes on average for the base station to exit the dormancy state.
 - If a power-related command for an NE needs to be executed on the base station but the NE is in the dormancy state, the base station will first trigger the NE to exit the dormancy state and then execute the command. It takes less than 10 minutes on average for the base station to exit the dormancy state.
- Dormancy pending is triggered when the **PND DORMANCY** command is executed or when software is downloaded and activated. If the main control board is powered off before the pending duration elapses, the pending will be resumed after the main control board is powered on and will continue until the pending duration elapses.
- If a switchover between the active and standby main control boards occurs, the pending duration needs to be reconfigured.
- The base station triggers a 30-minute pending period by default after startup. During this period, dormancy cannot take effect.
- When the **ULD FILE** command is executed to upload files from the RF module after dormancy takes effect, the execution will fail with a message indicating that the dormancy is in effect and will not be woken up. The command can be successfully executed if the files to be uploaded are one-click logs or logs of RF modules not in the dormancy state.
- When the **EXP DEVFILE** command is executed to export the device archive file after dormancy takes effect, the command can be successfully executed and the RF modules will not wake up from dormancy. However, the exported device archive file does not include that of RF modules in the dormancy state. It is recommended that the file be exported at a time when dormancy is not in effect.

- During the wakeup of the RF module from the super dormancy state, the real-time value may not be displayed when you query the RF-related information. The real-time value is displayed only after the wakeup is complete. For example, the value of **TX optical power** is **NULL** or the buffered value in the **DSP SFP** command output during the wakeup.
- When the RF module exits super dormancy, cell reestablishment will be triggered. Multiple cell setup attempts may be initiated before the cell is successfully established. In this case, the value of the **VS.Cell.UnavailTime.Sys**, **VS.Cell.HSDPA.UnavailTime**, **VS.Cell.HSUPA.UnavailTime**, **VS.Cell.UnavailTime.Sys.lubFail**, **L.Cell.Unavail.Dur.Sys**, or **N.Cell.Unavail.Dur.System** counter is not 0. If cell-related licenses have expired, the dormant cells will fail to be reestablished. If the number of cells configured on a BBP exceeds its specification, the dormant cells will fail to be reestablished when the RF module exits the dormancy state. Therefore, if new cells are added for capacity expansion or other purposes during the dormancy period, the new cells should be manually bound to specified BBPs.
- If the dormancy conditions are met during a base station version upgrade, the RF module will enter the dormancy state only after the upgrade is complete. If an upgrade is performed when the RF module is in the dormancy state, dormancy pending will be triggered. Assume that the main control board of one side requires the RF module to be upgraded and the main control board of another side requires the RF module to exit the dormancy state when the RF module is shared using SDR technology in separate-MPT scenarios. In this case, cells on the side where the RF module is required to exit the dormancy state may fail to be reestablished, and alarms related to these cells are reported. These cells become normal after the upgrade is complete.
- Some alarms reported for a cell before it enters dormancy will not be cleared after the cell enters dormancy.
 - GSM/UMTS/LTE cells: If a fault that causes an alarm is rectified during the specified dormancy period, the alarm persists and will be cleared after the cell exits dormancy.
 - NR cells: If a cell becomes unavailable due to a CU fault (such as an NG interface fault), the cell will not enter dormancy. If the CU fault is rectified during the specified dormancy period for the RF module, the related alarm will be cleared and the cell will enter dormancy. If the cell becomes unavailable due to a DU fault, the alarm persists after the fault is rectified and will be cleared after the cell exits dormancy.
- If an RF module serving an NR hyper cell or combined cell does not meet the dormancy conditions, all RF modules serving the cell exit the dormancy state. Assume that this cell is a sector split cell. All RF modules serving another NR hyper cell or combined cell that shares this RF module with this cell may exit the dormancy state accordingly.
- Assume that the local and neighboring base stations are in the dormancy state, and an upgrade is performed for the local base station, not for the neighboring base stations. The local base station reverts to the normal state after the upgrade, and CEUs in weak coverage areas may access the available cells served by the local base station. This affects KPIs related to the aspects such as access and service drop of the local base station. It is recommended that the **PND DORMANCY** command be executed on both the base station to

be upgraded and its neighboring base stations within the upgrade time window.

- Before an RF module enters the super dormancy state, new UEs are prohibited from accessing an NR cell during NR carrier shutdown. If UEs attempt to be handed over to the NR cell, the values of the counters related to handover preparation failures (such as **N.HO.IntraFreq.FailOut.PrepFailure**) slightly increase. Such counters are not used for calculating Huawei-defined KPIs outgoing intra- and inter-frequency handover success rates, but may be used for calculating user-defined KPIs. When dormancy is set to take effect on the hour, no UEs can be handed over to the deactivated NR cell. Therefore, even slight increases in the counter values may significantly affect the user-defined outgoing handover success rates. If super dormancy is scheduled to take effect on the hour, you are advised to set the dormancy start time to 5 minutes earlier and the lengths of penalty timers applied to handover failures to 30s. After penalty timers for handover failures are set, UEs may experience service drops due to delayed handovers. The penalty timers for handover failures are specified by the **gNBMobilityCommParam.NecHoPrepFailPunishTimer** and **gNBMobilityCommParam.OptHoPrepFailPunishTimer** parameters.
- Assuming that the neighboring base stations are in the dormancy state, CEUs in weak coverage areas may access the cells served by the local base station. This affects KPIs related to the aspects such as access and service drop of the local base station. For example, the value of **N.NsaDc.DRB.AbnormRel** may increase.
- When the multi-RRU cell dormancy compatibility function is enabled, RF module dormancy can take effect in multi-RRU cell scenarios. As dormancy can take effect in more scenarios, the energy consumption of the entire base station decreases and the user-perceived rate may decrease. If this function is disabled, there will be no such impacts.
- When the multi-RRU cell dormancy compatibility function is enabled and dormancy functions configured for RF modules are different, the specified dormancy functions will take effect based on the configurations.
- When the multi-RRU cell dormancy compatibility function is enabled and the dormancy periods configured for RF modules are different, the specified dormancy functions will take effect only during the period in which multiple RF modules meet the dormancy requirements.
- When the **DORM_MNTMODE_CTRL_SW** option of the **EQUIPMENT.EQMDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter is selected, RF modules are allowed to enter the dormancy state. Modifying the configuration of a base station in the dormancy state may trigger dormancy exit. You are advised to start dormancy pending before configuration modification.

For scheduled RF module super dormancy:

When scheduled RF module super dormancy takes effect:

- For NR
 - If NR UEs are transferred to LTE cells, performance counters on the LTE and NR sides are affected. The affected performance counters are the same as those described in **6.1.2.2 Impacts** for LTE and NR intelligent carrier shutdown.

- If NR UEs are transferred to NR cells, see the network impact of intelligent carrier shutdown in *Energy Conservation and Emission Reduction in 5G RAN Feature Documentation* for the impact on performance counters and KPIs.
- For UMTS
 - UMTS scheduled RF module dormancy causes changes in the values of counters related to inter-frequency handovers, inter-RAT handovers, and redirections.
 - During UE transfer, new UE access and incoming handovers are prohibited, which may cause changes in the values of counters related to access and service drops.
- For GSM
 - When a GSM cell exits dormancy, its TRXs will be reestablished, which may decrease the availability of the cell and its TRXs.
 - If the **BSCEXSOFTPARA.IgnoreCbcSwitch** parameter is set to **ON**, the values of the **CB.BSC.RPT.FAIL** and **CB.BSCRPT.RESTART.NUM** counters may increase because TRXs are reestablished upon the cell's exit from dormancy.

For NR load-based RF module super dormancy:

- NR load-based RF module super dormancy causes changes in the values of counters related to inter-frequency handovers and redirections.
- If a UE attempts to be handed over to a capacity-layer cell that is about to enter the dormancy state during UE transfer (which last 30s at most), the source cell of the handover does not initiate handover execution. In this case, the values of the counters related to outgoing and incoming handover attempts increase in the source cell and capacity-layer cell, respectively; the values of the counters related to handover executions and successful handovers do not increase. As Huawei-defined KPIs **Inter-Frequency Handover Out Success Rate (CU)** and **Intra-RAT Handover In Success Rate (CU)** are calculated using the numbers of outgoing and incoming handover executions as their denominators, these KPIs are not affected in this scenario. However, if the user-defined KPIs are calculated using the numbers of outgoing and incoming handover attempts as their denominators, these KPIs may be slightly affected in this scenario.

For LNR load-based RF module super dormancy:

- When LNR load-based RF module super dormancy takes effect, performance counters on the LTE and NR sides are affected. The affected performance counters are the same as those described in [6.1.2.2 Impacts](#) for LTE and NR intelligent carrier shutdown.
- In inter-base-station SA networking, X2 link setup may be triggered for energy saving purposes. This occupies the specifications of EN-DC X2 links and affects the experience of NSA UEs. The probability of ALM-29225 Maximum Number of X2 Interfaces Reached being reported also increases. Therefore, this function is not recommended when the X2 specifications are less than required.
- In inter-base-station SA networking, if the **LNR_INTER_BS_PWR_SAVE_SW** option of the **EnodebPwrSavingAlgo.PwrSavingAlgoSwitch** parameter or the **LNR_SA_GNB_X2_SETUP_SW** option of the

eNodeBOamParam.InterfaceSonConfigSwitch parameter is selected but the **X2_SETUP_FOR_SA_SW** option of the **gNBInterfaceParam.X2ConfigSw** parameter is deselected on the gNodeB side, ALM-29204 X2 Interface Fault and ALM-25888 SCTP Link Fault may be reported.

- In inter-base-station SA networking, the configurations of the neighboring NR cell with overlapping coverage areas on the LTE side must be the same as those on the NR side, such as the cell ID, base station ID, and operator information. Otherwise, the SA X2 link may fail to be set up.

For LTE load-based RF module super dormancy:

When LTE load-based RF module super dormancy is triggered, inter-frequency handovers are performed for UEs in the capacity-layer cell, increasing the values of counters listed in [Table 7-49](#) and [Table 7-50](#). In addition, to promptly decrease the number of UEs in each capacity-layer cell, the eNodeB prohibits UEs from accessing the cells and broadcasts the barred state of the cells through SIB1. However, a small number of UEs may still send RRC connection setup requests to access the cells, causing an increase in the **L.RRC.SetupFail.Rej** counter value and a decrease in the **RRC Setup Success Rate** value. The number of such access failures is indicated by the **L.RRC.SetupFail.ResFail.CellBarred** counter.

Table 7-49 Performance counters of capacity-layer cells

Networking	Counter Description	Counter Name
Macro networking	Number of intra-eNodeB inter-frequency outgoing handover attempts in a cell	L.HHO.IntraeNB.InterFreq.PrepAttOut
	Number of intra-eNodeB inter-frequency outgoing handover executions in a cell	L.HHO.IntraeNB.InterFreq.ExecAttOut
	Number of successful intra-eNodeB inter-frequency outgoing handovers in a cell	L.HHO.IntraeNB.InterFreq.ExecSuccessOut
Inter-eNodeB networking	Number of inter-eNodeB inter-frequency outgoing handover attempts in a cell	L.HHO.IntereNB.InterFreq.PrepAttOut
	Number of inter-eNodeB inter-frequency outgoing handover executions in a cell	L.HHO.IntereNB.InterFreq.ExecAttOut
	Number of successful inter-eNodeB inter-frequency outgoing handovers in a cell	L.HHO.IntereNB.InterFreq.ExecSuccessOut

Table 7-50 Performance counters of basic-layer cells

Networking	Counter Description	Counter Name
Macro networking	Number of intra-eNodeB intra-duplex-mode incoming handover attempts in a cell	L.HHO.IntraeNB.PrepAttIn
	Number of intra-eNodeB intra-duplex-mode incoming handover executions in a cell	L.HHO.IntraeNB.ExecAttIn
	Number of successful intra-eNodeB intra-duplex-mode incoming handovers in a cell	L.HHO.IntraeNB.ExecSucIn
Inter-eNodeB networking	Number of inter-eNodeB intra-duplex-mode incoming handover attempts in a cell	L.HHO.IntereNB.PrepAttIn
	Number of inter-eNodeB intra-duplex-mode incoming handover executions in a cell	L.HHO.IntereNB.ExecAttIn
	Number of successful inter-eNodeB intra-duplex-mode incoming handovers in a cell	L.HHO.IntereNB.ExecSucIn

For GSM load-based RF module super dormancy:

- GSM load-based RF module super dormancy causes changes in the values of counters related to handovers to neighboring cells. If no neighboring cell is available or signal quality in the neighboring cell is poor, call drops may occur.
- The load-based dormancy algorithm uses the **GCELLDORMANCY.DormancyCellStrTime** and **GCELLDORMANCY.DormancyCellStpTime** parameters on both the BSC and eGBTS. Therefore, the system time of the BSC and that of the eGBTS must be the same. If they are different, the corresponding cell will enter load-based dormancy earlier or later than expected.
- When a GSM cell exits dormancy, its TRXs will be reestablished, which may decrease the availability of the cell and its TRXs.
- If the **BSCEXSOFTPARA.IgnoreCbcSwitch** parameter is set to **ON**, the values of the **CB.BSC.RPT.FAIL** and **CB.BSCRPT.RESTART.NUM** counters may increase because TRXs are reestablished upon the cell's exit from dormancy.

For UMTS load-based RF module super dormancy:

- UMTS load-based RF module dormancy causes changes in the values of counters related to inter-frequency handovers and redirections.

- During UE transfer, new UE access and incoming handovers are prohibited, which may cause changes in the values of counters related to access and service drops.
- If the **UCELLDORMANCYPOLICY.ForcibleTriggerUeNumThld** parameter is set to a value other than **0** and the cell serving CS services enters the energy saving state during UE transfer, voice services will be interrupted.
- The load-based dormancy algorithm uses the **UCELLDORMANCYPOLICY.StartTime** and **UCELLDORMANCYPOLICY.StopTime** parameters on both the RNC and NodeB. Therefore, the system time of the RNC and that of the NodeB must be the same. If they are different, the corresponding cell will enter load-based dormancy earlier or later than expected.

For GU load-based RF module super dormancy:

- When GU load-based RF module dormancy is in effect, performance counters on the GSM and UMTS sides are affected. The affected performance counters are the same as those described in [6.3.2.2 Impacts](#) under [6.3 GUL Joint Carrier Shutdown](#).
- It is not recommended that GSM load-based RF module super dormancy be enabled for GSM cells that serve as basic-layer cells for a UMTS cell that covers the same area as these GSM cells. Otherwise, in extreme cases, the UMTS cell and these neighboring GSM cells are in dormancy in the same period, resulting in coverage unavailability and consequently UE access failures in the area in this period.
- During UE transfer, new UE access and incoming handovers are prohibited, which may cause changes in the values of counters related to access and service drops.
- If the **UCELLDORMANCYPOLICY.ForcibleTriggerUeNumThld** parameter is set to a value other than **0** and the cell serving CS services enters the energy saving state during UE transfer, voice services will be interrupted.
- The load-based dormancy algorithm uses the **UCELLDORMANCYPOLICY.StartTime** and **UCELLDORMANCYPOLICY.StopTime** parameters on both the RNC and NodeB. Therefore, the system time of the RNC and that of the NodeB must be the same. If they are different, the corresponding cell will enter load-based dormancy earlier or later than expected.

7.4.3 Requirements

7.4.3.1 Licenses

RAT	Feature ID	Feature Name	Model	NE	Sales Unit
NR	MRFD-200261	RF Module Super Deep Dormancy (NR)	NR0SRFM SDD00	gNode B	per RU
LTE TDD	MRFD-200231	RF Module Super Deep Dormancy (LTE TDD)	LT4SRFMS DD00	eNode B	per RU

RAT	Feature ID	Feature Name	Model	NE	Sales Unit
LTE FDD	MRFD-2 01221	RF Module Super Deep Dormancy (LTE FDD)	LT1SRFMU DY01	eNode B	per RU
UMTS	MRFD-2 11211	RF Module Super Deep Dormancy (UMTS)	LQW9RFM SDD01	NodeB	per RU
GSM	MRFD-2 11201	RF Module Super Deep Dormancy (GSM)	LGB3JZXM	eGBTS	per RU

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

CAUTION

- For an RF module shared by multiple RATs using SDR technology, it can enter super dormancy only when the licenses for all RATs of the RF module are acquired.
- It is recommended that the RF unit working mode (specified by the **RRU.RS (LTE eNodeB, 5G gNodeB)** parameter) be set to a single RAT for a single-mode RF module. If this parameter is set to a value indicating multiple RATs for an RF module and RAT introduction has been performed on the base station, the RF module can enter the dormancy state only when the licenses of newly introduced RATs are loaded.

7.4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Virtual grid model building and maintenance	VG_MODEL_AL LOW_BUILD_FL AG option of the NRCellFreqRel ation.Aggregat ionAttribute parameter and the NR_IF_HO_ME AS_QTY_PRED_ FUN_SW option of the NRCellSmartM ultiCarr.NrMult iCarrierAlgoSw itch parameter	<i>Virtual Grid-based Multi- Frequency Coordination in 5G RAN Feature Documenta tion</i>	To enable precise dormancy and calculate overlap rates based on grids, virtual grid model building and maintenance must be enabled. To further improve the accuracy of grid models, see <i>Virtual Grid-based Multi-Frequency Coordination in 5G RAN Feature Documentation</i> .
LTE FDD LTE TDD	Virtual grid function	SMART_CARRIE R_SELECTION_ SW option of the MultiCarrUnifi edSch.MultiCar rierUnifiedSchS w parameter	<i>Multi- carrier Unified Scheduling in eRAN Feature Documenta tion</i>	To use grid models to calculate the cell co-coverage proportion for precise dormancy, virtual grid function needs to be enabled. To further improve the accuracy of grid models, see <i>Multi-carrier Unified Scheduling in eRAN Feature Documentation</i> .

7.4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD High-frequency NR TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	The time period configured for timing carrier shutdown cannot overlap with the period configured for NR load-based RF module super dormancy, LNR load-based RF module super dormancy, or optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module super dormancy of the local cell. You are advised to preferentially plan the periods during which timing carrier shutdown takes effect.
LTE FDD LTE TDD	High speed specified policy management	HSCellSleepInterfAvoidSwitch option of the HighSpdAdaptionPara.InterfAvoidMode parameter	<i>High Speed Mobility in eRAN Feature Documentation</i>	The high-speed cell sleeping function in high speed specified policy management does not work with LNR load-based RF module dormancy.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Deployment Position	NRDUCell.DeployPosition set to DEPLOY_IAB	None	If this parameter is set to DEPLOY_IAB , the following functions are not supported: NR load-based RF module super dormancy, LNR load-based RF module super dormancy, and optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module super dormancy.
Low-frequency NR TDD	Supplementary Cell Indicator	NRDUCell.SupplementaryCellInd	None	If the NRDUCell.DuplexMode parameter is set to CELL_TDD and the value of supplementary cell indicator is INTRA_FREQ_ASSISTANT_CELL , the following functions cannot be enabled: NR load-based RF module super dormancy, LNR load-based RF module super dormancy, and optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module super dormancy.

RAT	Function Name	Function Switch	Reference	Description
NR FDD	MBS broadcast	MBS_BROADCAST_SWITCH option of the NRCellFeatureSwitch.MbsSwitch parameter	None	When the MBS broadcast switch is on, the following functions cannot be enabled: NR load-based RF module super dormancy, LNR load-based RF module super dormancy, and optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module super dormancy.
LTE FDD LTE TDD	Multi Cell Share Mode	Cell.MultiCellShareMode set to SPLIT	None	Split cells cannot be configured as capacity-layer cells for LTE load-based RF module super dormancy in precise dormancy mode.
LTE TDD	Intelligent Multi-Beam of 8T8R	None	<i>Intelligent Multi-Beam of 8T8R (TDD) in eRAN Feature Documentation</i>	Cells with Intelligent Multi-Beam of 8T8R enabled cannot be configured as capacity-layer cells for LTE load-based RF module super dormancy in precise dormancy mode.
Low-frequency NR TDD High-frequency NR TDD	UBRU	ADD UBRU	None	If a UBRU is configured, RF module super dormancy cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	DRS	ADD DRS	None	If a DRS is configured, RF module super dormancy cannot be enabled.
GSM UMTS LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	GNSS	ADD GNSS	None	RF modules configured with GNSS do not support RF module super dormancy.
GSM UMTS LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	RRU	RRU.TP (LTE eNodeB, 5G gNodeB)	None	If this parameter is set to a value other than TRUNK , RF module super dormancy cannot be enabled.
LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	RRU Chain	RRUCHAIN.AT (LTE eNodeB, 5G gNodeB)	None	If AT in an RRUCHAIN MO of an RRU chain where the RF module is located is set to HYBRIDPORT , RF module super dormancy cannot be enabled for this RF module.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	Intra-mode Dual-site Ring Topology Mode	EQUIPMENT.IMD_SRTMODE (LTE eNodeB, 5G gNodeB)	None	If this parameter is set to INSUBBRACK or BETWEENSUBBRACK , RF module super dormancy cannot be enabled.
GSM UMTS LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	SDR Connection Switch	EQUIPMENT.SDR_CONNSW (LTE eNodeB, 5G gNodeB)	None	If this parameter is set to OFF or ADJUSTING , RF module super dormancy cannot be enabled.
GSM UMTS LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	RF HA group	ADD RFHAGRP	None	RF modules configured with master/backup mode do not support RF module super dormancy.

RAT	Function Name	Function Switch	Reference	Description
GSM UMTS LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	Switch	ALMPORT.SW <i>(LTE eNodeB, 5G gNodeB)</i>	None	If the port switch of an RF module is turned on, RF module super dormancy cannot be enabled for this RF module.
GSM UMTS LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	Monitoring module	None	None	If monitoring modules (EMU/TCU/FMU/CCU/MEB/PMU) are managed by an RF module, RF module super dormancy cannot be enabled for this RF module.
LTE FDD LTE TDD	Intra-RAT level-by-level energy saving	INTRA_RAT_GRAD_PWR_SAVING_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	If the MULTI_RRU_CELL_DORM_COMPAT_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter is selected, the INTRA_RAT_GRAD_PWR_SAVING_SW option of this parameter must be deselected.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Static Multiple Beam	Number of SectorSplitCell MOs ≥ 2	<i>Massive MIMO Basics (FDD) in eRAN Feature Documentation</i> <i>Smart 8T8R (FDD) in eRAN Feature Documentation</i>	Cells with Static Multiple Beam enabled in FDD massive MIMO scenarios cannot be configured as capacity-layer cells for LTE load-based RF module super dormancy in precise dormancy mode. Cells with Static Multiple Beam enabled in FDD Smart 8T8R scenarios cannot be configured as capacity-layer cells for LTE load-based RF module super dormancy in precise dormancy mode.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	SFN cell	Cell.MultiRruCellMode set to SFN	<i>SFN in eRAN Feature Documentation</i>	A cell cannot be configured as an SFN cell when it serves as a capacity-layer cell for LTE load-based RF module super dormancy in precise dormancy mode (that is, when the LTE_DORMANCY_BY_LOAD_PE_SW option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter is selected and the CellPwrSavingAlgo.DormancyCellType parameter is set to CAPACITY_CELL) and the CellAuxEnbBind or CellAuxDuBind MO is configured for the cell.
LTE FDD LTE TDD	Inter-eNodeB SFN based on eNodeB coordination	Number of CellAuxEnbBind MOs ≥ 1	<i>SFN in eRAN Feature Documentation</i>	Cells with inter-eNodeB SFN based on eNodeB coordination enabled do not support LTE load-based RF module super dormancy in non-precise dormancy mode.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Inter-eNodeB adaptive SFN/SDMA based on eNodeB coordination	Number of CellAuxEnbBind MOs ≥ 1	<i>SFN in eRAN Feature Documentation</i>	Cells with inter-eNodeB adaptive SFN/SDMA based on eNodeB coordination enabled do not support LTE load-based RF module super dormancy in non-precise dormancy mode.
LTE FDD LTE TDD	Auxiliary eNodeB Cell Distributed Unit Binding Relationship	Number of CellAuxDuBind MOs ≥ 1	None	Cells with Auxiliary eNodeB Cell Distributed Unit Binding Relationship configured do not support LTE load-based RF module super dormancy in non-precise dormancy mode.
NR FDD Low-frequency NR TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SWITCH option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC in 5G RAN Feature Documentation</i>	In low latency and high reliability scenarios, the following functions cannot be enabled: NR load-based RF module super dormancy, LNR load-based RF module super dormancy, and optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module super dormancy.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Dynamic MIMO layer number decrease	DYNAMIC_MIMO_LAYER_DEC_SW option of the gNodeBAlgo.DedicatedNetworkFunctionSw parameter	None	Dynamic MIMO layer number decrease does not work with optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module super dormancy.
NR FDD Low-frequency NR TDD High-frequency NR TDD	NR load-based RF module super dormancy	INTRA_NR_DORMANCY_BY_LOAD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	NR load-based RF module super dormancy does not work with optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module super dormancy.
LTE TDD LTE FDD NR FDD Low-frequency NR TDD	LNR load-based RF module super dormancy	LNR_DORMANCY_BY_LOAD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	LNR load-based RF module super dormancy does not work with optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module super dormancy.
LTE FDD	Out-of-band relay enhancement	OUT_OF_BAND_RELAY_ENH_SW option of the CellRelayAlgoPara.RelayEnhSw parameter	<i>RuralCow Solution</i>	When out-of-band relay enhancement is enabled, RF module super dormancy must be disabled.

7.4.3.2.3 Function Impacts

When RF module super dormancy and another energy saving function (such as symbol power saving, channel shutdown, deep dormancy, fast deep dormancy, or carrier shutdown) are both enabled, RF module super dormancy preferentially takes effect.

RF module super dormancy and RF module timing dormancy cannot be both enabled.

RF module super dormancy and iSDU-based wakeup dormancy for RF modules cannot be both enabled.

In the same period, load-based RF module dormancy takes precedence over scheduled RF module dormancy. When load conditions are not met, neither of these functions can take effect. When load conditions are met, these functions take effect based on the dormancy switch (specified by the **RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB)** parameter) and the energy saving period configured using one of the following cell-level MOs: **NRDUCellPowerSaving**, **CellPwrSavingPolicy**, **GCELLDORMANCY**, and **UCELLDORMANCYPOLICY**. The function activation is irrelevant to the dormancy period (specified by the **RRU.DORMANCYSTARTTIME (LTE eNodeB, 5G gNodeB)** and **RRU.DORMANCYSTOPTIME (LTE eNodeB, 5G gNodeB)** parameters) and the energy saving policy specified by the **BRDPWRSVGPOLICY.POLICYID (LTE eNodeB, 5G gNodeB)** parameter.

In the same period, super dormancy takes precedence over deep dormancy. If O&M protection is enabled for super dormancy due to operations (such as log extraction), deep dormancy instead of super dormancy takes effect when conditions are met.

When RF modules are in deep dormancy, they cannot switch directly to super dormancy. The decision to enter super dormancy can only be made after these RF modules exit deep dormancy.

In the overlapping period, if the deep dormancy period is earlier than the super dormancy period, deep dormancy instead of super dormancy takes effect when conditions are met.

Cells do not provide services after entering dormancy, and related features do not take effect in these cells.

The following tables list other function impacts.

- **Table 7-51** lists the function impacts of scheduled RF module deep dormancy.
- **Table 7-52** lists the function impacts of NR load-based RF module super dormancy.
- In addition to the function impacts of NR load-based RF module super dormancy (except the impact relationship with intra-RAT ANR), LNR load-based RF module super dormancy has the function impacts listed in **Table 7-52** and **Table 7-53**.
- **Table 7-54** lists the function impacts of LTE load-based RF module super dormancy.
- The GSM-related content in **Table 7-51** describes the function impacts of GSM load-based RF module super dormancy.
- The UMTS-related content in **Table 7-51** describes the function impacts of UMTS and GU load-based RF module super dormancy.

Table 7-51 Function impacts of scheduled RF module super dormancy

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Carrier aggregation	None	<i>Carrier Aggregation in eRAN Feature Documentation</i>	- Cells in the RF module dormancy state do not participate in CA. In CA-group-based configuration mode, after such a cell in a CA group exits this energy saving state, this cell cannot be configured as an SCell for an existing CA UE until the UE accesses the network again. - After a cell exits the RF module dormancy state, it does not participate in CA if the number of routes is insufficient. The route for CA will be released for a cell in the RF module dormancy state. After this cell exits this state, it may not obtain a route if the number of routes is insufficient and therefore, cannot

RAT	Function Name	Function Switch	Reference	Description
				participate in CA.
LTE FDD LTE TDD	Automatic baseband resource configuration	eNodeBResModeAlgo.BbpResAutoConfigSw	None	When the eNodeBResModeAlgo.BbpResAutoConfigSw parameter is set to ON , RF module super dormancy does not take effect.
LTE TDD	Clock synchronization detection	ClkDetect.ClkAsyncDetectSwitch	<i>Synchronization in eRAN Feature Documentation</i>	If clock out-of-synchronization occurs during the dormancy period, this issue may fail to be detected.
Low-frequency NR TDD	Clock out-of-synchronization detection	CLK_DETECT_SW option of the gNodeBParam.ClkOutofSyncDetectSwitch parameter	<i>Synchronization in 5G RAN Feature Documentation</i>	If clock out-of-synchronization occurs during the dormancy period, this issue may fail to be detected.
Low-frequency NR TDD	Air-interface-based network-wide synchronization deviation detection	GAP_BASED_SYNC_DIFF_DET_SW option of the gNodeBParam.SyncDiffDetectSwitch parameter	<i>Synchronization in 5G RAN Feature Documentation</i>	If sequence detection is performed in all cells of a to-be-detected base station during the period when RF module super dormancy takes effect, the sequence detection fails. In this case, the possible clock fault of the base station may fail to be detected.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module super dormancy takes precedence over energy saving based on flexible frequency-domain scheduling when both functions are enabled. Energy saving based on flexible frequency-domain scheduling does not take effect when RF module super dormancy is in effect.
NR FDD Low-frequency NR TDD	TTI-level channel shutdown	TTI_RF_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module super dormancy takes precedence over TTI-level channel shutdown when both functions are enabled. TTI-level channel shutdown does not take effect when RF module super dormancy is in effect.
NR FDD Low-frequency NR TDD	Intelligent power control energy saving	SMART_POWER_CTRL_ENERGY_SAVE_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module super dormancy takes precedence over intelligent power control energy saving when both functions are enabled. Intelligent power control energy saving does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Low-power RF channel dynamic shutdown	LOW_POWER_RF_SHUT_DOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module super dormancy takes precedence over low-power RF channel dynamic shutdown when both functions are enabled. Low-power RF channel dynamic shutdown does not take effect when RF module super dormancy is in effect.
LTE FDD LTE TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over energy saving based on flexible frequency-domain scheduling when both functions are enabled. Energy saving based on flexible frequency-domain scheduling does not take effect when RF module super dormancy is in effect.
NR FDD Low-frequency NR TDD	Cell-level MRO	MRO_SCENARIO_IDENT_SW option of the gNBMrScenarioIdentifySwitch parameter and the INTRA_FREQ_MRO_SW option of the NRCellMr.MroSwitch parameter	<i>MRO in 5G RAN Feature Documentation</i>	When both RF module super dormancy and intra-frequency MRO are enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, MRO may take effect later than expected.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	UE-level MRO	UE_MRO_SWITCH option of the NRCellMro.MroSwitch parameter	<i>MRO in 5G RAN Feature Documentation</i>	When RF module super dormancy and UE-level MRO are both enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, the penalty mechanism accuracy of the UE-level MRO is affected.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Intelligent power control energy saving	- Single-cell intelligent power control energy saving: INTEL_INTRA_CELL_PC_SW option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch - Multi-cell coordinated intelligent power control energy saving: INTEL_INTRA_CELL_PC_SW and INTEL_MULTICELL_COOR_PC_SW options of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over intelligent power control energy saving when both functions are enabled. Intelligent power control energy saving does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Dynamic carrier shutdown phase 2	CellPwrSav ngAlgo.Carri erDynShutd ownPh2Sw set to CAPACITY_C ELL	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over dynamic carrier shutdown phase 2 when both functions are enabled. Dynamic carrier shutdown phase 2 does not take effect when RF module super dormancy is in effect.
LTE FDD LTE TDD	CMAS	CellAlgoSwi tch.CmasSwi tch	<i>CMAS in eRAN Feature Documentation</i>	The following applies when cells enter RF module super dormancy: • With PWS message-triggered exit of cells from energy saving disabled, CMAS warning notifications cannot be properly sent in LTE cells in the dormancy state. However, UEs can still receive CMAS warning notifications from other cells. • With PWS message-triggered exit of cells from energy saving enabled, cells exit the dormancy state upon receipt of a PWS message.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	CMAS	gNB SibConfig.SibType set to SIB_TYPE_8	<i>CMAS in 5G RAN Feature Documentation</i>	When an RF module enters the super dormancy state, the NR cells it serves also enter the dormancy state. In this case, CMAS warning notifications cannot be properly sent in these cells. However, UEs can still receive CMAS warning notifications from other cells.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	ETWS	None	<i>ETWS in eRAN Feature Documentation</i>	The following applies when cells enter RF module super dormancy: • With PWS message-triggered exit of cells from energy saving disabled, ETWS warning notifications cannot be sent in LTE cells in the dormancy state. However, UEs can still receive ETWS warning notifications from other cells. • With PWS message-triggered exit of cells from energy saving enabled, cells exit the dormancy state upon receipt of a PWS message.
LTE FDD	Downlink intelligent joint power control	DL_INTEL_JOINT_PC_SW option of the CellDisSchExtParam.DisSchExtSwitch parameter	<i>Smart 8T8R (FDD) in eRAN Feature Documentation</i>	Downlink intelligent joint power control does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Downlink intelligent joint AMC and power control	DL_INTEL_JOINT_AMC_AND_PC_SW option of the CellDisSchExtParam.DisSchExtSwitch parameter	<i>Smart 8T8R (FDD) in eRAN Feature Documentation</i>	Downlink intelligent joint AMC and power control does not take effect when RF module super dormancy is in effect.
LTE FDD LTE TDD	RF channel intelligent shutdown	CellRfShutdown.RfShutdownSwitch	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over RF channel intelligent shutdown when both functions are enabled. RF channel intelligent shutdown does not take effect when RF module super dormancy is in effect.
LTE FDD LTE TDD	Intelligent power-off of carriers in the same coverage	CellShutdown.CellShutdownSwitch set to ON	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over intelligent power-off of carriers in the same coverage when both functions are enabled. Intelligent power-off of carriers in the same coverage does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Multi-carrier coordinated energy saving	CellShutDown.CellShutdownSwitch set to ON_MULTI_CARRIER_HIER_SHUTDOWN , ON_MULTICARRIER_HIERMANUAL_SHUTDOWN , or ON_CARRIER_PRECISE_SHUTDOWN	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over multi-carrier coordinated energy saving when both functions are enabled. Multi-carrier coordinated energy saving does not take effect when RF module super dormancy is in effect.
LTE FDD LTE TDD	Dynamic carrier shutdown	CARRIER_DYNAMIC_SHUTDOWN_SWITCH option of the CellShutDown.CarrierShutDownEnhancedSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over dynamic carrier shutdown when both functions are enabled. Dynamic carrier shutdown does not take effect when RF module super dormancy is in effect.
LTE FDD LTE TDD	Dynamic voltage adjustment	CellAlgoSwitch.DynAdjustVoltageSwitch	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over dynamic voltage adjustment when both functions are enabled. Dynamic voltage adjustment does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Low power consumption mode (LTE)	CellLowPower.LowPwrSwitch	<i>Green Site</i>	RF module super dormancy takes precedence over low power consumption mode when both functions are enabled. Low power consumption mode does not take effect when RF module super dormancy is in effect.
LTE FDD LTE TDD	Cell Switch-off Based on Automatic Co-coverage Identification	CellShutdown.CellShutdownSwitch set to ON_CO_COV_AUTO_IDE NT	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over Cell Switch-off Based on Automatic Co-coverage Identification when both functions are enabled. Cell Switch-off Based on Automatic Co-coverage Identification does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Basic symbol power saving	FDD: SymbolShutdownSwitch.PowerSaveSwitch parameter TDD: TddSymbolsShutdownSwitch option of the ENodeBAlgoSwitch.PowerSaveSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over basic symbol power saving when both functions are enabled. Basic symbol power saving does not take effect when RF module super dormancy is in effect.
LTE FDD LTE TDD	CRS dynamic muting	CRS_DYNAMIC_IC_MUTING_SW option of the SymbolPowerSaving.SymbolShutdownEnhancedSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over CRS dynamic muting when both functions are enabled. CRS dynamic muting does not take effect when RF module super dormancy is in effect.
LTE FDD LTE TDD	Automatic cell shutdown	CellAutoShutdown.CellAutoShutdownSwitch	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over automatic cell shutdown when both functions are enabled. Automatic cell shutdown does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	RF channel dynamic muting	RF_CHN_DYNAMIC_MUTING_SW option of the CellRfChnDynamicMuting.RfChannelDynamicMutingAlgorithmSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over RF channel dynamic muting when both functions are enabled. RF channel dynamic muting does not take effect when RF module super dormancy is in effect.
NR FDD Low-frequency NR TDD High-frequency NR TDD	Symbol power saving	NRDUCellAIAlgorithmSwitch.SymbolShutdownSwitch	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module super dormancy takes precedence over symbol power saving when both functions are enabled. Symbol power saving does not take effect when RF module super dormancy is in effect.
NR FDD Low-frequency NR TDD High-frequency NR TDD	Power saving scheduling	POWER_SAVING_SCHEDULE_SW option of the NRDUCellAIAlgorithmSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module super dormancy takes precedence over power saving scheduling when both functions are enabled. Power saving scheduling does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD High-frequency NR TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module super dormancy takes precedence over RF channel intelligent shutdown when both functions are enabled. RF channel intelligent shutdown does not take effect when RF module super dormancy is in effect.
NR FDD Low-frequency NR TDD High-frequency NR TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module super dormancy takes precedence over timing carrier shutdown when both functions are enabled. Timing carrier shutdown does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Intelligent carrier shutdown in precise mode	• INTRA_GNB_MULT I_CARR_S D_SW or INTER_GNB_MULT I_CARR_S D_SW option of the NRDUCel lAlgoSwitch.Power SavingSwitch parameter • MULTI_CARR_PRECISE_SHD N_SW option of the NRDUCel lAlgoSwitch.Power SavingSwitch parameter selected	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module super dormancy takes precedence over intelligent carrier shutdown when both functions are enabled. Intelligent carrier shutdown does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD High-frequency NR TDD	Intelligent carrier shutdown in non-precise mode	INTRA_GNB_MULT I_CARR_S D_SW or INTER_GNB_MULT I_CARR_S D_SW option of the NRDUCel lAlgoSwitch.Power SavingSwitch parameter MULTI_CARR_PRECISE_SHD N_SW option of the NRDUCel lAlgoSwitch.Power SavingSwitch parameter deselected	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module super dormancy takes precedence over intelligent carrier shutdown when both functions are enabled. Intelligent carrier shutdown does not take effect when RF module super dormancy is in effect.
NR FDD Low-frequency NR TDD	Uplink symbol power saving	UL_SYMBOL_SHUTDOWN_SW option of the NRDUCel lAlgoSwitch.Power SavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	When RF module super dormancy takes effect, uplink symbol power saving produces no energy saving gains.
NR FDD Low-frequency NR TDD	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCel lAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation in 5G RAN Feature Documentation</i>	A carrier exits a power sharing group once it enters the super dormancy state.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Ultra-long holdover for sites with abnormal clock sources	CLK_OTASY NC_PROC_SW option of the TASM.MSCLKDIAGPROC ESSSW parameter	<i>Synchronization in 5G RAN Feature Documentation</i>	Assume that ultra-long hold switch is turned on but ultra-long hold enhancement switch is turned off. If sequence detection is performed in all cells of a to-be-detected base station during the period when RF module super dormancy takes effect, the sequence detection fails. In this case, the base station cannot enter the ultra-long holdover state even if its clock is faulty, or it exits the ultra-long holdover state as sequence detection cannot be performed for a long time due to energy saving. If ultra-long hold enhancement switch is turned on, RF module super dormancy does not take effect for the base station in the ultra-long holdover state or its reference neighboring base stations.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Intelligent PIM interference avoidance	PIM_INTERF_SMART_AVOID_SW option of the PimInterfSmartAvoid.PimAlgoSwitch parameter	<i>Interference Detection and Suppression in eRAN Feature Documentation</i>	RF module super dormancy is terminated during PIM interference detection.
NR FDD	Intelligent PIM interference avoidance	PIM_INTERF_SMART_AVOID_SW option of the NRDUCellPimIntrfAvd.PimAlgoSwitch parameter	<i>Interference Avoidance in 5G RAN Feature Documentation</i>	RF module super dormancy is terminated during PIM interference detection.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Fast carrier shutdown	For capacity-layer cells: • Intra-base-station multi-carrier scenarios: INTRA_GNB_MULT I_CARR_S D_SW option of the NRDUCel lAlgoSwitch.Power SavingSwitch parameter selected; inter-base-station multi-carrier scenarios: INTER_GNB_MULT I_CARR_S D_SW option of the NRDUCel lAlgoSwitch.Power SavingSwitch parameter selected • FAST_CARRIER_SHUTDOWN_SW option of the NRDUCel lAlgoSwit	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Fast carrier shutdown does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
		ch.PowerSavingExtSwitch parameter For basic-layer cells: FAST_CARR_SHUT_SERV_ASSUR_SW option of the NRDUCellAI goSwitch.PowerSavingExtSwitch parameter		
NR FDD Low-frequency NR TDD	Deep symbol power saving	DEEP_SYMBOL_SHUTDOWN_SW option of the NRDUCellAI goSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Deep symbol power saving does not take effect when RF module super dormancy is in effect.
LTE FDD LTE TDD	Deep symbol power saving	DEEP_SYMBOL_SHUTDOWN_SW option of the SymbolPowerSaving.SymbolShutdownEnhancedSw parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	Deep symbol power saving does not take effect when RF module super dormancy is in effect.
Low-frequency NR TDD	NR multi-band power adaptive scheduling	NR_MULTI_BAND_PWR_ADAPT_SWITCH option of the NRDUCellAI goSwitch.DynamicPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	A carrier exits a power adaptive scheduling group once it enters the super dormancy state.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	SOC level-based low power consumption mode	SOCLVL_BASED_LOW_POWER_SW option of the CellLowPower.LowPowerConsumeModeExtSw parameter	<i>Green Site</i>	RF module super dormancy takes precedence over SOC level-based low power consumption mode and SOC level-based low power consumption mode does not take effect when both functions are enabled.
NR FDD	Intelligent MCS optimization for uplink SU-MIMO	UL_SU_MCS_INTEL_OPT_SW option of the NRDUCellAiAlgo.AiAmcAlgoSwitch parameter	<i>Smart 8T8R (FDD) in 5G RAN Feature Documentation</i>	When enabled together with RF module super dormancy, intelligent MCS optimization for uplink SU-MIMO offers fewer gains or even fails to take effect.
NR FDD Low-frequency NR TDD	Dynamic RF channel shutdown	RF_CHN_DYNAMIC_SHUTDOWN_SW option of the NRDUCellAiAlgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Dynamic RF channel shutdown does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD NR FDD	LTE-NR multi-dimensional scheduling optimization	• LTE: LNR_MULTTI_D_OPT_SCH_SW option of the GigaBandCell.GigaBandCellAlgoSw parameter • NR: LNR_MULTTI_D_OPT_SCH_SW option of the NRDUCel lGigaBand.GigaBandCellAlgoSw parameter	<i>LTE FDD and NR Inter-RAT Unified Scheduling</i>	LTE-NR multi-dimensional scheduling optimization does not take effect for cells in the dormancy state.
LTE FDD LTE TDD	Load simulation	CellSimuLoad MO configured	None	Scheduled RF module super dormancy does not take effect when at least one CellSimuLoad MO is configured.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD High-frequency NR TDD	Cell load simulation	STRNRDUCELLSIMULATELOAD	None	- If the SIMU_LOAD_DORMANCY_IMP_SW option of the gNB PwrSavingAlgo.PwrSavingsAlgoSwitch parameter is selected, and cell load simulation takes effect, the RF module serving the local cell is not allowed to enter the dormancy state, but the RF modules serving the cells not enabled with cell load simulation are allowed to enter the dormancy state. - If the SIMU_LOAD_DORMANCY_IMP_SW option of the gNB PwrSavingAlgo.PwrSavingsAlgoSwitch parameter is deselected, and cell load simulation takes effect, all RF modules of the base station are not allowed to enter the

RAT	Function Name	Function Switch	Reference	Description
				dormancy state.
LTE FDD	8T beam mode shift for energy saving	BEAM_MODE_SHIFT_SW option of the SectorSplitGroup.SplitCellGroupOptions parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over 8T beam mode shift for energy saving when both functions are enabled.
LTE FDD	32T beam mode shift for energy saving	BEAM_MODE_SHIFT_SW option of the SectorSplitGroup.SplitCellGroupOptions parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over 32T beam mode shift for energy saving when both functions are enabled.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	Intelligent Detection for Fronthaul Optical Paths	None	<i>Intelligent Detection for Fronthaul Optical Paths</i>	• If the BBU and RF module are not in the dormancy state when optical path detection is started, the detection process will delay the time at which they enter that state. • If the BBU and RF module are in the dormancy state when optical path detection is started, the following logic applies: – In a scenario where wakeup from dormancy is supported, starting optical path detection will cause the BBU and RF module to automatically exit the dormancy state. They will not re-enter the state within 30 minutes after the detection process starts. Once

RAT	Function Name	Function Switch	Reference	Description
				30 minutes elapse, they will evaluate whether to re-enter the dormancy state. - In a scenario where wakeup from dormancy is not supported, optical path detection cannot be started.
LTE FDD	Static Multiple Beam 2.0	SMB2_SW option of the SectorSplitGroup.SplitCellGroupOptSwitch parameter	<i>Smart 8T8R (FDD) in eRAN Feature Documentation</i>	RF module super dormancy preferentially takes effect when it is enabled together with Static Multiple Beam 2.0.
LTE FDD	LNR CRS muting	LNR_CRSMUTE_SW option of the CellAlgoExtSwitch.LnrZeroBufferzoneSw parameter	<i>LTE and NR Zero Bufferzone</i>	When LNR CRS muting and RF module super dormancy are both enabled, RF module super dormancy preferentially takes effect. LNR CRS muting does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Initial transmission power reduction	INIT_TRANS_POWER_REDUCE_SW option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	When initial transmission power reduction and RF module super dormancy are both enabled, RF module super dormancy preferentially takes effect.
NR FDD Low-frequency NR TDD	Initial transmission power reduction	INIT_TRANS_POWER_REDUCE_SW option of the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Initial transmission power reduction does not take effect when RF module super dormancy is in effect.
LTE FDD LTE TDD	Retransmission power reduction	RETRANS_POWER_REDUCE_SW option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	When retransmission power reduction and RF module super dormancy are both enabled, RF module super dormancy preferentially takes effect.
NR FDD Low-frequency NR TDD	Retransmission power reduction	RETRANS_POWER_REDUCE_SW option of the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Retransmission power reduction does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Uplink TTI-level RF channel dynamic muting	UL_TTI_CHN_DYN_MUTING_SW option of the CellRfChnDynamicMuting.RfChannelDynMutingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	When uplink TTI-level RF channel dynamic muting and RF module super dormancy are both enabled, RF module super dormancy preferentially takes effect.
NR FDD	Uplink TTI-level RF channel dynamic muting	UL_TTI_CHN_DYN_MUTING_SW option of the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Uplink TTI-level RF channel dynamic muting does not take effect when RF module super dormancy is in effect.
LTE FDD	Port sparseness	PORT_SPARSENESS_SW option of the CellPwrSavingAlgo.PowerSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over port sparseness when both functions are enabled.
LTE FDD	GSM and LTE Buffer Zone Optimization	CellAlgoSwitch.AvoidInterfSwitch	<i>GSM and LTE Buffer Zone Optimization</i>	Cells in the RF module super dormancy state do not participate in GSM and LTE Buffer Zone Optimization.
LTE FDD	GSM and LTE Spectrum Concurrency	SpectrumCloud.SpectrumCloudSwitch set to GL_SPECTRUM_CONCURRENCY	<i>GSM and LTE Spectrum Concurrency</i>	Cells in the RF module super dormancy state do not participate in GSM and LTE Spectrum Concurrency.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	GSM and LTE Spectrum Concurrency Phase 2	GL_SPCT_CO NCURRENCY _PH2_SW option of the SpectrumCl oud.Spectru mCloudEnhS witch parameter	<i>GSM and LTE Spectrum Concurrency</i>	Cells in the RF module super dormancy state do not participate in GSM and LTE Spectrum Concurrency Phase 2.
LTE FDD	UMTS and LTE Spectrum Sharing Based on DC-HSDPA (LTE FDD)	SpectrumCl oud.Spectru mCloudSwit ch set to DC_HSDPA_ BASED_UL_S PECTRUM_S HR	<i>UMTS and LTE Spectrum Sharing Based on DC- HSDPA</i>	RF module super dormancy cannot take effect in cells where UMTS and LTE Spectrum Sharing Based on DC-HSDPA is enabled.
LTE FDD	UMTS and LTE Spectrum Sharing (LTE FDD)	SpectrumCl oud.Spectru mCloudSwit ch set to UL_SPECTRU M_SHARING	<i>UMTS and LTE Spectrum Sharing</i>	UMTS and LTE Spectrum Sharing does not take effect when RF module super dormancy is in effect.
LTE FDD	UMTS and LTE Spectrum Sharing Phase 2 (LTE FDD)	UL_SPECTRU M_SHR_PH2 _SW option of the SpectrumCl oud.Spectru mCloudEnhS witch parameter	<i>UMTS and LTE Spectrum Sharing</i>	UMTS and LTE Spectrum Sharing Phase 2 does not take effect when RF module super dormancy is in effect.
LTE FDD	UMTS and LTE Zero Bufferzone (LTE FDD)	UMTS_LTE_Z ERO_BUFFE R_ZONE_SW option of the ULZeroBufFe rZone.ZeroB ufZoneSwitc h parameter	<i>UMTS and LTE Zero Bufferzone</i>	UMTS and LTE Zero Bufferzone does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
UMTS	Multi-Carrier Switch off Based on Power Backup	NODEBPOWEROUTAGE./SDSW	<i>Green BTS in RAN Feature Documentation</i>	RF module super dormancy takes precedence over Multi-Carrier Switch off Based on Power Backup when both functions are enabled. Multi-Carrier Switch off Based on Power Backup does not take effect when RF module super dormancy is in effect.
UMTS	UMTS low power consumption mode	NODEBALGP ARA.LOWP WRCONSUM MPMODESW	<i>Green BTS in RAN Feature Documentation</i>	RF module super dormancy takes precedence over low power consumption mode when both functions are enabled. Low power consumption mode does not take effect when RF module super dormancy is in effect.
UMTS	VAM Power Saving	NODEBALGP ARA.VAMP WRSVSW	<i>Green BTS in RAN Feature Documentation</i>	RF module super dormancy takes precedence over VAM Power Saving when both functions are enabled. VAM Power Saving does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
UMTS	RF Channel Muting Based Power Saving	ULOCCELLAL GPARA.RFC HANNELMU TPWRSAVS W	<i>Green BTS in RAN Feature Documentation</i>	RF module super dormancy takes precedence over RF Channel Muting Based Power Saving when both functions are enabled. RF Channel Muting Based Power Saving does not take effect when RF module super dormancy is in effect.
UMTS	UMTS RF channel dynamic muting	ULOCCELLAL GPARA.RFC HNDYNMUT INGMODE set to BASICPWRS AV or DEEPPWRS AV	<i>Green BTS in RAN Feature Documentation</i>	RF module super dormancy takes precedence over UMTS RF channel dynamic muting when both functions are enabled. UMTS RF channel dynamic muting does not take effect when RF module super dormancy is in effect.
UMTS	Time Division Carrier Off Based Power Saving	ULOCCELLAL GPARA.TDP WRSVSW	<i>Green BTS in RAN Feature Documentation</i>	RF module super dormancy takes precedence over Time Division Carrier Off Based Power Saving when both functions are enabled. Time Division Carrier Off Based Power Saving does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
UMTS	Dynamic Power Saving Based on Maximum Power Restriction	ULOCCELLAL GPARA.PWR RSTRBASED DYNENRSAV MODE set to BASICPWRS AV or DEEPPWRS AV	<i>Green BTS in RAN Feature Documentation</i>	RF module super dormancy takes precedence over Dynamic Power Saving Based on Maximum Power Restriction when both functions are enabled. Dynamic Power Saving Based on Maximum Power Restriction does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
UMTS	UMTS Adaptive Power Saving	ULOCCELLAL GPARA.ADP PWRSVSW	<i>Green BTS in RAN Feature Documentation</i>	RF module super dormancy takes precedence over UMTS Adaptive Power Saving when both functions are enabled. UMTS Adaptive Power Saving does not take effect when RF module super dormancy is in effect. UMTS Adaptive Power Saving may dynamically adjust power parameters based on air interface changes and update the parameter values on the GUI at around 23:35 each day. If the cell is in dormancy at that time, it exits dormancy, which affects dormancy gains.

RAT	Function Name	Function Switch	Reference	Description
UMTS	Multi-Carrier Switch off Based on Traffic Load	• UCELLDYNSHUTDOWN.DynShutdownSwitch set to ON_1 • UCELLDYNSHUTDOWN.DynShutdownType set to CONDITIONALSHUTDOWN	<i>Green BTS in RAN Feature Documentation</i>	It is not recommended that the carrier shutdown period set on the base station controller overlap the RF module super dormancy period of the local cell. If the periods overlap, RF module super dormancy preferentially takes effect. Multi-Carrier Switch off Based on Traffic Load does not take effect when RF module super dormancy is in effect.
UMTS	Multi-Carrier Switch off Based on QoS	• UCELLDYNSHUTDOWN.DynShutdownSwitch set to ON_1 • UCELLDYNSHUTDOWN.DynShutdownType set to QOSSHUTDOWN	<i>Green BTS in RAN Feature Documentation</i>	It is not recommended that the carrier shutdown period set on the base station controller overlap the RF module super dormancy period of the local cell. If the periods overlap, RF module super dormancy preferentially takes effect. Multi-Carrier Switch off Based on QoS does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
UMTS	Forced cell shutdown	• UCELLDYNSHUTDOWN.DynShutdownSwitch set to ON_1 • UCELLDYNSHUTDOWN.DynShutdownType set to FORCESHUTDOWN	<i>Green BTS in RAN Feature Documentation</i>	It is not recommended that the carrier shutdown period set on the base station controller overlap the RF module super dormancy period of the local cell. If the periods overlap, RF module super dormancy preferentially takes effect. Forced cell shutdown does not take effect when RF module super dormancy is in effect.
UMTS	RRU Intelligent Dormancy	NODEBALGPARA.RRUINTELLIGENTDORMANCY	<i>Green BTS in RAN Feature Documentation</i>	It is not recommended that the carrier shutdown period set on the base station controller overlap the RF module super dormancy period of the local cell. If the periods overlap, RF module super dormancy preferentially takes effect. RRU Intelligent Dormancy does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
UMTS	Uplink CoMP (Joint Reception)	ULOCCELL.UL COMPSW	<i>Uplink CoMP in RAN Feature Documentation</i>	After RF module super dormancy takes effect, coordinating links cannot be set up for the corresponding cells, which affects Uplink CoMP (Joint Reception) gains.
UMTS	Uplink CoMP (Joint Reception) Based on Coordinated BBU	ULOCCELL.UL COMPSW	<i>Uplink CoMP in RAN Feature Documentation</i>	After RF module super dormancy takes effect, coordinating links cannot be set up for the corresponding cells, which affects Uplink CoMP (Joint Reception) Based on Coordinated BBU gains.
UMTS	HetNet Uplink CoMP (Joint Reception)	ULOCCELL.UL COMPSW	<i>Uplink CoMP in RAN Feature Documentation</i>	After RF module super dormancy takes effect, coordinating links cannot be set up for the corresponding cells, which affects HetNet Uplink CoMP (Joint Reception) gains.
UMTS	Coverage Expansion Under Interference	ULOCCELL.UL COVEXPANSION	<i>Coverage Expansion Under Interference in RAN Feature Documentation</i>	After RF module super dormancy takes effect, coordinating links cannot be set up for the corresponding cells, which affects Coverage Expansion Under Interference gains.

RAT	Function Name	Function Switch	Reference	Description
UMTS	Inter-NodeB Uplink CoMP (Joint Reception)	ULOCCELL.IN TERNODEBU LCOMP	<i>Inter-NodeB Uplink CoMP in RAN Feature Documentation</i>	After RF module super dormancy takes effect, coordinating links cannot be set up for the corresponding cells, which affects Inter-NodeB Uplink CoMP (Joint Reception) gains.
UMTS	Inter-NodeB HS-DPCCH CoMP (Joint Reception)	ULOCCELL.IN TERNODEB HSDPCCHC OMP	<i>Inter-NodeB Uplink CoMP in RAN Feature Documentation</i>	After RF module super dormancy takes effect, coordinating links cannot be set up for the corresponding cells, which affects Inter-NodeB HS-DPCCH CoMP (Joint Reception) gains.
UMTS	Sleeping Cell Detection and Recovery	NODEBALGP ARA.SLEEP INGCELLDET ECTOPTION	<i>Sleeping Cell Detection and Recovery in RAN Feature Documentation</i>	8:00 to 8:30 is the key monitoring period for differential detection. If the dormancy function is enabled during this period, differential detection for the corresponding cell becomes inaccurate.

RAT	Function Name	Function Switch	Reference	Description
UMTS	Adaptive Interference Rejection	NODEBBASE BANDEQMP ARA.ULBASE BANDEQM WORKMODE set to AIR	<i>Adaptive Interference Rejection in RAN Feature Documentation</i>	When both Adaptive Interference Rejection and dormancy are enabled, Adaptive Interference Rejection is more likely to be triggered.
UMTS	Machine Learning based PCPICH Power Optimization	ULOCCELLAL GPARA.SMA RTPCPICHO PTSWITCH	<i>Machine Learning based PCPICH Power Optimization in RAN Feature Documentation</i>	Machine Learning based PCPICH Power Optimization may dynamically adjust power parameters based on air interface changes and update the parameter values on the GUI at around 23:35 each day. If the cell is in dormancy at that time, it exits dormancy, which affects dormancy gains.
UMTS	HSDPA Scheduler Pool	NODEBALGP ARA.HSDPA SCHPOOLS W	<i>HSDPA Scheduler Pool in RAN Feature Documentation</i>	After RF module dormancy is enabled, HSDPA Scheduler Pool cannot take effect as the baseband equipment can contain only one BBP.

RAT	Function Name	Function Switch	Reference	Description
UMTS	HSUPA Scheduler Pool	NODEBALGP ARA.HSUPA SCHPOOLS W	<i>HSUPA Scheduler Pool in RAN Feature Documentation</i>	After RF module dormancy is enabled, HSUPA Scheduler Pool cannot take effect as the baseband equipment can contain only one BBP.
UMTS	Inter-Dependence of BBU Uplink Resource	NODEBALGP ARA.ULRES WORKMODE not set to NORMAL	<i>Inter-Dependence of BBU Uplink Resource in RAN Feature Documentation</i>	After RF module dormancy is enabled, Inter-Dependence of BBU Uplink Resource cannot take effect as the baseband equipment can contain only one BBP.
UMTS	Hitless Cell Rehoming	NODEBALGP ARA.DLAUT ORESHUFFLE ESW	<i>Hitless Cell Rehoming in RAN Feature Documentation</i>	After RF module dormancy is enabled, Hitless Cell Rehoming cannot take effect as the baseband equipment can contain only one BBP.
UMTS	DB-HSDPA	NODEBMULTICELLGRP. MULTICELLGRP RPTYPE set to HSDPA , and cells added to a group identified by the NODEBMULTICELLGRP MO operating in different frequency bands	<i>Multi-Carrier HSDPA in RAN Feature Documentation</i>	Due to networking restrictions, RF modules operating in mid- and low-frequency bands are not recommended to enter dormancy simultaneously. An RF module serving DB-HSDPA cells cannot enter dormancy.

RAT	Function Name	Function Switch	Reference	Description
UMTS	DB-HSDPA +MIMO	ULOCCELL.HSDPA_DBMIMO	<i>Multi-Carrier HSDPA in RAN Feature Documentation</i>	Due to networking restrictions, RF modules operating in mid- and low-frequency bands are not recommended to enter dormancy simultaneously. An RF module serving DB-HSDPA+MIMO cells cannot enter dormancy.
UMTS	Inter-NodeB DB-HSDPA	INTERNBDB_HSDPA option of the UCELLALGO_SWITCH.HspaPlusSwitch parameter	<i>Multi-Carrier HSDPA in RAN Feature Documentation</i>	Due to networking restrictions, RF modules operating in mid- and low-frequency bands are not recommended to enter dormancy simultaneously. An RF module serving inter-NodeB DB-HSDPA cells cannot enter dormancy.

RAT	Function Name	Function Switch	Reference	Description
UMTS	UMTS SOC level-based low power consumption mode	NODEBALGP ARA.LOWP WRCONS MEMODEEX TSW	<i>Green Site</i>	RF module super dormancy takes precedence over UMTS SOC level-based low power consumption mode when both functions are enabled. UMTS SOC level-based low power consumption mode does not take effect when RF module super dormancy is in effect.
UMTS	Cell Switch-off Based on Automatic Co-coverage Identification (UMTS)	ULOCCELLAL GPARA.CELL SWITCHOFF BSDONCCIS W	<i>Cell Switch-off Based on Automatic Co-coverage Identification</i>	It is not recommended that the carrier shutdown period set on the base station controller overlap the RF module super dormancy period of the local cell. If the periods overlap, RF module super dormancy preferentially takes effect. Cell Switch-off Based on Automatic Co-coverage Identification (UMTS) does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
UMTS	NCR cell	ULOCCELL.NC RCELL	<i>RuralCow Solution</i>	If NCR cells are configured on an RF module, RF module super dormancy cannot take effect on that module.
GSM	Baseband frequency hopping	GCELLHOPT P.FHMODE set to BaseBand_F H or Hybrid_FH	<i>Frequency Hopping in GBSS Feature Documentati on</i>	RF modules involved in the same baseband frequency hopping group must enter or exit the super dormancy state at the same time.
GSM	Soft-Synchronized Network	TASM.FLAG	<i>Soft- Synchronized Network in GBSS Feature Documentati on</i>	Cells on which Soft-Synchronized Network is being performed can be instructed to enter dormancy, but cells in the dormancy state cannot use Soft-Synchronized Network.
GSM	Multi-site Cell	GLOCELL.LO CELLTYPE set to MULTISITE_ CELL	<i>Multi-site Cell in GBSS Feature Documentati on</i>	Multiple RF modules serving a multi-site cell can enter the dormancy state independently.
GSM	Active Backup Power Control	GBTSENERG YMGTPARA. BAKPWRS VMETHOD set to BYPWR	<i>Intelligent Power Consumption Decrease in GBSS Feature Documentati on</i>	RF modules with reduced power can enter the super dormancy state, and RF modules whose TRXs are shut down cannot be woken up from the super dormancy state.

RAT	Function Name	Function Switch	Reference	Description
GSM	TRX Power Amplifier Intelligent Shutdown	• BSCDSTP A.BSCDY NSWITCH TRXPAAL LOW • GCELLBA SICPARA. DYNOpen TrxPower • GTRXDEV .CPS	<i>Intelligent Power Consumption Decrease in GBSS Feature Documentation</i>	RF module super dormancy takes precedence over TRX Power Amplifier Intelligent Shutdown when both functions are enabled.
GSM	GSM Low Power Consumption Mode	GBTSENERG YMGTPARA. LOWPWRCO NSUMPMO DESW	<i>Intelligent Power Consumption Decrease in GBSS Feature Documentation</i>	RF modules whose TRXs are shut down cannot be woken up from the super dormancy state.
GSM	Dynamic Cell Power Off	GCELLDYNT URNOFF.TU RNOFFENAB LE set to ENABLE	<i>Dynamic Cell Power Off in GBSS Feature Documentation</i>	RF module super dormancy takes precedence over Dynamic Cell Power Off when both functions are enabled.
GSM	Dynamic BCCH TRX Timeslot Shutdown	BTSOTHPAR A.BCCHTRXT sShutdownSw	<i>Dynamic BCCH TRX Timeslot Shutdown in GBSS Feature Documentation</i>	When a cell enters the dormancy state, all timeslots on the BCCH TRX are shut down, and Dynamic BCCH TRX Timeslot Shutdown does not take effect.
GSM	Cell Frequency Scan	None	<i>Cell Frequency Scan in GBSS Feature Documentation</i>	Frequency scan prevents cells from entering the super dormancy state. Cells already in the super dormancy state will exit this state when frequency scan is performed.

RAT	Function Name	Function Switch	Reference	Description
GSM	Multi-Sector BCCH Sharing	GCELLBASIC PARA.MultiSectorBcchSharingSw	<i>Multi-Sector BCCH Sharing in GBSS Feature Documentation</i>	Each sector of a cell can enter the dormancy state independently.
GSM	BTS Test Function	None	<i>Operation and Maintenance in GBSS Feature Documentation</i>	BTS Test Function prevents cells from entering the super dormancy state. Cells already in the super dormancy state will exit this state when BTS Test Function is performed.
GSM	GSM SOC level-based low power consumption mode	• IPWRM.S OCLVLAD JSW set to ON • GBTSENE RGYMG T PARA.SO CLVLPWR SAVPOLI CY set to LOW_PWR_AT_SOCLVL1, LOW_PWR_AT_SOCLVL2, or LOW_PWR_AT_SOCLVL3	<i>Green Site</i>	RF modules whose TRXs are shut down cannot be woken up from the super dormancy state.
GSM	GSM and LTE Buffer Zone Optimization	CellAlgoSwitch.AvoidInterfSwitch	<i>GSM and LTE Buffer Zone Optimization</i>	Cells in the RF module super dormancy state do not participate in GSM and LTE Buffer Zone Optimization.

RAT	Function Name	Function Switch	Reference	Description
GSM	GSM and LTE Spectrum Concurrency	GCELLGLSPE CCLOUD.GL SpecConcur Sw	<i>GSM and LTE Spectrum Concurrency</i>	Cells in the RF module super dormancy state do not participate in GSM and LTE Spectrum Concurrency.
GSM	GSM and LTE Spectrum Concurrency Phase 2	GCELLGLSPE CCLOUD.GL SpecConcur Enh	<i>GSM and LTE Spectrum Concurrency</i>	Cells in the RF module super dormancy state do not participate in GSM and LTE Spectrum Concurrency Phase 2.
UMTS	UMTS and LTE Spectrum Sharing Based on DC-HSDPA (UMTS)	ULOCCELL.DC HSDPASHAR EDWITHLTE	<i>UMTS and LTE Spectrum Sharing Based on DC- HSDPA</i>	RF module super dormancy cannot take effect in cells where UMTS and LTE Spectrum Sharing Based on DC-HSDPA is enabled.
UMTS	UMTS and LTE Spectrum Sharing (UMTS)	ULOCCELL.UL SPECTRUMS HARINGSWI TCH	<i>UMTS and LTE Spectrum Sharing</i>	UMTS and LTE Spectrum Sharing does not take effect when RF module super dormancy is in effect.
UMTS	UMTS and LTE Spectrum Sharing Phase 2 (UMTS)	ULOCCELL.UL SPECTRUMS HARINGPHA SE2	<i>UMTS and LTE Spectrum Sharing</i>	UMTS and LTE Spectrum Sharing Phase 2 does not take effect when RF module super dormancy is in effect.
UMTS	UMTS and LTE Zero Bufferzone (UMTS)	ULOCCELL.UL OBUFFERZO NE	<i>UMTS and LTE Zero Bufferzone</i>	UMTS and LTE Zero Bufferzone does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios	DUAL_PLAN_E_TOB_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	<i>Base Station Equipment Reliability</i>	When RF module super dormancy is enabled, RF modules enter the super dormancy state, which causes an active/standby BBU switchover failure. Therefore, RF module super dormancy and "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" cannot be both enabled.

Table 7-52 Function impacts of NR load-based RF module super dormancy

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Clock out-of-synchronization detection	CLK_DETECT_SW option of the gNodeBParam.ClkOutofsyncDetSwitch parameter	<i>Synchronization in 5G RAN Feature Documentation</i>	If clock out-of-synchronization occurs during the dormancy period, this issue may fail to be detected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Air-interface-based network-wide synchronization deviation detection	GAP_BASED_SYNC_DIFF_DET_SW option of the gNodeBParam.SyncDiffDetectSwitch parameter	<i>Synchronization in 5G RAN Feature Documentation</i>	If sequence detection is performed in all cells of a to-be-detected base station during the period when RF module super dormancy takes effect, the sequence detection fails. In this case, the possible clock fault of the base station may fail to be detected.
NR FDD Low-frequency NR TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module super dormancy takes precedence over energy saving based on flexible frequency-domain scheduling when both functions are enabled. Energy saving based on flexible frequency-domain scheduling does not take effect when RF module super dormancy is in effect.
NR FDD Low-frequency NR TDD	TTI-level channel shutdown	TTI_RF_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module super dormancy takes precedence over TTI-level channel shutdown when both functions are enabled. TTI-level channel shutdown does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Intelligent power control energy saving	SMART_POWER_CTRL_ENERGY_SAVE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module super dormancy takes precedence over intelligent power control energy saving when both functions are enabled. Intelligent power control energy saving does not take effect when RF module super dormancy is in effect.
Low-frequency NR TDD	Low-power RF channel dynamic shutdown	LOW_POWER_RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module super dormancy takes precedence over low-power RF channel dynamic shutdown when both functions are enabled. Low-power RF channel dynamic shutdown does not take effect when RF module super dormancy is in effect.
NR FDD Low-frequency NR TDD	Intra-RAT ANR	NR_NR_ANR_DEL_SW and NR_NR_ANR_SW options of the NRCCellAlgoSwitch.AnrSwitch parameter	<i>ANR in 5G RAN Feature Documentation</i>	When automatic removal of neighboring cells in ANR takes effect and the neighbor relationships with the basic-layer cells are automatically removed for capacity-layer cells, load-based RF module super dormancy does not take effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Basic VoNR functions	VONR_SW option of the NRCellAlgoSwitch.VonrSwitch parameter	<i>VoNR in 5G RAN Feature Documentation</i>	Load-based RF module super dormancy does not take effect if some VoNR UEs fail to be handed over from a capacity-layer cell during UE transfer.
NR FDD Low-frequency NR TDD	Cell-level MRO	MRO_SCENARIO_IDENT_SW option of the gNBMrScenarioIdentifySwitch parameter and the INTRA_FREQ_MRO_SW option of the NRCellMr.MroSwitch parameter	<i>MRO in 5G RAN Feature Documentation</i>	When both RF module super dormancy and intra-frequency MRO are enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, MRO may take effect later than expected.
NR FDD Low-frequency NR TDD	UE-level MRO	UE_MRO_SWITCH option of the NRCellMr.MroSwitch parameter	<i>MRO in 5G RAN Feature Documentation</i>	When RF module super dormancy and UE-level MRO are both enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, the penalty mechanism accuracy of the UE-level MRO is affected.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	CMAS	gNB SibConfig.SibType set to SIB_TYPE_8	<i>CMAS in 5G RAN Feature Documentation</i>	When an RF module enters the super dormancy state, the NR cells it serves also enter the dormancy state. In this case, CMAS warning notifications cannot be properly sent in these cells. However, UEs can still receive CMAS warning notifications from other cells.
NR FDD Low-frequency NR TDD High-frequency NR TDD	Symbol power saving	NRDUCellAI goSwitch.SymbolShutdownSwitch	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module super dormancy takes precedence over symbol power saving when both functions are enabled. Symbol power saving does not take effect when RF module super dormancy is in effect.
NR FDD Low-frequency NR TDD High-frequency NR TDD	Power saving scheduling	POWER_SAVING_SCHED ULE_SW option of the NRDUCellAI goSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module super dormancy takes precedence over power saving scheduling when both functions are enabled. Power saving scheduling does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD High-frequency NR TDD	RF channel intelligent shutdown	RF_SHUTDOWNSW option of the NRDUCellAlgorithmSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module super dormancy takes precedence over RF channel intelligent shutdown when both functions are enabled. RF channel intelligent shutdown does not take effect when RF module super dormancy is in effect.
NR FDD Low-frequency NR TDD	Intelligent carrier shutdown in precise mode	INTRAGNB_MULTICARRIER_SHUTDOWN or INTERGNB_MULTICARRIER_SHUTDOWN option of the NRDUCellAlgorithmSwitch.PowerSavingSwitch parameter MULTICARRIER_PRECISE_SHUTDOWN option of the NRDUCellAlgorithmSwitch.PowerSavingSwitch parameter selected	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module super dormancy takes precedence over intelligent carrier shutdown when both functions are enabled. Intelligent carrier shutdown does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD High-frequency NR TDD	Intelligent carrier shutdown in non-precise mode	• INTRA_GNB_MULT I_CARR_S D_SW or INTER_GNB_MULT I_CARR_S D_SW option of the NRDUCell AlgoSwitch.Power SavingSwitch parameter • MULTI_CARR_PRECISE_SHD N_SW option of the NRDUCell AlgoSwitch.Power SavingSwitch parameter deselected	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module super dormancy takes precedence over intelligent carrier shutdown when both functions are enabled. Intelligent carrier shutdown does not take effect when RF module super dormancy is in effect.
NR FDD Low-frequency NR TDD	Uplink symbol power saving	UL_SYMBOL_SHUTDOWN_SW option of the NRDUCell AlgoSwitch.Power SavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	When RF module super dormancy takes effect, uplink symbol power saving produces no energy saving gains.
NR FDD Low-frequency NR TDD	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCell AlgoSwitch.DynamicPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation in 5G RAN Feature Documentation</i>	A carrier exits a power sharing group once it enters the super dormancy state.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Ultra-long holdover for sites with abnormal clock sources	CLK_OTASYNC_PROC_SW option of the TASM.MSCLKDIAGPROC ESSSW parameter	<i>Synchronization in 5G RAN Feature Documentation</i>	Assume that ultra-long hold switch is turned on but ultra-long hold enhancement switch is turned off. If sequence detection is performed in all cells of a to-be-detected base station during the period when RF module super dormancy takes effect, the sequence detection fails. In this case, the base station cannot enter the ultra-long holdover state even if its clock is faulty, or it exits the ultra-long holdover state as sequence detection cannot be performed for a long time due to energy saving. If ultra-long hold enhancement switch is turned on, RF module super dormancy does not take effect for the base station in the ultra-long holdover state or its reference neighboring base stations.

RAT	Function Name	Function Switch	Reference	Description
NR FDD	Intelligent PIM interference avoidance	PIM_INTERF_S SMART_AVOI D_SW option of the NRDUCellPimIntrfAvd.PimAlgoSwitch parameter	<i>Interference Avoidance in 5G RAN Feature Documentation</i>	RF module super dormancy is terminated during PIM interference detection.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Fast carrier shutdown	For capacity-layer cells: • Intra-base-station multi-carrier scenarios: INTRA_GNB_MULT I_CARR_S D_SW option of the NRDUCell AlgoSwitch.Power SavingSwitch parameter selected; inter-base-station multi-carrier scenarios: INTER_GNB_MULT I_CARR_S D_SW option of the NRDUCell AlgoSwitch.Power SavingSwitch parameter selected • FAST_CARRIER_SHUTDOWN_SW option of the NRDUCell AlgoSwit	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Fast carrier shutdown does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
		ch.PowerSavingExtSwitch parameter For basic-layer cells: FAST_CARR_SHUT_SERV_ASSUR_SW option of the NRDUCellAIgoSwitch.PowerSavingExtSwitch parameter		
NR FDD Low-frequency NR TDD	Deep symbol power saving	DEEP_SYMBOL_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Deep symbol power saving does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD High-frequency NR TDD	Cell load simulation	STRNRDUCELLSIMULATELOAD	None	• If the SIMU_LOAD_DORMANCY_IMP_SW option of the gNB PwrSavingsAlgo.PwrSavingsAlgoSwitch parameter is selected, and cell load simulation takes effect, the RF module serving the local cell is not allowed to enter the dormancy state, but the RF modules serving the cells not enabled with cell load simulation are allowed to enter the dormancy state. • If the SIMU_LOAD_DORMANCY_IMP_SW option of the gNB PwrSavingsAlgo.PwrSavingsAlgoSwitch parameter is deselected, and cell load simulation takes effect, all RF modules of the base station are not allowed to enter the

RAT	Function Name	Function Switch	Reference	Description
				dormancy state.
Low-frequency NR TDD	NR multi-band power adaptive scheduling	NR_MULTI_BAND_PWR_ADAPT_SCH_SW option of the NRDUCellIgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	A carrier exits a power adaptive scheduling group once it enters the super dormancy state.
NR FDD Low-frequency NR TDD	Dynamic RF channel shutdown	RF_CHN_DYN_SHUTDOWN_SW option of the NRDUCellIgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Dynamic RF channel shutdown does not take effect when RF module super dormancy is in effect.
NR FDD	LTE-NR multi-dimensional scheduling optimization	LNR_MULTI_DIM_OPT_SCH_SW option of the NRDUCellGigaBand.GigaBandCellAlgorithmSw parameter	<i>LTE FDD and NR Inter-RAT Unified Scheduling</i>	LTE-NR multi-dimensional scheduling optimization does not take effect for cells in the dormancy state.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Speed turbo UE identification	gNodeBParam.StUelIdentSw	None	When any of the following configuration operations is performed and the configured function takes effect on the base station, the served cells already in the dormancy state remain in this state but other served cells will not enter this state, resulting in a decrease in the energy saving gains: - Any option of the gNodeBParam.StUelIdentSw parameter is selected. - The gNBAsAppParam.AppDomainName parameter is set. - The gNBAsAppParam.Applv4 parameter is set. - The gNBAsAppParam.Applv6 parameter is set. The following methods can be used to address the preceding issue: - Select the SPEED_TURBO
NR FDD Low-frequency NR TDD	gNodeB acceleration service application	gNBAsAppParam.AppDomainName gNBAsAppParam.Applv4 gNBAsAppParam.Applv6	None	

RAT	Function Name	Function Switch	Reference	Description
				_PWR_SAVING_SW option of the gNB PwrSavin gAlgo.PwrSavin gAlgoSwitch parameter. Dormancy can take effect for cells served by the base station in some of the preceding scenarios within the period specified by the gNB PwrSavin gAlgo.SpeedTurboPwrSavStartTime and gNB PwrSavin gAlgo.SpeedTurboPwrSavEndTime parameters, increasing the energy saving gains. • Select the STATIC_STATIC_PWR_SAVING_SW option of the gNB PwrSavin gAlgo.PwrSavin gAlgoSwitch parameter. Dormancy can take effect for cells served by the base station in some of the preceding scenarios within the period

RAT	Function Name	Function Switch	Reference	Description
				specified by the gNBPwrSavin gAlgo.StaticST PwrSavStartTime and gNBPwrSavin gAlgo.StaticST PwrSavEndTime parameters, increasing the energy saving gains.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD High-frequency NR TDD	Intelligent Detection for Fronthaul Optical Paths	None	<i>Intelligent Detection for Fronthaul Optical Paths</i>	• If the BBU and RF module are not in the dormancy state when optical path detection is started, the detection process will delay the time at which they enter that state. • If the BBU and RF module are in the dormancy state when optical path detection is started, the following logic applies: – In a scenario where wakeup from dormancy is supported, starting optical path detection will cause the BBU and RF module to automatically exit the dormancy state. They will not re-enter the state within 30 minutes after the detection

RAT	Function Name	Function Switch	Reference	Description
				process starts. Once 30 minutes elapse, they will evaluate whether to re-enter the dormancy state. - In a scenario where wakeup from dormancy is not supported, optical path detection cannot be started.
NR FDD Low-frequency NR TDD	Initial transmission power reduction	INIT_TRANS_POWER_REDUCE_SW option of the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Initial transmission power reduction does not take effect when RF module super dormancy is in effect.
NR FDD Low-frequency NR TDD	Retransmission power reduction	RETRANS_POWER_REDUCE_SW option of the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Retransmission power reduction does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD	Uplink TTI-level RF channel dynamic muting	UL_TTI_CHN_DYN_MUTING_SW option of the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Uplink TTI-level RF channel dynamic muting does not take effect when RF module super dormancy is in effect.
NR FDD Low-frequency NR TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag set to HIGH_SPEED	<i>High Speed Mobility in 5G RAN Feature Documentation</i>	When the high-speed cell function in high speed specified policy management is enabled, you are advised not to enable NR or LNR load-based RF module super dormancy.
Low-frequency NR TDD	Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios	DUAL_PLANE_TOB_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	<i>Base Station Equipment Reliability</i>	When RF module super dormancy is enabled, RF modules enter the super dormancy state, which causes an active/standby BBU switchover failure. Therefore, RF module super dormancy and "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" cannot be both enabled.

In addition to the function impacts of NR load-based RF module super dormancy (except the impact relationship with intra-RAT ANR), LNR load-based RF module super dormancy has the following function impacts.

Table 7-53 Function impacts of LNR load-based RF module super dormancy

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Work mode	Cell.<i>WorkMode</i>	None	• An LTE FDD cell with the Cell.<i>WorkMode</i> parameter set to DL_ONLY or LAA cannot be configured as an overlapping neighboring cell of an NR cell. • An LTE TDD cell with the Cell.<i>WorkMode</i> parameter set to LAA cannot be configured as an overlapping neighboring cell of an NR cell.
LTE FDD LTE TDD	Cell System Information Switch	CellSiMap.<i>SiSwitch</i>	None	A cell whose ForbidCellSiSwitch option of the CellSiMap.<i>SiSwitch</i> parameter is selected cannot be configured as a basic-layer cell.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	eMTC introduction	EMTC_SWITCH option of the CellEmtcAlgo.EmtcAlgoSwitch parameter	<i>eMTC in eRAN Feature Documentation</i>	If a capacity-layer cell serves eMTC UEs, the eMTC UEs may fail to be handed over, causing a failure for the capacity-layer cell to enter dormancy. An eMTC-only cell (specified by the EMTC_SWITCH and EMTC_ONLY_CELL_SWITCH options of the CellEmtcAlgo.EmtcAlgoSwitch parameter) cannot be configured as a basic-layer cell.
LTE FDD	Static Multiple Beam	SectorSplitGroup.BeamTemplateId set to a value greater than 1	<i>Massive MIMO Basics (FDD) in eRAN Feature Documentation</i>	For LTE FDD, if a Static Multiple Beam cell is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SWITCH option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in NR cells, the user-perceived rate may decrease in the split cell that serves as a basic-layer cell.

RAT	Function Name	Function Switch	Reference	Description
LTE TDD	Intelligent Multi-Beam of 8T8R	None	<i>Intelligent Multi-Beam of 8T8R (TDD) in eRAN Feature Documentation</i>	For LTE TDD, if a cell enabled with Intelligent Multi-Beam of 8T8R is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in NR cells, the user-perceived rate may decrease in the split cell that serves as a basic-layer cell.
LTE FDD	Static Multiple Beam	None	<i>Smart 8T8R (FDD) in eRAN Feature Documentation</i>	For LTE FDD, if a Static Multiple Beam cell is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in NR cells, the user-perceived rate may decrease in the split cell that serves as a basic-layer cell.

RAT	Function Name	Function Switch	Reference	Description
LTE TDD	Intra-frequency split	None	<i>Soft Split Resource Duplex (TDD) in eRAN Feature Documentation</i>	For LTE TDD, if an intra-frequency split cell is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in NR cells, the user-perceived rate may decrease in the split cell that serves as a basic-layer cell.
LTE FDD	RuralLink antenna solution (LTE)	None	<i>RuralLink Solution</i>	For LTE FDD, if a split cell established through the RuralLink antenna solution (LTE) is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in NR cells, the user-perceived rate may decrease in the split cell that serves as a basic-layer cell.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	LumiLink solution (LTE)	• SectorSplitGroup.BeamTemplateId set to 0 • SectorSplitGroup.BeamDeploymentMode set to MODE_5 • SectorSplitCell.CellBeamIndex set to 0, 1, or 2	<i>LumiLink Solution</i>	For LTE FDD, if a split cell established through the LumiLink solution (LTE) is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in NR cells, the user-perceived rate may decrease in the split cell that serves as a basic-layer cell.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	RAN sharing	RAN sharing with common carrier: ENodeBSharingMode.ENodeBSharingMode set to SHARED_FREQ RAN sharing with dedicated carrier: ENodeBSharingMode.ENodeBSharingMode set to SEPARATED_FREQ Hybrid RAN sharing: ENodeBSharingMode.ENodeBSharingMode set to HYBRID_SHARED	<i>RAN Sharing in eRAN Feature Documentation</i>	In RAN sharing scenarios, the PLMN IDs of a capacity-layer cell must be a subset of the PLMN IDs of its co-coverage basic-layer cells. Otherwise, the capacity-layer cell cannot enter the dormancy state.
LTE FDD LTE TDD	Dynamic voltage adjustment	CellAlgoSwitch.DynAdjVltSwitch	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over dynamic voltage adjustment when both functions are enabled. Dynamic voltage adjustment does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	NSA anchor carrier shutdown	NSA_ANCHOR_CARRIER_SHUTDOWN_SW option of the CellShutdownEnhancedSw parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	In NSA or NSA and SA hybrid networking, it is recommended that the INTER_RAT_GRAD_PWR_SAVING_SW option of the gNBpwrSavingAlgorithm.PwrSavingAlgorithmSwitch parameter be selected when NSA anchor carrier shutdown is enabled for the LTE basic-layer cell and NSA anchor cells and LNR load-based RF module super dormancy is enabled for NR capacity-layer cells and NSA anchor cells. • If LNR load-based RF module super dormancy takes effect first, the fact that NSA anchor carrier shutdown takes effect in NSA anchor cells does not prevent LNR load-based RF module super dormancy from taking effect in NR capacity-layer cells. • If NSA anchor carrier shutdown

RAT	Function Name	Function Switch	Reference	Description
				takes effect first, carriers are shut down in NSA anchor cells. If NR capacity-layer cells do not have other neighboring LTE cells, LNR load-based RF module super dormancy does not take effect.
LTE FDD	LTE-NR multi-dimensional scheduling optimization	LNR_MULTI_D_OPT_SCH_SW option of the GigaBandCell . GigaBandCellAlgoSw parameter	<i>LTE FDD and NR Inter-RAT Unified Scheduling</i>	LTE-NR multi-dimensional scheduling optimization does not take effect for cells in the dormancy state.
LTE FDD	8T beam mode shift for energy saving	BEAM_MODE_SHIFT_SW option of the SectorSplitGroup . SplitCellGroupOptSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over 8T beam mode shift for energy saving when both functions are enabled.
LTE FDD	32T beam mode shift for energy saving	BEAM_MODE_SHIFT_SW option of the SectorSplitGroup . SplitCellGroupOptSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over 32T beam mode shift for energy saving when both functions are enabled.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	Intelligent Detection for Fronthaul Optical Paths	None	<i>Intelligent Detection for Fronthaul Optical Paths</i>	• If the BBU and RF module are not in the dormancy state when optical path detection is started, the detection process will delay the time at which they enter that state. • If the BBU and RF module are in the dormancy state when optical path detection is started, the following logic applies: – In a scenario where wakeup from dormancy is supported, starting optical path detection will cause the BBU and RF module to automatically exit the dormancy state. They will not re-enter the state within 30 minutes after the detection process starts. Once

RAT	Function Name	Function Switch	Reference	Description
				30 minutes elapse, they will evaluate whether to re-enter the dormancy state. - In a scenario where wakeup from dormancy is not supported, optical path detection cannot be started.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Ultra-long holdover for sites with abnormal clock sources	CLK_OTASY NC_PROC_SW option of the TASM.MSCLKDIAGPROC ESSSW parameter	<i>Synchronization in 5G RAN Feature Documentation</i>	Assume that ultra-long hold switch is turned on but ultra-long hold enhancement switch is turned off. If sequence detection is performed in all cells of a to-be-detected base station during the period when RF module super dormancy takes effect, the sequence detection fails. In this case, the base station cannot enter the ultra-long holdover state even if its clock is faulty, or it exits the ultra-long holdover state as sequence detection cannot be performed for a long time due to energy saving. If ultra-long hold enhancement switch is turned on, RF module super dormancy does not take effect for the base station in the ultra-long holdover state or its reference neighboring base stations.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Static Multiple Beam 2.0	SMB2_SW option of the SectorSplitGroup.SplitCellGroupOptSwitch parameter	<i>Smart 8T8R (FDD) in eRAN Feature Documentation</i>	RF module super dormancy preferentially takes effect when it is enabled together with Static Multiple Beam 2.0.
LTE FDD	LNR CRS muting	LNR_CRSMUTE_SW option of the CellAlgoExtSwitch.LnrZeroBufferzoneSw parameter	<i>LTE and NR Zero Bufferzone</i>	When LNR CRS muting and RF module super dormancy are both enabled, RF module super dormancy preferentially takes effect. LNR CRS muting does not take effect when RF module super dormancy is in effect.
LTE FDD	Uplink TTI-level RF channel dynamic muting	UL_TTI_CHNDYN_MUTING_SW option of the CellRfChnDynamicMuting.RfChannelDynMutingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	When uplink TTI-level RF channel dynamic muting and RF module super dormancy are both enabled, RF module super dormancy preferentially takes effect.
NR FDD	Uplink TTI-level RF channel dynamic muting	UL_TTI_CHNDYN_MUTING_SW option of the NRDUCellPowerSaveAlgo.PowerSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Uplink TTI-level RF channel dynamic muting does not take effect when RF module super dormancy is in effect.

Table 7-54 Function impacts of LTE load-based RF module super dormancy

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Carrier aggregation	None	<i>Carrier Aggregation in eRAN Feature Documentation</i>	- Cells in the RF module super dormancy state do not participate in CA. In CA-group-based configuration mode, after such a cell in a CA group exits this energy saving state, this cell cannot be configured as an SCell for an existing CA UE until the UE accesses the network again. - After a cell exits the RF module dormancy state, it does not participate in CA if the number of routes is insufficient. The route for CA will be released for a cell in the RF module dormancy state. After this cell exits this state, it may not obtain a route if the number of routes is insufficient and therefore, cannot

RAT	Function Name	Function Switch	Reference	Description
				participate in CA.
LTE FDD LTE TDD	Automatic baseband resource configuration	eNodeBResModeAlgo.BbpResAutoConfigSw	None	When the eNodeBResModeAlgo.BbpResAutoConfigSw parameter is set to ON , RF module super dormancy does not take effect.
LTE TDD	Clock synchronization detection	ClkDetect.ClkAsyncDetectSwitch	<i>Synchronization in eRAN Feature Documentation</i>	If clock out-of-synchronization occurs during the dormancy period, this issue may fail to be detected.
LTE FDD LTE TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over energy saving based on flexible frequency-domain scheduling when both functions are enabled. Energy saving based on flexible frequency-domain scheduling does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Work mode	Cell.WorkMode	None	- An LTE FDD cell with the Cell.WorkMode parameter set to DL_ONLY or LAA cannot be configured as a basic-layer cell. - An LTE TDD cell with the Cell.WorkMode parameter set to LAA cannot be configured as a basic-layer cell.
LTE FDD LTE TDD	Cell System Information Switch	CellSiMap.SiSwitch	None	A cell whose ForbidCellSiSwitch option of the CellSiMap.SiSwitch parameter is selected cannot be configured as a basic-layer cell.
LTE FDD LTE TDD	eMTC introduction	EMTC_SWITCH option of the CellEmtcAlgorithm.EmtcAlgorithmSwitch parameter	<i>eMTC in eRAN Feature Documentation</i>	If a capacity-layer cell serves eMTC UEs, the eMTC UEs may fail to be handed over, causing a failure for the capacity-layer cell to enter dormancy. An eMTC-only cell (specified by the EMTC_SWITCH and EMTC_ONLY_CELL_SWITCH options of the CellEmtcAlgorithm.EmtcAlgorithmSwitch parameter) cannot be configured as a basic-layer cell.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Static Multiple Beam	SectorSplitGroup.BeamTemplateId set to a value greater than 1	<i>Massive MIMO Basics (FDD) in eRAN Feature Documentation</i>	For LTE FDD, if a Static Multiple Beam cell is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in LTE cells, the user-perceived rate may decrease in the split cell that serves as a basic-layer cell.
LTE TDD	Intelligent Multi-Beam of 8T8R	None	<i>Intelligent Multi-Beam of 8T8R (TDD) in eRAN Feature Documentation</i>	For LTE TDD, if a cell enabled with Intelligent Multi-Beam of 8T8R is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in LTE cells, the user-perceived rate may decrease in the split cell that serves as a basic-layer cell.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Static Multiple Beam	None	<i>Smart 8T8R (FDD) in eRAN Feature Documentation</i>	For LTE FDD, if a Static Multiple Beam cell is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in LTE cells, the user-perceived rate may decrease in the split cell that serves as a basic-layer cell.
LTE FDD	Downlink intelligent joint power control	DL_INTEL_JOINT_PC_SW option of the CellDISchExtParam.DISchExtSwitch parameter	<i>Smart 8T8R (FDD) in eRAN Feature Documentation</i>	Downlink intelligent joint power control does not take effect when RF module super dormancy is in effect.
LTE FDD	Downlink intelligent joint AMC and power control	DL_INTEL_JOINT_AMC_AND_PC_SW option of the CellDISchExtParam.DISchExtSwitch parameter	<i>Smart 8T8R (FDD) in eRAN Feature Documentation</i>	Downlink intelligent joint AMC and power control does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE TDD	Intra-frequency split	None	<i>Soft Split Resource Duplex (TDD) in eRAN Feature Documentation</i>	For LTE TDD, if an intra-frequency split cell is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in LTE cells, the user-perceived rate may decrease in the split cell that serves as a basic-layer cell.
LTE FDD	RuralLink antenna solution (LTE)	None	<i>RuralLink Solution</i>	For LTE FDD, if a split cell established through the RuralLink antenna solution (LTE) is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in LTE cells, the user-perceived rate may decrease in the split cell that serves as a basic-layer cell.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	LumiLink solution (LTE)	• SectorSplitGroup.BeamTemplateId set to 0 • SectorSplitGroup.BeamDeploymentMode set to MODE_5 • SectorSplitCell.CellBeamIndex set to 0, 1, or 2	<i>LumiLink Solution</i>	For LTE FDD, if a split cell established through the LumiLink solution (LTE) is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in LTE cells, the user-perceived rate may decrease in the split cell that serves as a basic-layer cell.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Basic functions of mobility management in connected mode	None	<i>Mobility Management in Connected Mode in eRAN Feature Documentation</i>	LTE load-based RF module super dormancy triggers inter-frequency handover. An inter-frequency blind handover to a basic-layer cell fails if the cell is in the dormancy state due to energy saving. Therefore, measurement-based inter-frequency handovers are recommended for cells where LTE load-based RF module super dormancy is enabled.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	RAN sharing	RAN sharing with common carrier: ENodeBSharingMode.ENodeBSharingMode set to SHARED_FREQ RAN sharing with dedicated carrier: ENodeBSharingMode.ENodeBSharingMode set to SEPARATED_FREQ Hybrid RAN sharing: ENodeBSharingMode.ENodeBSharingMode set to HYBRID_SHARED	<i>RAN Sharing in eRAN Feature Documentation</i>	In RAN sharing scenarios, the PLMN IDs of a capacity-layer cell must be a subset of the PLMN IDs of its co-coverage basic-layer cells. Otherwise, the capacity-layer cell cannot enter the dormancy state.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	CMAS	CellAlgoSwitch.CmasSwitch	<i>CMAS in eRAN Feature Documentation</i>	The following applies when cells enter RF module super dormancy: • With PWS message-triggered exit of cells from energy saving disabled, CMAS warning notifications cannot be properly sent in LTE cells in the dormancy state. However, UEs can still receive CMAS warning notifications from other cells. • With PWS message-triggered exit of cells from energy saving enabled, cells exit the dormancy state upon receipt of a PWS message.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	ETWS	None	<i>ETWS in eRAN Feature Documentation</i>	The following applies when cells enter RF module super dormancy: • With PWS message-triggered exit of cells from energy saving disabled, ETWS warning notifications cannot be sent in LTE cells in the dormancy state. However, UEs can still receive ETWS warning notifications from other cells. • With PWS message-triggered exit of cells from energy saving enabled, cells exit the dormancy state upon receipt of a PWS message.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Intelligent power control energy saving	- Single-cell intelligent power control energy saving: INTEL_INTRA_CELL_PC_SW option of the CellPwrSavingAlgo or PwrSavingAlgoSwitch - Multi-cell coordinated intelligent power control energy saving: INTEL_INTRA_CELL_PC_SW and INTEL_MULTICELL_COOR_PC_SW options of the CellPwrSavingAlgo or PwrSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over intelligent power control energy saving when both functions are enabled. Intelligent power control energy saving does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Dynamic carrier shutdown phase 2	CellPwrSavingsAlgo.CarrierDynShutdownPh2Switch set to CAPACITY_CELL	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over dynamic carrier shutdown phase 2 when both functions are enabled. Dynamic carrier shutdown phase 2 does not take effect when RF module super dormancy is in effect.
LTE FDD LTE TDD	RF channel intelligent shutdown	CellRfShutdown.RfShutdownSwitch	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over RF channel intelligent shutdown when both functions are enabled. RF channel intelligent shutdown does not take effect when RF module super dormancy is in effect.
LTE FDD LTE TDD	Intelligent power-off of carriers in the same coverage	CellShutdown.CellShutdownSwitch set to ON	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over intelligent power-off of carriers in the same coverage when both functions are enabled. Intelligent power-off of carriers in the same coverage does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Multi-carrier coordinated energy saving	CellShutdown.CellShutdownSwitch set to ON_MULTI_CARRIER_HIER_SHUTDOWN , ON_MULTICARRIER_HIERMANUAL_SHUTDOWN , or ON_CARRIER_PRECISE_SHUTDOWN	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over multi-carrier coordinated energy saving when both functions are enabled. Multi-carrier coordinated energy saving does not take effect when RF module super dormancy is in effect.
LTE FDD LTE TDD	Dynamic carrier shutdown	CARRIER_DYNAMIC_SHUTDOWN_SWITCH option of the CellShutdown.CarrierShutdownEnhancedSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over dynamic carrier shutdown when both functions are enabled. Dynamic carrier shutdown does not take effect when RF module super dormancy is in effect.
LTE FDD LTE TDD	Dynamic voltage adjustment	CellAlgorithm.DynAdjustVoltageSwitch	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over dynamic voltage adjustment when both functions are enabled. Dynamic voltage adjustment does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Low power consumption mode (LTE)	CellLowPower.LowPwrSwitch	<i>Green Site</i>	RF module super dormancy takes precedence over low power consumption mode when both functions are enabled. Low power consumption mode does not take effect when RF module super dormancy is in effect.
LTE FDD LTE TDD	Cell Switch-off Based on Automatic Co-coverage Identification	CellShutdown.CellShutdownSwitch set to ON_CO_COV_AUTO_IDENT	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over Cell Switch-off Based on Automatic Co-coverage Identification when both functions are enabled. Cell Switch-off Based on Automatic Co-coverage Identification does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Basic symbol power saving	FDD: SymbolShutdownSwitch.PowerSaveSwitch parameter TDD: TddSymbolsShutdownSwitch option of the ENodeBAlgoSwitch.PowerSaveSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over basic symbol power saving when both functions are enabled. Basic symbol power saving does not take effect when RF module super dormancy is in effect.
LTE FDD LTE TDD	CRS dynamic muting	CRS_DYNAMIC_MUTING_SW option of the SymbolPowerSaving.SymbolShutdownEnhancedSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over CRS dynamic muting when both functions are enabled. CRS dynamic muting does not take effect when RF module super dormancy is in effect.
LTE FDD LTE TDD	Automatic cell shutdown	CellAutoShutdown.CellAutoShutdownSwitch	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over automatic cell shutdown when both functions are enabled. Automatic cell shutdown does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Dynamic RF channel shutdown	RF_CHN_DYN_MUTING_SW option of the CellRfChnDynaMuting.RfChannelDynMutatingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over RF channel dynamic muting when both functions are enabled. RF channel dynamic muting does not take effect when RF module super dormancy is in effect.
LTE FDD	Intelligent PIM interference avoidance	PIM_INTERF_SMART_AVOID_SW option of the PimInterfSmartAvoid.PimInterfSmartAvoidAlgoSwitch parameter	<i>Interference Detection and Suppression in eRAN Feature Documentation</i>	RF module super dormancy is terminated during PIM interference detection.
LTE FDD LTE TDD	Load simulation	CellSimuLoad MO configured	None	LTE load-based RF module super dormancy does not take effect when at least one CellSimuLoad MO is configured.
LTE FDD LTE TDD	Deep symbol power saving	DEEP_SYMBOL_SHUTDOWN_SW option of the SymbolPwrSaving.SymbolPwrSavingEnhancedSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	Deep symbol power saving does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	High speed specified policy management	HSCellSleepInterfAvoidSwitch option of the HighSpdAdaptionPara.InterfAvoidMode parameter	<i>High Speed Mobility in eRAN Feature Documentation</i>	LTE load-based RF module super dormancy is not recommended for cells where high speed specified policy management is enabled. This is to avoid UE performance loss caused by slow wakeup.
LTE FDD LTE TDD	SOC level-based low power consumption mode	SOCLVL_BASED_LOW_POWER_SW option of the CellLowPower.LowPwrConsumeModeExtSw parameter	<i>Green Site</i>	RF module super dormancy takes precedence over SOC level-based low power consumption mode and SOC level-based low power consumption mode does not take effect when both functions are enabled.
LTE FDD	LTE-NR multi-dimensional scheduling optimization	LNR_MULTI_DIM_OPT_SCHEDULE_SW option of the GigaBandCell.GigaBandCellAlgoSw parameter	<i>LTE FDD and NR Inter-RAT Unified Scheduling</i>	LTE-NR multi-dimensional scheduling optimization does not take effect for cells in the shutdown state.
LTE FDD	8T beam mode shift for energy saving	BEAM_MODE_SHIFT_SW option of the SectorSplitGroup.SplitCellGroupOptSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over 8T beam mode shift for energy saving when both functions are enabled.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	32T beam mode shift for energy saving	BEAM_MODE_SHIFT_SW option of the SectorSplitGroupSplitCellGroupOptions parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over 32T beam mode shift for energy saving when both functions are enabled.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Intelligent Detection for Fronthaul Optical Paths	None	<i>Intelligent Detection for Fronthaul Optical Paths</i>	• If the BBU and RF module are not in the dormancy state when optical path detection is started, the detection process will delay the time at which they enter that state. • If the BBU and RF module are in the dormancy state when optical path detection is started, the following logic applies: – In a scenario where wakeup from dormancy is supported, starting optical path detection will cause the BBU and RF module to automatically exit the dormancy state. They will not re-enter the state within 30 minutes after the detection process starts. Once

RAT	Function Name	Function Switch	Reference	Description
				30 minutes elapse, they will evaluate whether to re-enter the dormancy state. - In a scenario where wakeup from dormancy is not supported, optical path detection cannot be started.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Specified service carrier	Cell.SpecificCellFlag	<i>WBB in eRAN Feature Documentation</i>	WBB and non-emergency-call MBB UEs can only be handed over to the corresponding specified-service cells. LTE load-based RF module super dormancy enables UEs to be transferred out of a capacity-layer cell when the trigger conditions are met. If the number of UEs in the cell after the transfer is greater than the value of the CellPwrSavingPolicy.ForcibleTriggerUeNumThld parameter, LTE load-based RF module super dormancy cannot take effect; if the number of UEs in the cell after the transfer is less than or equal to the value of this parameter, LTE load-based RF module super dormancy can take effect and the remaining UEs in the cell are directly released.
LTE FDD	Static Multiple Beam 2.0	SMB2_SW option of the SectorSplitGroup.SplitCellGroupOptSwitch parameter	<i>Smart 8T8R (FDD) in eRAN Feature Documentation</i>	RF module super dormancy preferentially takes effect when it is enabled together with Static Multiple Beam 2.0.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	LNR CRS muting	LNR_CRS_MUTE_SW option of the CellAlgoExtSwitch.LnrZeroBufferzoneSw parameter	<i>LTE and NR Zero Bufferzone</i>	When LNR CRS muting and RF module super dormancy are both enabled, RF module super dormancy preferentially takes effect. LNR CRS muting does not take effect when RF module super dormancy is in effect.
LTE FDD	Initial transmission power reduction	INIT_TRANSDUCE_SW option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	When initial transmission power reduction and RF module super dormancy are both enabled, RF module super dormancy preferentially takes effect.
LTE FDD LTE TDD	Retransmission power reduction	RETRANS_POWER_REDUCE_SW option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	When retransmission power reduction and RF module super dormancy are both enabled, RF module super dormancy preferentially takes effect.
LTE FDD	Uplink TTI-level RF channel dynamic muting	UL_TTI_CHANNEL_DYNAMIC_MUTING_SW option of the CellRfChnDynamicMuting.RfChannelDynMutingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	When uplink TTI-level RF channel dynamic muting and RF module super dormancy are both enabled, RF module super dormancy preferentially takes effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Port sparseness	PORT_SPARSENESS_SW option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	RF module super dormancy takes precedence over port sparseness when both functions are enabled.
LTE FDD	GSM and LTE Buffer Zone Optimization	CellAlgoSwitch.AvoidInterfSwitch	<i>GSM and LTE Buffer Zone Optimization</i>	Cells in the RF module super dormancy state do not participate in GSM and LTE Buffer Zone Optimization.
LTE FDD	GSM and LTE Spectrum Concurrency	SpectrumCloud.SpectrumCloudSwitch set to GL_SPECTRUM_CONCURRENCY	<i>GSM and LTE Spectrum Concurrency</i>	Cells in the RF module super dormancy state do not participate in GSM and LTE Spectrum Concurrency.
LTE FDD	GSM and LTE Spectrum Concurrency Phase 2	GL_SPCT_CONCURRENCY_PH2_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter	<i>GSM and LTE Spectrum Concurrency</i>	Cells in the RF module super dormancy state do not participate in GSM and LTE Spectrum Concurrency Phase 2.
LTE FDD	UMTS and LTE Spectrum Sharing Based on DC-HSDPA (LTE FDD)	SpectrumCloud.SpectrumCloudSwitch set to DC_HSDPA_BASED_UL_SPECTRUM_SHR	<i>UMTS and LTE Spectrum Sharing Based on DC-HSDPA</i>	RF module super dormancy cannot take effect in cells where UMTS and LTE Spectrum Sharing Based on DC-HSDPA is enabled.
LTE FDD	UMTS and LTE Spectrum Sharing (LTE FDD)	SpectrumCloud.SpectrumCloudSwitch set to UL_SPECTRUM_SHARING	<i>UMTS and LTE Spectrum Sharing</i>	UMTS and LTE Spectrum Sharing does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	UMTS and LTE Spectrum Sharing Phase 2 (LTE FDD)	UL_SPECTRUM_SHR_PH2_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter	<i>UMTS and LTE Spectrum Sharing</i>	UMTS and LTE Spectrum Sharing Phase 2 does not take effect when RF module super dormancy is in effect.
LTE FDD	UMTS and LTE Zero Bufferzone (LTE FDD)	UMTS_LTE_ZERO_BUFFER_ZONE_SW option of the ULZeroBufferZone.ZeroBufferZoneSwitch parameter	<i>UMTS and LTE Zero Bufferzone</i>	UMTS and LTE Zero Bufferzone does not take effect when RF module super dormancy is in effect.

7.4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

Boards

Only the UMPT series main control boards support this function, except for the UMPTb board.

Only UBBP series boards support this function, except for the UBBPei series boards and UBBPem.

The UBRU does not support RF module super dormancy.

RF Modules

Table 7-55 lists the RF modules that are compatible with RF module super dormancy.

Table 7-55 RF modules that are compatible with RF module super dormancy

RF Module	BOM Number
AAU5336	02314KUT

RF Module	BOM Number
AAU5336w	02314QVY, 02314RBD
AAU5356	02314RCY, 02314VTK, 02314YCL, 02315HGF, 02315PAQ
AAU5366	All BOM numbers are supported.
AAU5368	All BOM numbers are supported.
AAU5368md	All BOM numbers are supported.
AAU5636m	02314JMD, 02314LKN, 02314RCB, 02314WMM
AAU5656	02314TFM
AAU5668	All BOM numbers are supported.
AAU5668m	All BOM numbers are supported.
AAU5685md	All BOM numbers are supported.
AAU5767t	All BOM numbers are supported.
AAU5768	All BOM numbers are supported.
AAU5785t	All BOM numbers are supported.
AAU5832a	All BOM numbers are supported.
AAU5832f	All BOM numbers are supported.
AAU5836f	All BOM numbers are supported.
RRU5522	All BOM numbers are supported.
RRU5522t	All BOM numbers are supported.
RRU5522w	All BOM numbers are supported.
RRU5525t	All BOM numbers are supported.
RRU5526	02315GWS, 02315HHK
RRU5526et	All BOM numbers are supported.
RRU5526t	All BOM numbers are supported.
RRU5526w	All BOM numbers are supported.
RRU5527et	All BOM numbers are supported.
RRU5527t	All BOM numbers are supported.
RRU5529et	All BOM numbers are supported.
RRU5827	All BOM numbers are supported.
RRU5829	All BOM numbers are supported.
RRU5867	All BOM numbers are supported.

RF Module	BOM Number
RRU5871	All BOM numbers are supported.
RRU5872	All BOM numbers are supported.
RRU5920	All BOM numbers are supported.
RRU5920w	All BOM numbers are supported.

Among high-frequency modules, only the HAAU5233w and HAAU5235w support scheduled RF module super dormancy and NR load-based RF module super dormancy.

Table 7-56 lists the RF modules that are compatible with wake-up algorithm optimization for RF module super dormancy.

Table 7-56 RF modules compatible with wake-up algorithm optimization for RF module super dormancy

RF Module	BOM Number
AAU5685md	02315FWJ

7.4.3.4 Networking

- An RF module shared by multiple RATs can enter dormancy only when the dormancy conditions of all involved RATs are met.
- In multi-operator scenarios, the procedure of entering this function for both primary and secondary operators is determined by the networking of the primary operator.
- Requirements of NR load-based RF module super dormancy:
In NSA networking, if the capacity layer and basic layer are deployed at different sites, information is transmitted over the Xn interface. If no Xn interface is deployed between the gNodeBs enabled with this function, IP addresses need to be allocated for Xn setup. Otherwise, NR load-based RF module super dormancy cannot take effect in inter-base-station scenarios.
- Requirements of LNR load-based RF module super dormancy:
 - X2 links need to be set up in inter-base-station SA networking. X2 links can be set up automatically or manually. Automatic setup is recommended. For details about X2 configurations, see *X2 Self-Management in SA Networking*.
 - In NSA networking, the NSA DC capability must be enabled for all basic- and capacity-layer cells. To do so, select the **NSA_DC_CAPABILITY_SWITCH** option of the **NsaDcMgmtConfig.NsaDcAlgoSwitch** parameter on the eNodeB side and turn on the switch controlled by the **NRCellAlgoSwitch.NsaDcSwitch** parameter on the gNodeB side.

- In NSA and SA hybrid networking, the configuration and activation procedures of this function are the same as those in NSA networking. The load information is transmitted between the eNodeB and gNodeB over the X2 interface.
- All activated UMTS/LTE/NR cells of the base station have been bound to BBPs.

 NOTE

BBP binding is optional for LTE cells in the standby mode (indicated by the **Cell.EuCellStandbyMode** parameter with the value of **STANDBY**).

The following binding methods for LTE/NR are for reference:

- Method 1: automatic binding
 - LTE side: Set the **eNodeBResModeAlgo.BbpResAutoConfigSw** parameter to **ON_BIND** to enable automatic baseband resource configuration.
 - NR side: Set the **gNodeBParam.AutoCellBindBasebandEqmSw** parameter to **ON** to enable automatic cell binding to baseband equipment.
- Method 2: manual binding
 - LTE side: Run the **MOD EUCELLSECTOREQM** command to bind specified BBPs to cells.
 - NR side: Run the **MOD NRDUCELLTRP** command to bind specified BBPs to cells.

Binding method for UMTS: Set the **ULOCCELL.ULBASEBANDEQMID** and **ULOCCELL.DLBASEBANDEQMID** parameters to bind specified BBPs to cells in the uplink and downlink, respectively. Each piece of uplink or downlink baseband equipment can contain only one BBP.

- GU load-based RF module dormancy and GU inter-RAT scheduled RF module dormancy require information exchange between the GSM and UMTS networks. If the base station controllers are a BSC and an RNC, operators must add the physical connections for the lur-g interface between the BSC and the RNC. For details on how to configure and enable the lur-g interface, see the following paths:
 - BSC6910: section "Configuring the lur-g Interface (IP Transmission Mode)" in *BSC6910 GU Initial Configuration Guide* in *BSC6910 GU Product Documentation*.
 - BSC6960: sections "Configuring the lur-g Interface (over IP)" in *BSC6960 GSM Initial Configuration Guide* in *BSC6960 GSM Product Documentation* and "Configuring the lur Interface (in Transmission Resource Pool Mode)" in *BSC6960 UMTS Initial Configuration Guide* in *BSC6960 UMTS Product Documentation*.
- In the RRU cascading or RRU ring networking (with **RRUCHAIN.TT (LTE eNodeB, 5G gNodeB)** set to **RING**) scenario, it is recommended that the **RRUCHAIN.CR (LTE eNodeB, 5G gNodeB)** parameter be set to a value other than **AUTO**.
- If the physical networking needs to be adjusted or optical modules need to be replaced while RF modules are in dormancy, operators need to run the **PND DORMANCY** command to suspend RF module dormancy first, and adjust the

physical networking or replace optical modules only after all RF modules exit dormancy. This prevents the RF module status from becoming abnormal.

RF module supper dormancy is not supported in any of the following scenarios:

- BBU
 - Configured in a CI-free interconnection networking (the entire base station cannot enter dormancy)
 - Configured in a Cloud BB networking (the entire base station cannot enter dormancy)
- RF modules
 - A ring topology is used with more than one RF module cascaded on the ring.
 - A branch chain/ring topology is used. That is, the **RRU.TP (LTE eNodeB, 5G gNodeB)** parameter is set to **BRANCH**.
 - An intra-RAT dual-site ring topology is used. (In this topology, the entire base station cannot enter dormancy.) That is, the **EQUIPMENT.IMDSRTMODE (LTE eNodeB, 5G gNodeB)** parameter is set to a value other than **NONE**.
 - Active/standby RF groups are configured using the **RFHAGRP** MO.
- RF modules (with external devices)
 - Configured with a UBRU module (the entire base station cannot enter dormancy)
 - Configured with a DRS module
 - Has the alarm port switch turned on
 - Configured with remote electromechanical devices, such as the EMU, TCU, FMU, CCU, and MEB
 - Configured with a GNSS

7.4.3.5 Cells

- LNR or NR load-based super dormancy cannot be enabled for the NR SUL cell (with the **NRDUCell.DuplexMode** parameter set to **CELL_SUL**) alone. The SUL cell enters the dormancy state along with the NR TDD cell.
- NR SUL cells do not support optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module super dormancy.
- SDL cells (with **NRDUCell.DuplexMode** set to **CELL_SDL**) do not support LNR load-based RF module dormancy.
- SDL cells cannot serve as basic-layer cells. If an SDL cell is configured as a basic-layer cell, dormancy-related functions cannot take effect in the corresponding capacity-layer cells.
- Because the RNC and NodeB must collaborate to determine whether a UMTS RF module can enter super dormancy, all UMTS cells on the RF module must be in the logical-cell active state.
- RF module super dormancy cannot take effect in cells where UMTS and LTE Spectrum Sharing Based on DC-HSDPA is enabled.

7.4.3.6 Others

- The working mode of RF modules (specified by the **RRU.RS (5G gNodeB, LTE eNodeB)** parameter) is set to **NO**, **TDL**, **LO**, **UO**, **GO**, or any combination of these values.
- The NE where the RF module is located must be in a normal maintenance mode. The maintenance mode can be the NE maintenance mode, the maintenance mode of all APPs, or the maintenance mode of the RF module. If the **DORM_MNTMODE_CTRL_SW** option of the **EQUIPMENT.EQMDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter is selected, this requirement does not apply.
- It is recommended that new and standard optical modules with a rate of 25 Gbit/s or higher be installed on the AAU and the corresponding BBP. Optical modules with a rate of 10 Gbit/s or lower cannot be installed on the RF modules and the corresponding BBP.
 - By default, only standard optical modules can be installed on the RF module and the corresponding BBP. (Whether an optical module is a standard one can be queried using the **DSP HWCAP** command.) Non-standard optical modules may be designed differently by their vendors and may not support super dormancy even if the **RF_SDC_SW** option of the **EQUIPMENT.EQMDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter is selected.
 - For the AAU5356/AAU5366/AAU5368md/AAU5668m/AAU5768/AAU5836f and other RF modules (contact Huawei technical support for such RF modules) using non-standard 50 Gbit/s optical modules, the **RF_SDC_SW** option of the **EQUIPMENT.EQMDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter needs to be selected for super dormancy to take effect. For other RF modules using non-standard 50 Gbit/s optical modules, super dormancy cannot take effect even if the **RF_SDC_SW** option is selected.
 - By default, only Huawei ALDs are supported by RF modules.
 - If the ALDs used for the RF module are non-Huawei ALDs and super dormancy needs to take effect, select the **RF_ALD_SW** option of the **EQUIPMENT.EQMDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter.
 - Non-Huawei ALDs may be designed differently according to their vendors and may not support super dormancy even if the **RF_ALD_SW** option is selected. Therefore, Huawei ALDs are recommended.
- Super dormancy is not supported when the base station is configured with primary power supply.
- To implement reliability protection during dormancy, both the OSS and NEs must be upgraded to versions that support reliability protection.
- Active devices (such as active WDM devices, semi-active WDM devices, and switches) cannot be used on the fronthaul link between the BBU and RF modules. In LNR separate-MPT scenarios, active devices can be used on one fronthaul link.
- When LNR or NR load-based super dormancy is enabled and the NR capacity-layer cell is a MOCN-enabled cell, ensure that the operator group set for the basic-layer cells (co-coverage neighboring LTE or NR cells) contains the

operator group of the capacity-layer cell. MOCN refers to multi-operator core network.

- In high-frequency NR TDD scenarios, it is recommended that the software versions of the base stations serving low-frequency basic-layer cells be upgraded to V100R020C10 or later. If such base stations are in V100R020C00 or an earlier version, some exit conditions will become invalid.
- For GSM or UMTS dormancy scenarios, only the BSC6910 and BSC6960 support this function, and the BSC must be upgraded to the required version.
- For GSM super dormancy scenarios, only the eGBTS supports this function, and the GBTS does not support this function.

7.4.4 Operation and Maintenance

7.4.4.1 Data Configuration

7.4.4.1.1 Data Preparation

[Table 7-57](#) describes the parameter shared by all sub-functions. The parameters that need to be separately set for each sub-function are as follows:

- [Table 7-58](#) describes the parameters used for activating scheduled RF module super dormancy/NR local-cell-load-based RF module super dormancy.
- [Table 7-59](#) describes the parameters used for activating NR load-based RF module super dormancy.
- [Table 7-60](#) and [Table 7-61](#) describe the parameters used for activating LNR load-based RF module super dormancy.
- [Table 7-62](#) describes the parameters used for activating LTE load-based RF module super dormancy.
- [Table 7-63](#) describes the parameters used for activating GSM load-based RF module super dormancy.
- [Table 7-64](#) describes the parameters used for activating UMTS load-based RF module super dormancy.
- [Table 7-65](#) and [Table 7-66](#) describe the parameters used for activating GU load-based RF module super dormancy.

Table 7-57 Parameter shared by all sub-functions

Parameter Name	Parameter ID	Setting Notes
Equipment Dormancy Optional Feature Switch	EQUIPMENT.EQMDORO PTFEATSW (LTE eNodeB, 5G gNodeB)	When the RF_SDC_SW option is selected and the RF module enters the super dormancy state, the optical module enters the energy saving state. After the RF module exits the dormancy state, the optical module returns to the normal working state. Using an optical module incompatible with Huawei base stations can result in abnormal connections between the BBP and RF module after the RF module exits the dormancy state, potentially damaging the BBP. This is due to the uncertain compatibility and quality of such optical modules. It is recommended that the default deselected state of the RF_SDC_SW option of this parameter be retained.
Equipment Dormancy Optional Feature Switch	EQUIPMENT.EQMDORO PTFEATSW (LTE eNodeB, 5G gNodeB)	When the RF_SDFFD_SW option is selected, the fronthaul link is no longer detected after three consecutive failures, and RF module super dormancy does not take effect. If this option is selected, the fronthaul link is detected forcibly. It is recommended that the default deselected state of the RF_SDFFD_SW option of this parameter be retained.

Parameter Name	Parameter ID	Setting Notes
Equipment Dormancy Optional Feature Switch	EQUIPMENT.EQMDORO PTFEATSW (LTE eNodeB, 5G gNodeB)	When the DOR_LM_SW option is selected and an NE is connected to the NetEco, dormancy-related functions can take effect. Once an RF module enters the dormancy state, the maintenance link between the NetEco and the power supply that transmits signals through the RF module will be disconnected. When this option is selected and an NE is connected to the RPCU, dormancy-related functions can take effect. However, the upper-level board connected to the RPCU cannot enter the dormancy state. (RAT-specific dormancy-related functions can take effect on the upper-level RF module connected to the RPCU.) Once the RF module supplying power to the RPCU enters the dormancy state, the power supply link of the RPCU is interrupted, ALM-26918 CXU Maintenance Link Failure is reported for the RPCU, and ALM-26504 RF Unit CPRI Interface Error is reported for the upper-level RF module connected to the RPCU. It is recommended that the default deselected state of the DOR_LM_SW option of this parameter be retained.

Parameter Name	Parameter ID	Setting Notes
Equipment Dormancy Optional Feature Switch	EQUIPMENT.EQMDORO PTFEATSW (LTE eNodeB, 5G gNodeB)	When the RF_ALD_SW option is selected, the RET device of the ALD is in the energy saving state when the RF module enters the super dormancy state, and returns to its normal state after the RF module exits the super dormancy state. It is recommended that this option be deselected because the compatibility and quality of non-standard ALDs cannot be ensured.
Equipment Dormancy Optional Feature Switch	EQUIPMENT.EQMDORO PTFEATSW (LTE eNodeB, 5G gNodeB)	Select the DORM_MNTMODE_CTRL_SW option if the NE is in the non-normal maintenance mode (including the NE maintenance mode, the maintenance mode of all APPs, and the maintenance mode of the RF module) and dormancy can take effect.
Equipment Dormancy Optional Feature Switch	EQUIPMENT.EQMDORO PTFEATSW (LTE eNodeB, 5G gNodeB)	Select the RF_SDWAO_SW option to shorten the service recovery duration for RF modules. You are advised to perform configuration under the guidance of Huawei engineers.

Parameter Name	Parameter ID	Setting Notes
Equipment Dormancy Optional Feature Switch	EQUIPMENT.EQMDORO PTFEATSW (5G gNodeB, LTE eNodeB)	Deselect the GU_DC_SW option, which is the default setting. If the option is deselected and a configuration command is executed to trigger dormancy pending or dormancy pending is enabled on the local end that works in GSM or UMTS, then dormancy pending will take effect on the entire base station. In this case, the dormancy duration of the entire base station is shortened. This option must also be deselected for the NE to which the RF module belongs if the RRU.RS (LTE eNodeB, 5G gNodeB) parameter is configured to include GSM or UMTS and the RRU.DORMANCYSW (5G gNodeB, LTE eNodeB) parameter is set to ON, ON_SUPER, or ON_DEEP_SUPER.

Table 7-58 Parameters used for activating scheduled RF module super dormancy/NR local-cell-load-based RF module super dormancy

Parameter Name	Parameter ID	Setting Notes
Dormancy Switch	RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB)	To enable scheduled RF module super dormancy, set this parameter to ON_SUPER or ON_DEEP_SUPER . The former value is recommended.

Parameter Name	Parameter ID	Setting Notes
Dormancy Start Time	<i>RRU.DORMANCYSTARTTIME (LTE eNodeB, 5G gNodeB)</i>	You are advised to set this parameter to the beginning of a period with no services. You are advised not to set the dormancy start time to the top of the hour to prevent significant fluctuation in the N.HO.IntraFreq.FailOut.PreparationFailure counter value. For example, if super dormancy needs to start at 00:00:00, you are advised to set the <i>RRU.DORMANCYSTARTTIME (LTE eNodeB, 5G gNodeB)</i> parameter to 23:55:00.
Dormancy Stop Time	<i>RRU.DORMANCYSTOPTIME (LTE eNodeB, 5G gNodeB)</i>	You are advised to set this parameter to the end of a period with no services. The stop time specified by <i>RRU.DORMANCYSTOPTIME (LTE eNodeB, 5G gNodeB)</i> must be later than the start time specified by <i>RRU.DORMANCYSTARTTIME (LTE eNodeB, 5G gNodeB)</i> plus the waiting time specified by <i>RRU.DORMANCYTIMER (LTE eNodeB, 5G gNodeB)</i>. If the start time is the same as the stop time, scheduled RF module super dormancy does not take effect. If the stop time is 1s earlier than the start time (for example, when the configured dormancy period is 00:00:00–23:59:59), RF module super dormancy takes effect all day long.
Dormancy Waiting Timer	<i>RRU.DORMANCYTIMER (LTE eNodeB, 5G gNodeB)</i>	The default value is recommended.

Parameter Name	Parameter ID	Setting Notes
RRU Dormancy Optional Feature Switch	RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)	To enable the function of waking up RF modules at any time from super dormancy, select the ANYTIME_WAKEUP_SW option of this parameter. After this switch is turned on, an RF module enters dormancy within a specified period every day. If, on a particular day, the RF module should not enter dormancy during that period due to service fluctuations or other factors, turn off this switch in advance to ensure service continuity.
Max Random Dormancy Waiting Time	EQUIPMENT.MAXRAMDORMANCYWAITTIME (LTE eNodeB, 5G gNodeB)	The default value is recommended. When the dormancy conditions are met, the system waits for a random duration based on this parameter before entering super dormancy. This prevents a large number of sites from entering super dormancy simultaneously and avoids the resulting impact on core network signaling.
Custom Dormancy Standard Switch	RRU.CUSTOMDORMANCYSTDSW (LTE eNodeB, 5G gNodeB)	The default value is recommended.
Dormancy Standard	RRU.DORMANCYSTD (LTE eNodeB, 5G gNodeB)	The default value is recommended. When the RRU.CUSTOMDORMANCYSTDSW (5G gNodeB, LTE eNodeB) parameter is set to ON , it is recommended that the current parameter be set to include the RATs requiring dormancy.

Parameter Name	Parameter ID	Setting Notes
Power Saving Policy ID	RRU.PWRSVGPOLICYID (LTE eNodeB, 5G gNodeB)	When the dormancy switch is turned on and this parameter is set to a non-zero value, the dormancy period is determined by the configured power saving policy. Otherwise, the dormancy period is determined by Dormancy Start Time and Dormancy Stop Time .
Operation Type	OPERATIONTYPE	The value PENDING is recommended.
Pending Duration	PENDINGDURATION	None
Necessary HO Prepare Fail Punish Timer	gNBMobilityCommParam.NecHoPrepFailPunish-Timer	It is recommended that this parameter be set to 30 in scheduled RF module super dormancy scenarios.
Optimized HO Prepare Fail Punish Timer	gNBMobilityCommParam.OptHoPrepFailPunish-Timer	It is recommended that this parameter be set to 30 in scheduled RF module super dormancy scenarios.
Policy ID	BRDPWRSVGPOLICY.POLICYID (LTE eNodeB, 5G gNodeB)	When the dormancy switch is turned on and the RRU.PWRSVGPOLICYID (LTE eNodeB, 5G gNodeB) parameter is set to a non-zero value, set this parameter to specify the dormancy type and waiting time.
Power Saving Type	BRDPWRSVGPOLICY.POWERSAVINGTYPE (LTE eNodeB, 5G gNodeB)	It is recommended that this parameter be set to RF_DORMANCY .
Dormancy Waiting Timer	BRDPWRSVGPOLICY.DORMANCYTIMER (LTE eNodeB, 5G gNodeB)	The default value is recommended.

Parameter Name	Parameter ID	Setting Notes
Policy ID	BRDPWRSVGPOLICY-TIME.POLICYID (LTE eNodeB, 5G gNodeB)	When the dormancy switch is turned on and the RRU.PWRSVGPOLICYID (LTE eNodeB, 5G gNodeB) parameter is set to a non-zero value, set this parameter to configure the corresponding dormancy period.
Time ID	BRDPWRSVGPOLICY-TIME.TIMEID (LTE eNodeB, 5G gNodeB)	None
Start Time	BRDPWRSVGPOLICY-TIME.STARTTIME (LTE eNodeB, 5G gNodeB)	You are advised to set this parameter to the beginning of a period with no services. You are advised not to set the dormancy start time to the top of the hour to prevent significant fluctuations in the values of performance counters such as N.HO.IntraFreq.FailOut.Prepare Failure. For example, if super dormancy needs to start at 00:00:00, you are advised to set the BRDPWRSVGPOLICY-TIME.STARTTIME (LTE eNodeB, 5G gNodeB) parameter to 23:55:00.

Parameter Name	Parameter ID	Setting Notes
Stop Time	BRDPWRSVGPOLICY-TIME.STOPTIME (LTE eNodeB, 5G gNodeB)	You are advised to set this parameter to the end of a period with no services. The stop time specified by BRDPWRSVGPOLICY-TIME.STOPTIME (LTE eNodeB, 5G gNodeB) must be later than the start time specified by BRDPWRSVGPOLICY-TIME.STARTTIME (LTE eNodeB, 5G gNodeB) plus the waiting time specified by BRDPWRSVGPOLICY.DORMANCYTIMER (LTE eNodeB, 5G gNodeB). If the start time is the same as the stop time, scheduled RF module super dormancy does not take effect. If the stop time is 1s earlier than the start time (for example, when the configured dormancy period is 00:00:00–23:59:59), RF module super dormancy takes effect all day long.
Multi Period Power Saving Type	BRDPWRSVGPOLICY-TIME.MPPWRSVGTYPE (LTE eNodeB, 5G gNodeB)	The default value is recommended.
RRU Dormancy Optional Feature Switch	RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)	Super dormancy can take effect for RF modules configured with alarm ports and monitoring devices such as PMUs only if the RM_LM_SW option is selected. When RF module super dormancy is entered, the maintenance links of the monitoring devices configured for the RF module will be disconnected. It is recommended that the default deselected state of the RM_LM_SW option of this parameter be retained.

Parameter Name	Parameter ID	Setting Notes
Power Saving Algorithm Switch	EnodebPwrSavingAlgo.PwrSavingAlgoSwitch	Select the MULTI_RRU_CELL_DORM_COMPAT_SW option of this parameter when the multi-RRU cell dormancy compatibility function needs to be enabled for RF module dormancy to take effect in multi-RRU cells, such as SFN cells.
Power Saving Algorithm Switch	gNBPwrSavingAlgo.PwrSavingAlgoSwitch	Assume that the cell load simulation function is enabled. To forbid the RF module serving the local cell to enter dormancy and allow the RF modules serving the cells not enabled with cell load simulation to enter dormancy, select the SIMU_LOAD_DORMANCY_IMP_SW option.
Power Saving Algorithm Switch	NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch	Select the NR_DORMANCY_JUDGE_THRESHOLD_OPT_SW option of this parameter if optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module super dormancy is required.
Power Saving Policy Index	NRDUCellPowerSaving.PowerSavingPolicyIndex	Set this parameter based on the network plan if optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module super dormancy is required.

Parameter Name	Parameter ID	Setting Notes
Start Time	NRDUCellPowerSaving.StartTime	Set these parameters based on the network plan if optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module super dormancy is required. In addition, the following requirements must be met: - For the same policy of the same energy saving type, the interval between the stop time and the start time must be longer than 20 minutes. If the stop time is earlier than or equal to the start time, the stop time is interpreted as occurring on the next day. For example: If the start time is 8:00 a.m. and the stop time is 7:00 a.m., the power saving function is enabled from 8:00 a.m. to 7:00 a.m. on the next day. - For different policies of the same energy saving type, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving type needs to be enabled after policy A of this type has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A. - The period of RF module timing dormancy cannot overlap with that specified when the NRDUCellPowerSaving.PowerSavingType
Stop Time	NRDUCellPowerSaving.StopTime	

Parameter Name	Parameter ID	Setting Notes
		parameter is set to TIMING_CARRIER_SHUTDOWN for the local cell. The period must be the same as that specified by the RRU.DORMANCYSTARTTIME (LTE eNodeB, 5G gNodeB) and RRU.DORMANCYSTOPTIME (LTE eNodeB, 5G gNodeB) parameters.
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	Set this parameter to NR_DORMANCY_JUDGE_THRESHOLD_OPT if optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module super dormancy is required.
Downlink PRB Threshold	NRDUCellPowerSaving.DlPrbThld	You are advised to set this parameter to its recommended value when optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module super dormancy is enabled.
Uplink PRB Threshold	NRDUCellPowerSaving.UlPrbThld	You are advised to set this parameter to its recommended value when optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module super dormancy is enabled.
UE Number Threshold	NRDUCellPowerSaving.UserNumThld	You are advised to set this parameter to its recommended value when optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module super dormancy is enabled. In addition, this parameter must be set to a value less than or equal to 150 .

Parameter Name	Parameter ID	Setting Notes
Evaluation Period Granularity	NRDUCellPowerSaving.EvalPeriodGranularity	Set this parameter to MINUTE_LEVEL if optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module super dormancy is enabled.
Triggering Evaluation Period	NRDUCellPowerSaving.ShutdownTrigJudgePeriod	You are advised to set this parameter to its recommended value when optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module super dormancy is enabled.
Cell Load Filtering Mode	gNBPwrSavingAlgo.CellLoadFilterMode	This parameter specifies the filtering mode used in uplink/downlink PRB load measurement when optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module super dormancy is enabled. You are advised to set this parameter to its recommended value.

Parameter Name	Parameter ID	Setting Notes
SUL User Number Threshold	NRDUCellPowerSaving.SulUserNumThld	You are advised to set this parameter to its recommended value. - This parameter is invalid when it is set to 65535. - If the FLEX_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter is deselected and the cell has a NRDUCellPowerSaving record with power saving type set to LTE and NR intelligent carrier shutdown, optimized evaluation of NR RF module dormancy, NR load-based RF module dormancy, or LN load-based RF module dormancy, then the NRDUCellPowerSaving.SulUserNumThld parameter of this record must be set to 65535.
HandOver Switch	UCORRMALGOSWITCH.HoSwitch	- To enable UMTS inter-frequency handovers, select the HO_INTER_FREQ_HARD_HO_SWITCH option. - To enable inter-RAT handovers for UMTS, select the HO_INTER_RAT_CS_OUT_SWITCH and HO_INTER_RAT_PS_OUT_SWITCH options.
GSM Cell Index	U2GNCELL.GSMCellIndex	To enable inter-RAT handovers for UMTS, configure a basic-layer cell as a co-coverage inter-RAT neighboring cell of the capacity-layer cell.

Parameter Name	Parameter ID	Setting Notes
Inter-freq Target Frequency Trigger RSCP THD	UCELLMCLDR.TargetFreqThdRscp	You are advised to set this parameter to its recommended value.
Inter-freq Target Frequency Trigger Ec/No THD	UCELLMCLDR.TargetFreqThdEcNo	You are advised to set this parameter to its recommended value.
Cell Power Saving Inter Plmn Handover Allowed Switch	URNCCCELLDORMANCY-PARA.CellInterPlmnHoSwitch	Set this parameter to ON .
Intra RAT Inter Plmn Ho Allowed	UOPERATORSHARING-MODE.InterPlmnHoAllowedIntraRat	Set this parameter to YES .
Inter RAT Inter Plmn Ho Allowed	UOPERATORSHARING-MODE.InterPlmnHoAllowedInterRat	Set this parameter to YES .
Cell Power Saving Ignore CBS Switch	URNCCCELLDORMANCY-PARA.CellIgnoreCbsSwitch	Use the recommended value. CBS is generally used to send emergency and important information. Therefore, CBS takes precedence over dormancy, and this parameter is set to OFF by default. To increase the dormancy priority, set this parameter to ON . However, with this setting, the cell may already be in the dormancy state when CBS messages need to be sent, and waking it up takes time. As a result, CBS message broadcasting in the cell is delayed.
CBS Service Hold Duration	URNCCCELLDORMANCY-PARA.CbsServiceHoldDuration	Use the recommended value.

Parameter Name	Parameter ID	Setting Notes
Ignore CBC Switch	BSEXSOFTPARA.IgnoreCbcSwitch	Use the recommended value. CBC services are generally used to send emergency and important information. Therefore, CBC takes precedence over dormancy, and this parameter is set to OFF by default. To increase the dormancy priority, set this parameter to ON . However, with this setting, the cell may already be in the dormancy state when CBC messages need to be sent, and waking it up takes time. As a result, CBC message broadcasting in the cell is delayed.
CBC Service Hold Duration	BSEXSOFTPARA.CbcServiceHoldDuration	Use the recommended value.

Table 7-59 Parameters used for activating NR load-based RF module super dormancy

Parameter Name	Parameter ID	Setting Notes
Dormancy Switch	RRU.DORMANCYSW (5G gNodeB, LTE eNodeB)	To enable NR load-based RF module dormancy, set this parameter to ON_SUPER or ON_DEEP_SUPER . The former value is recommended.
RRU Dormancy Optional Feature Switch	RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)	Select the ANYTIME_WAKEUP_SW option of this parameter if NR load-based RF module dormancy needs to be enabled.

Parameter Name	Parameter ID	Setting Notes
Power Saving Switch	NRDUCellAlgoSwitch.PowerSavingSwitch	It is recommended that the INTRA_NR_DORMANCY_BY_LOAD_SW option of this parameter be selected if NR load-based RF module dormancy needs to be enabled.
Power Saving Extension Switch	NRDUCellAlgoSwitch.PowerSavingExtSwitch	To enable NR load-based precise dormancy, both the INTRA_NR_DORMANCY_BY_LOAD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter and the PRECISE_DORMANCY_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter must be selected. To calculate the overlap rate based on grids, select the INTRA_NR_DORMANCY_BY_LOAD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter and the PRECISE_DORMANCY_SW and GRID_MODE_SW options of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter.. The precise mode is not supported in high-frequency TDD.
Power Saving Policy Index	NRDUCellPowerSaving.PowerSavingPolicyIndex	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Start Time	NRDUCellPowerSaving.StartTime	Set these parameters based on the network plan and the following requirements: - For the same policy of the same energy saving type, the interval between the stop time and the start time must be longer than 20 minutes. If the stop time is earlier than or equal to the start time, the stop time is interpreted as occurring on the next day. For example: If the start time is 8:00 a.m. and the stop time is 7:00 a.m., the power saving function is enabled from 8:00 a.m. to 7:00 a.m. on the next day. - For different policies of the same energy saving type, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving type needs to be enabled after policy A of this type has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A. - The period of RF module super dormancy cannot overlap with that specified when the NRDUCellPowerSaving.PowerSavingType parameter is set to TIMING_CARRIER_SHUTDOWN for the local cell.
Stop Time	NRDUCellPowerSaving.StopTime	
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	Set this parameter to INTRA_NR_DORMANCY_BY_LOAD .

Parameter Name	Parameter ID	Setting Notes
Downlink PRB Threshold	NRDUCellPowerSaving.DlPrbThld	You are advised to set this parameter to its recommended value.
Downlink PRB Offset	NRDUCellPowerSaving.DlPrbOffset	You are advised to set this parameter to its recommended value.
Uplink PRB Threshold	NRDUCellPowerSaving.UlPrbThld	You are advised to set this parameter to its recommended value.

Parameter Name	Parameter ID	Setting Notes
SUL User Number Threshold	NRDUCellPowerSaving.SulUserNumThld	You are advised to set these parameters to their recommended values.
SUL User Number Offset	NRDUCellPowerSaving.SulUserNumOffset	- These parameters are invalid when they are set to 65535. - If NRDUCellPowerSaving.SulUserNumThld is not set to 65535, the sum of NRDUCellPowerSaving.SulUserNumThld and NRDUCellPowerSaving.SulUserNumOffset must be less than 65535, and NRDUCellPowerSaving.SulUserNumThld must be less than NRDUCellPowerSaving.UserNumThld. - If the FLEX_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter is deselected and the cell has a NRDUCellPowerSaving record with power saving type set to LTE and NR intelligent carrier shutdown, optimized evaluation of NR RF module dormancy, NR load-based RF module dormancy, or LN load-based RF module dormancy, then the NRDUCellPowerSaving.SulUserNumThld parameter of this record must be set to 65535.
Neighboring Cell Uplink PRB Offset	NRDUCellPowerSaving.NCellULPrbOffset	You are advised to set this parameter to its recommended value.

Parameter Name	Parameter ID	Setting Notes
UE Number Threshold	NRDUCellPowerSav- ing.UserNumThld	You are advised to set this parameter to its recommended value. The maximum value is 150 .
User Number Offset	NRDUCellPowerSav- ing.UserNumOffset	You are advised to set this parameter to its recommended value.
Evaluation Period Granularity	NRDUCellPowerSav- ing.EvalPeriodGranularity	You are advised to set this parameter to its recommended value. For high-frequency TDD, this parameter must be set to MINUTE_LEVEL .
Triggering Evaluation Period	NRDUCellPowerSav- ing.ShutdownTrigJudgePer- iod	You are advised to set this parameter to its recommended value.
Exit Evaluation Period	NRDUCellPowerSav- ing.ExitEvaluationPeriod	You are advised to set this parameter to its recommended value.

Parameter Name	Parameter ID	Setting Notes
Second-Level Trigger Evaluation Period	NRDUCellPowerSaving.SecondLevelTrigEvalPrd	You are advised to set this parameter to its recommended value. • This parameter can be configured and take effect only when the NRDUCellPowerSaving.EvalPeriodGranularity parameter is set to SECOND_LEVEL. • When the NRDUCellPowerSaving.PowerSavingType parameter is set to INTRA_NR_DORMANCY_BY_LOAD, and the NRDUCellPowerSaving.EvalPeriodGranularity parameter is set to SECOND_LEVEL, the NRDUCellPowerSaving.SecondLevelTrigEvalPrd parameter must be set to a value greater than or equal to 6. • The smaller the value of this parameter, the higher the probability of triggering power saving. The larger the value of this parameter, the lower the probability.

Parameter Name	Parameter ID	Setting Notes
Second-Level Exit Evaluation Period	NRDUCellPowerSaving.SecondLevelExitEvalPrd	You are advised to set this parameter to its recommended value. - This parameter can be configured and take effect only when the NRDUCellPowerSaving.EvalPeriodGranularity parameter is set to SECOND_LEVEL. - The smaller the value of this parameter, the higher the probability of triggering exit from power saving and the smaller the power saving gains. The larger the value of this parameter, the lower the probability.
UL PRB Protect Thld for Power Saving	NRDUCellPowerSaving.ProtectUlPrbThld	You are advised to set this parameter to its recommended value.
DL PRB Protect Thld for Power Saving	NRDUCellPowerSaving.ProtectDlPrbThld	You are advised to set this parameter to its recommended value.
Uplink PRB Protection Offset	NRDUCellPowerSaving.ProtectUlPrbOffset	You are advised to set this parameter to its recommended value.
Downlink PRB Protection Offset	NRDUCellPowerSaving.ProtectDlPrbOffset	You are advised to set this parameter to its recommended value.
mmWave User Number Threshold	NRDUCellPowerSaving.MmwaveUserNumThld	You are advised to set this parameter to its recommended value.
Power Saving Penalty Duration	NRDUCellPowerSaving.PwrSavingPenaltyTime	You are advised to set this parameter to its recommended value.

Parameter Name	Parameter ID	Setting Notes
Power Saving Cell Flag	NRCellRelation.PowerSavingCellFlag	This parameter specifies the flag of a neighboring cell in load-based dormancy. Set this parameter based on the network plan. When the NRDUCellPowerSaving.PowerSavingType parameter is set to INTRA_NR_DORMANCY_BY_LOAD, and the NRDUCellPowerSaving.EvalPeriodGranularity parameter is set to SECOND_LEVEL, the number of corresponding NRCellRelation MOs with NRCellRelation.PowerSavingCellFlag set to COVERAGE must be less than or equal to 16.
Overlap Rate	NRCellRelation.OverlapRate	This parameter specifies the degree of the overlap between the capacity layer and the basic layer. When NR load-based precise dormancy is enabled, set this parameter based on the co-coverage degree of respective neighboring cells.
No Removal Flag	NRCellRelation.NoRmvFlag	When automatic removal of neighboring cells in ANR is enabled (with the NR_NR_ANR_DEL_SW option of the NRCellAlgoSwitch.AnrSwitch parameter selected), it is recommended that the NRCellRelation.NoRmvFlag parameter be set to FORBID_ANR_RMV for the mutual neighbor relationships between the capacity- and basic-layer cells.

Parameter Name	Parameter ID	Setting Notes
PLMN Reserved Flag	NRExternalNCellPlmn.PlmnReservedFlag	In multi-operator sharing scenarios, set this parameter for capacity-layer cells based on the network plan.
Cell Load Filtering Mode	gNBPwrSavingAlgo.CellLoadFilterMode	This parameter specifies the filtering mode used in uplink/downlink PRB load measurement for load-based dormancy. You are advised to set this parameter to its recommended value.
RRU Dormancy Optional Feature Switch	RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)	Super dormancy can take effect for RF modules configured with alarm ports and monitoring devices such as PMUs only if the RM_LM_SW option is selected. When RF module super dormancy is entered, the maintenance links of the monitoring devices configured for the RF module will be disconnected. It is recommended that the default deselected state of the RM_LM_SW option of this parameter be retained.
Power Saving Algorithm Switch	gNBPwrSavingAlgo.PwrSavingAlgoSwitch	Assume that the cell load simulation function is enabled. To forbid the RF module serving the local cell to enter dormancy and allow the RF modules serving the cells not enabled with cell load simulation to enter dormancy, select the SIMU_LOAD_DORMANCY_IMP_SW option.

Table 7-60 Parameters on the LTE side used for activating LNR load-based RF module super dormancy

Parameter Name	Parameter ID	Setting Notes
Power Saving Algorithm Switch	CellPwrSavingAlgo.PwrSavingAlgoSwitch	Select the LNR_LOAD_BASED_DORMANCY_SW option of this parameter if LNR load-based RF module dormancy needs to be enabled.
Aggregation Attribute	NrNRRelationship.AggregationAttribute	Set the CARRIER_SHUTDOWN_OVERLAP_FLAG option of this parameter to the recommended value.
Power Saving Algorithm Switch	EnodebPwrSavingAlgo.PwrSavingAlgoSwitch	- Select the MULTI_RRU_CELL_DORM_COMPAT_SW option of this parameter when the multi-RRU cell dormancy compatibility function needs to be enabled for RF module dormancy to take effect in multi-RRU cells, such as SFN cells. - Select the SPLIT_CELL_RF_DORMANCY_SW option of this parameter when an LTE sector split cell needs to serve as a basic cell for load-based dormancy.
Interface SON Configuration Switch	eNodeBOamParam.InterfaceSonConfigSwitch	To enable the eNodeB to set up X2 links with gNodeBs that do not have NSA cells, select the LNR_SA_GNB_X2_SETUP_SW option.

Parameter Name	Parameter ID	Setting Notes
Power Saving Algorithm Switch	EnodebPwrSavingAlgo.PwrSavingAlgoSwitch	To enable inter-base-station energy saving between LTE and SA NR cells in LTE and NR scenarios, select the LNR_INTER_BS_PWR_SAVE_SW option. The LNR_SA_GNB_X2_SETUP_SW option of the eNodeBOamParam.InterfaceSonConfigSwitch parameter must also be selected.

Table 7-61 Parameters on the NR side used for activating LNR load-based RF module super dormancy

Parameter Name	Parameter ID	Setting Notes
Dormancy Switch	RRU.DORMANCYSW (5G gNodeB, LTE eNodeB)	To enable LNR load-based RF module dormancy, set this parameter to ON_SUPER or ON_DEEP_SUPER . The former value is recommended.
RRU Dormancy Optional Feature Switch	RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)	Select the ANYTIME_WAKEUP_SW option of this parameter if LNR load-based RF module dormancy needs to be enabled.
Power Saving Switch	NRDUCellAlgoSwitch.PowerSavingSwitch	Select the LNR_DORMANCY_BY_LOAD_SW option of this parameter.
Start Time	NRDUCellPowerSaving.StartTime	Set this parameter based on the network plan.
Stop Time	NRDUCellPowerSaving.StopTime	Set this parameter based on the network plan.
Power Saving Type	NRDUCellPowerSaving.PowerSavingType	Set this parameter to LNR_DORMANCY_BY_LOAD .
UE Number Threshold	NRDUCellPowerSaving.UserNumThld	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Downlink PRB Threshold	NRDUCellPowerSaving.DlPrbThld	Set this parameter based on the network plan.
Uplink PRB Threshold	NRDUCellPowerSaving.UlPrbThld	Set this parameter based on the network plan.
E-UTRAN Downlink PRB Threshold	NRDUCellPowerSaving.EutranDlPrbThld	Set this parameter based on the network plan.
Neighboring Cell Uplink PRB Threshold	NRDUCellPowerSaving.NCellUlPrbThld	Set this parameter based on the network plan.
E-UTRAN Downlink PRB Offset	NRDUCellPowerSaving.EutranDlPrbOffset	Set this parameter based on the network plan.
Neighboring Cell Uplink PRB Offset	NRDUCellPowerSaving.NCellUlPrbOffset	Set this parameter based on the network plan.
E-UTRAN Carrier Shutdown Ratio	NRDUCellPowerSaving.EutranCarrShutdownRatio	Set this parameter based on the network plan.
E-UTRAN Carrier Shutdown Offset	NRDUCellPowerSaving.EutranCarrShutdownExitOfs	Set this parameter based on the network plan.
E-UTRAN NR User Number Threshold	NRDUCellPowerSaving.EutranNrUserNumThld	Set this parameter based on the network plan. If this parameter is set to 65535 , this parameter and the NRDUCellPowerSaving.EutranNrUserNumThldOfs parameter are invalid.
E-UTRAN NR User Number Threshold Offset	NRDUCellPowerSaving.EutranNrUserNumThldOfs	Set this parameter based on the network plan.
E-UTRAN NR User Downlink PRB Threshold	NRDUCellPowerSaving.EutranNrUserDlPrbThld	Set this parameter based on the network plan.
E-UTRAN NR User Uplink PRB Threshold	NRDUCellPowerSaving.EutranNrUserUlPrbThld	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
E-UTRAN NR User Downlink PRB Threshold Offset	NRDUCellPowerSaving.EutranNrUserDlPrbThldOfs	Set this parameter based on the network plan.
E-UTRAN NR User Uplink PRB Threshold Offset	NRDUCellPowerSaving.EutranNrUserUlPrbThldOfs	Set this parameter based on the network plan.
SUL User Number Threshold	NRDUCellPowerSaving.SulUserNumThld	You are advised to set this parameter to its recommended value. - This parameter is invalid when it is set to 65535. - If the FLEX_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter is deselected and the cell has a NRDUCellPowerSaving record with power saving type set to LTE and NR intelligent carrier shutdown, optimized evaluation of NR RF module dormancy, NR load-based RF module dormancy, or LN load-based RF module dormancy, then the NRDUCellPowerSaving.SulUserNumThld parameter of this record must be set to 65535.
Power Saving Penalty Duration	NRDUCellPowerSaving.PwrSavingPenaltyTime	Set this parameter based on the network plan.
Evaluation Period Granularity	NRDUCellPowerSaving.EvalPeriodGranularity	You are advised to set this parameter to its recommended value.
Triggering Evaluation Period	NRDUCellPowerSaving.ShutdownTrigJudgePeriod	Set this parameter based on the network plan.
Exit Evaluation Period	NRDUCellPowerSaving.ExitEvaluationPeriod	You are advised to set this parameter to its recommended value.

Parameter Name	Parameter ID	Setting Notes
Second-Level Trigger Evaluation Period	NRDUCellPowerSaving.SecondLevelTrigEvalPrd	You are advised to set this parameter to its recommended value. • This parameter can be configured and take effect only when the NRDUCellPowerSaving.EvalPeriodGranularity parameter is set to SECOND_LEVEL. • When the NRDUCellPowerSaving.PowerSavingType parameter is set to LNR_DORMANCY_BY_LOAD, and the NRDUCellPowerSaving.EvalPeriodGranularity parameter is set to SECOND_LEVEL, the NRDUCellPowerSaving.SecondLevelTrigEvalPrd parameter must be set to a value greater than or equal to 6. • The smaller the value of this parameter, the higher the probability of triggering power saving. The larger the value of this parameter, the lower the probability.

Parameter Name	Parameter ID	Setting Notes
Second-Level Exit Evaluation Period	NRDUCellPowerSaving.SecondLevelExitEvalPrd	You are advised to set this parameter to its recommended value. - This parameter can be configured and take effect only when the NRDUCellPowerSaving.EvalPeriodGranularity parameter is set to SECOND_LEVEL. - The smaller the value of this parameter, the higher the probability of triggering exit from power saving and the smaller the power saving gains. The larger the value of this parameter, the lower the probability.
Power Saving Overlapping Indicator	NRCellEutraNRelation.PowerSavingOverlapInd	Configure the LNR_CARR_SHUTDOWN_OVERLAP_IND option based on the network plan. This parameter is mandatory only in SA networking.
Cell Load Filtering Mode	gNBPwrSavingAlgo.CellLoadFilterMode	This parameter specifies the filtering mode used in uplink/downlink PRB load measurement for load-based dormancy. You are advised to set this parameter to its recommended value.
Power Saving Algorithm Switch	gNBPwrSavingAlgo.PwrSavingAlgoSwitch	Select the INTER_RAT_GRAD_PWR_SAVING_SW option of this parameter if inter-RAT level-by-level energy saving is required.

Parameter Name	Parameter ID	Setting Notes
RRU Dormancy Optional Feature Switch	RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)	Super dormancy can take effect for RF modules configured with alarm ports and monitoring devices such as PMUs only if the RM_LM_SW option is selected. When RF module super dormancy is entered, the maintenance links of the monitoring devices configured for the RF module will be disconnected. It is recommended that the default deselected state of the RM_LM_SW option of this parameter be retained.
Power Saving Algorithm Switch	gNBPwrSavingAlgo.PwrSavingAlgoSwitch	When the LNR_LOAD_DORM_PRIM_OP_RECOG_SW option is selected, primary operator recognition for LNR load-based dormancy in MOCN scenarios is enabled. Only neighboring LTE cells served by the same operator as the primary operator of an NR cell can be configured as basic-layer cells for LNR load-based dormancy in the NR cell.
Power Saving Algorithm Switch	gNBPwrSavingAlgo.PwrSavingAlgoSwitch	Assume that the cell load simulation function is enabled. To forbid the RF module serving the local cell to enter dormancy and allow the RF modules serving the cells not enabled with cell load simulation to enter dormancy, select the SIMU_LOAD_DORMANCY_IMP_SW option.
X2 Config Switch	gNBInterfaceParam.X2ConfigSw	To allow X2 link setup in SA scenarios, select the X2_SETUP_FOR_SA_SW option.

Parameter Name	Parameter ID	Setting Notes
Power Saving Algorithm Switch	gNBPwrSavingAlgo.PwrSavingAlgoSwitch	- To enable energy saving for SA NR cells in LTE-NR inter-base-station energy saving scenarios, select the SA_LNR_INTER_BS_PWR_SAVE_SW option. The X2_SETUP_FOR_SA_SW option of the gNBInterfaceParam.X2ConfigSw parameter must also be selected. - When the SA_LNR_INTER_BS_PWR_SAVE_SW option is deselected, the NRCellEutraNRelation.EnodebId parameter must have the same value in all entries where the LNR_CARR_SHUTDOWN_OVERLAP_IND option of the NRCellEutraNRelation.PowerSavingOverlapInd parameter is set to Yes.

 NOTE

If the load-based super dormancy period configured for an NR DU cell overlaps with the scheduled super dormancy period configured for the RRU, the base station determines whether to enter the dormancy state based on the entry conditions of load-based super dormancy.

Table 7-62 Parameters used for activating LTE load-based RF module super dormancy

Parameter Name	Parameter ID	Setting Notes
Dormancy Switch	RRU.DORMANCYSW	To enable LTE load-based RF module dormancy, set this parameter to ON_SUPER or ON_DEEP_SUPER . The former value is recommended.

Parameter Name	Parameter ID	Setting Notes
RRU Dormancy Optional Feature Switch	RRUEXTINFO.RRUDOROPTFEATSW (<i>LTE eNodeB, 5G gNodeB</i>)	Select the ANYTIME_WAKEUP_SW option of this parameter if LTE load-based RF module dormancy needs to be enabled.
Power Saving Algorithm Switch	CellPwrSavingAlgo.PwrSavingAlgoSwitch	It is recommended that the INTRA_LTE_DORMANCY_BY_LOAD_SW option of this parameter be selected if LTE load-based dormancy needs to be enabled. To enable LTE load-based precise dormancy, deselect the INTRA_LTE_DORMANCY_BY_LOAD_SW option and then select the LTE_DORMANCY_BY_LOAD_PE_SW option of this parameter. To calculate overlap rates based on grids, select both the LTE_DORMANCY_BY_LOAD_PE_SW and OVERLAP_GRID_MODE_SW options of this parameter.
Policy Index	CellPwrSavingPolicy.PolicyIndex	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Start Time	CellPwrSavingPolicy.StartTime	Set these parameters based on the network plan and the following requirements: - For the same policy of the same energy saving type, the interval between the stop time and the start time must be longer than 20 minutes. If the end time is earlier than the start time, the end time is interpreted as occurring on the next day. For example: If the start time is 8:00 a.m. and the stop time is 7:00 a.m., the power saving function is enabled from 8:00 a.m. to 7:00 a.m. on the next day. - For different policies of the same energy saving type, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving type needs to be enabled after policy A of this type has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A.
Stop Time	CellPwrSavingPolicy.StopTime	
Power Saving Type	CellPwrSavingPolicy.PowerSavingType	- To enable LTE load-based dormancy, set this parameter to INTRA_LTE_DORMANCY_BY_LOAD. - To enable LTE load-based precise dormancy, set this parameter to LTE_PRECISE_DORMANCY_BY_LOAD.

Parameter Name	Parameter ID	Setting Notes
Trigger Judge Period	CellPwrSavingPolicy.TriggerJudgePeriod	You are advised to set this parameter to its recommended value.
Downlink PRB Threshold	CellPwrSavingPolicy.DlPrbThld	You are advised to set this parameter to its recommended value.
Downlink PRB Offset	CellPwrSavingPolicy.DlPrbOffset	You are advised to set this parameter to its recommended value.
Uplink PRB Threshold	CellPwrSavingPolicy.UlPrbThld	You are advised to set this parameter to its recommended value.
Uplink PRB Offset	CellPwrSavingPolicy.UlPrbOffset	You are advised to set this parameter to its recommended value.
UE Number Threshold	CellPwrSavingPolicy.UENumThld	You are advised to set this parameter to its recommended value.
Forcible Trigger UE Number Threshold	CellPwrSavingPolicy.ForcibleTriggerUeNumThld	You are advised to set this parameter to its recommended value.

Parameter Name	Parameter ID	Setting Notes
Exit Judge Period	CellPwrSavingPolicy.ExitJudgePeriod	You are advised to set this parameter to its recommended value. • If the CellPwrSavingPolicy.PowerSavingType parameter is set to INTRA_LTE_DORMANCY_BY_LOAD or LTE_PRECISE_DORMANCY_BY_LOAD, and the CellPwrSavingPolicy.EvalInterval parameter is set to EVAL_INTERVAL_60S, the CellPwrSavingPolicy.ExitJudgePeriod parameter must be set to an integer multiple of 12. • The smaller the value of this parameter, the higher the probability of triggering exit from power saving and the smaller the power saving gains. The larger the value of this parameter, the lower the probability.

Parameter Name	Parameter ID	Setting Notes
Evaluation Interval	CellPwrSavingPolicy.EvalInterval	You are advised to set this parameter to its recommended value. - Set this parameter to EVAL_INTERVAL_5S if a second-level evaluation period is required. In this case, the cell may be less likely to exit power saving given the same exit evaluation period (specified by CellPwrSavingPolicy.ExitJudgePeriod), and may be more likely to exit power saving given a shorter exit evaluation period. - Set this parameter to EVAL_INTERVAL_60S if a minute-level evaluation period is required. In this case, the CellPwrSavingPolicy.ExitJudgePeriod parameter must be set to an integer multiple of 12.
Penalty Time	CellPwrSavingPolicy.PenaltyTime	You are advised to set this parameter to its recommended value.
Inter-Frequency Mobility Load Balancing Threshold	CellMLB.InterFreqMlbThd	You are advised to set this parameter to its recommended value.
Load Offset	CellMLB.LoadOffset	You are advised to set this parameter to its recommended value.
Overlap Indicator Extension	EutranInterFreqN-Cell.OverlapIndicatorExtension	Set this parameter based on the network plan. If this parameter is set to POWER_SAVING_OVERLAP_IND , the neighboring cell has overlapping coverage with the serving cell.

Parameter Name	Parameter ID	Setting Notes
Overlap Rate	EutranInterFreqN-Cell.OverlapRate	In precise dormancy mode, set this parameter to the overlap rate between the capacity layer and the basic layer.
Power Saving Algorithm Switch	EnodebPwrSavingAlgo.PwrSavingAlgoSwitch	Select the INTRA_RAT_GRAD_PWR_SAVING_SW option of this parameter if intra-RAT level-by-level energy saving is required.
Cell Load Filtering Mode	EnodebPwrSavingAlgo.CellLoadFilterMode	This parameter specifies the filtering mode used in uplink/downlink PRB load measurement for load-based dormancy. You are advised to set this parameter to its recommended value.
Inter-eNodeB Information Exchange Mode	EnodebPwrSavingAlgo.InterEnbInfoExchangeMode	You are advised to set this parameter to its recommended value. • If this parameter is set to SLOW_EXCHANGE_MODE, the slow exchange mode is used, that is, information is exchanged at a long interval. • If this parameter is set to FAST_EXCHANGE_MODE, the fast exchange mode is used, that is, information is exchanged at a short interval. • You are advised to set this parameter to FAST_EXCHANGE_MODE if the CellPwrSavingPolicy.EvalInterval parameter is set to EVAL_INTERVAL_5S.

Parameter Name	Parameter ID	Setting Notes
Dormancy Cell Type	CellPwrSavingAlgo.DormancyCellType	In precise dormancy mode, set this parameter to CAPACITY_CELL for capacity-layer cells and BASIC_CELL for basic-layer cells.
Uplink PRB Protection Threshold	CellPwrSavingPolicy.ProtectUlPrbThld	Use the recommended value for a capacity-layer cell. This parameter does not need to be set for a basic-layer cell.
Downlink PRB Protection Threshold	CellPwrSavingPolicy.ProtectDlPrbThld	Use the recommended value for a capacity-layer cell. This parameter does not need to be set for a basic-layer cell.
Uplink PRB Protection Offset	CellPwrSavingPolicy.ProtectUlPrbOffset	Use the recommended value for a capacity-layer cell. This parameter does not need to be set for a basic-layer cell.
Downlink PRB Protection Offset	CellPwrSavingPolicy.ProtectDlPrbOffset	Use the recommended value for a capacity-layer cell. This parameter does not need to be set for a basic-layer cell.
Power Saving Algorithm Switch	EnodebPwrSavingAlgo.PwrSavingAlgoSwitch	To enable optimization for cell PLMN reservation scenarios in multi-operator sharing, select the CELL_PLMN_RSVD_OPT_SW option.
Aggregation Attribute	EutranExternalCellPlmn.AggregationAttribute	In multi-operator sharing scenarios, set this parameter based on the network plan. To set the operator as a reserved operator, select the PLMN_RESERVED_FLAG option of this parameter.

Parameter Name	Parameter ID	Setting Notes
Configuration Adaptation Opt Sw	eNodeBOamParam.ConfigAdaptOptSwitch	In multi-operator sharing scenarios, set this parameter based on the network plan. To enable the cell reservation status to be automatically broadcast to neighboring base stations through X2 signaling messages, select the RSVD_FOR_OP_X2_BROADCAST_SW option of this parameter.
RRU Dormancy Optional Feature Switch	RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)	Super dormancy can take effect for RF modules configured with alarm ports and monitoring devices such as PMUs only if the RM_LM_SW option is selected. When RF module super dormancy is entered, the maintenance links of the monitoring devices configured for the RF module will be disconnected. It is recommended that the default deselected state of the RM_LM_SW option of this parameter be retained.
Power Saving Algorithm Switch	EnodebPwrSavingAlgo.PwrSavingAlgoSwitch	• Select the MULTI_RRU_CELL_DORM_COMPAT_SW option of this parameter when the multi-RRU cell dormancy compatibility function needs to be enabled for RF module dormancy to take effect in multi-RRU cells, such as SFN cells. • Select the SPLIT_CELL_RF_DORMANCY_SW option of this parameter when an LTE sector split cell needs to serve as a basic cell for load-based dormancy.

Table 7-63 Parameters used for activating GSM load-based RF module super dormancy

Parameter Name	Parameter ID	Setting Notes
Dormancy Switch	RRU.DORMANCYSW (5G gNodeB, LTE eNodeB)	Set this parameter to ON_SUPER if GSM load-based RF module super dormancy needs to be enabled.
RRU Dormancy Optional Feature Switch	RRUEXTINFO.RRUDOROPTFEATSW	To enable GSM load-based RF module super dormancy, select the ANYTIME_WAKEUP_SW option.
Enable Cell Dormancy	GCELLOTHPARA.DormancyEnable	To enable GSM load-based RF module super dormancy, set this parameter to ENABLE .
Co-Coverage Cell Index 1	GCELLOTHPARA.CoCoverageCellId1	When super dormancy is to be enabled for a non-concentric cell, at least one co-coverage cell must be configured and its coverage area must cover the cell with the Enable Cell Dormancy parameter set to ENABLE . If these conditions are not met due to incorrect planning, the current cell may enter dormancy unexpectedly, leading to coverage deterioration. If this occurs, indicators such as the paging success rate, access success rate, handover success rate, and service drop rate will deteriorate, and the traffic volume will decrease.
Co-Coverage Cell Index 2	GCELLOTHPARA.CoCoverageCellId2	
Co-Coverage Cell Index 3	GCELLOTHPARA.CoCoverageCellId3	
Co-Coverage Cell Index 4	GCELLOTHPARA.CoCoverageCellId4	
Dormancy Policy Index	GCELLDORMANCY.DormancyPolicyIndex	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Dormancy Cell Start Time	GCELLDORMANCY.DormancyCellStrTime	Set these parameters based on the network plan and the following requirements: - For the same policy of the same energy saving type, the interval between the stop time and the start time must be longer than 20 minutes. If the stop time is earlier than or equal to the start time, the stop time is interpreted as occurring on the next day. For example: If the start time is 8:00 a.m. and the stop time is 7:00 a.m., the power saving function is enabled from 8:00 a.m. to 7:00 a.m. on the next day. - For different policies of the same energy saving type, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving type needs to be enabled after policy A of this type has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A.
Dormancy Cell Stop Time	GCELLDORMANCY.DormancyCellStpTime	
Dormancy Load Detection Times	GCELLDORMANCY.DormancyCellLoadStatTm	Retain the default value.
Dormancy Cell Occupied Channel Num	GCELLDORMANCY.DormancyCellChanNum	Retain the default value.
Dormancy Load Threshold	GCELLDORMANCY.DormancyLoadThrD	Retain the default value.
Wake-up Load Threshold	GCELLDORMANCY.WakeUpLoadThrD	Retain the default value.

Parameter Name	Parameter ID	Setting Notes
Dormancy Penalty Time	GCELLDORMANCY.DormancyPenaltyTime	Retain the default value.
Ignore CBC Switch	BSCEXSOFTPARA.IgnoreCbcSwitch	Use the recommended value. CBC services are generally used to send emergency and important information. Therefore, CBC takes precedence over dormancy, and this parameter is set to OFF by default. To increase the dormancy priority, set this parameter to ON . However, with this setting, the cell may already be in the dormancy state when CBC messages need to be sent, and waking it up takes time. As a result, CBC message broadcasting in the cell is delayed.
CBC Service Hold Duration	BSCEXSOFTPARA.CbcServiceHoldDuration	Use the recommended value.
Type of Cell Load Calculation	GCELLCHMGAD.LOADSTATYPE	Retain the default value.

Table 7-64 Parameters used for activating UMTS load-based RF module super dormancy

Parameter Name	Parameter ID	Setting Notes
Dormancy Switch	RRU.DORMANCYSW	To enable UMTS load-based RF module super dormancy, set this parameter to ON_SUPER or ON_DEEP_SUPER . The former value is recommended.
RRU Dormancy Optional Feature Switch	RRUEXTINFO.RRUDOROPTFEATSW	To enable UMTS load-based RF module super dormancy, select the ANYTIME_WAKEUP_SW option.

Parameter Name	Parameter ID	Setting Notes
Dormancy Algorithm Switch	ULOCCELLALGPARA.DORMANCYALGOSW	To enable UMTS load-based RF module super dormancy, select the DORMANCY_BY_LOAD_SW option.
Dormancy Cell Type	UCELL.DormancyCellType	Set this parameter to CAPACITY_CELL for capacity-layer cells. Set this parameter to BASIC_CELL for basic-layer cells.
Blind Handover Flag	UINTERFREQNCCELL.Blin dHoFlag	Set this parameter to TRUE when a basic-layer cell is configured as a co-coverage inter-band neighboring cell of the capacity-layer cell.
HandOver Switch	UCORRMALGOSWITCH. HoSwitch	To enable UMTS inter-frequency handovers, select the HO_INTER_FREQ_HARD_HO_SWITCH option.
CELL ID	UCELLDORMANCYPOLICY.CellId	Set this parameter based on the network plan and the following requirements: When UCELL.DormancyCellType is set to BASIC_CELL , UCELLDORMANCYPOLICY.CellId cannot be configured for the cell specified by UCELL.CellId . In other words, no energy saving policy can be configured for basic-layer cells.
Power Saving Policy Index	UCELLDORMANCYPOLICY.PolicyIndex	Set this parameter based on the network plan.
Dormancy Type	UCELLDORMANCYPOLICY.DormancyType	Set this parameter to INTRA_DORMANCY_BY_LOAD .

Parameter Name	Parameter ID	Setting Notes
Start Time	UCELLDORMANCYPOLICY.StartTime	Set these parameters based on the network plan and the following requirements: - For the same policy of the same energy saving type, the interval between the stop time and the start time must be longer than 20 minutes. If the stop time is earlier than or equal to the start time, the stop time is interpreted as occurring on the next day. For example: If the start time is 8:00 a.m. and the stop time is 7:00 a.m., the power saving function is enabled from 8:00 a.m. to 7:00 a.m. on the next day. - For different policies of the same energy saving type, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving type needs to be enabled after policy A of this type has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A.
Stop Time	UCELLDORMANCYPOLICY.StopTime	
Trigger Judge Period	UCELLDORMANCYPOLICY.TriggerJudgePeriod	Use the recommended value.

Parameter Name	Parameter ID	Setting Notes
Exit Judge Period	UCELLDORMANCYPOLICY.ExitJudgePeriod	Use the recommended value.
Penalty Time	UCELLDORMANCYPOLICY.PenaltyTime	Use the recommended value.
Forcible Trigger UE Number Threshold	UCELLDORMANCYPOLICY.ForcibleTriggerUeNumThld	Use the recommended value.
Cell Power Saving Total User Number Threshold	UCELLDORMANCYPOLICY.TotalUserNumThd	Use the recommended value.
Cell Power Saving Hsdpa User Number Threshold	UCELLDORMANCYPOLICY.HsdpaUserNumThd	Use the recommended value.
Cell Power Saving Hsupa User Number Threshold	UCELLDORMANCYPOLICY.HsupaUserNumThd	Use the recommended value.
Cell Power Saving Neighbour Cell Load Remain Threshold	UCELLDORMANCYPOLICY.NCellLdrRemainThd	Use the recommended value.
Considering User on Common Channel for Dormancy	UCELLDORMANCYPOLICY.DormancyCchUserSwitch	Use the recommended value.
Multi-carrier Cell Power Saving HSDPA User Number Threshold	UCELLDORMANCYPOLICY.McHsdpaUserNumThd	Use the recommended value.
Cell Power Saving Inter Plmn Handover Allowed Switch	URNCELLDORMANCYPARA.CellInterPlmnHoSwitch	Set this parameter to ON .
Intra RAT Inter Plmn Ho Allowed	UOPERATORSHARING-MODE.InterPlmnHoAllowedIntraRat	Set this parameter to YES .

Parameter Name	Parameter ID	Setting Notes
Cell Power Saving Ignore CBS Switch	URNCELLDORMANCY-PARA.<i>CellIgnoreCbsSwitch</i>	Use the recommended value. CBS is generally used to send emergency and important information. Therefore, CBS takes precedence over dormancy, and this parameter is set to OFF by default. To increase the dormancy priority, set this parameter to ON . However, with this setting, the cell may already be in the dormancy state when CBS messages need to be sent, and waking it up takes time. As a result, CBS message broadcasting in the cell is delayed.
CBS Service Hold Duration	URNCELLDORMANCY-PARA.<i>CbsServiceHoldDuration</i>	Use the recommended value.

Table 7-65 Parameters on the UMTS side used for activating GU load-based RF module super dormancy

Parameter Name	Parameter ID	Setting Notes
Dormancy Switch	RRU.<i>DORMANCYSW</i>	To enable GU load-based RF module super dormancy, set this parameter to ON_SUPER or ON_DEEP_SUPER . The former value is recommended.
RRU Dormancy Optional Feature Switch	RRUEXTINFO.<i>RRUDOROPTFEATSW</i>	To enable GU load-based RF module super dormancy, select the ANYTIME_WAKEUP_SW option.

Parameter Name	Parameter ID	Setting Notes
Dormancy Algorithm Switch	ULOCCELLALGPARA.DORMANCYALGOSW	To enable GU load-based RF module super dormancy, select the DORMANCY_BY_LOAD_SW option.
Power Saving Policy Index	UCELLDORMANCYPOLICY.PolicyIndex	Set this parameter based on the network plan.
Dormancy Type	UCELLDORMANCYPOLICY.DormancyType	Set this parameter to INTER_GU_DORMANCY_BY_LOAD .

Parameter Name	Parameter ID	Setting Notes
Start Time	UCELLDORMANCYPOLICY.StartTime	Set these parameters based on the network plan and the following requirements: - For the same policy of the same energy saving type, the interval between the stop time and the start time must be longer than 20 minutes. If the stop time is earlier than or equal to the start time, the stop time is interpreted as occurring on the next day. For example: If the start time is 8:00 a.m. and the stop time is 7:00 a.m., the power saving function is enabled from 8:00 a.m. to 7:00 a.m. on the next day. - For different policies of the same energy saving type, the interval between the stop time of one policy and the start time of another must be longer than 5 minutes. For example, if policy B of an energy saving type needs to be enabled after policy A of this type has been enabled, the start time of policy B must be over 5 minutes later than the stop time of policy A.
Stop Time	UCELLDORMANCYPOLICY.StopTime	
Trigger Judge Period	UCELLDORMANCYPOLICY.TriggerJudgePeriod	Use the recommended value.

Parameter Name	Parameter ID	Setting Notes
Exit Judge Period	UCELLDORMANCYPOLICY.ExitJudgePeriod	Use the recommended value.
Penalty Time	UCELLDORMANCYPOLICY.PenaltyTime	Use the recommended value.
Forcible Trigger UE Number Threshold	UCELLDORMANCYPOLICY.ForcibleTriggerUeNumThld	Use the recommended value.
Cell Power Saving Total User Number Threshold	UCELLDORMANCYPOLICY.TotalUserNumThd	Use the recommended value.
Considering User on Common Channel for Dormancy	UCELLDORMANCYPOLICY.DormancyCchUserSwitch	Use the recommended value.
Overlap Flag	U2GNCELL.NIRATOverlap	Set this parameter to TRUE when a basic-layer cell is configured as a co-coverage inter-RAT neighboring cell of the capacity-layer cell.
Inter RAT Inter Plmn Ho Allowed	UOPERATORSHARING-MODE.InterPlmnHoAllowedInterRat	Set this parameter to YES .
Switch for Inter-RAT Common Measurement	UMBSCCRRM.IRATCommonMeasSwitch	Turn on this switch if inter-RAT common measurement needs to be started.
HandOver Switch	UCORRMALGOSWITCH.HoSwitch	To enable inter-RAT handovers for UMTS, select the HO_INTER_RAT_CS_OUTPUT_SWITCH and HO_INTER_RAT_PS_OUTPUT_SWITCH options.
Total Trans Power Threshold for Cell Power Saving	UCELLDORMANCYPOLICY.IRATShutdownDLLoadThd	You are advised to set this parameter to its recommended value.
Total Receive Power Threshold for Cell Power Saving	UCELLDORMANCYPOLICY.IRATShutdownULLoadThd	You are advised to set this parameter to its recommended value.
Load Thre for Neigh GSM Cell for Cell Power Saving	UCELLDORMANCYPOLICY.IRATshutdownGSM-LoadThd	You are advised to set this parameter to its recommended value.

Parameter Name	Parameter ID	Setting Notes
Load Hyst Thre for Neigh GSM Cell for Cell Power Saving	UCELLDORMANCYPOLICY.IRATshutdownGSM-LoadHyst	You are advised to set this parameter to its recommended value.
Cell Power Saving Ignore CBS Switch	URNCCELLDORMANCY-PARA.CellIgnoreCbsSwitch	Use the recommended value. CBS is generally used to send emergency and important information. Therefore, CBS takes precedence over dormancy, and this parameter is set to OFF by default. To increase the dormancy priority, set this parameter to ON . However, with this setting, the cell may already be in the dormancy state when CBS messages need to be sent, and waking it up takes time. As a result, CBS message broadcasting in the cell is delayed.
CBS Service Hold Duration	URNCCELLDORMANCY-PARA.CbsServiceHoldDuration	Use the recommended value.

Table 7-66 Parameters on the GSM side used for activating GU load-based RF module super dormancy

Parameter Name	Parameter ID	Setting Notes
Information Exchange Switch at Iur-g	GCELLOTHEXT.IURGINFOCTRL	Set this parameter to YES when the GSM base station controller is BSC6910.
Inter-RAT Cell Dynamic Shutdown	GCELLSOFT.IRATShutdownSwitch	Set this parameter to ON .

7.4.4.1.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After

Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [7.4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

For details about configuring intra-base-station neighboring cells, see *SA Mobility Management in Connected Mode in 5G RAN Feature Documentation*.

Activating scheduled RF module super dormancy/NR local-cell-load-based RF module super dormancy

(Recommended) Scenario 1: Enabling scheduled RF module super dormancy and specifying multiple dormancy periods

```
//Enabling scheduled RF module super dormancy
//Configuring a board energy saving policy
ADD BRDPWRSVGPOLICY: POLICYID=1, POWERSAVINGTYPE=RF_DORMANCY;
//Setting the time for the board energy saving policy
ADD BRDPWRSVGPOLICYTIME: POLICYID=1, TIMEID=1, STARTTIME=00&10&00, STOPTIME=03&00&00;
ADD BRDPWRSVGPOLICYTIME: POLICYID=1, TIMEID=2, STARTTIME=04&00&00, STOPTIME=06&00&00;
//Configuring the energy saving policy for an RF module
MOD RRU: CN=0, SRN=60, SN=0, RT=AIRU, DORMANCYSW=ON_SUPER, PWRVGPOLICYID=1;
//Running the MOD RRUEXTINFO command to enable RF module super dormancy if the RRUEXTINFO MO
exists
MOD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=ANYTIME_WAKEUP_SW-1;
//Running the ADD RRUEXTINFO and PND DORMANCY commands to enable RF module super dormancy if
the RRUEXTINFO MO does not exist. Executing the ADD RRUEXTINFO command temporarily suspends base
station dormancy, preventing the base station from entering the dormancy state. If the PND DORMANCY
command is not executed, the base station can enter dormancy only after the pending duration (30
minutes) elapses and the entry conditions are met.
ADD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=ANYTIME_WAKEUP_SW-1;
PND DORMANCY: OPERATIONTYPE=UNPENDING;
//(Optional) Changing the lengths of the penalty timers applied to resource-related handovers to 30
seconds when the handover success rate decreases
MOD GNBMOBILITYCOMMPARAM: NecHoPrepFailPunishTimer=30, OptHoPrepFailPunishTimer=30;
//(Optional) Turning on the multi-RRU cell dormancy compatibility switch
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=MULTI_RRU_CELL_DORM_COMPAT_SW-1;
//(Optional) Enabling optimized evaluation of NR RF module dormancy
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=0,
PowerSavingAlgoSwitch=NR_DORMANCY_JUDGE_THLD_OPT_SW-1;
//(Optional) Configuring an energy saving policy for optimized evaluation of NR RF module dormancy
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=11,
PowerSavingType=NR_DORMANCY_JUDGE_THLD_OPT, StartTime=00&00&00, StopTime=06&00&00,
DlPrbThld=5, UserNumThld=2, EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=5,
UlPrbThld=5;
//(Optional, in SUL co-deployment scenarios) Modifying the energy saving policy
MOD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=11,
PowerSavingType=NR_DORMANCY_JUDGE_THLD_OPT, SulUserNumThld=4;
```

Scenario 2: Enabling scheduled RF module super dormancy and specifying a dormancy period

```
//Enabling scheduled RF module super dormancy and specifying a dormancy period
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=ON_SUPER, DORMANCYSTARTTIME=00&10&00,
DORMANCYSTOPTIME=06&00&00, DORMANCYTIMER=5, CUSTOMDORMANCYSTDSW=OFF,
PWRVGPOLICYID=0;
//Running the MOD RRUEXTINFO command to enable RF module super dormancy if the RRUEXTINFO MO
exists
MOD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=ANYTIME_WAKEUP_SW-1;
```

```
//Running the ADD RRUEXTINFO and PND DORMANCY commands to enable RF module super dormancy if
the RRUEXTINFO MO does not exist. Executing the ADD RRUEXTINFO command temporarily suspends base
station dormancy, preventing the base station from entering the dormancy state. If the PND DORMANCY
command is not executed, the base station can enter dormancy only after the pending duration (30
minutes) elapses and the entry conditions are met.
ADD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=ANYTIME_WAKEUP_SW-1;
PND DORMANCY: OPERATIONTYPE=UNPENDING;
//(Optional) Changing the lengths of the penalty timers applied to resource-related handovers to 30
seconds when the handover success rate decreases
MOD GNBMOBILITYCOMMPARAM: NecHoPrepFailPunishTimer=30, OptHoPrepFailPunishTimer=30;
//(Optional) Turning on the multi-RRU cell dormancy compatibility switch
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=MULTI_RRU_CELL_DORM_COMPAT_SW-1;
//(Optional) Enabling optimized evaluation of NR RF module dormancy
MOD NRDUCELLPWRSERVEALGO: NrDuCellId=0,
PowerSavingAlgoSwitch=NR_DORMANCY_JUDGE_THLD_OPT_SW-1;
//(Optional) Configuring an energy saving policy for optimized evaluation of NR RF module dormancy
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=11,
PowerSavingType=NR_DORMANCY_JUDGE_THLD_OPT, StartTime=00&00&00, StopTime=06&00&00,
DlPrbThld=5, UserNumThld=2, EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=5,
UlPrbThld=5;
//(Optional, in SUL co-deployment scenarios) Modifying the energy saving policy
MOD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=11,
PowerSavingType=NR_DORMANCY_JUDGE_THLD_OPT, SulUserNumThld=4;
```

The following configurations are required in scheduled UMTS cell dormancy scenarios:

```
//RNC side:
//(Optional) Allowing all CBS cells of the RNC to enter scheduled dormancy when CBS exists in UMTS cells
SET URNCCCELLDORMANCYPARA: LogicRncId=0, CellIgnoreCbsSwitch=ON, CbsServiceHoldDuration=5;

//Inter-frequency handover
//Supporting measurement-based handovers (from cell 0 in dormancy to cell 1 operating in a different
frequency band)
ADD UINTERFREQNCELL: RNCId=0, CellId=0, NCellRncId=0, NCellId=1;
//Enabling inter-frequency hard handover
SET UCORRMALGOSWITCH: HoSwitch=HO_INTER_FREQ_HARD_HO_SWITCH-1;

//Inter-RAT handover
//Selecting the HO_INTER_RAT_CS_OUT_SWITCH and HO_INTER_RAT_PS_OUT_SWITCH options
SET UCORRMALGOSWITCH:
HoSwitch=HO_INTER_RAT_CS_OUT_SWITCH-1&HO_INTER_RAT_PS_OUT_SWITCH-1;
//Supporting measurement-based inter-RAT handovers (from cell 0 in dormancy to GSM cell 1)
ADD U2GNCCELL: RNCId=0, CellId=0, GSMCellIndex=1;

//(Optional) Multi-operator sharing scenarios
//RNC side:

//Inter-frequency handover
//Performing the following configuration if cell 1 is intended to serve as the basic-layer cell of cell 0 when
they belong to different operators
SET URNCCCELLDORMANCYPARA: LogicRncId=0, CellInterPlmnHoSwitch=ON;
SET UOPERATORSHARINGMODE: LogicRncId=0, InterPlmnHoAllowedIntraRat=YES;

//Inter-RAT handover
//Allowing UEs to be handed over between inter-RAT inter-operator cells
SET UOPERATORSHARINGMODE: LogicRncId=0, InterPlmnHoAllowedInterRat=YES;
```

(Optional) GSM capacity-layer cell with CBC enabled

```
//BSC side:
//Setting IgnoreCbcSwitch to ON if all CBC cells under the BSC are allowed to enter load-based dormancy.
Otherwise, the default value OFF is recommended.
SET BSCEXSOFTPARA: IgnoreCbcSwitch=ON, CbcServiceHoldDuration=5;
```

(Optional) UMTS capacity-layer cell with CBS enabled

```
//RNC side:
//Setting CellIgnoreCbsSwitch to ON if all CBS cells under the RNC are allowed to enter load-based
```

dormancy. Otherwise, the default value OFF is recommended.
SET URNCCCELLDORMANCYPARA: LogicRncId=0, CellIgnoreCbsSwitch=ON, CbsServiceHoldDuration=5;

Activating NR load-based RF module super dormancy

//Turning on the RF module-level switches on the NR side. If the start time and stop time are the same, only load-based RF module super dormancy takes effect, while scheduled RF module super dormancy does not take effect.
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=ON_SUPER, DORMANCYSTARTTIME=00&00&00, DORMANCYSTOPTIME=00&00&00, DORMANCYTIMER=5, CUSTOMDORMANCYSTDSW=OFF;
//Running the MOD RRUEXTINFO command to enable RF module super dormancy if the RRUEXTINFO MO exists
MOD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=ANYTIME_WAKEUP_SW-1;
//Running the ADD RRUEXTINFO and PND DORMANCY commands to enable RF module super dormancy if the RRUEXTINFO MO does not exist. Executing the ADD RRUEXTINFO command temporarily suspends base station dormancy, preventing the base station from entering the dormancy state. If the PND DORMANCY command is not executed, the base station can enter dormancy only after the pending duration (30 minutes) elapses and the entry conditions are met.
ADD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=ANYTIME_WAKEUP_SW-1;
PND DORMANCY: OPERATIONTYPE=UNPENDING;
//Turning on the switch on the NR DU cell side
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=INTRA_NR_DORMANCY_BY_LOAD_SW-1;
//Configuring an energy saving policy in NR TDD low-frequency and NR FDD scenarios
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=10, PowerSavingType=INTRA_NR_DORMANCY_BY_LOAD, StartTime=00&00&00, StopTime=06&00&00, DlPrbThld=10, DlPrbOffset=10, UserNumThld=20, UserNumOffset=20, EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=5, ExitEvaluationPeriod=1, PwrSavingPenaltyTime=30, UlPrbThld=25, NCellUlPrbOffset=25;
//Configuring an energy saving policy in NR TDD high-frequency scenarios
ADD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=10, PowerSavingType=INTRA_NR_DORMANCY_BY_LOAD, StartTime=00&00&00, StopTime=06&00&00, DlPrbThld=10, DlPrbOffset=10, UserNumThld=20, UserNumOffset=20, EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=5, ExitEvaluationPeriod=1, PwrSavingPenaltyTime=30, UlPrbThld=25, NCellUlPrbOffset=25, MmwaveUserNumThld=20;
//Setting PowerSavingCellFlag, and (recommended) setting NoRmvFlag to FORBID_ANR_RMV for the bidirectional neighbor relationships between capacity- and basic-layer cells when automatic removal of neighboring cells in ANR takes effect
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=0, CellId=10, PowerSavingCellFlag=COVERAGE, NoRmvFlag=FORBID_ANR_RMV;
MOD NRCELLRELATION: NrCellId=10, Mcc="302", Mnc="220", gNBId=0, CellId=0, PowerSavingCellFlag=CAPACITY, NoRmvFlag=FORBID_ANR_RMV;
//((Optional) Configuring the precise dormancy switch and energy saving policy. The precise mode is not supported in high-frequency TDD.
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingExtSwitch=PRECISE_DORMANCY_SW-1;
MOD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=10, PowerSavingType=INTRA_NR_DORMANCY_BY_LOAD, ProtectDlPrbThld=50, ProtectDlPrbOffset=20, ProtectUlPrbThld=50, ProtectUlPrbOffset=20;
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=0, CellId=10, OverlapRate=40;
//((Optional) Turning on the grid mode switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingExtSwitch=GRID_MODE_SW-1;
//Setting the cell load filtering mode
MOD GNBPWRSAVINGALGO: CellLoadFilterMode=FAST_FILTER_MODE;
//((Optional) Changing the lengths of the penalty timers applied to resource-related handovers to 30 seconds when the handover success rate decreases
MOD GNBMOBILITYCOMMPARAM: NecHoPrepFailPunishTimer=30, OptHoPrepFailPunishTimer=30;

((Optional) Multi-operator sharing scenarios

//Setting the PlmnReservedFlag parameter in the NR external-cell PLMN list for the capacity-layer cell based on the network plan
ADD NREXTERNALNCELLPLMN: Mcc="302", Mnc="220", gNBId=0, CellId=10, SharedMcc=xx, SharedMnc=xx, PlmnReservedFlag=FALSE;

((Optional) In SUL co-deployment scenarios

//Modifying the energy saving policy
MOD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=10, PowerSavingType=INTRA_NR_DORMANCY_BY_LOAD, SulUserNumThld=4, SulUserNumOffset=4;

Activating LNR load-based RF module super dormancy

```
//LTE side:
//Turning on the LNR load-based RF module dormancy switch
MOD CELLPWRSAVINGALGO: LocalCellId=0, PwrSavingAlgoSwitch=LNR_LOAD_BASED_DORMANCY_SW-1;
//(Optional) Turning on the LTE-NR SA gNodeB X2 setup switch in inter-base-station SA networking
MOD ENODEBOAMPARAM: InterfaceSonConfigSwitch=LNR_SA_GNB_X2_SETUP_SW-1;
//(Optional) Turning on LTE-NR SA inter-base-station energy saving switch in inter-base-station SA
networking
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=LNR_INTER_BS_PWR_SAVE_SW-1;
//(Optional, only in NSA networking, NSA and SA hybrid networking, or inter-base-station SA networking)
Manually configuring all the neighbor relationships between LTE and NR cells with overlapping coverage
areas with the parameters set as follows: MCC is 460, MNC is 01, and gNodeB ID is 441233
MOD NRNRELATIONSHIP: LocalCellId=0, Mcc="460", Mnc="01", GnodebId=441233, CellId=1,
AggregationAttribute=CARRIER_SHUTDOWN_OVERLAP_FLAG-1;
//(Optional) Turning on the multi-RRU cell dormancy compatibility switch
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=MULTI_RRU_CELL_DORM_COMPAT_SW-1;
//(Optional) Turning on the RF module dormancy switch for the LTE sector split cell that serves as the basic-
layer cell for load-based dormancy
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=SPLIT_CELL_RF_DORMANCY_SW-1;
//NR side:
//Turning on the RF module-level switches on the NR side. If the start time and stop time are the same,
only load-based RF module super dormancy takes effect, while scheduled RF module super dormancy does
not take effect.
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=ON_SUPER, DORMANCYSTARTTIME=00&00&00,
DORMANCYSTOPTIME=00&00&00, DORMANCYTIMER=5, CUSTOMDORMANCYSTDSW=OFF;
//Running the MOD RRUEXTINFO command to enable RF module super dormancy if the RRUEXTINFO MO
exists
MOD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=ANYTIME_WAKEUP_SW-1;
//Running the ADD RRUEXTINFO and PND DORMANCY commands to enable RF module super dormancy if
the RRUEXTINFO MO does not exist. Executing the ADD RRUEXTINFO command temporarily suspends base
station dormancy, preventing the base station from entering the dormancy state. If the PND DORMANCY
command is not executed, the base station can enter dormancy only after the pending duration (30
minutes) elapses and the entry conditions are met.
ADD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=ANYTIME_WAKEUP_SW-1;
PND DORMANCY: OPERATIONTYPE=UNPENDING;
//Turning on the LNR load-based RF module super dormancy switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, PowerSavingSwitch=LNR_DORMANCY_BY_LOAD_SW-1;
//(Optional, required only in SA networking) Configuring all neighbor relationships in the overlapping
coverage area with the parameters set as follows: MCC is 460, MNC is 01, and eNodeB ID is 441282
MOD NRCELLEUTRANRELATION: NrCellId=1, Mcc="460", Mnc="01", EnodebId=441282, CellId=0,
PowerSavingOverlapInd=LNR_CARR_SHUTDOWN_OVERLAP_IND-1;
//Adding energy saving configurations
ADD NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=0, StartTime=00&00&00,
StopTime=06&00&00, PowerSavingType=LNR_DORMANCY_BY_LOAD, DIPrbThld=5, UIPrbThld=5,
UserNumThld=20, PwrSavingPenaltyTime=30, EutranDIPrbThld=30, EutranDIPrbOffset=30,
NCellUIPrbThld=30, NCellUIPrbOffset=30, EutranCarrShutdownRatio=90, EutranCarrShutdownExitOfs=70,
EutranNrUserNumThld=20, EutranNrUserNumThldOfs=20, EutranNrUserDIPrbThld=15,
EutranNrUserDIPrbThldOfs=15, EutranNrUserUIPrbThld=15, EutranNrUserUIPrbThldOfs=15,
EvalPeriodGranularity=MINUTE_LEVEL, ShutdownTrigJudgePeriod=1, ExitEvaluationPeriod=1;
//Setting the cell load filtering mode
MOD GNBPWRSAVINGALGO: CellLoadFilterMode=FAST_FILTER_MODE;
//(Optional, in SUL co-deployment scenarios) Modifying the energy saving policy
MOD NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=0,
PowerSavingType=LNR_DORMANCY_BY_LOAD, SulUserNumThld=4;
//(Optional) Turning on the inter-RAT level-by-level energy saving switch
MOD GNBPWRSAVINGALGO: PwrSavingAlgoSwitch=INTER_RAT_GRAD_PWR_SAVING_SW-1;
//(Optional) Changing the lengths of the penalty timers applied to resource-related handovers to 30
seconds when the handover success rate decreases
MOD GNBMOBILITYCOMMPARAM: NecHoPrepFailPunishTimer=30, OptHoPrepFailPunishTimer=30;
//(Optional) Turning on the switch for primary operator recognition for LNR load-based dormancy
MOD GNBPWRSAVINGALGO: PwrSavingAlgoSwitch=LNR_LOAD_DORM_PRIM_OP_RECOG_SW-1;
//(Optional) Turning on the SA X2 setup switch in inter-base-station SA networking
MOD GNBINTERFACEPARAM: X2ConfigSw=X2_SETUP_FOR_SA_SW-1;
//(Optional) Turning on LTE-NR SA inter-base-station energy saving switch in inter-base-station SA
networking
MOD GNBPWRSAVINGALGO: PwrSavingAlgoSwitch=SA_LNR_INTER_BS_PWR_SAVE_SW-1;
```

Activating LTE load-based RF module super dormancy

//Turning on the RF module-level switches on the LTE side. If the start time and stop time are the same, only load-based RF module super dormancy takes effect, while scheduled RF module super dormancy does

```
not take effect.
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=ON_SUPER, DORMANCYSTARTTIME=00&00&00,
DORMANCYSTOPTIME=00&00&00, DORMANCYTIMER=5, CUSTOMDORMANCYSTDSW=OFF;
//Running the MOD RRUEXTINFO command to enable RF module super dormancy if the RRUEXTINFO MO
exists
MOD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=ANYTIME_WAKEUP_SW-1;
//Running the ADD RRUEXTINFO and PND DORMANCY commands to enable RF module super dormancy if
the RRUEXTINFO MO does not exist. Executing the ADD RRUEXTINFO command temporarily suspends base
station dormancy, preventing the base station from entering the dormancy state. If the PND DORMANCY
command is not executed, the base station can enter dormancy only after the pending duration (30
minutes) elapses and the entry conditions are met.
ADD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=ANYTIME_WAKEUP_SW-1;
PND DORMANCY: OPERATIONTYPE=UNPENDING;
//Turning on the switch on the LTE cell side
MOD CELLPWRSAVINGALGO: LocalCellId=0, PwrSavingAlgoSwitch=INTRA_LTE_DORMANCY_BY_LOAD_SW-1;
//Adding an energy saving policy
ADD CELLPWRSAVINGPOLICY: LocalCellId=0, PolicyIndex=0,
PowerSavingType=INTRA_LTE_DORMANCY_BY_LOAD, StartTime=00&00&00, StopTime=06&00&00,
DlPrbThld=10, DlPrbOffset=10, UlPrbThld=10, UlPrbOffset=10, UENumThld=10, ExitJudgePeriod=24,
EvalInterval=EVAL_INTERVAL_60S;
//Configuring the overlap indicator extension for an inter-frequency neighboring cell
MOD EUTRANINTERFREQNCELL: LocalCellId=0, Mcc="460", Mnc="00", eNodeBId=720800, CellId=1,
OverlapIndicatorExtension=POWER_SAVING_OVERLAP_IND-1;
//(Optional) Turning on the intra-RAT level-by-level energy saving switch
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=INTRA_RAT_GRAD_PWR_SAVING_SW-1;
//(Optional) Setting the inter-eNodeB information exchange mode
MOD ENODEBPWRSAVINGALGO: InterEnbInfoExchangeMode=SLOW_EXCHANGE_MODE;
//Setting the cell load filtering mode
MOD ENODEBPWRSAVINGALGO: CellLoadFilterMode=FAST_FILTER_MODE;
//(Optional) Turning on the RF module dormancy switch for the LTE sector split cell that serves as the basic-
layer cell for load-based dormancy
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=SPLIT_CELL_RF_DORMANCY_SW-1;
//(Optional) Setting the inter-frequency MLB threshold and load offset
MOD CELLMLB: LocalCellId=0, InterFreqMlbThd=40, LoadOffset=8;
```

Activating LTE load-based precise dormancy

```
//Turning on the RF module-level switches on the LTE side. If the start time and stop time are the same,
only load-based RF module super dormancy takes effect, while scheduled RF module super dormancy does
not take effect.
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=ON_SUPER, DORMANCYSTARTTIME=00&00&00,
DORMANCYSTOPTIME=00&00&00, DORMANCYTIMER=5, CUSTOMDORMANCYSTDSW=OFF;
//Running the MOD RRUEXTINFO command to enable RF module super dormancy if the RRUEXTINFO MO
exists
MOD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=ANYTIME_WAKEUP_SW-1;
//Running the ADD RRUEXTINFO and PND DORMANCY commands to enable RF module super dormancy if
the RRUEXTINFO MO does not exist. Executing the ADD RRUEXTINFO command temporarily suspends base
station dormancy, preventing the base station from entering the dormancy state. If the PND DORMANCY
command is not executed, the base station can enter dormancy only after the pending duration (30
minutes) elapses and the entry conditions are met.
ADD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=ANYTIME_WAKEUP_SW-1;
PND DORMANCY: OPERATIONTYPE=UNPENDING;
//Turning on the switch for LTE load-based precise dormancy and setting the dormancy cell type
MOD CELLPWRSAVINGALGO: LocalCellId=0, PwrSavingAlgoSwitch=LTE_DORMANCY_BY_LOAD_PE_SW-1,
DormancyCellType=CAPACITY_CELL;
MOD CELLPWRSAVINGALGO: LocalCellId=1, PwrSavingAlgoSwitch=LTE_DORMANCY_BY_LOAD_PE_SW-1,
DormancyCellType=BASIC_CELL;
//(Optional) Turning on the overlap grid mode switch
MOD CELLPWRSAVINGALGO: LocalCellId=0, PwrSavingAlgoSwitch=OVERLAP_GRID_MODE_SW-1;
MOD CELLPWRSAVINGALGO: LocalCellId=1, PwrSavingAlgoSwitch=OVERLAP_GRID_MODE_SW-1;
//Adding an energy saving policy
ADD CELLPWRSAVINGPOLICY: LocalCellId=0, PolicyIndex=0,
PowerSavingType=LTE_PRECISE_DORMANCY_BY_LOAD, StartTime=00&00&00, StopTime=06&00&00,
DlPrbThld=10, DlPrbOffset=10, UlPrbThld=10, UlPrbOffset=10, UENumThld=10, ExitJudgePeriod=24,
EvalInterval=EVAL_INTERVAL_60S, ProtectDlPrbThld=50, ProtectDlPrbOffset=20, ProtectUlPrbThld=50,
ProtectUlPrbOffset=20;
//Setting the overlap rate for an inter-frequency neighboring cell
MOD EUTRANINTERFREQNCELL: LocalCellId=0, Mcc="460", Mnc="00", eNodeBId=720800, CellId=1,
OverlapRate=40;
//(Optional) Turning on the intra-RAT level-by-level energy saving switch
```



```
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=INTRA_RAT_GRAD_PWR_SAVING_SW-1;  
//(Optional) Setting the inter-eNodeB information exchange mode  
MOD ENODEBPWRSAVINGALGO: InterEnbInfoExchangeMode=SLOW_EXCHANGE_MODE;  
//Setting the cell load filtering mode  
MOD ENODEBPWRSAVINGALGO: CellLoadFilterMode=FAST_FILTER_MODE;  
//(Optional) Turning on the RF module dormancy switch for the LTE sector split cell that serves as the basic-  
layer cell for load-based dormancy  
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=SPLIT_CELL_RF_DORMANCY_SW-1;  
//(Optional) Setting the inter-frequency MLB threshold and load offset  
MOD CELLMLB: LocalCellId=0, InterFreqMlbThd=40, LoadOffset=8;
```

(Optional) Multi-operator sharing scenarios

```
//Enabling optimization for cell PLMN reservation scenarios  
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=CELL_PLMN_RSVD_OPT_SW-1;  
//Setting PLMN_RESERVED_FLAG in the external-cell PLMN list based on the network plan  
ADD EUTRANEXTERNALCELLPLMN: Mcc="460", Mnc="11", eNodeBId=222, CellId=1, ShareMcc="460",  
ShareMnc="20", AggregationAttribute=PLMN_RESERVED_FLAG-1;  
//Turning on the switch for broadcast of operator-specific cell reservation status over the X2 interface based  
on the network plan  
MOD ENODEBOAMPARAM: ConfigAdaptOptSwitch=RSVD_FOR_OP_X2_BROADCAST_SW-1;
```

Activating GSM load-based RF module super dormancy

```
//eGBTS side:  
//Turning on the RF module-level switches on the GSM side. If the start time and stop time are the same,  
only load-based RF module super dormancy takes effect, while scheduled RF module super dormancy does  
not take effect.  
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=ON_SUPER, DORMANCYSTARTTIME=00&00&00,  
DORMANCYSTOPTIME=00&00&00, DORMANCYTIMER=5, CUSTOMDORMANCYSTDSW=OFF;  
//Running the MOD RRUEXTINFO command to enable RF module super dormancy if the RRUEXTINFO MO  
exists  
MOD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=ANYTIME_WAKEUP_SW-1;  
//Running the ADD RRUEXTINFO and PND DORMANCY commands to enable RF module super dormancy if  
the RRUEXTINFO MO does not exist. Executing the ADD RRUEXTINFO command temporarily suspends base  
station dormancy, preventing the base station from entering the dormancy state. If the PND DORMANCY  
command is not executed, the base station can enter dormancy only after the pending duration (30  
minutes) elapses and the entry conditions are met.  
ADD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=ANYTIME_WAKEUP_SW-1;  
PND DORMANCY: OPERATIONTYPE=UNPENDING;  
  
//BSC side:  
//Turning on the switch for the GSM cell. No co-coverage cell needs to be configured for a concentric cell.  
However, for a non-concentric cell, at least one co-coverage cell must be configured, and its coverage area  
must cover the cell expected to enter dormancy.  
SET GCELLOTH PARA: IDTYPE=BYID, CELLID=123, DormancyEnable=ENABLE, CoCoverageCellId1=1,  
CoCoverageCellId2=2, CoCoverageCellId3=3, CoCoverageCellId4=4;  
//Adding an energy saving policy  
ADD GCELLDORMANCY: IDTYPE=BYID, CELLID=123, DormancyPolicyIndex=0,  
DormancyCellStrTime=00&00&00, DormancyCellStpTime=06&00&00, DormancyCellLoadStatTm=5,  
DormancyLoadThrD=50, DormancyCellChanNum=6, WakeUpLoadThrD=80, DormancyPenaltyTime=30;
```

(Optional) GSM cell with CBC enabled

```
//BSC side:  
//Setting IgnoreCbcSwitch to ON if all CBC cells under the BSC are allowed to enter load-based dormancy.  
Otherwise, the default value OFF is recommended.  
SET BSCEXSOFTPARA: IgnoreCbcSwitch=ON, CbcServiceHoldDuration=5;
```

Activating UMTS load-based RF module super dormancy

```
//NodeB side:  
//Turning on the RF module-level switches on the UMTS side. If the start time and stop time are the same,  
only load-based RF module super dormancy takes effect, while scheduled RF module super dormancy does  
not take effect.  
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=ON_SUPER, DORMANCYSTARTTIME=00&00&00,  
DORMANCYSTOPTIME=00&00&00, DORMANCYTIMER=5, CUSTOMDORMANCYSTDSW=OFF;  
//Running the MOD RRUEXTINFO command to enable RF module super dormancy if the RRUEXTINFO MO  
exists  
MOD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=ANYTIME_WAKEUP_SW-1;
```

```
//Running the ADD RRUEXTINFO and PND DORMANCY commands to enable RF module super dormancy if
the RRUEXTINFO MO does not exist. Executing the ADD RRUEXTINFO command temporarily suspends base
station dormancy, preventing the base station from entering the dormancy state. If the PND DORMANCY
command is not executed, the base station can enter dormancy only after the pending duration (30
minutes) elapses and the entry conditions are met.
ADD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=ANYTIME_WAKEUP_SW-1;
PND DORMANCY: OPERATIONTYPE=UNPENDING;
//Turning on the switch for the UMTS cell, after which the base station enters the dormancy-pending state
SET ULOCELLPWRSAV PARA: ULOCELLID=0, DORMANCYALGOSW=DORMANCY_BY_LOAD_SW-1;
//Canceling the dormancy-pending state of the base station
PND DORMANCY: OPERATIONTYPE=UNPENDING;

//RNC side:
//Configuring cell 0 as the capacity-layer cell and cell 1 as the basic-layer cell
MOD UCELL: LogicRncId=0, CellId=0, DormancyCellType=CAPACITY_CELL;
MOD UCELL: LogicRncId=0, CellId=1, DormancyCellType=BASIC_CELL;
//Setting cell 1 as an inter-frequency co-coverage neighbor of cell 0 with BlindHoFlag set to TRUE
ADD UINTERFREQNCELL: RNCId=0, CellId=0, NCellRncId=0, NCellId=1, BlindHoFlag=TRUE;
//((Optional) Setting cell 1 as an inter-frequency co-coverage neighbor of cell 0 using the MOD command if
the neighbor relationship has been configured
MOD UINTERFREQNCELL: RNCId=0, CellId=0, NCellRncId=0, NCellId=1, BlindHoFlag=TRUE;
//Enabling inter-frequency hard handover
SET UCORRMALGOSWITCH: HoSwitch=HO_INTER_FREQ_HARD_HO_SWITCH-1;
//Adding an energy saving policy for capacity-layer cell 0
ADD UCELLDORMANCYPOLICY: LogicRncId=0, CellId=0, PolicyIndex=0,
DormancyType=INTRA_DORMANCY_BY_LOAD, StartTime=00&00&00, StopTime=06&00&00,
TriggerJudgePeriod=5, ExitJudgePeriod=24, PenaltyTime=30, ForcibleTriggerUeNumThld=0,
TotalUserNumThld=1, HsdpaUserNumThld=0, HsupaUserNumThld=0, NCellLdrRemainThld=20,
DormancyCchUserSwitch=ON, McHsdpaUserNumThld=2;
```

(Optional) Multi-operator sharing scenarios

```
//RNC side:
//Performing the following configuration if cell 1 is intended to serve as the basic-layer cell of cell 0 when
they belong to different operators
SET URNCCCELLDORMANCYPARA: LogicRncId=0, CellInterPlmnHoSwitch=ON;
SET UOPERATORSHARINGMODE: LogicRncId=0, InterPlmnHoAllowedIntraRat=YES;
```

(Optional) Capacity-layer cell with CBS enabled

```
//RNC side:
//Setting CellIgnoreCbsSwitch to ON if all CBS cells under the RNC are allowed to enter load-based
dormancy. Otherwise, the default value OFF is recommended.
SET URNCCCELLDORMANCYPARA: LogicRncId=0, CellIgnoreCbsSwitch=ON, CbsServiceHoldDuration=5;
```

Activating GU load-based RF module super dormancy

```
//NodeB side:
//Turning on the RF module-level switches on the UMTS side. If the start time and stop time are the same,
only load-based RF module super dormancy takes effect, while scheduled RF module super dormancy does
not take effect.
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=ON_SUPER, DORMANCYSTARTTIME=00&00&00,
DORMANCYSTOPTIME=00&00&00, DORMANCYTIMER=5, CUSTOMDORMANCYSTDSW=OFF;
//Running the MOD RRUEXTINFO command to enable RF module super dormancy if the RRUEXTINFO MO
exists
MOD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=ANYTIME_WAKEUP_SW-1;
//Running the ADD RRUEXTINFO and PND DORMANCY commands to enable RF module super dormancy if
the RRUEXTINFO MO does not exist. Executing the ADD RRUEXTINFO command temporarily suspends base
station dormancy, preventing the base station from entering the dormancy state. If the PND DORMANCY
command is not executed, the base station can enter dormancy only after the pending duration (30
minutes) elapses and the entry conditions are met.
ADD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=ANYTIME_WAKEUP_SW-1;
PND DORMANCY: OPERATIONTYPE=UNPENDING;
//Turning on the switch for the UMTS cell, after which the base station enters the dormancy-pending state
SET ULOCELLPWRSAV PARA: ULOCELLID=0, DORMANCYALGOSW=DORMANCY_BY_LOAD_SW-1;
//Canceling the dormancy-pending state of the base station
PND DORMANCY: OPERATIONTYPE=UNPENDING;

//RNC side:
//Turning on the switch for inter-RAT common measurement
```

```
SET UMBSCRRM: LogicRncId=0, IRATCommonMeasSwitch=ON;
//Selecting the HO_INTER_RAT_CS_OUT_SWITCH and HO_INTER_RAT_PS_OUT_SWITCH options
SET UCORRMALGOSWITCH:
HoSwitch=HO_INTER_RAT_CS_OUT_SWITCH-1&HO_INTER_RAT_PS_OUT_SWITCH-1;
//Setting GSM cell 1 as the basic-layer cell of UMTS cell 0 (capacity-layer cell)
ADD U2GNCELL: RNCId=0, CellId=0, GSMCellIndex=1, NIRATOverLap=TRUE;
//Optional Setting GSM cell 1 as the basic-layer cell of UMTS cell 0 (capacity-layer cell) using the MOD
command if the neighbor relationship has been configured
MOD U2GNCELL: RNCId=0, CellId=0, GSMCellIndex=1, NIRATOverLap=TRUE;
//Adding an energy saving policy for capacity-layer cell 0
ADD UCELLDORMANCYPOLICY: LogicRncId=0, CellId=0, PolicyIndex=1,
DormancyType=INTER_GU_DORMANCY_BY_LOAD, StartTime=00&00&00, StopTime=06&00&00,
TriggerJudgePeriod=5, ExitJudgePeriod=24, PenaltyTime=30, ForcibleTriggerUeNumThld=0,
TotalUserNumThd=2, DormancyCchUserSwitch=ON, IRATShutdownDLLoadThd=30,
IRATShutdownULLoadThd=10, IRATshutdownGSMLoadThd=25, IRATshutdownGSMLoadHyst=20;

//BSC side:
//Enabling the function of information exchange on the lur-g interface (only on the BSC6910, not on the
BSC6960)
SET GCELLOTHEXT: IDTYPE=BYID, CELLID=1, IURGINFOCTRL=YES;
//Enabling inter-RAT cell dynamic shutdown on the BSC side
SET GCELLSOFT: IDTYPE=BYID, CELLID=1, IRATShutdownSwitch=ON;
```

(Optional) Multi-operator sharing scenarios

```
//RNC side:
//Allowing UEs to be handed over between inter-RAT inter-operator cells
SET UOPERATORSHARINGMODE: LogicRncId=0, InterPlmnHoAllowedInterRat=YES;
```

(Optional) Capacity-layer cell with CBS enabled

```
//RNC side:
//Setting CellIgnoreCbsSwitch to ON if all CBS cells under the RNC are allowed to enter load-based
dormancy. Otherwise, the default value OFF is recommended.
SET URNCCELLDORMANCYPARA: LogicRncId=0, CellIgnoreCbsSwitch=ON, CbsServiceHoldDuration=5;
```

Setting parameters shared by preceding sub-functions

```
//(Optional) Turning on the RF module super dormancy compatibility switch if super dormancy is required
in scenarios using non-standard optical modules
SET EQUIPMENT: EQMDOROPTFEATSW=RF_SDC_SW-1;
//(Optional) Turning on the switch controlling "forced fronthaul link detection when RF module super
dormancy is to be enabled" if fronthaul link detection is required
SET EQUIPMENT: EQMDOROPTFEATSW=RF_SDFD_SW-1;
//(Optional) Turning on the dormancy loose mode switch if the NE is connected to the NetEco or an RPCU
and super dormancy is allowed
SET EQUIPMENT: EQMDOROPTFEATSW=DOR_LM_SW-1;
//(Optional) Turning on the ALD compatibility switch for RF module super dormancy if super dormancy is
allowed in scenarios using non-standard ALDs
SET EQUIPMENT: EQMDOROPTFEATSW=RF_ALD_SW-1;
//(Optional) Turning on the dormancy and maintenance mode control switch
SET EQUIPMENT: EQMDOROPTFEATSW=DORM_MNTMODE_CTRL_SW-1;
//(Optional) Turning on the switch controlling wake-up algorithm optimization for RF module super
dormancy if the service recovery duration for RF modules needs to be shortened
SET EQUIPMENT: EQMDOROPTFEATSW=RF_SDWAO_SW-1;

//(Optional) Enabling super dormancy pending
PND DORMANCY: OPERATIONTYPE=PENDING, PENDINGDURATION=30;
```

Optimization Command Examples

```
//Turning on the simulated load dormancy impact switch
MOD GNBPWRSAVINGALGO: PwrSavingAlgoSwitch=SIMU_LOAD_DORMANCY_IMP_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

Deactivating scheduled RF module super dormancy

```
//(Optional) Disabling optimized evaluation of NR RF module dormancy
MOD NRDUCELLPWRSAVEALGO: NrDuCellId=0,
PowerSavingAlgoSwitch=NR_DORMANCY_JUDGE_THLD_OPT_SW-0;
//Removing the energy saving configuration of optimized evaluation of NR RF module dormancy
RMV NRDUCELLPOWERSAVING: NrDuCellId=0, PowerSavingPolicyIndex=11;
//(Optional) Turning off the multi-RRU cell dormancy compatibility switch
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=MULTI_RRU_CELL_DORM_COMPAT_SW-0;
//Disabling RF module super dormancy
MOD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=ANYTIME_WAKEUP_SW-0;
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=OFF;
```

Deactivating NR load-based RF module super dormancy

```
//Disabling RF module super dormancy
MOD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=ANYTIME_WAKEUP_SW-0;
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=OFF;
//Turning off the switches for precise dormancy. The precise mode is not supported in high-frequency TDD.
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingExtSwitch=GRID_MODE_SW-0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingExtSwitch=PRECISE_DORMANCY_SW-0;
//Turning off the switch for NR load-based RF module super dormancy on the NR DU cell side
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, PowerSavingSwitch=INTRA_NR_DORMANCY_BY_LOAD_SW-0;
```

Deactivating LNR load-based RF module super dormancy

```
//Disabling RF module super dormancy
MOD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=ANYTIME_WAKEUP_SW-0;
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=OFF;
//LTE side:
//(Optional) Turning off the multi-RRU cell dormancy compatibility switch
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=MULTI_RRU_CELL_DORM_COMPAT_SW-0;
//(Optional) Turning off the LTE-NR SA inter-base-station energy saving switch in inter-base-station SA networking
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=LNR_INTER_BS_PWR_SAVE_SW-0;
//(Optional) Turning off the LTE-NR SA gNodeB X2 setup switch in inter-base-station SA networking
MOD ENODEBOAMPARAM: InterfaceSonConfigSwitch=LNR_SA_GNB_X2_SETUP_SW-0;
//Turning off the LNR load-based RF module super dormancy switch
MOD CELLPWRSAVINGALGO: LocalCellId=0, PwrSavingAlgoSwitch=LNR_LOAD_BASED_DORMANCY_SW-0;
//(Optional, required only when the neighbor relationship between LTE and NR cells with overlapping coverage areas is manually configured in NSA networking, NSA and SA hybrid networking, or inter-base-station SA networking) Removing the neighbor relationship for energy saving in the overlapping coverage areas
MOD NRNRELATIONSHIP: LocalCellId=0, Mcc="460", Mnc="01", GnodebId=441233, CellId=1,
AggregationAttribute=CARRIER_SHUTDOWN_OVERLAP_FLAG-0;
//NR side:
//(Optional) Turning off the switch for primary operator recognition for LNR load-based dormancy
MOD GNBPWRSAVINGALGO: PwrSavingAlgoSwitch=LNR_LOAD_DORM_PRIM_OP_RECOG_SW-0;
//(Optional) Turning off the LTE-NR SA inter-base-station energy saving switch in inter-base-station SA networking
MOD GNBPWRSAVINGALGO: PwrSavingAlgoSwitch=SA_LNR_INTER_BS_PWR_SAVE_SW-0;
//(Optional) Turning off the SA X2 setup switch in inter-base-station SA networking
MOD GNBINTERFACEPARAM: X2ConfigSw=X2_SETUP_FOR_SA_SW-0;
//Turning off the LNR load-based RF module super dormancy switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, PowerSavingSwitch=LNR_DORMANCY_BY_LOAD_SW-0;
//(Optional, required only in SA networking) Removing the overlapping relationship (between the local cell and the neighboring LTE cell) for energy saving
MOD NRCELLEUTRANRELATION: NrCellId=1, Mcc="460", Mnc="01", EnodebId=441282, CellId=0,
PowerSavingOverlapInd=LNR_CARR_SHUTDOWN_OVERLAP_IND-0;
//Removing the energy saving policy
RMV NRDUCELLPOWERSAVING: NrDuCellId=1, PowerSavingPolicyIndex=0;
```

Deactivating LTE load-based RF module super dormancy

```
//Disabling RF module super dormancy
MOD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=ANYTIME_WAKEUP_SW-0;
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=OFF;
//Turning off the switch on the LTE cell side
MOD CELLPWRSAVINGALGO: LocalCellId=0, PwrSavingAlgoSwitch=INTRA_LTE_DORMANCY_BY_LOAD_SW-0;
//Optional Turning off the intra-RAT level-by-level energy saving switch
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=INTRA_RAT_GRAD_PWR_SAVING_SW-0;
```

Deactivating LTE load-based precise dormancy

```
//Disabling RF module super dormancy
MOD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=ANYTIME_WAKEUP_SW-0;
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=OFF;
//Optional Turning off the overlap grid mode switch
MOD CELLPWRSAVINGALGO: LocalCellId=0, PwrSavingAlgoSwitch=OVERLAP_GRID_MODE_SW-0;
MOD CELLPWRSAVINGALGO: LocalCellId=1, PwrSavingAlgoSwitch=OVERLAP_GRID_MODE_SW-0;
//Turning off the switch for LTE load-based precise dormancy
MOD CELLPWRSAVINGALGO: LocalCellId=0, PwrSavingAlgoSwitch=LTE_DORMANCY_BY_LOAD_PE_SW-0,
DormancyCellType=CAPACITY_CELL;
MOD CELLPWRSAVINGALGO: LocalCellId=1, PwrSavingAlgoSwitch=LTE_DORMANCY_BY_LOAD_PE_SW-0,
DormancyCellType=CAPACITY_CELL;
//Removing the energy saving policy
RMV CELLPWRSAVINGPOLICY: LocalCellId=0, PolicyIndex=0;
//Optional Turning off the intra-RAT level-by-level energy saving switch
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=INTRA_RAT_GRAD_PWR_SAVING_SW-0;
```

(Optional) Multi-operator sharing scenarios

```
//Turning off the switch for broadcast of operator-specific cell reservation status over the X2 interface based
on the network plan
MOD ENODEBOAMPARAM: ConfigAdaptOptSwitch=RSVD_FOR_OP_X2_BROADCAST_SW-0;
//Disabling optimization for cell PLMN reservation scenarios
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=CELL_PLMN_RSVD_OPT_SW-0;
```

Deactivating GSM load-based RF module super dormancy

```
//eGBTS side:
//Disabling RF module super dormancy
MOD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=ANYTIME_WAKEUP_SW-0;
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=OFF;

//BSC side:
//Removing the energy saving policy
RMV GCELLDORMANCY: IDTYPE=BYID, CELLID=123, DormancyPolicyIndex=0;
//Disabling cell dormancy allowance
SET GCELLOTHPARA: IDTYPE=BYID, CELLID=123, DormancyEnable=DISABLE;
```

Deactivating UMTS load-based RF module super dormancy

```
//NodeB side:
//Disabling RF module super dormancy
MOD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=ANYTIME_WAKEUP_SW-0;
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=OFF;
//Turning off the switch for the UMTS cell, after which the base station enters the dormancy-pending state
SET ULOCELLPWRSVAPARA: ULOCELLID=0, DORMANCYALGOSW=DORMANCY_BY_LOAD_SW-0;
//Canceling the dormancy-pending state of the base station
PND DORMANCY: OPERATIONTYPE=UNPENDING;

//RNC side:
//Removing the energy saving policy
RMV UCELLDORMANCYPOLICY: LogicRncId=0, CellId=0, PolicyIndex=0;
```

Deactivating GU load-based RF module super dormancy

```
//NodeB side:
//Disabling RF module super dormancy
MOD RRUEXTINFO: CN=0, SRN=60, SN=0, RRUDOROPTFEATSW=ANYTIME_WAKEUP_SW-0;
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=OFF;
```

```
//Turning off the switch for the UMTS cell, after which the base station enters the dormancy-pending state
SET ULOCCELLPWRSVAPARA: ULOCCELLID=0, DORMANCYALGOSW=DORMANCY_BY_LOAD_SW-0;
//Canceling the dormancy-pending state of the base station
PND DORMANCY: OPERATIONTYPE=UNPENDING;

//RNC side:
//Removing the energy saving policy
RMV UCELLDORMANCYPOLICY: LogicRncId=0, CellId=0, PolicyIndex=1;
```

Performing deactivation using parameters shared by preceding sub-functions

```
//(Optional) Turning off the RF module super dormancy compatibility switch if super dormancy is required
in scenarios using non-standard optical modules
SET EQUIPMENT: EQMDOROPTFEATSW=RF_SDC_SW-0;
//(Optional) Turning off the switch controlling "forced fronthaul link detection when RF module super
dormancy is to be enabled" if fronthaul link detection is required
SET EQUIPMENT: EQMDOROPTFEATSW=RF_SDFFD_SW-0;
//(Optional) Turning off the dormancy loose mode switch if the NE is connected to the NetEco or an RPCU
and super dormancy is allowed
SET EQUIPMENT: EQMDOROPTFEATSW=DOR_LM_SW-0;
//(Optional) Turning off the ALD compatibility switch for RF module super dormancy if super dormancy is
allowed in scenarios using non-standard ALDs
SET EQUIPMENT: EQMDOROPTFEATSW=RF_ALD_SW-0;
//(Optional) Turning off the dormancy and maintenance mode control switch
SET EQUIPMENT: EQMDOROPTFEATSW=DORM_MNTMODE_CTRL_SW-0;
//(Optional) Turning off the switch controlling wake-up algorithm optimization for RF module super
dormancy if RF modules are required to wake up and restore services based on the original mechanism
SET EQUIPMENT: EQMDOROPTFEATSW=RF_SDWAO_SW-0;
```

7.4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.4.4.2 Activation Verification

The activation of this function can be verified using the following common methods:

- You can run the **DSP BRDDORMANCY** command to observe the current or latest RF module dormancy status.
 - Run the **DSP BRDDORMANCY** command with the **QUERYTYPE** parameter set to **BOARD** and the **QUERYMODE** parameter set to **CURRENT**. The values displayed in the query results are periodically refreshed and may not always reflect the most recent updates.
 - If the value of **Board Dormancy State** in the command output is **Super Dormancy**, RF module super dormancy has taken effect.
 - If the value of **Board Dormancy State** in the command output is **Not in the Dormancy State**, check the value of **Reason for Not Entering Super Dormancy** to determine the reason why the feature does not take effect. [Table 7-67](#) lists the possible reasons.
 - Run the **DSP BRDDORMANCY** command with the **QUERYTYPE** parameter set to **BOARD**, the **QUERYMODE** parameter set to **HISTORY**, and the **PWRSAVINGFEATURE** parameter set to **SUPER_DORMANCY**. You can determine the possible reasons why this function does not take

effect based on the values of **Latest Exit Cause** and **Latest Disabled Cause** within the specified period. [Table 7-67](#) lists the possible reasons. [Table 7-68](#) describes the meanings of **Latest Exit Cause**, **Latest Disabled Cause**, **Latest Exit Time**, and **Latest Disabled Time**.

Latest Exit Cause and **Latest Disabled Cause** indicate the reasons for the most recent exit and failure of the function. These reasons can be queried after the effective period has ended. For example, if the effective period is from 00:00:00 to 06:00:00, **Latest Exit Cause** and **Latest Disabled Cause** record the reasons for the most recent exit and failure of the function from 00:00:00 to 06:00:00. These reasons can be queried by an operation performed within the time from 06:00:00 to 23:59:59.

- Only scheduled RF module super dormancy supports the query of the **Latest Exit Cause**, **Latest Disabled Cause**, **Latest Exit Time**, and **Latest Disabled Time** fields by running the **DSP BRDDORMANCY** command.
- The collection period is 10 minutes. If the causes change frequently, the latest causes may not be updated in a timely manner.
- In separate-MPT scenarios where the system time is not strictly aligned, the cause for an exit from dormancy is not recorded if the remaining time is less than 1 minute.
- When the **QUERYMODE** parameter is set to **HISTORY**, values of **Process Being Protected**, **Not All NE Dormancy Notifications Received**, and **Carriers Existed on the RF Module** are not included in the command output.

Table 7-67 Possible reasons why board dormancy does not take effect

Cause	Description
Maintenance Link Abnormal	The RF module is disconnected, and the BBU cannot obtain the dormancy state of the RF module.
Board Unavailable	The link is normal, but the board is unavailable.
Dormancy Capability Not Supported by the Hardware	• The RF module does not support super dormancy. • The BBP does not support super dormancy.

Cause	Description
Configuration Conditions Not Met	- The dormancy switch is turned off. - No eGBTS, NodeB, eNodeB, or gNodeB application is available. - When the customized dormancy standard switch is turned on, the dormancy RATs do not include all RATs of the local end. - The RF module is blocked. - When the DORM_MNTMODE_CTRL_SW option of the EQUIPMENT.EQMDOROPTFEATSW (LTE eNodeB, 5G gNodeB) parameter is deselected, the following conditions related to the maintenance mode are not met: - The RF module must be in the normal state. - The local NE must be in the normal state. - All the APPs of the local NE must be in the normal state. - In co-SDR scenarios, the local NE is not in the normal state. - The RF module is powered by the iSDU, and the customized timing power-off switch is turned on for the power supply port on the iSDU (by setting the PDPATTR.CTSSW (LTE eNodeB, 5G gNodeB) parameter to ON).
Requirements Not Met by the Peer NE of the Separate-MPT Base Station	In separate-MPT scenarios, the local end meets the entry conditions, but the peer end does not. The reasons why dormancy does not take effect at the peer end can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD . Table 7-67 lists the possible reasons.
License Insufficiency	- The license items for super dormancy are insufficient. - The license items affecting basic cell service capabilities (such as the cell access capability) are insufficient.
Impacted Features Exist	- RF module deep dormancy is in effect. - RF module deep dormancy is in the process of taking effect. - RF module deep dormancy is in the exit process.

Cause	Description
Not Within Dormancy Period	- The current time is beyond the configured scheduled dormancy period, and load-based dormancy is not configured for some cells. - Time synchronization fails, and the NE does not enter the dormancy state. You can run the DSP TIMESRC command to check whether a time source is available.
Base Station in Dormancy Suspension	- Dormancy pending is triggered due to the execution of the PND DORMANCY command. - Dormancy pending is triggered due to base station reset or configuration verification. - Dormancy pending is triggered due to reliability protection.
Upgrading	- The base station is being upgraded. - The base station is in the protection period for successful upgrade. The protection period lasts less than 15 minutes.
Process Being Protected	- After the RF module link recovers, it takes about 5 minutes for the related alarm to be cleared. The cell will enter the dormancy state after the alarm is cleared. - The base station instructs the cell to enter dormancy, and then instructs the RF module to enter dormancy 30s later. - Random dormancy waiting is ongoing. The random dormancy waiting time can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to SITE. - Less than 5 minutes have passed since the RF module exited dormancy. - The RF module is entering the dormancy state.

Cause	Description
Reliability Requirements Not Met	- The reliability requirements, such as those for solder joints or temperature difference, are not met. - The BBU CPRI status is stable for less than 1 minute. - Faults occur on components such as the current RF module, chain/ring of the current RF module, and related ports or NE, preventing the RF module from entering or exiting the dormancy state. - The base station is faulty. For example, the current RF module or the NE is faulty and does not meet the conditions for entering the dormancy state. - In separate-MPT scenarios, the peer NE is stable for less than 5 minutes. - In RRU cascading scenarios, the RRUCHAIN.CR (LTE eNodeB, 5G gNodeB) parameter is set to AUTO but the maximum rates supported by cascaded RRUs differ. - In RRU ring scenarios, the RRUCHAIN.CR (LTE eNodeB, 5G gNodeB) parameter is set to AUTO but the maximum rates supported by the ports connected to the master and slave chains are different.
Conditions Not Met for the Fronthaul Link	- Active devices (such as active WDM devices, semi-active WDM devices, and switches) are used on the fronthaul link. - The fronthaul link is unstable. For example, the optical power is unstable. - The optical module has reliability risks. For example, RX_LOS error is reported.

Cause	Description
Service Conditions Not Met	• A possible reason why scheduled RF module super dormancy does not take effect is that some activated cells under the base station are not bound to BBPs. • Service faults exist and the dormancy entry conditions are not met. • In separate-MPT scenarios, peer services are faulty and the conditions for entering the dormancy mode are not met. • New RF modules and eGBTS/NodeB/eNodeB/gNodeB applications are in the protection period after being started and do not meet the dormancy entry conditions. • For load-based RF module super dormancy, a dual-band RF module has one of the bands serving as the basic layer and the other serving as the capacity layer and therefore does not meet the dormancy entry conditions. LTE side: • For load-based super dormancy, run the DSP CELLPWRSAVINGPOLICY command to query the reason why the cell does not meet the requirements. For details, see Table 7-74. • Multi-RRU cells, such as SFN cells, do not meet the dormancy entry conditions, as the MULTI_RRU_CELL_DORM_COMPAT_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter is deselected. NR side: • The ultra-long holdover function of the system clock is in effect. • Different dormancy periods are configured for the RF modules bound to a combined-RRU cell or multi-TRP cell (such as an SFN cell, hyper cell, or combined cell), or the current time is beyond the period configured for such a cell. • For load-based super dormancy, run the DSP NRDUCELLPOWERSAVING command to query the reason why the cell does not meet the requirements. For details, see Table 7-72 or Table 7-73. UMTS side: • Multiple RF modules are configured for a cell, and their dormancy periods are different or the current time falls outside the configured periods.

Cause	Description
	- Multiple RF modules are configured for a cell group, and their dormancy periods are different or the current time falls outside the configured periods. - Run the DSP ULOCELLPWRS command to query the reason why the cell does not meet the dormancy entry conditions. For details, see Table 7-70. GSM side: - For load-based super dormancy, run the DSP GTRX command to query the reason why the cell does not meet the requirements. For details, see Table 7-69.
Not All NE Dormancy Notifications Received	The dormancy notifications of some NEs are not received. For example, in a separate-MPT base station, the peer end does not meet the requirements, causing the RF module to fail to enter the dormancy state. The reasons why dormancy does not take effect at the peer end can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD . Table 7-67 lists the possible reasons.
Optical Module Compatibility Requirement Not Met	- The BBP or RF module uses non-standard optical modules. - There is a configured optical module with unknown information (for example, the information is not obtained when the BBP starts up) on the BBP or RF module. - There is an unconfigured optical module with unknown information on the BBP or RF module.
ALD Compatibility Not Met	The RF module uses a non-standard ALD.

Cause	Description
Dormancy Capability Not Supported by the Networking	- Cloud BB networking is used. - The following conditions are met when the DOR_LM_SW option of the EQUIPMENT.EQMDOROPTFEATSW parameter is deselected: - The connected electromechanical components are configuration-free. - An RPCU is configured. - A ring topology is used, with more than one RF module cascaded on the ring. - Monitoring devices, such as a PMU or an alarm port, are configured for RF modules.
Carriers Existed on the RF Module	None
Protection Due to Ongoing Device Maintenance and Test	- Device-related maintenance and test operations are being performed. - Protection will be canceled within 30 minutes after device-related maintenance and test operations are completed. Dormancy will be allowed after the protection is canceled.
Protection Due to Ongoing Service Maintenance and Test	- Service-related maintenance and test operations are being performed. - Protection will be canceled within 30 minutes after service-related maintenance and test operations are completed. Dormancy will be allowed after the protection is canceled.
RF Module Not Idle	In cascading scenarios, the lower-level RF module or CXU is not in the super dormancy state.
Other Reason	For example, a plug-and-play module is used.

Table 7-68 Time and causes for the most recent exit and failure of a function

Command Output	Description
Latest Exit Cause	Indicates the cause of the most recent exit of a function during its effective period.

Command Output	Description
Latest Disabled Cause	Indicates the cause of the most recent failure of a function to take effect during its effective period.
Latest Exit Time	Indicates the recorded time of the latest exit of a function during its effective period.
Latest Disabled Time	Indicates the recorded time of the latest failure for a function to take effect during its effective period.

- Query the cell dormancy status using the following methods if the outputs of the preceding commands indicate that RF module super dormancy does not take effect:
 - Querying the cell dormancy status on the GSM side:
 - On the BSC, run the **DSP GCELLSTAT** command. If the value of **Cell Dormancy Status** is **In the Dormancy State** in the command output, the frequency band where the main BCCH TRX is located is in the dormancy state.
 - On the BSC, run the **DSP GTRXSTAT** command. If the value of **TRX in Dormancy** is **Yes** in the command output, the TRX is in the dormancy state.
 - On the eGBTS, run the **DSP GLOCELL** command.
 - If the value of **Local Cell Status** is **DORMANCY** in the command output, the RF resources of the local cell are in the dormancy state.
 - If the value of **Cell Status** is **DORMANCY** in the command output, the logical cell is in the dormancy state.
 - On the eGBTS, run the **DSP GTRX** command. If the value of **TRX Dormancy State** is **Dormancy** in the command output, the TRX is in the dormancy state. The current dormancy state and the reason why dormancy does not take effect can also be queried. The values displayed in the query results are periodically refreshed and may not always reflect the most recent updates.

Table 7-69 Possible reasons why GSM load-based or scheduled RF module super dormancy does not take effect

Cause	Description
Switch Turned Off	None

Cause	Description
License Insufficient	The license items for dormancy are insufficient.
Service Maintenance Protection	- Service-related maintenance and test operations are being performed. - Protection will be canceled within 30 minutes after service-related maintenance and test operations are completed. Dormancy will be allowed after the protection is canceled.
Abis Fault	The Abis interface between the BSC and the base station is faulty.
BSC Version Not Support	The BSC is not upgraded to a required version.
Impacted Feature in Effect	A cell with CBC enabled cannot enter dormancy in the following scenarios: - If BSEXSOFTPARA.IgnoreCbcSwitch is set to OFF for a cell, the cell cannot enter dormancy. - If BSEXSOFTPARA.IgnoreCbcSwitch is set to ON for a cell, the cell can enter dormancy. However, when CBC is triggered, it is preferentially guaranteed, and the cell cannot enter dormancy during the duration specified by BSEXSOFTPARA.CbcServiceHoldDuration.
Abnormal TRX Service	- An alarm, such as ALM-28010 Carrier Configuration Error, ALM-28006 Radio Signaling Link Disconnected, or ALM-12310 Radio Link Abnormal Alarm, is reported when a TRX is faulty due to reasons such as abnormal frequency, bandwidth, or power. - The TRX channel is faulty.
Local Cell Unavailable	ALM-28016 GSM Local Cell Unusable is reported when a cell becomes unavailable due to reasons such as RF resource faults or a mismatch between the base station software version and the RF unit.
Local Cell Capability Decline	ALM-28017 GSM Local Cell Capability Decline is reported when the cell capability declines due to reasons such as abnormal cell data configuration, insufficient CPRI bandwidth, and insufficient license capacity.
BSC Under Dynamic Configuration Protection	Operations that affect services, such as changing cell frequencies or system information, are performed on the BSC.

Cause	Description
Alarms on the BSC	Alarms that affect services are reported on the BSC, such as ALM-21816 Cell Logical Channel Failure Alarm, ALM-22008 PTP BVC Faulty, and ALM-21824 Logical Object Binding Error.
Under Penalty	- The current cell is under penalty due to a migration failure. The default penalty duration is 30 minutes. You can configure the penalty duration using the GCELLDORMANCY.DormancyPenaltyTime parameter. - The dynamic configuration of the current cell is under penalty. The penalty duration is fixed at 30 minutes.
Co-coverage Cell Unavailable	The co-coverage cell of the dormant cell is unavailable.
Co-coverage Cell Load Condition Not Met	The load on the co-coverage cell of the dormant cell is too high, failing to meet the dormancy entry threshold.
Local Cell Load Condition Not Met	The load on the dormant cell is high, and the number of channels occupied by this cell does not meet a certain threshold.
RF Dormancy Basic Capb Condition Not Met	The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD . Table 7-67 lists the possible reasons.
Triggering Time Not Reached	The current time falls outside the specified energy-saving effective period.
Others	- The GSM cell meets the dormancy entry conditions but does not receive an instruction to enter dormancy. The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD. Table 7-67 lists the possible reasons. - The base station startup time is less than 15 minutes, and the system is not stable. - Internal software reasons cause the function not to take effect.

- Querying the cell dormancy status on the UMTS side:
 - Run the **DSP ULOCCELL** command. If the value of **Local Cell Status** is **Local Cell Unavailable-RF Module Dormancy**, the cell has entered dormancy.
 - Run the **DSP ULOCCELLPWRSV** command to query the reason why the cell does not enter dormancy. The values displayed in the query results are periodically refreshed and may not always reflect the most recent updates. After the reset, the reason why the cell does not enter dormancy within 15 minutes is incorrect. [Table 7-70](#) describes the possible reasons why RF module super dormancy does not take effect.

You can also run the **DSP ULOCCELLPWRSV** command to query the **Cause of Latest Dormancy Failure**, **Time of Latest Dormancy Failure**, **Cause of Latest Dormancy Exit**, and **Time of Latest Dormancy Exit** in the specified period.

[Table 7-24](#) describes the meanings of **Cause of Latest Dormancy Failure**, **Time of Latest Dormancy Failure**, **Cause of Latest Dormancy Exit**, and **Time of Latest Dormancy Exit**.

Cause of Latest Dormancy Failure and **Cause of Latest Dormancy Exit** indicate the reasons for the most recent failure and exit of the function. These reasons can be queried after the effective period has ended. For example, if the effective period is from 00:00:00 to 06:00:00, **Cause of Latest Dormancy Failure** and **Cause of Latest Dormancy Exit** record the reasons for the most recent failure and exit of the function from 00:00:00 to 06:00:00. These reasons can be queried by an operation performed within the time from 06:00:00 to 23:59:59.

Table 7-70 Possible reasons why UMTS scheduled RF module super dormancy and UMTS or GU load-based RF module super dormancy do not take effect

Cause	Description
RNC Version Not Support	The RNC is not upgraded to a required version.
Suspended Due to RNC Configuration	During dormancy, an RNC MML command affecting dormancy is executed, leading to dormancy suspension. After 30 minutes, dormancy suspension is canceled and dormancy is allowed.
RNC Fault	Alarms that affect dormancy activation exist on the RNC, such as ALM-22202 UMTS Cell Unavailable.

Cause	Description
RNC Service Maintenance Protection	- Service-related maintenance and test operations are being performed on the RNC. - Protection will be canceled within 30 minutes after service-related maintenance and test operations on the RNC side are completed. Dormancy will be allowed after the protection is canceled.
RNC High Priority Feature In Effect	A cell with CBS enabled cannot enter dormancy in the following scenarios: - If URNCELLDORMANCY-PARA.CellIgnoreCbsSwitch is set to OFF for a cell, the cell cannot enter dormancy. - If URNCELLDORMANCY-PARA.CellIgnoreCbsSwitch is set to ON for a cell, the cell can enter dormancy. However, when CBS is triggered, it is preferentially guaranteed, and the cell cannot enter dormancy during the duration specified by URNCELLDORMANCY-PARA.CbsServiceHoldDuration.
Service Fault	- Alarms that affect dormancy activation exist on the NodeB, such as ALM-28206 Local Cell Capability Decline. - The lub interface is faulty. For example, the value of Local Cell Status is Local Cell Available-NCP Fault in the DSP ULOCELL command output.
RF Module Capability Not Support	The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD . Table 7-67 lists the possible reasons.
Baseband Capability Not Support	The uplink or downlink baseband equipment contains multiple BBPs or contains BBPs that do not support RF module super dormancy.
Local Cell Unavailable	None
Logical Cell Not Activated for Non-Power-Saving Reasons	None

Cause	Description
Mutually Exclusive or Impacted Feature	The cell is enabled with mutually exclusive or impacted features/functions as described in 7.4.3.2.2 Mutually Exclusive Functions or 7.4.3.2.3 Function Impacts , respectively.
Service Maintenance Protection	- Service-related maintenance and test operations are being performed on the NodeB. - Once the operations on the NodeB side are completed, the protection will be canceled 30 minutes later, and then dormancy will be allowed.
Under Penalty	The cell enters the penalty state when UE transfer is not completed. The cell can enter dormancy only after the penalty state is cleared. The specific penalty time is as follows: 30 minutes for scheduled RF module super dormancy - Value of the UCELLDORMANCYPOLICY.PenaltyTime parameter on the RNC for UMTS or GU load-based RF module super dormancy
Cell in Cell Group Not Meet Conditions	A multi-cell group is present. In such situations, all cells in the group must meet the dormancy entry conditions. Multi-cell group configurations involve: NODEBMULTICELLGRP MO NODEBEXSECCELLGRP MO - A local cell for which UCELL.DLASYMFILTER is set to TRUE and cells with UCELL.DLASYMFILTERNFCELLID specified
Not In Dormancy Period	The current time falls outside the specified energy-saving effective period.
Load Conditions Not Met	The load of the capacity-layer cell does not meet the conditions.
No Configured Power Saving Policy	No energy saving policy for UMTS or GU load-based dormancy is configured for the capacity-layer cell.
No Configured Co-coverage Neighboring Cell	No inter-band or inter-RAT co-coverage neighboring cells are configured for the capacity-layer cell.

Cause	Description
Number of UEs Not Meet Threshold Conditions	The total number of UEs, number of FACH UEs, number of HSDPA UEs, number of HSUPA UEs, or number of multi-carrier UEs in the capacity-layer cell does not meet the energy-saving policy configuration requirements.
No Available Co-coverage Neighboring Cell	Inter-band or inter-RAT co-coverage neighboring cells are unavailable.
Co-coverage Neighboring Cell Not Meet Load Conditions	The load of inter-band or inter-RAT co-coverage neighboring cells does not meet the dormancy entry conditions. Specifically: • The inter-band or inter-RAT co-coverage neighboring cells are heavily loaded or overloaded. • The load margin of the inter-band or inter-RAT co-coverage neighboring cells does not meet the energy-saving policy configuration requirements. • The load of the inter-RAT co-coverage neighboring cells does not meet the energy-saving policy configuration requirements. • The load of inter-RAT co-coverage neighboring cells cannot be obtained.
Services of Co-coverage Neighboring Cells Not Support	In GU load-based RF module super dormancy scenarios, data services configured for capacity-layer cells are not supported by inter-RAT co-coverage neighboring cells.
Process Being Protected	• The cell setup process is protected to ensure configuration consistency between the base station and the base station controller. • After the logical cell is successfully activated, a 3-minute protection timer is started. Dormancy is allowed only after the protection timer expires. • Other energy saving features are being activated or deactivated.

Cause	Description
Carrier Shutdown Reliability Protected	- When RF module dormancy is enabled and the UCELLDYNSHUTDOWN.DynShutdownSwitch parameter is set on the RNC side to enable dynamic cell shutdown, the cell enters the carrier shutdown reliability protection state if any RNC or NodeB configuration affecting dormancy is modified during dynamic cell shutdown. The cell can enter dormancy only after carrier shutdown is exited. - When RF module dormancy is enabled and carrier shutdown features are enabled on the NodeB side, the cell enters the carrier shutdown reliability protection state if any RNC configuration affecting dormancy is modified during dynamic cell shutdown. The cell can enter dormancy only after carrier shutdown is exited. The carrier shutdown features on the NodeB side are as follows: - Multi-Carrier Switch off Based on Power Backup (NODEBPOWEROUTAGE.ISDSW) - UMTS Low Power Consumption Mode (NODEBALGPARA.LOWPWRCONSUMPMODESW) - UMTS SOC Level-Based Low Power Consumption Mode (NODEBALGPARA.LOWPWRCONSUMEMODEEXTSW)
PIM Detection Protected	- When PIM interference detection is enabled for a UMTS cell by setting ULOCCELLALGPARA.PIMDETECTIONSW to ON and the cell is in the dormancy state, it exits dormancy when LTE or NR PIM interference detection is started. - If a massive MIMO module is shared by UMTS and LTE or NR cells and the UMTS cell is in the dormancy state, it exits dormancy when LTE or NR PIM interference characteristic separation training is started.
Other Reason	- The conditions for entering the energy saving function are not met throughout the triggering evaluation period. - Other reasons, such as internal software issues, cause the function not to take effect.
NULL	The service meets the dormancy conditions or the dormancy feature is not enabled.

Table 7-71 Time and causes for the most recent exit and failure of the function in the **DSP ULOCELLPWRSAP** command output

Command Output	Description
Cause of Latest Dormancy Failure	Indicates the cause of the most recent dormancy failure during the effective period.
Time of Latest Dormancy Failure	Indicates the recorded time of the most recent dormancy failure during the effective period.
Cause of Latest Dormancy Exit	Indicates the cause of the most recent dormancy exit during the effective period.
Time of Latest Dormancy Exit	Indicates the recorded time of the most recent dormancy exit during the effective period.

- LTE side: Run the **DSP CELL** command. If the value of the **Cell power save state** parameter is **CELL_PWRSAP_RF_DEEP_SLEEP** in the command output, the cell has entered the dormancy state.
- NR side: Run the **DSP NRDUCELL** command. If the **Power Saving State** parameter has a value of **Dormancy of All RF Modules:Enabled** in the command output, the cell has entered the dormancy state.

In addition to the preceding observation methods, the following functions can be observed using the **DSP NRDUCELLPOWERSAVING** command: optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module super dormancy (referred to as optimized evaluation of NR RF module dormancy) and load-based RF module super dormancy. The values displayed in the query results are periodically refreshed and may not always reflect the most recent updates.

- The **DSP NRDUCELLPOWERSAVING** command can be used to query the status of NR load-based RF module super dormancy and optimized evaluation of NR RF module dormancy, as well as the reasons why these functions do not take effect.
 - If the value of the **Power Saving Type** parameter is **Intra-NR Load-based Dormancy** and the value of the **Current State** parameter is **Enabled**, NR load-based RF module super dormancy has taken effect. If the value of the **Power Saving Type** parameter is **NR RF Module Dormancy Evaluation Opt** and the value of the **Current State** parameter is **Enabled**, optimized evaluation of NR RF module dormancy has taken effect.
 - If the value of the **Power Saving Type** parameter is **Intra-NR Load-based Dormancy** or **NR RF Module Dormancy Evaluation Opt** and the value of the **Current State** parameter is **Disabled**, check the **Current Disabled Cause** parameter value to determine the reason why this function does not take effect. [Table 7-72](#) lists the possible reasons.

Table 7-72 Possible reasons why NR load-based RF module super dormancy or optimized evaluation of NR RF module dormancy does not take effect

Cause	Description
Cell in Base Station Not Bound to BBP	None
Switch Turned Off	None
RF Group Configuration Condition Not Met	The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD . Table 7-67 lists the possible reasons.
Cell Unavailable	The cell is blocked or deactivated, fails to be set up, or is in other unavailable states.
Protection Due to Ongoing Maintenance and Test	- Service-related maintenance and test operations are being performed. - Protection will be canceled within 30 minutes after service-related maintenance and test operations are completed. Dormancy will be allowed after the protection is canceled.
Reliability Requirements Not Met	Alarms that affect the dormancy function exist, for example, the High-Risk Function Enable alarm exists.
Triggering Time Not Reached	The current time falls outside the specified energy-saving effective period.
Parameter Inconsistency Between Cells	(High-frequency TDD) The parameters in the RF module super dormancy policies are inconsistent among different cells.
Ultra-Long Holdover in Progress	None
Impacted Feature Taken Effect	None

Cause	Description
Neighboring Cell Not Configured	- No inter-frequency neighboring cells are configured for the serving cell. - Inter-frequency neighboring cells are configured for the serving cell, but there is no neighboring cell for which Power Saving Cell Flag is set to COVERAGE. - In multi-operator sharing scenarios: - The PLMN IDs of the normal operators of the capacity-layer cell are not a subset of the PLMN IDs of normal operators of basic-layer cells. - The PLMN IDs of the reserved operators of the capacity-layer cell are not a subset of the PLMN IDs of normal or reserved operators of basic-layer cells.
In Penalty State	- The cell is in the penalty state after exiting the dormancy state within the specified effective period. It can enter this state again after the penalty stops. The penalty duration is specified by the NRDUCellPowerSaving.PwrSavingPenaltyTime parameter. - The cell is in the penalty state because some UEs failed to be transferred out of the cell. The penalty lasts 30 minutes. The causes of UE transfer failures include UE capability mismatch and missing configuration of neighboring frequency relationships.
Xn Interface Unavailable	The Xn interface is abnormal in multi-carrier inter-base-station scenarios.
Neighboring Cell Unavailable	- A neighboring cell for which Power Saving Cell Flag is set to COVERAGE is unavailable. - A neighboring cell for which Power Saving Cell Flag is set to COVERAGE is a hyper cell or combined cell, and some TRPs of this cell are unavailable. - A neighboring cell for which Power Saving Cell Flag is set to COVERAGE is an NSA cell.
Downlink PRB Threshold Not Reached	For NR load-based RF module super dormancy, the total downlink PRB usage of the capacity- and basic-layer cells does not reach the downlink PRB usage threshold for triggering this function. For optimized evaluation of NR RF module dormancy, the downlink PRB usage of the current NR cell does not reach the downlink PRB usage threshold for triggering this function.

Cause	Description
Uplink PRB Threshold Not Reached	For NR load-based RF module super dormancy, the total uplink PRB usage of the capacity- and basic-layer cells does not reach the uplink PRB usage threshold for triggering this function. For optimized evaluation of NR RF module dormancy, the uplink PRB usage of the current NR cell does not reach the uplink PRB usage threshold for triggering this function.
Basic Cell PRB Usage Limit Exceeded	For NR load-based precise dormancy, the uplink or downlink PRB usage of a basic-layer cell exceeds the congestion threshold.
UE Number Threshold Not Reached	None
Triggering Decision Period Not Met	The current time falls within the specified effective period, but the cell does not meet the dormancy entry conditions throughout the period specified by the NRDUCellPowerSaving.ShutdownTrigJudgePeriod parameter.
Multi-Carrier Not Coordinated	Not all the activated cells sharing the same RF module meet the entry conditions for NR load-based RF module dormancy or optimized evaluation of NR RF module dormancy.
RF Dormancy Basic Capb Condition Not Met	The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD . Table 7-67 lists the possible reasons.
Others	- The local NR cell meets the dormancy entry conditions but does not receive an instruction to enter dormancy. The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD. Table 7-67 lists the possible reasons. - In the current period, the conditions for this feature to take effect are met, but the conditions for LNR load-based RF module super dormancy are not met. - For optimized evaluation of NR RF module dormancy, the dormancy entry conditions are not met throughout the triggering evaluation period. - Internal software reasons cause the function not to take effect.

- You can also run the **DSP NRDUCELLPOWERSAVING** command to query the status of LNR load-based RF module super dormancy and the reason why it does not take effect. The values displayed in the query results are periodically refreshed and may not always reflect the most recent updates.
 - If the value of the **Power Saving Type** parameter is **LNR Load-based Dormancy** and the value of the **Current State** parameter is **Enabled**, this function has taken effect.
 - If the value of the **Power Saving Type** parameter is **LNR Load-based Dormancy** and the value of the **Current State** parameter is **Disabled**, check the **Current Disabled Cause** parameter value to determine the reason why this function does not take effect. [Table 7-73](#) lists the possible reasons.

Table 7-73 Possible reasons why LNR load-based RF module super dormancy does not take effect

Cause	Description
Cell in Base Station Not Bound to BBP	None
Switch Turned Off	The function switch for the NR cell has not been turned on.
RF Group Configuration Condition Not Met	The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD . Table 7-67 lists the possible reasons.
Cell Unavailable	The NR cell is blocked or deactivated, fails to be set up, or is unavailable due to other reasons.
Protection Due to Ongoing Maintenance and Test	• Service-related maintenance and test operations are being performed. • Protection will be canceled within 30 minutes after service-related maintenance and test operations are completed. Dormancy will be allowed after the protection is canceled.
Reliability Requirements Not Met	Alarms that affect the dormancy function exist, for example, the High-Risk Function Enable alarm exists.
Triggering Time Not Reached	The current time falls outside the specified effective period for the function.
Ultra-Long Holdover in Progress	None

Cause	Description
Impacted Feature Taken Effect	None
Neighboring Cell Not Configured	• In SA networking, the neighbor relationships with LTE cells are not configured for the NR cell, the overlap indicator is not configured for the neighboring LTE cells of the NR cell, or the NR cell fails to obtain information about its neighboring LTE cells. • In SA networking: – The PLMN IDs of the normal operators of the NR cell are not a subset of the PLMN IDs of normal operators of neighboring LTE cells. – The PLMN IDs of the reserved operators of the NR cell are not a subset of the PLMN IDs of normal or reserved operators of neighboring LTE cells.
Obtaining NCell Configurations	• In NSA networking or NSA and SA hybrid networking, the neighbor relationships with LTE cells are not configured for the NR cell, the overlap indicator is not configured for the neighboring LTE cells of the NR cell, the X2 link is faulty, or the NR cell fails to obtain information about its neighboring LTE cells. • In SA networking, neighboring LTE cell information is being subscribed to. • In NSA networking or NSA and SA hybrid networking: – The PLMN IDs of the normal operators of the NR cell are not a subset of the PLMN IDs of normal operators of neighboring LTE cells. – The PLMN IDs of the reserved operators of the NR cell are not a subset of the PLMN IDs of normal or reserved operators of neighboring LTE cells.
In Penalty State	• The cell enters the penalty state after exiting dormancy within the specified effective period. It can re-enter dormancy once the penalty state ends. The penalty duration is specified by the NRDUCellPowerSaving.PwrSavingPenaltyTime parameter. • The cell is in the penalty state because some UEs failed to be transferred out of the cell. The penalty lasts 30 minutes.

Cause	Description
Neighboring Cell Unavailable	- The function switches of the neighboring LTE cells are not turned on, the licenses for the neighboring LTE cells are insufficient, the neighboring LTE cells are in barred state or not activated, no neighboring LTE cell is an anchor cell, or the LTE cells are in other unavailable states. - If NSA anchor carrier shutdown is enabled in neighboring LTE cells, it is recommended that the INTER_RAT_GRAD_PWR_SAVING_SW option of the gNBpwrSavingAlgo.PwrSavingAlgoSwitch parameter be selected for the NR cell. This will prevent the NR cell from exiting the LNR load-based RF module super dormancy state in this case. For details about NSA anchor carrier shutdown, see <i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>. - The LTE sector split cell is configured as a basic-layer cell, and the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter is deselected.
UE Number Threshold Not Reached	- The downlink PRB usage in the NR cell has reached the trigger threshold, but the number of UEs remains below the trigger threshold. - The downlink PRB usage and the number of UEs in the NR cell have not reached their respective trigger thresholds.
Downlink PRB Threshold Not Reached	The number of UEs in the NR cell has reached the trigger threshold, but the downlink PRB usage remains below the trigger threshold.
Uplink PRB Threshold Not Reached	The number of UEs in the NR cell has reached the trigger threshold, but the uplink PRB usage remains below the trigger threshold.
Neighboring Cell Downlink PRB Exceeded Threshold	The current downlink PRB usage of neighboring cells exceeds the configured downlink PRB usage threshold. That is, the proportion of neighboring LTE cells meeting the PRB usage threshold does not meet the requirement specified by the NRDUCellPowerSaving.EutranCarrShutdownRatio parameter.

Cause	Description
Neighboring Cell Uplink PRB Exceeded Threshold	The current uplink PRB usage of neighboring cells exceeds the configured uplink PRB usage threshold, that is, the proportion of neighboring LTE cells meeting the PRB usage threshold does not meet the requirement specified by the NRDUCellPowerSaving.EutranCarrShutdownRatio parameter.
NR UEs in NCell Exceeding Threshold	The number of NR UEs in neighboring LTE cells of an NR cell exceeds the threshold. That is, the proportion of neighboring LTE cells meeting the NR UE number threshold does not meet the requirement specified by the NRDUCellPowerSaving.EutranCarrShutdownRatio parameter.
Neighboring Cell NR UE Downlink PRB Exceeded Threshold	The current downlink PRB usage of NR UEs exceeds the configured downlink PRB threshold. That is, the proportion of neighboring LTE cells meeting the NR UE downlink PRB usage threshold does not meet the requirement specified by the NRDUCellPowerSaving.EutranCarrShutdownRatio parameter.
Neighboring Cell NR UE Uplink PRB Exceeded Threshold	The current uplink PRB usage of NR UEs exceeds the configured uplink PRB usage threshold. That is, the proportion of neighboring LTE cells meeting the NR UE uplink PRB usage threshold does not meet the requirement specified by the NRDUCellPowerSaving.EutranCarrShutdownRatio parameter.
Triggering Decision Period Not Met	The current time falls within the check period specified by the NRDUCellPowerSaving.ShutdownTrigJudgePeriod parameter, but the thresholds for function triggering have not been met.
UE Being Transferred or Transfer Failed	None
Multi-Carrier Not Coordinated	Not all the activated cells that share the same RF module meet the conditions for entering the LNR load-based RF module dormancy state.
RF Dormancy Basic Capb Condition Not Met	The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD . Table 7-67 lists the possible reasons.

Cause	Description
Other Reason	- In the current period, the conditions for this function to take effect are met, but the conditions for NR load-based RF module dormancy are not met. - The current time is within the effective period specified for the LNR load-based RF module dormancy function, but the gNodeB has not obtained data such as the PRB usage and UE number. - The local NR cell meets the conditions for entering the dormancy state, but it does not receive an instruction to enter the dormancy state. The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD. Table 7-67 lists the possible reasons. - Other reasons, such as the internal software reasons, cause the function not to take effect. - The configurations are incorrect. For example, in multi-operator sharing scenarios, the neighbor relationships with LTE cells served by the LTE primary operators are not configured for the NR cell.

- You can also run the **DSP CELLPWRSAVINGPOLICY** command to query the status of LTE load-based super dormancy and the reason why it does not take effect. The values displayed in the query results are periodically refreshed and may not always reflect the most recent updates.
 - If the value of **Power Saving Type** is **INTRA_LTE_DORMANCY_BY_LOAD** or **LTE_PRECISE_DORMANCY_BY_LOAD** and the value of **Current State** is **ENABLED**, this function has taken effect.
 - If the value of **Power Saving Type** is **INTRA_LTE_DORMANCY_BY_LOAD** or **LTE_PRECISE_DORMANCY_BY_LOAD** and the value of **Current State** is other than **ENABLED**, check [Table 7-74](#) and [Table 7-75](#) for the possible reasons.

Table 7-74 Possible reasons why LTE load-based super dormancy does not take effect

Cause	Description
CELL_IN_BS_NOT_BOUND_TO_BBP	None
SWITCH_OFF	None
RF_GROUP_CFG_COND_NOT_MET	The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD . Table 7-67 lists the possible reasons.

Cause	Description
CELL_UNAVAILABLE	The cell is blocked or deactivated, fails to be set up, or is unavailable due to other reasons.
IN_MAINTENANCE_TEST_PROTECTION	- Service-related maintenance and test operations are being performed. - Protection will be canceled within 30 minutes after service-related maintenance and test operations are completed. Dormancy will be allowed after the protection is canceled.
RELIABILITY_NOT_MET	Alarms that affect the dormancy function exist, for example, the High-Risk Function Enable alarm exists.
NOT_IN_SHUTDOWN_TIME	The current time falls outside the specified energy-saving effective period.
IN_PENALTY_STATE	- The cell enters the penalty state after exiting dormancy within the specified effective period. It can re-enter dormancy once the penalty state ends. The penalty time is specified by the CellPwrSavingPolicy.PenaltyTime parameter. - If some UEs in a cell fail to be transferred, the cell is in the penalty state for energy saving. The penalty time is specified by the CellPwrSavingPolicy.PenaltyTime parameter.
HIGH_PRIORITY_TAKE_EFFECT	None

Cause	Description
NEIGHBORING_CELL_NOT_CONFIG	- No inter-frequency neighboring cells are configured for the current capacity-layer cell. - The current capacity-layer cell has been configured with inter-frequency neighboring cells, but the number of neighboring cells with the POWER_SAVING_OVERLAP_IND option selected is 0. - In multi-operator sharing scenarios, the operators of the local cell must be a subset of the operators of its co-coverage neighboring basic cells. If optimization for cell PLMN reservation scenarios is enabled by selecting the CELL_PLMN_RSVD_OPT_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter: - If the operators of the local cell are non-reserved operators, the operators must be a subset of non-reserved operators of its co-coverage neighboring basic cells. - If the operators of the local cell are reserved operators, the operators must be a subset of non-reserved operators or reserved operators of its co-coverage neighboring basic cells.
X2_INTERFACE_UNAVAILABLE	The X2 interface is abnormal when the capacity-layer cell and basic-layer cells are provided by different base stations.
NEIGHBORING_CELL_UNAVAILABLE	- A neighboring cell with the POWER_SAVING_OVERLAP_IND option selected is unavailable. - The LTE sector split cell is configured as a basic-layer cell, and the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter is deselected.
TRIGGER_JUDGE_PERIOD_NOT_MET	The current time falls within the specified effective period, but the cell does not meet the dormancy entry conditions throughout the period specified by the CellPwrSavingPolicy.TriggerJudgePeriod parameter.
NOT_IN_UE_NUM_THRESHOLD	None
NOT_IN_DL_PRB_THRESHOLD	None
NOT_IN_UL_PRB_THRESHOLD	None

Cause	Description
MULTI_CARR_NOT_COORDINATED	- After the multi-RRU cell dormancy compatibility switch is turned on (by selecting the MULTI_RRU_CELL_DORM_COMPAT_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter), not all RF modules serving the same cell as the RF module meet the dormancy conditions. - All cells in a sector split group are enabled with load-based dormancy and meet the trigger conditions.
RF_DORM_BASE_CAP_CONFIG_NOT_MEET	The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD . Table 7-67 lists the possible reasons.
OTHERS	- LTE cells meet the dormancy entry conditions but the local cell does not receive an instruction to enter dormancy. The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD. Table 7-67 lists the possible reasons. - Internal software reasons cause the function not to take effect.

Table 7-75 Possible reasons why LTE load-based precise dormancy does not take effect

Cause	Description
CELL_IN_BS_NOT_BOUND_TO_BBP	None
SWITCH_OFF	None
RF_GROUP_CONFIG_CONDITION_NOT_MET	The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD . Table 7-67 lists the possible reasons.
CELL_UNAVAILABLE	The cell is blocked or deactivated, fails to be set up, or is unavailable due to other reasons.
IN_MAINTENANCE_TEST_PROTECTION	- Service-related maintenance and test operations are being performed. - Protection will be canceled within 30 minutes after service-related maintenance and test operations are completed. Dormancy will be allowed after the protection is canceled.

Cause	Description
RELIABILITY_NOT_MET	Alarms that affect the dormancy function exist, for example, the High-Risk Function Enable alarm exists.
NOT_IN_SHUTDOWN_TIME	The current time falls outside the specified energy-saving effective period.
IN_PENALTY_STATE	- The cell enters the penalty state after exiting dormancy within the specified effective period. It can re-enter dormancy once the penalty state ends. The penalty time is specified by the CellPwrSavingPolicy.PenaltyTime parameter. - If some UEs in a cell fail to be transferred, the cell is in the penalty state for energy saving. The penalty time is specified by the CellPwrSavingPolicy.PenaltyTime parameter.
HIGH_PRIORITY_FEATURE_TAKEN_EFFECT	None
NEIGHBORING_CELL_NOT_CONFIG	- No inter-frequency neighboring cells are configured or obtained for the current capacity-layer cell. - The LTE_DORMANCY_BY_LOAD_PE_SW option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter is deselected for all inter-frequency neighboring cells. - The CellPwrSavingAlgo.DormancyCellType parameter is not set to BASIC_CELL. - The EutranInterFreqNCell.OverlapRate parameter is set to a value less than 20 for the inter-frequency neighboring cell. - In multi-operator sharing scenarios, the operators of the local cell must be a subset of the operators of its co-coverage neighboring basic cells. If optimization for cell PLMN reservation scenarios is enabled by selecting the CELL_PLMN_RSVD_OPT_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter: - If the operators of the local cell are non-reserved operators, the operators must be a subset of non-reserved operators of its co-coverage neighboring basic cells. - If the operators of the local cell are reserved operators, the operators must be a subset of non-reserved operators or reserved operators of its co-coverage neighboring basic cells.

Cause	Description
NEIGHBORING_CELL_UNAVAILABLE	- There are basic- and capacity-layer cells served by the same RRU. - The basic-layer cells are unavailable. - The LTE sector split cell is configured as a basic-layer cell, and the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter is deselected.
TRIG_JUDGE_PERIOD_NOT_MET	The current time falls within the specified effective period, but the cell does not meet the dormancy entry conditions throughout the period specified by the CellPwrSavingPolicy.TriggerJudgePeriod parameter.
NOT_IN_UE_NUM_THRESHOLD	None
NOT_IN_DL_PRB_THRESHOLD	None
NOT_IN_UL_PRB_THRESHOLD	None
BASIC_CELL_PRB_USAGE_EXCEED	The uplink or downlink PRB usage of a basic-layer cell exceeds the congestion threshold.
MULTI_CARR_NOT_COORDINATED	After the multi-RRU cell dormancy compatibility switch is turned on (by selecting the MULTI_RRU_CELL_DORM_COMPAT_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter), not all RF modules serving the same cell as the RF module meet the dormancy conditions.
RF_DORM_BASE_CAP_COVD_NOT_MET	The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD . Table 7-67 lists the possible reasons.
OTHERS	- The local LTE cell meets the dormancy entry conditions but does not receive an instruction to enter dormancy. The reasons why dormancy does not take effect can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to BOARD. Table 7-67 lists the possible reasons. - Internal software reasons cause the function not to take effect.

- In the case of LTE load-based RF module super dormancy, use the following methods to query the cell's operator reservation status:
 - Run the **DSP EUTRANEXTERNALCELL** command to query the external E-UTRAN cell reservation status of the operator.
 - If the value of **Cell Reserved for Operator Use** is **NOT_RESERVED**, the cell is not reserved for the operator.
 - If the value of **Cell Reserved for Operator Use** is **RESERVED**, the cell is reserved for the operator.
 - Run the **DSP CELLCNOPERATORSTATUS** command to query the intra-base-station cell reservation status of the operator.
 - If the value of **Cell Operator Reserved State** is **NULL**, the cell is not activated or the PLMN ID of the operator is not on the list broadcast through SIB1 of the cell.
 - If the value of **Cell Operator Reserved State** is **Not Reserved**, the cell is not reserved for the operator.
 - If the value of **Cell Operator Reserved State** is neither **NULL** nor **Not Reserved**, the cell is reserved for the operator.

7.4.4.3 Network Monitoring

On the live network, the one-week energy consumption data obtained before and after the energy saving feature is enabled can be compared to evaluate the energy saving gains of this feature. It is expected that the energy consumption decreases significantly after this feature is enabled. The specific data and observation methods are as follows:

- Energy consumption of the RF module. Check the **VS.EnergyCons.BTSBoard (LTE eNodeB, 5G gNodeB)** counter on the MAE-Access.
- Unavailability duration of the NR cell in the energy saving state. Check the **N.Cell.Unavail.Dur.EnergySaving** counter on the MAE-Access.
- Duration of the NR cell in the dormancy state. Check the **N.Cell.Unavail.Dur.RFModuleDeepDormancy** counter on the MAE-Access.
- Unavailability duration of the LTE cell in the energy saving state. Check the **L.Cell.Unavail.Dur.EnergySaving** counter on the MAE-Access.
- Duration of the LTE cell in the dormancy state. Check the **L.Cell.Unavail.Dur.RFModuleDeepDormancy** counter on the MAE-Access.
- Duration of the UMTS cell in the dormancy state. Check the **VS.Cell.UnavailTime.RFModuleDormancy** counter on the MAE-Access.
- Duration of the GSM TRX in the dormancy state. Check the **TRX.DORMANCY.DUR** counter on the MAE-Access.
- Duration of the RF module in the super dormancy state. Check the **VS.RADIOEQM.SuperDormancy.Dur (LTE eNodeB, 5G gNodeB)** counter on the MAE-Access.

 NOTE

- There is delay in determining whether to enter or exit the energy saving state. As a result, the duration measured by the performance counters may deviate from the actual duration by approximately 1 minute.
- The sampling period is not synchronous with the reporting period, so a deviation of one sampling period may occur.
- During super dormancy, energy consumption cannot be measured in real time. Instead, the average value from equipment tests is reported. However, this value can be influenced by factors such as ambient temperature and optical module type, leading to potential energy consumption deviations.
- MAE-Access performance counters are accurate to three decimal places. When the MAE-Access displays short-measurement-period performance counters, the precision of the displayed energy consumption may deviate from the actual value.
- For UMTS RF module super dormancy, the RNC and NodeB must cooperate to measure the **VS.Cell.UnavailTime.RFModuleDormancy** counter. If the RNC becomes disconnected from the NodeB, the measured duration may deviate from the actual value by approximately 1 to 5 minutes.

7.5 Frequency Band-based Energy Saving of Wideband Modules

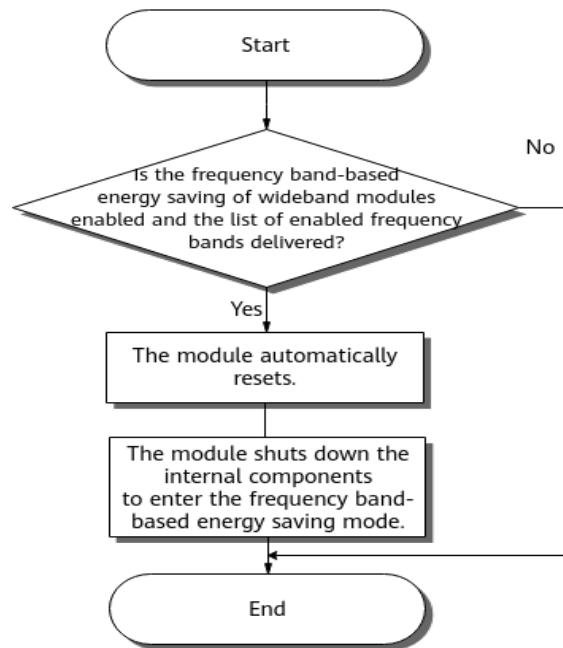
7.5.1 Principles

Frequency band-based energy saving of wideband modules refers to that when a wideband module (an RF module supporting multiple frequency bands) is configured with only one or two frequency bands (the wideband module has unused frequency bands), some digital processing units can be blocked to save energy and reduce the power consumption of the RF module. This function is controlled by the **RRUEXTINFO.FREQBANDCTRLSW (5G gNodeB, LTE eNodeB)** parameter.

7.5.1.1 Entering the Frequency Band-based Energy Saving of Wideband Modules

Figure 7-31 shows the dormancy process of the frequency band-based energy saving of wideband modules.

Figure 7-31 Workflow of the frequency band-based energy saving of wideband modules



1. Set the **RRUEXTINFO.FREQBANDCTRLSW (5G gNodeB, LTE eNodeB)** parameter to **ON**, which enables the frequency band-based energy saving of wideband modules, and set the **RRUEXTINFO.FREQBANDCTRLFLAG (5G gNodeB, LTE eNodeB)** parameter to the list of enabled frequency bands of the module.
2. The module automatically resets to make the parameter modification take effect.
3. After the module is reset, some digital processing units are shut down to reduce the power consumption of the module and enter the frequency band-based energy saving mode.

7.5.1.2 Exiting the Frequency Band-based Energy Saving of Wideband Modules

If **RRUEXTINFO.FREQBANDCTRLSW (5G gNodeB, LTE eNodeB)** is changed to **OFF**, the wideband module exits the frequency band-based energy saving mode. After the module is restarted, it recovers to the normal working state.

7.5.2 Network Analysis

7.5.2.1 Benefits

It is recommended that the frequency band-based energy saving of wideband modules be enabled in low-traffic scenarios for a long time (more than one week) to reduce the power consumption of the base station.

If partial frequency bands are enabled for the same module and the enabled and disabled frequency bands share the same PA, the energy saving gains decrease.

7.5.2.2 Impacts

- The impacts between this function and other functions are described in [7.5.3.2.3 Function Impacts](#).
- After **RRUEXTINFO.FREQBANDCTRLSW (5G gNodeB, LTE eNodeB)** is turned on, the board is automatically reset to make the modification take effect. The service interruption duration is less than 5 minutes.
 - Cells cannot be set up on a frequency band that has been disabled. In this case, you need to enable the specific frequency band of the **RRUEXTINFO.FREQBANDCTRLFLAG (5G gNodeB, LTE eNodeB)** parameter.

7.5.3 Requirements

7.5.3.1 Licenses

This function is not under license control.

7.5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.5.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
GSM UMTS LTE FDD LTE TDD NR FDD Low-frequency NR TDD	Frequency band control flag	RRUEXTINFO.FREQBANDCTRLFLAG (5G gNodeB, LTE eNodeB)	<i>Multi-RAT Coordinated Energy Saving</i>	When the frequency band enable switch (RRUEXTINFO.FREQBANDCTRLSW (5G gNodeB, LTE eNodeB)) is turned on, at least one bit of the frequency band control flag must be enabled, and the frequency band value corresponding to the bit cannot be 0 .

RAT	Function Name	Function Switch	Reference	Description
GSM UMTS LTE FDD LTE TDD NR FDD Low-frequency NR TDD	RRU type	RRU.RT (5G gNodeB, LTE eNodeB)	None	When the frequency band enable switch (RRUEXTINFO.FREQBAN DCTRLSW (5G gNodeB, LTE eNodeB)) is turned on, RRU.RT (5G gNodeB, LTE eNodeB) must be set to MRRU .

7.5.3.2.2 Mutually Exclusive Functions

None

7.5.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Ultra-long holdover for sites with abnormal clock sources	CLK_OTASY NC_PROC_SW option of the TASM.MSCLKDIAGPROC ESSSW parameter	<i>Synchronization in 5G RAN Feature Documentation</i>	- Assume that the Ultra-Long Hold Switch is turned on but the Ultra-Long Hold Enhancement Switch is turned off. If sequence detection is performed in all cells on a to-be-detected base station during the period when "frequency band-based energy saving of wideband modules" takes effect, the sequence detection fails. In this case, the base station cannot enter the ultra-long holdover state even if its clock is faulty, or it exits the ultra-long holdover state as sequence detection cannot be performed for a long time due to energy saving. - If the Ultra-Long Hold Enhancement Switch is turned on,

RAT	Function Name	Function Switch	Reference	Description
				"frequency band-based energy saving of wideband modules" does not take effect for the base station in the ultra-long holdover state or its reference neighboring base stations.

7.5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

Boards

No requirements

RF Modules

Only the following RF modules support this function:

RRU5517et (part number: 02314RPD), RRU5517t (part numbers: 02314ERQ and 02314PEF), RRU5527t (part number: 02314WTK), and RRU5513t (part number: 02313NWQ)

7.5.3.4 Others

None

7.5.4 Operation and Maintenance

7.5.4.1 When to Use

It is recommended that the frequency band-based energy saving of wideband modules be enabled in low-traffic scenarios for a long time (more than one week).

7.5.4.2 Data Configuration

7.5.4.2.1 Data Preparation

Table 7-76 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
FreqBand Control Switch	RRUEXTINFO.FREQBAN DCTRLSW (5G gNodeB, LTE eNodeB)	It is recommended that this switch be turned on to enable the frequency band-based energy saving of wideband modules in low-traffic scenarios for a long time (more than one week).
FreqBand Control Flag	RRUEXTINFO.FREQBAN DCTRLFLAG (5G gNodeB, LTE eNodeB)	It is used to indicate the list of enabled frequency bands and enable required frequency bands.
Frequency Band1	RRUEXTINFO.FREQBAN D1 (5G gNodeB, LTE eNodeB)	It must be set for enabled frequency bands of the RF module according to the frequency bands defined in 3GPP specifications.
Frequency Band2	RRUEXTINFO.FREQBAN D2 (5G gNodeB, LTE eNodeB)	It must be set for enabled frequency bands of the RF module according to the frequency bands defined in 3GPP specifications.
Frequency Band3	RRUEXTINFO.FREQBAN D3 (5G gNodeB, LTE eNodeB)	It must be set for enabled frequency bands of the RF module according to the frequency bands defined in 3GPP specifications.
Frequency Band4	RRUEXTINFO.FREQBAN D4 (5G gNodeB, LTE eNodeB)	It must be set for enabled frequency bands of the RF module according to the frequency bands defined in 3GPP specifications.
Frequency Band5	RRUEXTINFO.FREQBAN D5 (5G gNodeB, LTE eNodeB)	It must be set for enabled frequency bands of the RF module according to the frequency bands defined in 3GPP specifications.
Frequency Band6	RRUEXTINFO.FREQBAN D6 (5G gNodeB, LTE eNodeB)	It must be set for enabled frequency bands of the RF module according to the frequency bands defined in 3GPP specifications.

7.5.4.2.2 Using MML Commands

Before using MML commands, refer to [7.5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency,

and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

```
//If RRUEXTINFO exists, run the MOD RRUEXTINFO command to enable the frequency band-based energy saving of wideband modules.  
MOD RRUEXTINFO: CN=0, SRN=60, SN=0, FREQBAND1=3, FREQBANDCTRLSW=ON,  
FREQBANDCTRLFLAG=FREQBAND1-1;  
//If RRUEXTINFO does not exist, run the ADD RRUEXTINFO command to enable the frequency band-based energy saving of wideband modules.  
ADD RRUEXTINFO: CN=0, SRN=60, SN=0, FREQBAND1=3, FREQBANDCTRLSW=ON,  
FREQBANDCTRLFLAG=FREQBAND1-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling the frequency band-based energy saving of wideband modules  
MOD RRUEXTINFO: CN=0, SRN=60, SN=0, FREQBANDCTRLSW=OFF;
```

7.5.4.2.3 Using the MAE-Deployment

- Fast batch activation
This function can be batch activated using the Feature Operation and Maintenance function of the MAE-Deployment. For detailed operations, see the following section in the MAE-Deployment product documentation or online help: **MAE-Deployment Operation and Maintenance > MAE-Deployment Guidelines > Enhanced Feature Management > Feature Operation and Maintenance**.
- Single/Batch configuration
This function can be activated for a single base station or a batch of base stations on the MAE-Deployment. For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.5.4.3 Activation Verification

Run the **DSP RRUEXTINFO** command to check whether the frequency band-based energy saving of wideband modules takes effect. If the value of **Band Enable Flag** is **DISABLE** in the command output, this function takes effect. If the value of **Band Enable Flag** is **ENABLE** in the command output, this function does not take effect.

7.5.4.4 Network Monitoring

On the live network, the one-week energy consumption data obtained before and after the power saving feature is enabled can be compared to evaluate the power saving gains. It is expected that the energy consumption decreases significantly after this feature is enabled.

Check the **VS.EnergyCons.BTSBoard (5G gNodeB, LTE eNodeB)** counter (which indicates the power consumption of a board) on the MAE-Access to observe the power consumption of RF modules.

7.6 pRRU Deep Dormancy

In indoor scenarios such as shopping malls, stadiums, and office buildings, there are no service requirements in fixed periods. With this function, **certain pRRUs can be shut down and enter deep dormancy during these periods, reducing energy consumption**, as shown in [Figure 7-32](#).

Once a pRRU enters deep dormancy, it disables components, including its power amplifier (PA), RF processing unit (that is, radio-on-a-chip, ROC), and some functions of the digital intermediate frequency (IF) processing unit, to reduce base station energy consumption. However, essential devices, for example, interface modules, remain enabled.

Figure 7-32 Working principles of pRRU deep dormancy

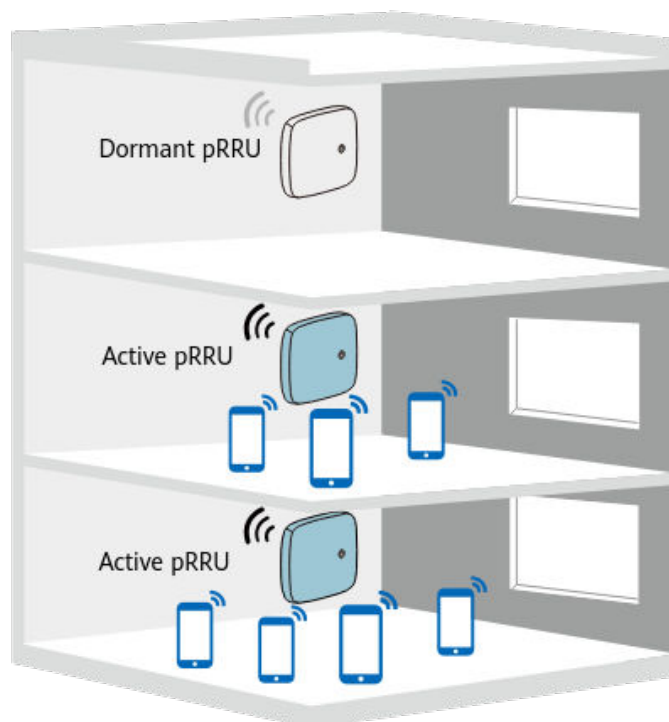
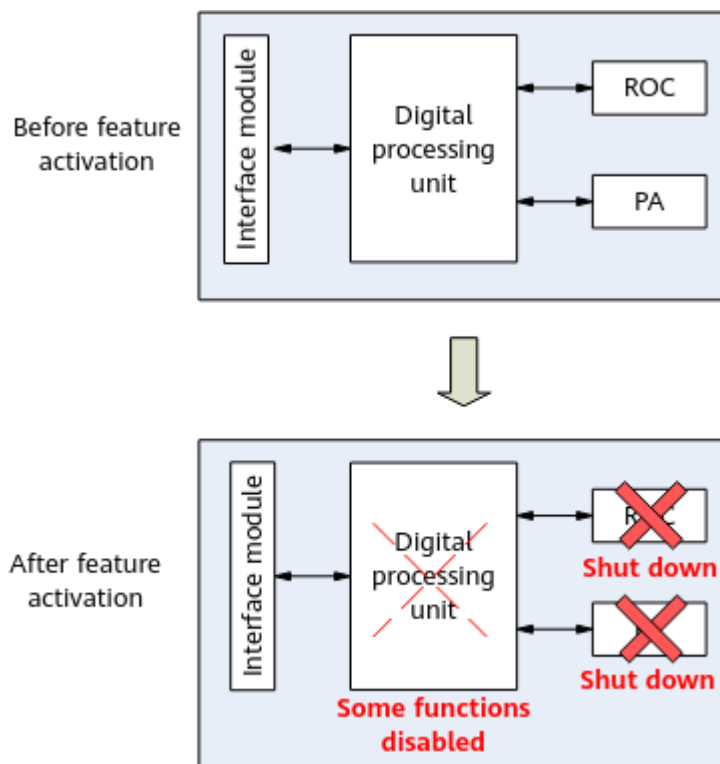


Figure 7-33 Component shutdown in pRRU deep dormancy



7.6.1 Principles

pRRU deep dormancy can be triggered in any of the following modes:

- **In scheduled mode**, deep dormancy is triggered within a pre-defined time period, irrespective of the network load. This mode applies when there is a fixed time period every day with no services running indoors.
- **In intelligent mode**, dynamic entry into or exit from deep dormancy is triggered based on the network load with the basic indoor coverage guaranteed. This mode applies when the period with low network load changes.

7.6.1.1 pRRU Deep Dormancy in Scheduled Mode

This function applies to UMTS, LTE, and NR.

pRRU deep dormancy in scheduled mode is enabled by setting the **RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB)** parameter to **ON**. The deep dormancy period is specified by the **RRU.DORMANCYSTARTTIME (LTE eNodeB, 5G gNodeB)** and **RRU.DORMANCYSTOPTIME (LTE eNodeB, 5G gNodeB)** parameters.

- Specific pRRUs enter the deep dormancy state 2 minutes after the dormancy start time specified by the **RRU.DORMANCYSTARTTIME (LTE eNodeB, 5G gNodeB)** parameter arrives. The 2 minutes is referred to as a transitional period, during which the pRRUs can work properly after pRRU deep dormancy is triggered.
- These pRRUs exit the deep dormancy state when the dormancy stop time specified by the **RRU.DORMANCYSTOPTIME (LTE eNodeB, 5G gNodeB)**

parameter arrives. For RF modules other than pRRU5x5x4B series, they will restart upon the exit from deep dormancy. After the restart, they will restore to the normal working state. For pRRU5x5x4B series, they will not restart upon the exit from deep dormancy. Instead, the file system will be loaded and the pRRUs will restore to the normal working state.

 CAUTION

You are advised not to perform the following operations during the specified dormancy period: (1) base station reconfiguration; (2) license installation, license activation, or software version download; (3) FFT spectrum scanning using the Web LMT; (4) hardware self-diagnosis using the **STR HWTST** command; (5) introduction of a new RAT to the base station. Run the **PND DORMANCY** command and wait 10 minutes before performing any of the preceding operations.

A pRRU in the deep dormancy state (an energy saving state) does not transmit power. When a pRRU enters this state, it does not trigger the following alarms, assuming there are no other faults:

- Node alarm
 - ALM-29248 RF Out of Service
- NR alarms
 - ALM-29840 gNodeB Out of Service
 - ALM-29841 NR Cell Unavailable
 - ALM-29870 NR DU Cell TRP Unavailable
 - ALM-29871 NR DU Cell TRP Capability Degraded
- LTE alarms
 - ALM-29240 Cell Unavailable
 - ALM-29243 Cell Capability Degraded
 - ALM-29249 eNodeB Out of Service
- UMTS alarms
 - ALM-28203 Local Cell Unusable (NodeB)
 - ALM-28206 Local Cell Capability Decline (NodeB)
 - ALM-26787 RHUB-pRRU CPRI Interface Error (NodeB)
 - ALM-22202 UMTS Cell Unavailable (RNC)
 - ALM-22214 NodeB Unavailable (RNC)
 - ALM-22206 UMTS Cell Setup Failed (RNC)

The **VS.Cell.UnavailTime.Sys** counter is not measured.

 NOTE

- Hardware faults may occur if the pRRU temperature differential is excessively large before and after the pRRUs enter the dormancy state. To prevent such risks, the pRRUs will not enter the deep dormancy state if their temperature differential within 24 hours exceeds a specified threshold.
- In UMTS, if the traffic model of an indoor building changes, for example, when the opening hours of a shopping mall change, the period when UEs use the indoor wireless network also changes. In this case, the start time and end time of deep dormancy need to be adjusted to accommodate these changes. The **VS.CellDCHUEs** counter can be used to help determine the periods when the network has no UEs to serve. Based on this information, the parameters for the start time and end time of pRRU deep dormancy can be reconfigured accordingly.

7.6.1.2 pRRU Deep Dormancy in Intelligent Mode

This function applies only to LTE.

In intelligent mode, when there is a light load in a cell, the eNodeB instructs some of the pRRUs serving the cell to enter the deep dormancy state and uses the remaining pRRUs to ensure the basic indoor coverage. In this way, energy consumption is reduced. The eNodeB automatically selects which pRRUs should enter the deep dormancy state.

 CAUTION

You are advised not to perform base station reconfiguration, license installation, license activation, or software version download during the specified dormancy period. Run the **PND DORMANCY** command and wait 10 minutes before performing any of the preceding operations.

To ensure basic coverage after pRRUs enter dormancy, the downlink EARFCN of a carrier needs to be specified by the **EnodebMpruEs.MpruDormancyDLEarfcn** parameter for this carrier to be retained as a basic coverage carrier.

- If the **EnodebMpruEs.MpruDormancyDLEarfcn** parameter is set to a value other than **-1**, pRRU deep dormancy in intelligent mode is enabled.
 - If the **EnodebMpruEs.MpruDormancyDLEarfcn** parameter is set to **-2**, the base station automatically selects the downlink EARFCN of a low-frequency carrier for this carrier to be retained as a basic coverage carrier and selects candidate cells for dormancy based on the downlink EARFCN.
 - In other scenarios, the EARFCN specified by the **EnodebMpruEs.MpruDormancyDLEarfcn** parameter is used as the downlink EARFCN of the retained basic coverage carrier, and candidate cells for dormancy are selected based on the downlink EARFCN.

The base station checks whether each of its served cells has co-pRRU inter-frequency neighboring cells in sequence to select candidate cells for dormancy.

- If a logical cell has co-pRRU inter-frequency neighboring cells, the base station checks whether the EARFCN of the cell is the basic EARFCN to be retained.

- If it is, the base station adds the cell to the candidate cell list for dormancy.
- If it is not, the base station does not add the cell to the candidate cell list for dormancy.
- If a logical cell does not have co-pRRU inter-frequency neighboring cells, the base station directly adds the cell to the candidate cell list for dormancy.
- If the **EnodebMpruEs.MpruDormancyDIEarfcn** parameter is set to **-1**, pRRU deep dormancy in intelligent mode is disabled.

 NOTE

The **PRRU_LTE_SW** option of the **Equipment.EqmDorOptFeatSw** parameter must be selected to flexibly control the RAT in which pRRU deep dormancy in intelligent mode is enabled.

7.6.1.2.1 pRRU Data Sampling and Processing

After pRRU deep dormancy in intelligent mode is enabled, the eNodeB collects measurement reports (MRs) and sounding reference signal (SRS) measurement data. A maximum of 200 samples can be collected for each pRRU. When 200 samples are collected or the collection has lasted for seven days, the eNodeB stops the initial data collection. The pRRU deep dormancy function does not take effect until the initial data collection is complete. After an eNodeB reset, 200 samples must be collected or the collection must last for seven days before the eNodeB stops the initial data collection.

If a cell is deleted during initial data collection, the collected data will be discarded. However, if a cell is deactivated during initial data collection, the collected data is reserved for 24 hours. The reserved data can still be used if the cell is activated within 24 hours but will be otherwise discarded (In this case, data collection re-starts after the cell is re-activated.)

When the initial data collection is complete, the eNodeB processes the collected data and determines which pRRUs should enter the dormancy state based on the processing result.

Data update starts after the initial data collection. The data will be updated once a day. The earliest 5% of the collected data will be overwritten for each pRRU. After seven days of data update (which is referred to as a data update period), the eNodeB processes the data. The processing result is used as a basis for the eNodeB to determine which pRRUs should enter the dormancy state in the next period. The data is collected at the non-energy-saving time.

 NOTE

To implement SRS measurements, **SRSCfg.SrsCfgInd** must be configured.

7.6.1.2.2 Selection of pRRUs for Entering the Deep Dormancy State

The pRRU selection principle is that the eNodeB should determine the minimum number of pRRUs to be retained to keep the minimum downlink level of the cell greater than the value of the **EnodebMpruEs.MpruDormancyMinimumRsrp** parameter. All the remaining pRRUs can be selected to enter the deep dormancy state.

After the sampling, the eNodeB selects pRRUs as follows:

- At each sampling time, the eNodeB retains the minimum number of pRRUs that can meet the minimum downlink level requirement of the cell.
- For all sampling moments, the eNodeB retains the pRRUs that most frequently meet the preceding requirement.

The pRRUs that should enter the deep dormancy state for an eNodeB can be selected after the eNodeB determines the pRRUs to be retained for all cells served by the eNodeB.

7.6.1.2.3 eNodeB Actions

During the period specified by the **EnodebMpruEs.MpruDormancyStartTime** and **EnodebMpruEs.MpruDormancyStopTime** parameters, the eNodeB shuts down some pRRUs if the downlink PRB usage of the cell is less than the value of the **EnodebMpruEs.MpruDormancyDlPrbUsageThld** parameter.

To ensure accurate calculation of the license control item for pRRU deep dormancy in intelligent mode, perform the following operations. Otherwise, the eNodeB reports ALM-26819 Data Configuration Exceeding Licensed Limit.

- Set **RRU.DORMANCYSW** to **ON**.
- Set **RRU.DORMANCYSTARTTIME** and **RRU.DORMANCYSTOPTIME** to the same value.

When there are multiple inter-frequency LTE cells served by pRRUs that will be shut down, if the eNodeB serves LTE cells on other frequencies than the downlink frequency where the MPRU enters dormancy, the eNodeB sets these inter-frequency cells to the barred state and hands over RRC_CONNECTED UEs from these inter-frequency cells to the local cell. When the handovers are complete, the eNodeB increases the CRS power by 3 dB for the local cell. Then, the eNodeB instructs the selected pRRUs to enter the deep dormancy state and cancels the barred state of the inter-frequency cells.

If the eNodeB operates only on one LTE frequency, the eNodeB increases the CRS power by 3 dB for the local cell. Then, the eNodeB instructs the selected pRRUs to enter the deep dormancy state.

If the CRS power cannot be increased by 3 dB due to a small power headroom in the local cell, the eNodeB selects pRRUs for deep dormancy as it does in the scenario where the CRS power is not increased by 3 dB.

NOTE

Hardware faults may occur if the pRRU temperature variation is excessively large before and after the pRRUs enter the dormancy state. To prevent such risks, the pRRUs will not enter the deep dormancy state if their temperature variation within 24 hours exceeds a specified threshold.

7.6.1.2.4 Leaving

If any of the following conditions is met, the pRRUs in the deep dormancy state exit this state and the eNodeB restores the CRS power of the cell to the original values:

- The period specified by the **EnodebMpruEs.MpruDormancyStartTime** and **EnodebMpruEs.MpruDormancyStopTime** parameters ends.

- The downlink PRB usage of the cell is greater than the **ENodebMpruEs.MpruDormancyDlPrbUsageThld** parameter value plus a fixed offset (5% if CRS power is increased or 20% if CRS power is not increased).
- A retained active pRRU of the cell becomes unavailable.

 NOTE

When the pRRUs exit the deep dormancy state because a retained active pRRU of the cell becomes unavailable, the eNodeB will select pRRUs again for entry into the deep dormancy state.

7.6.2 Network Analysis

7.6.2.1 Benefits

pRRU deep dormancy reduces energy consumption when the network is under no load.

To prevent possible hardware faults caused by an excessively large temperature differential, this function may not take effect or may be terminated when the temperature differential of RF modules is excessively large. Energy saving gains may decrease in this case.

Scheduled Mode

You are advised to enable pRRU deep dormancy in scheduled mode on indoor LampSite base stations when there are no services at night. This function helps reduce device power consumption in such indoor space as shopping malls, office buildings, and subway stations. For details about counters related to power consumption, see [7.6.4.3 Network Monitoring](#).

The overall effect of using pRRU deep dormancy in scheduled mode depends on how long there are no services and on the size of the area with no services.

- When the duration with no services is longer, more power can be saved.
- When the area with no services is larger, more pRRUs can enter the deep dormancy state.

Intelligent Mode

You are advised to enable pRRU deep dormancy in intelligent mode on indoor LampSite base stations when there is contiguous light traffic and the indoor coverage integrity concerns customers. This helps reduce device power consumption.

7.6.2.2 Impacts

The impacts between this function and other functions are described in [7.6.3.2.3 Function Impacts](#).

- Scheduled mode
 - The signal strength in an area with some pRRUs in the deep dormancy state will decrease because these pRRUs do not transmit signals. UE

- service quality may deteriorate, and there may be service drops and access failures.
- When normal cells are in the same CA group as an MPRU aggregation cell, if all of the pRRUs of the MPRU aggregation cell enter the deep dormancy state, the normal cells will report ALM-29243 Cell Capability Degraded alarms. In this case, CA cannot provide maximum gains.
 - If a batch processing task of the OSS contains a NodeBFunction, eNodeBFunction, or gNodeBFunction MML command that affects cell setup or a Node MML command, and an NE of the base station is in the deep dormancy state, then the OSS executes the related configurations only after the base station exits deep dormancy. This increases the waiting time before configuration execution.
 - If an XML configuration file for a base station contains a NodeBFunction, eNodeBFunction, or gNodeBFunction MML command that affects cell setup or a Node MML command (including a single power-related command), and an NE of the base station is in the deep dormancy state, then the base station automatically exits the deep dormancy state before command execution.
- Intelligent mode
 - When pRRU deep dormancy in intelligent mode takes effect, the downlink level of indoor coverage at certain cell edges may slightly decrease but is still greater than or equal to the value of the **EnodebMpruEs.MpruDormancyMinimumRsrp** parameter. As a result, the downlink peak throughput of the cells may decrease.
 - In UL or LNR co-MPT scenarios, if pRRUs need to enter the deep dormancy state on the LTE side, the corresponding UMTS or NR carriers on the dormant pRRUs will become unavailable. As a result, the radio signal quality on the UMTS or NR side may deteriorate in the coverage area, resulting in service drops or access failures for UMTS or NR UEs.
 - If a batch processing task of the OSS contains an eNodeBFunction or Node MML command that affects cell setup, and an NE of the base station is in the deep dormancy state, then the OSS executes related configurations after the base station exits the deep dormancy state. This increases the waiting time before configuration execution.
 - If an XML configuration file for a base station contains an eNodeBFunction or Node MML command that affects cell setup (including a single power-related command), and an NE of the base station is in the deep dormancy state, then the base station automatically exits the deep dormancy state and then executes the commands.

7.6.3 Requirements

7.6.3.1 Licenses

RAT	Feature ID	Feature Name	License Control Item Name	Model	Sales Unit
UM TS	WRFD-200205	pRRU Deep Dormancy	pRRU Deep Dormancy (per pRRU)	LQW9PRRU DD01	per pRRU
LTE FDD	LOFD-130202	pRRU Deep Dormancy	pRRU Deep Dormancy	LT1SDORM AN00	per pRRU
LTE TDD	TDLOFD-131202	pRRU Deep Dormancy	pRRU Deep Dormancy (per pRRU)	LT4SPRRUD D00	per pRRU
NR	FOFD-030216	pRRU Deep Dormancy	pRRU Deep Dormancy (per pRRU)	NR0SDORM AN10	per pRRU

If a LampSite base station is RFA-capable, there are no license requirements for the RFA mode. For example, in the LTE FDD + GSM (RFA via MERC) scenario, this function can take effect as long as an FDD license item is available. No additional license items are required for RFA. RFA is short for Radio Frequency Access, and MERC is short for Multi Extended Radio Frequency Card.

If RAN sharing or MOCN is enabled, the licensed resource is allocated between the primary and secondary operators using the following method:

Method 1: Some license control items, such as **Dynamic CE Function (per NodeB)**, can be allocated only through the common group. For details about NodeB license allocation among multiple operators, see *License Management in RAN Feature Documentation*.

A license for this feature is required for each RAT in a multimode base station.

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (LoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

7.6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.6.3.2.1 Prerequisite Functions

None

7.6.3.2.2 Mutually Exclusive Functions

All functions are compatible with pRRU deep dormancy in scheduled mode.

The following table describes the functions that are mutually exclusive with pRRU deep dormancy in intelligent mode.

RAT	Function Name	Function Switch	Reference
LTE FDD LTE TDD	TDOA-based positioning	TDOA_SWITCH option of the CellAlgoSwitch.LbsSwitch parameter	None
NB-IoT	NB-IoT network deployment	None	None

7.6.3.2.3 Function Impacts

The following table describes the functions that have impact relationships with pRRU deep dormancy in scheduled mode.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	CRS dynamic muting	CRS_DYNAMIC_MUTING_SW option of the SymbolPwrSaving.SymbolShutdownEnhancedSw parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	CRS dynamic muting does not take effect when pRRU deep dormancy in scheduled mode takes effect and all pRRUs enter the dormancy state.
LTE FDD LTE TDD	pRRU deep dormancy in intelligent mode	EnodebMpruEs.MpruDormancyDLEarfcn	<i>Multi-RAT Coordinated Energy Saving</i>	Only pRRU deep dormancy in scheduled mode takes effect when it is enabled together with pRRU deep dormancy in intelligent mode.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Carrier aggregation	None	<i>Carrier Aggregation in eRAN Feature Documentation</i>	- Cells in the energy saving state due to pRRU deep dormancy do not participate in CA. In CA-group-based configuration mode, after such a cell in a CA group exits this energy saving state, this cell cannot be configured as an SCell for an existing CA UE until the UE accesses the network again. - After a cell exits pRRU deep dormancy in scheduled mode, it does not participate in CA if the number of routes is insufficient. The routes for CA will be released for a cell that has entered pRRU deep dormancy in scheduled mode. After this cell exits this function, it may not obtain a route if the number of routes is insufficient and therefore, cannot participate in CA.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Dynamic carrier shutdown phase 2	CellPwrSavin gAlgo.Carrier DynShutdown nPh2Sw set to CAPACITY_CELL	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	When dynamic carrier shutdown phase 2 and pRRU deep dormancy in scheduled mode are both enabled, pRRU deep dormancy in scheduled mode will take effect but dynamic carrier shutdown phase 2 will not.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Virtual Grid-based Multi-Frequency Coordination	• RSRP prediction model – VG_MODEL_ALLOW_BUILD_FLAG option of the NRCellFReqRelationAggregationAttribute parameter – NR_IF_HO_MEAS_QTY_PRED_FUNCTION option of the NRCellSmartMultiCarrierArr.NrMultiCarrierAlgorithmSwitch parameter • Downlink intra-frequency spectral efficiency prediction model – DL_INTERF_SE_MODEL_BUILD_ALLO	<i>Virtual Grid-based Multi-Frequency Coordination</i> <i>Virtual Grid-based Multi-Frequency Coordination</i> in <i>5G RAN Feature Documentation</i>	After pRRU deep dormancy takes effect, the building of virtual grid models will be suspended and existing virtual grid models will be brought offline and cannot be used.

RAT	Function Name	Function Switch	Reference	Description
		W option of the NRCell Smart MultiC arr.<i>ModelBuildi ngSwit ch</i> parame ter - DL_INT RA_FRE Q_SPCT _EFF_P RED_S W option of the NRCell Smart MultiC arr.<i>NrM ultiCarr ierAlgo Switch</i> parame ter • Intra- frequency SRS beam prediction model INTRAF_S RS_MODE L_BUILD_A LLOW option of the NRCellSm artMultiC arr.<i>Model BuildingS witch</i> parameter		

RAT	Function Name	Function Switch	Reference	Description
NR FDD	Time-frequency two-dimensional unified scheduling optimization	TIME_FREQ_2D_UNIF_OPT_SCH_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgorithmSwitch parameter	<i>Multi-Frequency Smart Aggregation in 5G RAN Feature Documentation</i>	Time-frequency two-dimensional unified scheduling optimization does not take effect for cells in the deep dormancy state.
Low-frequency NR TDD	CA Performance Boost Based on Multi-dimensional Optimization	CA_PRFM_BOOST_BSD_ON_MD_OPT_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgorithmSwitch parameter	<i>Multi-Frequency Smart Aggregation in 5G RAN Feature Documentation</i>	CA Performance Boost Based on Multi-dimensional Optimization does not take effect for cells in the deep dormancy state.
Low-frequency NR TDD	pRRU-level precise symbol power saving	DL_PRECISE_SYMBOL_SHUTDOWN_SW option of the NRDUCellAlgorithmSwitch.PowerSavingExtensionSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	pRRU-level precise symbol power saving does not take effect when pRRU deep dormancy in scheduled mode is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD NR FDD	LTE-NR multi-dimensional scheduling optimization	• LTE: LNR_MULTI_D_OPT_SCH_SW option of the GigaBandCell.GigaBandCellAlgoSw parameter • NR: LNR_MULTI_D_OPT_SCH_SW option of the NRDUCellGigaBand.GigaBandCellAlgoSw parameter	<i>LTE FDD and NR Inter-RAT Unified Scheduling</i>	LTE-NR multi-dimensional scheduling optimization does not take effect for cells in the dormancy state.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	CMAS	CellAlgoSwitch.CmasSwitch	<i>CMAS in eRAN Feature Documentation</i>	The following applies when cells enter pRRU deep dormancy: • With PWS message-triggered exit of cells from energy saving disabled, CMAS warning notifications cannot be properly sent in LTE cells in the dormancy state. However, UEs can still receive CMAS warning notifications from other cells. • With PWS message-triggered exit of cells from energy saving enabled, cells exit the dormancy state upon receipt of a PWS message.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	ETWS	None	<i>ETWS in eRAN Feature Documentation</i>	The following applies when cells enter pRRU deep dormancy: • With PWS message-triggered exit of cells from energy saving disabled, ETWS warning notifications cannot be sent in LTE cells in the dormancy state. However, UEs can still receive ETWS warning notifications from other cells. • With PWS message-triggered exit of cells from energy saving enabled, cells exit the dormancy state upon receipt of a PWS message.

The following table describes the functions that have impact relationships with pRRU deep dormancy in intelligent mode.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Intelligent power-off of carriers in the same coverage	CellShutdown n.CellShutdownSwitch	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	When pRRU deep dormancy and intelligent power-off of carriers in the same coverage are both enabled, the function whose triggering conditions are first met preferentially takes effect. The other function can take effect only after the function in effect is terminated.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	RF channel intelligent shutdown	CellRfShutdown.RfShutdownSwitch	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	When pRRU deep dormancy in intelligent mode and RF channel intelligent shutdown both meet their triggering conditions, only the former function takes effect.
LTE FDD LTE TDD	Dynamic carrier shutdown	CARRIER_DYNAMIC_SHUTDOWN_SW option of the CellShutdown.CarrierShutdownEnhancedSw parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	When pRRU deep dormancy in intelligent mode and dynamic carrier shutdown are both enabled, the function whose triggering conditions are first met preferentially takes effect. The other function can take effect only after the function in effect is terminated.
LTE FDD LTE TDD	Smart carrier selection based on virtual grids	SMART_CARRIER_SELECTION_SW option of the MultiCarrUnifiedSch.MultiCarrierUnifiedSchSw parameter	<i>Multi-carrier Unified Scheduling in eRAN Feature Documentation</i>	Entering pRRU deep dormancy affects the progress of virtual grid model building.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Carrier aggregation	None	<i>Carrier Aggregation in eRAN Feature Documentation</i>	- Cells in the energy saving state due to pRRU deep dormancy do not participate in CA. In CA-group-based configuration mode, after such a cell in a CA group exits this energy saving state, this cell cannot be configured as an SCell for an existing CA UE until the UE accesses the network again. - After a cell exits the pRRU deep dormancy state, it does not participate in CA if the number of routes is insufficient. The routes for CA will be released for a cell that has entered pRRU deep dormancy. After this cell exits this function, it may not obtain a route if the number of routes is insufficient and therefore, cannot participate in CA.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Mobility load balancing	CellAlgoSwitch.MlbAlgoSwitch	<i>Intra-RAT Mobility Load Balancing in eRAN Feature Documentation</i>	When a cell providing basic coverage in pRRU deep dormancy in intelligent mode serves as a candidate cell for mobility load balancing, the load of this cell increases due to load balancing. As a result, the probability of exiting the deep dormancy state increases and the energy saving gains may be affected.
LTE FDD	Virtual 4T4R	Virtual4T4RSwitch option of the CellAlgoSwitch.EmimoSwitch parameter	<i>Virtual 4T4R (FDD) Virtual 4T4R (FDD) in eRAN Feature Documentation</i>	Virtual 4T4R no longer takes effect when some pRRUs enter the dormancy state.
LTE TDD	Downlink D-MIMO	CellAlgoSwitch.DMIMOAlgorithmSwitch	<i>D-MIMO (TDD) in eRAN Feature Documentation</i>	Downlink D-MIMO no longer takes effect when some pRRUs enter the dormancy state.
LTE FDD	UL coordinated channel shutdown	CellRfShutdown.MultiRatJointChnShutdownSw	<i>Multi-RAT Coordinated Energy Saving</i>	When both pRRU deep dormancy in intelligent mode and UL coordinated channel shutdown meet their triggering conditions, only the former takes effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Multi-carrier coordinated energy saving	CellShutdown n.CellShutdownSwitch set to ON_MULTI_CARRIER_HIER_SHUTDOWN or ON_MULTICARRIER_HIERMANUAL_SHUTDOWN	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	When pRRU deep dormancy in intelligent mode and multi-carrier coordinated energy saving are both enabled, the function whose triggering conditions are first met preferentially takes effect. The other function can take effect only after the function in effect is terminated.
LTE FDD LTE TDD	Dynamic carrier shutdown phase 2	CellPwrSavingAlgo.CarrierDynShutdownPh2Sw set to CAPACITY_CELL	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	When pRRU deep dormancy in intelligent mode and dynamic carrier shutdown phase 2 are both enabled, the function whose triggering conditions are first met preferentially takes effect. The other function can take effect only after the function in effect is terminated.
LTE FDD LTE TDD	pRRU deep dormancy in scheduled mode	RRU.DORMANCYSW	<i>Multi-RAT Coordinated Energy Saving</i>	Only pRRU deep dormancy in scheduled mode takes effect when it is enabled together with pRRU deep dormancy in intelligent mode.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	LTE FDD and NR Uplink Spectrum Sharing	SpectrumCloud.SpectrumCloudSwitch set to LTE_NR_UPLINK_SPECTRUM_SHR	<i>LTE FDD and NR SUL Dynamic Spectrum Sharing</i>	For DBS3900 LampSite and DBS5900 LampSite, when pRRU deep dormancy in intelligent mode is enabled together with LTE FDD and NR Uplink Spectrum Sharing, the demodulation performance of the common PUCCH in NR cells deteriorates.
LTE FDD NR FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	LTE: SpectrumCloud.SpectrumCloudSwitch set to LTE_NR_SPECTRUM_SHR NR: LTE_NR_FDD_SPCT_SHR_SW option of the NRDUCellAlgorithmSwitch.SpectrumCloudSwitch parameter	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>	For DBS3900 LampSite and DBS5900 LampSite, when pRRU deep dormancy in intelligent mode is enabled together with LTE FDD and NR Flash Dynamic Spectrum Sharing, the demodulation performance of the common PUCCH in NR cells deteriorates.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	CMAS	CellAlgoSwitch.CmasSwitch	<i>CMAS in eRAN Feature Documentation</i>	The following applies when cells enter pRRU deep dormancy: • With PWS message-triggered exit of cells from energy saving disabled, CMAS warning notifications cannot be properly sent in LTE cells in the dormancy state. However, UEs can still receive CMAS warning notifications from other cells. • With PWS message-triggered exit of cells from energy saving enabled, cells exit the dormancy state upon receipt of a PWS message.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	ETWS	None	<i>ETWS in eRAN Feature Documentation</i>	The following applies when cells enter pRRU deep dormancy: • With PWS message-triggered exit of cells from energy saving disabled, ETWS warning notifications cannot be sent in LTE cells in the dormancy state. However, UEs can still receive ETWS warning notifications from other cells. • With PWS message-triggered exit of cells from energy saving enabled, cells exit the dormancy state upon receipt of a PWS message.

7.6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

DBS3900 LampSite and DBS5900 LampSite

Scenarios of BBU+DCU3900 for RF signal input are not compatible with this function.

Scenarios of BBU+DCU3910 for RF signal input are not compatible with UMTS pRRU deep dormancy in scheduled mode.

Boards

- UMTS:
All UMTS-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

- LTE:
 - pRRU deep dormancy in scheduled mode: It is supported by all LTE-capable main control boards and baseband processing units. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
 - pRRU deep dormancy in intelligent mode: It is supported only by the UMPT and UBBP boards.
- NR:

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All pRRUs except the pRRU3901 support pRRU deep dormancy. Only the pRRU3902, pRRU3907, pRRU3912, pRRU3917, and 5900 series pRRUs support UMTS pRRU deep dormancy in scheduled mode.

7.6.3.4 Networking

Scheduled Mode

pRRU deep dormancy in scheduled mode has the following requirements:

- For pRRU deep dormancy to take effect on a multimode base station, it must be enabled for all RATs. For example:
 - For a multimode LampSite base station where LTE and NR use separate MPTs but share RF modules, pRRU deep dormancy must be enabled for both NR and LTE with the same start time and stop time so that this function can take effect.
 - For a multimode LampSite base station where LTE and NR use separate MPTs and RF modules, when CPRI MUX networking is used, pRRU deep dormancy must be enabled for both NR and LTE with the same start time and stop time so that this function can take effect; when CPRI MUX networking is not used, pRRU deep dormancy needs to be enabled on at least one side.
- For multimode base stations, each RAT must be upgraded to a version that supports pRRU deep dormancy in scheduled mode. This prevents the function from failing to take effect and alarms from being generated after function activation.
- pRRU deep dormancy is not recommended if the pRRUs serve NB-IoT carriers. If this function is used, data transmission of NB-IoT UEs may be affected.
- pRRU deep dormancy is not supported in pRRU cascading scenarios.

Intelligent Mode

pRRU deep dormancy in intelligent mode has the following requirements:

- To enable pRRU deep dormancy in intelligent mode, pRRUs with embedded antennas need to be used for indoor coverage. pRRU deep dormancy in intelligent mode is not recommended if pRRUs connected to external

antennas are used for indoor coverage. This is because the cell coverage cannot be ensured after some pRRUs enter the dormancy state.

- If a base station operates on multiple LTE frequencies and cells on these frequencies cover the same area, it is recommended that the cell on the frequency providing basic coverage be configured as a target neighboring cell of other inter-frequency cells. If an inter-frequency cell is an LAA cell or a cell involved in downlink CA, the cell providing basic coverage does not need to be configured as a neighboring cell of the inter-frequency cells.
- The bandwidth of the cell on the frequency providing basic coverage must be 20 or 15 MHz.
- This function is not applicable to base stations with multiple frequencies not covering the same area in indoor, densely populated, and heavily loaded scenarios.
- For a separate-MPT UL multimode base station or LNR multimode base station, pRRU deep dormancy in scheduled mode for the entire base station must be configured on the UMTS or NR side, with the dormancy period being the same as that set for deep dormancy in intelligent mode on the LTE side.
- For a co-MPT UL base station, pRRU deep dormancy is controlled by the LTE side.
- pRRU deep dormancy is not supported in pRRU cascading scenarios.

 NOTE

It is recommended that the **eNodeBResModeAlgo.BbpResAutoConfigSw** parameter be set to **ON_BIND** to enable automatic baseband resource configuration, and that the **gNodeBParam.AutoCellBindBasebandEqmSw** parameter be set to **ON** to enable automatic binding of cells to baseband equipment. Under other settings, baseband resource fragments may cause activation failures of some cells when pRRUs exit the deep dormancy state.

7.6.3.5 Cells

pRRU deep dormancy in intelligent mode requires that the cell bandwidth be higher than 10 MHz.

7.6.3.6 Others

You are not advised to enable both pRRU deep dormancy in scheduled mode and pRRU deep dormancy in intelligent mode during the same period. Otherwise, indoor coverage problems may occur.

For configuration reliability protection during deep dormancy to take effect, both the OSS and NEs must be upgraded to versions that support this function. For the required OSS versions, contact Huawei engineers.

7.6.4 Operation and Maintenance

7.6.4.1 Data Configuration

7.6.4.1.1 Data Preparation

Table 7-77 describes the parameters used for activation of pRRU deep dormancy in scheduled mode.

Table 7-77 Parameters used for activation of pRRU deep dormancy in scheduled mode

Parameter Name	Parameter ID	Setting Notes
Dormancy Switch	RRU.DORMANCY SW (LTE eNodeB, 5G gNodeB)	Set this parameter to ON when pRRU deep dormancy is required.
Dormancy Start Time	RRU.DORMANCY STARTTIME (LTE eNodeB, 5G gNodeB)	You are advised to set this parameter to the beginning of a period with no services.
Dormancy Stop Time	RRU.DORMANCY STOPTIME (LTE eNodeB, 5G gNodeB)	You are advised to set this parameter to the end of a period with no services. It takes some time for a pRRU in the deep dormancy state to restore its working state. To ensure that the network resumes normal services at the time point <i>A</i> , you are advised to set the dormancy stop time to 3 minutes earlier than <i>A</i> . For example, if the network is expected to restore services at 06:00:00, you are advised to set the dormancy stop time to 05:57:00.
Operation Type	OPERATIONTYPE	The value PENDING is recommended.
Pending Duration	PENDINGDURATION	Set this parameter based on the network plan.

Table 7-78 describes the parameters used for activation of pRRU deep dormancy in intelligent mode.

Table 7-78 Parameters used for activation of pRRU deep dormancy in intelligent mode

Parameter Name	Parameter ID	Setting Notes
MPRU Dormancy Downlink EARFCN	EnodebMpruEs.MpruDormancyDIEarfcn	It is recommended that this parameter be set to a frequency in the 2300 MHz band for TDD and to a frequency in the 1800 MHz band for FDD if carriers of multiple frequencies are configured for an eNodeB with a pRRU serving multiple carriers. The value -2 is recommended if a pRRU serves only one carrier. With this setting, the basic frequency is automatically selected.
MPRU Dormancy Start Time	EnodebMpruEs.MpruDormancyStartTime	When pRRU deep dormancy is used, it is recommended that this parameter be set to the start time of low-traffic hours at night.
MPRU Dormancy Stop Time	EnodebMpruEs.MpruDormancyStopTime	When pRRU deep dormancy is used, it is recommended that this parameter be set to the stop time of low-traffic hours.
MPRU Dormancy Minimum RSRP	EnodebMpruEs.MpruDormancyMinimumRsrp	Use the recommended value.
MPRU Dormancy Downlink PRB Usage Threshold	EnodebMpruEs.MpruDormancyDlPrbUsageThld	Use the recommended value.
SRS Configuration Indicator	SRSCfg.SrsCfgInd	Set this parameter to BOOLEAN_TRUE for a cell when pRRU deep dormancy is used.
Operation Type	OPERATIONTYPE	It is recommended that this parameter be set to PENDING when pRRU deep dormancy is used.
Pending Duration	PENDINGDURATION	None
Equipment Dormancy Optional Feature Switch	EQUIPMENT.EQMDOROPTFEATSW	When pRRU deep dormancy in intelligent mode is used, deselect the PPRU_NR_SW option and select the PPRU_LTE_SW option.
Dormancy Switch	RRU.DORMANCYSW	Turn on this switch when pRRU deep dormancy in intelligent mode is used.
Dormancy Start Time	RRU.DORMANCYSTARTTIME	The default value is recommended.

Parameter Name	Parameter ID	Setting Notes
Dormancy Stop Time	RRU.DORMANCY STOPTIME	Set this parameter to the same value as Dormancy Start Time when pRRU deep dormancy in intelligent mode is used.

7.6.4.1.2 Using MML Commands

Before using MML commands, refer to [7.6.3 Requirements](#) and complete the parameter configurations for related functions based on the impact and mutually exclusive relationships between the functions, as well as the actual network scenario.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Scenario 1:

```
//Activating pRRU deep dormancy in scheduled mode  
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=ON, DORMANCYSTARTTIME=00&00&00,  
DORMANCYSTOPTIME=06&00&00, CUSTOMDORMANCYSTDSW=OFF;  
//(Optional) Enabling deep dormancy pending  
PND DORMANCY: OPERATIONTYPE=PENDING, PENDINGDURATION=30;
```

Scenario 2:

```
//Activating pRRU deep dormancy in intelligent mode  
MOD ENODEBMPRUES: MpruDormancyDIEarfcn=38950, MpruDormancyStartTime=20&00&00,  
MpruDormancyStopTime=08&00&00, MpruDormancyMinimumRsrp=-115,  
MpruDormancyDIPrbUsageThld=10;  
//Setting SRS information for pRRU deep dormancy in intelligent mode to take effect  
MOD SRSCFG: LocalCellId=0, SrsCfgInd=BOOLEAN_TRUE;  
//(Optional) Enabling deep dormancy pending  
PND DORMANCY: OPERATIONTYPE=PENDING, PENDINGDURATION=30;  
//Activating LTE intelligent dormancy control  
SET EQUIPMENT: EQMDOROPTFEATSW=PRRU_LTE_SW-1&PRRU_NR_SW-0;  
//Setting pRRU deep dormancy in scheduled mode  
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=ON, DORMANCYSTARTTIME=00&00&00,  
DORMANCYSTOPTIME=00&00&00, CUSTOMDORMANCYSTDSW=OFF;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

Scenario 1:

```
//Deactivating pRRU deep dormancy in scheduled mode  
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=OFF;
```

Scenario 2:

```
//Deactivating pRRU deep dormancy in scheduled mode  
MOD RRU: CN=0, SRN=60, SN=0, DORMANCYSW=OFF;
```

```
//Deactivating LTE intelligent dormancy control  
SET EQUIPMENT: EQMDOROPTFEATSW=PRRU_LTE_SW-0;  
//Deactivating pRRU deep dormancy in intelligent mode  
MOD ENODEBMPRUES: MpruDormancyDLEarfcn=-1;
```

7.6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.6.4.2 Activation Verification

- You can use the following method to check whether an NE or base station is in the deep dormancy pending state:

Run the **DSP BRDDORMANCY** command with the **QUERYTYPE** parameter set to **SITE**.

- If the value of the **NE Dormancy Pending Status** parameter is **PENDING** in the command output, deep dormancy pending is triggered for the NE.
- If the value of the **Site Dormancy Pending Status** parameter is **PENDING** in the command output, dormancy pending is triggered for an NE of the base station.

- Perform the following operations to query the RRU dormancy status to check whether pRRU deep dormancy has taken effect.

Scenario 1: Querying the real-time pRRU deep dormancy status

```
//Querying the real-time dormancy status  
DSP BRDDORMANCY: QUERYTYPE=BOARD, QUERYMODE=CURRENT;
```

- If the value of **Board Dormancy State** is **Deep Dormancy** for the boards corresponding to the RF modules in the output of the preceding command, pRRU deep dormancy in scheduled mode has taken effect.
- If the value of **Board Dormancy State** is **Not in the Dormancy State** in the output of the preceding command, you can check the value of **Reason for Not Entering Deep Dormancy** to determine the possible reason. [Table 7-79](#) lists the details. The value of **Reason for Not Entering Deep Dormancy** is periodically refreshed and may not be updated in a timely manner.

Scenario 2: Querying the reason for the exit from pRRU deep dormancy and its failure to take effect within the latest specified period

```
//Querying whether pRRU deep dormancy takes effect within the latest specified period  
DSP BRDDORMANCY:  
QUERYTYPE=BOARD,QUERYMODE=HISTORY,PWRSAVINGFEATURE=DEEP_DORMANCY;
```

In the output of the preceding command, the value of **Latest Exit Cause** or **Latest Disabled Cause** indicates the possible reason for the exit from pRRU deep dormancy or its failure to take effect, as listed in [Table 7-79](#). In addition, the value of **Latest Exit Time** or **Latest Disabled Time** indicates the specific time for the latest exit or the latest failure to take effect.

Latest Exit Cause, **Latest Exit Time**, **Latest Disabled Cause**, and **Latest Disabled Time** are used to query the time and reasons for the latest exit and the latest failure to take effect within the specified period. The query may be performed outside the specified period. For example, if the specified period is

from 00:00:00 to 06:00:00, **Latest Exit Cause** or **Latest Disabled Cause** indicates the reason for the latest exit or the latest failure to take effect within the period from 00:00:00 to 06:00:00. This reason, together with the time the latest exit or application failure occurred, can be queried even by an operation performed within the time from 06:00:00 to 23:59:59.

 NOTE

For **Latest Disabled Cause** or **Latest Exit Cause**:

- The collection period is 10 minutes. If the reason changes frequently, the latest reason may not be updated in a timely manner.
- In separate-MPT scenarios, the system time is not strictly aligned. If the remaining time is less than 1 minute upon the exit from dormancy, the exit cause is not recorded.

Table 7-79 Possible reasons for the failure of pRRU deep dormancy to take effect

Cause Value	Description
Maintenance Link Abnormal	The RF module is disconnected, and the BBU cannot obtain the dormancy state of the RF module.
Board Unavailable	Links are normal, and the RRU is starting up.
Dormancy Capability Not Supported by the Hardware	The RF module does not support deep dormancy.
Configuration Conditions Not Met	• The dormancy switch is turned off. • No UMTS, eNodeB, gNodeB, or RFA application is available. • When the customized dormancy standard switch is turned on, the dormancy RATs do not include all RATs of the local end.
License Insufficiency	• The deep dormancy license items of an RAT are insufficient. • The license items affecting basic cell service capabilities (such as the cell access capability) are insufficient.
Not Within Dormancy Period	• The current time is beyond the configured dormancy period. • Time synchronization fails, and the NE does not enter the dormancy state. You can run the DSP TIMESRC command to check whether a time source is available.

Cause Value	Description
Base Station in Dormancy Suspension	• Dormancy pending is triggered due to the execution of the PND DORMANCY command. • Dormancy pending is triggered due to base station reset or configuration verification. • Dormancy pending is triggered due to reliability protection.
Upgrading	• The base station is being upgraded. • The base station is in the protection period for successful upgrade. The protection period lasts less than 15 minutes.
Process Being Protected	• After the RF module link recovers, it takes about 5 minutes for the related alarm to be cleared. The cell will enter the dormancy state after the alarm is cleared. • The base station instructs the cell to enter the dormancy state, and then instructs the RF module to enter the dormancy state 30s later. • Random dormancy waiting is ongoing. The random dormancy waiting time can be queried by running the DSP BRDDORMANCY command with the QUERYTYPE parameter set to SITE. • Less than 5 minutes have passed since the RF module exited dormancy.
Reliability Requirements Not Met	• The reliability requirements, such as those for solder joints or temperature difference, are not met. • The current RF module is faulty and does not meet the conditions for entering the dormancy state.
Service Conditions Not Met	Service faults exist and the conditions for entering the dormancy state are not met.
Not All NE Dormancy Notifications Received	The dormancy notifications of some NEs are not received. For example, in a separate-MPT base station, the peer end does not meet the requirements, and the RF module does not enter the dormancy state.
Carriers Existed on the RF Module	None

Cause Value	Description
Protection Due to Ongoing Device Maintenance and Test	- Device-related maintenance and test operations are being performed. - The base station is in the protection period for device-related maintenance and test operations. The protection will be canceled within 30 minutes after the operations are completed. Dormancy will be allowed after the protection is canceled.
Protection Due to Ongoing Service Maintenance and Test	- Service-related maintenance and test operations are being performed. - The base station is in the protection period for service-related maintenance and test operations. The protection will be canceled within 30 minutes after the operations are completed. Dormancy will be allowed after the protection is canceled.
Others	- The plug-and-play module is used, which does not meet the conditions for taking effect. - The internal software is faulty, or the evaluation period defined in the software is not reached.

pRRU Deep Dormancy in Scheduled Mode

- Perform the following operation to query the LTE cell status to check whether pRRU deep dormancy has taken effect:

```
//Querying the cell status
DSP CELL: LocalCellId=1;
```

If the value of the **Cell power save state** parameter is **RF Module Regular Time Sleep Mode** or **Certain RF Modules in Sleeping State** in the preceding command output, this function has taken effect. Alarms described in [7.6.1.1 pRRU Deep Dormancy in Scheduled Mode](#) are still suppressed within 10 minutes after pRRU deep dormancy stops. If commands such as **BLK BRD** are executed to block a pRRU, the value of the **Work Status** parameter is still **RRU deep dormancy is enabled** in the **DSP CELL** command output.

Feature query using the MAE-Deployment returns results on a per eNodeB basis. If none of the pRRUs of an eNodeB is enabled with pRRU deep dormancy, the feature status on the MAE-Deployment is **OFF**. If pRRU deep dormancy is enabled on any pRRUs of an eNodeB, the feature status on the MAE-Deployment is **ON**.

- Perform the following operation to query the NR DU cell status to check whether pRRU deep dormancy has taken effect:

Run the **DSP NRDUCELL** command. If the **Power Saving State** parameter indicates **Dormancy of All RF Modules:Enabled** or **Dormancy of Some RF Modules:Enabled**, pRRU deep dormancy has taken effect

NOTE

Within 10 minutes after the pRRUs exit the deep dormancy state, the following alarms are still not reported unless they are caused by other faults:

- ALM-29840 gNodeB Out of Service
- ALM-29841 NR Cell Unavailable
- ALM-29870 NR DU Cell TRP Unavailable
- ALM-29871 NR DU Cell TRP Capability Degraded

pRRU Deep Dormancy in Intelligent Mode

- Perform the following operation to query the dormancy status:
 - Run the **DSP ENODEBMPRUDORMANCY** command to query which pRRUs are in the deep dormancy state. If the value of **Dormancy Flag** is **ENABLED** in the command output, the pRRU enters the deep dormancy state. If the value of **Dormancy Flag** is **DISABLED** in the command output, the pRRU fails to enter the deep dormancy state.
 - Run the **DSP INTELPRRUDORMANTSTAT** command to query the status of pRRU deep dormancy in intelligent mode. [Table 7-80](#) lists the command output.

Table 7-80 pRRU deep dormancy in intelligent mode

Intel pRRU Dormancy Status	Description
ENABLED	pRRU deep dormancy in intelligent mode is enabled.
SWITCH_OFF	The switch for pRRU deep dormancy in intelligent mode is turned off.
CELL_NOT_AVAILABLE	The cell status is abnormal.
SRS_NOT_CFG	The SRS configuration switch is turned off or the SRS configuration is incorrect.
CELL_CFG_NOT_SUPPORTED	• The cell is an uplink cell. • The cell is an NB-IoT cell.
HARDWARE_NOT_SUPPORTED	None
NOT_IN_EFFECTIVE_TIME	None
PRB_USAGE_NOT_MET	None
IN_PUNISH_TIME	The cell is in the penalty state after exiting the dormancy state.
OTHER_FEATURE_EFFECTIVE	Mutually exclusive features exist.
NO_DORMANCY_PRRU	None

Intel pRRU Dormancy Status	Description
UNDER_COVERAGE_EVAL	Basic indoor coverage is being evaluated.
OTHER_REASON	There are other internal causes.

7.6.4.3 Network Monitoring

It is recommended that counter values be collected separately for energy saving periods and other periods of the day to prevent data in non-energy-saving periods from affecting the function gain evaluation.

- You can monitor the base station energy consumption using the counters described in [Table 7-81](#). On the live network, the one-week base station energy consumption data obtained before and after pRRU deep dormancy is enabled can be compared to assess the energy saving gains of this function. It is expected that the base station energy consumption decreases significantly after this function is enabled.

Table 7-81 Counters related to base station energy consumption

Counter ID	Counter Name	Counter Description
1593835637	VS.EnergyCons.BTS.Adding.LTE	Sum of energy consumption of all boards in LTE
1593835804	VS.EnergyCons.BTS.Adding.NR	Sum of energy consumption of all boards in NR
1593835959	VS.EnergyCons.BTS.Adding.Service.LTE	Sum of power consumption of all boards (LTE services)
1593835961	VS.EnergyCons.BTS.Adding.Service.NR	Sum of power consumption of all boards (NR services)
1593835628	VS.EnergyCons.BTS.Adding.UMTS	The sum of power consumption of all boards in UMTS

- You can monitor the energy consumption of baseband processing units and RF modules using the counters described in [Table 7-82](#). For a multimode LampSite base station supporting LTE or NR, it is recommended that RF module energy consumption counters be used to evaluate the energy saving effect.

Table 7-82 Counters related to board energy consumption

Counter ID	Counter Name	Counter Description
1593835836	VS.EnergyCons.BTSBoard (LTE eNodeB, 5G gNodeB)	Power consumption of board
1593835835	VS.EnergyCons.BU	Power consumption of baseband unit

- For 5G, you can also use the following counters for observation:
 - N.Cell.Unavail.Dur.RFModuleDeepDormancy:** The duration of cell unavailability due to pRRU deep dormancy can be obtained by observing this counter on the MAE-Access.
 - N.Cell.Unavail.Dur.EnergySaving:** The duration indicated by this counter has a deviation of less than 3 minutes from the pRRU deep dormancy period configured by the operator. This is because evaluation of the time and other triggering conditions is required before entry into the pRRU dormancy state and operations such as pRRU restart and cell reestablishment are performed upon an exit from the dormancy state.

7.7 Adaptive BBU Deep Dormancy

7.7.1 Principles

When the RF modules have entered the deep/super dormancy state on a network under light or no load, cells served by the BBP also enter the dormancy state, and the BBP is running with no services, wasting base station energy.

To address this, adaptive BBU deep dormancy is introduced. **When all cells served by a BBP have entered the dormancy state, the BBP adaptively enters the deep dormancy state based on the configured period for dormancy to take effect on the corresponding RF module and load of the RF module, reducing the energy consumption of the base station.**

This function is configured as follows:

- Enable BBP deep dormancy for the corresponding RAT.
 - For UMTS, select the **BBP_DEEP_DORMANCY_SW** option of the **NODEBALGPARA.DORMANCYSWITCH** parameter.
 - For LTE TDD, select the **TDD_BBP_DEEP_DORMANCY_SW** option of the **EnodebPwrSavingAlgo.DormancySwitch** parameter.
 - For LTE FDD, select the **FDD_BBP_DEEP_DORMANCY_SW** option of the **EnodebPwrSavingAlgo.DormancySwitch** parameter.
 - For NR, select the **BBP_DEEP_DORMANCY_SW** option of the **gNBPwrSavingAlgo.DormancySwitch** parameter.
- Enable adaptive BBP deep dormancy by setting the **BBP.PWRSVGSW (5G gNodeB, LTE eNodeB)** parameter to **BBP_DEEP_DORMANCY_ADAPT**.

Adaptive BBU deep dormancy also provides the following functions:

- Dormancy pending and pending cancellation
If operations on the base station (such as upgrades) require all BBPs to remain active, dormancy pending can be enabled. Dormancy pending can be canceled by running the **PND DORMANCY** command with the **OPERATIONTYPE** parameter set to **UNPENDING**. For details, see [7.10 Dormancy Pending and Pending Cancellation](#).
- Reliability protection during dormancy
The reliability protection involves license reliability protection, configuration reliability protection, reliability protection against faults, and O&M reliability protection during dormancy. For details, see [7.11 Reliability Protection for Deep Dormancy and Super Dormancy](#).

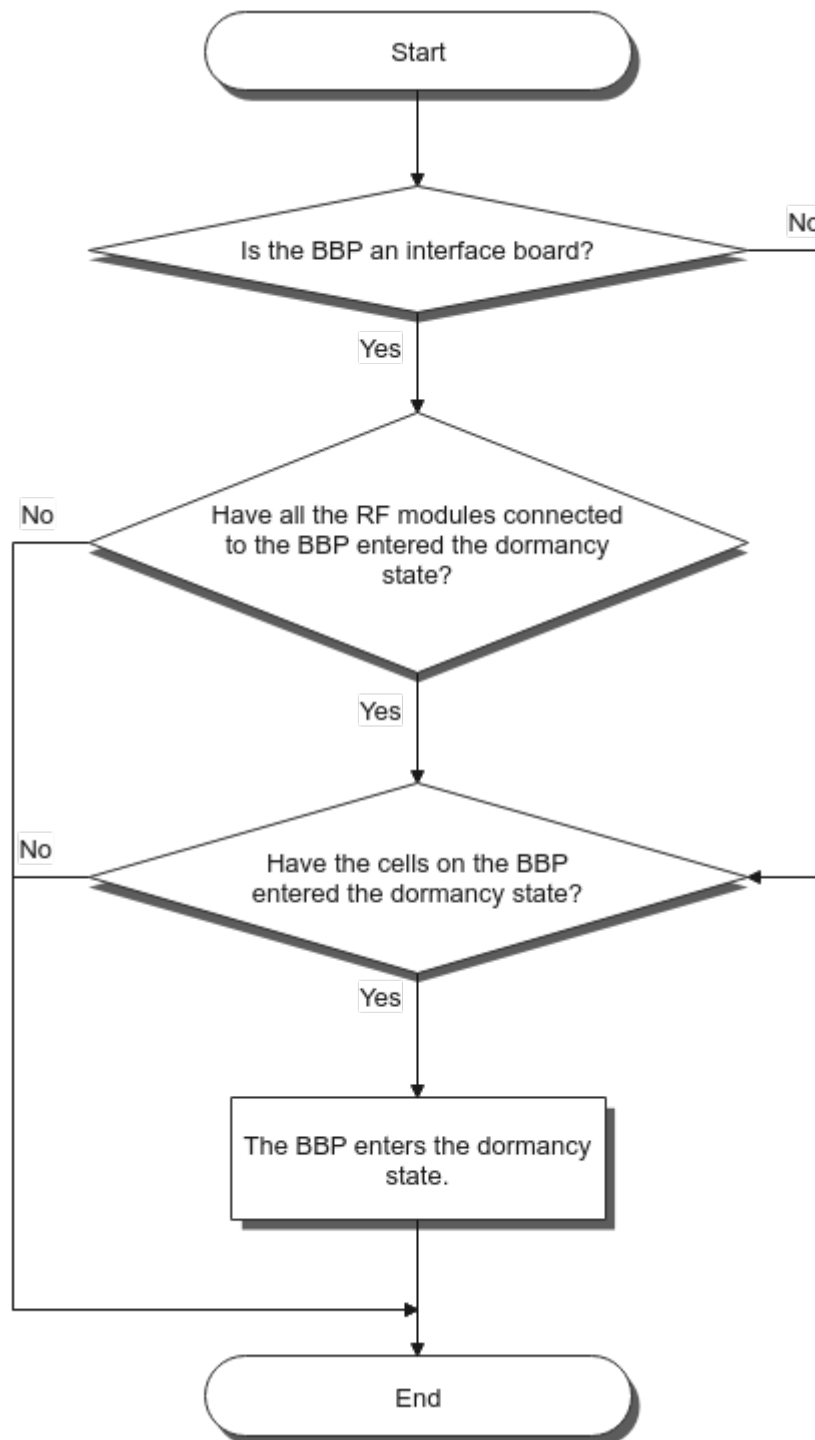
 NOTE

If a BBP works in multiple RATs, it can enter the dormancy state only when the corresponding function switches are turned on and specific conditions are met for each RAT.

7.7.1.1 Entering the Deep Dormancy State

[Figure 7-34](#) shows the dormancy process after adaptive BBU deep dormancy is enabled.

Figure 7-34 Adaptive BBU deep dormancy flowchart



1. Check whether the BBP is an interface board.
 - If yes, the procedure goes to 2.
 - If not, the procedure goes to 3.
2. Check whether all the RF modules connected to the BBP have entered the RF module dormancy state (including deep dormancy and super dormancy).
If all the RF modules have entered the dormancy state, the procedure goes to 3.

3. Check whether all the cells on the BBP have entered the dormancy state.
If so, the procedure goes to [4](#).
4. The BBP enters the adaptive deep dormancy state.

7.7.1.2 Exiting the Deep Dormancy State

If an RF module connected to the BBP needs to exit the load-based deep/super dormancy state as the load conditions are no longer met, the BBP starts the deep dormancy exit process. After the BBP exits the deep dormancy state, the RF module exits the deep/super dormancy state, and cell services are restored.

In other scenarios, the BBP starts to exit the deep dormancy state 5 minutes before the time for the RF module connected to the BBP to exit the dormancy state arrives. After the BBP exits the deep dormancy state, the RF module exits the dormancy state, and cell services are restored.

7.7.2 Network Analysis

7.7.2.1 Benefits

It is recommended that adaptive BBU deep dormancy be enabled together with RF module deep/super dormancy to reduce base station energy waste. To ensure that the actual operation duration meets the typical lifecycle management requirement, the BBP can enter the dormancy state when its reliability requirements (such as temperature differential) are met. If the BBP is reset during the dormancy period for certain purposes such as upgrade, the BBP and the cells will revert to the normal state, and the energy saving gains decrease.

7.7.2.2 Impacts

The impacts between this function and other functions are described in [7.7.3.2.3 Function Impacts](#).

- Commands such as **STR HWTST** and **STR HWOLTST** cannot be executed to start BBP hardware status check after a BBP enters the deep dormancy state.
- Commands such as **DSP SFP** and **DSP CPRIPORT** cannot be executed to query information about the BBP or RF modules connected to the BBP after a BBP enters the deep dormancy state.
- Tracing or monitoring tasks that have been started before the BBP enters the deep dormancy state may automatically stop after the BBP enters the deep dormancy state. In the deep dormancy state, such tasks may fail to be started, and those that are held may not report any data.
- Base station reconfiguration during the BBP dormancy period is not recommended. If reconfiguration is required, the dormancy function must be disabled first. Cell reconfigurations performed when the BBP is in the dormancy state take effect only after the BBP exits the dormancy state. If new configurations do not meet the conditions for the BBP to enter the dormancy state, the BBP exits the state.
- If a batch processing task of the OSS contains a NodeBFunction, eNodeBFunction, or gNodeBFunction MML command that affects cell setup or a Node MML command, and an NE of the base station is in the deep

dormancy state, the OSS executes the related configurations only after the base station exits deep dormancy. This increases the waiting time before configuration execution. On average, the base station takes less than 8 minutes to exit deep dormancy.

- When the **ULD FILE** command is executed to upload files from the BBP or the RF module connected to the BBP after BBP dormancy takes effect, the execution will fail with a message indicating that dormancy is in effect and will not stop. The command can be successfully executed if the files to be uploaded are one-click logs or logs of BBPs not in the dormancy state.
- Assume that the local and neighboring base stations are in the deep dormancy state, and an upgrade is performed for the local base station, not for the neighboring base stations. The local base station reverts to the normal state after the upgrade, and CEUs in weak coverage areas may access the available cells served by the local base station. This affects KPIs related to the aspects such as access and service drop of the local base station. It is recommended that the **PND DORMANCY** command be executed on both the base station to be upgraded and its neighboring base stations within the upgrade time window.
- After the BBP enters the deep dormancy state or is in the process of exiting this state, the following alarms caused by deep dormancy of the BBP are not reported.
 - Node alarms
 - ALM-26214 Board Powered Off
 - NR alarms
 - ALM-29841 NR Cell Unavailable
 - ALM-29844 NR DU Cell Unavailable
 - ALM-29870 NR DU Cell TRP Unavailable
 - ALM-29871 NR DU Cell TRP Capability Degraded
 - LTE alarms
 - ALM-29240 Cell Unavailable
 - ALM-29243 Cell Capability Degraded
 - UMTS alarms
 - ALM-28203 Local Cell Unusable
 - ALM-28206 Local Cell Capability Decline
 - ALM-22202 UMTS Cell Unavailable (RNC)
 - ALM-22206 UMTS Cell Setup Failed (RNC)
 - ALM-22214 NodeB Unavailable (RNC)
 - ALM-22215 UMTS Cell Max DL Power Mismatch (RNC)
 - ALM-22203 UMTS Cell Common Channel Faulty (RNC)

- ALM-22207 UMTS Cell Reconfiguration Failed (RNC)
- ALM-22208 UMTS Cell Common Channel Setup Failed (RNC)
- ALM-22209 UMTS Cell Common Channel Reconfiguration Failed (RNC)
- ALM-22217 UMTS Cell HSDPA Function Fault (RNC)
- ALM-22218 UMTS Cell HSUPA Function Fault (RNC)
- ALM-22219 UMTS Cell EFACH Function Fault (RNC)
- ALM-22220 UMTS Cell MIMO Function Faulty (RNC)
- ALM-22222 UMTS Cell ERACH Function Fault (RNC)
- ALM-22223 UMTS Cell EPCH Function Fault (RNC)
- ALM-22237 UMTS Cell DC-HSUPA Function Fault (RNC)
- ALM-22241 UMTS Cell MC-HSDPA Function Fault (RNC)
- If a power-related command needs to be executed on a base station where an NE is in deep dormancy, the base station first exits deep dormancy and then executes the command. On average, the base station takes less than 8 minutes to exit deep dormancy.
- If the dormancy entry conditions are met during a base station version download or activation, the BBP will enter the dormancy state only after the upgrade is complete. If a version download or activation is performed when the BBP is in the deep dormancy state, the BBP will exit the deep dormancy state first.
- After a BBP enters the deep dormancy state, it exits the dormancy state if the associated RF module dormancy switch is turned off.
- When the **DORM_MNTMODE_CTRL_SW** option of the **EQUIPMENT.EQMDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter is selected, the BBP is allowed to enter the dormancy state. Modifying the configuration of a base station when dormancy is in effect may trigger dormancy exit. You are advised to start dormancy pending before configuration modification.
- When the **BBP_DEEP_DORMANCY_SW** option of the **NODEBALGPARA.DORMANCYSWITCH** parameter is selected, the UE allocation policy is automatically adjusted to increase the likelihood of the BBP entering the dormancy state. As a result, the load of the main control board and BBP supporting UMTS changes.

7.7.3 Requirements

7.7.3.1 Licenses

RAT	Feature ID	Feature Name	Model	NE	Sales Unit
UMTS	MRFD-221212	BBU Deep Dormancy (UMTS)	LQW9BBUDD01	NodeB	per NodeB
LTE TDD	MRFD-181233	BBU Deep Dormancy (LTE TDD)	LT4SBBUDDTD	eNodeB	per eNodeB
LTE FDD	MRFD-211221	BBU Deep Dormancy (LTE FDD)	LT1S0BBUDD01	eNodeB	per eNodeB
NR	MRFD-181263	BBU Deep Dormancy (NR)	NR0S0BBUDD00	gNodeB	per gNodeB

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

7.7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.7.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference
UMTS LTE FDD LTE TDD NR FDD Low-frequency NR TDD	RF module deep dormancy	RRU.DORMANCYSW (5G gNodeB, LTE eNodeB)	<i>Multi-RAT Coordinated Energy Saving</i>

RAT	Function Name	Function Switch	Reference
UMTS LTE FDD LTE TDD NR FDD Low-frequency NR TDD	RF module super dormancy	RRU.DORMANCYSW <i>(5G gNodeB, LTE eNodeB)</i>	<i>Multi-RAT Coordinated Energy Saving</i>
UMTS LTE FDD LTE TDD NR FDD Low-frequency NR TDD	pRRU deep dormancy	RRU.DORMANCYSW <i>(5G gNodeB, LTE eNodeB)</i>	<i>Multi-RAT Coordinated Energy Saving</i>

 NOTE

The BBP can enter the dormancy state only when the RF module connected to the BBP enters the dormancy state triggered by one of the preceding functions.

7.7.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
LTE TDD LTE FDD	Co-processing resources	ADD COPROGRES	None	BBP deep dormancy cannot be enabled on an eNodeB configured with co-processing resources.
NR FDD	Dual-plane function in intra-frequency dual-network redundancy networking	DUAL_PLANE_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	The dual-plane function in intra-frequency dual-network redundancy networking cannot be enabled together with adaptive BBU deep dormancy on a gNodeB.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios	DUAL_PLANE_TOB_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	<i>Base Station Equipment Reliability</i>	The dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios cannot be enabled together with adaptive BBU deep dormancy on a gNodeB.
NR FDD Low-frequency NR TDD	Primary/Backup function in intra-frequency dual-network redundancy networking	PRIMARY_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	The primary/backup function in intra-frequency dual-network redundancy networking cannot be enabled together with adaptive BBU deep dormancy on a gNodeB.
Low-frequency NR TDD	Dual-BBU backup in dual-network redundancy networking	DUAL_BBU_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	Dual-BBU backup in dual-network redundancy networking cannot be enabled together with adaptive BBU deep dormancy on a gNodeB.
NR FDD Low-frequency NR TDD	Primary/Secondary BBP	NRDUCellTrp.SecondaryBasebandEqmld	None	BBP deep dormancy cannot be enabled on a gNodeB configured with the primary/secondary BBP.
NR FDD Low-frequency NR TDD	BBP collaboration	NRDUCellTrp.BbResMutualAidSw	None	BBP collaboration cannot be enabled together with BBP deep dormancy on a gNodeB.
Low-frequency NR TDD	IAB	NRDUCell.DeployPosition	None	IAB cannot be enabled together with BBP deep dormancy on a gNodeB.

RAT	Function Name	Function Switch	Reference	Description
UMTS LTE FDD LTE TDD NR FDD Low-frequency NR TDD	BBP baseband processing link	ADD BBPLNK	None	BBP deep dormancy cannot be enabled for a BBP configured with a baseband processing link.
NR FDD Low-frequency NR TDD	BBU universal baseband radio unit	ADD UBRU	None	BBP deep dormancy cannot be enabled for a BBU configured with a universal baseband radio unit (UBRU).
UMTS LTE FDD LTE TDD NR FDD Low-frequency NR TDD	BBU ring topology	RRUCHAIN.TT <i>(5G gNodeB, LTE eNodeB)</i>	None	If a ring topology is used for BBU networking, BBP deep dormancy cannot be enabled.

7.7.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
UMTS	Adaptive tilt adjustment	None	None	After BBP dormancy is enabled, the adaptive tilt adjustment function for UMTS does not take effect. The NodeB sends the failure cause to the RNC and records the antenna adjustment information in SON logs. It is recommended that energy saving be prioritized and adaptive tilt adjustment be disabled.
LTE FDD LTE TDD NR FDD Low-frequency NR TDD	LTE smart power lock NR smart power lock	LTE: CellSmartPwrLock.SmartPowerLockMode parameter set to AVERAGE_POWER_CONTROL NR: NRDUCellSmartPwrLock.SmartPowerLockSw parameter set to AVERAGE_POWER_CONTROL_SW .	None	LTE smart power lock and NR smart power lock do not take effect if adaptive BBU deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD NR FDD Low-frequency NR TDD	Intelligent Detection for Fronthaul Optical Paths	None	<i>Intelligent Detection for Fronthaul Optical Paths</i>	- If the BBU and RF module are not in the dormancy state when optical path detection is started, the detection process will delay the time at which they enter that state. - If the BBU and RF module are in the dormancy state when optical path detection is started, the following logic applies: - In a scenario where wakeup from dormancy is supported, starting optical path detection will cause the BBU and RF module to automatically exit the dormancy state. They will not re-enter the state within 30 minutes after the detection process starts. Once 30 minutes elapse, they will evaluate whether to re-enter the dormancy state. - In a scenario where wakeup from dormancy is not supported, optical path detection cannot be started.
LTE FDD LTE TDD	Inter-cell cable sequence detection	STR INTERCELLCROSS FEEDERTST	<i>MIMO in eRAN Feature Documentation</i>	Once inter-cell cable sequence detection begins, the task of adaptive BBU deep dormancy is suspended and resumes upon completion of the detection process.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Baseband processor shutdown	- Adaptive baseband shutdown BBP.CELLDEPLOY set to ADAPT_BBP_SHUTDOWN - Power saving first BBP.CELLDEPLOY set to POWERSAVING - Baseband shutdown for energy saving BB_SHUTDOWN_POWERSAVING_SW option of the gNodeBParam.BaseStationDeepDormancySw parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Adaptive baseband shutdown, power saving first, and baseband shutdown for energy saving do not take effect if adaptive BBU deep dormancy is in effect.

7.7.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

BBUs

Only the BBU5900 and BBU5910 support this function.

Base Station Models

- 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
- DBS3900 LampSite and DBS5900 LampSite for LTE and NR

Boards

The main control board must be a UMPTe, UMPTg, UMPTga, UMPTha, or UMPThb series board.

The BBP must be a UBBPg (excluding UBBPg1) or UBBPh series board that supports UMTS, LTE TDD, LTE FDD, or NR.

RF Modules

This function serves as a dormancy function for BBU boards and has no special requirements for RF modules. The RF module requirements of its prerequisite functions need to be considered.

7.7.3.4 Networking

- HEI port interconnection is not supported.
- Cloud BB networking is not supported.
- A ring topology (with **RRUCHAIN.TT (LTE eNodeB, 5G gNodeB)** set to **RING**) is not supported if it includes more than one cascaded RF module.
- CI interconnection between the root and leaf BBUs is not supported.
- The effective working modes of the BBP do not include GSM.

7.7.3.5 Cells

For LTE cells, the function is not supported in A+P scenarios.

For NR cells, UEM deployment and PDCP moved downward are not supported.

All activated cells of the base station have been bound to the one and only BBP.

In LTE or NR, at least one activated cell works in the mode specified by the **BBP.BBWS (LTE eNodeB, 5G gNodeB)** parameter for the BBP.

In UMTS, at least one local cell works in the mode specified by the **BBP.BBWS (LTE eNodeB, 5G gNodeB)** parameter for the BBP.

7.7.3.6 Others

- Certificates cannot be deployed on the BBP.
- The NE where the BBP is located must be in the normal maintenance mode (including the maintenance mode of the base station, all APPs, and RF modules associated with the BBP). If the **DORM_MNTMODE_CTRL_SW** option of the **EQUIPMENT.EQMDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter is selected, this requirement does not apply.
- BBP deep dormancy does not take effect when the BBU power is insufficient.
- The CGPS or RGPS clock cannot be configured.
- The ambient temperature of the BBU must be stable. If the BBU is installed indoors, the temperature and humidity of the equipment room are usually controllable. If the BBU is installed in an outdoor cabinet, the cabinet door must be closed to reduce temperature and humidity fluctuations.
- By default, the BBP supports only standard optical modules.
- When the base station is connected to the NetEco, BBP deep dormancy cannot take effect. To allow dormancy to take effect in such scenarios, you can select the **DOR_LM_SW** option of the **Equipment.EQMDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter. After the BBP enters the dormancy

state, the maintenance link between the NetEco and the power supply that transmits signals through the BBP will be disconnected.

- When the base station is connected to the RPCU, BBP deep dormancy cannot take effect. To allow dormancy to take effect in such scenarios, you can select the **DOR_LM_SW** option of the **Equipment.EQMDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter. However, the upper-level board connected to the RPCU cannot enter the dormancy state.
- When monitoring devices, such as alarm ports and PMUs, are configured for RF modules, BBP deep dormancy cannot take effect. To allow dormancy to take effect in such scenarios, you can select the **RM_LM_SW** option of the **RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter. After the BBP enters the deep dormancy state, the maintenance links of the monitoring devices configured for the RF modules will be disconnected.
- When the base station is configured with an RF HA group, BBP deep dormancy cannot take effect.

7.7.4 Operation and Maintenance

7.7.4.1 When to Use

It is recommended that adaptive BBP deep dormancy be enabled in the following scenario to achieve better energy saving gains:

RF module deep/super dormancy is enabled for all RF modules connected to the BBP.

You are advised to select a site where the load of all cells on the BBP is low within the same period of time, so that all cells meet the load conditions for entering the dormancy state. In this way, the BBP can be shut down.

The temperature in the equipment room is stable. Therefore, this function is recommended for BBUs in indoor equipment rooms.

7.7.4.2 Data Configuration

7.7.4.2.1 Data Preparation

Table 7-83 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Dormancy Switch	NODEBALGPARA.DORMANCY SWITCH	Select the BBP_DEEP_DORMANCY_SW option of this parameter.
Dormancy Switch	EnodebPwrSavingAlgo.DormancySwitch	Select the TDD_BBP_DEEP_DORMANCY_SW option of this parameter.

Parameter Name	Parameter ID	Setting Notes
Dormancy Switch	EnodebPwrSavingAlgo.DormancySwitch	Select the FDD_BBP_DEEP_DORMANCY_SW option of this parameter.
Dormancy Switch	gNB PwrSavingAlgo.DormancySwitch	Select the BBP_DEEP_DORMANCY_SW option of this parameter.
Power Saving Switch	BBP.PWRSVGSW (LTE eNodeB, 5G gNodeB)	Set this parameter to BBP_DEEP_DORMANCY_ADAPT .
Equipment Dormancy Optional Feature Switch	Equipment.EQMDOROPTFEATSW (LTE eNodeB, 5G gNodeB)	When the BBP_DDC_SW option is selected, BBP deep dormancy can take effect even if the base station uses non-standard optical modules. When a BBP enters the deep dormancy state, the optical module enters the power saving state. After the BBP exits the dormancy state, the optical module returns to the normal working state. When non-standard optical modules are used, the connections between the BBP and RF modules may be abnormal and the BBP may even be damaged after the exit from the dormancy state, because the compatibility and quality of non-standard optical modules cannot be ensured. It is recommended that the default deselected state of the BBP_DDC_SW option of this parameter be retained.
Equipment Dormancy Optional Feature Switch	Equipment.EQMDOROPTFEATSW (LTE eNodeB, 5G gNodeB)	The dormancy function can take effect on the NE connected to the NetEco only when the DOR_LM_SW option is selected. If the BBP enters the dormancy state, the maintenance link between the NetEco and the power supply that transmits signals through the BBP will be disconnected. It is recommended that the default deselected state of the DOR_LM_SW option of this parameter be retained.

Parameter Name	Parameter ID	Setting Notes
Equipment Dormancy Optional Feature Switch	EQUIPMENT.EQMDORO PTFEATSW (LTE eNodeB, 5G gNodeB)	Select the DORM_MNTMODE_CTRL_SW option if the NE where the BBP is located is in the normal maintenance mode (including the maintenance mode of the base station, all APPs, and RF modules associated with the BBP) and dormancy is allowed.
RRU Dormancy Optional Feature Switch	RRUEXTINFO.RRUDORO PTFEATSW (LTE eNodeB, 5G gNodeB)	BBU deep dormancy can take effect when the RF module is configured with alarm ports and monitoring devices such as PMUs only if the RM_LM_SW option is selected. When the BBP enters the deep dormancy state, the maintenance links of the monitoring devices configured for the RF module will be disconnected. It is recommended that the default deselected state of the RM_LM_SW option of this parameter be retained.

NOTE

This function requires that the baseband working standard (specified by the **BBP.BBWS (LTE eNodeB, 5G gNodeB)** parameter) of the BBP be UMTS, LTE TDD, LTE FDD, NR, or a combination of these RATs.

7.7.4.2.2 Using MML Commands

Before using MML commands, refer to [7.7.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

```
//Enabling adaptive BBP deep dormancy
MOD BBP: CN=0, SRN=0, SN=2, PWRSVGSW=BBP_DEEP_DORMANCY_ADAPT;
//(Optional) Enabling adaptive BBP deep dormancy for a NodeB if the UMTS option of the Base Band Work
Standard parameter for a BBP is selected, after which the base station enters the dormancy-pending state
SET NODEBALGPARA: DORMANCY_SWITCH=BBP_DEEP_DORMANCY_SW-1;
//Canceling the dormancy-pending state of the base station
PND DORMANCY: OPERATIONTYPE=UNPENDING;
//(Optional) Enabling adaptive BBP deep dormancy for an eNodeB if the LTE_TDD option of the Base Band
Work Standard parameter for a BBP is selected
MOD ENODEBPWRSAVINGALGO: DormancySwitch=TDD_BBP_DEEP_DORMANCY_SW-1;
//(Optional) Enabling adaptive BBP deep dormancy for an eNodeB if the LTE_FDD option of the Base Band
```

```
Work Standard parameter for a BBP is selected
MOD ENODEBPWRSAVINGALGO: DormancySwitch=FDD_BBP_DEEP_DORMANCY_SW-1;
//(Optional) Enabling adaptive BBP deep dormancy for a gNodeB if the NR option of the Base Band Work
Standard parameter for a BBP is selected
MOD GNBWPWRSAVINGALGO: DormancySwitch=BBP_DEEP_DORMANCY_SW-1;
//(Optional) Turning on the dormancy and maintenance mode control switch
SET EQUIPMENT: EQMDOROPTFEATSW=DORM_MNTMODE_CTRL_SW-1;
//(Optional) Selecting the BBP_DDC_SW option for BBP deep dormancy to take effect even if the base
station uses non-standard optical modules. In this case, the connections between the BBP and RF modules
may be abnormal and the BBP may even be damaged after the exit from the dormancy state, because the
compatibility and quality of non-standard optical modules cannot be ensured.
SET EQUIPMENT: EQMDOROPTFEATSW=BBP_DDC_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling BBP deep dormancy
MOD BBP: CN=0, SRN=0, SN=2, PWRSVGSW=OFF;
//(Optional) Disabling adaptive BBP deep dormancy for a NodeB if the UMTS option of the Base Band
Work Standard parameter for a BBP is selected, after which the base station enters the dormancy-pending
state
SET NODEBALGPARA: DORMANCYSWITCH=BBP_DEEP_DORMANCY_SW-0;
//Canceling the dormancy-pending state of the base station
PND DORMANCY: OPERATIONTYPE=UNPENDING;
//(Optional) Disabling adaptive BBP deep dormancy for an eNodeB if the LTE_TDD option of the Base Band
Work Standard parameter for a BBP is selected
MOD ENODEBPWRSAVINGALGO: DormancySwitch=TDD_BBP_DEEP_DORMANCY_SW-0;
//(Optional) Disabling adaptive BBP deep dormancy for an eNodeB if the LTE_FDD option of the Base Band
Work Standard parameter for a BBP is selected
MOD ENODEBPWRSAVINGALGO: DormancySwitch=FDD_BBP_DEEP_DORMANCY_SW-0;
//(Optional) Disabling adaptive BBP deep dormancy for a gNodeB if the NR option of the Base Band Work
Standard parameter for a BBP is selected
MOD GNBWPWRSAVINGALGO: DormancySwitch=BBP_DEEP_DORMANCY_SW-0;
```

7.7.4.2.3 Using the MAE-Deployment

- Fast batch activation
This function can be batch activated using the Feature Operation and Maintenance function of the MAE-Deployment. For detailed operations, see the following section in the MAE-Deployment product documentation or online help: **MAE-Deployment Operation and Maintenance > MAE-Deployment Guidelines > Enhanced Feature Management > Feature Operation and Maintenance**.
- Single/Batch configuration
This function can be activated for a single base station or a batch of base stations on the MAE-Deployment. For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.7.4.3 Activation Verification

Run the following command to check whether adaptive BBP deep dormancy is in effect:

Run the **DSP BRDDORMANCY** command with the **Query Type** parameter set to **BOARD** and the **QUERYMODE** parameter set to **CURRENT** to query the status of adaptive BBP deep dormancy. The values displayed in the query results are periodically refreshed and may not be updated in a timely manner.

- If the value of **Board Dormancy State** in **Display BBP board power saving status** is **DORMANCY**, adaptive BBP deep dormancy has taken effect.
//Querying the BBP dormancy status
DSP BRDDORMANCY: QUERYTYPE=BOARD, QUERYMODE=CURRENT;
- If the value of **Board Dormancy State** in **Display BBP board power saving status** is **Not in the Dormancy State**, check the value of **Reason for Not Entering Deep Dormancy**, as described in [Table 7-84](#).

Run the **DSP BRDDORMANCY** command with the **QUERYTYPE** parameter set to **BOARD**, the **QUERYMODE** parameter set to **HISTORY**, and the **PWRSAVINGFEATURE** parameter set to **DEEP_DORMANCY**. You can determine the possible reasons why this function does not take effect based on the values of **Latest Exit Cause** and **Latest Disabled Cause** within the specified period. [Table 7-84](#) lists the possible reasons.

[Table 7-85](#) describes the meanings of **Latest Exit Cause**, **Latest Disabled Cause**, **Latest Exit Time**, and **Latest Disabled Time**.

Latest Exit Cause and **Latest Disabled Cause** indicate the reasons for the most recent exit and failure of the function. These reasons can be queried after the effective period has ended. For example, if the effective period is from 00:00:00 to 06:00:00, **Latest Exit Cause** and **Latest Disabled Cause** record the reasons for the most recent exit and failure of the function from 00:00:00 to 06:00:00. These reasons can be queried by an operation performed within the time from 06:00:00 to 23:59:59.

- The effective period of adaptive BBP deep dormancy is the intersection of the dormancy periods configured for the RF modules connected to the BBP.
- The collection period is 10 minutes. If the reason changes frequently, the latest reason may not be updated in a timely manner.
- In separate-MPT scenarios, where the system time is not strictly aligned, the reason for an exit from dormancy is not recorded if the remaining time is less than 1 minute.
- When the **QUERYMODE** parameter is set to **HISTORY**, the value of **Process Being Protected** is not displayed.

Table 7-84 Possible reasons why adaptive BBP deep dormancy does not take effect

Cause Value	Description
Dormancy Capability Not Supported by the Hardware	• The BBU hardware capability does not support the function. • The BBP hardware capability does not support the function. • The hardware capability of the main control board does not support the function. • The RF module hardware capability does not support the function. • The CXU hardware capability does not support the function.
Optical Module Compatibility Requirement Not Met	• The BBP uses non-standard optical modules. • The information about the optical modules used by the BBP is unknown. For example, the information is not obtained when the BBP starts up.
Dormancy Capability Not Supported by the Networking	• A ring topology is used for the RF units connected to the BBP and includes more than one cascaded RF module. • HEI port interconnection is used. • Cloud BB is configured. • The RF module connected to the BBP is configured with monitoring devices such as a PMU or an alarm port, or the RF module is connected to an RGPS device. • The BBP is connected to a CXU such as a CGPS device. • The following conditions are met when the DOR_LM_SW option of the EQUIPMENT.EQMDOROPTFEATSW parameter is deselected: – The connected electromechanical components are configuration-free. – An RPCU is configured. • The root and leaf BBUs are interconnected over CI ports. • The RF module that the BBP is serving or directly connected to belongs to an RF HA group.

Cause Value	Description
Configuration Conditions Not Met	- Switches of the BBP.PWRSVGSW (LTE eNodeB, 5G gNodeB), EnodebPwrSavingAlgo.DormancySwitch, gNBpwrSavingAlgo.DormancySwitch, and NODEBALGPARA.DORMANCYSWITCH parameters are turned off. - The BBP is blocked. - The NodeB baseband resources are blocked. - When the DORM_MNTMODE_CTRL_SW option of the EQUIPMENT.EQMDOROPTFEATSW (LTE eNodeB, 5G gNodeB) parameter is deselected, the NE where the BBP is located is in the non-normal maintenance mode (including the maintenance mode of the NE, all APPs, and RF modules associated with the BBP). - The baseband working standards of the BBP do not overlap with the APP. - The effective working modes of the BBP include GSM. - The RF modules associated with the BBP are blocked. - For LTE or NR, there are no activated cells in the RAT corresponding to the Work Mode configured for the BBP. - For UMTS, no local cell is configured for the BBP.
License Insufficiency	- The deep dormancy license items of an RAT are insufficient. - The license file expires. - The licenses are insufficient or have expired, affecting the function.
Impacted Features Exist	LTE cells such as SFN and relay cells do not support the function.
Requirements Not Met by the Peer NE of the Separate-MPT Base Station	- In separate-MPT scenarios, the peer LTE or NR version does not support BBP deep dormancy. - In a separate-MPT multimode base station, the peer configuration does not meet the requirements. - In a separate-MPT multimode base station, an alarm related to the peer is reported. - In a separate-MPT multimode base station, the peer mode uses the BBP for CPRI MUX topology.

Cause Value	Description
BBP Not Idle	- The connected RF module does not enter the deep, super, or scheduled dormancy state. - The cells deployed on the BBP have not entered the dormancy state. - Coordination services of the peer RAT are using the BBP. - In A+P scenarios, the BBP is required to transmit antenna data. 5 minutes has not elapsed since the latest exit from the dormancy state. - The smart power lock function is using this BBP. - There are UMTS UEs that have not been released served by the BBP.
Reliability Requirements Not Met	- The reliability requirements, such as those for solder joints or temperature difference, are not met. - The ambient temperature does not meet the requirements. (It is required that the ambient temperature of the BBU remain stable.) - The BBP cannot be detected.
Base Station in Dormancy Suspension	- Dormancy pending is triggered due to the execution of the PND DORMANCY command. - Dormancy pending is triggered due to base station reset or configuration verification. - Dormancy pending is triggered due to reliability protection.
Upgrading	- The base station is being upgraded. - The base station is in the protection period for successful upgrade. The protection period lasts less than 15 minutes.
Device Fault or Configuration Conflict	- An RF module connected to the BBP is faulty. - The BBP is faulty or unusable. - The configurations of the BBP conflict. - The BBP reports a certificate-related alarm, such as the device certificate invalid alarm and the certificate about to expire alarm.
BBU Power Insufficiency	- The power module provides insufficient power supply capability. - The CANBUS is faulty or the power cannot be obtained.

Cause Value	Description
Process Being Protected	- When the base station is reset, the BBP enters the dormancy state after a delay of 30 minutes. - After the BBP is reset, it enters the dormancy state after a delay of 15 minutes. - The requirements must be met throughout 5 minutes before the BBP enters the dormancy state. - The dormancy is in the preprocessing phase. - The APP is reset.
Protection Due to Ongoing Device Maintenance and Test	- Device-related maintenance and test operations are being performed. - The base station is in the protection period for device-related maintenance and test operations. The protection will be canceled within 30 minutes after the operations are completed. Dormancy will be allowed after the protection is canceled.
Other Reason	- Certificates are deployed on the BBP. - The ESN of the main control board cannot be obtained due to other exceptions. For example, the main control board has been in the dormancy state for more than 23.5 hours. - Other reasons, such as the internal software reasons, cause the function not to take effect.

Table 7-85 Time and causes for the most recent exit and failure of a function

Command Output of DSP BRDDORMANCY	Description
Latest Exit Cause	Indicates the cause for the latest exit of a feature during its effective period.
Latest Disabled Cause	Indicates the cause for the latest application failure of a feature during its effective period.
Latest Exit Time	Indicates the recorded time of the latest exit of a feature during its effective period.
Latest Disabled Time	Indicates the recorded time of the latest application failure of a feature during its effective period.

7.7.4.4 Network Monitoring

On the live network, the one-week energy consumption data obtained before and after the energy saving feature is enabled can be compared to evaluate the energy saving gains. It is expected that the energy consumption decreases significantly after this feature is enabled. The specific data and observation methods are as follows:

- Check the **VS.EnergyCons.BU (LTE eNodeB, 5G gNodeB)** counter (which indicates the energy consumption of the BBU) to see the energy consumption of the BBU in a base station.
- Check the **VS.BBUBoard.BBP.DeepDormancy.Dur (LTE eNodeB, 5G gNodeB)** counter (which indicates the BBP deep dormancy duration) on the MAE-Access to see the effective duration of adaptive BBP deep dormancy.
- Check the **VS.Energycons.DiffDurBbpDeepDormancy.BTSBoard (LTE eNodeB, 5G gNodeB)** counter (which indicates the difference in board energy consumption during BBP dormancy) to see the impact of enabling adaptive BBP deep dormancy on the **VS.EnergyCons.BTSBoard (LTE eNodeB, 5G gNodeB)** counter (which indicates the energy consumption of the board).
Energy consumption difference = Actual energy consumption of the board during BBP dormancy – Reported energy consumption of the board during BBP dormancy

NOTE

- There is delay in making decisions on whether to enter or exit the energy saving state. Therefore, the duration measured by the performance counters has an approximately 1-minute deviation from the actual duration.
- The sampling period is not synchronous with the reporting period, so a deviation of one sampling period may occur.

7.8 Main Control Board Dynamic Energy Saving

7.8.1 Principles

When the transmission rates over the main control board ports are low, the traffic volume is also low. Configuring the main control board with full-load resources in such conditions will lead to energy waste.

To address this, main control board dynamic energy saving is introduced. When the transmission rates over the main control board ports are low, this function can be enabled to dynamically adjust the port transmission rates, reducing the base station's energy consumption.

This function is controlled by the **MPT_DYN_ENG_SAV_FORBID_SW** option of the **EQUIPMENT.EQMDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter.

- If this option is selected, main control board dynamic energy saving is forbidden from taking effect.
- If this option is deselected, main control board dynamic energy saving is allowed to take effect.

By default, this option is deselected, allowing main control board dynamic energy saving to take effect.

7.8.2 Network Analysis

7.8.2.1 Benefits

Main control board dynamic energy saving can dynamically adjust the transmission rates over the main control board ports, reducing the base station's energy consumption.

7.8.2.2 Impacts

- When main control board dynamic energy saving is in effect, the maximum and average board CPU usages (**VS.BBUBoard.CPULoad.Max (LTE eNodeB, 5G gNodeB)** and **VS.BBUBoard.CPULoad.Mean (LTE eNodeB, 5G gNodeB)**) of the main control board may increase.
- When main control board dynamic energy saving is in effect, the maximum and average user-plane CPU usage of a board in a gNodeB (**VS.NRBoard.UPlane.CPULoad.Max** and **VS.NRBoard.UPlane.CPULoad.Avg**) may increase.

7.8.3 Requirements

7.8.3.1 Licenses

None

7.8.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.8.3.2.1 Prerequisite Functions

None

7.8.3.2.2 Mutually Exclusive Functions

None

7.8.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TD High-frequency NR TD	Primary/Backup function in intra-frequency dual-network redundancy networking	PRIMARY_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	The primary/backup function in intra-frequency dual-network redundancy networking and main control board dynamic energy saving cannot take effect together.

7.8.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
- DBS3900 LampSite and DBS5900 LampSite

Boards

Only the UMPThb series boards support this function.

7.8.3.4 Networking

Co-transmission through backplane interconnection on a separate-MPT MBTS is not supported. For details about this scenario, see *Common Transmission*.

The same RAT cannot be deployed on multiple main control boards.

BBU interconnection is not supported.

7.8.3.5 Others

None

7.8.4 Operation and Maintenance

7.8.4.1 Data Configuration

7.8.4.1.1 Data Preparation

Table 7-86 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Equipment Dormancy Optional Feature Switch	EQUIPMENT.EQMDORO PTFEATSW (LTE eNodeB, 5G gNodeB)	Deselect the MPT_DYN_ENG_SAV_FORBID_SW option of this parameter to enable main control board dynamic energy saving.

7.8.4.1.2 Using MML Commands

Before using MML commands, refer to [7.8.2.2 Impacts](#) and [7.8.3.2 Functions](#) and complete the parameter configurations for related functions based on the impact and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

```
//Enabling main control board dynamic energy saving
SET EQUIPMENT: EQMDOROPTFEATSW=MPT_DYN_ENG_SAV_FORBID_SW-0;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling main control board dynamic energy saving
SET EQUIPMENT: EQMDOROPTFEATSW=MPT_DYN_ENG_SAV_FORBID_SW-1;
```

7.8.4.1.3 Using the MAE-Deployment

- Fast batch activation
This function can be batch activated using the Feature Operation and Maintenance function of the MAE-Deployment. For detailed operations, see the following section in the MAE-Deployment product documentation or online help: **MAE-Deployment Operation and Maintenance > MAE-Deployment Guidelines > Enhanced Feature Management > Feature Operation and Maintenance.**

- Single/Batch configuration

This function can be activated for a single base station or a batch of base stations on the MAE-Deployment. For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.8.4.2 Activation Verification

You can run the **DSP ETHPORT** command to check the transmission rate over a main control board port through the value of **Local Speed** when **Port Status** is **Up**.

You can run the following command to verify whether main control board dynamic energy saving has taken effect.

Run the **DSP BRDDORMANCY** command with the **Query Type** parameter set to **BOARD** and the **QUERYMODE** parameter set to **CURRENT** to query the current status of main control board dynamic energy saving. The values displayed in the query results are periodically refreshed and may not always reflect the most recent updates.

- If the value of **Board Dormancy State** in **Display MPT board power saving status** is **Main Ctrl Brd Dyn Energy Saving**, main control board dynamic energy saving has taken effect.
//Querying status of main control board dynamic energy saving
DSP BRDDORMANCY: QUERYTYPE=BOARD, QUERYMODE=CURRENT;
- If the value of **Board Dormancy State** in **Display MPT board power saving status** is **Not in the Dormancy State**, check the possible reasons as described in [Table 7-87](#).

Table 7-87 Possible reasons why main control board dynamic energy saving does not take effect

Cause	Description
Hardware capability not supported	The main control board hardware capability does not support the function.
Networking not supported	Co-transmission through backplane interconnection on a separate-MPT MBTS Same RAT deployed on multiple main control boards BBU interconnection
Configuration conditions not met	The switch is not turned on.

Cause	Description
Impacted features	Other impacted features, such as primary/backup function in intra-frequency dual-network redundancy networking, have taken effect.
Unsatisfied transmission rate	The transmission rates of the main control board ports are higher than 10 Gbit/s.
Device fault or configuration conflict on the base station	Board Software Program Error or Board Overload alarm is reported, the availability status of the board is abnormal, or other faults or conflicts.
Process being protected	- After the base station is reset, it enters the energy saving state after a delay of about 30 minutes. - The APP of the main control board is reset.
Other reason	Other reasons, such as internal software issues, cause the function not to take effect.

7.8.4.3 Network Monitoring

On the live network, the one-week values of the **VS.EnergyCons.BU (LTE eNodeB, 5G gNodeB)** counter (which indicates the energy consumption of the BBU) obtained before and after the energy saving feature is enabled can be compared to evaluate the energy saving gains. It is expected that the energy consumption decreases after this feature is enabled.

7.9 RHUB Intelligent Shutdown

7.9.1 Principles

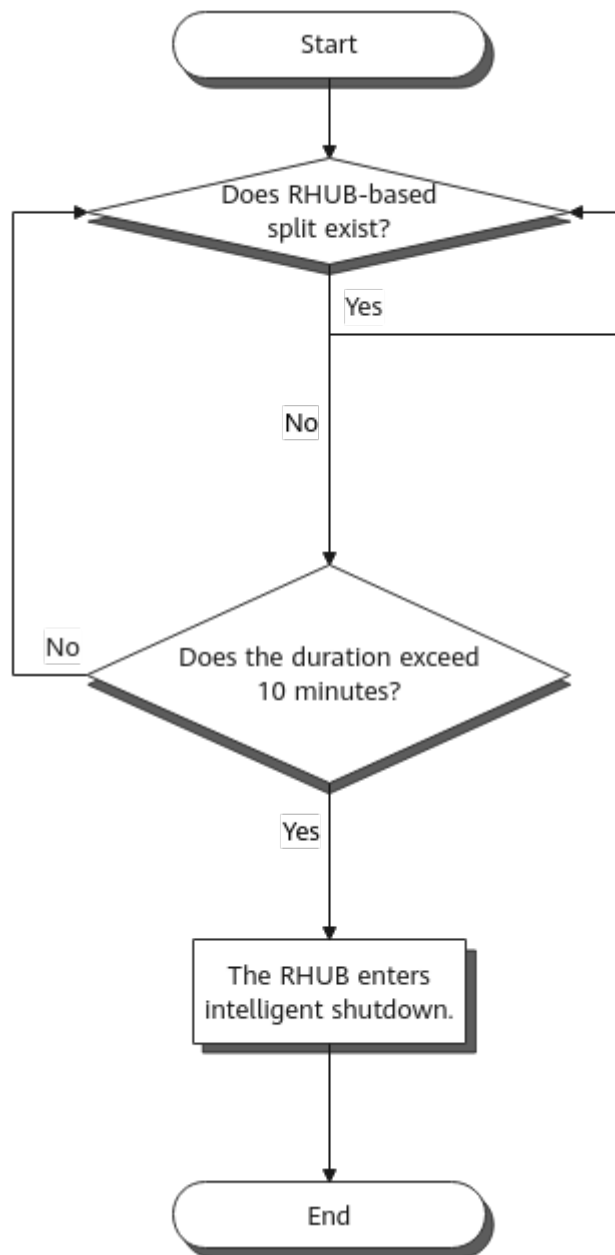
An RHUB contains two sets of independent data processing units. In scenarios without RHUB-based split, only one set is required to meet service requirements, making the other set redundant and causing energy waste. To address this, the RHUB intelligent shutdown function is introduced to reduce device energy consumption by disabling the redundant data processing unit. For more details on RHUB-based split, see *LampSite Pro Solution Description in 3900 & 5900 Series Base Station Product Documentation*.

This function is controlled by the **RHUB_SHUTDOWN_SW** option of the **EQUIPMENT.EQMDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter.

7.9.1.1 Entering RHUB Intelligent Shutdown

Figure 7-35 shows the RHUB intelligent shutdown procedure after this function is enabled.

Figure 7-35 RHUB intelligent shutdown procedure



1. The system checks whether RHUB-based split exists.
 - If so, the system proceeds to [2](#).
 - If not, the system continues to check for RHUB-based split.
2. The system checks whether the duration without RHUB-based split exceeds 10 minutes.
 - If so, the system proceeds to [3](#).
 - If not, the system continues to check for RHUB-based split.
3. The RHUB enters intelligent shutdown.

7.9.1.2 Exiting RHUB Intelligent Shutdown

When RHUB intelligent shutdown is in effect, if capacity expansion operations are performed on the RHUB and RHUB-based split is triggered accordingly, the expanded cells cannot be activated. To address this issue, the **RHUB_SHUTDOWN_SW** option of the **EQUIPMENT.EQMDOROPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter must be deselected to trigger an automatic RHUB reset in the preceding scenario. This enables the RHUB to exit shutdown and allows RHUB-based split to take effect. However, resetting an RHUB will interrupt services in all cells served by the RHUB.

7.9.2 Network Analysis

7.9.2.1 Benefits

It is recommended that RHUB intelligent shutdown be enabled in scenarios without RHUB-based split to reduce base station energy consumption.

7.9.2.2 Impacts

The impacts between this function and other functions are described in [7.9.3.2.3 Function Impacts](#).

When an RHUB is about to exit intelligent shutdown, it must be reset, which will interrupt services in all served cells.

7.9.3 Requirements

7.9.3.1 Licenses

None

7.9.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.9.3.2.1 Prerequisite Functions

None

7.9.3.2.2 Mutually Exclusive Functions

None

7.9.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
UMTS LTE FDD LTE TDD NR FDD Low- freque ncy NR TDD	RHUB hardware testing	Commands such as STR HWTST and STR HWOLTST	None	After an RHUB enters the shutdown state, commands such as STR HWTST and STR HWOLTST cannot be used to initiate any RHUB hardware status check.

7.9.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

BBUs

Only the BBU5900 and BBU5910 support this function.

Base Station Models

DBS5900 LampSite

Boards

No requirements

RF Modules

Only the RHUB5775 supports this function.

7.9.3.4 Networking

In separate-MPT scenarios, this function takes effect only when it is enabled on both main control boards.

7.9.3.5 Others

None

7.9.4 Operation and Maintenance

7.9.4.1 Data Configuration

7.9.4.1.1 Data Preparation

Table 7-88 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Equipment Dormancy Optional Feature Switch	EQUIPMENT.EQMDOROPTFEATSW (LTE eNodeB, 5G gNodeB)	Select the RHUB_SHUTDOWN_SW option.

7.9.4.1.2 Using MML Commands

Before using MML commands, refer to [7.9.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

```
//Turning on the RHUB intelligent shutdown switch
SET EQUIPMENT: EQMDOROPTFEATSW=RHUB_SHUTDOWN_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the RHUB intelligent shutdown switch
SET EQUIPMENT: EQMDOROPTFEATSW=RHUB_SHUTDOWN_SW-0;
```

7.9.4.2 Activation Verification

To check whether RHUB intelligent shutdown has taken effect, run the **DSP RHUB** command to query the running status of the RHUB.

- If the value of the **Shutdown Status** field in the command output is **ENABLE**, RHUB intelligent shutdown has taken effect.
//Querying the RHUB shutdown status
DSP RHUB;;
- If the value of the **Shutdown Status** field in the command output is **DISABLE**, the **Not Shutdown Cause** field will display the cause why the function does not take effect. [Table 7-89](#) lists the possible causes.

Table 7-89 Possible causes for RHUB intelligent shutdown not taking effect

Cause	Description
HARDWARE_NOT_SUPPORT	The RHUB hardware does not support the function.
CFG_CONDITION_NOT_MET	- The RHUB intelligent shutdown switch is turned off. - In separate-MPT scenarios, the RHUB intelligent shutdown switch is turned on only for some main control boards.
IN_EFFECT	Detection is ongoing, but the current duration does not meet the requirement for the duration without RHUB-based split to exceed 10 minutes.
ON_USE	RHUB-based split exists.
OTHERS	Other reasons, such as internal software issues, cause the function not to take effect.

7.9.4.3 Network Monitoring

On the live network, the one-week energy consumption data obtained before and after the energy saving function is enabled can be compared to evaluate the energy saving gains. It is expected that the energy consumption decreases significantly after this function is enabled. The specific data and observation methods are as follows:

Energy consumption of the BBU. Check the **VS.EnergyCons.BTSBoard (LTE eNodeB, 5G gNodeB)** counter.

NOTE

- There is delay in making decisions on whether to enter or exit the energy saving state. Therefore, the duration measured by the performance counters has an approximately 1-minute deviation from the actual duration.
- The sampling period is not synchronous with the reporting period, so a deviation of one sampling period may occur.

7.10 Dormancy Pending and Pending Cancellation

These functions work on the basis of RF module deep dormancy, RF module fast deep dormancy, RF module super dormancy, adaptive BBU deep dormancy, and pRRU deep dormancy.

If operations on the base station (such as upgrades) require all boards to remain active, dormancy pending can be enabled in the following situations:

- The **PND DORMANCY** command is executed with the **OPERATIONTYPE** parameter set to **PENDING**. Under this setting, dormancy pending is performed for a period specified by the **PENDINGDURATION** parameter. In separate-MPT scenarios, if the main control boards of two sides are

configured with different pending durations, the longer one takes effect for the base station.

- Dormancy is automatically pending for 30 minutes for the base station if the base station restarts; the GBTSTFunction, NodeBFunction, eNodeBFunction, or gNodeBFunction MML commands that affect cell setup are executed; or the Node MML commands (except the **MOD RRU** command with only parameters related to RF module dormancy specified) are executed.
- In separate-MPT co-RF-module scenarios, if dormancy is enabled on the local side and cell reestablishment is triggered (by reconfiguration, exit from the dormancy state, CPRI link intermittent disconnection, BBP restart, license update, or other reasons) on the peer side, dormancy will be automatically pending for 30 minutes on the local side.

Dormancy pending enabled in the preceding scenarios can be canceled by running the **PND DORMANCY** command with the **OPERATIONTYPE** parameter set to **UNPENDING**. In LNR separate-MPT scenarios, this command is used to cancel dormancy pending only on the local NE. The dormancy function of the local NE is also affected by the pending states of intra-base-station NEs.

During the specified pending duration, the base station does not enter the dormancy state. If the base station is in the dormancy state, it automatically exits this state. The pending duration can be queried using the **DSP BRDDORMANCY** command.

Pending State Query

The pending state of an NE or base station can be queried by running the **DSP BRDDORMANCY** command with the **QUERYTYPE** parameter set to **SITE**.

- If the value of the **NE Dormancy Pending Status** parameter is **PENDING** in the command output, dormancy pending is triggered for the NE.
- If the value of the **Site Dormancy Pending Status** parameter is **PENDING** in the command output, dormancy pending is triggered for an NE of the base station.

7.11 Reliability Protection for Deep Dormancy and Super Dormancy

These functions work on the basis of RF module deep dormancy, RF module super dormancy, adaptive BBU deep dormancy, and pRRU deep dormancy.

Adaptive BBU deep dormancy requires RF module dormancy. For details about the dependency requirements, see [7.7.3.2.1 Prerequisite Functions](#). Therefore, only the reliability protection for RF module dormancy is described below, and that for BBP dormancy is not detailed.

7.11.1 License Reliability

To improve the license key file (LKF) installation reliability, the base station is awoken and dormancy pending is triggered when the LKF is being installed during dormancy.

- If the LKF to be installed contains no error, exit from dormancy is triggered for the entire base station after the LKF is successfully installed.
- If the LKF to be installed contains a minor error, the LKF installation fails and exit from dormancy is triggered for the entire base station. After the exit is complete, the LKF can be installed again.
- If the LKF to be installed contains a major error, the LKF installation fails and exit from dormancy is triggered for the entire base station.

 NOTE

- In all scenarios, a 30-minute dormancy pending period starts after the base station exits dormancy.
- During an ongoing dormancy exit process, LKF installation operations will fail.
- For details about minor and major errors during LKF installation, see *License Management*.

If the base station lacks license items that could cause cell establishment failures after exiting dormancy, it will not enter dormancy until the license insufficiency is resolved.

If **ALM-26817 License on Trial** or **ALM-26815 Licensed Feature Entering Keep-Alive Period** is reported and the involved license item may cause a cell establishment failure after the exit from dormancy, the base station will not enter dormancy during the remaining grace period of less than 24 hours until the license insufficiency is resolved.

If license items that affect basic cell service capabilities (such as cell access capability) are insufficient, the base station will not enter dormancy.

For UMTS or GU load-based RF module deep/super dormancy:

- To improve RNC license reliability, if RNC license function control items are modified during dormancy, the base station is awoken and a 30-minute dormancy pending period starts.
 - If the cell-level command **ADD UCELLLICENSE**, **MOD UCELLLICENSE**, or **RMV UCELLLICENSE** is executed, the RNC instructs the NodeB to exit dormancy.
 - If the NodeB-level command **ADD UNODEBLICENSE**, **MOD UNODEBLICENSE**, or **RMV UNODEBLICENSE** is executed, the RNC instructs the NodeB to exit dormancy.
- To improve NodeB license reliability, the following reliability protection principles are applied:
 - If license items that affect basic cell service capabilities (such as CEs) are insufficient, the base station will not enter dormancy.
 - If the base station lacks license items that could cause the RNC cell to report alarms after exiting from dormancy, the base station will not enter dormancy.
 - If the base station is in the dormancy state and there are insufficient license items that affect basic cell service capabilities, the base station will exit dormancy.
 - If the base station is in the dormancy state and there are insufficient license items that may cause the RNC cell to report an alarm after exiting from dormancy, the base station will exit dormancy.

7.11.2 Configuration Reliability

NOTE

Configuration reliability protection during dormancy requires coordination between the OSS and NE. Therefore, both the OSS and NE must be upgraded to versions that support this function.

Delivering configuration data during the dormancy period may cause the following issues:

- The result of dynamic verification performed before configuration data is imported into the database is incorrect, and therefore the accuracy of configuration data cannot be ensured.
- The alarms generated when the configuration data fails to take effect cannot be reported in time.

To address these issues, configuration reliability protection during dormancy is introduced. With this function, the base station is awakened and dormancy pending is triggered before or during configuration delivery. In this case, dynamic verification can be performed before configuration data is imported into the database, and the alarms generated when the configurations fail to take effect can be reported in time. The details for different configuration scenarios are described as follows:

- A single MML command
 - If the base station receives a power-related command and an NE of the base station is in the dormancy state, exit from the dormancy state is triggered for the base station (with a forced 15-minute dormancy-pending period). After the exit process is complete, the base station executes the command, and the execution progress is reported. After the command is executed, a 30-minute common dormancy-pending period is automatically triggered for the base station.
 - If the base station receives a gNodeBFunction, eNodeBFunction, NodeBFunction, or GBTSTFunction MML command that affects cell setup, or Node MML command (except power-related commands), and an NE of the base station is in the dormancy state, the base station directly executes the command and starts to exit dormancy. After the command is executed, a 30-minute common dormancy-pending period is automatically triggered for the base station.
 - If the base station controller receives an RNC or BSC MML command that affects cell setup and an NE of the base station is in the dormancy state, the base station controller directly executes the command and instructs the base station to exit dormancy. After the command is executed, a 30-minute protection period is triggered. Dormancy will be allowed after the protection is canceled.

 NOTE

- The base station first exits dormancy and then executes a power-related command received during dormancy. If the command execution times out, the output of this command indicates that the command execution fails, displaying **The system is busy performing a time-consuming operation. Please try again later.** If a non-power-related command is delivered during this process, it needs to be queued. If the command execution times out, the output of this command indicates that the command execution fails, displaying **The system is busy performing a time-consuming operation. Please try again later.**
- The **PND DORMANCY** command can be used only to cancel common dormancy pending (triggered by the **PND DORMANCY** command or other causes). Forcible dormancy pending cannot be canceled using this command.
- If the base station is only in the forcible dormancy pending state and the **PND DORMANCY** command is executed with the **OPERATIONTYPE** parameter set to **UNPENDING** to cancel dormancy pending, the command output indicates a cancellation failure, displaying **The system is forced to suspend dormancy, and the suspension cannot be canceled by running this command.**
- If the base station is in both forcible dormancy pending and common dormancy pending states and the **PND DORMANCY** command is executed with the **OPERATIONTYPE** parameter set to **UNPENDING** to cancel dormancy pending, the command output indicates a cancellation failure, displaying **The common suspension of dormancy has been canceled successfully. The forcible suspension of dormancy cannot be canceled by running this command.**
- An XML configuration file
If the base station receives an XML script file, it checks whether the script file contains a gNodeBFunction, eNodeBFunction, NodeBFunction, or GBTSFunction MML command that affects cell setup, or a Node MML command (except the **MOD RRU** command with only parameters related to RF module dormancy specified).
 - If it does not, the base station directly executes the commands in the XML script file.
 - If it does, the base station first exits dormancy (with a forced 15-minute dormancy period) and then executes the commands in the XML script file. After the commands are executed, a 30-minute common dormancy pending period is automatically triggered for the base station.
- Batch processing (batch scripts, batch configurations, or incremental configurations)
The MAE-Access or MAE-Deployment checks whether the batch scripts, batch configurations, or incremental configurations contain gNodeBFunction, eNodeBFunction, NodeBFunction, or GBTSFunction MML commands that affect cell setup or Node MML commands (for the MAE-Deployment, except the **MOD RRU** command with only parameters related to RF module dormancy specified and the **MOD BBP** command with only parameters related to BBP dormancy specified).
 - If they do not, the batch scripts, batch configurations, or incremental configurations are directly executed.
 - If they do, the MAE-Access or MAE-Deployment wakes up the base stations in advance. After the base stations exit dormancy and enter forcible dormancy pending (which lasts eight hours), the MAE-Access or MAE-Deployment performs batch processing. A 30-minute common

dormancy pending period is automatically triggered when the base station executes a gNodeBFunction, eNodeBFunction, NodeBFunction, or GBTSFunction MML command that affects cell setup or all Node MML commands.

After batch processing is complete, the MAE-Access or MAE-Deployment instructs the base station to cancel forcible dormancy pending.

If the MAE-Access or MAE-Deployment batch scripts, batch configurations, or incremental configurations contain an RNC or BSC MML command that affects cell setup and an NE of the base station is in the dormancy state, the MAE-Access or MAE-Deployment directly executes the command and instructs the base station to exit dormancy. After the command is executed, a 30-minute protection period is triggered. Dormancy will be allowed after the protection is canceled.

7.11.3 Reliability in Fault Scenarios

If the base station has a module fault or service fault that could cause cell unavailability or fault transfer after a module enters dormancy, the following risks may occur. Therefore, RF modules that have reported any of the following alarms will not enter dormancy and will exit dormancy when the fault occurs.

- The cell cannot be set up when the cell exits the dormancy state. For example, ALM-29240 Cell Unavailable generated due to incorrect bandwidth configuration.
- The alarm is cleared and then reported again. For example, ALM-26247 Configuration Failure caused by RF module equipment alarms or PRACH root sequence conflicts.
- Service unavailability alarms cannot be reported in a timely manner during the dormancy period. For example, ALM-29213 eNodeB S1 Control Plane Transmission Interruption.

Faults that affect the dormancy reliability include device faults, configuration conflicts, and resource conflicts.

- When a device fault occurs on a module, this module does not enter dormancy.
- When a service fault occurs, all modules of the base station do not enter dormancy. For a separate-MPT multimode base station, neither the local nor peer RAT enters dormancy.

The preceding alarms should be cleared for the dormancy function to take effect.

 NOTE

- The base station determines whether to enter dormancy based on the actual faults. When a base station is faulty, alarms may not be reported due to alarm filtering, the alarm correlation mechanism, or alarm masking. For details about the alarm reporting mechanism, see *Fault Management*. For example, **ALM-26238 RRU Network Topology Type and Configuration Mismatch** will prevent the RF modules on the chain/ring from entering dormancy. If its root alarms (such as **ALM-26231 BBU CPRI Optical Module or Electrical Port Not Ready** or **ALM-26311 Inter-BBU Optical Module Not in Position**) are generated on the base station, **ALM-26238 RRU Network Topology Type and Configuration Mismatch** is suppressed by these root alarms and therefore is not displayed on the alarm console. However, its impact on dormancy still remains.
- For UMTS, the base station controller and base station need to synchronize alarm information. If any of the following unstable scenarios occurs, the RF modules enter dormancy. After the modules exit dormancy, alarms may be generated.
 - Alarms are generated during the dormancy entry process.
 - The dormancy conditions are met within 2 minutes after RF module dormancy is enabled.
 - The base station controller is upgraded or reset.

7.11.4 O&M Reliability

Software management operations will automatically trigger dormancy pending. If the system is in the dormancy state when pending is triggered, the system automatically exits dormancy and will not enter the dormancy state again within the pending period. If the system has not entered the dormancy state when pending is triggered, the system will not enter the dormancy state within the pending period. The pending duration varies according to software management actions, and has three levels: 24-hour pending for software download, 12-hour pending for activation and rollback, and 1-hour pending for actions such as software supplementation, hot patch-related operations, data reconfigurations, and changes in loading control rights.

Cell/User tracing and monitoring: If module dormancy takes effect during tracing and monitoring, no data is reported for the traced object. After an exit from the dormancy state, tracing data reporting automatically recovers.

Device tracing and monitoring:

- During the dormancy period, device tracing and monitoring tasks cannot be started. In special scenarios, such as for FFT scanning, dormancy is exited instead.
- If module dormancy takes effect after a device tracing and monitoring task is started, the task is automatically terminated.

Other O&M:

- If O&M commands (including test commands) are started before the dormancy state is entered, dormancy is prohibited during the command execution. The protection will be canceled within 30 minutes after the command execution is completed. Dormancy will be allowed after the protection is canceled.
- If O&M commands (including test commands) are executed when dormancy is in effect, one of the following situations may occur:
 - The command execution fails, and the dormancy state is exited, with the message **Power saving has taken effect and is now being stopped for**

the test object. Run this command 10 minutes later displayed. Such commands include **STR VSWRTEST**.

- The command execution fails, and the dormancy state is not exited. Such commands include **STR HWOLTST**.
- The command execution succeeds, and the dormancy state is not exited. Such commands include **EXP DEVFILE**.

7.12 Reliability Protection During Fast Deep Dormancy

7.12.1 License Reliability

- If the LKF to be installed contains no error, exit from dormancy is triggered for the entire base station after the LKF is successfully installed.
- If the LKF to be installed contains a minor error, the LKF installation fails and exit from dormancy is triggered for the entire base station. After the exit is complete, the LKF can be installed again.
- If the LKF to be installed contains a major error, the LKF installation fails and exit from dormancy is triggered for the entire base station.

NOTE

- In all scenarios, a 30-minute dormancy pending period starts after the base station exits dormancy.
- During an ongoing dormancy exit process, LKF installation operations will fail.
- For details about minor and major errors during LKF installation, see *License Management*.

7.12.2 Configuration Reliability

NOTE

Configuration reliability protection during dormancy requires coordination between the OSS and NE. Therefore, both the OSS and NE must be upgraded to versions that support this function.

Delivering configuration data during the dormancy period may cause the following issues:

- The result of dynamic verification performed before configuration data is imported into the database is incorrect, and therefore the accuracy of configuration data cannot be ensured.
- The alarms generated when the configuration data fails to take effect cannot be reported in time.

To address these issues, configuration reliability protection during dormancy is introduced. With this function, the base station is awakened and dormancy pending is triggered before or during configuration delivery. In this case, dynamic verification can be performed before configuration data is imported into the database, and the alarms generated when the configurations fail to take effect can be reported in time. The details for different configuration scenarios are described as follows:

- A single MML command
 - If the base station receives a power-related command and an NE of the base station is in the dormancy state, exit from the dormancy state is triggered for the base station (with a forced 15-minute dormancy-pending period). After the exit process is complete, the base station executes the command, and the execution progress is reported. After the command is executed, a 30-minute common dormancy-pending period is automatically triggered for the base station.
 - If the base station receives a gNodeBFunction or eNodeBFunction that affects cell setup, or a Node MML command (except power-related commands), and an NE of the base station is in the dormancy state, the base station directly executes the command and starts to exit dormancy. After the command is executed, a 30-minute common dormancy-pending period is automatically triggered for the base station.
- An XML configuration file

If the base station receives an XML script file, it checks whether the script file contains a gNodeBFunction or eNodeBFunction MML command that affects cell setup, or a Node MML command (except the **MOD RRU** command with only parameters related to RF module dormancy specified).

 - If it does not, the base station directly executes the commands in the XML script file.
 - If it does, the base station first exits dormancy (with a forced 15-minute dormancy period) and then executes the commands in the XML script file. After the commands are executed, a 30-minute common dormancy pending period is automatically triggered for the base station.
- Batch processing (batch scripts, batch configurations, or incremental configurations)

The MAE-Access or MAE-Deployment checks whether the batch scripts, batch configurations, or incremental configurations contain gNodeBFunction or eNodeBFunction MML commands that affect cell setup or Node MML commands (for the MAE-Deployment, except the **MOD RRU** command with only parameters related to RF module dormancy specified and the **MOD BBP** command with only parameters related to BBP dormancy specified).

 - If they do not, the batch scripts, batch configurations, or incremental configurations are directly executed.
 - If they do, the MAE-Access or MAE-Deployment wakes up the base stations in advance. After the base stations exit dormancy and enter forcible dormancy pending (which lasts eight hours), the MAE-Access or MAE-Deployment performs batch processing. A 30-minute common dormancy pending period is automatically triggered when the base station executes a gNodeBFunction or eNodeBFunction MML command that affects cell setup or all Node MML commands.

After batch processing is complete, the MAE-Access or MAE-Deployment instructs the base station to cancel forcible dormancy pending.

7.12.3 O&M Reliability

Software management operations automatically trigger dormancy pending. If the system is already in the dormancy state when pending is triggered, it automatically exits dormancy and will not re-enter dormancy during the pending

period. If the system is not in the dormancy state when pending is triggered, it will not enter dormancy during the pending period. The pending duration varies depending on the software management action and has three levels: 24-hour pending for software download, 12-hour pending for activation and rollback, and 1-hour pending for operations such as software supplementation, hot patch-related operations, data reconfigurations, and changes in loading control rights.

Cell/User tracing and monitoring: If module dormancy takes effect during tracing and monitoring, no data is reported for the traced object. After the module exits dormancy, tracing-data reporting automatically recovers.

Device tracing and monitoring:

- During the dormancy period, device tracing and monitoring tasks cannot be started. In special scenarios, such as for FFT scanning, dormancy is exited instead.
- If module dormancy takes effect after a device tracing and monitoring task is started, the task is automatically terminated.

Other O&M:

- If O&M commands (including test commands) are started before the dormancy state is entered, dormancy is prohibited during command execution. The protection will be canceled within 30 minutes after the command execution is completed, after which dormancy is allowed again.
- If O&M commands (including test commands) are executed when dormancy is in effect, one of the following situations may occur:
 - The command execution fails, and the dormancy state is exited, with the message **Power saving has taken effect and is now being stopped for the test object. Run this command 10 minutes later** displayed. Such commands include **STR VSWRTEST**.
 - The command execution fails, and the dormancy state is not exited. Such commands include **STR HWOLTST**.
 - The command execution succeeds, and the dormancy state is not exited. Such commands include **EXP DEVFILE**.

8 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

9 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

10 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

11 Reference Documents

- *Power Control in eRAN Feature Documentation*
- *MIMO in eRAN Feature Documentation*
- *Energy Conservation and Emission Reduction in eRAN Feature Documentation*
- *ETWS in eRAN Feature Documentation*
- *CMAS in eRAN Feature Documentation*
- *Dynamic Power Sharing Between LTE Carriers in eRAN Feature Documentation*
- *Multi-carrier Unified Scheduling in eRAN Feature Documentation*
- *Carrier Aggregation in eRAN Feature Documentation*
- *Massive MIMO Basics (FDD) in eRAN Feature Documentation*
- *eMTC in eRAN Feature Documentation*
- *Smart 8T8R (FDD) in eRAN Feature Documentation*
- *URLLC in 5G RAN Feature Documentation*
- *CMAS in 5G RAN Feature Documentation*
- *Synchronization in 5G RAN Feature Documentation*
- *Fault Management*
- *Green Site*
- *EPC-based NSA Basics*
- *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*
- *LTE FDD and NR Dynamic Spectrum Sharing*
- *Virtual Grid-based Multi-Frequency Coordination in 5G RAN Feature Documentation*
- *ANR in 5G RAN Feature Documentation*
- *SA Mobility Management in Connected Mode in 5G RAN Feature Documentation*
- *Energy Conservation and Emission Reduction in 5G RAN Feature Documentation*
- *Remote Interference Management (Low-Frequency TDD) in 5G RAN Feature Documentation*

- *CoMP in 5G RAN Feature Documentation*
- *Beam Management in 5G RAN Feature Documentation*
- *Smart 8T8R (FDD) in 5G RAN Feature Documentation*
- *RedCap in 5G RAN Feature Documentation*
- *Power Sharing in RAN Feature Documentation*
- *UMTS and LTE Dynamic Power Sharing in RAN Feature Documentation*
- *Green BTS in RAN Feature Documentation*
- *LTE and NR Power Allocation*
- *NR Inter-Carrier Dynamic Power Allocation in 5G RAN Feature Documentation*
- *License Control Item Lists (FDD), License Control Item Lists (CIoT), and License Control Item Lists (TDD) in eRAN Feature Documentation*
- *License Control Item Lists in 5G RAN Feature Documentation*
- *Dynamic BCCH TRX Timeslot Shutdown in GBSS Feature Documentation*
- *Distributed Antenna Solutions (Low-Frequency TDD) in 5G RAN Feature Documentation*
- *Interference Detection and Suppression in eRAN Feature Documentation*
- *MAE-Optimization in MAE Product Documentation (EulerOS, TaiShan)*
- *LTE FDD and NR Hybrid Dynamic Spectrum Sharing*
- *Massive MIMO Enhancements (FDD) in eRAN Feature Documentation*
- *Co-MM (Low-Frequency TDD) in 5G RAN Feature Documentation*
- *ANR Management in eRAN Feature Documentation*
- *WBB in eRAN Feature Documentation*
- *LTE and NR Power Allocation in eRAN Feature Documentation*
- *Multi-Frequency Smart Aggregation in 5G RAN Feature Documentation*
- *Downlink Scheduling in 5G RAN Feature Documentation*
- *Interference Avoidance in 5G RAN Feature Documentation*
- *Massive MIMO Basics (TDD) in eRAN Feature Documentation*
- *Hyper Cell in 5G RAN Feature Documentation*
- *Cell Combination in 5G RAN Feature Documentation*
- *Super Uplink in 5G RAN Feature Documentation*
- *SA Mobility Management in Connected Mode in 5G RAN Feature Documentation*
- *CA Experience Improvement in 5G RAN Feature Documentation*
- *LTE FDD and NR Inter-RAT Unified Scheduling*
- *NB-IoT Basics (FDD) in eRAN Feature Documentation*
- *RuralLink Solution*
- *LumiLink Solution*
- *X2 Self-Management in SA Networking*
- *Intelligent Detection for Fronthaul Optical Paths*
- *LTE and NR Zero Bufferzone*
- *UMTS and LTE Spectrum Sharing Based on DC-HSDPA*

- *Base Station Equipment Reliability*
- *Common Transmission*
- *Cell Switch-off Based on Automatic Co-coverage Identification*
- *RuralCow Solution*
- *NSA Mobility Management*
- *UMTS and LTE Spectrum Sharing*
- *UMTS and LTE Zero Bufferzone*
- *GSM and LTE Buffer Zone Optimization*
- *GSM and LTE Spectrum Concurrency*
- *Multi-Carrier HSDPA in RAN Feature Documentation*
- *HSDPA Scheduler Pool in RAN Feature Documentation*
- *HSUPA Scheduler Pool in RAN Feature Documentation*
- *Inter-Dependence of BBU Uplink Resource in RAN Feature Documentation*
- *Adaptive Interference Rejection in RAN Feature Documentation*
- *Sleeping Cell Detection and Recovery in RAN Feature Documentation*
- *Inter-NodeB Uplink CoMP in RAN Feature Documentation*
- *Hitless Cell Rehomming in RAN Feature Documentation*
- *Coverage Expansion Under Interference in RAN Feature Documentation*
- *Machine Learning based PCPICH Power Optimization in RAN Feature Documentation*
- *Call Admission Control in RAN Feature Documentation*
- *Uplink CoMP in RAN Feature Documentation*
- *CBS in RAN Feature Documentation*
- *License Management in RAN Feature Documentation*
- *OCNS in RAN Feature Documentation*
- *Frequency Hopping in GBSS Feature Documentation*
- *Soft-Synchronized Network in GBSS Feature Documentation*
- *Cell Frequency Scan in GBSS Feature Documentation*
- *Cell Broadcast in GBSS Feature Documentation*
- *Multi-site Cell in GBSS Feature Documentation*
- *Multi-Sector BCCH Sharing in GBSS Feature Documentation*
- *Dynamic Cell Power Off in GBSS Feature Documentation*
- *Operation and Maintenance in GBSS Feature Documentation*
- *BSC6960 GSM Initial Configuration Guide in BSC6960 GSM Product Documentation*
- *BSC6960 UMTS Initial Configuration Guide in BSC6960 UMTS Product Documentation*
- *BSC6910 GU Initial Configuration Guide in BSC6910 GU Product Documentation*
- *LampSite Pro Solution Description in 3900 & 5900 Series Base Station Product Documentation*